

AOS-CX 10.06 Command-Line Interface Guide

6100 Switch Series



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This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

Aruba 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in **Support and other resources**.

CLI access

Access the CLI through the following interfaces:

Console port

Connect the management port on the switch to your computer using a serial cable and then use terminal emulation software to reach the switch from the computer. Typically, the console port is used when first installing the switch and performing initial configuration tasks.

Management port (out-of-band connection)

Connect the management port on the switch to your network, and then use SSH client software to reach the switch from a computer connected to the same network. This requires that a DHCP server is installed on the network.

In the switch factory default state, the management port and SSH on the management VRF (`mgmt`) are enabled.

Data port (in-band connection)

Connect a data port on the switch to your network, and then use SSH client software to reach the switch from a computer connected to the same network. Management traffic ingresses and egresses switch data ports with rest of the traffic on the network, therefore it can be affected by traffic congestion and other issues impacting the network.

Getting CLI help

Procedure

- To show the available commands that you can execute in the current command context, enter the `?` symbol.

For example:

```
switch# ?  
  boot          Reboot all or part of the system  
  checkpoint     Checkpoint information  
switch#
```

The `?` symbol does not display on the screen when you enter it.

The commands that are available to you depend on your authority and the command context. In a given command context, you can only list and execute the commands available in that context.

- To show the available parameters for a command, enter the command followed by a space and then enter the `?` symbol.

For example:

```
switch(config)# access-list ?  
  all          All access-lists  
  ip           Internet Protocol v4 (IPv4)
```

```
ipv6      Internet Protocol v6 (IPv6)
log-timer Set ACL log timer length (frequency)
mac       Ethernet MAC Protocol
switch(config)# access-list
```

After the CLI displays the information, it automatically displays the text you entered before you entered the ? symbol.

If there is no <cr> symbol at the end the command help output, the command is not complete as displayed. You must specify one of the listed parameters.

The <cr> symbol alone in the command help output indicates that there are no additional parameters and that you must press the enter key to complete the command. For example:

```
switch# list ?
<cr>
switch# list
```

The <cr> symbol at the end of the command help output indicates that the parameters preceding the <cr> are optional and you can enter the command as is displayed. For example:

```
switch# configure ?
terminal    Configuration terminal (default)
<cr>
switch# configure
```

- To show information about a parameter for a command, enter the command and parameter followed by a space, then enter the ? symbol.

For example:

```
switch(config)# access-list log-timer ?
<30-300>    Specify value (in seconds)
default     Default value (300 seconds)
switch(config)# access-list log-timer
```

Authority levels

In command descriptions, the authority level indicates the user role that is required to execute a command:

Administrators

Users with the role: administrators

Users with administrator rights can execute any command.

Operators

Users with the role: operators.

Users with operator rights can execute commands in the operator context (>) only.

Auditors

Users with the role: auditors.

Users with auditor rights can execute commands in the auditor context (auditor>) only.

Local user group members with execution rights for a command

You can create up to 29 user-defined local user groups on the switch. Each group can be defined to allow execution of up to 1024 specific CLI commands.

Command contexts

The command context determines the following:

- Which parts of the switch can be managed
- Which commands are available to users with the appropriate authority

Command contexts have a parent-child tree structure in which contexts might themselves contain nested contexts.

Operator context (>)

The operator context enables you to execute commands to view—but not change—the configuration.

The operator context requires the least user privilege to execute commands.

In command descriptions, this context is listed as: Operator (>)

Switch prompt example

```
switch>
```

Authority

Operators or Administrators

Showing the available commands in this context

At the command prompt, enter the ? symbol.

Navigating to the operator context (>)

To navigate to the operator command context (>), do one of the following:

Procedure

- Log in to the switch CLI with a user ID that has the `operator-group` role.
- From the manager context (#), enter the `disable` command.

Auditor context

When you log in to the switch as user with auditor rights, you have access to the auditor command context only.

Users with auditor rights have access to a limited set of commands. For more information about auditors, see the *Security Guide* for your switch and software version.

Switch prompt example

```
auditor>
```

Showing the available commands in this context

At the command prompt, enter the ? symbol.

Manager context (#)

From the manager context (#), you can execute commands that do not require saving changes to the configuration.

In command descriptions, this context is listed as: Manager (#)

Switch prompt example

```
switch#
```

Authority

Administrators or local user group members with execution rights for this command.

Showing the available commands in this context

At the command prompt, enter the `?` symbol.

Access to manager context commands from descendant contexts

The `do` command enables you to access commands from the manager context while you are in a child or descendent context, such as `config` or `config-if`.

For example, to execute the `clear` command from the `config` context, enter the following: `do clear`

The `show` command can be executed from configuration contexts as well as the manager context, so using the `do` command with the `show` command is deprecated. Support for `do show` might be discontinued in a future software release.

Navigating to the manager context (#)

To navigate to the manager command context (#), do one of the following:

Procedure

- Log in to the switch CLI with a user ID that has the `administrators` role.
- From the operator context (>), enter the `enable` command.

You must have administrator authority to enter the `enable` command.

```
switch> enable
switch#
```

- From the configuration context (`config`), enter either the `exit` or the `end` command.

For example:

```
switch(config)# exit
switch#
```

- From any child or descendent context, enter the `end` command.

For example:

```
switch(config-vlan-100)# end
switch#
```

Global configuration context (config)

From the global configuration context (`config`), you can execute commands that change the configuration of the switch.

In command descriptions, this context is listed as: `config`

Switch prompt example

```
switch(config)#
```

Authority

Administrators or local user group members with execution rights for this command.

Showing the available commands in this context

At the command prompt, enter the **?** symbol.



TIP: You can use the `do` command to execute some manager context commands—such as the `clear` command—from the global configuration context.

Navigating to the `config` context

To navigate to the `config` command context, do one of the following:

Procedure

- From the manager context (`#`), enter the `configure terminal` command:

```
switch# configure terminal  
switch(config)#
```

- From a child configuration context, enter the `exit` command.

For example:

```
switch(config-vlan-100)# exit  
switch(config)#
```

Other configuration command contexts

All other configuration command contexts are descendants of the global configuration command context (`config`).

From these command contexts, you can execute commands that apply to that specific context, such as an interface or a VLAN.

Switch prompt examples

- `switch(config-if)#`
- `switch(config-router)#`
- `switch(config-vlan-100)#`

Authority

Administrators or local user group members with execution rights for this command.

Showing the available commands in this context

At the command prompt, enter the **?** symbol.

Support for range contexts

Some switch features enable you to use a single command to apply configuration settings to multiple items. You specify the multiple items by creating a type of command context called a `range context`. Then you can execute commands that are applied to every item in the range. For example:

```
switch(config)# interface 1/1/1-1/1/5  
switch(config-if-<1/1/1-1/1/5>)# no shutdown
```

You can use a range context to specify multiple items for the following:

Physical interfaces

- Command example: `interface 1/1/1-1/1/8,1/1/10,1/1/12`
- Switch prompt example: `switch(config-if-<1/1/1-1/1/8,1/1/10,1/1/12>)#`

LAG interfaces

- Command example: `interface lag 1-10`
- Switch prompt example: `switch(config-if-lag-<1-10>)#`

Loopback interfaces

- Command example: `interface loopback 1-10`
- Switch prompt example: `switch(config-if-loopback-<1-10>)#`

VLAN interfaces

- Command example: `interface vlan 1,2,3-6`
- Switch prompt example: `switch(config-vlan-if-<1,2,3-6>)#`

VLANs

- Command example: `vlan 1-10,15,20-25`
- Switch prompt example: `switch(config-vlan-<1-10,15,20-25>)#`

Commands entered in a range context are applied to each item in the range individually:

- Each item in the range has its own entry in the output of `show running-config` commands. For example, you can configure a range of interfaces as follows:

```
switch(config)# interface 1/1/1-1/1/5  
switch(config-if-<1/1/1-1/1/5>)# no shutdown
```

In the output for the `show running-config` command, the interfaces are displayed individually:

```
switch(config-if-<1/1/1-1/1/5>)# show running-config  
Current configuration:  
...  
interface 1/1/1  
    no shutdown  
interface 1/1/2  
    no shutdown  
interface 1/1/3  
    no shutdown  
interface 1/1/4  
    no shutdown
```

```
interface 1/1/5
    no shutdown
...
switch(config-if-<1/1/1-1/1/5>) #
```

- If you specify a range context for interfaces, you cannot execute commands that create a context within the range context. For example, you cannot execute the `vrrp` command from an interface range context, even though you can execute the command from the `config-if` context for a single interface.
- If error is encountered during the execution of a command for an item in the range, the error message returned includes a prefix that identifies the item to which the error applies. However command execution does not stop until the command is attempted on all the items in the range.
- The range context is created only if every item in the range is successfully created or already exists in configuration. If an error occurs during the creation of an item in a range, the items that are created successfully are added to the configuration, but the range context is not created. The switch prompt.

For example, in the following sequence:

1. VLANs 1 through 100 are created successfully, so the switch prompt reflects the range of VLANs:
`switch(config-vlan-<1-100>) #`
2. The command `interface vlan 95-105` fails for VLANs 101 through 105, so the range context is not created and the switch prompt remains in the global configuration context: `switch(config) #`
3. The configuration includes all the VLANs and VLAN interfaces that are created successfully.

```
switch(config) # vlan 1-100
switch(config-vlan-<1-100>) # exit
switch(config) # interface vlan 95-105
VLAN 101 should be created before creating interface VLAN101.
VLAN 102 should be created before creating interface VLAN102.
VLAN 103 should be created before creating interface VLAN103.
VLAN 104 should be created before creating interface VLAN104.
VLAN 105 should be created before creating interface VLAN105.
switch(config) # show running-config
Current configuration:
...
vlan 1-100
interface vlan95
interface vlan96
interface vlan97
interface vlan98
interface vlan99
interface vlan100
...
switch(config) #
```

- If the `no` form of the command can be used to remove an item from the configuration, you can use a range context with the `no` form of the command to remove multiple items from the configuration.

For example, you can remove VLANs 95 through 100 from the configuration by entering: `no vlan 95-100`

Rules for range contexts

- For interfaces that use the *member/slot/port* notation, items in the range must be specified in ascending order.
- Contiguous items in the range are represented by the smallest and largest values separated by a hyphen.

For example:

- Command: `interface 1/1/1-1/1/8`
Switch prompt: `switch(config-if-<1/1/1-1/1/8>) #`
 - Command: `vlan 1-10`
Switch prompt: `switch(config-vlan-<1-10>) #`
- Noncontiguous items in the range must be separated by commas.

For example:

- Command: `interface 1/1/1-1/1/8,1/1/10,1/1/12`
Switch prompt: `switch(config-if-<1/1/1-1/1/8,1/1/10,1/1/12>) #`
 - Command: `vlan 1-10,15,20-25`
Switch prompt: `switch(config-vlan-<1-10,15,20-25>) #`
- The switch prompt is truncated to 50 characters.

Command history

You can use the **up arrow** key or **Ctrl+P** to display the previous command in the session history, if any.

You can use the **down arrow** key or **Ctrl+N** to display the next command in the session history, if any.

You can use the `show history` command to show a numbered list of the commands executed during this session. You use the command numbers to specify commands to repeat using the `repeat` command. The `show history` and `repeat` commands are not saved in the history buffer.

The commands saved in the history command buffer are in the same format in which you entered the commands. If you enter an incomplete command, the command saved in the history command buffer is also an incomplete one.

If you execute the same command repeatedly, the switch saves only the earliest record. However, if you execute the same command in different formats, the switch saves them as different commands.

For example, if you execute the `show startup-config` command repeatedly, the system saves only one command in the history command buffer. If you execute the command in the format of `show start` and `show startup` respectively, the system saves them as two commands.

Command completion

The CLI supports both command abbreviation and command completion:

- If you enter enough letters to match a valid command, the CLI accepts the command.
For example, you can enter `con` instead of `configure` to navigate from the manager context to the global configuration context.

```
switch# con
switch(config) #
```

- If you enter part of a command word and then press the **Tab** key, one of the following occurs:

- If you have entered enough letters to match a valid command, the CLI displays the remainder of the word.
 - If you have not entered enough letters to match a valid command, the CLI does not complete the command.
- If you press the **Tab** key a second time, the CLI displays commands that match the letters you entered. For example:

```
switch(config)# c1
class clear clock
switch(config)# cl
```

- If you press the **Tab** key twice after a completed word, the CLI displays the command options. For example, if you enter the word `clock` followed by a space and then press the **Tab** key twice, the CLI displays the commands available in that command context that start with that word, and then displays the prompt—including the characters you entered—enabling you to complete the command without retyping.

```
switch(config)# clock
date          datetime      time    timezone
switch(config)# clock
```

Pipe (|) support in show commands

The pipe (|) command is a CLI session command that filters the output of show `show` commands according to the criteria specified by the parameter `include`, `exclude`, `count`, `begin`, or `redirect`.

- The pipe (|) command is supported for use with the `show` command only.
- You can use multiple pipe commands with a single `show` command.
For example: `show running-config | include "vlan" | exclude "vlan2" | count`
- You can use the pipe command with the `page` command.
- Command completion by pressing the **Tab** key is not supported for pipe commands.

Command syntax notation conventions

Convention	Usage
example-text	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.

Table Continued

Convention	Usage
Any of the following: <code><example-text></code> <code><example-text></code> <i>example-text</i> <i>example-text</i>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: - For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. - For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or	Ellipsis:
...	- In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. - In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment. The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the `interface` context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where `<VLAN-ID>` is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

```
member/slot/port
```

On the 6100 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface `1/1/4` in software is associated with physical port 4 on the switch.

boot

Syntax

boot

Description

Presents you with the boot menu prompt. You can then specify which boot profile: primary, secondary, or Service OS console.

Command context

Service OS (svos>)

Authority

Administrators or local user group members with execution rights for this command.

Example

Presenting the boot menu prompt:

```
SVOS> boot

ServiceOS Information:
  Version:      PL.01.07.0004-internal
  Build Date:   2020-11-23 18:07:42 PST
  Build ID:     ServiceOS:PL.01.07.0004-internal:133137f635df:202011231807
  SHA:         133137f635dff5778bf3e109eb75825b68d64789

Boot Profiles:

0. Service OS Console
1. Primary Software Image [PL.10.06.0001]
2. Secondary Software Image [PL.10.08.0000-168-g3089099c34e6]

Select profile(primary):
```

cat

Syntax

cat <FILENAME/DIRECTORY-NAME>

Description

Prints the contents of a file to the console. The Service OS does not allow command output redirection, so this command is only useful for reading short text files.

Command context

Service OS (SVOS>)

Parameters

<FILENAME/DIRECTORY-NAME>

Shows the contents of the specified file or directory.

Authority

Administrators or local user group members with execution rights for this command.

Example

Showing the contents of /nos/hosts:

```
SVOS> cat /nos/hosts
127.0.0.1          localhost.localdomain    localhost
SVOS>
```

cd path

Syntax

cd path

Description

Changes the current working directory.

Command context

Service OS (SVOS>)

Authority

Administrators or local user group members with execution rights for this command.

Example

Changing the current working directory:

```
cd /
```

config-clear

Syntax

config-clear

Description

Configures the switch to set all configuration settings to factory default when the switch is restarted. The next time the switch starts, the current `startup-config` is renamed to `startup-config-fixme`, and a new `startup-config` is created with factory default settings.



NOTE: Using this command is not the same as performing zeroization, which securely erases the entire primary storage and other devices, and not just the configuration.

Command context

Service OS (SVOS>)

Authority

Administrators or local user group members with execution rights for this command.

Example

Configuring the system to clear the switch configuration:

```
SVOS> config-clear
```

The switch configuration will be cleared.

Continue (y/n)? **y**

The system has been configured to clear the startup-config on the next boot. Please execute the 'boot' command to complete this action.

```
SVOS>
```

cp

Syntax

```
cp [options] <SOURCE-FILENAME/SOURCE-DIRECTORY> <DESTINATION-FILENAME/DESTINATION-DIRECTORY>
```

Description

Copies files or directories.

Command context

Service OS (SVOS>)

Parameters

[options]

Selects the options for the command.

-d, -P

Specifies the preservation of symlinks (default if **-R**).

-a

Same as **-dpR**.

R, -r

Specifies recursiveness, all files, and subdirectories are copied.

-L

Specifies the following of all symlinks.

-H

Specifies the following of symlinks on command line.

-p

Specifies the preservation of file attributes if possible.

-f

Specifies the overwriting of a file or directory.

-i

Specifies the prompting before an overwrite.

-l, -s

Specifies the creation of (sym) links.

<SOURCE-FILENAME/SOURCE-DIRECTORY>

Specifies the name of the source file or directory.

<DESTINATION-FILENAME/DESTINATION-DIRECTORY>

Specifies the name of the destination file or directory.

Authority

Administrators or local user group members with execution rights for this command.

Example

Copying /home/customers directory to the /home/clients directory:

```
SVOS> cp /home/customers /home/clients
```

du

Syntax

```
du [options] <FILENAME/DIRECTORY-NAME>...
```

Description

Shows estimated disk space used for each file or directory or both.

Command context

Service OS (SVOS>)

Parameters

[options]

Selects the options for the command.

-a

Show file sizes.

-L

Shows all symlinks.

-H

Shows symlinks on a command line.

-d, N

Shows limited output to directories (and files with -a) of depth less than N.

-c

Shows the total disk space usage of all files or directories or both.

-l

Shows the count sizes if hard linked.

-s

Shows only a total for each argument.

-x

Does not show directories on different file systems.

-h

Show sizes in human readable format (1K, 243M, and 2G).

-m

Show sizes in megabytes.

-k

Show sizes in kilobytes (default).

<FILENAME/DIRECTORY-NAME>

Specifies the file or directory or both for displaying a size estimate.

Authority

Administrators or local user group members with execution rights for this command.

Example

Estimating disk space for the /nos directory:

```
SVOS> du -ah /nos
196.4M /nos/primary.swi
196.4M /nos
SVOS>
```

erase zeroize

Syntax

```
erase zeroize
```

Description

Securely erases any user data contained on the eMMC or other storage devices on the management module.



NOTE: Back up all data before running this command or all user/config data will be lost.

Command context

Service OS (SVOS>)

Authority

Administrators or local user group members with execution rights for this command.

Example

Erasing user data:

```
SVOS> SVOS> erase --help
Usage: erase zeroize

Securely erases storage devices on the management module.
SVOS>
```
SVOS> erase zeroize
#####WARNING#####
This will securely erase all customer data and reset the switch
to factory defaults. This will initiate a reboot and render the
switch unavailable until the zeroization is complete.
This should take several minutes to one hour to complete.
#####WARNING#####

Continue (y/n)? y
reboot: Restarting system

ServiceOS Information:
 Version: PL.01.07.0004-internal
 Build Date: 2020-11-23 18:07:42 PST
 Build ID: ServiceOS:PL.01.07.0004-internal:133137f635df:202011231807
 SHA: 133137f635dff5778bf3e109eb75825b68d64789

Preparing for zeroization

Storage zeroization
WARNING: DO NOT POWER OFF UNTIL
ZEROIZATION IS COMPLETE
This should take several minutes
to one hour to complete

Restoring files
```

## exit

### Syntax

exit

### Description

Logs the user out from the SVOS> prompt.

### Command context

Service OS (SVOS>)

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Logging the user out from the `SVOS>` prompt:

```
SVOS> exit

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 RESTRICTED RIGHTS LEGEND
Confidential computer software. Valid license from Hewlett Packard Enterprise
Development LP required for possession, use or copying. Consistent with FAR
12.211 and 12.212, Commercial Computer Software, Computer Software
Documentation, and Technical Data for Commercial Items are licensed to the
U.S. Government under vendor's standard commercial license.

To reboot without logging in, enter 'reboot' as the login user name.

ServiceOS login:
```

# format

## Syntax

`format`

## Description

Configures the primary storage device with the correct partition and file system formatting. This command removes all pre-existing data on the primary storage device.

## Command context

Service OS (`SVOS>`)

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Configuring the primary storage device with the correct partition and file system formatting:

```
SVOS> format
#####WARNING#####
The following action will cause all data on
the primary storage device to be lost. After
formatting has completed, a reboot will be
initiated to complete storage initialization.
#####WARNING#####

Continue? (y/n): y

Working...This may take a few minutes...
```

# identify

## Syntax

`identify`

## Description

## Command context

Service OS (svos>)

## Authority

Administrators or local user group members with execution rights for this command.

## Example

# ls

## Syntax

`ls [<OPTIONS>] [<FILE-NME>]`

## Description

This command lists directory contents.

## Command context

Service OS (svos>)

## Parameters

### <OPTIONS>

Specifies options for the command.

**-1**

Shows one-column output.

**-a**

Shows entries which start with a period (.).

**-A**

Shows output similar to **-a**, but excludes a period (.) and a double period (..).

**-C**

Shows output list by columns.

**-x**

Shows output list by lines.

**-d**

Shows listing of directory entries instead of contents

**-L**

Follows symlinks.

- H**  
Follows symlinks on the command line.
- R**  
Recurse.
- P**  
Appends a slash (/) to directory entries.
- F**  
Appends an indicator to entries. An indicator can be as an asterisk (\*) or slash (/) or equal sign (=) or at sign (@) or pipe (|).
- l**  
Shows the output in a long listing format.
- i**  
Shows the list inode numbers.
- n**  
Shows a list of numeric UIDs and GIDs instead of names.
- s**  
Shows a list of allocated blocks.
- e**  
Shows in one column a list with the full date and time.
- h**  
Shows list sizes in human readable format (1K, 243M, 2G) with a one-column output.
- r**  
Shows in one column a sort in reverse order.
- S**  
Shows in one column a sort by size.
- X**  
Shows in the output sort by extension.
- v**  
Shows in one column a sort by version.
- c**  
With **-l**, it shows a sort in one column by `ctime`.
- t**  
With **-l**, it shows a sort by `mtime`.
- u**  
With **-l**, sort by `atime`.
- C**  
With **-l**, it shows a sort in one column by `ctime`.

**-w <N>**

Assumes that the terminal has the number of columns wide as specified by <N>.

**--color[={always | never | auto}]**

Controls color in the output.

**<FILE-NAME>**

Specifies the name of the file to list.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Listing directory contents:

```
SVOS> ls -la /nos
drwxr-xr-x 3 0 0 4096 Nov 21 03:19 .
drwxr-xr-x 11 0 0 220 Nov 21 03:21 ..
drwx----- 2 0 0 16384 Nov 21 03:20 lost+found
-rwxr-xr-x 1 0 0 205957424 Nov 21 03:19 primary.swi
SVOS>
```

## md5sum

### Syntax

md5sum [-c | -s | -w] [<FILE-NAME>]

### Description

This command computes and checks the MD5 message digest.

### Command context

Service OS (svos>)

### Parameters

**[-c | -s | -w]**

Selects the options for the command.

**-c**

Specifies to check the sums against the list in files.

**-s**

Specifies not output anything, status code shows success.

**-w**

Specifies to warn about improperly formatted checksum lines.

**<FILE-NAME>**

Specifies the file name to run the checksum against.

### Authority

Administrators or local user group members with execution rights for this command.

## Example

Computing and checking the MD5 message digest for /nos/primary.swi:

```
SVOS> md5sum /nos/primary.swi
93ffc89e7ec357854704d8e450c4b7ab /nos/primary.swi
SVOS>
```

## mkdir

### Syntax

```
mkdir [-m | -p] [<DIRECTORY-NAME>]
```

### Description

This command makes directories.

### Command context

Service OS (SVOS>)

### Parameters

**[-m | -p]**

Specifies the options for the command.

**-m**

Specifies the mode.

**-p**

Specifies to make parent directories as needed with no errors for pre-existing directories.

**<DIRECTORY-NAME>**

Specifies the directory to create.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Making the dir directory:

```
SVOS> mkdir dir
```

## mount

### Syntax

```
mount <DEVICE>
```

### Description

This command mounts the eMMC partitions to the following locations: /coredump, /logs, /nos, /selftest, and mounts the USB device to /mnt/usb.

Users can mount USB flash drives formatted as either FAT16 or FAT32 with a single partition.

## Command context

Service OS (svos>)

## Parameters

<DEVICE>

Specifies the device to be mounted. Supported device options include `all` and `usb`.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Mounting all of the eMMC partitions:

```
SVOS> mount all
SVOS> mount usb
```

# mv

## Syntax

```
mv [-f | -i | -n] <TARGET-DIRECTORY>
```

## Description

This command moves (renames) files.

## Command context

Service OS (svos>)

## Parameters

**-f**

Specifies not to prompt before overwriting.

**-i**

Specifies to prompt before overwriting.

**-n**

Specifies to not overwrite an existing file.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Moving the file named myfile:

```
SVOS> mv myfile
```

# password

## Syntax

password

## Description

Sets the admin user account password for both Service OS and AOS-CX once the user boots into AOS-CX and saves the configuration. This will overwrite the previous password if one exists. User input is masked with asterisks.

This command is not available if enhanced secure mode is set.

## Command context

Service OS (svos>)

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Setting the admin account password:

```
SVOS> password
Enter password:*****
Confirm password:*****
SVOS>
```

# pwd

## Syntax

pwd

## Description

Displays the current working directory.

## Command context

Service OS (svos>)

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Displaying the current working directory:

```
SVOS> pwd
/home
SVOS>
```



# reboot

## Syntax

reboot

## Description

Reboots the Management Module.

## Command context

Service OS (SVOS>)

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Rebooting the management module:

```
SVOS> reboot
reboot: Restarting system
```

# rm

## Syntax

rm [-f | -i | -R | -r] <FILE-NAME>

## Description

Removes files or directories.

## Command context

Service OS (SVOS>)

## Parameters

[-f | -i | -R | -r]

Selects the options for removing files or directories.

-f

Never prompt before removing files or directories.

-i

Always prompt before removing files or directories.

-R | -r

Recursive.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Removing the file named foo:

```
SVOS> rm foo
```

## rmmdir

### Syntax

```
rmmdir [-p] <DIRECTORY-NAME>
```

### Description

Removes empty directories.

### Command context

Service OS (SVOS>)

### Parameters

#### -p

Specifies to remove parent directories.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Removing the empty `foo` directory:

```
SVOS> rmmdir foo
SVOS>
```

## secure-mode

### Syntax

```
secure-mode <enhanced | standard | status>
```

### Description

Sets the secure mode to enhanced or standard secure mode. Also can display the current secure mode. A zeroization is required before switching between enhanced and standard secure modes.

The command also displays a message notifying the user that they are already in the targeted secure mode.

### Command context

Service OS (SVOS>)

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Setting the secure mode to enhanced or standard:

```
SVOS> secure-mode --help
Usage: secure-mode <enhanced | standard | status>
```

```

Set or retrieve the secure mode setting. Requires a zeroization to change modes.
SVOS>
...

...

SVOS> secure-mode enhanced
#####WARNING#####
This will set the switch into enhanced secure mode. Before
enhanced secure mode is enabled, the switch must securely erase
all customer data and reset the switch to factory defaults.
This will initiate a reboot and render the switch unavailable
until the zeroization is complete.
This should take several minutes to one hour to complete.
#####WARNING#####

Continue (y/n)? y
reboot: Restarting system

...

...

SVOS> secure-mode standard
#####WARNING#####
This will set the switch into standard secure mode. Before
standard secure mode is enabled, the switch must securely erase
all customer data and reset the switch to factory defaults.
This will initiate a reboot and render the switch unavailable
until the zeroization is complete.
This should take several minutes to one hour to complete.
#####WARNING#####

Continue (y/n)? y
reboot: Restarting system

...

...

SVOS> secure-mode standard
#####WARNING#####
Secure mode is already set to standard. Setting it again will
repeat the zeroization process. The switch must securely erase
all customer data and reset the switch to factory defaults.
This will initiate a reboot and render the switch unavailable
until the zeroization is complete.
This should take several minutes to one hour to complete.
#####WARNING#####

Continue (y/n)? y
reboot: Restarting system

...

...

SVOS> secure-mode status
enhanced secure mode is set.
SVOS>

```

## sh

### Syntax

sh

## Description

Launches a bash shell for support purposes. To quit bash, enter `exit`.

This command is not available if enhanced secure mode is set.

## Command context

Service OS (SVOS>)

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Launching a bash shell:

```
SVOS> sh
switch:/cli/fs/home#
```

# umount

## Syntax

```
umount <DEVICE>
```

## Description

Unmounts the eMMC partitions mounted to the following locations: `/coredump`, `/logs`, `/nos`, `/selftest`, and unmounts the USB device mounted to `/mnt/usb`.

## Command context

Service OS (SVOS>)

## Parameters

**<DEVICE>**

Specifies the device to be unmounted. Supported device options include `all` and `usb`.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Unmounting all devices:

```
SVOS> umount all
SVOS> umount usb
```

Unmounting a USB device:

```
SVOS> umount all
SVOS> umount usb
```

# update

## Syntax

```
update {primary | secondary} <IMAGE>
```

## Description

Verifies and installs a product image. The user can select the primary or secondary boot profile to update and the location of the file.

## Command context

Service OS (SVOS>)

## Parameters

**{primary | secondary}**

Selects either the primary or secondary image.

**<IMAGE>**

Specifies the image name.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Updating the software image using TFTP:



**NOTE:** The OOBM port is disabled on first boot and must be enabled using the `ip` command.

```
SVOS> ip dhcp
SVOS> ip show
Interface : Link Up
IP Address : 192.0.2.22
Subnet Mask : 255.255.200.20
Gateway : 10.0.24.1
SVOS> tftp -g -r XL.10.00.0001.swi -l image.swi 192.4.8.10
XL.10.00.0001.swi 100% |*****| 178M 0:00:00 ETA
SVOS> ls
image.swi
SVOS> update primary image.swi
Updating primary software image...
Verifying image...
Done
```

Update the software image using USB:



**NOTE:** This example assumes that the user has preloaded a USB flash drive with the image to be updated. The image name on the flash drive is not important.

```
SVOS> mount usb
SVOS> ls /mnt/usb
image.swi
SVOS> update primary /mnt/usb/image.swi
Updating primary software image...
Verifying image...
Done
```

# version

## Syntax

version

## Description

Displays the following build strings:

- Version.
- Build date.
- Build time.
- Build ID.
- SHA.

## Command context

Service OS (svos>)

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Displaying version build strings:

```
SVOS> version
ServiceOS Information:
 Version: GT.01.01.0001
 Build Date: 2017-07-19 14:52:31 PDT
 Build ID: ServiceOS:GT.01.01.0001:461519208911:201707191452
 SHA: 46151920891195cdb2267ea6889a3c6cbc3d4193
SVOS>
```



**NOTE:** On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

## ACL application

ACLs can be applied as follows:

| ACL type                    | IPv4+6 | MAC |
|-----------------------------|--------|-----|
| Direction                   | In     | In  |
| L2 interface (port)         | Yes    | Yes |
| L2 LAG                      | Yes    | Yes |
| VLAN                        | Yes    | Yes |
| Control plane (default VRF) | Yes    |     |



**NOTE:** The following match criteria is not supported. If this match criteria is attempted to be configured, an error message will be displayed and the action will not be completed.

TTL on IP ACLs

## access-list copy

### Syntax

```
access-list {ip|ipv6|mac} <ACL-NAME> copy <DESTINATION-ACL>
```

### Description

Copies an IPv4, IPv6, or MAC ACL to a new destination ACL or overwrites an existing ACL.

### Command context

config

### Parameters

**{ip|ipv6|mac}**

Specifies the type of ACL.

**<ACL-NAME>**

Specifies the name of the ACL to be copied.

**<DESTINATION-ACL>**

Specifies the name of the destination ACL.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Copying MY\_IP\_ACL to MY\_IP\_ACL2:

```
switch(config)# access-list ip MY_IP_ACL copy MY_IP_ACL2
switch(config-acl-ip)# exit
```

```
switch(config)# do show access-list
```

| Type     | Name                   |  |                        |
|----------|------------------------|--|------------------------|
| Sequence | Comment                |  |                        |
|          | Action                 |  | L3 Protocol            |
|          | Source IP Address      |  | Source L4 Port(s)      |
|          | Destination IP Address |  | Destination L4 Port(s) |
|          | Additional Parameters  |  |                        |

|      |                          |  |        |
|------|--------------------------|--|--------|
| IPv4 | MY_IP_ACL                |  |        |
| 1    | permit                   |  | udp    |
|      | any                      |  |        |
|      | 172.16.1.0/255.255.255.0 |  |        |
| 2    | permit                   |  | tcp    |
|      | 172.16.2.0/255.255.0.0   |  | > 1023 |
|      | any                      |  |        |
| 3    | permit                   |  | tcp    |
|      | 172.26.1.0/255.255.255.0 |  |        |
|      | any                      |  |        |
|      | dscp: AF11               |  |        |
|      | ack                      |  |        |
|      | syn                      |  |        |
| 4    | deny                     |  | any    |
|      | any                      |  |        |
|      | any                      |  |        |
|      | Hit-counts: enabled      |  |        |

|      |                          |  |        |
|------|--------------------------|--|--------|
| IPv4 | MY_IP_ACL2               |  |        |
| 1    | permit                   |  | udp    |
|      | any                      |  |        |
|      | 172.16.1.0/255.255.255.0 |  |        |
| 2    | permit                   |  | tcp    |
|      | 172.16.2.0/255.255.0.0   |  | > 1023 |
|      | any                      |  |        |
| 3    | permit                   |  | tcp    |
|      | 172.26.1.0/255.255.255.0 |  |        |
|      | any                      |  |        |
|      | dscp: AF11               |  |        |
|      | ack                      |  |        |
|      | syn                      |  |        |
| 4    | deny                     |  | any    |
|      | any                      |  |        |
|      | any                      |  |        |
|      | Hit-counts: enabled      |  |        |

Copying MY\_IPV6\_ACL to MY\_IPV6\_ACL2:

```
switch(config)# access-list ipv6 MY_IPV6_ACL copy MY_IPV6_ACL2
switch(config-acl-ip)# exit
```

```
switch(config)# do show access-list
```

| Type     | Name    |  |             |
|----------|---------|--|-------------|
| Sequence | Comment |  |             |
|          | Action  |  | L3 Protocol |



|       | Source IP Address                                                 | Source L4 Port(s)      |
|-------|-------------------------------------------------------------------|------------------------|
|       | Destination IP Address                                            | Destination L4 Port(s) |
|       | Additional Parameters                                             |                        |
| ----- |                                                                   |                        |
| IPv6  | MY_IPV6_ACL                                                       |                        |
| 1     | permit<br>any<br>2001::1/64                                       | udp                    |
| 2     | Permit all TCP ephemeral ports<br>permit<br>2001:2001::2:1<br>any | tcp<br>> 1023          |
| 3     | permit<br>2001:2011::1/64<br>any                                  | tcp                    |
| 4     | deny<br>any<br>any<br>Hit-counts: enabled                         | any                    |
| ----- |                                                                   |                        |
| IPv6  | MY_IPV6_ACL2                                                      |                        |
| 1     | permit<br>any<br>2001::1/64                                       | udp                    |
| 2     | Permit all TCP ephemeral ports<br>permit<br>2001:2001::2:1<br>any | tcp<br>> 1023          |
| 3     | permit<br>2001:2011::1/64<br>any                                  | tcp                    |
| 4     | deny<br>any<br>any<br>Hit-counts: enabled                         | any                    |

Copying MY\_MAC\_ACL to MY\_MAC\_ACL2:

```
switch(config)# access-list mac MY_MAC_ACL copy MY_MAC_ACL2
switch(config-acl-mac)# exit
```

```
switch(config)# do show access-list
```

| Type     | Name                                                                          |           |
|----------|-------------------------------------------------------------------------------|-----------|
| Sequence | Comment                                                                       |           |
|          | Action                                                                        | EtherType |
|          | Source MAC Address                                                            |           |
|          | Destination MAC Address                                                       |           |
|          | Additional Parameters                                                         |           |
| -----    |                                                                               |           |
| MAC      | MY_MAC_ACL                                                                    |           |
| 1        | permit<br>1122.3344.5566/ffff.ffff.0000<br>any                                | ipv6      |
| 2        | permit<br>aaaa.bbbb.cccc<br>1111.2222.3333<br>QoS Priority Code Point: 4      | any       |
| 3        | Permit all vlan-1 tagged Appletalk traffic<br>permit<br>any<br>any<br>VLAN: 1 | appletalk |
| 4        | deny<br>any                                                                   | any       |

```

any
Hit-counts: enabled

MAC MY_MAC_ACL2
1 permit ipv6
 1122.3344.5566/ffff.ffff.0000
 any
2 permit any
 aaaa.bbbb.cccc
 1111.2222.3333
 QoS Priority Code Point: 4
3 Permit all vlan-1 tagged Appletalk traffic
 permit appletalk
 any
 any
 VLAN: 1
4 deny any
 any
 any
Hit-counts: enabled

```

## access-list ip

### Syntax

Syntax to create an IPv4 ACL and enter its context. Plus syntax to remove an ACL:

```
access-list ip <ACL-NAME>
```

```
no access-list ip <ACL-NAME>
```

Syntax (within the ACL context) for creating or removing ACEs for protocols ah, gre, esp, igmp, ospf, pim (ip is available as an alias for any):

```

[<SEQUENCE-NUMBER>]
{permit|deny}
{any|ip|ah|gre|esp|igmp|ospf|pim|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]

```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for creating or removing ACEs for protocols sctp, tcp, udp:

```

[<SEQUENCE-NUMBER>]
{permit|deny}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]

```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for creating or removing ACEs for protocol icmp:

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{icmp}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] }
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] }
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]

no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for ACE comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>

no <SEQUENCE-NUMBER> comment
```

## Description

Creates an IPv4 Access Control List (ACL) comprised of one or more Access Control Entries (ACEs) ordered and prioritized by sequence number. The lowest sequence number is the highest prioritized ACE.

The `no` form of this command deletes the entire ACL, or deletes an ACE identified by sequence number, or deletes only the comment from the ACE identified by sequence number.

## Command context

config

The `access-list ip <ACL-NAME>` command takes you into the named ACL context where you enter the ACEs.

## Parameters

### <ACL-NAME>

Specifies the name of this ACL.

### <SEQUENCE-NUMBER>

Specifies a sequence number for the ACE. Range: 1 to 4294967295.

### {permit|deny}

Specifies whether to permit or deny traffic matching this ACE.

### <IP-PROTOCOL-NUM>

Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.

### {any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>} ] }

Specifies the source IPv4 address.

- `any` - specifies any source IPv4 address.
- `<SRC-IP-ADDRESS>` - specifies the source IPv4 host address.

- `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
- `<SUBNET-MASK>` - specifies the address bits to mask (dotted decimal notation).

**[`{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] }`]**

Specifies the destination IPv4 address.

- `any` - specifies any destination IPv4 address.
- `<DST-IP-ADDRESS>` - specifies the destination IPv4 host address.
- `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
- `<SUBNET-MASK>` - specifies the address bits to mask (dotted decimal notation).

**[`{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>`]**

Specifies the port, or port range. Port numbers are in the range of 0 to 65535.

- `eq <PORT>` - specifies the Layer 4 port.
- `gt <PORT>` - specifies any Layer 4 port greater than the indicated port.
- `lt <PORT>` - specifies any Layer 4 port less than the indicated port.
- `range <MIN-PORT> <MAX-PORT>` - specifies the Layer 4 port range.




---

**NOTE:** Upon application of the ACL, ACEs with L4 port ranges may consume more than one hardware entry.

---

#### **urg**

Specifies matching on the TCP Flag: Urgent.

#### **ack**

Specifies matching on the TCP Flag: Acknowledgment.

#### **psh**

Specifies matching on the TCP Flag: Push buffered data to receiving application.

#### **rst**

Specifies matching on the TCP Flag: Reset the connection.

#### **syn**

Specifies matching on the TCP Flag: Synchronize sequence numbers.

#### **fin**

Specifies matching on the TCP Flag: Finish connection.

#### **established**

Specifies matching on the TCP Flag: Established connection.

**[`icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}`]**

Specifies the ICMP type.

- `echo` - specifies an ICMP echo request packet.
- `echo-reply` - specifies an ICMP echo reply packet.
- `<ICMP-TYPE-VALUE>` - specifies an ICMP type value. Range: 0 to 255.

**[icmp-code <ICMP-CODE-VALUE>]**

Specifies the ICMP code value. Range: 0 to 255.

**dscp <DSCP-SPECIFIER>**

Specifies the Differentiated Services Code Point (DSCP), either a numeric `<DSCP-VALUE>` (0 to 63) or one of these keywords:

- `AF11` - DSCP 10 (Assured Forwarding Class 1, low drop probability)
- `AF12` - DSCP 12 (Assured Forwarding Class 1, medium drop probability)
- `AF13` - DSCP 14 (Assured Forwarding Class 1, high drop probability)
- `AF21` - DSCP 18 (Assured Forwarding Class 2, low drop probability)
- `AF22` - DSCP 20 (Assured Forwarding Class 2, medium drop probability)
- `AF23` - DSCP 22 (Assured Forwarding Class 2, high drop probability)
- `AF31` - DSCP 26 (Assured Forwarding Class 3, low drop probability)
- `AF32` - DSCP 28 (Assured Forwarding Class 3, medium drop probability)
- `AF33` - DSCP 30 (Assured Forwarding Class 3, high drop probability)
- `AF41` - DSCP 34 (Assured Forwarding Class 4, low drop probability)
- `AF42` - DSCP 36 (Assured Forwarding Class 4, medium drop probability)
- `AF43` - DSCP 38 (Assured Forwarding Class 4, high drop probability)
- `CS0` - DSCP 0 (Class Selector 0: Default)
- `CS1` - DSCP 8 (Class Selector 1: Scavenger)
- `CS2` - DSCP 16 (Class Selector 2: OAM)
- `CS3` - DSCP 24 (Class Selector 3: Signaling)
- `CS4` - DSCP 32 (Class Selector 4: Real time)
- `CS5` - DSCP 40 (Class Selector 5: Broadcast video)
- `CS6` - DSCP 48 (Class Selector 6: Network control)
- `CS7` - DSCP 56 (Class Selector 7)
- `EF` - DSCP 46 (Expedited Forwarding)

**ip-precedence <IP-PRECEDENCE-VALUE>**

Specifies an IP precedence value. Range: 0 to 7.

**tos <TOS-VALUE>**

Specifies the Type of Service value. Range: 0 to 31.

### fragment

Specifies a fragment packet.

### vlan <VLAN-ID>

Specifies VLAN tag to match on. 802.1Q VLAN ID.



**NOTE:** This parameter cannot be used in any ACL that will be applied to a VLAN.

### count

Keeps the hit counts of the number of packets matching this ACE.

### log

[<SEQUENCE-NUMBER>] comment <TEXT-STRING>

Adds a comment to an ACE. The `no` form removes only the comment from the ACE.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

- If the <IP-PROTOCOL-NUM> parameter is used instead of a protocol name, ensure that any needed ACE-definition parameters specific to the selected protocol are also provided.
- When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with `log` option is logged. Until the log-timer wait-period is over, any packets matching other ACL types do not create a log. At the end of the wait-period, the switch creates a summary log for all the ACLs that were matched, regardless of type.

## Examples

Creating an IPv4 ACL with four entries:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 10 permit udp any 172.16.1.0/24
switch(config-acl-ip)# 20 permit tcp 172.16.2.0/16 gt 1023 any
switch(config-acl-ip)# 30 permit tcp 172.26.1.0/24 any syn ack dscp 10
switch(config-acl-ip)# 40 deny any any any count
switch(config-acl-ip)# exit
```

```
switch(config)# show access-list
```

| Type | Name      | Sequence | Comment | Action | Source IP Address | Destination IP Address   | Additional Parameters | L3 Protocol | Source L4 Port(s) | Destination L4 Port(s) |
|------|-----------|----------|---------|--------|-------------------|--------------------------|-----------------------|-------------|-------------------|------------------------|
| IPv4 | MY_IP_ACL |          |         |        |                   |                          |                       |             |                   |                        |
|      |           | 10       | permit  |        | any               | 172.16.1.0/255.255.255.0 |                       | udp         |                   |                        |
|      |           | 20       | permit  |        | any               | 172.16.2.0/255.255.0.0   |                       | tcp         | > 1023            |                        |
|      |           | 30       | permit  |        | any               | 172.26.1.0/255.255.255.0 |                       | tcp         |                   |                        |
|      |           |          |         |        |                   | any                      |                       |             |                   |                        |

```

dscp: AF11
ack
syn
40 deny any
any
any
Hit-counts: enabled

```

Adding a comment to an existing IPv4 ACE:

```

switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 20 comment Permit all TCP ephemeral ports
switch(config-acl-ip)# exit

switch(config)# show access-list

```

| Type | Name      | Sequence | Comment                        | Action              | L3 Protocol | Source IP Address        | Source L4 Port(s) | Destination IP Address | Destination L4 Port(s) | Additional Parameters |
|------|-----------|----------|--------------------------------|---------------------|-------------|--------------------------|-------------------|------------------------|------------------------|-----------------------|
| IPv4 | MY_IP_ACL | 10       | permit                         | any                 | udp         | 172.16.1.0/255.255.255.0 |                   |                        |                        |                       |
|      |           | 20       | Permit all TCP ephemeral ports | permit              | tcp         | 172.16.2.0/255.255.0.0   | > 1023            |                        |                        |                       |
|      |           | 30       | permit                         | any                 | tcp         | 172.26.1.0/255.255.255.0 |                   |                        |                        |                       |
|      |           | 40       | deny                           | any                 | any         |                          |                   |                        |                        |                       |
|      |           |          |                                | dscp: AF11          |             |                          |                   |                        |                        |                       |
|      |           |          |                                | ack                 |             |                          |                   |                        |                        |                       |
|      |           |          |                                | syn                 |             |                          |                   |                        |                        |                       |
|      |           |          |                                | any                 |             |                          |                   |                        |                        |                       |
|      |           |          |                                | any                 |             |                          |                   |                        |                        |                       |
|      |           |          |                                | Hit-counts: enabled |             |                          |                   |                        |                        |                       |

Removing a comment from an existing IPv4 ACE:

```

switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# no 20 comment
switch(config-acl-ip)# exit

switch(config)# show access-list

```

| Type | Name      | Sequence | Comment | Action | L3 Protocol | Source IP Address        | Source L4 Port(s) | Destination IP Address | Destination L4 Port(s) | Additional Parameters |
|------|-----------|----------|---------|--------|-------------|--------------------------|-------------------|------------------------|------------------------|-----------------------|
| IPv4 | MY_IP_ACL | 10       | permit  | any    | udp         | 172.16.1.0/255.255.255.0 |                   |                        |                        |                       |
|      |           | 20       | permit  | any    | tcp         | 172.16.2.0/255.255.0.0   | > 1023            |                        |                        |                       |
|      |           | 30       | permit  | any    | tcp         | 172.26.1.0/255.255.255.0 |                   |                        |                        |                       |

```

any
dscp: AF11
ack
syn
40 deny any
any
any
Hit-counts: enabled

```

Adding an ACE (insert line 25) to an existing IPv4 ACL:

```

switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 25 permit icmp 172.16.2.0/16 any
switch(config-acl-ip)# exit

```

```

switch(config)# show access-list

```

| Type  | Sequence | Name | Comment   | Action                   | L3 Protocol | Source IP Address | Source L4 Port(s) | Destination IP Address | Destination L4 Port(s) | Additional Parameters |
|-------|----------|------|-----------|--------------------------|-------------|-------------------|-------------------|------------------------|------------------------|-----------------------|
| ----- |          |      |           |                          |             |                   |                   |                        |                        |                       |
| IPv4  |          |      | MY_IP_ACL |                          |             |                   |                   |                        |                        |                       |
|       | 10       |      | permit    |                          | udp         |                   |                   |                        |                        |                       |
|       |          |      | any       |                          |             |                   |                   |                        |                        |                       |
|       |          |      |           | 172.16.1.0/255.255.255.0 |             |                   |                   |                        |                        |                       |
|       | 20       |      | permit    |                          | tcp         |                   |                   |                        |                        |                       |
|       |          |      |           | 172.16.2.0/255.255.0.0   |             |                   |                   |                        |                        |                       |
|       |          |      | any       |                          | > 1023      |                   |                   |                        |                        |                       |
|       | 25       |      | permit    |                          | icmp        |                   |                   |                        |                        |                       |
|       |          |      |           | 172.16.2.0/255.255.0.0   |             | any               |                   |                        |                        |                       |
|       | 30       |      | permit    |                          | tcp         |                   |                   |                        |                        |                       |
|       |          |      |           | 172.26.1.0/255.255.255.0 |             |                   |                   |                        |                        |                       |
|       |          |      | any       |                          |             |                   |                   |                        |                        |                       |
|       |          |      |           | dscp: AF11               |             |                   |                   |                        |                        |                       |
|       |          |      | ack       |                          |             |                   |                   |                        |                        |                       |
|       |          |      | syn       |                          |             |                   |                   |                        |                        |                       |
|       | 40       |      | deny      |                          | any         |                   |                   |                        |                        |                       |
|       |          |      | any       |                          |             |                   |                   |                        |                        |                       |
|       |          |      | any       |                          |             |                   |                   |                        |                        |                       |
|       |          |      |           | Hit-counts: enabled      |             |                   |                   |                        |                        |                       |

Replacing an ACE in an existing IPv4 ACL:

```

switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 25 permit icmp 172.17.1.0/16 any
switch(config-acl-ip)# exit

```

```

switch(config)# show access-list

```

| Type  | Sequence | Name | Comment   | Action                   | L3 Protocol | Source IP Address | Source L4 Port(s) | Destination IP Address | Destination L4 Port(s) | Additional Parameters |
|-------|----------|------|-----------|--------------------------|-------------|-------------------|-------------------|------------------------|------------------------|-----------------------|
| ----- |          |      |           |                          |             |                   |                   |                        |                        |                       |
| IPv4  |          |      | MY_IP_ACL |                          |             |                   |                   |                        |                        |                       |
|       | 10       |      | permit    |                          | udp         |                   |                   |                        |                        |                       |
|       |          |      | any       |                          |             |                   |                   |                        |                        |                       |
|       |          |      |           | 172.16.1.0/255.255.255.0 |             |                   |                   |                        |                        |                       |
|       | 20       |      | permit    |                          | tcp         |                   |                   |                        |                        |                       |
|       |          |      |           | 172.16.2.0/255.255.0.0   |             |                   |                   |                        |                        |                       |
|       |          |      | any       |                          | > 1023      |                   |                   |                        |                        |                       |



[illegible]

## Removing an ACE from an IPv4 ACL:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# no 25
switch(config-acl-ip)# exit
```

```
switch(config)# show access-list
```

|          |                        |                        |
|----------|------------------------|------------------------|
| Type     | Name                   |                        |
| Sequence | Comment                |                        |
|          | Action                 | L3 Protocol            |
|          | Source IP Address      | Source L4 Port(s)      |
|          | Destination IP Address | Destination L4 Port(s) |
|          | Additional Parameters  |                        |

```
IPv4 MY_IP_ACL
10 permit udp
 any
 172.16.1.0/255.255.255.0
20 permit tcp
 172.16.2.0/255.255.0.0
 > 1023
 any
30 permit tcp
 172.26.1.0/255.255.255.0
 any
 dscp: AF11
 ack
 syn
40 deny any
 any
 any
Hit-counts: enabled
```

## Copy an IPv4 ACL:

```
switch(config)# access-list ip MY IP ACL copy MY IP ACL2
```

```
switch(config)# show access-list
```

|          |                        |                        |
|----------|------------------------|------------------------|
| Type     | Name                   |                        |
| Sequence | Comment                |                        |
|          | Action                 | L3 Protocol            |
|          | Source IP Address      | Source L4 Port(s)      |
|          | Destination IP Address | Destination L4 Port(s) |
|          | Additional Parameters  |                        |

```
IPv4 MY_IP_ACL
10 permit udp
 any
 172.16.1.0/255.255.255.0
20 permit tcp
 172.16.2.0/255.255.0.0
 > 1023
```

```

 any
30 permit
 172.26.1.0/255.255.255.0
 any
 dscp: AF11
 ack
 syn
40 deny
 any
 any
 Hit-counts: enabled

IPv4 MY_IP_ACL2
10 permit
 any
 172.16.1.0/255.255.255.0
20 permit
 172.16.2.0/255.255.0.0
 any
30 permit
 172.26.1.0/255.255.255.0
 any
 dscp: AF11
 ack
 syn
40 deny
 any
 any
 Hit-counts: enabled

```

#### Removing an IPv4 ACL:

```

switch(config)# no access-list ip MY_IP_ACL

switch(config)# show access-list
Type Name
Sequence Comment
 Action
 Source IP Address
 Destination IP Address
 Additional Parameters

IPv4 MY_IP_ACL2
1 permit
 any
 172.16.1.0/255.255.255.0
2 permit
 172.16.2.0/255.255.0.0
 any
3 permit
 172.26.1.0/255.255.255.0
 any
 dscp: AF11
 ack
 syn
4 deny
 any

```

```
any
Hit-counts: enabled
```

## access-list ipv6

### Syntax

Syntax to create an IPv6 ACL and enter its context. Plus syntax to remove an ACL:

```
access-list ipv6 <ACL-NAME>
```

```
no access-list ipv6 <ACL-NAME>
```

Syntax (within the ACL context) for creating or removing ACEs for protocols ah, gre, esp, ospf, pim (ipv6 is available as an alias for any):

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{any|ipv6|ah|gre|esp|ospf|pim|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]}
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]
```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for creating or removing ACEs for protocols sctp, tcp, udp:

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]
```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for creating or removing ACEs for protocol icmpv6:

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{icmpv6}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]]}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]]}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]
```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for ACE comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>
```

```
no <SEQUENCE-NUMBER> comment
```

## Description

Creates an IPv6 Access Control List (ACL). The ACL is made of one or more Access Control Entries (ACEs) ordered and prioritized by sequence number. The lowest sequence number is the highest prioritized ACE.

The `no` form of this command deletes the entire ACL, or deletes an ACE identified by sequence number, or deletes only the comment from the ACE identified by sequence number.

## Command context

`config`

The `access-list ipv6 <ACL-NAME>` command takes you into the named ACL context where you enter the ACEs.

## Parameters

**<ACL-NAME>**

Specifies the name of this ACL.

**<SEQUENCE-NUMBER>**

Specifies a sequence number for the ACE. Range: 1 to 4294967295.

**{permit|deny}**

Specifies whether to permit or deny traffic matching this ACE.

**<IP-PROTOCOL-NUM>**

Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.

**{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]}**

Specifies the source IPv6 address.

- `any` - specifies any source IPv6 address.
- `<SRC-IP-ADDRESS>` - specifies the source IPv6 host address.
  - `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 128.

**{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]}**

Specifies the destination IPv6 address.

- `any` - specifies any destination IPv6 address.
- `<DST-IP-ADDRESS>` - specifies the destination IPv6 host address.
  - `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 128.

**[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]**

Specifies the port, or port range. Port numbers are in the range of 0 to 65535.

- `eq <PORT>` - specifies the Layer 4 port.
- `gt <PORT>` - specifies any Layer 4 port greater than the indicated port.

- `lt <PORT>` - specifies any Layer 4 port less than the indicated port.
- `range <MIN-PORT> <MAX-PORT>` - specifies the Layer 4 port range.



**NOTE:** Upon application of the ACL, ACEs with L4 port ranges may consume more than one hardware entry.

---

**[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}]**

Specifies the ICMP type.

- `echo` - specifies an ICMP echo request packet.
- `echo-reply` - specifies an ICMP echo reply packet.
- `<ICMP-TYPE-VALUE>` - specifies an ICMP type value. Range: 0 to 255.

**[icmp-code <ICMP-CODE-VALUE>]**

Specifies the ICMP code value. Range: 0 to 255.

**dscp <DSCP-SPECIFIER>**

Specifies the Differentiated Services Code Point (DSCP), either a numeric `<DSCP-VALUE>` (0 to 63) or one of these keywords:

- `AF11` - DSCP 10 (Assured Forwarding Class 1, low drop probability)
- `AF12` - DSCP 12 (Assured Forwarding Class 1, medium drop probability)
- `AF13` - DSCP 14 (Assured Forwarding Class 1, high drop probability)
- `AF21` - DSCP 18 (Assured Forwarding Class 2, low drop probability)
- `AF22` - DSCP 20 (Assured Forwarding Class 2, medium drop probability)
- `AF23` - DSCP 22 (Assured Forwarding Class 2, high drop probability)
- `AF31` - DSCP 26 (Assured Forwarding Class 3, low drop probability)
- `AF32` - DSCP 28 (Assured Forwarding Class 3, medium drop probability)
- `AF33` - DSCP 30 (Assured Forwarding Class 3, high drop probability)
- `AF41` - DSCP 34 (Assured Forwarding Class 4, low drop probability)
- `AF42` - DSCP 36 (Assured Forwarding Class 4, medium drop probability)
- `AF43` - DSCP 38 (Assured Forwarding Class 4, high drop probability)
- `CS0` - DSCP 0 (Class Selector 0: Default)
- `CS1` - DSCP 8 (Class Selector 1: Scavenger)
- `CS2` - DSCP 16 (Class Selector 2: OAM)
- `CS3` - DSCP 24 (Class Selector 3: Signaling)
- `CS4` - DSCP 32 (Class Selector 4: Real time)
- `CS5` - DSCP 40 (Class Selector 5: Broadcast video)
- `CS6` - DSCP 48 (Class Selector 6: Network control)

- CS7 - DSCP 56 (Class Selector 7)
- EF - DSCP 46 (Expedited Forwarding)

**ip-precedence** <IP-PRECEDENCE-VALUE>

Specifies an IP precedence value. Range: 0-7.

**tos** <TOS-VALUE>

Specifies the Type of Service value. Range: 0-31.

**vlan** <VLAN-ID>

Specifies VLAN tag to match on. 802.1Q VLAN ID.



**NOTE:** This parameter cannot be used in any ACL that will be applied to a VLAN.

**count**

Keeps the hit counts of the number of packets matching this ACE.

**log**

[<SEQUENCE-NUMBER>] **comment** <TEXT-STRING>

Adds a comment to an ACE. The **no** form removes only the comment from the ACE.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

- If the <IP-PROTOCOL-NUM> parameter is used instead of a protocol name, ensure that any needed ACE-definition parameters specific to the selected protocol are also provided.
- When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with **log** option is logged. Until the log-timer wait-period is over, any packets matching other ACL types do not create a log. At the end of the wait-period, the switch creates a summary log all the ACLs that were matched, regardless of type.

## Examples

Creating an IPv6 ACL with four entries:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 10 permit udp any 2001::1/64
switch(config-acl-ipv6)# 20 permit tcp 2001:2001::2:1/128 gt 1023 any
switch(config-acl-ipv6)# 30 permit tcp 2001:2011::1/64 any
switch(config-acl-ipv6)# 40 deny any any any count
switch(config-acl-ipv6)# exit
```

```
switch(config)# do show access-list
```

| Type | Name | Sequence | Comment | Action                 | L3 Protocol            |
|------|------|----------|---------|------------------------|------------------------|
|      |      |          |         | Source IP Address      | Source L4 Port(s)      |
|      |      |          |         | Destination IP Address | Destination L4 Port(s) |
|      |      |          |         | Additional Parameters  |                        |

---

```
IPv6 MY_IPV6_ACL
10 permit udp
```

```

any
2001::1/64
20 permit tcp
2001:2001::2:1 > 1023
any
30 permit tcp
2001:2011::1/64
any
40 deny any
any
any
Hit-counts: enabled

```

Adding a comment to an existing IPv6 ACE:

```

switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 20 comment Permit all TCP ephemeral ports
switch(config-acl-ipv6)# exit

```

```

switch(config)# do show access-list

```

| Type | Name        | Sequence | Comment                        | Action | L3 Protocol | Source IP Address | Source L4 Port(s) | Destination IP Address | Destination L4 Port(s) | Additional Parameters |
|------|-------------|----------|--------------------------------|--------|-------------|-------------------|-------------------|------------------------|------------------------|-----------------------|
| IPv6 | MY_IPV6_ACL | 10       | permit                         | any    | udp         | 2001::1/64        |                   |                        |                        |                       |
|      |             | 20       | Permit all TCP ephemeral ports | permit | tcp         | 2001:2001::2:1    | > 1023            |                        |                        |                       |
|      |             | 30       | permit                         | any    | tcp         | 2001:2011::1/64   |                   |                        |                        |                       |
|      |             | 40       | deny                           | any    | any         |                   |                   |                        |                        |                       |
|      |             |          | any                            |        |             |                   |                   |                        |                        |                       |
|      |             |          | any                            |        |             |                   |                   |                        |                        |                       |
|      |             |          | Hit-counts: enabled            |        |             |                   |                   |                        |                        |                       |

Removing a comment from an existing IPv6 ACE:

```

switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# no 20 comment
switch(config-acl-ipv6)# exit

```

```

switch(config)# do show access-list

```

| Type | Name        | Sequence | Comment | Action | L3 Protocol | Source IP Address | Source L4 Port(s) | Destination IP Address | Destination L4 Port(s) | Additional Parameters |
|------|-------------|----------|---------|--------|-------------|-------------------|-------------------|------------------------|------------------------|-----------------------|
| IPv6 | MY_IPV6_ACL | 10       | permit  | any    | udp         | 2001::1/64        |                   |                        |                        |                       |
|      |             | 20       | permit  | any    | tcp         | 2001:2001::2:1    | > 1023            |                        |                        |                       |

```

30 permit tcp
 2001:2011::1/64
 any
40 deny any
 any
 any
Hit-counts: enabled

```

Adding an ACE to an existing IPv6 ACL:

```

switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 25 permit icmpv6 2001::1/64 any
switch(config-acl-ipv6)# exit

```

```

switch(config)# do show access-list

```

| Type | Name        | Sequence | Comment | Action              | L3 Protocol | Source IP Address | Source L4 Port(s) | Destination IP Address | Destination L4 Port(s) | Additional Parameters |
|------|-------------|----------|---------|---------------------|-------------|-------------------|-------------------|------------------------|------------------------|-----------------------|
| IPv6 | MY_IPV6_ACL | 10       | permit  | any                 | udp         | 2001::1/64        |                   |                        |                        |                       |
|      |             | 20       | permit  | 2001:2001::2:1      | tcp         |                   | > 1023            |                        |                        |                       |
|      |             | 25       | permit  | 2001::1/64          | icmpv6      |                   |                   |                        |                        |                       |
|      |             | 30       | permit  | 2001:2011::1/64     | tcp         |                   |                   |                        |                        |                       |
|      |             | 40       | deny    | any                 | any         |                   |                   |                        |                        |                       |
|      |             |          |         | any                 |             |                   |                   |                        |                        |                       |
|      |             |          |         | Hit-counts: enabled |             |                   |                   |                        |                        |                       |

Replacing an ACE in an existing IPv6 ACL:

```

switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 25 permit icmpv6 2001::2:1/64 any
switch(config-acl-ipv6)# exit

```

```

switch(config)# do show access-list

```

| Type | Name        | Sequence | Comment | Action         | L3 Protocol | Source IP Address | Source L4 Port(s) | Destination IP Address | Destination L4 Port(s) | Additional Parameters |
|------|-------------|----------|---------|----------------|-------------|-------------------|-------------------|------------------------|------------------------|-----------------------|
| IPv6 | MY_IPV6_ACL | 10       | permit  | any            | udp         | 2001::1/64        |                   |                        |                        |                       |
|      |             | 20       | permit  | 2001:2001::2:1 | tcp         |                   | > 1023            |                        |                        |                       |
|      |             | 25       | permit  | 2001::2:1/64   | icmpv6      |                   |                   |                        |                        |                       |
|      |             |          |         | any            |             |                   |                   |                        |                        |                       |



```

30 permit tcp
 2001:2011::1/64
 any
40 deny any
 any
 any
Hit-counts: enabled

```

Removing an ACE from an IPv6 ACL:

```

switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# no 25
switch(config-acl-ipv6)# exit

switch(config)# do show access-list
Type Name
Sequence Comment
Action
Source IP Address L3 Protocol
Destination IP Address Source L4 Port(s)
Additional Parameters Destination L4 Port(s)

IPv6 MY_IPV6_ACL
10 permit udp
 any
 2001::1/64
20 permit tcp
 2001:2001::2:1 > 1023
 any
30 permit tcp
 2001:2011::1/64
 any
40 deny any
 any
 any
Hit-counts: enabled

```

Removing an IPv6 ACL:

```

switch(config)# no access-list ipv6 MY_IPV6_ACL

switch(config)# do show access-list
Type Name
Sequence Comment
Action
Source IP Address L3 Protocol
Destination IP Address Source L4 Port(s)
Additional Parameters Destination L4 Port(s)

IPv6 MY_IPV6_ACL2
1 permit udp
 any
 2001::1/64
2 Permit all TCP ephemeral ports
 permit tcp
 2001:2001::2:1 > 1023
 any
3 permit tcp
 2001:2011::1/64
 any
4 deny any
 any

```

```
any
Hit-counts: enabled
```

## access-list log-timer

### Syntax

```
access-list log-timer {default|<INTERVAL>}
```

### Description

Sets the log timer interval for all ACEs that have the `log` parameter configured.

### Command context

```
config
```

### Parameters

#### **default**

Resets the log timer to its default 300 seconds.

#### **<INTERVAL>**

Specifies the log timer interval in seconds. Range: 30 to 300.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

- The first packet that matches an ACE with the `log` parameter within an ACL log timer window (configured with the `access-list log-timer` command) has its header contents extracted and sent to the configured logging destination, such as the console and syslog server. Each time the ACL log timer expires, a summary of all ACEs with `log` configured are sent to the logging destination. This capability allows throttling of logging ACL hits.
- If no further log messages are generated in the wait-period, the switch suspends the timer and resets itself to log as soon as a new match occurs.
- When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with the `log` option is logged. Any packets, matching other ACL types, do not create a log until the log-timer wait-period is over. At the end of the wait-period, a summary log is made of all the ACLs that were matched, regardless of type.
- You may see a minor discrepancy between the ACL logging statistics and the hit counts statistics due to the time required to record the log message.

### Examples



**NOTE:** Although these examples use debug logging, you can alternatively use event logging.

Enabling debug logging for the ACL logging module:

```
switch# debug acl log severity info
switch# show debug

module sub_module severity vlan port ip mac instance vrf
```

```

acl acl_log info -----

```

Setting the debug destination to console with the minimum security level of info:

```
switch# debug destination console severity info
switch# show debug destination
```

```

show debug destination

```

```
CONSOLE:info
```

Setting the access list log-timer to 30 seconds:

```
switch(config)# access-list log-timer 30
switch(config)# do show access-list log-timer
ACL log timer length (frequency): 30 seconds
```

Creating an IPv4 ACL with one entry with the log parameter:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# deny icmp 1.1.1.1 1.1.1.2 log
switch(config-acl-ip)# do show access-list
```

| Type     | Name                |                        |                        |
|----------|---------------------|------------------------|------------------------|
| Sequence | Comment             | Action                 | L3 Protocol            |
|          |                     | Source IP Address      | Source L4 Port(s)      |
|          |                     | Destination IP Address | Destination L4 Port(s) |
|          |                     | Additional Parameters  |                        |
| IPv4     | MY_IP_ACL           |                        |                        |
| 10       | deny                |                        | icmp                   |
|          | 1.1.1.1             |                        |                        |
|          | 1.1.1.2             |                        |                        |
|          | Logging: enabled    |                        |                        |
|          | Hit-counts: enabled |                        |                        |

Enabling interface 1/1/1 and applying the ACL:

Sending packets that will match the ACE and observe the ACL logging message on the console:

```
2017-10-10T20:13:36.044+00:00 ops-switchd[875]: debug|LOG_INFO|AMM|1/5|ACL|ACL_LOG|
List MY_IP_ACL, seq# 10 denied icmp 1.1.1.1 -> 1.1.1.2 type 8 code 0,
on vlan 1, port 1/1/1, direction in
```

When the access list log-timer expires, the summary message is printed on the console. The number 30 is the number of packets received during the last access list log-timer window.

```
2017-10-10T20:14:06.051+00:00 ops-switchd[875]: debug|LOG_INFO|AMM|1/5|ACL|ACL_LOG|
MY_IP_ACL on 1/1/1 (in): 30 10 deny icmp 1.1.1.1 1.1.1.2 log count
```

Resetting the ACL log timer to the default value:

```
switch(config)# access-list log-timer default
```

## access-list mac

### Syntax

```
access-list mac <ACL-NAME>
```

```
no access-list mac <ACL-NAME>
```

```

[<SEQUENCE-NUMBER>]
{permit|deny}
{any|<SRC-MAC-ADDRESS>[/<ETHERNET-MASK>]} }
{any|<DST-MAC-ADDRESS>[/<ETHERNET-MASK>]} }
{any|aarp|appletalk|arp|fcoe|fcoe-init|ip|ipv6|
 ipx-arp|ipx-non-arp|is-is|lldp|mpls-multicast|mpls-unicast|q-in-q|
 rbridge|trill|wake-on-lan|<NUMERIC-ETHERTYPE>}
[pcp <PCP-VALUE>] [vlan <VLAN-ID>] [count] [log]

no <SEQUENCE-NUMBER>

[<SEQUENCE-NUMBER>] comment <TEXT-STRING>

no <SEQUENCE-NUMBER> comment

```

## Description

Creates a MAC Access Control List (ACL). The ACL is made of one or more Access Control Entries (ACEs) ordered and prioritized by sequence numbers. The lowest sequence number is the highest prioritized ACE.

The **no** form of this command deletes the entire ACL, or deletes an ACE identified by sequence number, or deletes only the comment from the ACE identified by sequence number.

## Command context

config

The **access-list mac <ACL-NAME>** command takes you into the named ACL context where you enter the ACEs.

## Parameters

**<ACL-NAME>**

Specifies the name of this ACL.

**<SEQUENCE-NUMBER>**

Specifies a sequence number for the ACE. Range: 1 to 4294967295.

**{permit|deny}**

Specifies whether to permit or deny traffic matching this ACE.

**comment**

Specifies storing the remaining entered text as an ACE comment.

**{ any | <SRC-MAC-ADDRESS> [/ <ETHERNET-MASK> ] }**

Specifies the source host MAC address (xxxx.xxxx.xxxx), OUI, or the keyword `any`. You can optionally include the following:

<ETHERNET-MASK> - The address bits to mask (xxxx.xxxx.xxxx).

**{ any | <DST-MAC-ADDRESS> [/ <ETHERNET-MASK> ] }**

Specifies the destination host MAC address (xxxx.xxxx.xxxx), OUI, or the keyword `any`. You can optionally include the following:

<ETHERNET-MASK> - The address bits to mask (xxxx.xxxx.xxxx).

**{ any | aarp | appletalk | . . . | wake-on-lan | <NUMERIC-ETHERTYPE> }**

Specifies the protocol encapsulated in the Ethernet frame. The encapsulated protocol is identified by the EtherType Ethernet field. The EtherType is specified in one of the following three ways:

- `any` - any EtherType.
- <NUMERIC-ETHERTYPE> - the numerical EtherType protocol number. Range: 0x600 to 0xffff.
- One of these EtherType protocol name keywords:

aarp  
appletalk  
arp  
fcoe  
fcoe-init  
ip  
ipv6  
ipx-arpa  
ipx-non-arpa  
is-is  
lldp  
mpls-multicast  
mpls-unicast  
q-in-q  
rbridge  
trill  
wake-on-lan

**pcp <PCP-VALUE>**

Specifies 802.1Q QoS Priority Code Point value. Range: 0 to 7.

**vlan <VID>**

Specifies a VLAN ID. The VLAN ID must exist.



**NOTE:** This parameter cannot be used in any ACL that will be applied to a VLAN.

**count**

Keeps the hit counts of the number of packets matching this ACE.

log

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with `log` option is logged. Until the log-timer wait-period is over, any packets matching other ACL types do not create a log. At the end of the wait-period, the switch creates a summary log all the ACLs that were matched, regardless of type.

## Examples

Creating a MAC ACL with four entries:

```
switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 10 permit 1122.3344.5566/ffff.ffff.0000 any ipv6
switch(config-acl-ip)# 20 permit aaaa.bbbb.cccc 1111.2222.3333 any pcp 4
switch(config-acl-ip)# 30 permit any any appletalk vlan 40
switch(config-acl-ip)# 40 deny any any any count
switch(config-acl-ip)# exit
```

```
switch(config)# do show access-list
```

| Type | Name       | Sequence | Action                                                                   | EtherType |
|------|------------|----------|--------------------------------------------------------------------------|-----------|
|      |            |          | Source MAC Address<br>Destination MAC Address<br>Additional Parameters   |           |
| MAC  | MY_MAC_ACL |          |                                                                          |           |
|      |            | 10       | permit<br>1122.3344.5566/ffff.ffff.0000<br>any                           | ipv6      |
|      |            | 20       | permit<br>aaaa.bbbb.cccc<br>1111.2222.3333<br>QoS Priority Code Point: 4 | any       |
|      |            | 30       | permit<br>any<br>any<br>VLAN: 40                                         | appletalk |
|      |            | 40       | deny<br>any<br>any<br>Hit-counts: enabled                                | any       |

Adding a comment to an existing MAC ACE:

```
switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 30 comment Permit all vlan-40 tagged Appletalk traffic
switch(config-acl-ip)# exit
```

```
switch(config)# do show access-list
```

| Type | Name | Sequence | Action                                                                 | EtherType |
|------|------|----------|------------------------------------------------------------------------|-----------|
|      |      |          | Source MAC Address<br>Destination MAC Address<br>Additional Parameters |           |
|      |      |          |                                                                        |           |

```

MAC MY_MAC_ACL
10 permit 1122.3344.5566/ffff.ffff.0000 ipv6
 any
20 permit any
 aaaa.bbbb.cccc
 1111.2222.3333
 QoS Priority Code Point: 4
30 Permit all vlan-40 tagged Appletalk traffic
 permit appletalk
 any
 any
 VLAN: 40
40 deny any
 any
 any
 Hit-counts: enabled

```

Removing a comment from an existing MAC ACE:

```

switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-mac)# no 30 comment
switch(config-acl-mac)# exit

switch(config)# do show access-list
Type Name
Sequence Comment
 Action
 Source MAC Address
 Destination MAC Address
 Additional Parameters

MAC MY_MAC_ACL
10 permit 1122.3344.5566/ffff.ffff.0000 ipv6
 any
20 permit any
 aaaa.bbbb.cccc
 1111.2222.3333
 QoS Priority Code Point: 4
30 permit appletalk
 any
 any
 VLAN: 1
40 deny any
 any
 any
 Hit-counts: enabled

```

Adding an ACE to an existing MAC ACL:

```

switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 35 permit any aabb.cc11.1234 0xffee
switch(config-acl-ip)# exit

switch(config)# do show access-list
Type Name
Sequence Comment
 Action
 Source MAC Address
 Destination MAC Address
 Additional Parameters

MAC MY_MAC_ACL

```

```

10 permit
 1122.3344.5566/ffff.ffff.0000
 any
20 permit
 aaaa.bbbb.cccc
 1111.2222.3333
 QoS Priority Code Point: 4
30 permit
 any
 any
 VLAN: 1
35 permit
 any
 aabb.cc11.1234
40 deny
 any
 any
Hit-counts: enabled

```

Replacing an ACE in an existing MAC ACL:

```

switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 35 permit any aabb.cc11.1234 0xeeee
switch(config-acl-ip)# exit

```

```

switch(config)# do show access-list

```

| Type     | Name       |                               |                         |
|----------|------------|-------------------------------|-------------------------|
| Sequence | Comment    | Action                        | EtherType               |
|          |            | Source MAC Address            | Destination MAC Address |
|          |            | Additional Parameters         |                         |
| -----    |            |                               |                         |
| MAC      | MY_MAC_ACL |                               |                         |
| 10       | permit     | 1122.3344.5566/ffff.ffff.0000 | ipv6                    |
|          |            | any                           |                         |
| 20       | permit     | aaaa.bbbb.cccc                | any                     |
|          |            | 1111.2222.3333                |                         |
|          |            | QoS Priority Code Point: 4    |                         |
| 30       | permit     |                               | appletalk               |
|          |            | any                           |                         |
|          |            | any                           |                         |
|          |            | VLAN: 1                       |                         |
| 35       | permit     |                               | 0xeeee                  |
|          |            | any                           |                         |
|          |            | aabb.cc11.1234                |                         |
| 40       | deny       |                               | any                     |
|          |            | any                           |                         |
|          |            | any                           |                         |
|          |            | Hit-counts: enabled           |                         |

Removing an ACE from an MAC ACL:

```

switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# no 35
switch(config-acl-ip)# exit

```

```

switch(config)# do show access-list

```

| Type     | Name    |                    |           |
|----------|---------|--------------------|-----------|
| Sequence | Comment | Action             | EtherType |
|          |         | Source MAC Address |           |



|       | Destination MAC Address                                                  | Additional Parameters |
|-------|--------------------------------------------------------------------------|-----------------------|
| ----- |                                                                          |                       |
| MAC   | MY_MAC_ACL                                                               |                       |
| 10    | permit<br>1122.3344.5566/ffff.ffff.0000<br>any                           | ipv6                  |
| 20    | permit<br>aaaa.bbbb.cccc<br>1111.2222.3333<br>QoS Priority Code Point: 4 | any                   |
| 30    | permit<br>any<br>any<br>VLAN: 1                                          | appletalk             |
| 40    | deny<br>any<br>any<br>Hit-counts: enabled                                | any                   |

Removing a MAC ACL:

```
switch(config)# no access-list mac MY_MAC_ACL
```

```
switch(config)# do show access-list
```

| Type  | Name                                          | Sequence | Comment | Action                        | EtherType               |
|-------|-----------------------------------------------|----------|---------|-------------------------------|-------------------------|
|       |                                               |          |         | Source MAC Address            | Destination MAC Address |
|       |                                               |          |         | Additional Parameters         |                         |
| ----- |                                               |          |         |                               |                         |
| MAC   | MY_MAC_ACL2                                   |          |         |                               |                         |
|       | 1 permit                                      |          |         | 1122.3344.5566/ffff.ffff.0000 | any                     |
|       |                                               |          |         |                               | ipv6                    |
|       | 2 permit                                      |          |         | aaaa.bbbb.cccc                | any                     |
|       |                                               |          |         | 1111.2222.3333                |                         |
|       |                                               |          |         | QoS Priority Code Point: 4    |                         |
|       | 3 Permit all vlan-40 tagged Appletalk traffic |          |         | permit                        | appletalk               |
|       |                                               |          |         | any                           |                         |
|       |                                               |          |         | any                           |                         |
|       |                                               |          |         | VLAN: 1                       |                         |
|       | 4 deny                                        |          |         | any                           | any                     |
|       |                                               |          |         | any                           |                         |
|       |                                               |          |         | Hit-counts: enabled           |                         |

## access-list resequence

### Syntax

```
access-list {ip|ipv6|mac} <ACL-NAME> resequence <STARTING-SEQUENCE-NUMBER> <INCREMENT>
```

### Description

Resequences the ACE sequence numbers in an ACL.

## Command context

config

## Parameters

{ip|ipv6|mac}

Specifies the ACL type.

<ACL-NAME>

Specifies the ACL name.

<STARTING-SEQUENCE-NUMBER>

Specifies the starting sequence number.

<INCREMENT>

Specifies the sequence number increment.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Resequencing an IPv4 ACL to start at 1 with an increment of 1:

```
switch(config)# access-list ip MY_IP_ACL resequence 1 1
switch(config-acl-ip)# exit
```

```
switch(config)# do show access-list
```

| Type | Name      | Sequence | Comment | Action              | L3 Protocol | Source IP Address        | Source L4 Port(s) | Destination IP Address | Destination L4 Port(s) | Additional Parameters |
|------|-----------|----------|---------|---------------------|-------------|--------------------------|-------------------|------------------------|------------------------|-----------------------|
| IPv4 | MY_IP_ACL | 1        | permit  | any                 | udp         | 172.16.1.0/255.255.255.0 |                   |                        |                        |                       |
|      |           | 2        | permit  | any                 | tcp         | 172.16.2.0/255.255.0.0   | > 1023            |                        |                        |                       |
|      |           | 3        | permit  | any                 | tcp         | 172.26.1.0/255.255.255.0 |                   |                        |                        |                       |
|      |           |          |         | dscp: AF11          |             |                          |                   |                        |                        |                       |
|      |           |          |         | ack                 |             |                          |                   |                        |                        |                       |
|      |           |          |         | syn                 |             |                          |                   |                        |                        |                       |
|      |           | 4        | deny    | any                 | any         |                          |                   |                        |                        |                       |
|      |           |          |         | any                 |             |                          |                   |                        |                        |                       |
|      |           |          |         | Hit-counts: enabled |             |                          |                   |                        |                        |                       |

Resequencing an IPv6 ACL to start at 1 with an increment of 1:

```
switch(config)# access-list ipv6 MY_IPV6_ACL resequence 1 1
switch(config-acl-ip)# exit
```

```
switch(config)# do show access-list
```

| Type | Name | Sequence | Comment |
|------|------|----------|---------|
|------|------|----------|---------|

|       | Action                         | L3 Protocol            |
|-------|--------------------------------|------------------------|
|       | Source IP Address              | Source L4 Port(s)      |
|       | Destination IP Address         | Destination L4 Port(s) |
|       | Additional Parameters          |                        |
| ----- |                                |                        |
| IPv6  | MY_IPV6_ACL                    |                        |
| 1     | permit                         | udp                    |
|       | any                            |                        |
|       | 2001::1/64                     |                        |
| 2     | Permit all TCP ephemeral ports |                        |
|       | permit                         | tcp                    |
|       | 2001:2001::2:1                 | > 1023                 |
|       | any                            |                        |
| 3     | permit                         | tcp                    |
|       | 2001:2011::1/64                |                        |
|       | any                            |                        |
| 4     | deny                           | any                    |
|       | any                            |                        |
|       | any                            |                        |
|       | Hit-counts: enabled            |                        |

Resequencing a MAC ACL to start at 1 with an increment of 1:

```
switch(config)# access-list mac MY_MAC_ACL resequence 1 1
switch(config-acl-mac)# exit
```

```
switch(config)# do show access-list
```

| Type     | Name                                        |           |
|----------|---------------------------------------------|-----------|
| Sequence | Comment                                     | EtherType |
|          | Action                                      |           |
|          | Source MAC Address                          |           |
|          | Destination MAC Address                     |           |
|          | Additional Parameters                       |           |
| -----    |                                             |           |
| MAC      | MY_MAC_ACL                                  |           |
| 1        | permit                                      | ipv6      |
|          | 1122.3344.5566/ffff.ffff.0000               |           |
|          | any                                         |           |
| 2        | permit                                      | any       |
|          | aaaa.bbbb.cccc                              |           |
|          | 1111.2222.3333                              |           |
|          | QoS Priority Code Point: 4                  |           |
| 3        | Permit all vlan-40 tagged Appletalk traffic |           |
|          | permit                                      | appletalk |
|          | any                                         |           |
|          | any                                         |           |
|          | VLAN: 1                                     |           |
| 4        | deny                                        | any       |
|          | any                                         |           |
|          | any                                         |           |
|          | Hit-counts: enabled                         |           |

## access-list reset

### Syntax

```
access-list {all|ip <ACL-NAME>|ipv6 <ACL-NAME>|mac <ACL-NAME>} reset
```

### Description

Changes the user-specified ACL configuration to match the active ACL configuration. Use this command when a discrepancy exists between what the user configured and what is active and accepted by the system.

## Command context

config

## Parameters

**all**|**ip** <ACL-NAME>|**ipv6** <ACL-NAME>|**mac** <ACL-NAME>

Specifies **one** of the following:

- a reset of all ACLs.
- a reset of a named IPv4 ACL.
- a reset of a named IPv6 ACL.
- a reset of a named MAC ACL.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

The output of the `show access-list` command displays the active configuration of the product. The active configuration is the ACLs that have been configured and accepted by the system. The output of the `show access-list` command with the `configuration` parameter, displays the ACLs that have been configured. The output of this command may not be the same as what was programmed in hardware or what is active on the product.

If the active ACLs and user-configured ACLs are not the same, a warning message is displayed in the output of the `show` command. Modify the user-configured ACL until the warning message is no longer displayed or run the `access-list reset` command to change the user-specified configuration to match the active configuration.

## Examples

Apply an ACL with TCP acknowledgments (ACKs) on ingress, which is unsupported by hardware:

```
switch(config-acl)# 10 permit tcp 172.16.2.0/16 any ack
```

Displaying the user-specified configuration:

```
switch(config)# do show access-list commands
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active configuration.
access-list ip TEST_ACL
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active configuration.
interface 1/1/1
 apply access-list ip TEST_ACL in

switch(config)# do show access-list commands configuration
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active configuration.
access-list ip TEST_ACL
 10 permit tcp 172.16.2.0/255.255.0.0 any ack
! access-list ip TEST_ACL user configuration does not match active configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active configuration.
interface 1/1/1
 apply access-list ip TEST_ACL in

switch(config)# do show access-list
Type Name
Sequence Comment
Action L3 Protocol
```

```

Source IP Address Source L4 Port(s)
Destination IP Address Destination L4 Port(s)
Additional Parameters

% Warning: TEST_ACL user configuration does not match active configuration.
% run 'access-list TYPE NAME reset' to reset access-list to match active configuration.
IPv4 TEST_ACL

switch(config)# do show access-list configuration
Type Name
Sequence Comment
Action
Source IP Address Source L4 Port(s)
Destination IP Address Destination L4 Port(s)
Additional Parameters

% Warning: TEST_ACL user configuration does not match active configuration.
% run 'access-list TYPE NAME reset' to reset access-list to match active configuration.
IPv4 TEST_ACL
10
 permit
 172.16.2.0/255.255.0.0 tcp
 any
 ack

```

Resetting the user-specified configuration to match the active configuration.

```
switch(config)# access-list ip TEST_ACL reset
```

Displaying the updated user-specified configuration.

```

switch(config)# do show access-list commands
access-list ip TEST_ACL
interface 1/1/1
 apply access-list ip TEST_ACL in

switch(config)# do show access-list commands configuration
access-list ip TEST_ACL
interface 1/1/1
 apply access-list ip TEST_ACL in

switch(config)# do show access-list
Type Name
Sequence Comment
Action
Source IP Address Source L4 Port(s)
Destination IP Address Destination L4 Port(s)
Additional Parameters

IPv4 TEST_ACL

switch(config)# do show access-list configuration
Type Name
Sequence Comment
Action
Source IP Address Source L4 Port(s)
Destination IP Address Destination L4 Port(s)
Additional Parameters

IPv4 TEST_ACL

```

# apply access-list control-plane

## Syntax

```
apply access-list {ip|ipv6} <ACL-NAME> control-plane vrf <VRF-NAME>
```

```
no apply access-list {ip|ipv6} <ACL-NAME> control-plane vrf <VRF-NAME>
```

## Description

Applies an ACL to the specified VRF.

The `no` form of this command removes application of the ACL from the specified VRF.

## Command context

config

## Parameters

**ip|ipv6**

Specifies the ACL type: `ip` for IPv4, or `ipv6` for IPv6.

**<ACL-NAME>**

Specifies the ACL name.

**vrf <VRF-NAME>**

Specifies the VRF name.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

Only one ACL per type (`ip`, or `ipv6`) may be applied to a control plane VRF at a time. Therefore, using the `apply access-list control-plane` command on a VRF with an already-applied ACL of the same type, will replace the applied ACL.

## Examples

Applying `My_ip_ACL` to control plane traffic on the default VRF:

```
switch(config)# apply access-list ip My_ip_ACL control-plane vrf default
```

Replacing `My_ip_ACL` with `My_Replacement_ACL` on the default VRF:

```
switch(config)# apply access-list ip My_Replacement_ACL control-plane vrf default
```

Remove (unapply) the `My_Replacement_ACL` from the default VRF. Any other interfaces or VLANs with `My_Replacement_ACL` applied are unaffected.

```
switch(config)# no apply access-list ip My_Replacement_ACL control-plane vrf default
```

# apply access-list (to interface or LAG)

## Syntax

### Description

Applies an ACL to the interface (Individual front plane port) or Link Aggregation Group (LAG) identified by the current interface or LAG context.

The `no` form of this command removes application of the ACL from the current interface or LAG identified by the current interface or LAG context.

### Command context

`config-if`

`config-lag-if`

### Parameters

`ip|ipv6|mac`

Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC ACL.

`<ACL-NAME>`

Specifies the ACL name.

`in`

Selects the inbound (ingress) traffic direction.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

- Each ACL of a given type can be applied to the same interface or LAG once. Therefore, using the `apply access-list` command on an interface or LAG with an already-applied ACL of the same type will replace the currently applied ACL.
- An ACL can be applied to an individual front plane port or to a Link Aggregation Group (LAG).
- A port that is a member of a LAG with an applied ACL cannot have a different ACL applied to that member port.
- When the port membership of a LAG with an applied ACL is changed, the LAG ACL is automatically applied or removed from that port depending on the modification type.

### Examples

Applying `My_IP_ACL` to ingress traffic on interface range `1/1/10` to `1/1/12`:

```
switch(config)# int 1/1/10-1/1/12
switch((config-if-<1/1/10-1/1/12>)# apply access-list ip My_IP_ACL in
switch((config-if-<1/1/10-1/1/12>)# exit
```

Applying `MY_IPV6_ACL` to ingress traffic on interface `1/1/1` and to ingress traffic on LAG `100`:

```
switch(config)# interface 1/1/1
switch(config-if)# apply access-list ipv6 MY_IPV6_ACL in
switch(config-if)# exit
```

```
switch(config)# interface lag 100
switch(config-lag-if)# apply access-list ipv6 MY_IPV6_ACL in
switch(config-lag-if)# exit
switch(config)#
```

Applying MY\_MAC\_ACL to ingress traffic on interface 1/1/1 and ingress traffic on interface 1/1/2:

```
switch(config)# interface 1/1/1
switch(config-if)# apply access-list mac MY_MAC_ACL in
switch(config-if)# exit

switch(config)# interface 1/1/2
switch(config-if)# apply access-list mac MY_MAC_ACL in
switch(config-if)# exit
switch(config)#
```

## apply access-list (to VLAN)

### Syntax

```
apply access-list {ip|ipv6|mac} <ACL-NAME> in
```

```
no apply access-list {ip|ipv6|mac} <ACL-NAME> in
```

### Description

Applies an ACL to the VLAN identified by the current VLAN context.

The `no` form of this command removes application of the ACL from the VLAN identified by the current VLAN context.

### Command context

```
config-vlan
```

### Parameters

**ip|ipv6|mac**

Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC ACL.

**<ACL-NAME>**

Specifies the ACL name.

**in**

Selects the inbound (ingress) traffic direction.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

- Each ACL of a given type can be applied to the same VLAN once. Therefore, using the `apply access-list` command on a VLAN with an already-applied ACL of the same type, will replace the applied ACL.
- When an ACL is applied to a VLAN, it will create hardware entries on all stack members regardless of whether a VLAN member exists on any specific stack member.



## Examples

Applying My\_ip\_ACL to ingress traffic on VLAN range 20 to 25:

```
switch(config)# vlan 20-25
switch(config-vlan-<20-25>)# apply access-list ip My_ip_ACL in
```

Applying My\_ip\_ACL to ingress traffic on VLAN 10:

```
switch(config)# vlan 10
switch(config-vlan-10)# apply access-list ip My_ip_ACL in
```

Applying My\_ipv6\_ACL to ingress traffic on VLAN 10:

```
switch(config)# vlan 10
switch(config-vlan-10)# apply access-list ipv6 My_ipv6_ACL in
```

Applying My\_mac\_ACL to ingress traffic on VLAN 10:

```
switch(config)# vlan 10
switch(config-vlan-10)# apply access-list mac My_mac_ACL in
```

Replacing My\_ipv6\_ACL with My\_Replacement\_ACL on VLAN 10 (following the preceding examples):

```
switch(config)# vlan 10
switch(config-vlan-10)# apply access-list ipv6 My_Replacement_ACL in
```

Removing (unapplying) several ACLs on VLAN 10:

```
switch(config)# vlan 10
switch(config-vlan-10)# no apply access-list ipv6 My_Replacement_ACL in
switch(config-vlan-10)# no apply access-list mac My_mac_ACL in
```

## clear access-list hitcounts

### Syntax

```
clear access-list hitcounts { all | [{ip|ipv6|mac} <ACL-NAME>] [interface <IF-NAME>|
vlan <VLAN-ID>] [in] }
```

### Description

Clears the hit counts for ACLs with ACEs that include the `count` keyword.

### Command context

Operator (>) or Manager (#)

### Parameters

**all**

Selects all ACLs.

**ip|ipv6|mac**

Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC.

**<ACL-NAME>**

Specifies the ACL name.

**interface** *<IF-NAME>*

Specifies the interface name (port or LAG).

**vlan** *<VLAN-ID>*

Specifies the VLAN.

**in**

Selects the inbound (ingress) traffic direction.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Clearing the hit counts for My\_ip\_ACL applied to VLAN 10 (ingress):

```
switch# clear access-list hitcounts ip My_ip_ACL vlan 10 in
```

Clearing the hit counts for all ACLs:

```
switch# clear access-list hitcounts all
```

## clear access-list hitcounts control-plane

### Syntax

```
clear access-list hitcounts [{ip|ipv6} <ACL-NAME>] control-plane vrf <VRF-NAME>
```

### Description

Clears the hit counts for ACLs applied to the Control Plane VRF.

### Command context

Operator (>) or Manager (#)

### Parameters

**ip|ipv6**

Specifies the ACL type: ip for IPv4, or ipv6 for IPv6.

**<ACL-NAME>**

Specifies the ACL name.

**vrf <VRF-NAME>**

Specifies the VRF name.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Clearing the hit counts for an IPv4 ACL applied to the Control Plane default VRF:

```
switch# clear access-list hitcounts ip My_ipv4_ACL control-plane vrf default
```

Clearing the hit counts for an IPv6 ACL applied to the Control Plane `default` VRF:

```
switch# clear access-list hitcounts ipv6 My_ipv6_ACL control-plane vrf default
```

## show access-list

### Syntax

Syntax that filters by ACLs applied to an interface or VLAN:

```
show access-list [interface <IF-NAME>|vlan <VLAN-ID>] [ip|ipv6|mac]
 [in] [commands] [configuration]
```

Syntax that filters by the named ACL:

```
show access-list [ip|ipv6|mac] [<ACL-NAME>]
 [commands] [configuration]
```

### Description

Shows information about your defined ACLs and where they have been applied. When `show access-list` is entered without parameters, information for all ACLs is shown. The parameters filter the list of ACLs for which information is shown.

Available filtering includes:

- The content of a specific ACL.
- All ACLs of a specific type.
- All ACLs applied to a specific interface (port or LAG).
- All ACLs applied to a specific VLAN.
- All IPv4 or IPv6 ACLs applied to interface VLANs.

### Command context

Operator (>) or Manager (#)

### Parameters

**interface <IF-NAME>**

Specifies the interface name (port or LAG).

**vlan <VLAN-ID>**

Specifies the VLAN.

**ip|ipv6|mac**

Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC.

**in**

Selects the inbound (ingress) traffic direction.

**<ACL-NAME>**

Specifies the ACL name.

## commands

Specifies that the ACL definition is to be shown as the commands and parameters used to create it rather than in tabular form.

## configuration

Specifies that the user-configured ACLs be shown as entered, even if the ACLs are not active due to ACE-definition command issues or hardware issues. This parameter is useful if there is a mismatch between the entered configuration and the previous successfully programmed (active) ACLs configuration.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing an IPv4 ACL:

```
switch# show access-list ip MY_ACL
Type Name
Sequence Comment
 Action
 Source IP Address
 Destination IP Address
 Additional Parameters

IPv4 MY_ACL
10 permit any
 172.16.1.0/255.255.255.0
 udp
20 permit any
 172.16.2.0/255.255.0.0
 tcp
 > 1023
30 permit any
 172.26.1.0//255.255.255.0
 tcp
 syn
 ack
 dscp 10
40 deny any
 any
 any
 Hit-counts: enabled

```

Showing an IPv4 ACL as commands:

```
switch# show access-list ip MY_ACL commands
access-list ip MY_ACL
10 permit udp any 172.16.1.0/255.255.255.0
20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any
30 permit tcp 172.26.1.0/255.255.255.0 any syn ack dscp 10
40 deny any any any count
```

Showing IPv4 ACLs applied to VLAN 10, inbound:

```
switch# show access-list vlan 10 ip in
Type Name
Sequence Comment
 Action
 Source IP Address
 Destination IP Address
 L3 Protocol
 Source L4 Port(s)
 Destination L4 Port(s)
```

| Additional Parameters |           |                           |        |
|-----------------------|-----------|---------------------------|--------|
| IPv4                  | My_ip_ACL |                           |        |
|                       | 10        | permit                    | udp    |
|                       |           | any                       |        |
|                       |           | 172.16.1.0/255.255.255.0  |        |
|                       | 20        | permit                    | tcp    |
|                       |           | 172.16.2.0/255.255.0.0    | > 1023 |
|                       |           | any                       |        |
|                       | 30        | permit                    | tcp    |
|                       |           | 172.26.1.0//255.255.255.0 |        |
|                       |           | any                       |        |
|                       |           | syn                       |        |
|                       |           | ack                       |        |
|                       |           | dscp 10                   |        |
|                       | 40        | deny                      | any    |
|                       |           | any                       |        |
|                       |           | any                       |        |
| Hit-counts: enabled   |           |                           |        |

Showing IPv6 ACLs applied to LAG 128, inbound:

```
switch# show access-list interface lag128 ipv6 in
```

| Type                | Name        | Sequence | Comment | Action             | L3 Protocol | Source IP Address | Source L4 Port(s) | Destination IP Address | Destination L4 Port(s) | Additional Parameters |
|---------------------|-------------|----------|---------|--------------------|-------------|-------------------|-------------------|------------------------|------------------------|-----------------------|
| IPv6                | MY_IPV6_ACL |          |         |                    |             |                   |                   |                        |                        |                       |
|                     |             | 10       | permit  |                    | udp         |                   |                   |                        |                        |                       |
|                     |             |          | any     |                    |             |                   |                   |                        |                        |                       |
|                     |             |          |         | 2001::1/64         |             |                   |                   |                        |                        |                       |
|                     |             | 20       | permit  |                    | tcp         |                   |                   |                        |                        |                       |
|                     |             |          |         | 2001:2001::2:1/128 | > 1023      |                   |                   |                        |                        |                       |
|                     |             |          | any     |                    |             |                   |                   |                        |                        |                       |
|                     |             | 30       | permit  |                    | tcp         |                   |                   |                        |                        |                       |
|                     |             |          |         | 2001:2011::1/64    |             |                   |                   |                        |                        |                       |
|                     |             | 40       | deny    |                    | any         |                   |                   |                        |                        |                       |
|                     |             |          | any     |                    |             |                   |                   |                        |                        |                       |
|                     |             |          | any     |                    |             |                   |                   |                        |                        |                       |
| Hit-counts: enabled |             |          |         |                    |             |                   |                   |                        |                        |                       |

Showing an IPv6 ACL as commands:

```
switch# show access-list ipv6 MY_IPV6_ACL commands
access-list ipv6 MY_IPV6_ACL
 10 permit udp any 2001::1/64
 20 permit tcp 2001:2001::2:1/128 gt 1023 any
 40 deny any any any count
```

Showing a MAC ACL:

```
switch# show access-list mac MY_MAC_ACL
```

| Type | Name | Sequence | Comment | Action | EtherType | Source MAC Address | Destination MAC Address | Additional Parameters |
|------|------|----------|---------|--------|-----------|--------------------|-------------------------|-----------------------|
|      |      |          |         |        |           |                    |                         |                       |

```

MAC MY_MAC_ACL
 10 permit 1122.3344.5566/ffff.ffff.0000 ipv6
 any
 20 permit aaaa.bbbb.cccc any
 1111.2222.3333
 QoS Priority Code Point: 4
 30 deny any any any any
 any
 any
 Hit-counts: enabled

```

Showing a MAC ACL as commands:

```

switch# show access-list mac MY_MAC_ACL commands
access-list mac MY_MAC_ACL
 10 permit 1122.3344.5566/ffff.ffff.0000 any ipv6
 20 permit aaaa.bbbb.cccc 1111.2222.3333 any pc 4
 30 deny any any any count

```

## show access-list control-plane

### Syntax

```

show access-list [ip|ipv6] [<ACL-NAME>] control-plane [vrf <VRF-NAME>]
 [commands] [configuration]

```

### Description

Shows information about your defined ACLs that have been applied to the Control Plane. When `show access-list control-plane` is entered without parameters, information for all ACLs applied to the Control Plane is shown. The parameters filter the list of ACLs for which information is shown.

Available filtering includes:

- The content of a specific ACL that has been applied to the Control Plane.
- All ACLs of a specific type that have been applied to the Control Plane.
- All ACLs applied to the Control Plane for a specific VRF.

### Command context

Operator (>) or Manager (#)

### Parameters

**ip|ipv6**

Specifies the ACL type: `ip` for IPv4, or `ipv6` for IPv6.

**<ACL-NAME>**

Specifies the ACL name.

**vrf <VRF-NAME>**

Specifies the VRF name.

### [commands]

Specifies that the ACL definition is to be shown as the commands and parameters used to create it rather than in tabular form.

### [configuration]

Specifies that the user-configured ACLs be shown as entered, even if the ACLs are not active due to ACE-definition command issues or hardware issues. This parameter is useful if there is a mismatch between the entered configuration and the previous successfully programmed (active) ACLs configuration.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing an IPv4 ACL applied to the Control Plane default VRF:

```
switch# show access-list ip My_ipv4_ACL control-plane vrf default
Type Name
Sequence Comment
 Action
 Source IP Address
 Destination IP Address
 Additional Parameters

IPv4 My_ipv4_ACL
10 permit udp
 any
 172.16.1.0/24
20 permit tcp
 172.16.2.0/16
 any
 > 1023
30 permit tcp
 172.26.1.0/24
 any
 syn
 ack
 dscp 10
40 deny any
 any
 any
Hit-counts: enabled

```

Showing an IPv6 ACL applied to the Control Plane default VRF:

```
switch# show access-list ipv6 My_ipv6_ACL control-plane vrf default
Type Name
Sequence Comment
 Action
 Source IP Address
 Destination IP Address
 Additional Parameters

IPv6 My_ipv6_ACL
10 permit udp
 any
 2001::1/64
20 permit tcp
 2001:2001::2:1/128
 any
 > 1023

```

```

30 permit tcp
 2001:2011::1/64
40 deny any
 any
 any
Hit-counts: enabled

```

---

## show access-list hitcounts

### Syntax

```
show access-list hitcounts { [{ip|ipv6|mac} <ACL-NAME>] [interface <IF-NAME> |
 vlan <VLAN-ID>] [in]}
```

### Description

Shows the hit count of the number of times an ACL has matched a packet or frame for ACEs with the `count` keyword. For ACEs without the `count` keyword, a dash is shown in place of a hit count.

### Command context

Operator (>) or Manager (#)

### Parameters

**ip|ipv6|mac**

Specifies the ACL type: `ip` for IPv4, `ipv6` for IPv6, or `mac` for MAC.

**<ACL-NAME>**

Specifies the ACL name.

**interface <IF-NAME>**

Specifies the interface name (port or LAG).

**vlan <VLAN-ID>**

Specifies the VLAN.

**in**

Selects the inbound (ingress) traffic direction.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Usage

- ACL hit counts are aggregated across all:
  - physical interfaces to which the ACL is applied to on ingress,
  - VLANs to which the ACL is applied to on ingress.
- If an ACL with an ACE with the `count` keyword is applied to multiple physical interfaces or VLANs, the hit counts are aggregated. There is one aggregation for physical interfaces and another for VLANs.
- Accumulated hit counts for an applied ACL are cleared upon any modification of the ACL.



## Examples

Showing the hit counts for My\_ip\_ACL applied to port 1/1/2:

```
switch# show access-list hitcounts ip My_ip_ACL interface 1/1/2
Statistics for ACL My_ip_ACL (ipv4):
interface 1/1/2* (in):
 Hit Count Configuration
 - 10 permit udp any 172.16.1.0/24
 - 20 permit tcp 172.16.2.0/16 gt 1023 any
 - 30 permit tcp 172.26.1.0/24 any syn ack dscp 10
 0 40 deny any any any count
* access-list statistics are shared among each combination of context type
 (interface, VLAN, VRF) and direction (in, control-plane).
 Use 'access-list TYPE NAME copy' to create a new access-list for separate statistics.
```

Showing the hit counts for My\_ip\_ACL applied to VLAN 10:

```
switch# show access-list hitcounts ip My_ip_ACL vlan 10
Statistics for ACL My_ip_ACL (ipv4):
vlan 10* (in):
 Hit Count Configuration
 - 10 permit udp any 172.16.1.0/24
 - 20 permit tcp 172.16.2.0/16 gt 1023 any
 - 30 permit tcp 172.26.1.0/24 any syn ack dscp 10
 0 40 deny any any any count
* access-list statistics are shared among each combination of context type
 (interface, VLAN, VRF) and direction (in, control-plane).
 Use 'access-list TYPE NAME copy' to create a new access-list for separate statistics.
```

## show access-list hitcounts control-plane

### Syntax

```
show access-list hitcounts [{ip|ipv6} <ACL-NAME>] control-plane vrf <VRF-NAME>
```

### Description

Shows the hit count of the number of times an ACL (applied to the Control Plane) has matched a packet for ACEs with the `count` keyword. For ACEs without the `count` keyword, a dash is shown in place of a hit count.

### Command context

Operator (>) or Manager (#)

### Parameters

**ip|ipv6**

Specifies the ACL type: `ip` for IPv4, or `ipv6` for IPv6.

**<ACL-NAME>**

Specifies the ACL name.

**vrf <VRF-NAME>**

Specifies the VRF name.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Usage

- ACL hit counts are aggregated across all VRFs to which the ACL is applied to on ingress.
- Accumulated hit counts for an applied ACL are cleared upon any modification of the ACL.

## Examples

Showing the hit counts for an IPv4 ACL applied to the Control Plane default VRF:

```
switch# show access-list hitcounts ip My_ipv4_ACL control-plane vrf default
Statistics for ACL My_ipv4_ACL (ipv4):
VRF default* (control-plane):
 Hit Count Configuration
 - 10 permit udp any 172.16.1.0/255.255.255.0
 - 20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any
 - 30 permit tcp 172.26.1.0/255.255.255.0 any syn ack dscp 10
 8 40 deny any any any count
* access-list statistics are shared among each combination of
 context type (interface, VLAN, VRF) and direction (in, control-plane).
 use 'access-list TYPE NAME copy' to create a uniquely-named access-list.
```

## show capacities

### Syntax

```
show capacities <FEATURE>
```

### Description

Shows system capacities and their values for all features or a specific feature.

### Command context

Manager (#)

### Parameters

<FEATURE>

Specifies a feature. For example, aaa.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

Capacities are expressed in user-understandable terms. Thus they may not map to a specific hardware or software resource or component. They are not intended to define a feature exhaustively.

### Examples

Showing classifier-related capacities on the 6100:

```
switch# show capacities classifier

System Capacities: Filter Classifier
Capacities Name Value

Maximum number of Access Control Entries configurable in a system 4096
Maximum number of Access Control Lists configurable in a system 512
```

|                                                                |      |
|----------------------------------------------------------------|------|
| Maximum number of class entries configurable in a system       | 4096 |
| Maximum number of classes configurable in a system             | 512  |
| Maximum number of entries in an Access Control List            | 1024 |
| Maximum number of entries in a class                           | 1024 |
| Maximum number of entries in a policy                          | 64   |
| Maximum number of classifier policies configurable in a system | 512  |
| Maximum number of policy entries configurable in a system      | 4096 |

Showing all available capacities on the 6100:

```
switch# show capacities
```

| System Capacities:                                                                |       |
|-----------------------------------------------------------------------------------|-------|
| Capacities Name                                                                   | Value |
| Maximum number of Access Control Entries configurable in a system                 | 4096  |
| Maximum number of Access Control Lists configurable in a system                   | 512   |
| Maximum number of class entries configurable in a system                          | 4096  |
| Maximum number of classes configurable in a system                                | 512   |
| Maximum number of entries in an Access Control List                               | 1024  |
| Maximum number of entries in a class                                              | 1024  |
| Maximum number of entries in a policy                                             | 64    |
| Maximum number of classifier policies configurable in a system                    | 512   |
| Maximum number of policy entries configurable in a system                         | 4096  |
| Maximum number of clients supported for tracking the IP address in the system     | 128   |
| Maximum number of dynamic VLANs that can be allowed using MVRP                    | 256   |
| Maximum number of nexthops per IP ECMP group                                      | 1     |
| Maximum number of IP neighbors (IPv4+IPv6) supported in the system                | 1024  |
| Maximum number of IPv4 neighbors(# of ARP entries) supported in the system        | 1024  |
| Maximum number of IPv6 neighbors(# of ND entries) supported in the system         | 512   |
| Maximum number of L2 MAC addresses supported in the system                        | 8192  |
| Maximum number of L3 Groups for IP Tunnels and ECMP Groups                        | 1     |
| Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP groups | 1024  |
| Maximum number of configurable LAG ports                                          | 8     |
| Maximum number of members supported by a LAG port                                 | 8     |
| Maximum number of VLANs across ports allowed in loop-protect                      | 3328  |
| Maximum number of IGMP/MLD groups supported                                       | 512   |
| Maximum number of IGMP/MLD snooping groups supported                              | 512   |
| Maximum number of Mirror Sessions configurable in a system                        | 4     |
| Maximum number of enabled Mirror Sessions in a system                             | 4     |
| Maximum number of mstp instances configurable in a system                         | 16    |
| Maximum number of Clients that can be authenticated on a port                     | 32    |
| Maximum number of Device Profiles allowed to be created on the system             | 8     |
| Maximum number of Port Access Roles allowed to be created on the system           | 32    |
| Maximum number of MAC Address that can be authorized on a port                    | 32    |
| Maximum number of Port Access Role VLAN IDs allowed to be created on the system   | 50    |
| Maximum number of Port Access Role VLAN names allowed to be created on the system | 50    |
| Maximum number of RBAC rules per user group                                       | 1024  |
| Maximum number of RPVST VLANs configurable on the system                          | 16    |
| Maximum number of RPVST VPORTs supported in a system                              | 512   |
| Maximum number of SVIs supported in the system                                    | 16    |
| Maximum number of routes (IPv4+IPv6) on the system                                | 512   |
| Maximum number of IPv4 routes on the system                                       | 512   |
| Maximum number of IPv6 routes on the system                                       | 512   |
| Maximum number of VLANs supported in the system                                   | 512   |

Showing all available capacities for mirroring:

```
switch# show capacities mirroring
```

| System Capacities: Filter Mirroring                        |       |
|------------------------------------------------------------|-------|
| Capacities Name                                            | Value |
| Maximum number of Mirror Sessions configurable in a system | 4     |
| Maximum number of enabled Mirror Sessions in a system      | 4     |

Showing all available capacities for MSTP:

```
switch# show capacities mstp
```

| System Capacities: Filter MSTP                            |       |
|-----------------------------------------------------------|-------|
| Capacities Name                                           | Value |
| -----                                                     |       |
| Maximum number of mstp instances configurable in a system | 64    |

Showing all available capacities for VLAN count:

```
switch# show capacities vlan-count
```

| System Capacities: Filter VLAN Count            |       |
|-------------------------------------------------|-------|
| Capacities Name                                 | Value |
| -----                                           |       |
| Maximum number of VLANs supported in the system | 4094  |

## show capacities-status

### Syntax

```
show capacities-status <FEATURE>
```

### Description

Shows system capacities status and their values for all features or a specific feature.

### Command context

Manager (#)

### Parameters

<FEATURE>

Specifies the feature, for example `aaa` for which to display capacities, values, and status. Required.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Showing the system capacities status for all features:

```
switch# show capacities-status
```

| System Capacities Status                                                                       |       |         |
|------------------------------------------------------------------------------------------------|-------|---------|
| Capacities Status Name                                                                         | Value | Maximum |
| -----                                                                                          |       |         |
| Number of Access Control Entries currently configured                                          | 0     | 4096    |
| Number of Access Control Lists currently configured                                            | 0     | 512     |
| Number of class entries currently configured                                                   | 0     | 4096    |
| Number of classes currently configured                                                         | 0     | 512     |
| Number of policies currently configured                                                        | 0     | 512     |
| Number of policy entries currently configured                                                  | 0     | 4096    |
| Number of dynamic VLANs currently learnt using MVRP                                            | 0     | 256     |
| Number of IP neighbor (IPv4+IPv6) entries                                                      | 1     | 1024    |
| Number of IPv4 neighbor(ARP) entries                                                           | 1     | 1024    |
| Number of IPv6 neighbor(ND) entries                                                            | 0     | 512     |
| Number of L3 Groups for IP Tunnels and ECMP Groups currently configured                        | 0     | 1       |
| Number of L3 Destinations for Routes, Nexthops in ECMP groups and Tunnels currently configured | 0     | 1024    |
| Number of Mirror Sessions currently configured                                                 | 0     | 4       |
| Number of Mirror Sessions currently enabled                                                    | 0     | 4       |
| Number of mstp instances currently configured                                                  | 0     | 16      |
| Number of RPVST VLANs currently configured                                                     | 0     | 16      |
| Number of routes (IPv4+IPv6) currently configured                                              | 1     | 512     |
| Number of IPv4 routes currently configured                                                     | 1     | 512     |
| Number of IPv6 routes currently configured                                                     | 0     | 512     |
| Number of VLANs currently configured                                                           | 1     | 512     |

## show resources

### Syntax

```
show resources
```

### Description

Shows hardware resource consumption. Resource data is updated every 10 seconds.

Hardware resource consumption information is shown for:

- TCAM entries
- TCAM lookups
- Policers

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Usage

The widths for show resources can have features combined (IPv4 + IPv6) into one TCAM lookup. Therefore, the table widths for each ACL/classifier policy type are variable depending on what is applied. For example:

```
"Ingress IP Port ACL" = Ingress v4 Port ACLs + Ingress v6 Port ACLs
 = 1 TCAM entry + 4 TCAM entries
 = 5 TCAM entries
```

Widths per feature are as follows:

```
IPv4 ACL 2
MAC ACL 2
IPv6 ACL 4
IPv4 Class 2
IPv6 Class 4
```

### Examples

Showing hardware resource consumption on a 6100 switch:

```
switch# show resources
```

Resource Usage:

| Mod | Description | Used | Reserved | Free |
|-----|-------------|------|----------|------|
|     | Resource    |      |          |      |

---

|     |                               |     |     |      |
|-----|-------------------------------|-----|-----|------|
| 1/1 | Ingress IPv4 VLAN ACL Lookup  |     |     |      |
|     | Ingress TCAM Entries          | 4   | 128 |      |
|     | Ingress IPv6 VLAN ACL Lookup  |     |     |      |
|     | Ingress TCAM Entries          | 8   | 128 |      |
|     | Ingress IP CPURX Lookup       |     |     |      |
|     | Ingress TCAM Entries          | 126 | 128 |      |
|     | Ingress Policers              | 19  |     |      |
|     | Ingress IP Port Policy Lookup |     |     |      |
|     | Ingress TCAM Entries          | 2   | 128 |      |
|     | Ingress IP VLAN Policy Lookup |     |     |      |
|     | Ingress TCAM Entries          | 12  | 128 |      |
|     | Total                         |     |     |      |
|     | Ingress TCAM Entries          | 152 | 640 | 3448 |
|     | Ingress Lookups               | 5   |     | 27   |
|     | Ingress Policers              | 19  |     | 2029 |

## arp cache-limit

### Syntax

```
arp cache-limit <LIMIT>
```

### Description

Specifies the maximum number of entries in the ARP (Address Resolution Protocol) cache.

### Command context

```
config
```

### Parameters

**<LIMIT>**

Specifies the maximum number of entries in the ARP cache. Range: 4096 to 131072. Default: 131072.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch(config)# arp cache-limit 4097
```

## arp ipv4 mac

### Syntax

```
arp ipv4 <IPV4_ADDR> mac <MAC_ADDR>
```

```
no arp ipv4 <IPV4_ADDR> mac <MAC_ADDR>
```

### Description

Specifies a permanent static neighbor entry in the ARP table (for IPv4 neighbors).

The `no` form of this command deletes a permanent static neighbor entry from the ARP table.

### Command context

```
config-if
```

```
config-if-vlan
```

## Parameters

**ipv4** <IPV4-ADDR>

Specifies the IP address of the neighbor or the virtual IP address of the cluster in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. . Range: 4096 to 131072. Default: 131072.

**mac** <MAC-ADDR>

Specifies the MAC address of the neighbor or the multicast MAC address in IANA format (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Configuring a static ARP entry on a interface VLAN 10:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# arp ipv4 2.2.2.2 mac 01:00:5e:00:00:01
```

Removing a static ARP entry on interface VLAN10:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# no arp ipv4 2.2.2.2 mac 01:00:5e:00:00:01
```

# clear arp

## Syntax

```
clear arp [port <PORT-ID> | vrf {all-vrfs | <VRF-NAME>}]
```

## Description

Clears IPv4 and IPv6 neighbor entries from the ARP table. If you do not specify any parameters, ARP table entries are cleared for the default VRF.

## Command context

Manager (#)

## Parameters

**port** <PORT-ID>

Specifies a physical port on the switch. Format: member/slot/port. For example: 1/1/1. .

**all-vrfs**

Selects all VRFs.

<VRF-NAME>

Specifies the name of a VRF. Default: default.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Clearing all IPv4 and IPv6 neighbor ARP entries for the default VRF:



```
switch# clear arp
```

Clearing all ARP neighbor entries for a port ():

```
switch# clear arp 1/1/35
```

## ipv6 neighbor mac

### Syntax

```
ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
```

```
no ipv6 neighbor <IPV6-ADDR> mac <MAC-ADDR>
```

### Description

Specifies a permanent static neighbor entry in the ARP table (for IPv6 neighbors).

The `no` form of this command deletes a permanent static neighbor entry from the ARP table.

### Command context

```
config-if
```

### Parameters

**<IPV6-ADDR>>**

Specifies an IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

**mac <MAC-ADDR>>**

Specifies the MAC address of the neighbor (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F. Range: 4096 to 131072. Default: 131072.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Creates a static ARP entry on interface 1/1/1.

```
switch(config)# interface 1/1/1
switch(config-if)# arp ipv6 neighbor 2001:0db8:85a3::8a2e:0370:7334 mac 00:50:56:96:df:c8
```

## ip proxy-arp

### Syntax

```
ip proxy-arp
```

```
no ip proxy-arp
```

### Description

Enables proxy ARP for the specified Layer 3 interface. Proxy ARP is supported on Layer 3 physical interfaces, LAG interfaces, and VLAN interfaces. It is disabled by default. To enable proxy ARP on an interface, routing must be enabled on that interface.

The `no` form of this command disables proxy ARP for the specified interface.

### Command context

```
config-if
config-if-vlan
config-lag-vlan
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling proxy ARP on interface 1/1/1:

```
switch# interface 1/1/1
switch(config-if)# ip proxy-arp
```

Enabling proxy ARP on VLAN 3:

```
switch# interface vlan 3
switch(config-if-vlan)# ip proxy-arp
```

Enabling proxy ARP on a LAG 11:

```
switch(config)# int lag 11
switch(config-lag-if)# ip proxy-arp
```

Disabling proxy ARP on interface 1/1/1:

```
switch# interface 1/1/1
switch(config-if)# no ip proxy-arp
```

## show arp

### Syntax

```
show arp [vsx-peer]
```

### Description

Shows the entries in the ARP (Address Resolution Protocol) table.

### Command context

Manager (#)

### Parameters

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Administrators or local user group members with execution rights for this command.

## Usage

This command displays information about ARP entries, including the IP address, MAC address, port, and state.

When no parameters are specified, the `show arp` command shows all ARP entries for the default VRF (Virtual Router Forwarding) instance.

## Examples

```
switch# show arp
```

| IPv4 Address<br>Port | MAC               | State | Port   | Physical |
|----------------------|-------------------|-------|--------|----------|
| 192.168.1.2          | 00:50:56:96:7b:e0 |       | vlan10 | 1/1/29   |
| 192.168.1.3          | 00:50:56:96:7b:ac |       | vlan10 | 1/1/1    |

Total Number Of ARP Entries Listed- 2.

## show arp inspection

### Syntax

```
show arp inspection
```

### Description

Displays ARP inspection details.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# show arp inspection
 interface Interface
 statistics DAI statistics
 vlan Show VLAN configuration
```

## show arp inspection interface

### Syntax

```
show arp inspection interface
```

### Description

Displays the current configuration of dynamic ARP inspection on a VLAN or interface.

### Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

```
switch# show arp inspection interface
```

| Interface | Trust-State |
|-----------|-------------|
| 1/1/1     | Untrusted   |

```
switch# show arp inspection interface vsx-peer
```

| Interface | Trust-State |
|-----------|-------------|
| 1/1/1     | Untrusted   |
| lag100    | Trusted     |

```
switch# show arp inspection interface 1/1/1
```

| Interface | Trust-State |
|-----------|-------------|
| 1/1/1     | Untrusted   |

## show arp state

### Syntax

```
show arp state {all | failed | incomplete | permanent | reachable | stale} [vsx-peer]
```

### Description

Shows ARP (Address Resolution Protocol) cache entries that are in the specified state.

### Command context

Operator (>) or Manager (#)

### Parameters

#### all

Shows the ARP cache entries for all VRF (Virtual Router Forwarding) instances.

#### failed

Shows the ARP cache entries that are in `failed` state. The neighbor might have been deleted.

#### incomplete

Shows the ARP cache entries that are in `incomplete` state.

An incomplete state means that address resolution is in progress and the link-layer address of the neighbor has not yet been determined. A solicitation request was sent, and the switch is waiting for a solicitation reply or a timeout.

#### **permanent**

Shows the ARP cache entries that are in `permanent` state. ARP entries that are in a permanent state can be removed by administrative action only.

#### **reachable**

Shows the ARP cache entries that are in `reachable` state, meaning that the neighbor is known to have been reachable recently.

#### **stale**

Shows ARP cache entries that are in `stale` state.

ARP cache entries are in the `stale` state if the elapsed time is in excess of the ARP timeout in seconds since the last positive confirmation that the forwarding path was functioning properly.

#### **[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### **Authority**

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### **Examples**

```
switch# show arp state failed
```

| IPv4 Address | MAC | Port   | Physical Port | State  |
|--------------|-----|--------|---------------|--------|
| 192.168.1.4  |     | vlan10 |               | failed |

## **show arp summary**

### **Syntax**

```
show arp summary [all-vrfs | vrf <VRF-NAME>] [vsx-peer]
```

### **Description**

Shows a summary of the IPv4 and IPv6 neighbor entries on the switch for all VRFs or a specific VRF.

### **Command context**

Operator (>) or Manager (#)

### **Parameters**

#### **all-vrfs**

Selects all VRFs.

**vrf** <VRF-NAME>

Specifies the name of a VRF.

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing summary ARP information for all VRFs:

```
switch# show arp summary all-vrfs
```

| ARP Entry's State                | : IPv4 | IPv6 |
|----------------------------------|--------|------|
| Number of Reachable ARP entries  | : 2    | 0    |
| Number of Stale ARP entries      | : 0    | 0    |
| Number of Failed ARP entries     | : 2    | 2    |
| Number of Incomplete ARP entries | : 0    | 0    |
| Number of Permanent ARP entries  | : 0    | 0    |
| -----                            |        |      |
| Total ARP Entries: 6             | : 4    | 2    |
| -----                            |        |      |

Showing a summary of all IPv4 and IPv6 neighbor entries on the primary and secondary (peer) switches:

```
vsx-primary# show arp summary
```

| ARP Entry's State                | IPv4  | IPv6  |
|----------------------------------|-------|-------|
| Number of Reachable ARP entries  | 25858 | 32231 |
| Number of Stale ARP entries      | 0     | 1     |
| Number of Failed ARP entries     | 0     | 257   |
| Number of Incomplete ARP entries | 0     | 0     |
| Number of Permanent ARP entries  | 0     | 0     |
| -----                            |       |       |
| Total ARP Entries- 58347         | 25858 | 32489 |

```
vsx-primary# show arp summary vsx-peer
```

| ARP Entry's State                | IPv4  | IPv6  |
|----------------------------------|-------|-------|
| Number of Reachable ARP entries  | 25858 | 32168 |
| Number of Stale ARP entries      | 0     | 3     |
| Number of Failed ARP entries     | 0     | 317   |
| Number of Incomplete ARP entries | 0     | 0     |
| Number of Permanent ARP entries  | 0     | 0     |
| -----                            |       |       |

## show arp timeout

### Syntax

```
show arp timeout [<INTERFACE>] [vsx-peer]
```

### Description

Shows the age-out period for each ARP (Address Resolution Protocol) entry for a port, LAG, or VLAN interface.

### Command context

Operator (>) or Manager (#)

### Parameters

#### <INTERFACE>

Specifies a physical port, VLAN, or LAG on the switch. For physical ports, use the format `member/slot/port` (for example, 1/3/1).

#### [vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing ARP timeout information for a VLAN:

```
switch# show arp timeout vlan4
```

Showing ARP timeout information for a port:

```
switch# show arp timeout 1/1/1
```

ARP Timeout:

-----

| Port  | VRF     | Timeout |
|-------|---------|---------|
| 1/1/1 | default | 600     |

## show arp vrf

### Syntax

```
show arp {all-vrfs | vrf <VRF-NAME>} [vsx-peer]
```

## Description

Shows the ARP table for all VRF instances, or for the named VRF.

## Command context

Operator (>) or Manager (#)

## Parameters

### **all-vrfs**

Specifies all VRFs.

### **vrf <VRF-NAME>**

Specifies the name of a VRF. Length: 1 to 32 alphanumeric characters.

### **[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing ARP entries for VRF **test**.

```
switch# show arp vrf test
ARP IPv4 Entries:

IPv4 Address MAC Port Physical Port State VRF
10.20.30.40 00:50:56:bd:6a:c5 1/1/29 1/1/29 reachable test

Total Number Of ARP Entries Listed: 1.

```

```
switch# show arp all-vrfs
ARP IPv4 Entries:

IPv4 Address MAC Port Physical Port State VRF
192.168.120.10 00:50:56:bd:10:be 1/1/32 1/1/32 reachable red
10.20.30.40 00:50:56:bd:6a:c5 1/1/29 1/1/29 reachable test

Total Number Of ARP Entries Listed: 2.

```

Showing ARP entries for all VRFs.

```
switch# show arp all-vrfs
ARP IPv4 Entries:

IPv4 Address MAC Port Physical Port State VRF
192.168.120.10 00:50:56:bd:10:be 1/1/32 1/1/32 reachable red
10.20.30.40 00:50:56:bd:6a:c5 1/1/29 1/1/29 reachable test

Total Number Of ARP Entries Listed: 2.

```



## show ipv6 neighbors

### Syntax

```
show ipv6 neighbors {all-vrfs | vrf <VRF-NAME>} [vsx-peer]
```

### Description

Shows entries in the ARP table for all IPv6 neighbors for all VRFs or for a specific VRF.

When no parameters are specified, this command shows all ARP entries for the default VRF, and state information for `reachable` and `stale` entries only.

### Command context

Operator (>) or Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Parameters

#### **all-vrfs**

Specifies all VRFs.

#### **vrf <VRF-NAME>**

Specifies a VRF name. Length: 1 to 32 alphanumeric characters.

#### **[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Examples

```
switch# show ipv6 neighbors
IPv6 Entries:
```

| -----                                             |                   |          |               |           |
|---------------------------------------------------|-------------------|----------|---------------|-----------|
| IPv6 Address                                      | MAC               | Port     | Physical Port | State     |
| fe80::a21d:48ff:fe8f:2700                         | a0:1d:48:8f:27:00 | vlan2300 | 1/1/31        | reachable |
| fe80::f603:43ff:fe80:a600                         | f4:03:43:80:a6:00 | vlan2300 | 1/1/30        | reachable |
| -----                                             |                   |          |               |           |
| Total Number Of IPv6 Neighbors Entries Listed: 2. |                   |          |               |           |
| -----                                             |                   |          |               |           |

## show ipv6 neighbors state

### Syntax

```
show ipv6 neighbors state {all | failed | incomplete | permanent | reachable | stale} [vsx-peer]
```

## Description

Shows all IPv6 neighbor ARP (Address Resolution Protocol) cache entries, or those cache entries that are in the specified state.

## Command context

Operator (>) or Manager (#)

## Parameters

### **all**

Shows all ARP cache entries.

### **failed**

Shows ARP cache entries that are in `failed` state. The neighbor might have been deleted. Set the neighbor to be unreachable.

### **incomplete**

Shows ARP cache entries that are in `incomplete` state.

An incomplete state means that address resolution is in progress and the link-layer address of the neighbor has not yet been determined. This means that a solicitation request was sent, and you are waiting for a solicitation reply or a timeout.

### **permanent**

Shows ARP cache entries that are in `permanent` state.

### **reachable**

Shows ARP cache entries that are in `reachable` state, meaning that the neighbor is known to have been reachable recently.

### **stale**

Shows ARP cache entries that are in `stale` state.

ARP cache entries are in the `stale` state if the elapsed time is in excess of the ARP timeout in seconds since the last positive confirmation that the forwarding path was functioning properly.

### **[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ipv6 neighbors state all
```

| IPv6 Address              | MAC               | Port   | Physical Port | State     |
|---------------------------|-------------------|--------|---------------|-----------|
| 100::2                    | 48:0f:cf:af:f1:cc | lag1   | lag1          | reachable |
| 300::3                    | 48:0f:cf:af:33:be | vlan3  | 1/4/20        | reachable |
| fe80::4a0f:cfff:feaf:f1cc | 48:0f:cf:af:f1:cc | lag1   | lag1          | reachable |
| 200::3                    | 48:0f:cf:af:33:be | 1/4/11 | 1/4/11        | reachable |
| fe80::4a0f:cfff:feaf:33be | 48:0f:cf:af:33:be | vlan3  | 1/4/20        | reachable |

Total Number Of IPv6 Neighbors Entries Listed- 5.

---

## aruba-central

### Syntax

```
aruba-central

no aruba-central
```

### Description

Creates or enters the Aruba Central configuration context (`config-aruba-central`).

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Creating the Aruba Central configuration context:

```
switch(config)# aruba-central
switch(config-aruba-central)#
```

## disable

### Syntax

```
disable

no disable
```

### Description

Disables connection to Aruba Central server.

When the connection is disabled, the switch does not attempt to connect to the Aruba Central server or fetch central location from any of the three sources (CLI/Aruba Activate/DHCP). It also disconnects any active connection to the Aruba Central server.

The `no` form of this command enables connection to the Aruba Central server. The `no disable` command is identical to the `enable` command.

### Command context

```
config-aruba-central
```

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config-aruba-central)# disable
switch(config-aruba-central)#
```

# enable

## Syntax

enable

no enable

## Description

Enables connection to Aruba Central server. When the connection is enabled, the switch attempts to download the location of the Aruba Central server in one of the following ways at startup and after the connection is lost:

- Using command-line interface (CLI).
- Connecting to Aruba Activate server.
- Using DHCP options provided during ZTP.

DHCP servers are used to connect to Central on-premise management.

The `no` form of this command disables connection to the Aruba Central server. The `no enable` command is identical to the `disable` command.

## Command context

config-aruba-central

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config-aruba-central)# enable
switch(config-aruba-central)#
```

# location-override

## Syntax

location-override <location> [vrf <VRF-NAME>]  
no location-override

## Description

When `location` and `vrf` are configured, the switch overrides existing connections to Aruba Central. The switch attempts to establish connection to Aruba Central with the specified location and VRF with highest priority.

The `no` form of this command removes location override values from the Aruba Central configuration context.

### Command context

`config-aruba-central`

### Authority

Administrators or local user group members with execution rights for this command.

### Parameters

#### <location>

Specifies one of these values:

- <FQDN>: a fully qualified domain name.
- <IPv4>: an IPv4 address.
- <IPv6>: an IPv6 address.

#### vrf <VRF-NAME>

Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

### Examples

Configuring location override with location and VRF:

```
switch(config-aruba-central)# location-override aruba-central.com vrf default
switch(config-aruba-central)#
```

Configuring location override with location only:

```
switch(config-aruba-central)# location-override aruba-central.com
switch(config-aruba-central)#
```

Removing location override values from the Aruba Central configuration context:

```
switch(config-aruba-central)# no location-override
switch(config-aruba-central)#
```

## show aruba-central

### Syntax

`show aruba-central`

### Description

Shows information about Aruba Central connection and the status of the Activate server connection.

### Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Usage

This command shows the following information about the Aruba Network Analytics Engine agents that are available on the switch:

## Examples

Example of a switch that has the Aruba Central connection:

```
switch# show aruba-central
Central admin state :enabled

Central location :N/A
VRF for connection :N/A
Central connection status :N/A

Central source : dhcp
Central source connection status : connection_failure
Central source last connected on : N/A
System time synchronized from Activate : True

Activate server URL : 172.17.0.1
CLI location : N/A
CLI VRF : N/A
```

# show running-config current-context

## Syntax

```
show running-config current-context
```

## Description

Shows the running configuration for the current-context. If user is in the context of Aruba-Central(config-aruba-central), then Aruba Central running configuration is displayed.

## Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Shows the running configuration of Aruba Central:

```
switch(config-aruba-central)# show running-config current-context
aruba-central
 disable
```

## banner

### Syntax

```
banner {motd | exec} <DELIMITER>
no banner {motd | exec}
```

### Description

Enables the customization of the MOTD or the EXEC banner.

The `no` form of the command disables the MOTD or the EXEC banner.

### Command context

`config`

### Parameters

#### **exec**

Configures the banner displayed after the login prompt.

#### **motd**

Configures the banner displayed before the password prompt.

#### **<DELIMITER>**

Selects a character that will terminate the input string.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

This command enables the customization of two types of banners:

- **The MOTD banner.** The banner displayed on attempting to connect to a management interface.
- **The EXEC banner.** The banner displayed upon successful authentication.

You can create a banner that spans multiple lines. The maximum length of a banner is 4,095 characters. This requirement includes any non-visible characters. The minimum number of characters allowed is an empty string, which displays no banner.

End the banner text with a chosen delimiter character. A delimiter character can be any non-whitespace character that does not have special meaning to the CLI, such as the caret (^). A question mark (?) is not permitted. Question marks can however be included as part of the banner text.

### Examples

Configuring the banner displayed before the password prompt:



```
switch(config)# banner motd ^
Enter a new banner. Terminate the banner with the delimiter you have chosen.
(banner-motd)# This is an example of a banner text which a connecting user
(banner-motd)# will see before they are prompted for their password.
(banner-motd)#
(banner-motd)# As you can see it may span multiple lines and the input
(banner-motd)# will be terminated when the delimiter character is
(banner-motd)#encountered.^
```

Configuring the banner displayed after a user has logged on to the switch:

```
switch(config)# banner exec &
Enter a new banner. Terminate the banner with the delimiter you have chosen.
(banner-motd)# This is an example of a different banner text. This time
(banner-motd)# the banner entered will be displayed after a user has
(banner-motd)# authenticated.
(banner-motd)#
(banner-motd)# & This text will not be included because it comes after the '&'
```

Disabling the MOTD banner:

```
switch(config)# no banner motd
```

## show banner

### Syntax

```
show banner {motd | exec} [vsx-peer]
```

### Description

Displays the MOTD or EXEC banner message.

### Command context

config

### Parameters

#### **motd**

Displays the banner shown before the password prompt.

#### **exec**

Displays the banner shown after login.

#### **[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Displaying the MOTD banner shown before the password prompt:

```
switch# show banner motd
This is an example of a banner text which a connecting user
```

will see before they are prompted for their password.

As you can see it may span multiple lines and the input will be terminated when the delimiter character is encountered.

## boot set-default

### Syntax

```
boot set-default {primary | secondary}
```

### Description

Sets the default operating system image to use when the system is booted.

### Command context

Manager (#)

### Parameters

#### primary

Selects the primary network operating system image.

#### secondary

Selects the secondary network operating system image.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```

## boot system

### Syntax

```
boot system [primary | secondary | serviceos]
```

### Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

### Command context

Manager (#)

## Parameters

### **primary**

Selects the primary operating system image for this reboot and sets the configured default operating system image to `primary` for future reboots.

### **secondary**

Selects the secondary operating system image for this reboot and sets the configured default operating system image to `secondary` for future reboots.

### **serviceos**

Selects the service operating system for this reboot. Does not change the configured default operating system image. The service operating system acts as a standalone bootloader and recovery OS for switches running the ArubaOS-CX operating system and is used in rare cases when troubleshooting a switch.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the `show images` command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the `primary` or `secondary` optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.

You can use the `boot set-default` command to change the configured default operating system image.

- If you select `serviceos` as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the `boot system` command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the `boot system` command is aborted.

## Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:

```
switch# boot system

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#
```

## show boot-history

### Syntax

```
show boot-history [all]
```

### Description

Shows boot information. When no parameters are specified, shows the most recent information about the boot operation, and the three previous boot operations for the active management module. When the `all` parameter is specified, shows the boot information for the active management module.

### Command context

Manager (#)

### Parameters

**all**

Shows boot information for the active management module and all available line modules.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

### Index

The position of the boot in the history file. Range: 0 to3.

## Boot ID

A unique ID for the boot . A system-generated 128-bit string.

## Current Boot, up for *<SECONDS>* seconds

For the current boot, the `show boot-history` command shows the number of seconds the module has been running on the current software.

## Timestamp boot reason

For previous boot operations, the `show boot-history` command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values:

### *<DAEMON-NAME>* crash

The daemon identified by *<DAEMON-NAME>* caused the module to boot.

### Kernel crash

The operating system software associated with the module caused the module to boot.

### Reboot requested through database

The reboot occurred because of a request made through the CLI or other API.

### Uncontrolled reboot

The reason for the reboot is not known.

## Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=====

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
switch#
```

## diag cable-diagnostic

### Syntax

```
diag cable-diagnostic <IF-NAME>
```

### Description

Runs a cable diagnostic test on an interface.

### Command context

Manager (#)

### Parameters

#### **IF-NAME**

Specifies the name of the interface.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Running a cable diagnostic test on interfaces:

```
switch# diag cable-diagnostic 1/1/1
```

This command will cause a loss of link on the port under test and will take several seconds to complete.

Continue (y/n)? **y**

| Interface | MDI<br>Pair | Cable<br>Status | Distance<br>to Fault<br>(Meters) | MDI<br>Mode |
|-----------|-------------|-----------------|----------------------------------|-------------|
| 1/1/1     | 1-2         | good            | 5                                | mdi         |
|           | 3-6         | good            | 5                                | mdi         |
|           | 4-5         | good            | 5                                | mdi         |
|           | 7-8         | open            | 3                                |             |

```
switch# diag cable-diagnostic 1/1/2
```

This command will cause a loss of link on the port under test and will take several seconds to complete.

Continue (y/n)? **y**

| Interface | MDI<br>Pair | Cable<br>Status | Distance<br>to Fault<br>(Meters) | MDI<br>Mode |
|-----------|-------------|-----------------|----------------------------------|-------------|
| 1/1/2     | 1-2         | good            | 5                                | mdix        |
|           | 3-6         | good            | 5                                | mdix        |

|     |       |   |      |
|-----|-------|---|------|
| 4-5 | short | 1 |      |
| 7-8 | good  | 5 | mdix |



---

**NOTE:**

Running a cable diagnostic test will result in a brief interruption in connectivity on all tested ports.

---



## aaa authentication port-access captive-portal-profile

### Syntax

```
aaa authentication port-access captive-portal-profile <PROFILE-NAME>
no aaa authentication port-access captive-portal-profile <PROFILE-NAME>
```

### Description

Creates the specified captive portal profile (if it does not yet exist) and then enters its context. For existing captive portal profiles, this command enters the context of the specified captive portal profile.

The `no` form of this command deletes the specified captive portal profile.

### Command context

config

### Parameters

**<PROFILE-NAME>**

Specifies the captive portal profile name. Up to 64 characters.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating a captive portal profile named `employee` and entering its context for additional configuration:

```
switch(config)# aaa authentication port-access captive-portal-profile employee
switch(config-captive-portal)# url http://1.1.1.1/employee/captiveportal.php
switch(config-captive-portal)#
switch(config-captive-portal)# url-hash-key plaintext cjQrJ9#$erty
switch(config-captive-portal)#
switch(config-captive-portal)# exit
switch(config)#
```

Deleting the captive portal profile named `employee`:

```
switch(config)# no aaa authentication port-access captive-portal-profile employee
switch(config)#
```

## url

### Syntax

```
url <URL>
```

```
no url
```

### Description

Within the captive portal context, defines the captive portal URL.

The `no` form of this command deletes the captive portal URL.

### Command context

```
config-captive-portal
```

### Parameters

**<URL>**

Specifies the captive portal URL. Up to 1024 characters.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating a captive portal profile named `employee` and then setting its URL:

```
switch(config)# aaa authentication port-access captive-portal-profile employee
switch(config-captive-portal)# url http://1.1.1.1/employee/captiveportal.php
switch(config-captive-portal)#
switch(config-captive-portal)# exit
switch(config)#
```

Entering the captive portal profile `employee` and then deleting its URL:

```
switch(config)# aaa authentication port-access captive-portal-profile employee
switch(config-captive-portal)# no url
switch(config-captive-portal)#
switch(config-captive-portal)# exit
switch(config)#
```

## url-hash-key

### Syntax

```
url-hash-key {plaintext | ciphertext} <HASH-KEY>
```

```
no url-hash-key
```

### Description

Within the captive portal context, defines the captive portal hash key.

The `no` form of this command deletes the captive portal URL hash key.

## Command context

config-captive-portal

## Parameters

{plaintext | ciphertext}

Selects the URL hash key type as either `plaintext` or `ciphertext`.

<HASH-KEY>

Specifies the captive portal URL hash key. Up to 128 characters.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Creating a captive portal profile named `employee` and then setting its URL and URL hash key:

```
switch(config)# aaa authentication port-access captive-portal-profile employee
switch(config-captive-portal)# url http://1.1.1.1/employee/captiveportal.php
switch(config-captive-portal)#
switch(config-captive-portal)# url-hash-key plaintext cjQrJ9#$erty
switch(config-captive-portal)#
switch(config-captive-portal)# exit
switch(config)#
```

Entering the captive portal profile `employee` and then deleting its URL hash key:

```
switch(config)# aaa authentication port-access captive-portal-profile employee
switch(config-captive-portal)# no url-hash-key
switch(config-captive-portal)#
switch(config-captive-portal)# exit
switch(config)#
```

# show port-access captive-portal-profile

## Syntax

show port-access captive-portal-profile

## Description

Shows the configuration information for all captive portal profiles .

## Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Entering the `employee` captive portal profile context and then showing the profile configuration information:

```
switch# show port-access captive-portal-profile
```

Captive Portal Profile Configuration

|              |                                                |
|--------------|------------------------------------------------|
| Name         | : employee                                     |
| Type         | : local                                        |
| URL          | : http://1.1.1.1/employee/captiveportal.php    |
| URL Hash Key | : SWNGWyMeYubHPDgVIirpEUwNK5Uf+r1vmhBIncQPw1Y= |

# cdp

## Syntax

cdp

## Description

Configures CDP support globally on all active interfaces or on a specific interface. By default, CDP is enabled on all active interfaces.

When CDP is enabled, the switch adds entries to its CDP Neighbors table for any CDP packets it receives from neighboring CDP devices.

When CDP is disabled, the CDP Neighbors table is cleared and the switch drops all inbound CDP packets without entering the data in the CDP Neighbors table.

The `no` form of this command disables CDP support globally on all active interfaces or on a specific interface.

## Command context

config

config-if

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

*Not supported on the 6100 Switch Series.*

Enabling CDP globally:

```
switch(config)# cdp
```

Disabling CDP globally:

```
switch(config)# no cdp
```

Enabling CDP on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# cdp
```

Disabling CDP on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no cdp
```

## clear cdp counters

### Syntax

```
clear cdp counters
```

### Description

Clears CDP counters.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Clearing CDP counters:

```
switch(config) clear cdp counters
```

## clear cdp neighbor-info

### Syntax

```
clear cdp neighbor-info
```

### Description

Clears CDP neighbor information.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Clearing CDP neighbor information:

```
switch(config) clear neighbor-info
```

## show cdp

### Syntax

```
show cdp
```

### Description

Shows CDP information for all interfaces.

## Command context

config

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing CDP information:

```
switch(config)# show cdp
CDP Global Information
=====
CDP : Enabled
CDP Mode : Rx only
CDP Hold Time : 180 seconds
```

| Port   | CDP     |
|--------|---------|
| 1/1/1  | Enabled |
| 1/1/2  | Enabled |
| 1/1/3  | Enabled |
| 1/1/4  | Enabled |
| 1/1/5  | Enabled |
| 1/1/6  | Enabled |
| 1/1/7  | Enabled |
| 1/1/8  | Enabled |
| 1/1/9  | Enabled |
| 1/1/10 | Enabled |

## show cdp neighbor-info

### Syntax

```
show cdp neighbor-info <INTERFACE-ID>
```

### Description

Shows CDP information for all neighbors or for CDP information on a specific interface.

## Command context

config

## Parameters

<INTERFACE-ID>

Specifies an interface. Format: member/slot/port.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing all CDP neighbor information:

```
switch(config)# show cdp neighbor-info
Port Device ID Platform Capability
```

Showing CDP information for interface 1/1/1:

```
switch(config)# show cdp neighbor-info 1/1/1
Local Port : 1/1/1
MAC : 3c:a8:2a:7b:6b:2b
Device ID : SEPd4adbd2a30d6
Address : 2.71.0.230
Platform : Cisco IP Phone 3905
Duplex : full
Capability : host
Voice VLAN Support : Yes
Neighbor Port-ID : Port 1
```

## show cdp traffic

### Syntax

```
show cdp neighbor-info
```

### Description

Shows CDP statistics for each interface.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Showing CDP traffic statistics:

```
switch(config)# show cdp traffic
```

```
CDP Statistics
```

```
=====
```

| Port  | Transmitted Frames | Received Frames | Discarded Frames |
|-------|--------------------|-----------------|------------------|
| 1/1/1 | 0                  | 4               | 0                |
| 1/1/2 | 0                  | 0               | 0                |
| 1/1/3 | 0                  | 2               | 0                |
| 1/1/4 | 0                  | 0               | 0                |
| 1/1/5 | 0                  | 0               | 0                |



## checkpoint auto

### Syntax

```
checkpoint auto <TIME-LAPSE-INTERVAL>
```

### Description

Starts auto checkpoint mode. In auto checkpoint mode, the switch temporarily saves the runtime configuration as a checkpoint only for the specified time lapse interval. Configuration changes must be saved before the interval expires, otherwise the runtime configuration is restored from the temporary checkpoint.

### Command context

Manager (#)

### Parameters

**<TIME-LAPSE-INTERVAL>**

Specifies the time lapse interval in minutes. Range: 1 to 60.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

To save the runtime checkpoint permanently, run the `checkpoint auto confirm` command during the time lapse interval. The filename for the saved checkpoint is named `AUTO<YYYYMMDDHHMMSS>`. If the `checkpoint auto confirm` command is not entered during the specified time lapse interval, the previous runtime configuration is restored.

### Examples

Confirming the auto checkpoint:

```
switch# checkpoint auto 20
Auto checkpoint mode expires in 20 minute(s)
switch# WARNING Please "checkpoint auto confirm" within 2 minutes
switch# checkpoint auto confirm
checkpoint AUTO20170801011154 created
```

In this example, the runtime checkpoint was saved because the `checkpoint auto confirm` command was entered within the value set by the `time-lapse-interval` parameter, which was 20 minutes.

Not confirming the auto checkpoint:

```
switch# checkpoint auto 20
Auto checkpoint mode expires in 20 minute(s)
switch# WARNING Please "checkpoint auto confirm" within 2 minutes
WARNING: Restoring configuration. Do NOT add any new configuration.
Restoration successful
```

In this example, the runtime checkpoint was reverted because the `checkpoint auto confirm` command was not entered within the value set by the `time-lapse-interval` parameter, which was 20 minutes.

## checkpoint auto confirm

### Syntax

```
checkpoint auto confirm
```

### Description

Signals to the switch to save the running configuration used during the auto checkpoint mode. This command also ends the auto checkpoint mode.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

To save the runtime checkpoint permanently, run the `checkpoint auto confirm` command during the time lapse value set by the `checkpoint auto <TIME-LAPSE-INTERVAL>` command. The generated checkpoint name will be in the format `AUTO<YYYYMMDDHHMMSS>`. If the `checkpoint auto confirm` command is not entered during the specified time lapse interval, the previous runtime configuration is restored.

### Examples

Confirming the auto checkpoint:

```
switch# checkpoint auto confirm
```

## checkpoint diff

### Syntax

```
checkpoint diff {<CHECKPOINT-NAME1> | running-config | startup-config}
 {<CHECKPOINT-NAME2> | running-config | startup-config}
```

### Description

Shows the difference in configuration between two configurations. Compare checkpoints, the running configuration, or the startup configuration.

### Command context

Manager (#)

### Parameters

```
{<CHECKPOINT-NAME1> | running-config | startup-config}
```

Selects either a checkpoint, the running configuration, or the startup configuration as the baseline.

{<CHECKPOINT-NAME2> | running-config | startup-config}

Selects either a checkpoint, the running configuration, or the startup configuration to compare.

### Authority

Administrators or local user group members with execution rights for this command.

### Usability

The output of the `checkpoint diff` command has several symbols:

- The plus sign (+) at the beginning of a line indicates that the line exists in the comparison but not in the baseline.
- The minus sign (-) at the beginning of a line indicates that the line exists in the baseline but not in the comparison.

### Examples

In the following example, the configurations of checkpoints `cp1` and `cp2` are displayed before the `checkpoint diff` command, so that you can see the context of the `checkpoint diff` command.

```
Switch# checkpoint diff chkpt01 chkpt02
--- /tmp/chkpt011607564301327
+++ /tmp/chkpt021607564301353
@@ -1,7 +1,7 @@
!
!Version ArubaOS-CX PL.10.06.0100V
!export-password: default
-hostname Switch
+hostname Switch1
 user admin group administrators password ciphertext
AQBapTyg9tpaiAaTfSVV5eNdFzOORRvZ6CMpglh1P+LQUHQLYgAAAGAhmRqFbkNvrgy2SBVk7H8C5hvg/
Iib8rWYFZLEaSCrobNP9EwMu+hLNM0xmsh45yG8dncP7WkxjwrW4p4Qra6dVfr0EW8xh/lpQf8F/
2Wki20Lc9JLXiYge7ti0H6cVn+G
 radius-server tracking interval 60
 no usb

Switch1#
```

## checkpoint post-configuration

### Syntax

```
checkpoint post-configuration
```

```
no checkpoint post-configuration
```

### Description

Enables creation of system generated checkpoints when configuration changes occur. This feature is enabled by default.

The `no` form of this command disables system generated checkpoints.

### Command context

Manager (#)

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

System generated checkpoints are automatically created by default. Whenever a configuration change occurs, the switch starts a timeout counter (300 seconds by default). For each additional configuration change, the timeout counter is restarted. If the timeout expires with no additional configuration changes being made, the switch generates a new checkpoint.

System generated checkpoints are named with the prefix `CPC` followed by a time stamp in the format `<YYYYMMDDHHMMSS>`. For example: `CPC20170630073127`.

System checkpoints can be applied using the checkpoint rollback feature or copy command.

A maximum of 32 system checkpoints can be created. Beyond this limit, the newest system checkpoint replaces the oldest system checkpoint.

## Examples

Enabling system checkpoints:

```
switch(config)# checkpoint post-configuration
```

Disabling system checkpoints:

```
switch(config)# no checkpoint post-configuration
```

# checkpoint post-configuration timeout

## Syntax

```
checkpoint post-configuration timeout <TIMEOUT>
```

```
no checkpoint post-configuration timeout <TIMEOUT>
```

## Description

Sets the timeout for the creation of system checkpoints. The timeout specifies the amount of time since the latest configuration for the switch to create a system checkpoint.

The `no` form of this command resets the timeout to 300 seconds, regardless of the value of the *<TIMEOUT>* parameter.

## Command context

Manager (#)

## Parameters

**timeout** *<TIMEOUT>*

Specifies the timeout in seconds. Range: 5 to 600. Default: 300.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the timeout for system checkpoints to 60 seconds:

```
switch(config)# checkpoint post-configuration timeout 60
```

Resetting the timeout for system checkpoints to 300 seconds:

```
switch(config)# no checkpoint post-configuration timeout 1
```

## checkpoint rename

### Syntax

```
checkpoint rename <OLD-CHECKPOINT-NAME> <NEW-CHECKPOINT-NAME>
```

### Description

Renames an existing checkpoint.

### Command context

Manager (#)

### Parameters

**<OLD-CHECKPOINT-NAME>**

Specifies the name of an existing checkpoint to be renamed.

**<NEW-CHECKPOINT-NAME>**

Specifies the new name for the checkpoint. The checkpoint name can be alphanumeric. It can also contain underscores (\_) and dashes (-).



**NOTE:** Do not start the checkpoint name with `CPC` because it is used for system-generated checkpoints.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Renaming checkpoint `ckpt1` to `cfg001`:

```
switch# checkpoint rename ckpt1 cfg001
```

## checkpoint rollback

### Syntax

```
checkpoint rollback {<CHECKPOINT-NAME> | startup-config}
```

### Description

Applies the configuration from a pre-existing checkpoint or the startup configuration to the running configuration.

### Command context

Manager (#)

## Parameters

**<CHECKPOINT-NAME>**

Specifies a checkpoint name.

**startup-config**

Specifies the startup configuration.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Applying a checkpoint named `ckpt1` to the running configuration:

```
switch# checkpoint rollback ckpt1
Success
```

Applying a startup checkpoint to the running configuration:

```
switch# checkpoint rollback startup-config
Success
```

# copy checkpoint <CHECKPOINT-NAME> <REMOTE-URL>

## Syntax

```
copy checkpoint <CHECKPOINT-NAME> <REMOTE-URL> [vrf <VRF-NAME>]
```

## Description

Copies a checkpoint configuration to a remote location as a file. The configuration is exported in checkpoint format, which includes switch configuration and relevant metadata.

## Command context

Manager (#)

## Parameters

**<CHECKPOINT-NAME>**

Specifies the name of a checkpoint.

**<REMOTE-URL>**

Specifies the remote destination and filename using the syntax: `{tftp | sftp}://<IP-ADDRESS>[:<PORT-NUMBER>] [;blocksize=<BLOCKSIZE-VALUE>] /<FILE-NAME>`

**vrf <VRF-NAME>**

Specifies a VRF name.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Copying checkpoint configuration to remote file through TFTP:

```
switch# copy checkpoint ckpt1 tftp://192.168.1.10/ckptmeta vrf default
100.0%
Success
```

Copying checkpoint configuration to remote file through SFTP:

```
switch# copy checkpoint ckpt1 sftp://root@192.168.1.10/ckptmeta vrf default
The authenticity of host '192.168.1.10 (192.168.1.10)' can't be established.
ECDSA key fingerprint is SHA256:FtOm6Uxuxumil7VCwLnhz92H9LkjY+eURbdddOETy50.
Are you sure you want to continue connecting (yes/no)? yes
root@192.168.1.10's password:
sftp> put /tmp/ckptmeta ckptmeta
Uploading /tmp/ckptmeta to /root/ckptmeta
Warning: Permanently added '192.168.1.10' (ECDSA) to the list of known hosts.
Connected to 192.168.1.10.
Success
```

## copy checkpoint <CHECKPOINT-NAME> {running-config | startup-config}

### Syntax

```
copy checkpoint <CHECKPOINT-NAME> {running-config | startup-config}
```

### Description

Copies an existing checkpoint configuration to the running configuration or to the startup configuration.

### Command context

Manager (#)

### Parameters

<CHECKPOINT-NAME>

Specifies the name of an existing checkpoint.

{running-config | startup-config}

Selects whether the running configuration or the startup configuration receives the copied checkpoint configuration. If the startup configuration is already present, the command overwrites the startup configuration.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Copying **ckpt1** checkpoint to the running configuration:

```
switch# copy checkpoint ckpt1 running-config
Success
```

Copying **ckpt1** checkpoint to the startup configuration:

```
switch# copy checkpoint ckpt1 startup-config
Success
```

## copy checkpoint <CHECKPOINT-NAME> <STORAGE-URL>

### Syntax

```
copy checkpoint <CHECKPOINT-NAME> <STORAGE-URL>
```

### Description

Copies an existing checkpoint configuration to a USB drive. The file format is defined when the checkpoint was created.

### Command context

Manager (#)

### Parameters

<CHECKPOINT-NAME>

Specifies the name of the checkpoint to copy. The checkpoint name can be alphanumeric. It can also contain underscores (\_) and dashes (-).

<STORAGE-URL>>

Specifies the name of the target file on the USB drive using the following syntax: `usb: /<FILE>`

The USB drive must be formatted with the FAT file system.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Copying the `test` checkpoint to the `testCheck` file on the USB drive:

```
switch# copy checkpoint test usb:/testCheck
Success
```

## copy <REMOTE-URL> checkpoint <CHECKPOINT-NAME>

### Syntax

```
copy <REMOTE-URL> checkpoint <CHECKPOINT-NAME> [vrf <VRF-NAME>]
```

### Description

Copies a remote configuration file to a checkpoint. The remote configuration file must be in checkpoint format.

### Command context

Manager (#)

### Parameters

<REMOTE-URL>

Specifies a remote file using the following syntax: `{tftp | sftp}://<IP-ADDRESS>[:<PORT-NUMBER>][;blocksize=<BLOCKSIZE-VALUE>]/<FILE-NAME>>`



### <CHECKPOINT-NAME>

Specifies the name of the target checkpoint. The checkpoint name can be alphanumeric. It can also contain underscores (\_) and dashes (-). Required.



**NOTE:** Do not start the checkpoint name with `CPC` because it is used for system-generated checkpoints.

### **vrf** <VRF-NAME>

Specifies a VRF name. Default: `default`.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Copying a checkpoint format file to checkpoint `ckpt5` on the default VRF:

```
switch# copy tftp://192.168.1.10/ckptmeta checkpoint ckpt5
100.0%
100.0%
Success
```

## **copy <REMOTE-URL> {running-config | startup-config}**

### Syntax

```
copy <REMOTE-URL> {running-config | startup-config } [vrf <VRF-NAME>]
```

### Description

Copies a remote file containing a switch configuration to the running configuration or to the startup configuration.

### Command context

Manager (#)

### Parameters

#### <REMOTE-URL>

Specifies a remote file with the following syntax: {tftp | sftp}://<IP-ADDRESS>[:<PORT-NUMBER>] [;blocksize=<BLOCKSIZE-VALUE>] /<FILE-NAME>

#### {running-config | startup-config}

Selects whether the running configuration or the startup configuration receives the copied checkpoint configuration. If the startup configuration is already present, the command overwrites the startup configuration.

#### **vrf** <VRF-NAME>

Specifies the name of a VRF. Default: `default`.

### Authority

Administrators or local user group members with execution rights for this command.

## Usage

The switch copies only certain file types. The format of the file is automatically detected from contents of the file. The `startup-config` option only supports the JSON file format and checkpoints, but the `running-config` option supports the JSON and CLI file formats and checkpoints.

When a file of the CLI format is copied, it overwrites the running configuration. The CLI command does not clear the running configuration before applying the CLI commands. All of the CLI commands in the file are applied line-by-line. If a particular CLI command fails, the switch logs the failure and it continues to the next line in the CLI configuration. The event log (`show events -d hpe-config`) provides information as to which command failed.

## Examples

Copying a JSON format file to the running configuration:

```
switch# copy tftp://192.168.1.10/runjson running-config
100.0%
Configuration may take several minutes to complete according to configuration file size
--0%---10%---20%---30%---40%---50%---60%---70%---80%---90%---100%--
Success
```

Copying a CLI format file to the running configuration with an error in the file:

```
switch# copy tftp://192.168.1.10/runcli running-config
100.0%
Configuration may take several minutes to complete according to configuration file size
--0%---10%---20%---30%---40%---50%---60%---70%---80%---90%---100%--
Some of the configuration lines from the file were NOT applied. Use 'show
events -d hpe-config' for more info.
```

Copying a CLI format file to the startup configuration:

```
switch# copy tftp://192.168.1.10/startjson startup-config
100.0%
100.0%
Success
```

Copying an unsupported file format to the startup configuration:

```
switch# copy tftp://192.168.1.10/startfile startup-config
100.0%
100.0%
unsupported file format
```

**copy running-config {startup-config |  
checkpoint <CHECKPOINT-NAME>}**

## Syntax

```
copy running-config {startup-config | checkpoint <CHECKPOINT-NAME>}
```

## Description

Copies the running configuration to the startup configuration or to a new checkpoint. If the startup configuration is already present, the command overwrites the existing startup configuration.

## Command context

Manager (#)

## Parameters

### **startup-config**

Specifies that the startup configuration receives a copy of the running configuration.

### **checkpoint <CHECKPOINT-NAME>**

Specifies the name of a new checkpoint to receive a copy of the running configuration. The checkpoint name can be alphanumeric. It can also contain underscores (\_) and dashes (-).



**NOTE:** Do not start the checkpoint name with `CPC` because it is used for system-generated checkpoints.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Copying the running configuration to the startup configuration:

```
switch# copy running-config startup-config
Success
```

Copying the running configuration to a new checkpoint named `ckpt1`:

```
switch# copy running-config checkpoint ckpt1
Success
```

## **copy {running-config | startup-config} <REMOTE-URL>**

## Syntax

```
copy {running-config | startup-config} <REMOTE-URL> {cli | json} [vrf <VRF-NAME>]
```

## Description

Copies the running configuration or the startup configuration to a remote file in either CLI or JSON format.

## Command context

Manager (#)

## Parameters

### **{running-config | startup-config}**

Selects whether the running configuration or the startup configuration is copied to a remote file.

### **<REMOTE-URL>**

Specifies the remote file using the syntax: `{tftp | sftp}://<IP-ADDRESS>[:<PORT-NUMBER>][;blocksize=<BLOCKSIZE-VALUE>]/<FILE-NAME>`

### **{cli | json}**

Selects the remote file format: P: CLI or JSON.

### **vrf <VRF-NAME>**

Specifies the name of a VRF. Default: default.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Copying a running configuration to a remote file in CLI format:

```
switch# copy running-config tftp://192.168.1.10/runcli cli
100.0%
Success
```

Copying a running configuration to a remote file in JSON format:

```
switch# copy running-config tftp://192.168.1.10/runjson json
100.0%
Success
```

Copying a startup configuration to a remote file in CLI format:

```
switch# copy startup-config sftp://root@192.168.1.10/startcli cli
root@192.168.1.10's password:
sftp> put /tmp/startcli startcli
Uploading /tmp/startcli to /root/startcli
Connected to 192.168.1.10.
Success
```

Copying a startup configuration to a remote file in JSON format:

```
switch# copy startup-config sftp://root@192.168.1.10/startjson json
root@192.168.1.10's password:
sftp> put /tmp/startjson startjson
Uploading /tmp/startjson to /root/startjson
Connected to 192.168.1.10.
Success
```

## **copy {running-config | startup-config} <STORAGE-URL>**

### Syntax

```
copy {running-config | startup-config} <STORAGE-URL> {cli | json}
```

### Description

Copies the running configuration or a startup configuration to a USB drive.

### Command context

Manager (#)

### Parameters

**{running-config | startup-config}**

Selects the running configuration or the startup configuration to be copied to the switch USB drive.

**<STORAGE-URL>**

Specifies a remote file with the following syntax: `usb: /<file>`

```
{cli | json}
```

Selects the format of the remote file: CLI or JSON.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

The switch supports JSON and CLI file formats when copying the running or starting configuration to the USB drive. The USB drive must be formatted with the FAT file system.

The USB drive must be enabled and mounted with the following commands:

```
switch(config)# usb
switch(config)# end
switch# usb mount
```

### Examples

Copying a running configuration to a file named `runCLI` on the USB drive:

```
switch# copy running-config usb:/runCLI cli
Success
```

Copying a startup configuration to a file named `startCLI` on the USB drive:

```
switch# copy startup-config usb:/startCLI cli
Success
```

## copy startup-config running-config

### Syntax

```
copy startup-config running-config
```

### Description

Copies the startup configuration to the running configuration.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch# copy startup-config running-config
Success
```

## copy <STORAGE-URL> running-config

### Syntax

```
copy <STORAGE-URL> {running-config | startup-config | checkpoint <CHECKPOINT-NAME>}
```

## Description

This command copies a specified configuration from the USB drive to the running configuration, to a startup configuration, or to a checkpoint.

## Command context

Manager (#)

## Parameters

**<STORAGE-URL>**

Specifies the name of a configuration file on the USB drive with the syntax: `usb: /<FILE>`

**running-config**

Specifies that the configuration file is copied to the running configuration. The file must be in CLI, JSON, or checkpoint format or the copy will fail. the copy will not work.

**startup-config**

Specifies that the configuration file is copied to the startup configuration. The switch stores this configuration between reboots. The startup configuration is used as the operating configuration following a reboot of the switch. The file must be in JSON or checkpoint format or the copy will fail.

**checkpoint <CHECKPOINT-NAME>**

Specifies the name of a new checkpoint file to receive a copy of the configuration. The configuration file on the USB drive must be in checkpoint format.



**NOTE:** Do not start the checkpoint name with `CPC` because it is used for system-generated checkpoints.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

This command requires that the USB drive is formatted with the FAT file system and that the file be in the appropriate format as follows:

- **running-config:** This option requires the file on the USB drive be in CLI, JSON, or checkpoint format.
- **startup-config:** This option requires the file on the USB drive be in JSON or checkpoint format.
- **checkpoint <checkpoint-name>:** This option requires the file on the USB drive be in checkpoint format.

## Examples

Copying the file **runCli** from the USB drive to the running configuration:

```
switch# copy usb:/runCli running-config
Configuration may take several minutes to complete according to configuration
file size
--0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%--
Success
```

Copying the file **startUp** from the USB drive to the startup configuration:

```
switch# copy usb:/startUp startup-config
Success
```

Copying the file **testCheck** from the USB drive to the **abc** checkpoint:

```
switch# copy usb:/testCheck checkpoint abc
Success
```

## **erase {checkpoint <CHECKPOINT-NAME> | startup-config | all}**

### **Syntax**

```
erase {checkpoint <CHECKPOINT-NAME> | startup-config | all}
```

### **Description**

Deletes an existing checkpoint, startup configuration, or all checkpoints.

### **Command context**

Manager (#)

### **Parameters**

**checkpoint <CHECKPOINT-NAME>**

Specifies the name of a checkpoint.

**startup-config**

Specifies the startup configuration.

**all**

Specifies all checkpoints.

### **Authority**

Administrators or local user group members with execution rights for this command.

### **Examples**

Erasing checkpoint **ckpt1**:

```
switch# erase checkpoint ckpt1
```

Erasing the startup configuration:

```
switch# erase startup-config
```

Erasing all checkpoints:

```
switch# erase checkpoint all
```

## **show checkpoint <CHECKPOINT-NAME>**

### **Syntax**

```
show checkpoint <CHECKPOINT-NAME> [json]
```

### **Description**

Shows the configuration of a checkpoint.

## Command context

Manager (#)

## Parameters

**checkpoint** <CHECKPOINT-NAME>

Specifies the name of a checkpoint.

**[json]**

Specifies that the output is displayed in JSON format.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing the configuration of the ckpt1 checkpoint in non-JSON format:

```
switch# show checkpoint ckpt1
Checkpoint configuration:
!
!Version ArubaOS-CX PL.10.07.0000K-75-g55e5193
!export-password: default
lacp system-priority 65535
user admin group administrators password ciphertext
AQBapQjwibe36io0jFfde7ZzrHckncal1D+3n8XFTZKQdmYgAAADetYOeHSme93xzdD0uz6Vr9Kl+XBzB
+2GB0UBxSF7rv9N2x8KSgkqv7iqXVQ0Te6LkSMnH4BdNaT3Bf25qyvOqmr4YakO1V3rg8zAOADkPktQD8jo
THXflzwomoIzcmv/uX
cli-session
 timeout 0
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
interface lag 1
 no shutdown
 vlan access 1
interface lag 128
 no shutdown
 vlan access 1
interface lag 129
 shutdown
 vlan access 1
 lacp mode active
interface 1/1/1
 no shutdown
 lag 128
 lacp port-id 65535
interface 1/1/2
 no shutdown
 vlan access 1
interface 1/1/3
 no shutdown
 vlan access 1
interface 1/1/4
 no shutdown
 vlan access 1
interface 1/1/5
```



```

 no shutdown
 vlan access 1
interface 1/1/6
 no shutdown
 vlan access 1
interface 1/1/7
 no shutdown
 vlan access 1
interface 1/1/8
 no shutdown
 vlan access 1
interface 1/1/9
 no shutdown
 vlan access 1
interface 1/1/10
 no shutdown
 vlan access 1
interface 1/1/11
 no shutdown
 vlan access 1
interface 1/1/12
 no shutdown
 vlan access 1
interface 1/1/13
 no shutdown
 vlan access 1
interface 1/1/14
 no shutdown
 vlan access 1
interface 1/1/15
 no shutdown
 vlan access 1
interface 1/1/16
 no shutdown
 vlan access 1
interface vlan 1
 ip dhcp
snmp-server vrf default
!
!
!
!
!
https-server vrf default

```

Showing the configuration of the ckpt1 checkpoint in JSON format:

```

switch# show checkpoint ckpt1 json
Checkpoint configuration:
{
 "AAA_Server_Group": {
 "local": {
 "group_name": "local"
 },
 "none": {
 "group_name": "none"
 }
 },
 ...
 ...
 ...
 ...

```

## show checkpoint post-configuration

### Syntax

```
show checkpoint post-configuration
```

### Description

Shows the configuration settings for creating system checkpoints.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch# show checkpoint post-configuration
```

```
Checkpoint Post-Configuration feature
```

```

```

```
Status : enabled
```

```
Timeout (sec) : 300
```

## show checkpoint list

### Syntax

```
show checkpoint list {all | <START-DATE> <END-DATE>}
```

### Description

Shows a list of saved checkpoints.

### Command context

Manager (#)

### Parameters

**all**

Shows a detailed list of all saved checkpoints.

**<START-DATE>**

Specifies the starting date for the range of saved checkpoints to show. Format: YYYY-MM-DD.

**<END-DATE>**

Specifies the ending date for the range of saved checkpoints to show. Format: YYYY-MM-DD.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Showing the list of saved checkpoints:

```
switch# show checkpoint list
ckpt1
ckpt2
ckpt3
startup-config
AUTO20170308214100
```

Showing a detailed list of all saved system and user checkpoints:

```
switch# show checkpoint list all
NAME	TYPE	WRITER	DATE (UTC)	HARDWARE	IMAGE VERSION
CPC20171017195548	checkpoint	System	2020-10-17T19:55:48Z	00000	xx.10.0x.xxxxX-1-g7691be0
ckpt23	checkpoint	User	2020-10-17T20:09:00Z	00000	xx.10.0x.xxxxX-1-g7691be0
ckpt24	checkpoint	User	2020-10-17T20:09:09Z	00000	xx.10.0x.xxxxX-1-g7691be0
ckpt30	checkpoint	User	2020-10-17T20:10:36Z	00000	xx.10.0x.xxxxX-1-g7691be0
startup-config-backup	checkpoint	System	2020-10-17T20:17:56Z	00000	xx.10.0x.xxxxX-1-g7691be0
CPC20171017201712	checkpoint	System	2020-10-17T20:19:12Z	00000	xx.10.0x.xxxxAZ-10-g4c6b4446bd6
ckpt31	checkpoint	User	2020-10-17T20:19:24Z	00000	xx.10.0x.xxxxAZ-10-g4c6b4446bd6
startup-config	startup	User	2020-10-17T20:47:11Z	00000	xx.10.0x.xxxxAZ-10-g4c6b4446bd6
ckpt32	checkpoint	User	2020-10-17T20:50:24Z	00000	xx.10.0x.xxxxAZ-10-g4c6b4446bd6
CPC20171017205110	checkpoint	System	2020-10-17T20:51:10Z	00000	xx.10.0x.xxxxAZ-10-g4c6b4446bd6
```

Showing a detailed list of saved checkpoints for a specific date range:

```
switch# show checkpoint list date 2020-03-08 2020-03-12
NAME	TYPE	WRITER	DATE	HARDWARE	IMAGE VERSION
ckpt2	checkpoint	User	2020-03-08 18:10:01	0.0.0	
ckpt3	checkpoint	User	2020-03-09 23:11:02	0.0.0	
ckpt4	checkpoint	User	2020-03-11 00:00:03	0.0.0	
```

## write memory

### Syntax

```
write memory
```

### Description

Saves the running configuration to the startup configuration. It is an alias of the command `copy running-config startup-config`. If the startup configuration is already present, this command overwrites the startup configuration.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch# write memory
Success
```

## Classifier policy application

Classifier policies can be applied as follows:

| Policy type         | IPv4 | IPv6 |
|---------------------|------|------|
| Direction           | In   | In   |
| L2 interface (port) | Yes  | Yes  |
| L2 LAG              | Yes  | Yes  |
| VLAN                | Yes  | Yes  |



**NOTE:** Port policies and port-access client policies cannot be configured at the same time.

## apply policy (Context: config)

### Syntax

```
apply policy <POLICY-NAME> in
```

```
no apply policy <POLICY-NAME> in
```

### Description

Applies a policy to the global config context.

Only one policy can be globally applied at a time. Applying a policy globally again, replaces the previous globally applied policy.

The `no` form of this command removes application of the global policy.

### Command context

```
config
```

### Parameters

**<POLICY-NAME>**

Specifies the policy to apply.

**in**

Selects the inbound (ingress) traffic direction.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Applying policy global1 to the global config context:

```
switch(config)# apply policy global1 in
```

Removing application of policy global1 from the global config context:

```
switch(config)# no apply policy global1 in
```

## apply policy (Contexts: config-if, config-vlan)

### Syntax

Context: config-if:

```
apply policy <POLICY-NAME> {in}
```

```
no apply policy <POLICY-NAME> {in}
```

Context: config-vlan:

```
apply policy <POLICY-NAME> in
```

```
no apply policy <POLICY-NAME> in
```

### Description

Applies a policy to the current interface or VLAN context.

The **no** form of this command removes a policy from the interface or VLAN specified by the current context.

### Command context

config-if

config-vlan

### Parameters

**<POLICY-NAME>**

Specifies the policy to apply.

**in**

Selects the inbound (ingress) traffic direction.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage (applies to config-vlan context)

Only one policy may be applied to a VLAN at a time. Therefore, using the **apply policy** command on a VLAN with an already-applied policy of the same type, will replace the applied policy.

## Examples

Applying a policy to an interface (ingress):

```
switch(config)# interface 1/1/1
switch(config-if)# apply policy MY_POLICY in
```

Applying a policy to a VLAN (ingress):

```
switch(config)# vlan 10
switch(config-vlan-10)# apply policy MY_POLICY in
```

Removing a policy from a VLAN (ingress):

```
switch(config)# vlan 10
switch(config-vlan-10)# no apply policy MY_POLICY in
```

## class copy

### Syntax

```
class {ip|ipv6} <CLASS-NAME> copy <DESTINATION-CLASS>
```

### Description

Copies a class to a new destination class or overwrites an existing class. Copying a class copies all entries as well.

### Command context

config

### Parameters

{ip|ipv6} <CLASS-NAME>

Specifies the type and name of the class to be copied.

<DESTINATION-CLASS>

Specifies the name of the destination class.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Copying an IPv4 class. Copying a class with entries copies all its entries as well:

```
switch(config)# class ip MY_IP_CLASS copy MY_IP_CLASS2
switch(config)# do show class
```

| Type  | Sequence | Name         | Comment | Action                 | L3 Protocol            |
|-------|----------|--------------|---------|------------------------|------------------------|
|       |          |              |         | Source IP Address      | Source L4 Port(s)      |
|       |          |              |         | Destination IP Address | Destination L4 Port(s) |
|       |          |              |         | Additional Parameters  |                        |
| ----- |          |              |         |                        |                        |
| IPv4  |          | MY_IP_CLASS  |         |                        |                        |
|       | 11       | ignore       |         |                        | udp                    |
|       |          | any          |         |                        |                        |
|       | 21       | match        |         |                        | tcp                    |
|       |          | 192.168.0.1  |         |                        |                        |
|       |          | 192.168.0.2  |         |                        |                        |
| ----- |          |              |         |                        |                        |
| IPv4  |          | MY_IP_CLASS2 |         |                        |                        |
|       | 11       | ignore       |         |                        | udp                    |
|       |          | any          |         |                        |                        |

|    |             |     |
|----|-------------|-----|
| 21 | any         | tcp |
|    | match       |     |
|    | 192.168.0.1 |     |
|    | 192.168.0.2 |     |

Copying an IPv6 class:

```
switch(config)# class ipv6 MY_IPV6_CLASS copy MY_IPV6_CLASS2
switch(config)# do show class
```

| Type | Name           | Sequence | Comment | Action | L3 Protocol | Source IP Address | Source L4 Port(s) | Destination IP Address | Destination L4 Port(s) | Additional Parameters |
|------|----------------|----------|---------|--------|-------------|-------------------|-------------------|------------------------|------------------------|-----------------------|
| IPv6 | MY_IPV6_CLASS  | 2        | ignore  |        | udp         | any               |                   | any                    |                        |                       |
| IPv6 | MY_IPV6_CLASS2 | 2        | ignore  |        | udp         | any               |                   | any                    |                        |                       |

## class ip

### Syntax

Syntax to create an IPv4 class and enter its context. Plus syntax to remove a class:

```
class ip <CLASS-NAME>
```

```
no class ip <CLASS-NAME>
```

Syntax (within the class context) for creating or removing class entries for protocols ah, gre, esp, igmp, ospf, pim (ip is available as an alias for any):

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{any|ip|ah|gre|esp|igmp|ospf|pim|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]
```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocols sctp, tcp, udp:

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]
```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocol icmp:

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{icmp}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]
```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for class entry comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>
```

```
no <SEQUENCE-NUMBER> comment
```

## Description

Creates or modifies an IPv4 traffic class to match specified packets. Class is composed of one or more class entries ordered and prioritized by sequence numbers. With this command, the class can classify traffic based on IPv4 header information.

The `no` form of the command can be used to delete either an IPv4 traffic class (use `no` with the class command) or an individual IPv4 traffic class entry (use `no` with the sequence number).

## Command context

```
config
```

The `class ip <CLASS-NAME>` command takes you into the `config-class-ip` context where you enter the class entries.

## Parameters

**ip**

Specifies create or modify an IPv4 class.

**<CLASS-NAME>**

Specifies the name of this class.

**<SEQUENCE-NUMBER>**

Specifies a sequence number for the class entry. Optional. Range: 1-4294967295.

**{match|ignore}**

Creates a rule to match or ignore specified packets.

**<IP-PROTOCOL-NUM>**

Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.

**{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]} }**

Specifies the source IPv4 address.



- `any` - specifies any source IPv4 address.
- `<SRC-IP-ADDRESS>` - specifies the source IPv4 host address.
  - `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
  - `<SUBNET-MASK>` - specifies the address bits to mask (dotted decimal notation).

**{ `any` | `<DST-IP-ADDRESS>` [ / { `<PREFIX-LENGTH>` | `<SUBNET-MASK>` } ] }**

Specifies the destination IPv4 address.

- `any` - specifies any destination IPv4 address.
- `<DST-IP-ADDRESS>` - specifies the destination IPv4 host address.
  - `<PREFIX-LENGTH>` - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
  - `<SUBNET-MASK>` - specifies the address bits to mask (dotted decimal notation).

**[ { `eq` | `gt` | `lt` } `<PORT>` | `range` `<MIN-PORT>` `<MAX-PORT>` ]**

Specifies the port or port range. Port numbers are in the range of 0 to 65535.

- `eq` `<PORT>` - specifies the Layer 4 port.
- `gt` `<PORT>` - specifies any Layer 4 port greater than the indicated port.
- `lt` `<PORT>` - specifies any Layer 4 port less than the indicated port.
- `range` `<MIN-PORT>` `<MAX-PORT>` - specifies the Layer 4 port range.

#### **urg**

Specifies matching on the TCP Flag: Urgent.

#### **ack**

Specifies matching on the TCP Flag: Acknowledgment.

#### **psh**

Specifies matching on the TCP Flag: Push buffered data to receiving application.

#### **rst**

Specifies matching on the TCP Flag: Reset the connection.

#### **syn**

Specifies matching on the TCP Flag: Synchronize sequence numbers.

#### **fin**

Specifies matching on the TCP Flag: Finish connection.

#### **established**

Specifies matching on the TCP Flag: Established connection.

#### **dscp `<DSCP-SPECIFIER>`**

Specifies the Differentiated Services Code Point (DSCP), either a numeric `<DSCP-VALUE>` (0 to 63) or one of these keywords:

- AF11 - DSCP 10 (Assured Forwarding Class 1, low drop probability)
- AF12 - DSCP 12 (Assured Forwarding Class 1, medium drop probability)
- AF13 - DSCP 14 (Assured Forwarding Class 1, high drop probability)
- AF21 - DSCP 18 (Assured Forwarding Class 2, low drop probability)
- AF22 - DSCP 20 (Assured Forwarding Class 2, medium drop probability)
- AF23 - DSCP 22 (Assured Forwarding Class 2, high drop probability)
- AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability)
- AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability)
- AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability)
- AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability)
- AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability)
- AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability)
- CS0 - DSCP 0 (Class Selector 0: Default)
- CS1 - DSCP 8 (Class Selector 1: Scavenger)
- CS2 - DSCP 16 (Class Selector 2: OAM)
- CS3 - DSCP 24 (Class Selector 3: Signaling)
- CS4 - DSCP 32 (Class Selector 4: Realtime)
- CS5 - DSCP 40 (Class Selector 5: Broadcast video)
- CS6 - DSCP 48 (Class Selector 6: Network control)
- CS7 - DSCP 56 (Class Selector 7)
- EF - DSCP 46 (Expedited Forwarding)

**ip-precedence** *<IP-PRECEDENCE-VALUE>*

Specifies an IP precedence value. Range: 0 to 7.

**tos** *<TOS-VALUE>*

Specifies the Type of Service value. Range: 0 to 31.

**fragment**

Specifies a fragment packet.

**vlan** *<VLAN-ID>*

Specifies VLAN tag to match on. 802.1Q VLAN ID.



**NOTE:** This parameter cannot be used in any class that will be applied to a VLAN.

**count**

Keeps the hit counts of the number of packets matching this class entry.

[<SEQUENCE-NUMBER>] **comment** <TEXT-STRING>

Adds a comment to a class entry. The **no** form removes only the comment from the class entry.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

- Entering an existing <CLASS-NAME> value will cause the existing class to be modified, with any new <SEQUENCE-NUMBER> value creating an additional class entry, and any existing <SEQUENCE-NUMBER> value replacing the existing class entry with the same sequence number.
- If no sequence number is specified, a new class entry will be appended to the end of the class with a sequence number equal to the highest class entry currently in the list plus 10.
- If the <IP-PROTOCOL-NUM> parameter is used instead of a protocol name, ensure that any needed class entry-definition parameters specific to the selected protocol are also provided.

## Examples

Creating an IPv4 class with three entries:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# 10 match icmp any any
switch(config-class-ip)# 20 ignore udp any any
switch(config-class-ip)# 30 match tcp 192.168.0.1 192.168.0.2
switch(config-class-ip)# exit

switch(config)# do show class
```

| Type     | Name                   |  |                        |
|----------|------------------------|--|------------------------|
| Sequence | Comment                |  |                        |
|          | Action                 |  | L3 Protocol            |
|          | Source IP Address      |  | Source L4 Port(s)      |
|          | Destination IP Address |  | Destination L4 Port(s) |
|          | Additional Parameters  |  |                        |
| IPv4     | MY_IP_CLASS            |  |                        |
| 10       | match                  |  | icmp                   |
|          | any                    |  |                        |
|          | any                    |  |                        |
| 20       | ignore                 |  | udp                    |
|          | any                    |  |                        |
|          | any                    |  |                        |
| 30       | match                  |  | tcp                    |
|          | 192.168.0.1            |  |                        |
|          | 192.168.0.2            |  |                        |

Adding a comment to an existing IPv4 class entry:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# 30 comment myipClass
switch(config-class-ip)# exit

switch(config)# do show class
```

| Type     | Name                   |  |                        |
|----------|------------------------|--|------------------------|
| Sequence | Comment                |  |                        |
|          | Action                 |  | L3 Protocol            |
|          | Source IP Address      |  | Source L4 Port(s)      |
|          | Destination IP Address |  | Destination L4 Port(s) |
|          | Additional Parameters  |  |                        |
|          |                        |  |                        |

```
IPv4 MY_IP_CLASS
 10 match icmp
 any
 any
 20 ignore udp
 any
 any
 30 myipClass
 match tcp
 192.168.0.1
 192.168.0.2
```

Removing a comment from an existing IPv4 class entry:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# no 30 comment
switch(config-class-ip)# exit

switch(config)# do show class
Type Name
Sequence Comment
 Action
 Source IP Address
 Destination IP Address
 Additional Parameters

IPv4 MY_IP_CLASS
 10 match icmp
 any
 any
 20 ignore udp
 any
 any
 30 match tcp
 192.168.0.1
 192.168.0.2
```

Replacing an IPv4 class entry in an existing class:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# 10 match igmp any any
switch(config-class-ip)# exit

switch(config)# do show class
Type Name
Sequence Comment
 Action
 Source IP Address
 Destination IP Address
 Additional Parameters

IPv4 MY_IP_CLASS
 10 match igmp
 any
 any
 20 ignore udp
 any
 any
 30 match tcp
 192.168.0.1
 192.168.0.2
```

Removing an IPv4 class entry:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# no 10
switch(config-class-ip)# exit
```

```
switch(config)# do show class
```

| Type | Sequence | Name        | Comment | Action | L3 Protocol | Source IP Address | Source L4 Port(s) | Destination IP Address | Destination L4 Port(s) | Additional Parameters |
|------|----------|-------------|---------|--------|-------------|-------------------|-------------------|------------------------|------------------------|-----------------------|
| IPv4 | 20       | MY_IP_CLASS | ignore  |        | udp         |                   |                   |                        |                        |                       |
|      | 30       |             | match   |        | tcp         | 192.168.0.1       |                   | 192.168.0.2            |                        |                       |

Removing an IPv4 class. Removing a class with entries removes all its entries as well. If a class associated with a policy entry (or multiple policy entries) is removed, the corresponding entries are also removed.



**NOTE:** The corresponding entries are only removed if the class is unused by all policy entries.

```
switch(config)# no class ip MY_IP_CLASS
```

```
switch(config)# do show class
```

```
No Class found.
```

## class ipv6

### Syntax

Syntax to create an IPv6 class and enter its context. Plus syntax to remove a class:

```
class ipv6 <CLASS-NAME>
```

```
no class ipv6 <CLASS-NAME>
```

Syntax (within the class context) for creating or removing class entries for protocols ah, gre, esp, igmp, ospf, pim (ipv6 is available as an alias for any):

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{any|ipv6|ah|gre|esp|igmp|ospf|pim|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]
```

```
no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocols sctp, tcp, udp:

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
```

```

{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[{eq|gt|lt} <PORT>|range <MIN-PORT> <MAX-PORT>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]

no <SEQUENCE-NUMBER>

```

Syntax (within the class context) for creating or removing class entries for protocol icmpv6:

```

[<SEQUENCE-NUMBER>]
{permit|deny}
{icmpv6}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}]}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]

no <SEQUENCE-NUMBER>

```

Syntax (within the class context) for class entry comments:

```

[<SEQUENCE-NUMBER>] comment <TEXT-STRING>

no <SEQUENCE-NUMBER> comment

```

## Description

Creates or modifies an IPv6 traffic class to match specified packets. Class is composed of one or more class entries ordered and prioritized by sequence numbers. With this command, each class can classify traffic based on IPv6 header information.

The `no` form of the command deletes either an IPv6 traffic class (use `no` with the class command) or an individual IPv6 traffic class entry (use `no` with the sequence number).

## Command context

config

The `class ipv6 <CLASS-NAME>` command takes you into the `config-class-ipv6` command context where you enter the class entries.

## Parameters

### ipv6

Specifies create or modify an IPv6 class.

### <CLASS-NAME>

Specifies the name of this class.

### <SEQUENCE-NUMBER>

Specifies a sequence number for the class entry. Optional. Range: 1-4294967295.

### {match|ignore}

Creates a rule to match or ignore specified packets.

### <IP-PROTOCOL-NUM>

Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.

**{ any | <SRC-IP-ADDRESS> [ / { <PREFIX-LENGTH> | <SUBNET-MASK> } ] }**

Specifies the source IPv6 address.

- any - specifies any source IPv6 address.
- <SRC-IP-ADDRESS> - specifies the source IPv4 host address.
  - <PREFIX-LENGTH> - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
  - <SUBNET-MASK> - specifies the address bits to mask (dotted decimal notation).

**{ any | <DST-IP-ADDRESS> [ / { <PREFIX-LENGTH> | <SUBNET-MASK> } ] }**

Specifies the destination IPv4 address.

- any - specifies any destination IPv6 address.
- <DST-IP-ADDRESS> - specifies the destination IPv6 host address.
  - <PREFIX-LENGTH> - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32.
  - <SUBNET-MASK> - specifies the address bits to mask (dotted decimal notation).

**[ { eq | gt | lt } <PORT> | range <MIN-PORT> <MAX-PORT> ]**

Specifies the port or port range. Port numbers are in the range of 0 to 65535.

- eq <PORT> - specifies the Layer 4 port.
- gt <PORT> - specifies any Layer 4 port greater than the indicated port.
- lt <PORT> - specifies any Layer 4 port less than the indicated port.
- range <MIN-PORT> <MAX-PORT> - specifies the Layer 4 port range.

**urg, ack, psh, rst, syn, fin, established**

These TCP flag matching parameters are not supported.

**dscp <DSCP-SPECIFIER>**

Specifies the Differentiated Services Code Point (DSCP), either a numeric <DSCP-VALUE> (0 to 63) or one of these keywords:

- AF11 - DSCP 10 (Assured Forwarding Class 1, low drop probability)
- AF12 - DSCP 12 (Assured Forwarding Class 1, medium drop probability)
- AF13 - DSCP 14 (Assured Forwarding Class 1, high drop probability)
- AF21 - DSCP 18 (Assured Forwarding Class 2, low drop probability)
- AF22 - DSCP 20 (Assured Forwarding Class 2, medium drop probability)
- AF23 - DSCP 22 (Assured Forwarding Class 2, high drop probability)
- AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability)
- AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability)
- AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability)

- AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability)
- AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability)
- AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability)
- CS0 - DSCP 0 (Class Selector 0: Default)
- CS1 - DSCP 8 (Class Selector 1: Scavenger)
- CS2 - DSCP 16 (Class Selector 2: OAM)
- CS3 - DSCP 24 (Class Selector 3: Signaling)
- CS4 - DSCP 32 (Class Selector 4: Real time)
- CS5 - DSCP 40 (Class Selector 5: Broadcast video)
- CS6 - DSCP 48 (Class Selector 6: Network control)
- CS7 - DSCP 56 (Class Selector 7)
- EF - DSCP 46 (Expedited Forwarding)

**ip-precedence** *<IP-PRECEDENCE-VALUE>*

Specifies an IP precedence value. Range: 0 to 7.

**tos** *<TOS-VALUE>*

Specifies the Type of Service value. Range: 0 to 31.

**fragment**

Specifies a fragment packet.

**vlan** *<VLAN-ID>*

Specifies VLAN tag to match on. 802.1Q VLAN ID.



**NOTE:** This parameter cannot be used in any class that will be applied to a VLAN.

**count**

Keeps the hit counts of the number of packets matching this class entry.

[*<SEQUENCE-NUMBER>*] **comment** *<TEXT-STRING>*

Adds a comment to a class entry. The **no** form removes only the comment from the class entry.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

- If you enter an existing *<CLASS-NAME>* value, the existing class is modified with any new *<SEQUENCE-NUMBER>* value. This action creates an additional class entry. Any existing *<SEQUENCE-NUMBER>* value replaces the existing class entry with the same sequence number.
- If no sequence number is specified, a new class entry is appended to the end of the class with a sequence number equal to the highest class entry currently in the list plus 10.
- If the *<IP-PROTOCOL-NUM>* parameter is used instead of a protocol name, ensure that any needed class entry-definition parameters specific to the selected protocol are also provided.



## Examples

Creating an IPv6 class with two entries:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# 10 match icmpv6 any any
switch(config-class-ipv6)# 20 ignore udp any any
switch(config-class-ipv6)# exit

switch(config)# do show class
Type Name
Sequence Comment
 Action L3 Protocol
 Source IP Address Source L4 Port(s)
 Destination IP Address Destination L4 Port(s)
 Additional Parameters

IPv6 MY_IPV6_CLASS
 10 match icmpv6
 any
 any
 20 ignore udp
 any
 any
```

Adding a comment to an existing IPv6 class entry:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# 10 match icmpv6 any any
switch(config-class-ipv6)# 20 ignore udp any any
switch(config-class-ipv6)# 20 comment myipv6class
switch(config-class-ipv6)# exit

switch(config)# do show class
Type Name
Sequence Comment
 Action L3 Protocol
 Source IP Address Source L4 Port(s)
 Destination IP Address Destination L4 Port(s)
 Additional Parameters

IPv6 MY_IPV6_CLASS
 10 match icmpv6
 any
 any
 20 myipv6class
 ignore udp
 any
 any
```

Removing a comment from an existing IPv6 class entry:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# no 20 comment
switch(config-class-ipv6)# exit

switch(config)# do show class
Type Name
Sequence Comment
 Action L3 Protocol
 Source IP Address Source L4 Port(s)
 Destination IP Address Destination L4 Port(s)
 Additional Parameters
```

```

IPv6 MY_IPV6_CLASS
 10 match icmpv6
 any
 any
 20 ignore udp
 any
 any

```

Replacing an IPv6 class entry in an existing IPv6 class:

```

switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# 10 match any any 1020::
switch(config-class-ipv6)# exit

switch(config)# do show class
Type Name
Sequence Comment
 Action L3 Protocol
 Source IP Address Source L4 Port(s)
 Destination IP Address Destination L4 Port(s)
 Additional Parameters

IPv6 MY_IPV6_CLASS
 10 match any
 any
 1020::
 20 ignore udp
 any
 any

```

Removing an IPv6 class entry:

```

switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# no 10
switch(config-class-ipv6)# exit

switch(config)# do show class
Type Name
Sequence Comment
 Action L3 Protocol
 Source IP Address Source L4 Port(s)
 Destination IP Address Destination L4 Port(s)
 Additional Parameters

IPv6 MY_IPV6_CLASS
 20 ignore udp
 any
 any

```

Removing an IPv6 class. Removing a class with entries removes all its entries as well. If a class associated with a policy entry (or multiple policy entries) is removed, the corresponding entries are also removed.



**NOTE:** The corresponding entries are only removed if the class is unused by all policy entries.

```

switch(config)# no class ipv6 MY_IPV6_CLASS

switch(config)# do show class
No Class found.

```

# class resequence

## Syntax

```
class {ip|ipv6} <CLASS-NAME> resequence <STARTING-SEQUENCE-NUMBER> <INCREMENT>
```

## Description

Resequencing in an IPv4, or IPv6 class.

## Command context

config

## Parameters

**{ip|ipv6} <CLASS-NAME>**

Specifies the class where you want to resequence class entries.

**<STARTING-SEQUENCE-NUMBER>**

Specifies the sequence number to start resequencing from.

**<INCREMENT>**

Specifies how much to increment the sequence numbers by.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Resequencing an IPv4 class:

```
switch(config)# class ip MY_IP_CLASS resequence 1 10
switch(config)# do show class
Type Name
Sequence Comment
 Action
 Source IP Address L3 Protocol
 Destination IP Address Source L4 Port(s)
 Additional Parameters Destination L4 Port(s)

IPv4 MY_IP_CLASS
 1 match igmp
 any
 any
 11 ignore udp
 any
 any
 21 match tcp
 192.168.0.1
 192.168.0.2
```

Resequencing an IPv6 class:

```
switch(config)# class ipv6 MY_IPV6_CLASS resequence 1 1
switch(config-class-ipv6)# exit
switch(config)# do show class
Type Name
Sequence Comment
 Action
 Source IP Address L3 Protocol
 Source L4 Port(s)
```

|      | Destination IP Address<br>Additional Parameters | Destination L4 Port(s) |
|------|-------------------------------------------------|------------------------|
| IPv6 | MY_IPV6_CLASS                                   |                        |
|      | 1 match                                         | any                    |
|      | any                                             |                        |
|      | 1020::                                          |                        |
|      | 2 ignore                                        | udp                    |
|      | any                                             |                        |
|      | any                                             |                        |

## class reset

### Syntax

```
class { all | ip <CLASS-NAME> | ipv6 <CLASS-NAME>} reset
```

### Description

Changes the user-specified class configuration to match the active class configuration. Use this command when there is a discrepancy between what the user configured and what is active and accepted by the system.

### Command context

```
config
```

### Parameters

```
{ all | ip <CLASS-NAME> | ipv6 <CLASS-NAME>}
```

Specifies either `all` classes be reset or specifies the type (`ip` for IPv4, or `ipv6` for IPv6 ) and name of the class to be reset.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Resetting the user-specified configuration to the active configuration:

```
switch(config)# class all reset
```

## clear policy hitcounts

### Syntax

### Description

Clears the policy hit count statistics.

### Command context

Operator (>) or Manager (#)

## Parameters

**all**

Selects all policies.

**<POLICY-NAME>**

Specifies the policy name.

**interface <IF-NAME>**

Specifies the interface name.

**vlan <VLAN-ID>**

Specifies the VLAN.

**in**

Specifies the inbound (ingress) traffic direction.

**global**

Selects the globally applied policy.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Clearing hit counts for policy `MY_IPv6_Policy` applied to VLAN 10 (ingress):

```
switch# clear policy hitcounts My_IPv6_Policy vlan 10 in
```

Clearing hit counts for all policies:

```
switch# clear policy hitcounts all
```

# policy

## Syntax

`policy <POLICY-NAME>`

```
[<SEQUENCE-NUMBER>]
class {ip|ipv6} <CLASS-NAME>
 action {<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}
 [{<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}]

[<SEQUENCE-NUMBER>]
comment ...
```

## Description

Creates or modifies classifier policy and policy entries. A policy is made up of one or more policy entries ordered and prioritized by sequence numbers. Each entry has an IPv4/IPv6 class and zero or more policy actions associated with it.

A policy must be applied using the `apply` command.

The `no` form of the command can be used to delete either a policy (use `no` with the policy command) or an individual policy entry (use `no` with the sequence number).

## Command context

`config`

The `policy` command takes you into the `config-policy` context where you enter the policy entries.

## Parameters

**<POLICY-NAME>**

Specifies the name of the policy.

**<SEQUENCE-NUMBER>**

Specifies a sequence number for the policy entry. Optional. Range: 1 to 4294967295.

**comment**

Stores the remaining entered text as a policy entry comment.

**class {ip|ipv6} <CLASS-NAME>**

Specifies a type of class, `ip` for IPv4, `ipv6` for IPv6.

**<REMARK-ACTIONS>**

Remark actions can be any of the following options: { `cos <COS-VALUE>` | `ip-precedence <IP-PRECEDENCE-VALUE>` | `dscp <DSCP-VALUE>`} where:

**cos <COS-VALUE>**

Specifies the Class of Service (CoS) value.

**ip-precedence <IP-PRECEDENCE-VALUE>**

Specifies the numeric IP precedence value. Range: 0 to 7.

**dscp <DSCP-VALUE>**

Specifies a Differentiated Services Code Point (DSCP) value. Enter either a numeric value (0 to 63) or a keyword as follows:

- AF11 - DSCP 10 (Assured Forwarding Class 1, low drop probability)
- AF12 - DSCP 12 (Assured Forwarding Class 1, medium drop probability)
- AF13 - DSCP 14 (Assured Forwarding Class 1, high drop probability)
- AF21 - DSCP 18 (Assured Forwarding Class 2, low drop probability)
- AF22 - DSCP 20 (Assured Forwarding Class 2, medium drop probability)
- AF23 - DSCP 22 (Assured Forwarding Class 2, high drop probability)
- AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability)
- AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability)
- AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability)
- AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability)
- AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability)
- AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability)

- CS0 - DSCP 0 (Class Selector 0: Default)
- CS1 - DSCP 8 (Class Selector 1: Scavenger)
- CS2 - DSCP 16 (Class Selector 2: OAM)
- CS3 - DSCP 24 (Class Selector 3: Signaling)
- CS4 - DSCP 32 (Class Selector 4: Real time)
- CS5 - DSCP 40 (Class Selector 5: Broadcast video)
- CS6 - DSCP 48 (Class Selector 6: Network control)
- CS7 - DSCP 56 (Class Selector 7)
- EF - DSCP 46 (Expedited Forwarding)

#### <POLICE-ACTIONS>

Police actions can be the following {cir <RATE-BPS> exceed} where:

**cir kbps <RATE-KBPS>**

Specifies a Committed Information Rate value in Kilobits per second. Range: 1 to 4294967295.

**exceed**

Specifies action to take on packets that exceed the rate limit.

#### <OTHER-ACTIONS>

Other actions can be the following:

**drop**

Specifies drop traffic.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

- An applied policy will process a packet sequentially against policy entries in the list until the last policy entry in the list has been evaluated or the packet matches an entry.
- Entering an existing <POLICY-NAME> value will cause the existing policy to be modified, with any new <SEQUENCE-NUMBER> value creating an additional policy entry, and any existing <SEQUENCE-NUMBER> value replacing the existing policy entry with the same sequence number.
- If no sequence number is specified, a new policy entry will be appended to the end of the entry list with a sequence number equal to the highest policy entry currently in the list plus 10.

### Examples

Creating a policy with several entries:

```
switch(config)# policy MY_POLICY
switch(config-policy)# 10 class ipv6 MY_CLASS1 action dscp af21 action drop
switch(config-policy)# 20 class ip MY_CLASS3 action mirror 1
switch(config-policy)# exit
switch(config)# do show policy
 Name
Sequence Comment
```

```

 Class Type
 action

10 MY_POLICY
 MY_CLASS1 ipv6
 drop
 dscp AF21

20
 MY_CLASS3 ipv4
 mirror 1

```

Adding a comment to an existing policy entry:

```

switch(config)# policy MY_POLICY
switch(config-policy)# 20 comment MY_TEST_POLICY
switch(config-policy)# exit
switch(config)# do show policy
 Name
Sequence Comment
 Class Type
 action

 MY_POLICY
10 MY_CLASS1 ipv6
 drop
 dscp AF21

20 MY_TEST_POLICY
 MY_CLASS3 ipv4
 mirror 1

```

Removing a comment from an existing policy entry:

```

switch(config)# policy MY_POLICY
switch(config-policy)# no 20 comment
switch(config-policy)# exit
switch(config)# do show policy
 Name
Sequence Comment
 Class Type
 action

 MY_POLICY
10 MY_CLASS1 ipv6
 drop
 dscp AF21

20
 MY_CLASS3 ipv4
 mirror 1

```

Adding/Replacing a policy entry in an existing policy:

```

switch(config)# policy MY_POLICY
switch(config-policy)# 10 class ip MY_CLASS3 action drop action dscp af21
switch(config-policy)# exit
switch(config)# do show policy
 Name
Sequence Comment
 Class Type

```



```

 action

 MY_POLICY
10 MY_CLASS3 ipv4
 drop
 dscp AF21

 20
 MY_CLASS3 ipv4
 mirror 1

```

Removing a policy entry:

```

switch(config)# policy MY_POLICY
switch(config-policy)# no 10
switch(config-policy)# exit
switch(config)# do show policy
 Name
Sequence Comment
Class Type
 action

 MY_POLICY
20 MY_CLASS3 ipv4
 mirror 1

```

Removing a policy:

```

switch(config)# no policy MY_POLICY
switch(config)# do show policy
 Name
Sequence Comment
Class Type
 action

 MY_POLICY2
2 MY_CLASS3 ipv4
 mirror 1

```

## policy copy

### Syntax

```
policy <POLICY-NAME> copy <DESTINATION-POLICY>
```

### Description

Copies a policy to a new destination policy or overwrites an existing policy. Copying a policy copies all its entries as well.

### Command context

```
config
```

## Parameters

**<POLICY-NAME>**

Specifies the policy to be copied.

**<DESTINATION-POLICY>**

Specifies the name of the destination policy.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Copying a policy:

```
switch(config)# policy MY_POLICY copy MY_POLICY2
switch(config)# do show policy
 Name
Sequence Comment
 Class Type
 action

 MY_POLICY
2 my_class3 ipv4
 mirror 1

 MY_POLICY2
2 my_class3 ipv4
 mirror 1
```

# policy resequence

## Syntax

```
policy <POLICY-NAME> resequence <STARTING-SEQ-NUM> <INCREMENT>
```

## Description

Resequences numbering in a policy.

## Command context

config

## Parameters

**<POLICY-NAME>**

Specifies the policy where you want to resequence policy entries.

**<STARTING-SEQ-NUM>**

Specifies the sequence number to start resequencing from.

**<INCREMENT>**

Specifies how much to increment the sequence numbers by.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Resequencing a policy:

```
switch(config)# policy MY_POLICY resequence 1 1
switch(config)# do show policy
 Name
Sequence Comment
 Class Type
 action

 MY_POLICY
1 MY_CLASS3 ipv4
 drop
 dscp AF21
2
 MY_CLASS3 ipv4
 mirror 1
```

## policy reset

### Syntax

```
policy <POLICY-NAME> reset
```

### Description

Changes the user-specified policy configuration to match the active policy configuration. Use this command when a discrepancy exists between what the user configured and what is active and accepted by the system.

### Command context

```
config
```

### Parameters

**<POLICY-NAME>**

Specifies the policy to be reset.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Resetting a policy:

```
switch(config)# policy MY_POLICY reset
```

## show class

### Syntax

```
show class [ip | ipv6] [<CLASS-NAME>] [commands] [configuration]
```

## Description

Shows class configuration information.

## Command context

Operator (>) or Manager (#)

## Parameters

All parameters are optional.

[ip | ipv6 ]

Selects the class type for the display: ip for IPv4, ipv6 for IPv6.

<CLASS-NAME>

Specifies the class name.

## commands

Specifies whether to display output as the CLI commands showing the configured class entries.

## configuration

Specifies whether to display classes that have been configured by the user, even if they are not active due to issues with the command parameters or hardware issues. This parameter is useful during a mismatch between the entered configuration and the previous successfully programmed (active) classes.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing all class configuration:

```
switch# show class
Type Name
Sequence Comment
 action L3 Protocol
 Source IP address Source L4 Port(s)
 Destination IP address Destination L4 Port(s)
 Additional Parameters

ipv4 MY_IPV4_CLASS
 10 my first class entry comment
 match icmp
 192.168.0.1/255.255.255.0
 192.168.1.1/255.255.255.0
 VLAN: 1
 20 my second class entry comment
 ignore tcp
 10.100.0.10/255.255.255.0 < 3000
 10.100.1.10/255.255.255.0 > 2000
 VLAN: 1

```

Showing class configuration for the IPv4 class MY\_IPV4\_CLASS as CLI commands:

```
switch# show class ip MY_IPV4_CLASS commands
class ip "MY_IPV4_CLASS"
 10 match icmp 192.168.0.1/255.255.255.0 192.168.1.1/255.255.255.0 vlan 1
 10 comment my first class entry comment
```

```
20 ignore tcp 10.100.0.10/255.255.255.0 lt 3000 10.100.1.10/255.255.255.0 gt 2000 vlan 1
20 comment my second class entry comment
```

## show policy

### Syntax

Syntax that shows information for all policies:

```
show policy [commands] [configuration]
```

Syntax that filters by policies applied to an interface or VLAN:

```
show policy [interface <IF-NAME> [in] | vlan <VLAN-ID> [in]]
 [commands] [configuration]
```

Syntax that filters by the named policy:

```
show policy <POLICY-NAME> [commands] [configuration]
```

Syntax that filters by the globally applied policy:

```
show policy global [commands] [configuration]
```

Syntax that shows statistical information in the form of hit counts:

```
show policy hitcounts <POLICY-NAME> [interface <IF-NAME> [in] |
 vlan <VLAN-ID> [in]]
```

Syntax that shows statistical information in the form of hit counts for the globally applied policy:

```
show policy hitcounts global
```

### Description

Shows information about your defined policies and where they have been applied. When `show policy` is entered without parameters, information for all policies is shown. The parameters filter the list of policies for which information is shown.

Available filtering includes:

- The content of a specific policy.
- All policies applied to a specific interface.
- All policies applied to a specific VLAN.
- The globally applied policy.

To display policy statistics, use the `show policy hitcounts` form of this command.

### Command context

Operator (>) or Manager (#)

### Parameters

**interface <IF-NAME>**

Specifies the interface name.

**vlan <VLAN-ID>**

Specifies the VLAN.

**in**

Selects the inbound (ingress) traffic direction.

**<POLICY-NAME>**

Specifies the policy name.

**commands**

Causes the policy definition to be shown as the commands and parameters used to create it rather than in tabular form.

**configuration**

Causes the user-configured policies be shown as entered, even if the policies are not active due to policy-definition command issues or hardware issues. This parameter is useful if there is a mismatch between the entered configuration and the previous successfully programmed (active) policies configuration.

**global**

Selects the globally applied policy.

**hitcounts**

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing information for all policies:

```
switch# show policy
 Name
Sequence Comment
 Class Type
 action

 my_policy
10 This is my QOS class.
 class1 ipv4
 dscp af21
 drop

```

Showing a policy as commands:

```
switch# show policy commands
policy my_policy
10 class ip my_class1 action dscp af21 action drop
```

Showing the globally applied policy:

```
switch# show policy global commands
policy global1
10 class ip my_class1 action drop
apply policy my_policy in
```

Showing policy hit counts (statistics):

```
switch# show policy hitcounts my_policy
Statistics for Policy my_policy:
Interface 1/1/1* (in):
```

```

 Hit Count Configuration
10 class ip my_class1 action dscp af21 action drop
 20 10 match any any any count
* policy statistics are shared among each context type (interface, VLAN).
 Use 'policy NAME copy' to create a new policy for separate statistics.

```

Showing policy hit counts (statistics) for the globally applied policy:

```

switch# show policy hitcounts global
Statistics for Policy global1:
Global Policy:
 Hit Count Configuration
10 class ip my_class1 action mirror
 20 10 match any any any count
* policy statistics are shared among each context type (interface, vlan, VRF).
 use 'policy NAME copy' to create a uniquely-named policy

```

Showing policy hit counts with conform rate from my\_policy applied inbound:

```

switch# show policy hitcounts my_policy
Statistics for Policy my_policy:
Interface 1/1/1* (in):
 Hit Count Configuration
10 class ipv6 my_class1 action cir kbps 1024 exceed drop [1024 kbps conform]
 - 10 match any any any
* policy statistics are shared among each context type (interface, VLAN).
 Use 'policy NAME copy' to create a new policy for separate statistics.

```

## alias

### Syntax

```
alias <ALIAS-NAME> <COMMAND-STRING>
```

```
no alias <ALIAS-NAME>
```

### Description

Defines an alias for one or more CLI commands. The alias and its definition are valid only for the user that creates the alias.



**NOTE:** The alias name cannot be an existing token name.

The `no` form of this command removes the specified alias.

### Command context

config

### Parameters

**<ALIAS-NAME>**

Specifies the name of the alias you are defining.

**<COMMAND-STRING>**

Specifies one or more commands and their parameters. Separate commands with a semicolon (;).  
Length: 1 to 400 characters.

For commands that require user-supplied parameters, use `$1` through `$n`, in order, as placeholders. These parameters are replaced by the corresponding arguments from the command line, and must match the number of parameters required by the original command. For alias definitions that include multiple commands, continue numbering parameters through all commands. Do not restart numbering for each command.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Defining an alias:

```
switch(config)# alias srci show running-config interface $1
switch(config)# srci?
 shv Execute "show alias" to list the command list. Arguments to replace $1,
 $2 etc.
switch(config)# show alias
```

| Alias Name | Alias Definition                  |
|------------|-----------------------------------|
| srci       | show running-config interface \$1 |



Using alias in config context:

```
switch(config)# srci 1/1/1
interface 1/1/1
 no shutdown
 ip address 1.1.1.1/24
exit
```

Using alias in enable context:

```
switch# srci 1/1/1
interface 1/1/1
 no shutdown
 ip address 1.1.1.1/24
exit
```

Using alias in operator context:

```
switch> srci 1/1/1
interface 1/1/1
 no shutdown
 ip address 1.1.1.1/24
exit
```

Removing an alias:

```
switch(config)# no alias srci
switch(config)# show alias
Alias Name Alias Definition

```

## auto-confirm

### Syntax

```
auto-confirm
```

```
no auto-confirm
```

### Description

Specifies that the CLI automatically enters the affirmative response (y) to all confirmation prompts, enabling commands to execute without waiting for user confirmation.

The `no` form of this command sets auto-confirmation to the default value disabled.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

Some commands, such as `boot` command, prompt to confirm execution of the command or to save the current configuration. Typically, such commands display a confirmation message similar to the following:

```
Continue (y/n)?
```

This command is useful for automating switch configuration, but Hewlett Packard Enterprise recommends that you use the REST API instead of using CLI scripts to automate configuration operations.

When the switch reboots, auto-confirmation is set to the default (disabled).

### Example

```
switch# auto-confirm
```

## configure terminal

### Syntax

```
configure [terminal]
```

### Description

Enters the global configuration (`config`) context.

### Command context

Manager (#)

### Parameters

**terminal**

Configure from the terminal. This is the default parameter.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```
switch# configure terminal
switch(config)#
```

## disable

### Syntax

```
disable
```

### Description

Exits the manager context (#) and enters the operator context (>).

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch# disable
switch>
```

## do

### Syntax

```
do <COMMAND>
```

### Description

Executes a manager context (#) or operator context (>) command from a configuration (config) context.



**NOTE:** You can execute `exec` commands from a configuration context with or without using the `do` command.

### Command context

A configuration context such as `config` or `config-vlan-10`

### Parameters

<COMMAND>

Specifies the manager context (#) or the operator context (>) command to execute.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

You can execute `exec` commands from a configuration context with or without using the `do` command.

Use the `do` command to execute commands such as `clear`, `checkpoint`, `auto-confirm` and `show` commands while you are in a configuration context (such as `config` or `config-vlan-10`).

For all `exec` commands you can use with the `do` command, from the global configuration context (`config`), enter `do`, followed by a space, and then press the **tab** key twice.

### Examples

Clearing LLDP neighbors from the global configuration context:

```
switch(config)# clear lldp neighbors
switch(config)# do clear lldp neighbors
switch(config)#
```

## enable

### Syntax

```
enable
```

### Description

Exits the operator context (>) and enters the manager context (#).

## Command context

Operator (>)

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch> enable
switch#
```

# end

## Syntax

end

## Description

Exits the current context and enters the manager context (#).

## Command context

Any context

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch# configure terminal
switch(config)# vlan 10
switch(config-vlan-10)# vlan 22
switch(config-vlan-22)# end
switch#
```

# exit

## Syntax

exit

## Description

Exits the current context and enters its parent context.

## Command context

Any context

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch# configure terminal
switch(config)# vlan 10
switch(config-vlan-10)# vlan 22
switch(config-vlan-22)# exit
switch(config)# exit
switch#
```

## list

### Syntax

list

### Description

Shows a list of commands available from the current context.

### Command context

Any context

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

```
switch> list
 list
 enable
 exit
 show session-timeout
...
```

## page

### Syntax

page [<LINES>]

no page

### Description

Specifies the number of output lines the CLI displays before pausing to wait for a user to press a key. This value is the number of lines supported by the terminal session. This setting is not persistent and applies to the current session only.

The `page` command is enabled by default on the switch with the number of lines supported by the terminal. Change this default by using the `page` command to specify a different number of output lines.

The `no` form of this command sets the number of lines that are displayed to the default, which is the number of lines supported by the current terminal session.

## Command context

Manager (#)

## Parameters

<*LINES*>

Specifies the number to display before pausing. If not specified, the number of lines supported by the current terminal session is used. Range: 2-1000 lines.

Default: The number of lines supported by the current terminal session

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the page size to an unlimited number of lines:

```
switch# no page
switch#
```

Example output of a command after setting the page size to 10 lines:

```
switch# page 10
switch# list
 show hostname
 show domain-name
 list
 configure { terminal }
 disable
 exit
 end
 page
 page <2-1000>
-- MORE --, next page: Space, next line: Enter, quit: q
```

# Pipe (|) command

## Syntax

```
show running-config | begin 2 "vlan" | redirect "abc.txt"
show running-config | include "vlan" | exclude "vlan2" | count
show vlan | include "up" | include "VLAN100"
```

```
list | include "show" | exclude "show" | count
```

## Description

The pipe (|) command filters the output of `show` or `list` commands using the options `include`, `exclude`, `begin`, `count`, or `redirect`.

## Command context

Manager (#)

## Parameters

None

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

```
show {<SHOW-COMMAND-OPTIONS>}... [| {include <PATTERN-STRING> |
 exclude <PATTERN-STRING> |
 begin {<LINES-TO-DISPLAY>} <PATTERN-STRING>}}]...
 [| {count [<PATTERN-STRING>] |
 redirect [<FILE-NAME>}}]

list [| {include <PATTERN-STRING> |
 exclude <PATTERN-STRING> |
 begin {<LINES-TO-DISPLAY>} <PATTERN-STRING>}}]...
 [| {count [<PATTERN-STRING>] |
 redirect [<FILE-NAME>}}]
```

- The pipe (|) command is supported for use with the `show` and `list` commands only.
- You can use multiple pipe commands with a single `show` or `list` command.

## Examples

```
show running-config | redirect "abc.txt"
show running-config | begin 2 "vlan" | begin -2 "vlan" | begin "vlan"
show running-config | include "vlan" | exclude "vlan2" | count
show vlan | include "up" | include "VLAN100"
list | include "show" | exclude "show" | count
```

# session-timeout

## Syntax

```
session-timeout <MINUTES>
```

```
no session-timeout
```

## Description

Specifies the number of minutes a CLI session can be idle before the session is automatically terminated and the user is logged out.

The `no` form of this command sets the timeout to the default value of 30 minutes.

## Command context

```
config
```

## Parameters

<MINUTES>

Specifies the number of minutes the CLI session can remain idle. Specify 0 to configure CLI sessions to never time out. Range: 0 to 4320. Default: 30

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# session-timeout 15S
```

## repeat

### Syntax

```
repeat [id <POSITION>] [<COUNT>] [<DELAY>]
```

### Description

Repeatedly executes one or more commands. By default, the most recent command in the history is executed until you press **Ctrl+C**.

### Command context

Manager (#)

### Parameters

#### <POSITION>

Specifies the position of a command, or range of positions of multiple commands, in the history list as shown in the output of the `show history` command. <POSITION> can be a single number, a comma-separated list of numbers, or a range of numbers specified by the beginning and end of the range, separated by a hyphen.

If the `id` parameter is not specified, the `repeat` command executes the command that was entered most recently. Default: 1.

#### <COUNT>

Specifies number of times to execute the command or commands. Default: The command repeats an infinite number of times.

#### <DELAY>

Specifies the number of seconds to delay before executing the command. Default: 2

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```
switch# repeat id 1-4,7-8,10 count 2 delay 3
```

## show alias

### Syntax

```
show alias [vsx-peer]
```

### Description

Shows the command aliases that are defined on the switch.

### Command context

Manager (#)



## Parameters

### [vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch# show alias
Alias Name Alias Definition

hst hostname $1
int_config interface $1; no shutdown; ip address $2; lldp receive; mtu $3; exit

switch# show hst
Alias Name Alias Definition

hst hostname $1
```

# show history

## Syntax

```
show history [timestamp]
```

## Description

Shows a numbered list of the commands that have been executed during this CLI session. Commands are displayed in descending order, with the most recent command having the lowest number

## Command context

Manager (#)

## Parameters

### timestamp

Specifies that output include the time of execution of each command in the command history.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

Use the `show history` command to show the commands executed during this session. You use the command numbers to specify commands to repeat using the `repeat` command.

A command is not saved in the history if the command is one of the following:

- The same command as the command that was entered most recently.
- The `show history` command.
- The `repeat` command.

The commands saved in the history command buffer are in the same format in which you entered the commands. For example, if you execute the `show startup-config` command repeatedly, the system saves only one command in the history command buffer. If you execute the command in the format of `show start` and `show startup` respectively, the system saves them as two commands.

### Example

```
switch# show history
6 show vers
5 conf
4 int mgmt
3 ip static 10.1.1.2/24
2 no shut
1 end
switch# show history timestamp
6 Tue Jan 30 08:42:24 2018 show vers
5 Tue Jan 30 08:45:52 2018 conf
4 Tue Jan 30 08:45:54 2018 int mgmt
3 Tue Jan 30 08:46:00 2018 ip static 10.1.1.2/24
2 Tue Jan 30 08:46:02 2018 no shut
1 Tue Jan 30 08:46:02 2018 end
switch#
```

## show session-timeout

### Syntax

```
show session-timeout [vsx-peer]
```

### Description

Shows the configured session timeout value.

### Command context

Manager (#)

### Parameters

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```
switch# show session-timeout
session-timeout: 30 minutes (Default)
```

## cli-session

### Syntax

```
cli-session
```

```
no cli-session
```

### Description

Enters the CLI session context (shown in the switch prompt as `config-cli-session`) for the purpose of configuring CLI user session management. Session management enhances security by enforcing specific CLI user session requirements. The following information is provided at time of successful login:

- When applicable, the number of failed login attempts since the most recent successful login.
- The date, time, and location (console or IP address or hostname) of the most recent previous successful login.
- The count of successful logins within the past (configurable) time period.

For example:

```
switch login: admin
Password:
```

```
There were 3 failed login attempts since the last successful login
Last login: 2019-04-20 08:51:33 from the console
User "admin" has logged in 73 times in the past 30 days
```

The no form of this command disables concurrent CLI user session restrictions and reverts `timeout` and `tracking-range` to their default values.



**NOTE:** To ensure that enhanced security is maintained, it is recommended that you keep CLI user session management fully enabled by setting `max-per-user` to a nondefault value.



**NOTE:** The `cli-session` command applies only to SSH/console login connection types. It does not apply to other connection types such as REST.

### Command context

```
config
```

### Subcommands

These subcommands are available within the CLI session context.

**[no] max-per-user <SESSIONS>**

Specifies the maximum number of concurrent CLI sessions per user. The no form of this subcommand disables concurrent CLI user session restrictions. Default: Disabled (no value). Range: 1 to 5.



**NOTE:** When the same user name is configured for both local and remote authentication, both users, regardless of privilege level, are considered to be the same user for the purpose of counting concurrent CLI sessions. For example, with `max-per-user` set to 1 and user `admin1` configured for local and remote authentication, only the local user `admin1` or the remote user `admin1` can be logged in at any given moment. Both `admin1` users cannot be logged in simultaneously unless `max-per-user` is increased to at least 2.

**[no] timeout** <MINUTES>

Specifies the number of minutes a CLI session can be idle before the session is automatically terminated and the user is logged out. A value of 0 minutes disables the session timeout. The `no` form of this subcommand sets the timeout value to the default. Default 30: Range 0 to 4320.



**NOTE:** This subcommand is the recommended replacement for the `session-timeout` command.

**[no] tracking-range** <DAYS>

Specifies the maximum number of days to track CLI user session logins. The `no` form of this subcommand resets the value to its default. Default 30: Range 1 to 30.

**exit**

Exits the CLI session context.

**end**

Exits the CLI session context and then the config context.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring CLI user session settings for a maximum of one concurrent session, a 20-minute timeout, and tracking for a maximum of 25 days.

```
switch(config)# cli-session
switch(config-cli-session)# max-per-user 1
switch(config-cli-session)# timeout 20
switch(config-cli-session)# tracking-range 25
switch# exit
```

After successful earlier logins, logging in from the console without any intervening unsuccessful logins.

```
switch login: admin1
Password:
```

```
Last login: 2019-04-15 14:10:21 from the console
User 'admin1' has logged in 65 times in the past 25 days
```

Attempting to log in as `admin1` when already logged in as `admin1` from elsewhere.

```
switch login: admin1
Password:
Too many logins for 'admin1'
```

After successful earlier logins, attempting to log in twice with an invalid password, followed by a successful login.

```
switch login: admin1
Password:
```

```
Login incorrect
```

```
switch login: admin1
Password:
```

```
Login incorrect
switch login: admin1
Password:
```

```
There were 2 failed login attempts since the last successful login
Last login: 2019-04-15 17:22:45 from 192.168.1.1
User 'admin1' has logged in 72 times in the past 25 days
```

## clock date

### Syntax

```
clock date <DATE>
```

### Description

Sets the switch date.

### Command context

```
config
```

### Parameters

**<DATE>**

Specifies the date. Format: YYYY-MM-DD.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

This example sets the date to Dec 14, 2017.

```
switch(config)# clock date 2017-12-14
```

## clock datetime

### Syntax

```
clock datetime <DATE> <TIME>
```

### Description

Sets the switch date and time.

### Command context

```
config
```

### Parameters

**<DATE>**

Specifies the date. Format: YYYY-MM-DD.

**<TIME>**

Specifies the time in 24-hour clock format. Seconds are optional. Format: HH:MM or HH:MM:SS.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

This example sets the date and time to Dec 13, 2017 at 15:00.

```
switch(config)# clock datetime 2017-12-13 14:15:00
```

# clock time

## Syntax

```
clock time <TIME>
```

## Description

Sets the switch time.

## Command context

```
config
```

## Parameters

<TIME>

Specifies the time in 24-hour clock format. Seconds are optional. Format: HH:MM or HH:MM:SS.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

This example sets the time to 15:01:23.

```
switch(config)# clock time 15:01:23
```

# clock timezone

## Syntax

```
clock timezone <TIME-ZONE>
no clock timezone
```

## Description

Sets the time zone and its associated daylight savings time rule.

The `no` form of this command sets the time zone to the default value of UTC.

## Command context

```
config
```

## Parameters

**timezone** <TIME-ZONE>

Specifies the time zone, <TIME-ZONE>, using a name defined in the IANA time zone database. See [https://en.wikipedia.org/wiki/List\\_of\\_tz\\_database\\_time\\_zones](https://en.wikipedia.org/wiki/List_of_tz_database_time_zones).

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the time zone to Eastern Standard Time (EST):

```
switch(config)# clock timezone EST
```

# show clock

## Syntax

```
show clock
```

## Description

This command displays the current date, time, and time zone.

## Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

```
switch# show clock
Wed Nov 22 23:29:10 PDT 2017
System is configured for timezone : US/Pacific
```



## Classes of traffic

The different classes of traffic that can be individually configured are:

- `arp-broadcast`: Address Resolution Protocol packets with a broadcast destination MAC address.
- `arp-unicast`: Address Resolution Protocol packets with a switch system destination MAC address.
- `dhcp`: Dynamic Host Configuration Protocol packets. Also includes snooped DHCP packets if DHCP snooping is enabled.
- `icmp-broadcast-ipv4`: Internet Control Message Protocol packets with a broadcast or multicast destination IPv4 address.
- `icmp-multicast-ipv6`: Internet Control Message Protocol packets with a well-known multicast destination IPv6 address.
- `icmp-security-ipv6`: IPv6 Internet Control Message Protocol packets intercepted and inspected.
- `icmp-unicast-ipv4`: Internet Control Message Protocol packets with a destination IPv4 address owned by the switch
- `icmp-unicast-ipv6`: Internet Control Message Protocol packets with a destination IPv6 address owned by the switch.
- `igmp`: Internet Group Management Protocol packets.
- `lacp`: Link Aggregation Control Protocol packets with the destination MAC address 01:80:c2:00:00:02.
- `lldp`: Link Layer Discovery Protocol packets with the destination MAC address 01:80:c2:00:00:0e.
- `loop-protect`: Loop Protection packets with the destination MAC address 09:00:09:09:13:a6.
- `manageability`: Unicast IP packets addressed to the switch for specific protocols that do not have a dedicated CoPP class like HTTP, SSH, RADIUS.
- `mvrp`: Multiple VLAN Registration Protocol packets with the destination MAC address 01:80:c2:00:00:20 or 01:80:c2:00:00:21
- `ntp`: Network Time Protocol packets with a destination IP address owned by the switch.
- `stp`: Spanning Tree Protocol (STP) packets with the destination MAC address 01:80:c2:00:00:00 or Per-VLAN Spanning Tree (PVST) packets with the destination MAC address 01:00:0c:cc:cc:cd.
- `udld`: Unidirectional Link Detection packets with the destination MAC address 01:00:0c:cc:cc:cc or 00:e0:52:00:00:00, or Cisco Discovery Protocol packets with the destination MAC address 01:00:0c:cc:cc:cc.
- `unresolved-ip-unicast`: Packets to be software forwarded by the management processor.

To regulate any other traffic destined for the CPU, every CoPP policy has a class named `default` that can also be configured to regulate other traffic to the CPU or prevent other traffic from being delivered.



---

**NOTE:** All IPsec traffic received by the CPU will be regulated by the `ipsec` class regardless of the encapsulated protocol.

---

## apply copp-policy

### Syntax

```
apply copp-policy { <NAME> | default }

no apply copp-policy <NAME>
```

### Description

Applies a CoPP policy to the switch, replacing the policy that is in effect. There may be a brief interruption in traffic flow to the management processor while the switch implements the change.

Enter the `no apply copp-policy <NAME>` command with the name of a CoPP policy to unapply a CoPP policy and apply the default CoPP policy. This will only take effect if the specified policy is actively applied. Since there must always be a CoPP policy applied, this command effectively attempts to replace the applied CoPP policy with the default CoPP policy. The default CoPP policy cannot be unapplied using this command.

### Command context

```
config
```

### Parameters

**<NAME>**

Specifies the name of the policy to apply. Length: 1 to 64 characters.

**default**

Applies the default policy.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

If the new policy cannot be applied (for example, due to a lack of hardware resources), the previous policy remains in effect. Use the `show copp-policy` command to determine which policy is in effect.

### Examples

Applying a policy named `My_CoppPolicy`:

```
switch(config)# apply copp-policy My_CoppPolicy
```

Applying the default policy:

```
switch(config)# apply copp-policy default
```

Unapplying a policy named `My_CoppPolicy`:

```
switch(config)# no apply copp-policy My_CoppPolicy
```

# class

## Syntax

```
class <CLASS> {drop | priority <PRIORITY> rate <RATE>}
```

```
no class <CLASS> {drop | priority <PRIORITY> rate <RATE>}
```

## Description

Adds a class to a CoPP policy. If the class exists, the existing class is modified. Changes made to an active (applied) policy take effect immediately.

When adding or modifying a class in an active policy, CoPP immediately activates the change on the switch. In cases where insufficient hardware resources exist to support a class or its action, CoPP fails to activate the changed class on the switch. When this failure occurs, the active configuration on the switch will be out of sync with its definition. To diagnose and remedy this situation:

- Use the **show copp-policy** command to determine which classes are out of sync between the active policy and its definition.
- Use the **reset copp-policy** command to synchronize the active policy with its definition. This synchronization changes the classes in the definition to match the classes in the active policy.

The **no** form of this command removes the configuration for the class. Traffic for the class will be prioritized and regulated using the factory default configuration for the class. Use the **show copp-policy factory-default** command to display the factory default CoPP policy. To stop a class of traffic from reaching the processor, set the class action to drop.

## Command context

config-copp

## Parameters

**<CLASS>**

Specifies the **class** to add or edit.

**drop**

Drop packets matching the selected class.

**priority <PRIORITY>**

Specifies the priority for packets matching the selected class. Range: 0 to 6.

**rate <RATE>**

Specifies the maximum rate, in packets per second (pps), for packets matching the selected class. Range: 25 to 99999.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Adding a class to handle LACP traffic with priority of 2 and rate of 2000:

```
switch(config-copp)# class lacp priority 2 rate 2000
```

Modifying the class to drop LLDP packets:

```
switch(config-copp)# class lldp drop
```

Removing the class that handles LLDP packets.

```
switch(config-copp)# no class lldp
```

## clear copp-policy statistics

### Syntax

```
clear copp-policy statistics
```

### Description

Resets statistics for all CoPP classes to zero.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Displaying and then resetting statistics for all classes in the active policy:

## copp-policy

### Syntax

```
copp-policy {<NAME> | default [revert]}
```

```
no copp-policy <NAME>
```

### Description

Creates a CoPP policy and switches to the `config-copp` context for the policy. Or, if the specified policy exists, switches to the `config-copp` context for the policy. A predefined policy, named `default`, contains factory default classes and is applied to the switch at first startup. This policy cannot be deleted, but its configuration can be changed.

The `no` form of this command removes a CoPP policy. If a policy is active (applied), it cannot be removed. It must be replaced with another policy before it can be removed.

### Command context

```
config
```

### Parameters

**<NAME>**

Specifies the name of the policy to add or edit. Length: 1 to 64 characters. The name must not be a substring of any of the following reserved words: `default`, `factory-default`, `commands`, `configuration`, or `statistics`.

### **default**

Specifies the default CoPP policy. Use this default policy to configure the default policy.

### **revert**

Sets the default CoPP policy to its factory settings.

### **Authority**

Administrators or local user group members with execution rights for this command.

### **Examples**

Creating a policy named My\_CoppPolicy:

```
switch(config)# copp-policy My_CoppPolicy
switch(config-copp)#
```

Removing a policy named My\_CoppPolicy:

```
switch(config)# no copp-policy My_CoppPolicy
```

Setting the default policy to its factory settings:

```
switch(config)# copp-policy default revert
```

Unapplying the policy named My\_CoppPolicy:

```
switch(config)# no apply copp-policy My_CoppPolicy
```

## **default-class**

### **Syntax**

```
default-class priority <PRIORITY> rate <RATE>
```

### **Description**

Configures the default class that is automatically defined for all CoPP policies. The default class cannot be removed, but its configuration can be changed. The default class is applied to traffic that does not match any other class defined for a policy.

### **Command context**

```
config-copp
```

### **Parameters**

**priority** <PRIORITY>

Specifies the priority for packets matching the selected class. Range: 0 to 7.

**rate** <RATE>

Specifies the maximum rate, in packets per second (pps), for packets matching the selected class. Range: 25 to 99999.

### **Authority**

Administrators or local user group members with execution rights for this command.

## Example

Setting the default class to a priority of 2 and rate of 2000:

```
switch(config-copp)# default-class priority 2 rate 2000
```

## reset copp-policy

### Syntax

```
reset copp-policy { <NAME> | default }
```

### Description

Resets an active CoPP policy to match the settings that are currently in effect for the active policy on the switch. Changes made to the active policy that could not be activated are removed from the active policy. When the switch fails to add or modify a class in an active CoPP policy, it is possible the active policy settings on the switch may be out of sync with those defined in the policy.

### Command context

config

### Parameters

**<NAME>**

Specifies the name of the policy to reset. Length: 1 to 64 characters.

**default**

Resets the default policy to match its active settings.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Resetting a policy named My\_CoppPolicy:

```
switch# show copp-policy My_CoppPolicy
class drop priority rate pps hardware rate pps

igmp 6 5000 5000
lacp 2 2000 2000
default 1 6000 6000
switch# config terminal
switch(config)# copp-policy My_CoppPolicy
switch(config-copp)# class stp priority 4 rate 4000
switch(config-copp)# do show copp-policy My_CoppPolicy
class drop priority rate pps hardware rate pps

igmp 6 5000 5000
lacp 2 2000 2000
default 1 6000 6000
% Warning: user-specified classes in CoPP policy My_CoppPolicy do not match
active configuration.
switch(config-copp)# do show copp-policy My_CoppPolicy configuration
class drop priority rate pps applied

igmp 6 5000 yes
lacp 2 2000 yes
```

```

stp 4 4000 no
default 1 6000 yes
% Warning: user-specified classes in CoPP policy My_CoppPolicy do not match
active configuration.
switch(config-copp)# exit
switch(config)# reset copp-policy My_CoppPolicy
switch(config)# do show copp-policy My_CoppPolicy
class drop priority rate pps hardware rate pps

igmp 6 5000 5000
lacp 2 2000 2000
default 1 6000 6000

```

Resetting the default policy:

```
switch(config)# reset copp-policy default
```

## show copp-policy

### Syntax

```
show copp-policy [<NAME> | default] [commands] [configuration]
```

### Description

Shows CoPP policy settings for a specific CoPP policy. When entered without specifying either a name or the `default` parameter, shows all the CoPP policy settings that are active on the switch and have successfully been programmed into the hardware.

A warning is displayed if:

- The active and user-specified applications of a policy do not match.
- The active and user-specified configurations of a policy do not match.

### Command context

Operator (>) or Manager (#)

### Parameters

**<NAME>**

Specifies the name of the policy for which to display settings. Length: 1 to 64 characters.

**default**

Displays CoPP settings for the default policy.

**commands**

Displays output as CLI commands.

**configuration**

Displays user-specified CoPP settings and **not** the active settings.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Displaying the CoPP policies defined in the configuration and the active application:

```
switch# show copp-policy
applied copp_policy_name

My_CoppPolicy
applied default
switch#
```

Displaying the active configuration of all CoPP policies as CLI commands:

Displaying the default policy:

```
switch# show copp-policy default
class drop priority rate pps hardware rate pps

arp-broadcast 2 1250 1250
arp-unicast 3 825 825
<--OUTPUT OMITTED FOR BREVITY-->
default 2 4225 4225
```

## show copp-policy factory-default

### Syntax

```
show copp-policy factory-default [commands]
```

### Description

Display the configuration for the factory-default CoPP policy.

### Command context

Operator (>) or Manager (#)

### Parameters

#### commands

Displays output as CLI commands.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Displaying the factory-default policy:

```
switch# show copp-policy factory-default
class drop priority rate pps

arp-broadcast 2 1250
arp-unicast 3 825
<--OUTPUT OMITTED FOR BREVITY-->
default 2 4225
```

Displaying the active configuration of My\_CoppPolicy (My\_CoppPolicy is applied):



```
switch# config terminal
switch(config)# apply copp-policy My_CoppPolicy
switch(config)# do show copp-policy My_CoppPolicy
```

| class   | drop | priority | rate | pps | hardware | rate | pps |
|---------|------|----------|------|-----|----------|------|-----|
| igmp    |      | 6        | 5000 |     | 5000     |      |     |
| lacp    |      | 2        | 2000 |     | 2000     |      |     |
| default |      | 1        | 6000 |     | 6000     |      |     |

Displaying the active configuration of My\_CoppPolicy as CLI commands:

```
switch# show copp-policy My_CoppPolicy commands
copp-policy My_CoppPolicy
 class igmp priority 6 rate 5000
 class lacp priority 2 rate 2000
 default-class priority 1 rate 6000
apply copp-policy My_CoppPolicy
```

Displaying the user-specified configuration of My\_CoppPolicy:

```
switch# show copp-policy My_CoppPolicy configuration
```

| class   | drop | priority | rate | pps | applied |
|---------|------|----------|------|-----|---------|
| igmp    |      | 6        | 5000 |     | yes     |
| lacp    |      | 2        | 2000 |     | yes     |
| default |      | 1        | 6000 |     | yes     |

Displaying the user-specified configuration of My\_CoppPolicy as CLI commands:

```
switch# show copp-policy My_CoppPolicy commands configuration
copp-policy My_CoppPolicy
 class igmp priority 6 rate 5000
 class lacp priority 2 rate 2000
 default-class priority 1 rate 6000
apply copp-policy My_CoppPolicy
```

## show copp-policy statistics

### Syntax

```
show copp-policy statistics [class <CLASS> | default-class | non-zero]
```

### Description

Displays statistics for all classes, a single class, or all classes with non-zero statistics in the active CoPP policy.

### Command context

Operator (>) or Manager (#)

### Parameters

**<CLASS>**

Specifies the **class** for which to display statistics.

**default-class**

Displays statistics for the default class.

**non-zero**

Displays statistics for all classes with non-zero statistics.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Usage

If a single class is specified, the priority and rate that has been programmed in hardware for that class will be shown.

## Examples

Applying the `default` CoPP policy and displaying statistics for all classes in the actively applied policy:



**NOTE:** The rate displayed is the actual rate in hardware.

```
switch# config terminal
switch(config)# apply copp-policy default
switch(config)# exit
switch# show copp-policy statistics
Statistics for CoPP policy 'default':
Totals:
 packets passed : 1000 packets dropped : 1500
Class: default
 packets passed : 400 packets dropped : 600
Class: arp-broadcast
 packets passed : 500 packets dropped : 800
 <--OUTPUT OMITTED FOR BREVITY-->
```

Displaying statistics for the default class in the active policy:

```
switch(config)# show copp-policy statistics default-class
Statistics for CoPP policy 'default':
Class: default
Description: Default
 priority : 2
 rate (pps) : 4225

 packets passed : 400 packets dropped : 600
```

Displaying statistics for the class `arp-broadcast` in the actively applied policy:

```
switch# show copp-policy statistics class arp-broadcast
Statistics for CoPP policy 'default':
Class: arp-broadcast
Description: Address Resolution Protocol broadcast
 priority : 2
 rate (pps) : 1250

 packets passed : 500 packets dropped : 800
```

## show tech copp

### Syntax

```
show tech copp
```

## Description

Displays the output of all show commands supported by CoPP.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Capturing the command output into a local file:

```
switch# show tech copp local-file
Show Tech output stored in local-file. Please use 'copy show-tech local-file' to copy-out this file.

switch# copy show-tech local-file ?
 REMOTE_URL URL of syntax
 {tftp://|sftp://USER@}{IP|HOST}[:PORT][;blocksize=VAL]/FILE
 STORAGE_URL URL of syntax usb:/file
switch# copy show-tech local-file
```

## clear debug buffer

### Syntax

```
clear debug buffer
```

### Description

Clears all debug logs. Using the `show debug buffer` command will only display the logs generated after the `clear debug buffer` command.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Clearing all generated debug logs:

```
switch# show debug buffer

show debug buffer

2018-10-14:09:10:58.558710|lldpd|LOG_DEBUG|MSTR||LLDP|LLDP_CONFIG|No Port cfg changes
2018-10-14:09:10:58.558737|lldpd|LOG_DEBUG|MSTR||LLDP|LLDP_EVENT|lldpd_stats_run entered at time 8257199
2018-10-14:09:10:58.569317|lldpd|LOG_DEBUG|MSTR||LLDP|LLDP_CONFIG|No Port cfg changes
2018-10-14:09:11:21.881907|hpe-sysmond|LOG_INFO|MSTR||SYSMON|SYSMON_CONFIG|Sysmon poll interval changed to 32

switch# clear debug buffer
switch# show debug buffer

show debug buffer

2018-10-14:09:13:24.481407|hpe-sysmond|LOG_INFO|MSTR||SYSMON|SYSMON_CONFIG|Sysmon poll interval changed to 51
```

## debug {all | <MODULE-NAME>}

### Syntax

```
debug {all | <MODULE-NAME>} [<SUBMODULE-NAME>] [severity (emer|crit|alert|err|
notice|warning|info|debug)] {port <PORT-NAME> | vlan <VLAN-ID> | ip <IP-ADDRESS> |
mac <MAC-ADDRESS> | vrf <VRF-NAME> | instance <INSTANCE-ID>}
```

```
no debug {all | <MODULE-NAME>} [<SUBMODULE-NAME>] {port | vlan | ip | mac | vrf |
instance}
```

### Description

Enables debug logging for modules or submodules by name, with optional filtering by specific criteria.

The `no` form of this command disables debug logging.

## Command context

Manager (#)

## Parameters

**all**

Enables debug logging for all modules.

**<MODULE-NAME>**

Enables debug logging for a specific module. For a list of supported modules, enter the `debug` command followed by a space and a question mark (?).

**<SUBMODULE-NAME>**

Enables debug logging for a specific submodule. For a list of supported submodules, enter the `debug` **<MODULE-NAME>** command followed by a space and a question mark (?).

**severity (emer|crit|alert|err|notice|warning|info|debug)**

Selects the minimum severity log level for the destination. If a severity is not provided, the default log level is `debug`. Optional.

**emer**

Specifies storage of debug logs with a severity level of `emergency` only.

**crit**

Specifies storage of debug logs with severity level of `critical` and above.

**alert**

Specifies storage of debug logs with severity level of `alert` and above.

**err**

Specifies storage of debug logs with severity level of `error` and above.

**notice**

Specifies storage of debug logs with severity level of `notice` and above.

**warning**

Specifies storage of debug logs with severity level of `warning` and above.

**info**

Specifies storage of debug logs with severity level of `info` and above.

**debug**

Specifies storage of debug logs with severity level of `debug` (default).

**port**

Displays debug logs for the specified port, for example `1/1/1`.

**vlan <VLAN-ID>**

Displays debug logs for the specified VLAN. Provide a VLAN from 1 to 4094.

**ip <IP-ADDRESS>**

Displays debug logs for the specified IP Address.

**mac <MAC-ADDRESS>**

Displays debug logs for the specified MAC Address, for example `A:B:C:D:E:F`.

**vrf** <VRF-NAME>

Displays debug logs for the specified VRF.

**instance** <INSTANCE-ID>

Displays debug logs for the specified instance. Provide an instance ID from 1 to 255.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch# debug all
```

# debug db

## Syntax

```
debug db {all | sub-module} [level <MINIMUM-SEVERITY>] [filter]
```

```
no debug db {all | sub-module} [level <MINIMUM-SEVERITY>] [filter]
```

## Description

Enables or disables debug logging for a db module or submodules, with an option to filter by specific criteria.

The **no** form of this command disables debug logging for the db module or submodule.

## Command context

Manager (#)

## Parameters

**all**

Enables all submodules for the db log.

**sub-module**

Enables debug logging for supported submodules. Specify **rx** or **tx** debug logs.

**filter**

Specifies supported filters for the db log. Specify **table**, **column**, or **client**. Optional

**severity** (**emer**|**crit**|**alert**|**err**|**notice**|**warning**|**info**|**debug**)

Selects the minimum severity log level for the destination. If a severity is not provided, the default log level is **debug**. Optional.

**emer**

Specifies storage of debug logs with a severity level of **emergency** only.

**crit**

Specifies storage of debug logs with severity level of **critical** and above.

**alert**

Specifies storage of debug logs with severity level of **alert** and above.

**err**

Specifies storage of debug logs with severity level of `error` and above.

**notice**

Specifies storage of debug logs with severity level of `notice` and above.

**warning**

Specifies storage of debug logs with severity level of `warning` and above.

**info**

Specifies storage of debug logs with severity level of `info` and above.

**debug**

Specifies storage of debug logs with severity level of `debug` (default).

**Authority**

Administrators or local user group members with execution rights for this command.

**Usage**

DBlog is a high performance, configuration, and state database server logging infrastructure where a user can log the transactions which are sent or received by clients to the configuration and state database server. It can be enabled through the CLI and REST, and also supports filters where a user can filter out logs on the basis of table, column, or client. It is helpful for debugging when the user wants to debug an issue with a particular client, table, or column combination. It is not enabled by default. A combination of filters can also be applied to filter out messages based on table, column, and client.

There are three submodules for the "db" module:

1. `all`: When `all` is enabled, no filters are applied to any of the debug logs, even if other submodules are configured with filters.
2. `tx`: If enabled, only the replies and notifications sent out for the initial and incremental updates are logged.
3. `rx`: If enabled, only the transactions sent to the configuration and state database server are logged.

The keyword `all` may be used to enable or disable debug logging for all sub-modules. Also a combination of filters can be used to filter the message types.

If the table or client filter is applied, then the messages belonging to this specific table or client will be logged. The column filter can also be applied to further filter messages on a table, providing a mechanism to filter messages on a column. The table and client filter can be used in combination or separately, but column can only be used in conjunction with table.

**Examples**

Configuring all submodules with severity `debug`:

```
switch# debug db all severity debug
```

Configuring the `tx` submodule with table `Interface` filter and severity `debug`:

```
switch# debug db tx table Interface severity debug
```

Configuring the `rx` submodule with table `Interface column statistics` filter and severity `debug`:

```
switch# debug db rx table Interface column statistics severity debug
```

Disabling the `rx` submodule:

```
switch# no debug db rx
```

Disabling the `tx` submodule `table` Interface:

```
switch# no debug db tx table Interface
```

## debug destination

### Syntax

```
debug destination {syslog | file | console | buffer} [severity (emer|crit|alert|err|notice|warning|info|debug)]
```

```
no debug destination {syslog | file | console}
```

### Description

Sets the destination for debug logs and the minimum severity level for each destination

The `no` form of this command unsets the destination for debug logs.

### Command context

Manager (#)

### Parameters

**{syslog | file | console | buffer}**

Selects the destination to store debug logs. Required.

**syslog**

Specifies that the debug logs are stored in the `syslog`.

**file**

Specifies that debug logs are stored in `file`.

**console**

Specifies that debug logs are stored in `console`.

**buffer**

Specifies that debug logs are stored in `buffer` (default).

**severity (emer|crit|alert|err|notice|warning|info|debug)**

Selects the minimum severity log level for the destination. If a severity is not provided, the default log level is `debug`. Optional.

**emer**

Specifies storage of debug logs with a severity level of `emergency` only.

**crit**

Specifies storage of debug logs with severity level of `critical` and above.

**alert**

Specifies storage of debug logs with severity level of `alert` and above.

**err**

Specifies storage of debug logs with severity level of `error` and above.



**notice**

Specifies storage of debug logs with severity level of `notice` and above.

**warning**

Specifies storage of debug logs with severity level of `warning` and above.

**info**

Specifies storage of debug logs with severity level of `info` and above.

**debug**

Specifies storage of debug logs with severity level of `debug` (default).

**Authority**

Administrators or local user group members with execution rights for this command.

**Usage**

Events that have a severity equal to or higher than the configured severity level are stored in the designated destination. The product defaults to `buffer` for destination and `debug` as a severity level.

**Examples**

```
switch# debug destination syslog severity alert
switch# debug destination console severity info
switch# debug destination file severity warning
switch# debug destination buffer severity err
```

## show debug

**Syntax**

```
show debug
```

**Description**

Displays the enabled debug types.

**Command context**

Manager (#)

**Authority**

Administrators or local user group members with execution rights for this command.

**Examples**

```
switch# show debug

module sub_module severity vlan port ip mac instance vrf

all all err 1 1/1/1 10.0.0.1 1a:2b:3c:4d:5e:6f 2 default
```

# show debug buffer

## Syntax

```
show debug buffer [module <MODULE-NAME> | severity (emer|crit|alert|err|notice|warning|info|debug)]
```

## Description

Displays debug logs stored in the specified debug buffer with optional filtering by module or severity.

## Command context

Manager (#)

## Parameters

**<MODULE-NAME>**

Filters debug logs displayed by the specified module name.

**severity (emer|crit|alert|err|notice|warning|info|debug)**

Displays debug logs with a specified severity level. Defaults to debug. Optional.

**emer**

Displays debug logs with a severity level of emergency only.

**crit**

Displays debug logs with a severity level of critical and above.

**alert**

Displays debug logs with a severity level of alert and above.

**err**

Specifies storage of debug logs with severity level of error and above.

**notice**

Specifies storage of debug logs with severity level of notice and above.

**warning**

Displays debug logs with a severity level of warning and above.

**info**

Displays debug logs with a severity level of info and above.

**debug**

Displays debug logs with a severity level of debug (default).

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch# show debug buffer

show debug buffer

2017-03-06:06:51:15.089967|hpe-sysmond|SYSMON|SYSMON_CONFIG|LOG_INFO|Sysmon poll interval changed to 20
```

# show debug destination

## Syntax

```
show debug destination
```

## Description

Displays the configured debug destination and severity.

## Command context

Manager (#)

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch# show debug destination
```

```

show debug destination

```

```
CONSOLE:info
FILE:warning
```

## aaa authentication port-access allow-cdp-bpdu

### Syntax

```
aaa authentication port-access allow-cdp-bpdu

no aaa authentication port-access allow-cdp-bpdu
```

### Description

Allows all packets related to the CDP (Cisco Discovery Protocol) BPDU (Bridge Protocol Data Unit) on a secure port.

The `no` form of this command blocks the CDP BPDU on a secure port. On a nonsecure port, the command has no effect.

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Allowing a CDP BPDU on secure port 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-cdp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version ArubaOS-CX 10.0X.0000
led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
 enable
aaa authentication port-access dot1x authenticator
 enable
interface 1/1/1
 no shutdown
 vlan access 1
 aaa authentication port-access allow-cdp-bpdu
 aaa authentication port-access mac-auth
 enable
 aaa authentication port-access dot1x authenticator
 enable

switch(config-if)# do show port-access device-profile interface all
Port 1/1/1, Neighbor-Mac 00:0c:29:9e:d1:20
Profile Name : access_switches
```

```
LLDP Group :
CDP Group : aruba-ap_cdp
Role : test_ap_role
Status : In Progress
Failure Reason :
```

Blocking LLDP packet on secure port 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no aaa authentication port-access allow-cdp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version ArubaOS-CX 10.0X.0000
led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
 enable
interface 1/1/1
 no shutdown
 vlan access 1
 aaa authentication port-access mac-auth
 enable
```

## aaa authentication port-access allow-lldp-bpdu

### Syntax

```
aaa authentication port-access allow-lldp-bpdu
```

```
no aaa authentication port-access allow-lldp-bpdu
```

### Description

Allows all packets related to the LLDP BPDU (Bridge Protocol Data Unit) on a secure port.

The **no** form of this command blocks the LLDP BPDU on a secure port. On a nonsecure port, the command has no effect.

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Allowing an LLDP BPDU on secure port 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# aaa authentication port-access allow-lldp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version ArubaOS-CX 10.0X.0000
led locator on
```

```

!
!
vlan 1
aaa authentication port-access mac-auth
 enable
interface 1/1/1
 no shutdown
 vlan access 1
 aaa authentication port-access allow-lldp-bpdu
 aaa authentication port-access mac-auth
 enable

switch(config-if)# do show port-access device-profile interface all
Port 1/1/1, Neighbor-Mac 00:0c:29:9e:d1:20
 Profile Name : access_switches
 LLDP Group : 2920-grp
 CDP Group :
 Role : local_2920_role
 Status : Profile Applied
 Failure Reason :

```

Blocking LLDP BPDU on secure port 1/1/1:

```

switch(config)# interface 1/1/1
switch(config-if)# no aaa authentication port-access allow-lldp-bpdu
switch(config-if)# do show running-config
Current configuration:
!
!Version ArubaOS-CX 10.0X.0000led locator on
!
!
vlan 1
aaa authentication port-access mac-auth
 enable
interface 1/1/1
 no shutdown
 vlan access 1
 aaa authentication port-access mac-auth
 enable

```

## associate cdp-group

### Syntax

```
associate cdp-group <GROUP-NAME>
```

```
no associate cdp-group <GROUP-NAME>
```

### Description

Associates a CDP (Cisco Discovery Protocol) group with a device profile. A maximum of two CDP groups can be associated with a device profile.

The **no** form of this command removes a CDP group from a device profile.

### Command context

```
config-device-profile
```

## Parameters

### <GROUP-NAME>

Specifies the name of the CDP group to associate with this device profile. Range: 1 to 32 alphanumeric characters.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Associating the CDP group **my-cdp-group** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate cdp-group my-cdp-group
```

Removing the CDP group **my-cdp-group** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate cdp-group my-cdp-group
```

# associate lldp-group

## Syntax

```
associate lldp-group <GROUP-NAME>
```

```
no associate lldp-group <GROUP-NAME>
```

## Description

Associates an LLDP group with a device profile. A maximum of two LLDP groups can be associated with a device profile

The **no** form of this command removes an LLDP group from a device profile.

## Command context

```
config-device-profile
```

## Parameters

### <GROUP-NAME>

Specifies the name of the LLDP group to associate with the device profile. Range: 1 to 32 alphanumeric characters.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Associating the LLDP group **my-lldp-group** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate lldp-group my-lldp-group
```

Removing the LLDP group **my-lldp-group** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate lldp-group my-lldp-group
```

## associate mac-group

### Syntax

```
associate mac-group <GROUP-NAME>
```

```
no associate mac-group <GROUP-NAME>
```

### Description

Associates a MAC group with a device profile. A maximum of two MAC groups can be associated with a device profile.

The `no` form of this command removes a MAC group from a device profile.

### Command context

```
config-device-profile
```

### Parameters

**<GROUP-NAME>**

Specifies the name of the MAC group to associate with this device profile. Range: 1 to 32 alphanumeric characters.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Associating the MAC group `mac01-group` with the device profile `profile01`:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate mac-group mac01-group
```

Removing the MAC group `mac01-group` from the device profile `profile01`:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate mac-group mac01-group
```

## associate role

### Syntax

```
associate role <ROLE-NAME>
```

```
no associate role <ROLE-NAME>
```

### Description

Associates a role with a device profile. Only one role can be associated with a device profile. For information on how to configure a role, see the port access role information in the *Security Guide*.

The `no` form of this command removes a role from a device profile.



## Command context

config-device-profile

## Parameters

**<ROLE-NAME>**

Specifies the name of the role to associate with the device profile. Range: 1 to 64 alphanumeric characters.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Associating the role **my-role** with the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# associate role my-role
```

Removing the role **my-role** from the device profile **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no associate role my-role
```

# disable

## Syntax

disable

no disable

## Description

Disables a device profile.

The **no** form of this command enables a device profile.

## Command context

config-device-profile

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Disabling a device profile:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# disable
```

Enabling a device profile named profile01:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no disable
```

# enable

## Syntax

enable

no enable

## Description

Enables a device profile.

The **no** form of this command disables a device profile.

## Command context

config-device-profile

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling a device profile:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# enable
```

Disabling a device profile named profile01:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)# no enable
```

# ignore (for CDP groups)

## Syntax

```
ignore [seq <SEQ-NUM>] {platform <PLATFORM> | sw-version <SWVERSION> |
 voice-vlan-query <VLAN-ID>}
```

```
no ignore [seq <SEQ-ID>] {platform <PLATFORM> | sw-version <SWVERSION> |
 voice-vlan-query <VLAN-ID>}
```

## Description

Defines a rule to ignore devices for a CDP (Cisco Discovery Protocol) group. Up to 64 match/ignore rules can be defined for a group.

The **no** form of this command removes a rule for ignoring devices from a CDP group.

## Command context

config-cdp-group

## Parameters

### **seq** <SEQ-ID>

Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.

### **platform** <PLATFORM>

Specifies the hardware or model details of the neighbor. Range: 1 to 128 alphanumeric characters.

### **sw-version** <SWVERSION>

Specifies the software version of the neighbor. Range: 1 to 128 alphanumeric characters.

### **voice-vlan-query** <VLAN-ID>

Specifies the VLAN query value of the neighbor. Range: 1 to 65535.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Adding a rule to the CDP group **grp01** that ignores a device that transmits **PLATFORM01** in the platform TLV:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# ignore platform PLATFORM01
```

Adding a rule to the CDP group **grp01** that ignores a device that transmits **SWVERSION** in software version TLV:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# ignore sw-version SWVERSION
```

Removing the rule that matches the sequence number 25 from the CDP group named **grp01**.

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# no ignore seq 25
```

## ignore (for LLDP groups)

### Syntax

```
ignore [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
 vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}

no ignore [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
 vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
```

### Description

Defines a rule to ignore devices for an LLDP group. Up to 64 match/ignore rules can be defined for a group. The **no** form of this command removes a rule for ignoring devices from an LLDP group.

### Command context

config-lldp-group

## Parameters

**seq** <SEQ-ID>

Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.

**sys-desc** <SYS-DESC>

Specifies the LLDP system description type-length-value (TLV). Range: 1 to 256 alphanumeric characters.

**sysname** <SYS-NAME>

Specifies the LLDP system name TLV. Range: 1 to 64 alphanumeric characters.

**vendor-oui** <VENDOR-OUI>

Specifies the LLDP system vendor OUI TLV. Range: 1 to 6 alphanumeric characters.

**type** <KEY>

Specifies the vendor OUI subtype key. Optional.

**value** <VALUE>

Specifies the vendor OUI subtype value. Range: 1 to 256 alphanumeric characters.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Adding a rule to the LLDP group **grp01** that ignores a device that transmits **PLATFORM01** in the system description TLV:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# ignore sys-desc PLATFORM01
```

Removing the rule that matches the sequence number **25** from the LLDP group named **grp01**.

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# no match seq 25
```

# ignore (for MAC groups)

## Syntax

```
[seq <SEQ-ID>] ignore {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
```

```
no [seq <SEQ-ID>] ignore {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
```

## Description

Defines a rule to ignore devices for a MAC group based on the criteria of MAC address, MAC address mask, or MAC Organizational Unique Identifier (OUI). Up to 64 ignore rules can be defined for a group.

The **no** form of this command removes a rule for ignoring devices from a MAC group.

## Command context

config-mac-group

## Parameters

### **seq** <SEQ-ID>

Specifies the entry sequence ID of the rule to create or modify a MAC group. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence. Range: 1 to 4294967295.

### **mac** <MAC-ADDR>

Specifies the MAC address of the device to ignore.

### **mac-mask** <MAC-MASK>

Specifies the MAC address mask to ignore devices in that range. Supported MAC address masks: /32 and /40.

### **mac-oui** <MAC-OUI>

Specifies the MAC OUI to ignore devices in that range. Supports MAC OUI address of maximum length of 24 bits.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

To achieve the required configuration of matches for devices, it is recommended to first ignore the devices that you do not want to add. Then match the criteria for the rest of the devices that you want to add to the MAC group.

For example, if you want to ignore a specific device but add all the other devices that belong to a MAC OUI, then you must first configure the ignore criteria with a lower sequence number. And then configure match criteria with a higher sequence number.

## Examples

Adding a rule to the MAC group **grp01** to ignore a device based on MAC address, but match all other devices belonging to a MAC OUI:

```
switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac 1a:2b:3c:4d:5e:6f
switch(config-mac-group)# match mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
!
!
ssh server vrf
!
!
!
!
vlan 1
interface vlan 1
 no shutdown
 ip dhcp
```

```
mac-group grp01
 seq 10 ignore mac 1a:2b:3c:4d:5e:6f
 seq 20 match mac-oui 1a:2b:3c
...
```

Adding a rule to the MAC group **grp01** to ignore devices based on MAC address mask, but match all other devices belonging to a MAC OUI:

```
switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac-mask 1a:2b:3c:4d/32
switch(config-mac-group)# match mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
!
!
ssh server vrf
!
!
!
!
vlan 1
interface vlan 1
 no shutdown
 ip dhcp
mac-group grp01
 seq 10 ignore mac-mask 1a:2b:3c:4d/32
 seq 20 match mac-oui 1a:2b:3c
...
```

Adding a rule to the MAC group **grp01** that ignores a device based on complete MAC address:

```
switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac 1a:2b:3c:4d:5e:6f
```

Adding a rule to the MAC group **grp02** that ignores devices based on MAC mask:

```
switch(config)# mac-group grp01
switch(config-mac-group)# ignore mac-mask 1a:2b:3c:4d:5e/40
switch(config-mac-group)# ignore mac-mask 18:e3:ab:73/32
```

Adding a rule to the MAC group **grp03** that ignores devices based on MAC OUI:

```
switch(config)# mac-group grp03
switch(config-mac-group)# ignore mac-oui 81:cd:93
```

Adding a rule to the MAC group **grp01** that ignores devices with a sequence number and based on MAC address:

```
switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 ignore mac b2:c3:44:12:78:11
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
```

```

!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
 no shutdown
 ip dhcp
mac-group grp01
 seq 10 ignore mac b2:c3:44:12:78:11
...

```

Removing the rule from the MAC group **grp01** based on sequence number:

```

switch(config)# mac-group grp01
switch(config-mac-group)# no ignore seq 10
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
 no shutdown
 ip dhcp
mac-group grp01
...

```

Adding a rule to the MAC group **grp01** that ignores devices with MAC entry sequence number and based on MAC OUI:

```

switch(config)# mac-group grp01
switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 ignore mac b2:c3:44:12:78:11
switch(config-mac-group)# seq 20 ignore mac-oui 1a:2b:3c
switch(config-mac-group)# seq 30 ignore mac-mask 71:14:89:42/32
switch(config-mac-group)# exit
switch(config)#
switch(config)# ^Z

switch#
switch#
switch# show running-config
Current configuration:
!
!Version ArubaOS-CX PL.10.06.0002
!export-password: default
!
!
!
ssh server vrf default
vlan 1
spanning-tree
mac-group grp01
 seq 10 ignore mac b2:c3:44:12:78:11

```

```

 seq 20 ignore mac-oui 1a:2b:3c
 seq 30 ignore mac-mask 71:14:89:42/32
interface 1/1/1
 no shutdown
 vlan access 1
interface 1/1/2
 no shutdown
 vlan access 1
interface 1/1/3
 no shutdown
 vlan access 1
interface 1/1/4
 no shutdown
 vlan access 1
interface 1/1/5
 no shutdown
 vlan access 1
interface 1/1/6
 no shutdown
 vlan access 1
interface 1/1/7
 no shutdown
 vlan access 1
interface 1/1/8
 no shutdown
 vlan access 1
interface 1/1/9
 no shutdown
 vlan access 1
interface 1/1/10
 no shutdown
 vlan access 1
interface 1/1/11
 no shutdown
 vlan access 1
interface 1/1/12
 no shutdown
 vlan access 1
interface 1/1/13
 no shutdown
 vlan access 1
interface 1/1/14
 no shutdown
 vlan access 1
interface 1/1/15
 no shutdown
 vlan access 1
interface 1/1/16
 no shutdown
 vlan access 1
interface vlan 1
 ip dhcp
!
!
!
!
!
https-server vrf default

```

Removing the rule from the MAC group **grp01** based on sequence number and MAC OUI:

```

switch(config)# mac-group grp01
switch(config-mac-group)# no seq 20 ignore mac-oui 1a:2b:3c
switch(config-mac-group)# exit

```



```

switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
 no shutdown
 ip dhcp
mac-group grp01
 seq 10 ignore mac b2:c3:44:12:78:11
 seq 30 ignore mac-mask 71:14:89:f3/32
...

```

Removing the rule that matches the sequence number 25 from the MAC group named **grp01**.

```

switch(config)# mac-group grp01
switch(config-mac-group)# no ignore seq 25

```

## mac-group

### Syntax

```
mac-group <MAC-GROUP-NAME>
```

```
no mac-group <MAC-GROUP-NAME>
```

### Description

Creates a MAC group or modifies an existing MAC group. A MAC group is used to classify connected devices based on the MAC address details, such as mask or OUI.

A maximum of 32 MAC groups can be configured on the switch. A maximum of 2 MAC groups can be associated with a device profile. Each group accepts 64 match or ignore commands.

The **no** form of this command removes a MAC group.

### Command context

```
config
```

### Parameters

**<MAC-GROUP-NAME>**

Specifies the name of the MAC group to create or modify. The maximum number of characters supported is 32.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating a MAC group named **grp01**:

```

switch(config)# mac-group grp01
switch(config-mac-group)# exit

```

Removing a MAC group named grp01:

```
switch(config)# no mac-group grp01
```

## match (for CDP groups)

### Syntax

```
match [seq <SEQ-ID>] {platform <PLATFORM> | sw-version <SWVERSION> |
 voice-vlan-query <VLAN-ID>}

no match [seq <SEQ-ID>] {platform <PLATFORM> | sw-version <SWVERSION> |
 voice-vlan-query <VLAN-ID>}
```

### Description

Defines a rule to match devices for a CDP group. A maximum of 32 CDP groups can be configured on the switch. Up to 64 match or ignore rules can be defined for each group.

The **no** form of this command removes a rule for adding devices to a CDP group.

### Command context

config-cdp-group

### Parameters

**seq <SEQ-ID>**

Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.

**platform <PLATFORM>**

Specifies the hardware or model details of the neighbor. Range: 1 to 128 alphanumeric characters.

**sw-version <SWVERSION>**

Specifies the software version of the neighbor. Range: 1 to 128 alphanumeric characters.

**voice-vlan-query <VLAN-ID>**

Specifies the VLAN query value of the neighbor. Range: 1 to 65535.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Adding rules to match a Cisco device with a specific software version on VLAN 512 to the CDP group **grp01**:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# match platform CISCO
switch(config-cdp-group)# match sw-version 11.2(12)P
switch(config-cdp-group)# match voice-vlan-query 512
switch(config-cdp-group)# match seq 50 platform cisco sw-version 11.2(12)P voice-vlan-query 512
switch(config-cdp-group)# exit
switch(config)# do show running-config
```

```
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.000
!export-password: default
led locator on
```

```
!
!
vlan 1
port-access cdp-group grp01
 seq 10 match platform CISCO
 seq 20 match sw-version 11.2(12)P
 seq 30 match voice-vlan-query 512
 seq 50 match platform cisco sw-version 11.2(12)P voice-vlan-query 512
```

Removing a rule that matches the sequence number 25 from the CDP group named **grp01**:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# no match seq 25
```

Adding a rule that matches the value of vendor-OUI 000b86 to the CDP group named **grp01**:

```
switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# match vendor-oui 000b86
```

## match (for LLDP groups)

### Syntax

```
match [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
 vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}

no match [seq <SEQ-ID>] {sys-desc <SYS-DESC> | sysname <SYS-NAME> |
 vendor-oui <VENDOR-OUI> [type <KEY> [value <VALUE>]]}
```

### Description

Defines a rule to match devices for an LLDP group. Up to 64 match/ignore rules can be defined for a group. The **no** form of this command removes a rule.

### Command context

```
config-lldp-group
```

### Parameters

**seq** <SEQ-ID>

Specifies the ID of the rule to create or modify. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence.

**sys-desc** <SYS-DESC>

Specifies the LLDP system description type-length-value (TLV). Range: 1 to 256 alphanumeric characters.

**sysname** <SYS-NAME>

Specifies the LLDP system name TLV. Range: 1 to 64 alphanumeric characters.

**vendor-oui** <VENDOR-OUI>

Specifies the LLDP system vendor OUI TLV. Range: 1 to 6 alphanumeric characters.

**type** <KEY>

Specifies the vendor OUI subtype key.

**value** <VALUE>

Specifies the vendor OUI subtype value. Range: 1 to 256 alphanumeric characters.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Adding rules that match the LLDP system description **ArubaSwitch** and system name **Aruba** to the LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# match sys-desc ArubaSwitch
switch(config-lldp-group)# match sysname Aruba
switch(config)# do show running-config
```

Current configuration:

```
!
!Version ArubaOS-CX Virtual.10.0X.000
!export-password: default
led locator on
!
!
vlan 1
port-access lldp-group grp01
 seq 10 match sys-desc ArubaSwitch
 seq 20 match sysname Aruba
```

Removing a rule that matches the sequence number 25 from an LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# no match seq 25
```

Adding a rule that matches the value of vendor-OUI **000b86** with type of **1** to the LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# match vendor-oui 000b86 type 1
```

Adding a rule that matches the value of vendor-OUI **000c34** to the LLDP group named **grp01**:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)# match vendor-oui 000c34
```

## match (for MAC groups)

### Syntax

```
[seq <SEQ-ID>] match {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
```

```
no [seq <SEQ-ID>] match {mac <MAC-ADDR> | mac-mask <MAC-MASK> | mac-oui <MAC-OUI>}
```

### Description

Defines a rule to match devices for a MAC group based on the criteria of MAC address, MAC address mask, or MAC Organizational Unique Identifier (OUI). Up to 64 match rules can be defined for a group.



---

**NOTE:** You must not configure the following special MAC addresses:

- Null MAC—For example, 00:00:00:00:00:00 or 00:00:00/32
- Multicast MAC
- Broadcast MAC—For example, ff:ff:ff:ff:ff:ff
- System MAC

Although the switch accepts these addresses, it will not process these addresses for the local MAC match feature.

---

The `no` form of this command removes a rule for adding devices to a MAC group.

The number of clients that can onboard based on the match criteria is configured in the `aaa authentication port-access client-limit` command. For information about this command, see the *Security Guide* for your switch.

## Command context

`config-mac-group`

## Parameters

**seq** *<SEQ-ID>*

Specifies the entry sequence ID of the rule to create or modify a MAC group. If no ID is specified when adding a rule, an ID is automatically assigned in increments of 10 in the order in which rules are added. When more than one rule matches the command entered, the rule with the lowest ID takes precedence. Range: 1 to 4294967295.

**mac** *<MAC-ADDR>*

Specifies the MAC address of the device.

**mac-mask** *<MAC-MASK>*

Specifies the MAC address mask to add devices in that range. Supported MAC address masks: /32 and /40.

**mac-oui** *<MAC-OUI>*

Specifies the MAC OUI to add devices in that range. Supports MAC OUI address of maximum length of 24 bits.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Adding a device to the MAC group **grp01** based on complete MAC address:

```
switch(config)# mac-group grp01
switch(config-mac-group)# match mac 1a:2b:3c:4d:5e:6f
switch(config-mac-group)# exit
```

Adding devices to the MAC group **grp02** based on MAC mask:

```
switch(config)# mac-group grp01
switch(config-mac-group)# match mac-mask 1a:2b:3c:4d:5e/40
switch(config-mac-group)# match mac-mask 18:e3:ab:73/32
switch(config-mac-group)# exit
```

Adding devices to the MAC group **grp03** based on MAC OUI:

```
switch(config)# mac-group grp03
switch(config-mac-group)# match mac-oui 81:cd:93
switch(config-mac-group)# exit
```

Adding devices to the MAC group **grp01** with MAC entry sequence number and based on MAC address:

```
switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 match mac b2:c3:44:12:78:11
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
 no shutdown
 ip dhcp
mac-group grp01
 seq 10 match mac b2:c3:44:12:78:11
...
```

Removing devices from the MAC group **grp01** based on sequence number:

```
switch(config)# mac-group grp01
switch(config-mac-group)# no match seq 10
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
 no shutdown
 ip dhcp
mac-group grp01
...
```

Adding devices to the MAC group **grp01** with MAC entry sequence number and based on MAC address, MAC address mask, and MAC OUI:

```
switch(config)# mac-group grp01
switch(config-mac-group)# seq 10 match mac b2:c3:44:12:78:11
switch(config-mac-group)# seq 20 match mac-oui 1a:2b:3c
switch(config-mac-group)# seq 30 match mac-mask 71:14:89:f3/32
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
```

```

!
!
vlan 1
interface vlan 1
 no shutdown
 ip dhcp
mac-group grp01
 seq 10 match mac b2:c3:44:12:78:11
 seq 20 match mac-oui 1a:2b:3c
 seq 30 match mac-mask 71:14:89:f3/32
...

```

Removing devices from the MAC group **grp01** based on MAC OUI:

```

switch(config)# mac-group grp01
switch(config-mac-group)# no seq 20 match mac-oui 1a:2b:3c
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
 no shutdown
 ip dhcp
mac-group grp01
 seq 10 match mac b2:c3:44:12:78:11
 seq 30 match mac-mask 71:14:89:f3/32
...

```

Adding devices to the MAC group **grp03** with MAC entry sequence number and based on MAC address mask:

```

switch(config)# mac-group grp03
switch(config-mac-group)# seq 10 match mac-mask 10:14:a3:b7:55/40
switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
 no shutdown
 ip dhcp
mac-group grp03
 seq 10 match mac-mask 10:14:a3:b7:55/40
...

```

Removing devices from the MAC group **grp03** based on MAC address mask:

```

switch(config)# mac-group grp03
switch(config-mac-group)# no seq 10 match mac-mask 10:14:a3:b7:55/40

```

```

switch(config-mac-group)# exit
switch(config)# do show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.0001
!export-password: default
led locator on
!
!
vlan 1
interface vlan 1
 no shutdown
 ip dhcp
mac-group grp03
...

```

## port-access cdp-group

### Syntax

```

port-access cdp-group <CDP-GROUP-NAME>

no port-access cdp-group <CDP-GROUP-NAME>

```

### Description

Creates a CDP (Cisco Discovery Protocol) group or modifies an existing CDP group. A CDP Group is used to classify connected devices based on the CDP packet details advertised by the device. A maximum of 32 CDP groups can be configured on the switch. Each group accepts 64 match/ignore commands.

The `no` form of this command removes a CDP group.

### Command context

```
config
```

### Parameters

**<CDP-GROUP-NAME>**

Specifies the name of the CDP group to create or modify. The maximum number of characters supported is 32. Required.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating a CDP group named grp01:

```

switch(config)# port-access cdp-group grp01
switch(config-cdp-group)# match platform CISCO
switch(config-cdp-group)# match sw-version 11.2(12)P
switch(config-cdp-group)# match voice-vlan-query 512
switch(config-cdp-group)# seq 50 match platform cisco sw-version 11.2(12)P voice-vlan-query 512
switch(config-cdp-group)# exit
switch(config)# do show running-config

Current configuration:
!
!Version ArubaOS-CX Virtual.10.0X.000
!export-password: default

```



```
led locator on
!
!
vlan 1
port-access cdp-group grp01
 seq 10 match platform CISCO
 seq 20 match sw-version 11.2(12)P
 seq 30 match voice-vlan-query 512
 seq 50 match platform cisco sw-version 11.2(12)P voice-vlan-query 512
```

Removing a CDP group named grp01:

```
switch(config)# no port-access cdp-group grp01
```

## port-access device-profile

### Syntax

```
port-access device-profile <DEVICE-PROFILE-NAME>
```

```
no port-access device-profile <DEVICE-PROFILE-NAME>
```

### Description

Creates a new device profile and switches to the `config-device-profile` context. A maximum of 32 device profiles can be created.

The `no` form of this command removes a device profile.

### Command context

```
config
```

### Parameters

**<DEVICE-PROFILE-NAME>**

Specifies the name of a device profile. Range: 1 to 32 alphanumeric characters.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating a device profile named **profile01**:

```
switch(config)# port-access device-profile profile01
switch(config-device-profile)#
```

Removing a device profile named **profile01**:

```
switch(config)# no port-access device-profile profile01
```

## port-access device-profile mode block-until-profile-applied



**NOTE:** You must configure this mode in device profile only on standalone ports where there is no security configured and when you not want the port to be offline until one client is onboarded.

## Syntax

```
port-access device-profile
 mode block-until-profile-applied

no port-access device-profile mode block-until-profile-applied
```

## Description

Configures the switch to block the port until a profile match occurs for a device. This configuration is required when no security feature is enabled on the port.

You must enable this mode or security on the port for local MAC match feature to operate. You must not enable both features on the same port at the same time.



**NOTE:** You must not combine any other AAA configurations with the block-until-profile-applied mode.

The `no` form of this command removes a rule for adding devices to a MAC group.

## Command context

```
config-if
config-if-deviceprofile
```

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Configuring block-until-profile applied mode on port 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# port-access device-profile
switch(config-if-deviceprofile)# mode block-until-profile-applied
switch(config-if-deviceprofile)# end
```

# port-access lldp-group

## Syntax

```
port-access lldp-group <LLDP-GROUP-NAME>

no port-access lldp-group <LLDP-GROUP-NAME>
```

## Description

Creates an LLDP group or modifies an existing LLDP group. An LLDP group is used to classify connected devices based on the LLDP type-length-values (TLVs) advertised by the device. A maximum of 32 LLDP groups can be configured on the switch. Each group accepts 64 match/ignore commands.

The `no` form of this command removes an LLDP group.

## Command context

```
config
```

## Parameters

### <LLDP-GROUP-NAME>

Specifies the name of the LLDP group to create or modify. The maximum number of characters supported is 32. Required.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Creating an LLDP group named grp01:

```
switch(config)# port-access lldp-group grp01
switch(config-lldp-group)#
```

Removing an LLDP group named grp01:

```
switch(config)# no port-access lldp-group grp01
```

# show port-access device-profile

## Syntax

```
show port-access device-profile [[interface {all | <INTERFACE-ID>}
 [client-status <MAC-ADDR>]] | name <DEVICE-PROFILE-NAME>]
```

## Description

Shows the client status for a specific MAC address or profile name.

## Command context

Manager (#)

## Parameters

**interface {all | <INTERFACE-ID>}**

Select **all** for all interfaces or specify the name of an interface in the format: member/slot/port.

**client-status <MAC-ADDR>**

Specifies a MAC address (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F.

**name <DEVICE-PROFILE-NAME>**

Specifies the name of the device profile.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing the applied state of the device profiles:

```
switch# show port-access device-profile

Profile Name : accesspoints
LLDP Groups : 2920-grp
CDP Groups :
MAC Groups : 2920-mac-grp1,2920-iot-grp2
```

```

Role : local_role_1
State : Enabled

Profile Name : access_switches
LLDP Groups : 2920-grp
CDP Groups :
MAC Groups :
Role : local_2920_role
State : Enabled

Profile Name : iot_devices
LLDP Groups :
CDP Groups :
MAC Groups : iot_camera-grp1,iot_sensors-grp1
Role : local_2920_role
State : Enabled

Profile Name : lobbyaps
LLDP Groups :
CDP Groups : lobby_ap_cdp_grp
MAC Groups :
Role : test_ap_role
State : Disabled

```

Showing the applied state of the device profile on interface 1/1/3:

```

switch# show port-access device-profile interface 1/1/3 client-status 00:0c:29:9e:d1:20

Port 1/1/3, Neighbor-Mac 00:0c:29:9e:d1:20
 Profile Name : lobbyaps
 LLDP Group :
 CDP Group : aruba-ap_cdp
 MAC Group :
 Role : test_ap_role
 Status : Failed
 Failure Reason : Failed to apply MAC based VLAN

```

Showing the applied state of a specific device profile:

```

switch# show port-access device-profile name lldp-group

Profile Name : lldp-group
LLDP Groups :
CDP Groups :
MAC Groups : pc-behind-phone, lldp
Role : auth_role
State : Enabled

```

## ip dhcp

### Syntax

```
ip dhcp
```

### Description

Enables the DHCP client on the management interface enabling the interface to automatically obtain an IP address from a DHCP server on the network. By default, the DHCP client is enabled.

### Command context

```
config-if-mgmt
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

This example enables the DHCP client on the management interface.

```
switch(config)# interface mgmt
switch(config-if-mgmt)# ip dhcp
switch(config-if-mgmt)# no shutdown
```

If the interface is not enabled, you can enable it by entering the `no shutdown` command.

## show dhcp client vendor-class-identifier

### Syntax

```
show dhcp client vendor-class-identifier
```

### Description

Displays the Vendor Class Identifier (VCI) of the switch. The VCI is the string the switch sends to identify itself in the request to DHCP server used for Zero Touch Provisioning (ZTP).

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Example for a member of the Aruba 6300 Switch Series:

```
switch# show dhcp client vendor-class-identifier
Vendor Class Identifier: Aruba JL663A 6300M
```

Example for a member of the Aruba 8320 Switch Series:

```
switch# show dhcp client vendor-class-identifier
Vendor Class Identifier : Aruba JL479A 8320
```

Example for a member of the Aruba 8325 Switch Series:

```
switch# show dhcp client vendor-class-identifier
Vendor Class Identifier : Aruba JL635A 8325
```

Example for a member of the Aruba 8400 Switch Series:

```
switch# show dhcp client vendor-class-identifier
Vendor Class Identifier : Aruba JL375A 8400
```

## clear dhcpv4-snooping binding

### Syntax

```
clear dhcpv4-snooping binding {all | ip <IPV4-ADDR> vlan <VLAN-ID> |
 port <PORT-NUM> | vlan <VLAN-ID>}
```

### Description

Clears DHCPv4 snooping binding entries.

### Command context

Operator (>) or Manager (#)

### Parameters

**all**

Specifies that all DHCPv4 binding information is to be cleared.

**ip <IPV4-ADDR> vlan <VLAN-ID>**

Specifies the IPv4 address and VLAN for which all DHCPv4 binding information is to be cleared.

**port <PORT-NUM>**

Specifies the port number for which all DHCPv4 binding information is to be cleared.

**vlan <VLAN-ID>**

Specifies the VLAN for which all DHCPv4 binding information is to be cleared.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Clearing all DHCPv4 binding information for IP address 192.168.2.4 and VLAN 5:

```
switch(config)# clear dhcpv4-snooping binding ip 192.168.2.4 vlan 5
```

Clearing all DHCPv4 binding information for port 1/1/1:

```
switch(config)# clear dhcpv4-snooping binding port 1/1/1
```

Clearing all DHCPv4 binding information for VLAN 10:

```
switch(config)# clear dhcpv4-snooping binding vlan 10
```

Clearing all DHCPv4 binding information:

```
switch(config)# clear dhcpv4-snooping binding all
```

## clear dhcpv4-snooping statistics

### Syntax

```
clear dhcpv4-snooping statistics
```

### Description

Clears all DHCPv4 snooping statistics.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Clear all DHCPv4 snooping statistics:

```
switch# clear dhcpv4-snooping statistics
```

## dhcpv4-snooping

### Syntax

```
dhcpv4-snooping
```

```
no dhcpv4-snooping
```

### Description

Enables DHCPv4 snooping. DHCPv4 snooping is disabled by default. DHCP snooping is not supported on the management interface.

The no form of the command disables DHCPv4 snooping, flushing all the IP bindings learned since DHCPv4 snooping was enabled.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling DHCPv4 snooping:

```
switch(config)# dhcpv4-snooping
```

Disabling DHCPv4 snooping:

```
switch(config)# no dhcpv4-snooping
```



## dhcpv4-snooping (in config-vlan context)

### Syntax

```
dhcpv4-snooping
```

```
no dhcpv4-snooping
```

### Description

Enables DHCPv4 snooping in the `config-vlan` context. DHCPv4 snooping is disabled by default for all VLANs.

The `no` form of the command disables DHCPv4 snooping on the specified VLAN, flushing all the IP bindings learned for this VLAN since DHCPv4 snooping was enabled for this VLAN.

### Command context

```
config-vlan
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling DHCPv4 snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# dhcpv4-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Disabling DHCPv4 snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no dhcpv4-snooping
switch(config-vlan-100)# exit
switch(config)#
```

## dhcpv4-snooping allow-overwrite-binding

### Syntax

```
dhcpv4-snooping allow-overwrite-binding
```

```
no dhcpv4-snooping allow-overwrite-binding
```

### Description

Allows binding to be overwritten for the same IP address. When enabled, and a DHCP server offers a host an IP address that is already bound to an existing host in the binding table, the existing binding is overwritten for the new host if the new host is successfully able to acquire the same IP address. This overwriting is disabled by default, causing the DHCP server offers to be dropped.

The `no` form of the command disables DHCPv4 snooping overwrite binding.

### Command context

```
config
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling DHCPv4 snooping overwrite binding:

```
switch(config)# dhcpv4-snooping allow-overwrite-binding
```

Disabling DHCPv4 snooping overwrite binding:

```
switch(config)# no dhcpv4-snooping allow-overwrite-binding
```

# dhcpv4-snooping authorized-server

## Syntax

```
dhcpv4-snooping authorized-server <IPV4-ADDR> [vrf <VRF-NAME>]
```

```
no dhcpv4-snooping authorized-server <IPV4-ADDR> [vrf <VRF-NAME>]
```

## Description

Adds an authorized (trusted) DHCP server to a list of authorized servers for use by DHCPv4 snooping. This command can be issued multiple times, adding a maximum of 20 authorized servers per VRF. By default, with an empty list of authorized servers, all DHCP servers are considered to be trusted for DHCPv4 snooping purposes.



**NOTE:** The `mgmt` VRF cannot be used with this command.

The `no` form of this command deletes the specified DHCP server from the authorized list.

## Command context

`config`

## Parameters

**<IPV4-ADDR>**

Specifies the IPv4 address of the trusted DHCPv4 server.

**vrf <VRF-NAME>**

Specifies the VRF name.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

For authorized server lookup, the VRF is derived from the Switch Virtual Interface (SVI) configured for the incoming VLAN. If the SVI is not configured, the `default` VRF is assumed.

## Examples

Adding DHCP servers 192.168.2.2, 192.168.2.3, and 192.168.2.10 to the authorized server list:

```
switch(config)# dhcpv4-snooping authorized-server 192.168.2.2
switch(config)# dhcpv4-snooping authorized-server 192.168.2.3 vrf default
switch(config)# dhcpv4-snooping authorized-server 192.168.2.10 vrf default
```

Removing DHCP server 192.168.2.3 from the authorized server list:

```
switch(config)# no dhcpv4-snooping authorized-server 192.168.2.3 vrf default
```

## dhcpv4-snooping external-storage

### Syntax

```
dhcpv4-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
```

```
no dhcpv4-snooping external-storage
```

### Description

Configures external storage to be used for backing up IP bindings (used by DHCPv4 snooping) to a file. When configured, the switch stores all the IP bindings in an external storage file so that they are retained after the switch restarts. When the switch restarts, it reads the IP bindings from the configured external storage file to populate its local cache.

The no form of this command disables the saving of IP bindings in an external storage file.

### Command context

```
config
```

### Parameters

**volume <VOL-NAME>**

Specifies the name of the existing external storage volume where the IP bindings file will be saved. Before running the `dhcpv4-snooping external-storage volume` command, first create the external storage volume using command `external-storage <VOLUME-NAME>`. See *External storage commands* in the *Command-Line Interface Guide*.

**file <FILE-NAME>**

Specifies the file name to use for storing IP bindings. Maximum 255 characters.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring IP bindings storage in file `dsnoop_ipbindings` on existing volume `dhcp_snoop`:

```
switch(config)# dhcpv4-snooping external-storage volume dhcp_snoop file dsnoop_ipbindings
```

Disabling external storage:

```
switch(config)# no dhcpv4-snooping external-storage
```

## dhcpv4-snooping max-bindings

### Syntax

```
dhcpv4-snooping max-bindings <MAX-BINDINGS>
```

```
no dhcpv4-snooping max-bindings
```

### Description

Sets the maximum number of DHCP bindings allowed on the selected interface. For all interfaces on which this command is not run, the default max bindings applies.

The no form of the command reverts max bindings for the selected interface to its default.

### Command context

```
config-if
```

### Parameters

**<MAX-BINDINGS>**

Specifies the maximum number of DHCP bindings. You can use the `show capacities` command to see the maximum available for your switch model.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Set the DHCP max bindings to 512 on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# dhcpv4-snooping max-bindings 512
switch(config-if)# exit
switch(config)#
```

Revert DHCP max bindings to its default on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no dhcpv4-snooping max-bindings
switch(config-if)# exit
switch(config)#
```

## dhcpv4-snooping option 82

### Syntax

```
dhcpv4-snooping option 82 [remote-id {mac | subnet-ip | mgmt-ip}]
 [untrusted-policy {drop | keep | replace}]
```

```
no dhcpv4-snooping option 82
```

### Description

Configures the addition of option 82 DHCP relay information to DHCP client packets that are being forwarded on trusted ports. DHCP relay is enabled by default.

In the switch default state and when this command is entered without parameters (`dhcpv4-snooping option 82`), this default configuration is used:

```
dhcpv4-snooping option 82 remote-id mac untrusted-policy drop
```

When `remote-id` is omitted, its default (`mac`) is used. When `untrusted-policy` is omitted, its default (`drop`) is used.

The `no` form of this command disables DHCPv4 snooping option 82.

## Command context

`config`

## Parameters

### `remote-id`

Specifies what address to use as the remote ID for the `replace` option of `untrusted-policy`. Specify one of these address types:

#### `mac`

The default. Uses the switch MAC address as the remote ID.

#### `subnet-ip`

Uses the IP address of the client VLAN as the remote ID.

#### `mgmt-ip`

Uses the management interface IP address as the remote ID.

### `untrusted-policy`

Specifies what action to take for DHCP packets (with option 82) that are received on untrusted ports. Specify one of these actions:

#### `drop`

The default. Drop DHCP packets (with option 82) without forwarding them.

#### `keep`

Forward DHCP packets (with option 82).

#### `replace`

Replace the option 82 information in the DHCP packets with whatever is set for `remote-id` (one of: `mac`, `subnet-ip`, or `mgmt-ip`) and forward the packets.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring DHCPv4 snooping option 82 with the `keep` action:

```
switch(config)# dhcpv4-snooping option 82 untrusted-policy keep
```

Configuring DHCPv4 snooping option 82 with `mgmt-ip` as the `remote-id` and the `replace` action:

```
switch(config)# dhcpv4-snooping option 82 remote-id mgmt-ip untrusted-policy replace
```

Disabling DHCPv4 snooping option 82:

```
switch(config)# no dhcpv4-snooping option 82
```

## dhcpv4-snooping trust

### Syntax

```
dhcpv4-snooping trust
```

```
no dhcpv4-snooping trust
```

### Description

Enables DHCPv4 snooping trust on the selected port. Only server packets received on trusted ports are forwarded. All the ports are untrusted by default.

The no form of the command disables DHCPv4 snooping trust on the selected port.

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling DHCPv4 snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# dhcpv4-snooping trust
switch(config-if)# exit
switch(config)#
```

Disabling DHCPv4 snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no dhcpv4-snooping trust
switch(config-if)# exit
switch(config)#
```

## dhcpv4-snooping verify mac

### Syntax

```
dhcpv4-snooping verify mac
```

```
no dhcpv4-snooping verify mac
```

### Description

This command enables verification of the hardware address field in DHCP client packets. When enabled, the DHCP client hardware address field and the source MAC address must be the same for packets received on untrusted ports or else the packet is dropped. This DHCP snooping MAC verification is enabled by default.

The no form of the command disables DHCPv4 snooping MAC verification.

### Command context

```
config
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling DHCPv4 snooping MAC verification:

```
switch(config)# dhcpv4-snooping verify mac
```

Disabling DHCPv4 snooping MAC verification:

```
switch(config)# no dhcpv4-snooping verify mac
```

# show dhcpv4-snooping

## Syntax

```
show dhcpv4-snooping [vsx-peer]
```

## Description

Shows the DHCPv4 snooping configuration.

## Command context

Operator (>) or Manager (#)

## Parameters

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing the DHCPv4 snooping configuration:

```
switch(config)# show dhcpv4-snooping
```

DHCPv4-Snooping Information

|                         |       |                    |             |
|-------------------------|-------|--------------------|-------------|
| DHCPv4-Snooping         | : Yes | Verify MAC Address | : Yes       |
| Allow Overwrite Binding | : No  | Enabled VLANs      | : 1751-2000 |

Option 82 Configurations

|                     |        |           |       |
|---------------------|--------|-----------|-------|
| Untrusted Policy    | : drop | Insertion | : Yes |
| Option 82 Remote-id | : mac  |           |       |

External Storage Information

|                |     |
|----------------|-----|
| Volume Name    | :   |
| File Name      | :   |
| Inactive Since | : - |

```
Error : -
Authorized Server Configurations

VRF Authorized Servers

Port Information

Port Trust Max Bindings Static Bindings Dynamic Bindings

1/1/24 No 512 1 100
lag13 Yes 0 0 0
```

## show dhcpv4-snooping binding

### Syntax

```
show dhcpv4-snooping binding [vsx-peer]
```

### Description

Shows the DHCPv4 snooping binding configuration.

### Command context

Operator (>) or Manager (#)

### Parameters

#### [vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing the DHCPv4 snooping binding configuration:

```
switch(config)# show dhcpv4-snooping binding
```

| MacAddress        | IP       | VLAN | Interface | Time-Left |
|-------------------|----------|------|-----------|-----------|
| aa:b1:c1:dd:ee:ff | 10.2.3.4 | 1    | 1/1/2     | 582       |
| aa:b2:c2:dd:ee:ff | 10.2.3.5 | 1    | 1/1/2     | 584       |

## show dhcpv4-snooping statistics

### Syntax

```
show dhcpv4-snooping statistics [vsx-peer]
```



## Description

Shows the DHCPv4 snooping statistics.

## Command context

Operator (>) or Manager (#)

## Parameters

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing the DHCPv4 snooping statistics:

```
switch(config)# show dhcpv4-snooping statistics
```

| Packet-Type | Action  | Reason                        | Count |
|-------------|---------|-------------------------------|-------|
| server      | forward | from trusted port             | 5425  |
| client      | forward | to trusted port               | 3895  |
| server      | drop    | received on untrusted port    | 117   |
| server      | drop    | unauthorized server           | 214   |
| client      | drop    | destination on untrusted port | 78    |
| client      | drop    | untrusted option 82 field     | 85    |
| client      | drop    | bad DHCP release request      | 0     |
| client      | drop    | failed verify MAC check       | 5     |
| client      | drop    | failed on max-binding limit   | 15    |

## clear dhcpv6-snooping binding

### Syntax

```
clear dhcpv6-snooping binding {all | ip <IPV6-ADDR> vlan <VLAN-ID> |
 interface <IFNAME> | vlan <VLAN-ID>}
```

### Description

Clears DHCPv6 snooping binding entries.

### Command context

Operator (>) or Manager (#)

### Parameters

**all**

Specifies that all DHCPv6 binding information is to be cleared.

**ip <IPV6-ADDR> vlan <VLAN-ID>**

Specifies the IPv6 address and VLAN for which all DHCPv6 binding information is to be cleared.

**interface <IFNAME>**

Specifies the interface for which all DHCPv6 binding information is to be cleared.

**vlan <VLAN-ID>**

Specifies the VLAN for which all DHCPv6 binding information is to be cleared. Range: 1 to 4094.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Clearing all DHCPv6 binding information for 5000::1 vlan 1:

```
switch(config)# clear dhcpv6-snooping binding ip 5000::1 vlan 1
```

Clearing all DHCPv6 binding information for interface 1/1/10:

```
switch(config)# clear dhcpv6-snooping binding interface 1/1/10
```

Clearing all DHCPv6 binding information for VLAN 10:

```
switch(config)# clear dhcpv6-snooping binding vlan 10
```

Clearing all DHCPv6 binding information:

```
switch(config)# clear dhcpv6-snooping binding all
```

## clear dhcpv6-snooping statistics

### Syntax

```
clear dhcpv6-snooping statistics
```

### Description

Clears all DHCPv6 snooping statistics.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Clear all DHCPv6 snooping statistics:

```
switch# clear dhcpv6-snooping statistics
```

## dhcpv6-snooping

### Syntax

```
dhcpv6-snooping
```

```
no dhcpv6-snooping
```

### Description

Enables DHCPv6 snooping. DHCPv6 snooping is disabled by default. DHCPv6 snooping is not supported on the management interface.

The no form of the command disables DHCPv6 snooping, flushing all the IP bindings learned since DHCPv6 snooping was enabled.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling DHCPv6 snooping:

```
switch(config)# dhcpv6-snooping
```

Disabling DHCPv6 snooping:

```
switch(config)# no dhcpv6-snooping
```

## dhcpv6-snooping (in config-vlan context)

### Syntax

```
dhcpv6-snooping
```

```
no dhcpv6-snooping
```

### Description

Enables DHCPv6 snooping in the `config-vlan` context. DHCPv6 snooping is disabled by default for all VLANs.

The `no` form of the command disables DHCPv6 snooping on the specified VLAN, flushing all the IPv6 bindings learned for this VLAN since DHCPv6 snooping was enabled for this VLAN.

### Command context

```
config-vlan
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling DHCPv6 snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# dhcpv6-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Disabling DHCPv6 snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no dhcpv6-snooping
switch(config-vlan-100)# exit
switch(config)#
```

## dhcpv6-snooping authorized-server

### Syntax

```
dhcpv6-snooping authorized-server <IPV6-ADDR> [vrf <VRF-NAME>]
```

```
no dhcpv6-snooping authorized-server <IPV6-ADDR> [vrf <VRF-NAME>]
```

### Description

Adds an authorized (trusted) DHCPv6 server to a list of authorized servers for use by DHCPv6 snooping. This command can be issued multiple times, adding a maximum of 20 authorized servers per VRF. By default, with an empty list of authorized servers, all DHCPv6 servers are considered to be trusted for DHCPv6 snooping purposes.



**NOTE:** The `mgmt` VRF cannot be used with this command.



**NOTE:** Configure the link local IPv6 address instead of global IPv6 address of the DHCPv6 server as the authorized-server. For example:

```
switch(config)# dhcpv6-snooping authorized-server
fe80::2ca4:fa40:d4cd:bc2f
```

The no form of this command deletes the specified DHCPv6 server from the authorized list.

### Command context

config

### Parameters

**<IPV6-ADDR>**

Specifies the IPv6 address of the trusted DHCPv6 server.

**vrf <VRF-NAME>**

Specifies the VRF name.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

For authorized server lookup, the VRF is derived from the Switch Virtual Interface (SVI) configured for the incoming VLAN. If the SVI is not configured, the `default` VRF is assumed.

### Examples

Adding DHCP servers ABCD:5ACD::2000, and ABCD:5ACD::2010 to the authorized server list:

```
switch(config)# dhcpv6-snooping authorized-server ABCD:5ACD::2000 vrf default
switch(config)# dhcpv6-snooping authorized-server ABCD:5ACD::2010 vrf default
```

Removing DHCP server ABCD:5ACD::2000 from the authorized server list:

```
switch(config)# no dhcpv6-snooping authorized-server ABCD:5ACD::2000 vrf default
```

## dhcpv6-snooping external-storage

### Syntax

```
dhcpv6-snooping external-storage volume <VOL-NAME> file <FILE-NAME>
```

```
no dhcpv6-snooping external-storage
```

### Description

Configures external storage to be used for backing up IPv6 bindings (used by DHCPv6 snooping) to a file. When configured, the switch stores all the IP bindings in an external storage file so that they are retained after the switch restarts. When the switch restarts, it reads the IPv6 bindings from the configured external storage file to populate its local cache.

The no form of this command disables the saving of IPv6 bindings in an external storage file.

### Command context

config

## Parameters

**volume** <VOL-NAME>

Specifies the name of the existing external storage volume where the IPv6 bindings file will be saved. Before running the `dhcpv6-snooping external-storage volume` command, first create the external storage volume using command `external-storage <VOLUME-NAME>`. See *External storage commands* in the *Command-Line Interface Guide*.

**file** <FILE-NAME>

Specifies the file name to use for storing IPv6 bindings. Maximum 255 characters.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring IPv6 bindings storage in file `ipv6Bindings` on existing volume `dhcp_snoop`:

```
switch(config)# dhcpv6-snooping external-storage volume dhcp_snoop file ipv6Bindings
```

Disabling external storage:

```
switch(config)# no dhcpv6-snooping external-storage
```

# dhcpv6-snooping max-bindings

## Syntax

```
dhcpv6-snooping max-bindings <MAX-BINDINGS>
```

```
no dhcpv6-snooping max-bindings
```

## Description

Sets the maximum number of DHCPv6 bindings allowed on the selected interface. For all interfaces on which this command is not run, the default max bindings applies.

The `no` form of the command reverts max bindings for the selected interface to its default.

## Command context

```
config-if
```

## Parameters

<MAX-BINDINGS>

Specifies the maximum number of DHCPv6 bindings. You can use the `show capacities` command to see the maximum available for your switch model.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Set the DHCPv6 max bindings to 512 on interface `2/2/1`:

```
switch(config)# interface 2/2/1
switch(config-if)# dhcpv6-snooping max-bindings 512
```

```
switch(config-if)# exit
switch(config)#
```

Revert DHCPv6 max bindings to its default on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no dhcpv6-snooping max-bindings
switch(config-if)# exit
switch(config)#
```

## dhcpv6-snooping trust

### Syntax

```
dhcpv6-snooping trust
```

```
no dhcpv6-snooping trust
```

### Description

Enables DHCPv6 snooping trust on the selected interface. Only server packets received on trusted interfaces are forwarded. All the interfaces are untrusted by default.

The no form of the command disables DHCPv6 snooping trust on the selected interface.

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling DHCPv6 snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# dhcpv6-snooping trust
switch(config-if)# exit
switch(config)#
```

Disabling DHCPv6 snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no dhcpv6-snooping trust
switch(config-if)# exit
switch(config)#
```

## show dhcpv6-snooping

### Syntax

```
show dhcpv6-snooping [vsx-peer]
```

### Description

Shows the DHCPv6 snooping configuration.

## Command context

Operator (>) or Manager (#)

## Parameters

### [vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing the DHCPv6 snooping configuration:

```
switch(config)# show dhcpv6-snooping
```

### DHCPv6-Snooping Information

```
DHCPv6-Snooping : Yes Enabled VLANs : 1,5,7,100-110
```

### External Storage Information

```
Volume Name : dhcp_snoop
File Name : ip_binding
Inactive Since : 01:23:20 01/01/2019
Error : Failed to write external storage
```

### Authorized Server Configurations

| VRF     | Authorized Servers                      |
|---------|-----------------------------------------|
| default | 2001:0db8:85a3:0000:0000:8a2e:0370:7334 |
| default | 2002::2                                 |
| default | 2004::1                                 |
| red     | 2002::1                                 |
| red     | 2002::2                                 |
| red     | 2002::9                                 |
| green   | 5000::1                                 |
| green   | 5000::2                                 |
| green   | 5000::3                                 |
| green   | 5000::7                                 |
| green   | 5000::8                                 |

### Port Information

| Port     | Trust | Max Bindings | Static Bindings | Dynamic Bindings |
|----------|-------|--------------|-----------------|------------------|
| 1/1/2    | Yes   | 0            | 0               | 0                |
| 1/1/3    | Yes   | 0            | 3               | 0                |
| 1/1/5    | Yes   | 0            | 22              | 0                |
| 1/1/16   | No    | 100          | 0               | 20               |
| 10/10/10 | No    | 512          | 12              | 7                |
| lag120   | No    | 256          | 3               | 0                |



## show dhcpv6-snooping binding

### Syntax

```
show dhcpv6-snooping binding [vsx-peer]
```

### Description

Shows the DHCPv6 snooping binding configuration.

### Command context

Operator (>) or Manager (#)

### Parameters

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing the DHCPv6 snooping binding configuration:

```
switch# show dhcpv6-snooping binding
```

```
IP Binding Information
```

```
=====
```

| MAC-ADDRESS       | IPV6-ADDRESS                            | VLAN | INTERFACE | TIME-LEFT |
|-------------------|-----------------------------------------|------|-----------|-----------|
| -----             | -----                                   | ---- | -----     | -----     |
| 00:50:56:96:e4:cf | aaaa:bbbb:cccc:dddd:eeee:1234:5678:abcd | 1    | 1/1/1     | 584       |
| 00:50:56:96:04:4d | 1000::3                                 | 134  | 1/1/2     | 435       |
| 00:50:56:96:d8:3d | 2000:1000::4                            | 2002 | lag123    | 21234     |

## show dhcpv6-snooping statistics

### Syntax

```
show dhcpv6-snooping statistics [vsx-peer]
```

### Description

Shows the DHCPv6 snooping statistics.

### Command context

Operator (>) or Manager (#)

## Parameters

### [vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing the DHCPv6 snooping statistics:

```
switch(config)# show dhcpv6-snooping statistics
```

| Packet-Type | Action  | Reason                        | Count |
|-------------|---------|-------------------------------|-------|
| server      | forward | from trusted port             | 12    |
| client      | forward | to trusted port               | 20    |
| server      | drop    | received on untrusted port    | 5     |
| server      | drop    | unauthorized server           | 4     |
| client      | drop    | destination on untrusted port | 2     |
| client      | drop    | bad DHCP release request      | 5     |
| server      | drop    | relay reply on untrusted port | 2     |
| client      | drop    | failed on max-binding limit   | 5     |

## ip dns domain-list

### Syntax

```
ip dns domain-list <DOMAIN-NAME> [vrf default]
```

```
no ip dns domain-list <DOMAIN-NAME> [vrf default]
```

### Description

Configures one or more domain names that are appended to the DNS request. The DNS client appends each name in succession until the DNS server replies. Domains can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF.

The `no` form of this command removes a domain from the list.

### Command context

config

### Parameters

**list** <DOMAIN-NAME>

Specifies a domain name. Up to six domains can be added to the list. Length: 1 to 256 characters.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

This example defines a list with two entries: `domain1.com` and `domain2.com`.

```
switch(config)# ip dns domain-list domain1.com
switch(config)# ip dns domain-list domain2.com
```

This example removes the entry `domain1.com`.

```
switch(config)# no ip dns domain-list domain1.com
```

## ip dns domain-name

### Syntax

```
ip dns domain-name <DOMAIN-NAME> [vrf <VRF-NAME>]
```

```
no ip dns domain-name <DOMAIN-NAME> [vrf <VRF-NAME>]
```

## Description

Configures a domain name that is appended to the DNS request. The domain can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF. If a domain list is defined with the command **ip dns domain-list**, the domain name defined with this command is ignored.

The **no** form of this command removes the domain name.

## Command context

config

## Parameters

**<DOMAIN-NAME>**

Specifies the domain name to append to DNS requests. Length: 1 to 256 characters.

**vrf <VRF-NAME>**

Specifies a VRF name. Default: default.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the default domain name to domain.com:

```
switch(config)# ip dns domain-name domain.com
```

Removing the default domain name domain.com:

```
switch(config)# no ip dns domain-name domain.com
```

# ip dns host

## Syntax

```
ip dns host <HOST-NAME> <IP-ADDR> [vrf <VRF-NAME>]
```

```
no ip dns host <HOST-NAME> <IP-ADDR> [vrf <VRF-NAME>]
```

## Description

Associates a static IP address with a hostname. The DNS client returns this IP address instead of querying a DNS server for an IP address for the hostname. Up to six hosts can be defined. If no VRF is defined, the default VRF is used.

The **no** form of this command removes a static IP address associated with a hostname.

## Command context

config

## Parameters

**host <HOST-NAME>**

Specifies the name of a host. Length: 1 to 256 characters.

### **<IP-ADDR>**

Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

### **vrf <VRF-NAME>**

Specifies a VRF name. Default: default.

### **Authority**

Administrators or local user group members with execution rights for this command.

### **Examples**

This example defines an IPv4 address of 3.3.3.3 for host1.

```
switch(config)# ip dns host host1 3.3.3.3
```

This example defines an IPv6 address of b::5 for host 1.

```
switch(config)# ip dns host host1 b::5
```

This example defines removes the entry for host 1 with address b::5.

```
switch(config)# no ip dns host host1 b::5
```

## **ip dns server address**

### **Syntax**

```
ip dns server-address <IP-ADDR> [vrf <VRF-NAME>]
```

```
no ip dns server-address <IP-ADDR> [vrf <VRF-NAME>]
```

### **Description**

Configures the DNS name servers that the DNS client queries to resolve DNS queries. Up to six name servers can be defined. The DNS client queries the servers in the order that they are defined. If no VRF is defined, the default VRF is used.

The no form of this command removes a name server from the list.

### **Command context**

config

### **Parameters**

#### **<IP-ADDR>**

Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

#### **vrf <VRF-NAME>**

Specifies a VRF name. Default: default.

### **Authority**

Administrators or local user group members with execution rights for this command.

## Examples

This example defines a name server at 1.1.1.1.

```
switch(config)# ip dns server-address 1.1.1.1
```

This example defines a name server at a::1.

```
switch(config)# ip dns server-address a::1
```

This example removes a name server at a::1.

```
switch(config)# no ip dns server-address a::1
```

## show ip dns

### Syntax

```
show ip dns [vrf <VRF-NAME>]
```

### Description

Shows all DNS client configuration settings or the settings for a specific VRF.

### Command context

Manager (#)

### Parameters

**vrf** <VRF-NAME>

Specifies the VRF for which to show information. If no VRF is defined, the default VRF is used.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

## clear erps ring <ringid> instance <id>

### Syntax

```
clear erps ring <ringid> instance <id>
```

### Description

Removes the protection switching and triggers reversion both in revertive and non-revertive operation. This command will not change the configured revertive operation mode.

### Command context

Operator (>) or Manager (#)

### Parameters

#### <ringid>

Required, specifies the ID of the ring. Range: 1-239.

#### <id>

Required, specifies the ID of the ring instance. Range: 1-2.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Removes the protection switching and triggers reversion for ring 3, instance 2:

```
switch# clear erps ring 3 instance 2
```

## clear erps statistics

### Syntax

```
clear erps statistics [ring <id>] [instance <id>]
```

### Description

This command clears the ERPS statistics for a ring or a ring instance.

### Command context

Operator (>) or Manager (#)

## Parameters

**<ringid>**

Optional, specifies the ID of the ring. Range: 1-239.

**<id>**

Optional, specifies the ID of the ring instance. Range: 1-64.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Clear ERPS statistics for ring 1:

```
switch# clear erps statistics ring 1
```

Clear ERPS statistics for instance 1 of ring 1:

```
switch# clear erps statistics ring 1 instance 1
```

# erps ring

## Syntax

```
erps ring <ringid>
```

```
no erps ring <ringid>
```

## Description

This command creates an ERPS ring with a given ID.

The `no` form of this command removes all the configurations of the ring, including instances.

## Command context

```
config
```

## Parameters

**<ringid>**

Required, specifies the ID of the ring. Range: 1-239

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Create an ERPS ring:

```
switch(config)# erps ring 2
switch(config-ring-2)#
```

Remove an ERPS ring:



```
switch(config)# no erps ring 2
switch(config-ring-2)#
```

## erps ring <ringid> <port0|port1> interface

### Syntax

```
erps ring <ringid>
 <port0|port1> interface <ifname>
```

### Description

This command configures the ERPS ring member port. An L2 interface in the switch is associated to one of the two member ports of an ERPS ring. In case of an interconnection node, only port0 is applicable for the sub-ring.

The `no` form of this command removes the association of the ring port to the L2 interface on the switch.

### Command context

```
config-erps-ring-<ringid>
```

### Parameters

#### <ringid>

Required, specifies the ID of the ring. Range: 1-239

#### <port0>

Required, set port0 of the ring.

#### <port1>

Required, set port1 of the ring.

#### <ifname>

Required, interface name (string).

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configure the ERPS ring member port:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# port0 interface 1/1/1
```

Remove the association of the ring port to the L2 interface on the switch:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# no port0
```

## erps ring <ringid> description

### Syntax

```
erps ring <ringid>
 description <line>
```

## Description

This command adds descriptive information to help administrators and operators understand the purpose of a ring. 1-64 printable ASCII characters are allowed.

The `no` form of this command removes the ring instance description.

## Command context

`config-erps-ring-<ringid>`

## Parameters

**<ringid>**

Required, specifies the ID of the ring. Range: 1-239

**<line>**

Required, specifies the description text. Maximum length is 64 characters.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Add descriptive information to a ring:

```
switch(config)# erps ring 3
switch(config-erps-ring-3) description HPE RnD ring
```

Remove descriptive information from a ring:

```
switch(config)# erps ring 3
switch(config-erps-ring-3) no description
```

# erps ring <ringid> guard-interval

## Syntax

```
erps ring <ringid>
 guard-interval <10 milliseconds>
```

## Description

Guard timer is used in nodes recovering from a local failure to avoid loops due to earlier Signal Fail (SF) messages that may be in the ring.

The configuration specifies the guard timer duration in units of 10 ms. The timer period must be greater than the maximum expected forwarding delay in which an R-APS message traverses the entire ring. The default value is 50.

The `no` form of this command removes the configured value of the guard interval and sets it to the default value of 50.

## Command context

`config-erps-ring-<ringid>`

## Parameters

**<ringid>**

Required, specifies the ID of the ring. Range: 1-239

**<10 milliseconds>**

Required, specifies the guard timer duration in units of 10 ms. Default: 50.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Specify the guard timer duration:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# guard-interval 100
```

Remove the configured value of the guard interval and set it to the default value of 50:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# no guard-interval
```

# erps ring <ringid> hold-off-interval

## Syntax

```
erps ring <ringid>
 hold-off-interval <100 milliseconds>
```

## Description

Specifies hold-off interval in units of 100 ms. If specified, a defect is not reported immediately. Instead, the hold-off timer is started. On expiration of the timer, if the defect still exists, it is reported to protection switching. The default value for hold-off timer is 0.

The `no` form of this command removes the configured value of the hold-off interval and sets it to the default value of 0.

## Command context

config-erps-ring-<ringid>

## Parameters

**<ringid>**

Required, specifies the ID of the ring. Range: 1-239

**<100 milliseconds>**

Required, specifies the hold-off interval in units of 100 ms. Default: 0.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Specify the hold-off interval:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# hold-off-interval 100
```

Remove the configured value of the hold-off interval and set it to the default value of 0:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# no hold-off-interval
```

## erps ring <ringid> instance

### Syntax

```
erps ring <ringid>
 instance <id>
```

### Description

On a common ERPS network, a physical ring can be configured with a single ERPS ring, and only one blocked port can be specified in the ring. When the ERPS ring is in normal state, the blocked port prohibits all service packets from passing through. As a result, all service data is transmitted through one path over the ERPS ring, and the other link on the blocked port becomes idle, leading to ineffective use of bandwidth.

To improve link use efficiency, logical rings can be configured in the same physical ring in the ERPS multi-instance. A port may have different roles in different ERPS rings and different ERPS rings use different control VLANs.

An ERPS ring must be configured with an ERP instance, and each ERP instance specifies a range of VLANs. The topology calculated for a specific ERPS ring only takes effect in the ERPS ring. Different VLANs can use separate paths, implementing traffic load balancing and link backup.

The `no` form of this command removes the instance of the ring.

### Command context

```
config-erps-ring-<ringid>
```

### Parameters

**<ringid>**

Required, specifies the ID of the ring. Range: 1-239

**<id>**

Required, specifies the ERPS ring instance identifier. Range: 1-2.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Create a ring instance:

```
switch(config)# erps ring 3
switch(config-ring-3)# instance 2
```

Remove a ring instance:

```
switch(config)# erps ring 3
switch(config-ring-3)# no instance 2
```

## erps ring <ringid> instance <id> control-vlan

### Syntax

```
erps ring <ringid>
 instance <id> control-vlan <vid>
```

### Description

This command adds a control-channel VLAN to a ring instance. In an ERPS ring, the control VLAN should be used only to forward RAPS PDUs and not service packets. All the devices in an ERPS ring instance must be configured with the same control VLAN, and different ERPS ring instances must use different control VLANs.

The `no` form of this command removes the control-channel VLAN of the ring instance.

### Command context

```
config-erps-ring-<ringid>
```

### Parameters

**<ringid>**

Required, specifies the ID of the ring. Range: 1-239

**<id>**

Required, specifies the ERPS ring instance identifier. Range: 1-2.

**<vid>**

Required, VLAN ID. Range: 1-4094.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Add a control-channel VLAN to a ring instance:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 2
switch(config-erps-ring-3-inst-2) control-vlan 10
```

Remove the control-channel VLAN of the ring instance:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 2
switch(config-erps-ring-3-inst-2) no control-vlan
```

## erps ring <ringid> instance <id> description

### Syntax

```
erps ring <ringid>
 instance <id> description <line>
```

### Description

This command adds descriptive information to help administrators and operators understand the purpose of a ring instance. 1-64 printable ASCII characters are allowed.

The `no` form of this command removes the ring instance description.

### Command context

`config-erps-ring-<ringid>`

### Parameters

**<ringid>**

Required, specifies the ID of the ring. Range: 1-239

**<id>**

Required, specifies the ERPS ring instance identifier. Range: 1-2.

**<line>**

Required, descriptive information about the ring instance. 1-64 printable ASCII characters allowed.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Add ring instance description:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 2
switch(config-erps-ring-3-inst-2) description HPE RnD DataVlan
```

Remove ring instance description:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 2
switch(config-erps-ring-3-inst-2) no description
```

## erps ring <ringid> instance <id> enable

### Syntax

```
erps ring <ringid>
 instance <id> enable
```

### Description

This configuration enables protection switching on the given instance of the ring. It is disabled by default. The `no` form of this command disables protection switching on the given instance of the ring.

### Command context

`config-erps-ring-<ringid>`

### Parameters

**<ringid>**

Required, specifies the ID of the ring. Range: 1-239

**<id>**

Required, specifies the ERPS ring instance identifier. Range: 1-2.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enable protection switching on the given instance of the ring:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 2
switch(config-erps-ring-3-inst-2) enable
```

Disable protection switching on the given instance of the ring:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 2
switch(config-erps-ring-3-inst-2) no enable
```

# erps ring <ringid> instance <id> protected-vlans

## Syntax

```
erps ring <ringid>
 instance <id> protected-vlans <vid-list>
```

## Description

This command specifies the set of VLANs that are protected by this ring instance.

The `no` form of this command removes a set of VLANs that are protected by this ring instance.

## Command context

```
config-erps-ring-<ringid>
```

## Parameters

### <ringid>

Required, specifies the ID of the ring. Range: 1-239

### <id>

Required, specifies the ERPS ring instance identifier. Range: 1-2.

### <vd-list>

Required, range of VLANs to be protected by this ring instance. Range: 1-4094.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Specify a set of VLANs that are protected by this ring instance:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 2
switch(config-erps-ring-3-inst-2) protected-vlans 1,10-50
```

Remove a set of VLANs that are protected by this ring instance:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 2
switch(config-erps-ring-3-inst-2) no protected-vlans 11,13
```

## erps ring <ringid> instance <id> protection-switch {{manual|force} <port0>|<port1>}

### Syntax

```
erps ring <ringid> instance <id> protection-switch {{manual|force} <port0>|<port1>}
```

### Description

Blocks a specific ring interface in one of the two following ways:

- Force: The switch blocks a specific ring interface regardless of the protection switching state of the ring instance.
- Manual: The switch blocks a specific ring interface if no other protection switch event is active on the ring instance.



**NOTE:** The user can verify whether the protection-switch is successful by verifying the status of instance and port state over which this command is executed.

```
switch# erps ring 1 instance 1 protection-switch force port0
switch# show erps status
Status for ERPS Ring 1 Instance 1:
=====
Ring ID : 1
Instance ID : 1
Port0 : 1/1/5 (Block)
Port1 : 1/1/6 (Up)
Node Role (RPL) : Owner (port0)
Control VLAN : 50
Protected VLAN : 1-49
Subring (TCN) : No (No)
Revertive Operation : Revertive
MEG Level : 7
Transmission Interval : 5 sec
Guard Interval : 0 sec 500 ms
Hold-Off Interval : 0 sec 0 ms
WTR Interval : 1 min
Status : Forced-switch
Oper Down Reason : None
```

### Command context

Operator (>) or Manager (#)

### Parameters

**<ringid>**

Required, specifies the ID of the ring. Range: 1-239.

**<id>**

Required, specifies the ID of the ring instance. Range: 1-2.



### **manual**

A type of protection switch event in which the switch blocks a specific ring interface if no other protection switch event is active on the ring instance.

### **force**

A type of protection switch even in which the switch blocks a specific ring interface regardless of the protection switching state of the ring instance.

### **Authority**

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### **Examples**

Block ring 3, interface 2, port 0 if no other protection switch event is active on the ring instance:

```
switch# erps ring 3 instance 2 protection-switch manual port0
```

Block ring 3, instance 2, regardless of the protection switching state of the ring instance:

```
switch# erps ring 3 instance 2 protection-switch force port1
```

## **erps ring <ringid> instance <id> revertive**

### **Syntax**

```
erps ring <ringid> instance <id> revertive
```

### **Description**

Configures the default revertive mode of operation for an ERPS ring. In revertive operation, after the conditions causing protection switching are cleared, traffic channels are restored to the recovered link blocking the RPL. This configuration is meaningful only on the RPL node.

The `no` form of this command configures non-revertive mode of operation for an ERPS ring. In non-revertive operation, the traffic channels continue to use the RPL, if it has not failed, after conditions causing protection switching are cleared. This configuration is meaningful only on the RPL node.

### **Command context**

```
config-erps-ring-<ringid>
```

### **Parameters**

#### **<ringid>**

Required, specifies the ID of the ring. Range: 1-239

#### **<id>**

Required, specifies the ERPS ring instance identifier. Range: 1-2.

### **Authority**

Administrators or local user group members with execution rights for this command.

### **Examples**

Configuring the default revertive mode of operation for ERPS ring 3, instance 2:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 2
switch(config-erps-ring-3-inst-2)# revertive
```

Configuring non-revertive mode of operation for ERPS ring 3, instance 2:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 2
switch(config-erps-ring-3-inst-2)# no revertive
```

## erps ring <ringid> instance <id> role

### Syntax

```
erps ring <ringid>
 instance <id> role <rpl-owner|rpl-neighbor>
```

### Description

In ERPS, there is a central node called RPL Owner Node which blocks one of the ports to ensure that there is no loop formed for the Ethernet traffic. The link blocked by the RPL owner node is called the Ring Protection Link or RPL. The node at the other end of the RPL is known as RPL Neighbor Node. It uses R-APS control messages to coordinate the activities of switching on/off the RPL link.

This command specifies the role of the node as owner or neighbor.

The `no` form of this command removes the configuration of the node role from the instance.

### Command context

```
config-erps-ring-<ringid>
```

### Parameters

#### <ringid>

Required, specifies the ID of the ring. Range: 1-239

#### <id>

Required, specifies the ERPS ring instance identifier. Range: 1-2.

#### <rpl-owner>

Blocks traffic at one end of the RPL. The blocked end sends out periodic R-APS.

#### <rpl-neighbor>

Blocks traffic at one end of the RPL. The blocked end does not generate periodic R-APS.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Specify the role of the node as owner:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 2
switch(config-erps-ring-3-inst-2) role rpl-owner
```

Specify the role of the node as neighbor:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 3
switch(config-erps-ring-3-inst-2) role rpl-neighbour
```

Remove the configuration of the node role from the instance:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 2
switch(config-erps-ring-3-inst-2) no role
```

## erps ring <ringid> instance <id> rpl

### Syntax

```
erps ring <ringid>
 instance <id> rpl <port0|port1>
```

### Description

In ERPS, there is a central node called RPL Owner Node which blocks one of the ports to ensure that there is no loop formed for the Ethernet traffic. The link blocked by the RPL owner node is called the Ring Protection Link or RPL. The node at the other end of the RPL is known as RPL Neighbor Node. It uses R-APS control messages to coordinate the activities of switching the RPL link on and off.

This command specifies which of the ERPS ring ports is the RPL.

The `no` form of this command removes the RPL port configuration from the ERPS ring instance.

### Command context

config-erps-ring-<ringid>

### Parameters

#### <ringid>

Required, specifies the ID of the ring. Range: 1-239

#### <id>

Required, specifies the ERPS ring instance identifier. Range: 1-2.

#### <port0>

Required, configure port0 to be RPL port in this ERPS ring instance.

#### <port1>

Required, configure port1 to be RPL port in this ERPS ring instance.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configure port0 to be RPL port in this ERPS ring instance:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 2
switch(config-erps-ring-3-inst-2) role rpl-owner
switch(config-erps-ring-3-inst-2) rpl port0
```

Configure port1 to be RPL port in this ERPS ring instance:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 3
switch(config-erps-ring-3-inst-2) role rpl-neighbour
switch(config-erps-ring-3-inst-2) rpl port1
```

Remove the RPL port configuration from the ERPS ring Instance:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# instance 2
switch(config-erps-ring-3-inst-2) no rpl port0
```

## erps ring <ringid> meg-level

### Syntax

```
erps ring <ringid>
 meg-level <-0-7>
```

### Description

The R-APS messages transmitted by ERPS take the form of OAM PDUs as defined in G.8013. Each OAM PDU is transmitted at a specified level known as the Maintenance Entity Group (MEG) level. This command configures the level with which the ERPS packets must be transmitted.

The `no` form of this command removes the configured MEG level and sets it to the default value of 7.

### Command context

```
config-erps-ring-<ringid>
```

### Parameters

**<ringid>**

Required, specifies the ID of the ring. Range: 1-239

**<0-7>**

Required, specifies the meg-level. Range: 0-7. Default: 7.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Specify the meg-level:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# meg-level 4
```

Remove the configured meg-level and set it to the default value of 7:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# no meg-level
```

## erps ring <ringid> parent-ring

### Syntax

```
erps ring <ringid>
 parent-ring <ringid>
```

### Description

This command associates a sub-ring to a parent-ring and is required for the sub-ring to notify the parent-ring on change in topology.

The `no` form of this command removes the parent ring identifier.

### Command context

```
config-erps-ring-<ringid>
```

### Parameters

**<ringid>**

Required, specifies the ID of the ring. Range: 1-239

**<ringid>**

Required, specifies the ID of the parent-ring. Range: 1-239

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Associate a sub-ring to a parent-ring:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# parent-ring 2
```

Remove a parent-ring identifier:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# no parent-ring 2
```

## erps ring <ringid> sub-ring

### Syntax

```
erps ring <ringid>
 sub-ring
```

### Description

This command is to configure a sub-ring. If not specified, the ring is a major-ring.

The `no` form of this command removes the sub-ring configuration of the ring and configures it to be a major-ring.

### Command context

```
config-erps-ring-<ringid>
```

## Parameters

**<ringid>**

Required, specifies the ID of the ring. Range: 1-239

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configure a sub-ring:

```
switch(config)# erps ring 2
switch(config-erps-ring-2)# sub-ring
```

Remove the sub-ring configuration from ring 2 and configure it to be a major-ring:

```
switch(config)# erps ring 2
switch(config-erps-ring-2)# no sub-ring
```

# erps ring <ringid> tcn-propagation

## Syntax

```
erps ring <ringid>
 tcn-propagation
```

## Description

This command is to configure a sub-ring interconnection node to pass a topology change notification to the ring instance for the parent ring whenever the topology of the sub-ring changes. The parent ring instance performs a Forwarding Database (FDB) flush and sends a protocol message to ensure that other nodes on the parent ring also perform an FDB flush.

The **no** form of this command disables topology change notifications.

## Command context

config-erps-ring-<ringid>

## Parameters

**<ringid>**

Required, specifies the ID of the ring. Range: 1-239

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configure topology change notifications:

```
switch(config)# erps ring 2
switch(config-erps-ring-2)# tcn-propagation
```

Disable topology change notifications:

```
switch(config)# erps ring 2
switch(config-erps-ring-2)# no tcn-propagation
```

## erps ring <ringid> transmission-interval

### Syntax

```
erps ring <ringid>
 transmission-interval <seconds>
```

### Description

Specifies the R-APS periodic transmission interval in units of seconds. Default is 5 seconds.

The `no` form of this command removes the configured value of the transmission interval and sets it to the default value of 5 seconds.

### Command context

```
config-erps-ring-<ringid>
```

### Parameters

#### <ringid>

Required, specifies the ID of the ring. Range: 1-239

#### <seconds>

Required, specifies the R-APS periodic transmission interval in units of seconds. Range: 5 seconds.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Specify the R-APS periodic transmission interval as 10 seconds:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# transmission-interval 10
```

Remove the configured value of the transmission interval and set it to the default value of 5 seconds:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# no transmission-interval
```

## erps ring <ringid> wtr-interval

### Syntax

```
erps ring <ringid>
 wtr-interval <minutes>
```

### Description

The RPL owner node uses a delay timer before initiating an RPL block in case of both revertive mode of operation or before reverting to idle state after clearing operator commands (FS, MS).

The Wait to Restore (WTR) timer can be configured in 1-minute increments up to 12 minutes. The default value is 5 minutes. When recovering from an SF, the delay timer must be long enough to allow the recovering network to become stable. In the default revertive mode of operation, the WTR timer is used to prevent frequent operation of protection switching due to intermittent SF defects.

The `no` form of this command removes the configured value of the `wtr-interval` and sets it to the default value of 5 minutes.

### Command context

`config-erps-ring-<ringid>`

### Parameters

**<ringid>**

Required, specifies the ID of the ring. Range: 1-239

**<seconds>**

Required, specifies the `wtr-interval` in minutes. Range: 1-12. Default: 5.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Specify the `wtr-interval`:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# wtr-interval 7
```

Remove the configured value of the `wtr-interval` and set it to the default value of 5 minutes:

```
switch(config)# erps ring 3
switch(config-erps-ring-3)# no wtr-interval
```

## show erps statistics

### Syntax

```
show erps statistics [ring <ringid>] [instance <id> [port0|port1]]
```

### Description

This command displays ERPS statistics. The statistics can be displayed for the ring, the instance, or the instance ports.

### Command context

Operator (>) or Manager (#)

### Parameters

**<ringid>**

Optional, specifies the ID of the ring. Range: 1-239.

**<id>**

Optional, specifies the ID of the ring instance. Range: 1-2.

**<port0>**

Optional, specifies the ring member port 0.

**<port1>**

Optional, specifies the ring member port 1.



## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

```
switch# show erps statistics ring 1
```

```
Statistics for ERPS ring 1 instance 1:
```

```
=====
 Port0 Port1
 ---- ----
Local Failures 4 1

R-APS Port0 (Tx/Rx) Port1 (Tx/Rx)

NR 1/1 1/1
NR,RB 0/1 0/1
SF 1/0 1/0
MS 0/0 0/10
FS 30/0 0/0
```

```
Statistics for ERPS ring 1 instance 2:
```

```
=====
 Port0 Port1
 ---- ----
Local Failures 4 1
R-APS Port0 (Tx/Rx) Port1 (Tx/Rx)

NR 1/1 1/1
NR,RB 0/1 0/1
SF 1/0 1/0
MS 0/0 0/10
FS 30/0 0/0
```

```
switch# show erps statistics
```

```
Statistics for ERPS Ring 1 Instance 1 :
```

```
=====
 Port0 Port1
 ---- ----
Local Failures 4 1
R-APS Port0 (Tx/Rx) Port1 (Tx/Rx)

NR 33/9 33/9
NR,RB 58/0 58/0
SF 4/0 4/0
MS 0/0 0/0
FS 0/0 0/0
```

## show erps status

### Syntax

```
show erps status [ring <ringid>] [instance <id>]
```

### Description

This command displays detailed information about a specific ring or all instances of a ring.

The ring instance may be in one of the following states:

- **Idle:** The ring instance is operational.
- **Initializing:** The ring instance is not operational.
- **Protection:** Protection switching has been triggered by a local or remote link failure.
- **Pending:** Pending clearance of a previous protection switch.
- **Down:** Ring instance is not active.
- **Manual-switch:** Manual protection switching triggered by Admin-down.
- **Force-switch:** Forced protection switching triggered by admin.

A ring instance has the following reasons for "down" state:

- **Disabled:** Ring instance is administratively disabled.
- **Inconsistent Port Config:** The same port is configured as port0 and port1 or RPL port is configured by Admin-down.
- **Incomplete Port Config:** Only one or no ring port is configured.
- **Protected VLANs Not Configured:** Protected VLAN list is empty.
- **Control VLAN Not Configured:** Control VLAN is not configured.

The ring ports can be in one of the following states:

- **Up:** Port forwards control and data traffic.
- **Blocked:** Port blocks both control and data traffic.

## Command context

Operator (>) or Manager (#)

## Parameters

**<ringid>**

Optional, specifies the ID of the ring. Range: 1-239.

**<id>**

Optional, specifies the ID of the ring instance. Range: 1-2.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Show ERPS status for ring 1 and instance 1:

```
Status for ERPS Ring 1 Instance 1
=====
Ring ID : 1
Ring description : ring_1
Instance ID : 1
Instance description : inst_1
Port0 : 1/0/1 (Blocked)
Port1 : 1/0/2 (Up)
Node Role (RPL) : Owner (Port0)
```

```

Control VLAN : 100
Protected VLAN : None
Subring (TCN) : Yes (Yes)
Revertive Operation : Revertive
MEG Level : 1
Transmission Interval : 5 sec
Guard Interval : 500 ms
Hold-Off Interval : 1 sec
WTR Interval : 5 min
Status : Initializing
Oper Down Reason : Protected Vlans Not Configured

```

#### Show ERPS status for ring 1:

```
switch# show erps status ring 1
```

##### Status for ERPS Ring 1 Instance 1

```

=====
Ring ID : 1
Ring description : ring_1
Instance ID : 1
Instance description : inst_1
Port0 : 1/0/1 (Blocked)
Port1 : 1/0/2 (Up)
Node Role (RPL) : Owner (Port0)
Control VLAN : 100
Protected VLAN : 1-10
Subring (TCN) : Yes (Yes)
Revertive Operation : Non-Revertive
MEG Level : 1
Transmission Interval : 5 sec
Guard Interval : 500 ms
Hold-Off Interval : 1 sec
WTR Interval : 5 min
Status : Idle
Oper Down Reason : None

```

##### Status for ERPS Ring 1 Instance 2

```

=====
Ring ID : 1
Ring description : ring_1
Instance ID : 2
Instance description : inst_2
Port0 : 1/0/3 (Blocked)
Port1 : 1/0/4 (Up)
Node Role (RPL) : Owner (Port0)
Control VLAN : 110
Protected VLAN : 20-30
Subring (TCN) : No
Revertive Operation : Revertive
MEG Level : 1
Transmission Interval : 5 sec
Guard Interval : 500 ms
Hold-Off Interval : 1 sec
WTR Interval : 5 min
Status : Admin-Down
Oper Down Reason : None

```

# show erps summary

## Syntax

show erps summary

## Description

This command displays a summary of the ERPS configuration and state for the ERPS ring instances.

## Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

```
switch# show erps summary
```

```
ERPS Summary
=====
```

```
Flags: R - RPL, M - Major Ring, S - Sub Ring, T - TCN Enabled
 * - RPL port
```

```
Per-Instance Summary
=====
```

| Ring | Instance | Port0  | Port1  | Status        | Flags |
|------|----------|--------|--------|---------------|-------|
| ---- | -----    | -----  | -----  | -----         | ----- |
| 1    | 1        | 1/1/1  | *1/1/2 | Pending       | R,M   |
| 1    | 2        | 1/1/1  | 1/1/2  | Idle          | M     |
| 2    | 1        | *1/1/3 | -      | Protection    | R,S,T |
| 2    | 2        | 1/1/3  | -      | Admin-down    | S,T   |
| 3    | 1        | 1/1/4  | 1/1/5  | Manual-switch | M     |
| 3    | 2        | 1/1/4  | 1/1/5  | Force-switch  | M     |

## address

### Syntax

```
address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}
```

```
no address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}
```

### Description

Specifies the NAS IP address or hostname.

The `no` form of this command deletes an IP address or hostname.

### Command context

```
config-external-storage-<VOLUME-NAME>
```

### Parameters

**<IPV4-ADDR>**

Specifies the NAS server IPv4 address, Global.

**<IPV6-ADDR>**

Specifies the IPv6 address of the NAS server.

**<HOSTNAME>**

Specifies the hostname of the NAS server. String.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating the logfiles storage volume with IP address 10.1.1.1:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# address 10.1.1.1
```

Deleting an external storage volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no address 10.1.1.1
```

# directory

## Syntax

```
directory <DIRECTORY-NAME>
```

```
no directory <DIRECTORY-NAME>
```

## Description

Selects an existing directory on the external storage volume.

The `no` form of this command clears a directory of an external storage volume.

## Command context

```
config-external-storage-<VOLUME-NAME>
```

## Parameters

**<DIRECTORY-NAME>**

Specifies the external storage directory for mapping the volume.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Creating a volume named `logfiles` that is mapped under `/home` on the server:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# directory /home
```

Clearing the directory `/home`:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no directory /home
```

# disable

## Syntax

```
disable
```

```
no disable
```

## Description

Disables the external storage volume.

The `no` form of this command enables the external storage volume. This is identical to the `enable` command.

## Command context

```
config-external-storage-<VOLUME-NAME>
```

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Disabling a volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# disable
```

# enable

## Syntax

enable

no enable

## Description

Enables the external storage volume.

The `no` form of this command disables the external storage volume. This is identical to the `disable` command.

## Command context

config-external-storage-<VOLUME-NAME>

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Creating and then enabling a volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# enable
```

Disables the external storage volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# disable
```

# external-storage

## Syntax

external-storage <VOLUME-NAME>

no external-storage <VOLUME-NAME>

## Description

Creates or updates an external storage volume.

The `no` form of this command deletes an external storage volume.

### Command context

`config`

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating the logfiles storage volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)#
```

Deleting the logfiles storage volume:

```
switch(config)# no external-storage logfiles
```

## password

### Syntax

```
password {plaintext | ciphertext} <PASSWORD>
```

```
no password {plaintext | ciphertext} <PASSWORD>
```

### Description

Sets the password for logging in to a network attached storage server.

The `no` form of this command clears the password for logging in to a network attached storage server.

### Command context

`config-external-storage-<VOLUME-NAME>`

### Parameters

#### plaintext

Specifies the password be in plain text format.

#### ciphertext

Specifies the password be in ciphertext format.

#### <PASSWORD>

Specifies the password.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating a volume named logfiles with the password xxxx:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# password plaintext xxxx
```



Clearing the plaintext password xxxx:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no password plaintext xxxx
```

## show external-storage

### Syntax

```
show external-storage [<VOLUME-NAME>]
```

### Description

Shows external storage configuration and state for all volumes or for a specified volume.

### Command context

Operator (>) or Manager (#)

### Parameters

*<VOLUME-NAME>*

Specifies the external storage volume name that the show command will use.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch# show external-storage
```

|          | Address   | VRF | Username   | Type  | Directory | State        |
|----------|-----------|-----|------------|-------|-----------|--------------|
| nfsvol   | 10.1.1.1  | nas | ---        | NFSv3 | /home     | operational  |
| nfsfiles | 20.1.1.1  | nas | netstorage | NFSv4 | /netstor  | disabled     |
| scpdev   | nasserver | nas | scpstor    | SCP   | /scp      | unaccessible |

## show running-config external-storage

### Syntax

```
show running-config external-storage
```

### Description

Shows the running configuration of the external storage.

### Command context

Operator (>) or Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch# show running-config external-storage
```

```
external-storage nfsvol
 address 10.1.1.1
 vrf nas
 type nfsv4
 directoty /home
 enable
external-storage scpdev
 address 30.1.1.1
 vrf nas
 username switchuser
 password ciphertext xxx
 type scp
 directoty /home
 enable
```

## type

### Syntax

```
type {nfsv3 | nfsv4 | scp}
```

```
no type {nfsv3 | nfsv4 | scp}
```

### Description

Sets the network attached storage access type for reaching the external storage volume.

The `no` form of this command deletes an external storage volume.

### Command context

```
config-external-storage-<VOLUME-NAME>
```

### Parameters

#### **nfsv3**

Specifies the NFSv3 network access protocol.

#### **nfsv4**

Specifies the NFSv4 network access protocol.

#### **scp**

Specifies the SCP network access protocol.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Creating the logfiles volume using NFSV4:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# type nfsv4
```

Clearing the external storage access type:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no type nfsv4
```

## username

### Syntax

```
username <USER-NAME>
```

```
no username <USER-NAME>
```

### Description

Sets the username for logging in to a network attached storage server.

The `no` form of this command clears a username.

### Command context

```
config-external-storage-<VOLUME-NAME>
```

### Parameters

**<USER-NAME>**

Specifies the username.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating a volume named `logfiles` with the user name `nassuser`:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# username nasuser
```

Clearing the user name `nasuser` from accessing the `logfiles` volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no username nasuser
```

## vrf

### Syntax

```
vrf <VRF-NAME>
```

```
no vrf <VRF-NAME>
```

### Description

Setting a VRF to reach network attached storage.

The `no` form of this command clears access of a VRF to network attached storage.

### Command context

```
config-external-storage-<VOLUME-NAME>
```

## Parameters

**<VRF-NAME>**

Specifies the VRF name.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Creating the logfiles volume and setting a VRF named nas to access the network attached storage:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# vrf nas
```

Clearing access of a VRF named nas to the network attached storage:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no vrf nas
```

## active-gateway

### Syntax

```
active-gateway {ip | ipv6} [<IP-ADDRESS>] [mac <MAC-ADDRESS>]
```

```
no active-gateway {ip | ipv6} [<IP-ADDRESS>] [mac]
```

### Description

Configures an EVPN Anycast gateway that can be used on multiple VTEPs. The Active Gateway supports both IPv4 and IPv6 addresses. The Active Gateway MAC address used along with an IPv4 and IPv6 address must match on a given interface, for the EVPN Anycast Gateway solution to work as expected.

The `no` form of this command removes the active gateway for active-active routing.

### Command context

```
config-if-vlan
```

### Parameters

#### **ip**

Specifies the configuration of an IPv4 address.

#### **ipv6**

Specifies the configuration of an IPv6 address.

#### **<IP-ADDRESS>**

Specifies the IPv4 or IPv6 address.

- Syntax for IPv4: A.B.C.D
- Syntax for IPv6: A:B::C:D

#### **<MAC-ADDRESS>**

Specifies the Virtual MAC address. Syntax: xx:xx:xx:xx:xx:xx

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

Before configuring active gateway, confirm that an IP address is on the SVI that is in the same subnet as the active gateway IP you are trying to configure. If an active gateway IP does not have an SVI IP with the same subnet, the CLI allows the configuration, but the active gateway IP will not be programmed in the kernel, resulting the active gateway to be unreachable.

Active forwarding cannot be configured when ICMP redirect is enabled. Enter the `no ip icmp redirect` command for disabling ICMP redirect.

It is highly recommended that you use an IPv6 link-local address as a gateway (VIP) on the active gateway IPv6 configuration.

If VRRP or active forwarding is configured on an SVI, active gateway cannot be configured. Active gateway with overlapping networks is not allowed. Maximum of 16 unique virtual MACs are supported in a system.

The maximum number of supported active gateways per switch is 4,000. Since a maximum of 31 secondary IPv4 addresses can be configured on an SVI, 32 IPv4 active gateways (along with the primary IPv4 address) can be configured per SVI with IP multinetting support. This support is also the same for IPv6 addresses.



**NOTE:** Do not use peer system MAC address as an active-gateway VMAC. If same MAC address is used, the VSX synchronization will try to sync the configuration on secondary switch and cause traffic disruptions.

## Examples

Configuring active-gateway when the IP address is different from the SVI IP address on both VSX peers (valid for IPv6 and IPv4):

Switch 1:

```
Switch1(config)# int vlan 10
Switch1(config-if-vlan)# vrf attach VRF1
Switch1(config-if-vlan)# ip address 10.1.10.2/24
Switch1(config-if-vlan)# active-gateway ip mac 12:00:00:00:01:00
Switch1(config-if-vlan)# active-gateway ip 10.1.10.1
Switch1(config-if-vlan)# ipv6 address fd00:10:1:10::2/64
Switch1(config-if-vlan)# active-gateway ipv6 mac 12:00:00:00:01:00
Switch1(config-if-vlan)# active-gateway ipv6 fe80::1
```

Switch 2:

```
Switch2(config)# interface vlan10
Switch2(config-if-vlan)# vrf attach VRF1
Switch2(config-if-vlan)# ip address 10.1.10.3/24
Switch2(config-if-vlan)# active-gateway ip mac 12:00:00:00:01:00
Switch2(config-if-vlan)# active-gateway ip 10.1.10.1
Switch2(config-if-vlan)# ipv6 address fd00:10:1:10::3/64
Switch2(config-if-vlan)# active-gateway ipv6 mac 12:00:00:00:01:00
Switch2(config-if-vlan)# active-gateway ipv6 fe80::1
```

Configuring active-gateway when the IP address is the same as the SVI IP address on both VSX peers (valid for IPv4 only):

Switch 1:

```
Switch1(config)# int vlan 10
Switch1(config-if-vlan)# vrf attach VRF1
Switch1(config-if-vlan)# ip address 10.1.10.2/24
Switch1(config-if-vlan)# active-gateway ip mac 12:00:00:00:01:00
Switch1(config-if-vlan)# active-gateway ip 10.1.10.1
```

Switch 2:

```
Switch2(config)# int vlan 10
Switch2(config-if-vlan)# vrf attach VRF1
Switch2(config-if-vlan)# ip address 10.1.10.3/24
Switch2(config-if-vlan)# active-gateway ip mac 12:00:00:00:01:00
Switch2(config-if-vlan)# active-gateway ip 10.1.10.1
```

Configuring only the active gateway address:

```
switch(config-if-vlan) # ip address 192.168.1.250/24
switch(config-if-vlan) # active-gateway ip 192.168.1.251
```

Configuring only the active gateway IP MAC address:

```
switch2(config-if-vlan) # ip address 192.168.1.250/24
switch2(config-if-vlan) # active-gateway ip mac 00:00:00:01:00:01
```

Removing the active gateway for active-active routing (IPv6 and IPv4):

```
switch(config-if-vlan) # no active-gateway ip
switch(config-if-vlan) # no active-gateway ipv6
```

Removing the active gateway for active-active routing for an IP address:

```
switch(config-if-vlan) # no active-gateway ip 192.168.1.250
```

Removing the active gateway for active-active routing for virtual MAC addresses:

```
switch(config-if-vlan) # no active-gateway ip mac
```

## arp-suppression

*Not supported on the 8400 Switch Series.*

### Syntax

```
arp-suppression
```

```
no arp-suppression
```

### Description

Enables ARP suppression for EVPN-VXLAN globally across all Layer 2 VNIs configured on the VTEP. If the target address is present in the neighbor cache, the switch responds to the broadcast or unicast ARP request. ARP suppression is disabled by default. If the target IP/MAC is not present, the switch forwards arp request over the VXLAN data plane for neighbor resolution.

The `no` form of this command disables the ARP suppression.

### Command context

```
config-evpn
```

### Command context

```
config-evpn-vlan
```

### Examples

Configuring ARP suppression in EVPN:

```
switch(config-evpn) # arp-suppression
switch(config-evpn) # no arp-suppression
```

## evpn

### Syntax

evpn

no evpn

### Description

Specifies the EVPN context which provides the configurations for VLAN-based EVPN service mode. The **no** form of this command removes this configuration.

### Command context

config-evpn

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring the EVPN context:

```
switch(config)# evpn
switch(config-evpn)#
```

Removing the EVPN configuration context:

```
switch(config-evpn)# no evpn
```

## mac-move-detection count timer

### Syntax

mac-move-detection count <MAC-MOVE-COUNT> timer <MAC-MOVE-TIMER>

no mac-move-detection count <MAC-MOVE-COUNT> timer <MAC-MOVE-TIMER>

### Description

Configures EVPN MAC dampening for duplicate MAC and MAC-move count and timer across VTEPs.

The **no** form of this command resets the value of the count and timer to the default values of 5 and 180 seconds respectively.



**NOTE:** EVPN MAC dampening is always enabled. Links to the VTEPs must be always up for EVPN MAC dampening to be activated.

### Command context

config-evpn

### Parameters

**count** <MAC-MOVE-COUNT>

Specifies the number of MAC-moves for MAC dampening to take effect. Range: 2 to 10. Default: 5.



**timer** <MAC-MOVE-TIMER>

Specifies the MAC-move time limit in seconds for MAC dampening to take effect. Range: 1 to 1000 seconds. Default: 180 seconds.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring EVPN MAC dampening:

```
switch(config-evpn) # mac-move-detection count 6 timer 199
```

The above command dampens a MAC if the MAC moves six times within 199 seconds.

```
switch(config-evpn) # mac-move-detection count 8
```

The above command dampens a MAC if the MAC moves eight times within 180 seconds.

```
switch(config-evpn) # mac-move-detection timer 255
```

The above command dampens a MAC if the MAC moves five times within 255 seconds.

## nd-suppression

*Not supported on the 8400 Switch Series.*

### Syntax

nd-suppression

no nd-suppression

### Description

Enables ND suppression for EVPN-VXLAN globally. If the target address is present in the NDMD cache, the switch responds to the IPv6 multicast or unicast neighbor solicitation. ND suppression is disabled by default. This feature is not supported on 8400 platform.

The **no** form of this command disables the ND suppression.

### Command context

config-evpn

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring ND suppression in EVPN:

```
switch(config-evpn) # nd-suppression
switch(config-evpn) # no nd-suppression
```

## rd

### Syntax

```
rd {auto | <AS-NUMBER:ID> | <IP-ADDRESS:ID>}
no rd {auto | <AS-NUMBER:ID> | <IP-ADDRESS:ID>}
```

### Description

Specifies a unique number prepended to the advertised routes within the VLAN. It ensures support for overlapping IP addresses and MACs across different tenants. The default value is NULL. Route Distinguisher (RD) has to be manually configured by a user.

The **no** form of this command removes the currently configured value.

### Command context

config-evpn-vlan

### Parameters

#### auto

Specifies automatic route filtering.

#### <AS-NUMBER:ID>

Specifies the AS number. It can be a 1-byte or 4-byte value. If the AS number is a 2-byte value, the administrative number is a 4-byte value and if the AS number is 4-byte value, the administrative number is a 2-byte value.

#### <IP-ADDRESS:ID>

Specifies the IP address. It is a 4-byte value and the id is 2 bytes.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring Route Distinguisher for EVPN VLAN:

```
switch(config-evpn)# vlan 10
switch(config-evpn-vlan-10)# rd 6800:1
switch(config-evpn)# vlan 20
switch(config-evpn-vlan-20)# rd auto
```

## redistribute host-route

### Syntax

```
redistribute host-route
no redistribute host-route
```

### Description

Enables type-2 route advertisement to include the L3VNI, RT, and router MAC of the associated IP-VRF. It is applicable only in Symmetric routing where L3VNI is configured.

The `no` form of this command disables the redistribution of host routes.

### Command context

`config-evpn-vlan`

### Parameters

#### **host-route**

Specifies redistribution of host routes.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring Redistribute host-route in EVPN:

```
switch(config-evpn)# vlan 10
switch(config-evpn-vlan-10)# redistribute host-route
```

## redistribute local-svi

### Syntax

```
redistribute local-svi
```

```
no redistribute local-svi
```

### Description

Enables type-2 route advertisement for the local IP address and MAC address of the SVI interfaces corresponding to the EVPN-enabled VLANs.

The `no` form of this command disables type-2 route advertisement for the local IP address and MAC address of the SVI interfaces corresponding to the EVPN-enabled VLANs.

### Command context

`config-evpn`

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling type-2 route advertisement:

```
switch(config)# evpn
switch(config-evpn)# redistribute local-svi
```

Disabling type-2 route advertisement:

```
switch(config)# evpn
switch(config-evpn)# no redistribute local-svi
```

# route-target

## Syntax

```
route-target {import | export | both} {auto | <AS-NUMBER:ID> | <IP-ADDRESS:ID>}
no route-target {import | export | both} {auto | <AS-NUMBER:ID> | <IP-ADDRESS:ID>}
```

## Description

Controls the import and export of VPN routes only to the systems in the network for which routes are needed. The default value is NULL. Route Targets (RT) have to be manually configured by a user.

The `no` form of this command removes the currently configured value.

## Command context

config-evpn-vlan

## Parameters

### import

Configures the route-target to import EVPN routes.

### export

Configures the route-target to export EVPN routes.

### both

Configures the route-target to import and export EVPN routes.

### auto

Specifies automatic route filtering.

### <AS-NUMBER: ID>

Specifies the AS number. It can be a 1-byte or 4-byte value. If the AS number is a 2-byte value, the administrative number is a 4-byte value and if the AS number is 4-byte value, the administrative number is a 2-byte value.

### <IP-ADDRESS: ID>

Specifies the IP address. It is a 4-byte value and the id is 2 bytes.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring Route Targets for EVPN VLAN:

```
switch(config-evpn)# vlan 10
switch(config-evpn-vlan-10)# route-target import 6800:1
switch(config-evpn-vlan-10)# route-target export 6800:1
switch(config-evpn)# vlan 20
switch(config-evpn-vlan-20)# route-target both 6900:1
```

# route-target {evpn}

## Syntax

```
route-target {import | export | both} {auto | <AS-NUMBER:ID> | <IP-ADDRESS:ID>} evpn
no route-target {import | export | both} {auto | <AS-NUMBER:ID> | <IP-ADDRESS:ID>} evpn
```

## Description

Configures the route target (RT) for EVPN VRF to control the import and export of VPN routes only to the systems in the network for which routes are needed. The default value is NULL. Route targets have to be manually configured by a user.

The `no` form of this command removes the RT in EVPN VRF.

## Command context

`config-vrf`

## Parameters

### import

Imports the VRF routes that match the RT.

### export

Exports the RT in the VRF routes.

### both

Configures both import and export of routes for the VRF.

### auto

Specifies automatic route filtering.

### <AS-NUMBER: ID>

Specifies the AS number. It can be a 1-byte or 4-byte value. If the AS number is a 2-byte value, the administrative number is a 4-byte value. If the AS number is 4-byte value, the administrative number is a 2-byte value.

### <IP-ADDRESS: ID>

Specifies the IP address. It is a 4-byte value and the id is 2 byte value.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring Route Targets for EVPN VRF:

```
switch(config-vrf)# route-target import 6800:1 evpn
switch(config-vrf)# route-target export 6800:1 evpn
switch(config-vrf)# route-target both 6800:1 evpn
```

# show evpn evi

## Syntax

show evpn evi

## Description

Shows the information of EVPN instances.

## Command context

Operator (>) or Manager (#)

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing information for EVPN instances:

```
switch# show evpn evi

L2VNI : 100
 Route Distinguisher : 10.10.10.1:10
 VLAN : 10
 Status : Up
 RT Import : 1.1.1.1:1, 2.2.2.2:1, 3.3.3.3:1, 5:1
 RT Export : 4.4.4.4:61, 1000:21
 Local MACs : 30
 Remote MACs : 945
 Peer VTEPs : 8

L2VNI : 200
 Route Distinguisher :
 VLAN : 20
 Status : Down, No RD
 RT Import : 1.1.1.1:2, 2.2.2.2:2, 3.3.3.3:2, 4.4.4.4:2,
 5.5.5.5:2, 5:2
 RT Export : 4.4.4.4:62, 1000:22
 Local MACs :
 Remote MACs :
 Peer VTEPs :

L2VNI : 300
 Route Distinguisher : 10.10.10.1:30
 VLAN : 30
 Status : Up
 RT Import : 1.1.1.1:3, 2.2.2.2:3, 3.3.3.3:3, 5:3
 RT Export : 4.4.4.4:63, 1000:23
 Local MACs : 30
 Remote MACs : 945
 Peer VTEPs : 12

L3VNI : 1000
 Route Distinguisher : 10.10.10.1:1000
 VRF : vrf1000
 Status : Up
 RT Import : 1.1.1.1:4, 2.2.2.2:4, 3.3.3.3:4, 5:4
 RT Export : 4.4.4.4:64, 1000:24
 Local Type-5 Routes : 2
```

```
Remote Type-5 Routes : 3
Peer VTEPs : 6
```

## show evpn evi summary

### Syntax

```
show evpn evi summary
```

### Description

Shows the summary information for EVPN instances.

### Command context

Operator (>) or Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Showing summary information for EVPN instances:

```
switch# show evpn evi summary
```

| L2VNI | VLAN | Status            |
|-------|------|-------------------|
| 100   | 10   | Up                |
| 200   | 20   | Down, RT conflict |
| 210   | 21   | Up                |
| 220   | 22   | Down, No RT       |
| 230   | 23   | Down, No RT       |
| 240   | 24   | Up                |
| 250   | 25   | Up                |
| 260   | 26   | Up                |
| 270   | 27   | Up                |
| 280   | 28   | Up                |
| 290   | 29   | Up                |
| 310   | 31   | Up                |

| L3VNI | VRF     | Status                      |
|-------|---------|-----------------------------|
| 1000  | vrf1000 | Up                          |
| 1001  | vrf1001 | Down, RT conflict           |
| 1002  | vrf1002 | Down, No RD                 |
| 1004  | vrf1003 | Down, Administratively down |
| 1005  | vrf1003 | Up                          |

```
EVPN instances : 17
EVPN instances Up : 11
```

## show evpn evi <EVI-ID>

### Syntax

```
show evpn evi <EVI-ID>
```

### Description

Shows the information for the particular EVPN instance.

## Command context

Operator (>) or Manager (#)

## Parameter

**<EVI-ID>**

Specifies the EVPN instance ID.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing information for the particular EVPN instance:

```
switch# show evpn evi 100
L2VNI : 100
 Route Distinguisher : 10.10.10.1:10
 VLAN : 10
 Status : Up
 RT Import : 1.1.1.1:1, 2.2.2.2:1, 3.3.3.3:1, 5:1
 RT Export : 4.4.4.4:61, 1000:21
 Local MACs : 30
 Remote MACs : 945
 Peer VTEPs : 8
```

## show evpn evi detail

## Syntax

```
show evpn evi detail
```

## Description

Shows the detailed information for all EVPN instances.

## Command context

Operator (>) or Manager (#)

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing detailed information for EVPN instances:

```
switch# show evpn evi detail

L2VNI : 100
 Route Distinguisher : 10.10.10.1:10
 VLAN : 10
 Status : Up
 RT Import : 1.1.1.1:1, 2.2.2.2:1, 3.3.3.3:1, 5:1
 RT Export : 4.4.4.4:61, 1000:21
 Local MACs : 30
 Remote MACs : 307
 Peer VTEPs : 8

 Peer VTEPs Remote MACs
```



```

10.10.10.2 40
10.10.10.3 22
10.10.10.4 15
10.10.10.5 155
10.10.10.6 25
10.10.10.7 35
10.10.10.8 50
10.10.10.9 55

```

#### L2VNI : 200

```

Route Distinguisher :
VLAN : 20
Status : Down, No RD
RT Import : 1.1.1.1:2, 2.2.2.2:2, 3.3.3.3:2, 4.4.4.4:2,
 5.5.5.5:2, 5:2
RT Export : 4.4.4.4:62, 1000:22
Local MACs :
Remote MACs :
Peer VTEPs :

```

```

Peer VTEPs Remote MACs

```

#### L2VNI : 300

```

Route Distinguisher : 10.10.10.1:30
VLAN : 30
Status : Up
RT Import : 1.1.1.1:3, 2.2.2.2:3, 3.3.3.3:3, 5:3
RT Export : 4.4.4.4:63, 1000:23
Local MACs : 30
Remote MACs : 362
Peer VTEPs : 12

```

```

Peer VTEPs Remote MACs

```

```

10.10.10.2 60
10.10.10.3 12
10.10.10.4 13
10.10.10.5 15
10.10.10.6 15
10.10.10.7 35
10.10.10.8 53
10.10.10.9 45
10.10.10.10 11
10.10.10.11 12
10.10.10.12 35
10.10.10.13 56

```

#### L3VNI : 1000

```

Route Distinguisher : 10.10.10.1:1000
VRF : vrf1000
Status : Up
RT Import : 1.1.1.1:4, 2.2.2.2:4, 3.3.3.3:4, 5:4
RT Export : 4.4.4.4:64, 1000:24
Local Type-5 Routes : 2
Remote Type-5 Routes : 30
Peer VTEPs : 12

```

```

Peer VTEPs Remote Type-5 Routes

```

```

10.10.10.2 2
10.10.10.3 1
10.10.10.4 1

```

|             |   |
|-------------|---|
| 10.10.10.5  | 1 |
| 10.10.10.6  | 1 |
| 10.10.10.7  | 3 |
| 10.10.10.8  | 5 |
| 10.10.10.9  | 4 |
| 10.10.10.10 | 1 |
| 10.10.10.11 | 3 |
| 10.10.10.12 | 3 |
| 10.10.10.13 | 5 |

## show evpn evi <EVI-ID> detail

### Syntax

```
show evpn evi <EVI-ID> detail
```

### Description

Shows the detailed information for the particular EVPN instance.

### Command context

Operator (>) or Manager (#)

### Parameter

**<EVI-ID>**

Specifies the EVPN instance ID.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Showing detailed information for the particular EVPN instance:

```
switch# show evpn evi 100 detail
L2VNI : 100
 Route Distinguisher : 10.10.10.1:10
 VLAN : 10
 Status : Up
 RT Import : 1.1.1.1:1, 2.2.2.2:1, 3.3.3.3:1, 5:1
 RT Export : 4.4.4.4:61, 1000:21
 Local MACs : 30
 Remote MACs : 397
 Peer VTEPs : 8

 Peer VTEPs Remote MACs

 10.10.10.2 40
 10.10.10.3 22
 10.10.10.4 15
 10.10.10.5 155
 10.10.10.6 25
 10.10.10.7 35
 10.10.10.8 50
 10.10.10.9 55
```

## show evpn mac-ip

### Syntax

```
show evpn mac-ip
```

### Description

Show the information about the EVPN MAC-IP for the EVPN instances.

### Command context

Operator (>) or Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Showing information about the EVPN MAC-IP for the EVPN instances:

```
switch# show evpn mac-ip
Flags: Local (L), Remote (R)
```

| EVI | MAC               | IP      | Next-hop          | Seq-Num | Flags |
|-----|-------------------|---------|-------------------|---------|-------|
| 100 | 14:50:56:96:76:56 |         | vxlan1 (11.1.1.3) | 0       | R     |
| 100 | 14:50:56:96:76:56 | 3.3.4.5 | vxlan1 (11.1.1.3) | 0       | R     |
| 100 | 14:50:56:96:76:56 | 3.3.5.5 | vxlan1 (11.1.1.3) | 0       | R     |
| 100 | 24:50:56:96:76:56 | 3.3.3.2 | vxlan1 (11.1.1.3) | 1       | R     |
| 100 | 34:50:56:96:76:56 | 3.3.6.2 |                   | 2       | L     |
| 100 | 44:50:56:96:76:56 | 3.3.7.3 |                   | 2       | L     |
| 200 | 52:50:56:96:76:56 | 5.5.5.2 |                   | 0       | L     |

```
MACs : 5
Remote MACs : 2
```

## show evpn mac-ip evi

### Syntax

```
show evpn mac-ip evi <EVI-ID>
```

### Description

Show the information about the EVPN MAC-IP for the particular EVPN instance.

### Command context

Operator (>) or Manager (#)

### Parameter

<EVI-ID>

Specifies the EVPN instance ID.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing information about the EVPN MAC-IP for the particular EVPN instance:

```
switch# show evpn mac-ip evi 100
Flags: Local (L), Remote (R)
```

| MAC               | IP      | Next-hop          | Seq-Num | Flags |
|-------------------|---------|-------------------|---------|-------|
| 14:50:56:96:76:56 |         | vxlan1 (11.1.1.2) | 0       | R     |
| 14:50:56:96:76:56 | 3.3.4.5 | vxlan1 (11.1.1.3) | 0       | R     |
| 14:50:56:96:76:56 | 3.3.5.4 | vxlan1 (11.1.1.2) | 0       | R     |
| 24:50:56:96:76:56 | 3.3.3.2 | vxlan1 (11.1.1.3) | 1       | R     |
| 34:50:56:96:76:56 | 3.3.6.2 |                   | 2       | L     |
| 44:50:56:96:76:56 | 3.3.7.3 |                   | 2       | L     |

MACs : 4  
Remote MACs : 2

## show evpn vtep-neighbor

### Syntax

```
show evpn vtep-neighbor {all-vrfs | vrf <VRF-Name>}
```

### Description

Shows the remote VTEPs MAC-IP binding. The state of the peer VTEP denotes whether VXLAN tunnel to the VTEP is Up or Down.

### Command context

Operator (>) or Manager (#)

### Parameters

#### all-vrfs

Display information for all VRFs.

#### vrf <vrf-name>

Specify a VRF by VRF name (if no <VRF-NAME> is specified, the default VRF is implied).

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing EVPN VTEP neighbor information for all VRFs:

```
switch# show evpn vtep-neighbor all-vrfs
```

| VTEP-IP | L3VNI | MAC               | VRF     | State |
|---------|-------|-------------------|---------|-------|
| 2.2.2.2 | 1234  | 00:20:56:bd:27:bc | VRF1234 | Up    |
| 2.2.2.2 | 6789  | 00:20:56:bd:27:bc | VRF6789 | Up    |
| 3.3.3.3 | 1234  | 00:30:56:ef:aa:cc | VRF1234 | Down  |
| 4.4.4.4 | 6789  | 00:40:56:12:34:44 | VRF6789 | Up    |
| 5.5.5.5 | 6789  | 00:50:56:ab:11:ee | VRF6789 | Up    |

Showing EVPN VTEP neighbor information for the specified VRF name:

```
switch# show evpn vtep-neighbor vrf VRF1234
VTEP-IP L3VNI MAC VRF State

2.2.2.2 1234 00:20:56:bd:27:bc VRF1234 Up
3.3.3.3 1234 00:30:56:ef:aa:cc VRF1234 Down
```

## show running-config evpn

### Syntax

```
show running-config evpn
```

### Description

Shows all EVPN configurations.

### Command context

Operator (>) or Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Showing all EVPN configurations:

```
switch# show running-config evpn
evpn
 vlan 10
 rd 6800:1
 route-target import 6800:1
 route-target export 6800:1
 vlan 20
 rd 6900:1
 route-target import 6900:1
 route-target export 6900:1
```

## virtual-mac

### Syntax

```
virtual-mac <MAC-ADDR>
```

```
no virtual-mac
```

### Description

Configures the virtual MAC address for EVPN.

The `no` form of this command removes the virtual MAC address configuration.

### Command context

```
config
```

## Parameters

**<MAC-ADDR>**

Specifies the virtual MAC address.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring virtual MAC address for EVPN:

```
switch(config)# virtual-mac ab:12:33:33:03:22
switch(config)# no virtual-mac
```



**NOTE:** For EVPN symmetric IRB to work, `virtual-mac` must be configured and it must be unique for all the VTEPs involved in EVPN except for VSX nodes. In the VSX pair switches, same `virtual-mac` must be configured in both the VTEPs. In case of VSX VTEP, it is recommended to use the same value as the VSX `system-mac` value in order to simplify the troubleshooting.

# vlan

## Syntax

`vlan <ID>`

`no vlan <ID>`

## Description

Specifies the VLAN ID and enters the VLAN context under EVPN.

The `no` form of this command removes this configuration.

## Command context

`config-evpn-vlan`

## Parameters

**<ID>**

Specifies the VLAN ID. Range: 2 - 4040.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config-evpn)# vlan 10
switch(config-evpn-vlan-10)#
```

## Command for creating a fault monitoring profile

### Syntax

```
fault-monitor profile <profile-name>

no fault-monitor profile <profile-name>
```

### Description

Creates a fault monitoring profile. You can create maximum of 16 fault monitoring profiles to monitor different types of interfaces.

The `no` form of this command deletes the fault monitoring profile.

### Command context

config

### Parameters

**<profile-name>**

Specifies the fault monitor profile name. Profile name accepts the special characters and the name must not exceed 64 characters.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating a fault monitor profile:

```
switch(config)# fault-monitor profile noisy-ports
switch(config-fault-monitor-profile)#
```

Deleting a fault monitor profile:

```
switch(config)# no fault-monitor profile noisy-ports
switch(config)#
```

## Commands for enabling and disabling the faults

### Syntax

```
all | excessive-oversize-packets | excessive-jabbers | excessive-fragments |
excessive-crc-errors | excessive-tx-drops | excessive-link-flaps | excessive-broadcasts |
excessive-multicasts | excessive-collisions | excessive-late-collisions | excessive-alignment-errors

no all | excessive-oversize-packets | excessive-jabbers | excessive-fragments |
excessive-crc-errors | excessive-tx-drops | excessive-link-flaps | excessive-broadcasts |
excessive-multicasts | excessive-collisions | excessive-late-collisions | excessive-alignment-errors
```

## Description

Enables the faults for a profile.



**NOTE:** If user has not configured threshold, action or auto-enable parameters, then this command enables fault with the default values.

By default, the faults are disabled in a profile. Configuring threshold, action, or auto-enable will not enable the fault.

The `no` form of this command disables the faults for a profile.

## Command context

`config-fault-monitor-profile`

## Parameters

**all**

Enables all the faults for a profile.

**excessive-oversize-packets**

Enables excessive oversize packets fault for a profile.

**excessive-jabbers**

Enables excessive jabbers fault for a profile.

**excessive-fragments**

Enables excessive fragments fault for a profile.

**excessive-crc-errors**

Enables excessive CRC errors fault for a profile.

**excessive-tx-drops**

Enables excessive transmit fault for a profile.

**excessive-link-flaps**

Enables excessive link flaps fault for a profile.

**excessive-broadcasts**

Enables excessive broadcasts fault for a profile.

**excessive-multicasts**

Enables excessive multicasts fault for a profile.

**excessive-collisions**

Enables excessive collisions fault for a profile.

**excessive-late-collisions**

Enables excessive late collisions fault for a profile.

**excessive-alignment-errors**

Enables excessive alignment errors fault for a profile.

## Authority

Administrators or local user group members with execution rights for this command.



## Examples

Enabling the faults for a profile:

```
switch(config-fault-monitor-profile) # all
switch(config-fault-monitor-profile) # excessive-oversize-packets
switch(config-fault-monitor-profile) # excessive-jabbers
switch(config-fault-monitor-profile) # excessive-fragments
switch(config-fault-monitor-profile) # excessive-crc-errors
switch(config-fault-monitor-profile) # excessive-tx-drops
switch(config-fault-monitor-profile) # excessive-link-flaps
switch(config-fault-monitor-profile) # excessive-broadcasts
switch(config-fault-monitor-profile) # excessive-multicasts
switch(config-fault-monitor-profile) # excessive-collisions
switch(config-fault-monitor-profile) # excessive-late-collisions
switch(config-fault-monitor-profile) # excessive-alignment-errors
```

Disabling the faults for a profile:

```
switch(config-fault-monitor-profile) # no all
switch(config-fault-monitor-profile) # no excessive-oversize-packets
switch(config-fault-monitor-profile) # no excessive-jabbers
switch(config-fault-monitor-profile) # no excessive-fragments
switch(config-fault-monitor-profile) # no excessive-crc-errors
switch(config-fault-monitor-profile) # no excessive-tx-drops
switch(config-fault-monitor-profile) # no excessive-link-flaps
switch(config-fault-monitor-profile) # no excessive-broadcasts
switch(config-fault-monitor-profile) # no excessive-multicasts
switch(config-fault-monitor-profile) # no excessive-collisions
switch(config-fault-monitor-profile) # no excessive-late-collisions
switch(config-fault-monitor-profile) # no excessive-alignment-errors
```

## Commands for applying or changing the action and configuring auto-enable for a fault

### Syntax

```
[all | excessive-oversize-packets | excessive-jabbers | excessive-fragments |
excessive-crc-errors | excessive-tx-drops | excessive-link-flaps | excessive-broadcasts |
excessive-multicasts | excessive-collisions | excessive-late-collisions | excessive-alignment-errors]
[action {notify | notify-and-disable [auto-enable <timeout>]}]

no [all | excessive-oversize-packets | excessive-jabbers | excessive-fragments |
excessive-crc-errors | excessive-tx-drops | excessive-link-flaps | excessive-broadcasts |
excessive-multicasts | excessive-collisions | excessive-late-collisions | excessive-alignment-errors]
[action {notify | notify-and-disable [auto-enable]}]
```

### Description

Applies an action and configures the auto-enable timer for the fault. Default action: notify with auto-enable timer disabled.

The no form of this command removes the action and auto-enable timer for the fault.

### Command context

config-fault-monitor-profile

## Parameters

### notify

Specifies the action as `notify` for the faults. Notifies through events, DLOGs, and SNMP trap. It is enabled by default.

### notify-and-disable

Specifies the action as `notify-and-disable` for the faults. Notifies through events, DLOGs, and SNMP trap, and then disables the port.

### auto-enable <timeout>

Re-enables the port automatically after the specified time. Specifies the time value in seconds. Time range: 1-604800 seconds.



**NOTE:** The fault parameter values are saved in the database even after the fault is disabled from the profile. These values are used when you enable the fault in the profile again.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring `notify-and-disable` action for the faults:

```
switch(config-fault-monitor-profile)# all action notify-and-disable
switch(config-fault-monitor-profile)# excessive-oversize-packets action notify-and-disable
switch(config-fault-monitor-profile)# excessive-jabbers action notify-and-disable
switch(config-fault-monitor-profile)# excessive-late-collisions action notify-and-disable
switch(config-fault-monitor-profile)# excessive-alignment-errors action notify-and-disable
```

Configuring `notify` action for the faults:

```
switch(config-fault-monitor-profile)# all action notify
switch(config-fault-monitor-profile)# excessive-oversize-packets action notify
switch(config-fault-monitor-profile)# excessive-jabbers action notify
switch(config-fault-monitor-profile)# excessive-collisions action notify
```

Configuring `notify-and-disable` action with `auto-enable` timer for the faults:

```
switch(config-fault-monitor-profile)# excessive-oversize-packets action notify-and-disable auto-enable 80
switch(config-fault-monitor-profile)# excessive-jabbers action notify-and-disable auto-enable 100
switch(config-fault-monitor-profile)# excessive-collisions action notify-and-disable auto-enable 70
```

Changing the fault action to default:

```
switch(config-fault-monitor-profile)# no excessive-oversize-packets action
switch(config-fault-monitor-profile)# no excessive-jabbers action
switch(config-fault-monitor-profile)# no excessive-alignment-errors action
```

Removing the `auto-enable` timer configuration for the fault:

```
switch(config-fault-monitor-profile)# no all action notify-and-disable auto-enable
switch(config-fault-monitor-profile)# no excessive-jabbers action notify-and-disable auto-enable
switch(config-fault-monitor-profile)# no excessive-alignment-errors action notify-and-disable auto-enable
```

## Commands for configuring the threshold for the fault

### Syntax

```
excessive-jabbers | excessive-crc-errors | excessive-oversize-packets |
excessive-fragments | excessive-tx-drops | excessive-collisions |
```

```

excessive-late-collisions | excessive-alignment-errors {threshold value <value>}

no excessive-jabbers | excessive-crc-errors | excessive-oversize-packets |
excessive-fragments | excessive-tx-drops | excessive-collisions |
excessive-late-collisions | excessive-alignment-errors {threshold}

excessive-link-flaps {threshold count <count>}

no excessive-link-flaps {threshold}

excessive-broadcasts | excessive-multicasts {threshold [percent <value> | pps <value>]}

no excessive-broadcasts | excessive-multicasts {threshold}

```

## Description

Configures and changes the fault threshold value for the profile.

The `no` form of this command sets the threshold to its default value.

## Command context

`config-fault-monitor-profile`

## Parameters

**threshold value <value>**

Specifies the fault threshold value. Default threshold value: 25.

**threshold count <value>**

Specifies the fault threshold count. Default threshold count: 7.

**threshold percent <value>**

Specifies the fault threshold bandwidth percentage. Range: 1 to 100. Default threshold percent: 5.

**threshold pps <value>**

Specifies the fault threshold PPS. Range: 1-195312500.



**NOTE:** If excessive-broadcast or excessive-multicast faults are configured with the threshold higher than the `rate-limit` threshold, the following occurs:

- Fault reporting still happens as the port has actually received packets at a rate that violated its threshold.
- Traffic gets shaped as per `rate-limit` configuration and any packet exceeding the `rate-limit` threshold gets dropped.

These two fault-monitoring and `rate-limiting` features are orthogonal to one another.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring the excessive jabbers, oversize packets, CRC, fragments, and TX faults with threshold value:

```

switch(config-fault-monitor-profile) # excessive-jabbers threshold value 30
switch(config-fault-monitor-profile) # excessive-oversize-packets threshold value 40
switch(config-fault-monitor-profile) # excessive-crc-errors threshold value 35
switch(config-fault-monitor-profile) # excessive-fragments threshold value 50

```

```
switch(config-fault-monitor-profile)# excessive-tx-drops threshold value 20
switch(config-fault-monitor-profile)# excessive-collisions threshold value 40
switch(config-fault-monitor-profile)# excessive-late-collisions threshold value 30
switch(config-fault-monitor-profile)# excessive-alignment-errors threshold value 50
```

Configuring the excessive link flaps fault with threshold count:

```
switch(config-fault-monitor-profile)# excessive-link-flaps threshold count 10
```

Configuring the excessive broadcasts and multicasts faults with threshold:

```
switch(config-fault-monitor-profile)# excessive-broadcasts threshold percent 40
switch(config-fault-monitor-profile)# excessive-broadcasts threshold pps 400000
switch(config-fault-monitor-profile)# excessive-multicasts threshold percent 10
switch(config-fault-monitor-profile)# excessive-multicasts threshold pps 10000
```

Removing the configured threshold value for the faults:

```
switch(config-fault-monitor-profile)# no excessive-jabbers threshold
switch(config-fault-monitor-profile)# no excessive-oversize-packets threshold
switch(config-fault-monitor-profile)# no excessive-crc-errors threshold
switch(config-fault-monitor-profile)# no excessive-fragments threshold
switch(config-fault-monitor-profile)# no excessive-tx-drops threshold
switch(config-fault-monitor-profile)# no excessive-link-flaps threshold
switch(config-fault-monitor-profile)# no excessive-broadcasts threshold
switch(config-fault-monitor-profile)# no excessive-multicasts threshold
switch(config-fault-monitor-profile)# no excessive-late-collisions threshold
switch(config-fault-monitor-profile)# no excessive-alignment-errors threshold
```

## Command for applying a fault monitoring profile to a port

### Syntax

```
apply fault-monitor profile <profile-name>
```

```
no apply fault-monitor profile <profile-name>
```

### Description

Configures the fault monitoring profile to an interface. The profile is allowed to be modified after it is applied on a port.

The **no** form of this command removes the fault monitoring profile from an interface.

### Command context

```
config-if
```

### Parameters

**<profile-name>**

Specifies the fault monitor profile name.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring the fault monitoring profile to an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# apply fault-monitor profile noisy-ports
```

Configuring the fault monitoring profile to a specified range of interfaces:

```
switch(config)# interface 1/1/2-1/1/24
switch(config-if)# apply fault-monitor profile quiet-ports
```

## Command for configuring VSX synchronization for a fault monitoring profile

### Syntax

```
vsx-sync
```

```
no vsx-sync
```

### Description

Configures VSX synchronization for a fault monitoring profile.

The `no` form of this command removes the VSX synchronization for a fault monitoring profile.

### Command context

```
config-fault-monitor-profile
```

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Configuring VSX synchronization for a fault monitoring profile:

```
switch(config-fault-monitor-profile)# vsx-sync
```

## show fault-monitor profile

### Syntax

```
show fault-monitor profile
```

```
show fault-monitor profile <profile-name>
```

### Description

Shows the fault monitoring profile information. The command without the profile name displays all the available fault monitoring profiles.

### Parameters

<profile-name>

Specifies the fault monitor profile name.

### Authority

Administrators or local user group members with execution rights for this command.

## Example

Displaying the fault monitoring information for all the profiles available on the switch:

```
switch# show fault-monitor profile
```

```
Fault monitor profile: 'noisy-ports'
```

| Fault                      | Enabled | Threshold  | Action             | Auto Enable |
|----------------------------|---------|------------|--------------------|-------------|
| excessive-broadcasts       | yes     | 400000 pps | notify             | --          |
| excessive-multicasts       | yes     | 10 percent | notify             | --          |
| excessive-link-flaps       | yes     | 10         | notify             | --          |
| excessive-oversize-packets | yes     | 40         | notify-and-disable | 80          |
| excessive-jabbers          | yes     | 30         | notify-and-disable | 100         |
| excessive-fragments        | yes     | 50         | notify             | --          |
| excessive-crc-errors       | yes     | 35         | notify             | --          |
| excessive-tx-drops         | yes     | 20         | notify             | --          |
| excessive-collisions       | yes     | 40         | notify             | 70          |
| excessive-late-collisions  | yes     | 30         | notify-and-disable | --          |
| excessive-alignment-errors | yes     | 50         | notify-and-disable | 100         |

```
Fault monitor profile: 'quiet-ports'
```

| Fault                      | Enabled | Threshold  | Action             | Auto Enable |
|----------------------------|---------|------------|--------------------|-------------|
| excessive-broadcasts       | yes     | 300000 pps | notify             | --          |
| excessive-multicasts       | yes     | 100000 pps | notify             | --          |
| excessive-link-flaps       | yes     | 20         | notify-and-disable | 45          |
| excessive-oversize-packets | yes     | 25         | notify-and-disable | 20          |
| excessive-jabbers          | yes     | 30         | notify             | --          |
| excessive-fragments        | yes     | 60         | notify             | --          |
| excessive-crc-errors       | yes     | 25         | notify-and-disable | 30          |
| excessive-tx-drops         | yes     | 30         | notify             | --          |

Displaying the fault monitoring details for a particular profile:

```
switch# show fault-monitor profile noisy-ports
```

```
Fault monitor profile: 'noisy-ports'
```

| Fault                      | Enabled | Threshold  | Action             | Auto Enable |
|----------------------------|---------|------------|--------------------|-------------|
| excessive-broadcasts       | yes     | 400000 pps | notify             | --          |
| excessive-multicasts       | yes     | 10 percent | notify             | --          |
| excessive-link-flaps       | yes     | 10         | notify             | --          |
| excessive-oversize-packets | yes     | 40         | notify-and-disable | 80          |
| excessive-jabbers          | yes     | 30         | notify-and-disable | 100         |
| excessive-fragments        | yes     | 50         | notify             | --          |
| excessive-crc-errors       | yes     | 35         | notify             | --          |
| excessive-tx-drops         | yes     | 20         | notify             | --          |
| excessive-collisions       | yes     | 40         | notify             | 70          |
| excessive-late-collisions  | yes     | 30         | notify-and-disable | --          |
| excessive-alignment-errors | yes     | 50         | notify-and-disable | 100         |

## show interface fault-monitor profile

### Syntax

```
show interface [<IF-NAME>|<IF-RANGE>] fault-monitor profile
```

### Description

Shows the fault monitoring profile configured for the particular interface, range of interfaces, or all the interfaces.

### Parameters

**<IF-NAME>**

Specifies the interface name.

**<IF-RANGE>**

Specifies the interface range.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Displaying all the interfaces configured with the fault monitoring profiles:

```
switch# show interface fault-monitor profile
```

```

Port Fault Monitor Profile
```

```

1/1/1 noisy-ports
1/1/2 quiet-ports
1/1/4 quiet-ports
1/1/5 noisy-ports
1/1/6 noisy-ports
1/1/7 quiet-ports
```

Displaying the list or range of interfaces configured with the fault monitoring profiles:

```
switch# show interface 1/1/1-1/1/2,1/1/6 fault-monitor profile
```

```

Port Fault Monitor Profile
```

```

1/1/1 noisy-ports
1/1/2 quiet-ports
1/1/6 noisy-ports
```

## show interface fault-monitor status

### Syntax

```
show interface [<IF-NAME>|<IF-RANGE>] fault-monitor status
```

### Description

Shows the active fault monitoring profile configured for the particular interface, range of interfaces, or all the interfaces.

## Parameters

### <IF-NAME>

Specifies the interface name.

### <IF-RANGE>

Specifies the interface range.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Displaying active fault information for all the interfaces configured with the fault monitoring profiles:

```
switch# show interface fault-monitor status
```

| Port  | Fault                      | Fault Elapsed Time           | Port State | Time Left |
|-------|----------------------------|------------------------------|------------|-----------|
| 1/1/1 | excessive-broadcasts       | Tue Apr 14 14:29:09 UTC 2020 | down       | 60        |
|       | excessive-jabbers          | Tue Apr 15 14:29:09 UTC 2020 | --         | --        |
| 1/1/2 | excessive-oversize-packets | Tue Apr 16 14:29:09 UTC 2020 | down       | --        |

Displaying active fault information for the list or range of interfaces configured with the fault monitoring profiles:

```
switch# show interface 1/3/1,1/3/3 fault-monitor status
```

| Port  | Fault                | Occurring Since              | Port State | Time Left |
|-------|----------------------|------------------------------|------------|-----------|
| 1/1/4 | excessive-broadcasts | Tue Apr 14 14:29:09 UTC 2020 | down       | 60        |
|       | excessive-jabbers    | Tue Apr 15 14:29:09 UTC 2020 | --         | 100       |

## show running-config

## Syntax

```
show running-config
```

## Description

Shows the running configuration for the fault monitoring profile.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Displaying running configuration for the fault monitoring profiles:

```
switch# show running-config
fault-monitor profile noisy-ports
 excessive-broadcasts
 excessive-broadcasts threshold pps 10000
 excessive-multicasts
 excessive-multicasts threshold pps 10000
 excessive-link-flaps
 excessive-link-flaps threshold count 7
 excessive-oversize-packets
```



```

excessive-oversize-packets threshold value 40
excessive-jabbers
excessive-jabbers threshold value 30
excessive-fragments
excessive-fragments threshold value 50
excessive-crc-errors
excessive-crc-errors threshold value 35
excessive-tx-drops
excessive-tx-drops threshold value 20
excessive-collisions
excessive-collisions threshold value 40
excessive-late-collisions
excessive-late-collisions threshold value 30
excessive-alignment-errors
excessive-alignment-errors threshold value 50
interface 1/1/1
 apply fault-monitor profile noisy-ports

```

## show running-config all

### Syntax

```
show running-config all
```

### Description

Shows the running configuration with all options for the fault monitoring profile.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Displaying running configuration with options for the fault monitoring profile:

```

switch# show running-config all
fault-monitor profile noisy-ports
 excessive-broadcasts
 excessive-broadcasts threshold pps 10000
 excessive-broadcasts action notify-and-disable auto-enable 2000
 excessive-multicasts
 excessive-multicasts threshold pps 10000
 excessive-multicasts action notify
 excessive-link-flaps
 excessive-link-flaps threshold count 7
 excessive-link-flaps action notify-and-disable auto-enable 2000
 no excessive-oversize-packets
 excessive-oversize-packets threshold value 40
 excessive-oversize-packets action notify
 no excessive-jabbers
 excessive-jabbers threshold value 30
 excessive-jabbers action notify
 no excessive-fragments
 excessive-fragments threshold value 50
 excessive-fragments action notify
 no excessive-crc-errors
 excessive-crc-errors threshold value 35
 excessive-crc-errors action notify
 no excessive-late-collisions
 excessive-late-collisions threshold value 30
 excessive-late-collisions action notify

```

```
no excessive-collisions
excessive-collisions threshold value 40
excessive-collisions action notify
no excessive-tx-drops
excessive-tx-drops threshold value 20
excessive-tx-drops action notify
no excessive-alignment-errors
excessive-alignment-errors threshold value 50
excessive-alignment-errors action notify
```

## copy {primary | secondary} <REMOTE-URL>

### Syntax

```
copy {primary | secondary} <REMOTE-URL> [vrf <VRF-NAME>]
```

### Description

Uploads a firmware image to a TFTP or SFTP server.

### Command context

Manager (#)

### Parameters

**{primary | secondary}**

Selects the primary or secondary image profile to upload. Required

**<REMOTE-URL>**

Specifies the URL to receive the uploaded firmware using SFTP or TFTP. For information on how to format the remote URL, see [URL formatting for copy commands](#).

**vrf <VRF-NAME>**

Specifies a VRF name. Default: default.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

TFTP upload:

```
switch# copy primary tftp://192.0.2.0/00_10_00_0002.swi
100.0%
Verifying and writing system firmware...
```

SFTP upload:

```
switch# copy primary sftp://swuser@192.0.2.0/00_10_00_0002.swi
swuser@192.0.2.0's password:
Connected to 192.0.2.0.
sftp> put primary.swi XL_10_00_0002.swi
Uploading primary.swi to /users/swuser/00_10_00_0002.swi
primary.swi 100% 179MB 35.8MB/s 00:05
```

## copy {primary | secondary} <FIRMWARE-FILENAME>

### Syntax

```
copy {primary | secondary} <FIRMWARE-FILENAME>
```

## Description

Copies a firmware image to USB storage.

## Command context

Manager (#)

## Parameters

{**primary** | **secondary**}

Selects the primary or secondary image from which to copy the firmware. Required

## <FIRMWARE-FILENAME>

Specifies the name of the firmware file to create on the USB storage device. Prefix the filename with `usb:/`. For example: `usb:/firmware_v1.2.3.swi`

For information on how to format the path to a firmware file on a USB drive, see [USB URL](#).

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch# copy primary usb:/11.10.00.0002.swi
```

## copy primary secondary

### Syntax

```
copy primary secondary
```

### Description

Copies the firmware image from the primary to the secondary location.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch# copy primary secondary
The secondary image will be deleted.
```

```
Continue (y/n)? y
Verifying and writing system firmware...
```

## copy <REMOTE-URL>

### Syntax

```
copy <REMOTE-URL> {primary | secondary} [vrf <VRF-NAME>]
```

## Description

Downloads and installs a firmware image from a TFTP or SFTP server.

## Command context

Manager (#)

## Parameters

**<REMOTE-URL>**

Specifies the URL from which to download the firmware using SFTP or TFTP. For information on how to format the remote URL, see [URL formatting for copy commands](#).

**{primary | secondary}**

Selects the primary or secondary image profile for receiving the downloaded firmware. Required.

**vrf <VRF-NAME>**

Specifies the name of a VRF. Default: default.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

TFTP download:

```
switch# copy tftp://192.10.12.0/ss.10.00.0002.swi primary
The primary image will be deleted.

Continue (y/n)? y
100.0%
Verifying and writing system firmware...
```

SFTP download:

```
switch# copy sftp://swuser@192.10.12.0/ss.10.00.0002.swi primary
The primary image will be deleted.

Continue (y/n)? y
The authenticity of host '192.10.12.0 (192.10.12.0)' can't be established.
ECDSA key fingerprint is SHA256:L64khLwlyLgXlARKRMiwcAAK8oRaQ8C0oWP+PkGBXHY.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.10.12.0' (ECDSA) to the list of known hosts.
swuser@192.10.12.0's password:
Connected to 192.10.12.0.
Fetching /users/swuser/ss.10.00.0002.swi to ss.10.00.0002.swi.dnld
/users/swuser/ss.10.00.0002.swi 100% 179MB 25.6MB/s 00:07
Verifying and writing system firmware...
```

## copy secondary primary

### Syntax

copy secondary primary

### Description

Copies the firmware image from the secondary to the primary location.

## Command context

Manager (#)

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch# copy secondary primary
The primary image will be deleted.
```

```
Continue (y/n)? y
Verifying and writing system firmware...
```

```
switch# copy sftp://stor@192.22.1.0/im-switch.swi primary vrf default
The primary image will be deleted.
```

```
Continue (y/n)? y
The authenticity of host '192.22.1.0 (192.22.1.0)' can't be established.
ECDSA key fingerprint is SHA256:MyI1xbdKnehYut0NLfL69gDpNzCmZqBVvBaRR46m7o8.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.22.1.0' (ECDSA) to the list of known hosts.
stor@192.22.1.0's password:
Connected to 192.22.1.0.
sftp> get c8d5b9f-topflite.swi c8d5b9f-topflite.swi.dnld
Fetching /home/dr/im-switch.swi to c8d5b9f-topflite.swi.dnld
/home/dr/im-switch.swi 100% 226MB 56.6MB/s 00:04

Verifying and writing system firmware...
```

## copy <STORAGE-URL>

### Syntax

```
copy <STORAGE-URL> {primary | secondary}
```

### Description

Copies, verifies, and installs a firmware image from a USB storage device connected to the active management module.

## Command context

Manager (#)

## Parameters

<STORAGE-URL>

Specifies the name of the firmware file to copy from the USB storage device. Required. Prefix the filename with `usb:/`. For example, `usb:/firmware_v1.2.3.swi`. For information on how to format the path to a firmware file on a USB drive, see [USB URL](#).

{primary | secondary}

Selects the primary or secondary image profile for receiving the copied firmware.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch# copy usb:/11.10.00.0002.swi primary
The primary image will be deleted.

Continue (y/n)? y

Verifying and writing system firmware...
```

# URL formatting for copy commands

## TFTP URL

### Syntax

tftp://<IP-ADDR>[:<PORT-NUM>][;blocksize=<Value>]/<FILENAME>

### Examples

To specify a URL with:

- an IPv4 address: tftp://1.1.1.1/a.txt
- an IPv6 address: tftp://[2000::2]/a.txt
- a hostname: tftp://hpe.com/a.txt

To specify TFTP with:

- the port number of the server in the URL: tftp://1.1.1.1:12/a.txt
- the blocksize in the URL: tftp://1.1.1.1;blocksize=1462/a.txt  
The valid blocksize range is 8 to 65464.
- the port number of the server and blocksize in the URL: tftp://1.1.1.1:12;blocksize=1462/a.txt

To specify a file in a directory of URL: tftp://1.1.1.1/dir/a.txt

## SFTP URL

### Syntax

sftp://<USERNAME>@<IP-ADDR>[:<PORT-NUM>]/<FILENAME>

### Examples

To specify:

- A URL with an IPv4 address: sftp://user@1.1.1.1/a.txt
- A URL with an IPv6 address: sftp://user@[2000::2]/a.txt
- A URL with a hostname: sftp://user@hpe.com/a.txt
- SFTP port number of a server in the URL: sftp://user@1.1.1.1:12/a.txt

- A file in a directory of URL: `sftp://user@1.1.1.1/dir/a.txt`
- To specify a file with absolute path in the URL: `sftp://user@1.1.1.1/home/user/a.txt`

## USB URL

### Syntax

`usb: /<FILENAME>`

### Examples

To specify a file:

- In a USB storage device: `usb: /a.txt`
- In a directory of a USB storage device: `usb: /dir/a.txt`



## redundancy switchover

### Syntax

```
redundancy switchover
```

### Description

Causes the switch to immediately switch over to the Standby Management Module. This command must be executed from the Active Management Module and will fail if the Standby Management Module is in a failed state or not present.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

This example shows the redundancy switchover command on an active management module with a standby management module that is present.

```
switch#redundancy switchover
This command causes the switch to immediately switchover to the Standby Management Module.
Do you want to continue [y/n]?
```

This example shows the redundancy switchover command on an active management module with a standby management module that is absent.

```
switch#redundancy switchover
Standby Management Module not found, switchover request ignored.
```

This example shows the redundancy switchover command on a standby management module.

```
switch#redundancy switchover
Redundancy switchover must be performed from the Active Management Module, switchover request ignored.
```

## https-server rest access-mode

### Syntax

```
https-server rest access-mode {read-only | read-write}
```

### Description

Changes the REST API access mode. The default mode is `read-write`.

### Command context

`config`

### Parameters

#### **read-write**

Selects the read/write mode. Allows POST, PUT, and DELETE methods to be called on all configurable elements in the switch database.

#### **read-only**

Selects the read-only mode. Write access to most switch resources through the REST API is disabled.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

Setting the mode to `read-write` on the REST API allows POST, PUT, and DELETE methods to be called on all configurable elements in the switch database.

By default, REST APIs in the device are in the read-write mode. Some switch resources allow POST, PUT, and DELETE regardless of REST API mode. REST APIs that are required to support the Web UI or the Network Analytics Engine expose POST, PUT, or DELETE operations, even if the REST API access mode is set to read-only.

The REST API in read/write mode is intended for use by advanced programmers who have a good understanding of the system schema and data relationships in the switch database.



**CAUTION:** Because the REST API in read/write mode can access every configurable element in the database, it is powerful but must be used with extreme caution: No semantic validation is performed on the data you write to the database, and configuration errors can destabilize the switch.

### Example

```
switch(config)# https-server rest access-mode read-only
```

## https-server session close all

### Syntax

```
https-server session close all
```

### Description

Invalidates and closes all HTTPS sessions. All existing Web UI and REST sessions are logged out and all real-time notification feature WebSocket connections are closed.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

Typically, a user that has consumed the allowed concurrent HTTPS sessions and is unable to access the session cookie to log out manually must wait for the session idle timeout to start another session. This command is intended as a workaround to waiting for the idle timeout to close an HTTPS session. This command stops and starts the `hpe-restd` service, so using this command affects all existing REST sessions, Web UI sessions, and real-time notification subscriptions.

### Example

```
switch# https-server session close all
```

## https-server vrf



**NOTE:** Since the Aruba 6100 Switch Series only supports the `default` VRF, this command does not apply to the 6100.

### Syntax

```
https-server vrf <VRF-NAME>
```

```
no https-server vrf <VRF-NAME>
```

### Description

Configures and starts the HTTPS server on the specified VRF. HTTPS server features include the REST API and the web user interfaces.

The `no` form of the command stops any HTTPS servers running on the specified VRF and removes the HTTPS server configuration.

### Command context

config

### Parameters

**<VRF-NAME>**

Specifies the VRF name. Required. Length: Up to 32 alpha numeric characters.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

By using this command, you enable access to both the Web UI and to the REST API on the specified VRF. You can enable access on multiple VRFs.

When the HTTPS server is not configured and running, attempts to access the Web UI or REST API result in 404 Not Found errors.

The VRF you select determines from which network the Web UI and REST API can be accessed.

For example:

- If you want to enable access to the REST API and Web UI through the OOBM port (management IP address), specify the built-in management VRF (`mgmt`).
- If you want to enable access to the REST API and Web UI through the data ports (for "inband management"), specify the built-in default VRF (`default`).
- If you want to enable access to the REST API and Web UI through only a subset of data ports on the switch, specify other VRFs you have created.

Aruba Network Analytics Engine scripts run in the default VRF, but you do not have to enable HTTPS server access on the default VRF for the scripts to run. If the switch has custom Aruba Network Analytics Engine scripts that require access to the Internet, then for those scripts to perform their functions, you must configure a DNS name server on the default VRF.

## Examples

- To enable access on all ports on the switch, specify the default VRF:

```
switch(config)# https-server vrf default
```

- To enable access on the OOBM port (management interface IP address), specify the management VRF:

```
switch(config)# https-server vrf mgmt
```

- To enable access on ports that are members of the VRF named `vrfprogs`, specify `vrfprogs`:

```
switch(config)# https-server vrf vrfprogs
```

- To enable access on the management port and ports that are members of the VRF named `vrfprogs`, enter two commands:

```
switch(config)# https-server vrf mgmt
switch(config)# https-server vrf vrfprogs
```

## show https-server

### Syntax

```
show https-server [vsx-peer]
```

### Description

Shows the status and configuration of the HTTPS server. The REST API and web user interface are accessible only on VRFs that have the HTTPS server features configured.

## Command context

Manager (#)

## Authority

Administrators or local user group members with execution rights for this command.

## Parameters

### [vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Usage

Shows the configuration of the HTTPS server features.

## VRF

Shows the VRFs, if any, for which HTTPS server features are configured.

## REST Access Mode

Shows the configuration of the REST access mode:

### **read-write**

POST, PUT, and DELETE methods can be called on all configurable elements in the switch database. This is the default value.

### **read-only**

Write access to most switch resources through the REST API is disabled.

## Examples

```
switch# show https-server

HTTPS Server Configuration

VRF : default, mgmt
REST Access Mode : read-write

Max sessions per user : 6
Session timeout : 20
```

## ip icmp redirect

### Syntax

```
ip icmp redirect
no ip icmp redirect
```

### Description

Enables the sending of ICMPv4 and ICMPv6 redirect messages to the source host. Enabled by default. The `no` form of this command disables ICMPv4 and ICMPv6 redirect messages to the source host.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling ICMP redirect messages:

```
switch(config)# ip icmp redirect
```

Disabling ICMP redirect messages:

```
switch(config)# no ip icmp redirect
```

## ip icmp throttle

### Syntax

```
ip icmp throttle <packet-interval>
no ip icmp throttle
```

### Description

Used to configure the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages.

The `no` form of this command disables the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages.

### Command context

```
config
```

## Parameters

**<packet-interval>**

Specifies the ICMPv4/v6 packet interval in seconds. Default: 1 second. Range: 1-86400.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages:

```
switch(config)# ip icmp throttle 3000
```

Disabling the throttle parameter for both ICMPv4 and ICMPv6 error messages and redirect messages:

```
switch(config)# no ip icmp throttle
```

# ip icmp unreachable

## Syntax

```
ip icmp unreachable
no ip icmp unreachable
```

## Description

Enables the sending of ICMPv4 and ICMPv6 destination unreachable messages on the switch to a source host when a specific host is unreachable. The unreachable host address originates from the failed packet. Default setting.

The **no** form of this command disables the sending of ICMPv4 and ICMPv6 destination unreachable messages from the switch to a source host when a specific host is unreachable. This command does not prevent other hosts from sending an ICMP unreachable message.

## Command context

```
config
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling ICMPv4 and ICMPv6 destination unreachable messages to a source host:

```
switch(config)# ip icmp unreachable
```

Disabling ICMPv4 and ICMPv6 destination unreachable messages to a source host:

```
switch(config)# no ip icmp unreachable
```

For commands in the interface configuration context, the interface must be an L3 interface. The supported contexts includes: `config-if-vlan`.

## ip igmp

### Syntax

```
ip igmp {enable | disable}
```

### Description

Enables or disables IGMP on the current interface. IGMP is disabled by default.

### Command context

`config-if-vlan`

### Parameters

#### **enable**

Enable IGMP.

#### **disable**

Disable IGMP.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling IGMP on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp enable
```

Disabling IGMP on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp disable
```

## ip igmp apply access-list

### Syntax

```
ip igmp apply access-list <ACL-NAME>
```

```
no ip igmp apply access-list <ACL-NAME>
```



## Description

Configures the ACL on a particular interface to filter the IGMP join or leave packets based on rules set in the particular ACL name.

The `no` form of this command unconfigures the rules set for the ACL.



**NOTE:** This configuration will override the ACL associated with IGMP snooping on the corresponding L2 VLAN.

## Command context

`config-if-vlan`

## Parameters

### `access-list`

Associates an ACL with the IGMP.

### `<ACL-NAME>`

Specifies the name of the ACL.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

Existing classifier commands are used to configure the ACL. In case an IGMPv3 packet with multiple group addresses is received, it will only process the permitted group addresses based on the ACL rule set, and any existing joins will time out. If there is no match or if there is a deny rule match, the packet is dropped.

## Examples

Configuring the ACL on a VLAN to filter IGMP packets based on rules set in access list `mygroup`:

```
switch(config)# access-list ip mygroup
switch(config-acl-ip)# permit igmp any 239.1.1.1
switch(config-acl-ip)# exit
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp apply access-list mygroup
```

Configuring the ACL to remove the rules set in access list `mygroup`:

```
switch(config-if-vlan)# no ip igmp apply access-list mygroup
```

## `ip igmp last-member-query-interval`

### Syntax

```
ip igmp last-member-query-interval <INTERVAL-VALUE>
```

```
no ip igmp last-member-query-interval <INTERVAL-VALUE>
```

### Description

Configures an IGMP last member query interval value in seconds on an interface, depending on the command context you are in.

The `no` form of this command sets the value to a default of 1 second on an interface.

## Command context

config-if-vlan

## Parameters

<INTERVAL-VALUE>

Specifies an IGMP last-member-query-interval on the interface. Default: 1 second. Range: 1-2 seconds.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring an IGMP last member query interval of 2 on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp last-member-query-interval 2
switch(config-if-vlan)# no ip igmp last-member-query-interval
```

# ip igmp querier

## Syntax

```
ip igmp querier
no ip igmp querier
```

## Description

Configures an IGMP querier on an interface, depending on the command context you are in. This functionality will allow an interface to join in the querier-election process.

The `no` form of this command disables IGMP querier on an interface.

## Command context

config-if-vlan

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring an IGMP querier on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp querier
switch(config-if-vlan)# no ip igmp querier
```

# ip igmp querier interval

## Syntax

```
ip igmp querier interval <INTERVAL-VALUE>
no ip igmp querier interval
```

## Description

Configures the interval between IGMP queries on an interface, depending on the command context you are in.

The `no` form of this command sets the IGMP querier interval to the default value of 125 seconds on an interface.

## Command context

config-if-vlan

## Parameters

**<INTERVAL-VALUE>**

Specifies the IGMP querier interval in seconds on the interface. Default: 125 seconds. Range: 5-300.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring an IGMP querier interface interval of 100 on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp querier interval 100
```

Resetting an IGMP querier interval to the default value:

```
switch(config-if-vlan)# no ip igmp querier interval
```

# ip igmp querier query-max-response-time

## Syntax

```
ip igmp querier query-max-response-time <RESPONSE-TIME>
no ip igmp querier query-max-response-time <RESPONSE-TIME>
```

## Description

Configures the IGMP querier max response time value in seconds on an interface, depending on the command context you are in.

The `no` form of this command sets the querier max response time value to the default of 10 seconds on an interface.

## Command context

config-if-vlan

## Parameters

**<RESPONSE-TIME>**

Specifies the IGMP querier max response time value on the interface. Default: 10 seconds. Range: 10-128 seconds.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring the IGMP querier maximum response time of 50 for interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp query-max-response-time 50
```

Resetting an IGMP querier interval to the default value:

```
switch(config-if-vlan)# no ip igmp query-max-response-time
```

## ip igmp robustness

### Syntax

```
ip igmp robustness <VALUE>
no ip igmp robustness <VALUE>
```

### Description

Configures IGMP robustness on an interface, depending on the command context. The robustness parameter allows tuning for the expected packet loss on a subnet.

The **no** form of this command sets the robustness value to the default of 2 on an interface.

### Command context

config-if-vlan

### Parameters

<VALUE>

Specifies an IGMP robustness value on the interface. Default: 2. Range: 1-7.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring an IGMP robustness of 5 on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp robustness 5
```

Resetting the IGMP robustness to the default:

```
switch(config-if-vlan)# no ip igmp robustness
```

## ip igmp static-group

### Syntax

```
ip igmp static-group <MULTICAST-GROUP-IP>
no ip igmp static-group <MULTICAST-GROUP-IP>
```

## Description

Configures an IGMP static multicast group on an interface, depending on the command context you are in. You can configure a maximum of 32 IGMP static groups.

The `no` form of the command unconfigures IGMP static multicast group on an interface.

## Command context

config-if-vlan

## Parameters

**<MULTICAST-GROUP-IP>**

Specifies an IGMP static multicast group IP address on the interface. Format: A.B.C.D

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring an IGMP static group on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp static-group 239.1.1.1
```

Resetting an IGMP static group on an interface to the default (none):

```
switch(config-if)# no ip igmp static-group 239.1.1.10
```

# ip igmp version

## Syntax

ip igmp version <VERSION>

## Description

Configures the IGMP version on an interface, depending on the command context you are in.

## Command context

config-if-vlan

## Parameters

**<VERSION>**

Specifies the IGMP version on the interface. Select 2 for IGMPv2 (RFC2236). Select 3 for IGMPv3 (RFC3376). Values: 2 or 3.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring an IGMP version on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp version 2
```

## ip igmp version strict

### Syntax

```
ip igmp version <VERSION> strict
```

```
no ip igmp version <VERSION> strict
```

### Description

Configures an IGMP strict version on an interface, depending on the command context you are in. Drops packets that do not match the configured version.

The `no` form of the command removes the strict version configuration from the interface.

### Command context

```
config-if-vlan
```

### Parameters

**<VERSION>**

Specifies the IGMP version on the interface. Select 2 for IGMPv2 (RFC2236). Select 3 for IGMPv3 (RFC3376). Values: 2 or 3.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring the IGMP strict version to 2 on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ip igmp version 2 strict
```

Resetting the IGMP strict version to the default (none):

```
switch(config-if)# no ip igmp version 2 strict
```

## no ip igmp

### Syntax

```
no ip igmp
```

### Description

Disables all IGMP configurations on an interface, depending on the command context you are in.

### Command context

```
config-if-vlan
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Removes IGMP on interface VLAN 2:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ip igmp
```

## show ip igmp

### Syntax

```
show ip igmp [all-vrfs]
```

### Description

Shows IGMP configuration information and status, or shows information by VRF.

### Command context

Operator (>) or Manager (#)

### Parameters

#### all-vrfs

To show information for all VRFs, specify `all-vrfs`.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing IGMP configuration and status:

```
switch# show ip igmp

VRF Name : default
Interface : vlan2
IGMP Configured Version : 3
IGMP Operating Version : 3
Querier State : Querier
Querier IP [this switch] : 20.1.1.1
Querier Uptime : 1m 4s
Querier Expiration Time : 0m 1s
IGMP Snoop Enabled on VLAN : True
```

Showing IGMP information for all VRFs:

```
switch# show ip igmp all-vrfs

VRF Name : default
Interface : vlan5
IGMP Configured Version : 3
IGMP Operating Version : 2
Querier State : Querier
Querier IP [this switch] : 50.1.1.1
Querier Uptime : 1m 1s
Querier Expiration Time : 0m 4s
IGMP Snoop Enabled on VLAN : False
```

# show ip igmp counters

## Syntax

```
show ip igmp counters [all-vrfs]
```

## Description

Shows IGMP counter details, or shows counters by VRF.

## Command context

Operator (>) or Manager (#)

## Parameters

### all-vrfs

Specify `all-vrfs` to show information for all VRFs.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing IGMP counters:

```
switch# show ip igmp counters
```

IGMP Counters

Interface Name : vlan2  
VRF Name : default  
Membership Timeout : 0

|                                   | Rx    | Tx    |
|-----------------------------------|-------|-------|
|                                   | ----- | ----- |
| V1 All Hosts Queries              | 0     | 0     |
| V2 All Hosts Queries              | 0     | 12    |
| V3 All Hosts Queries              | 0     | 0     |
| V2 Group Specific Queries         | 0     | 0     |
| V3 Group Specific Queries         | 0     | 0     |
| Group And Source Specific Queries | 0     | 0     |
| V3 Member Reports                 | 0     | N/A   |
| V2 Member Reports                 | 0     | N/A   |
| V1 Member Reports                 | 0     | N/A   |
| V2 Member Leaves                  | 0     | N/A   |
| Packets dropped by ACL            | 0     | N/A   |

Showing IGMP counters for the default VRF:

```
switch# show ip igmp counters vrf default
```

IGMP Counters

Interface Name : vlan2  
VRF Name : default  
Membership Timeout : 0

|                      | Rx    | Tx    |
|----------------------|-------|-------|
|                      | ----- | ----- |
| V1 All Hosts Queries | 0     | 0     |
| V2 All Hosts Queries | 0     | 12    |



|                                   |   |     |
|-----------------------------------|---|-----|
| V3 All Hosts Queries              | 0 | 0   |
| V2 Group Specific Queries         | 0 | 0   |
| V3 Group Specific Queries         | 0 | 0   |
| Group And Source Specific Queries | 0 | 0   |
| V3 Member Reports                 | 0 | N/A |
| V2 Member Reports                 | 0 | N/A |
| V1 Member Reports                 | 0 | N/A |
| V2 Member Leaves                  | 0 | N/A |
| Packets dropped by ACL            | 0 | N/A |

## show ip igmp group

### Syntax

```
show ip igmp group <GROUP-IP> [source <SOURCE-IP>] [all-vrfs]
```

### Description

Shows IGMP joined group information for the specified group, or shows joined group source and display information by VRF.

### Command context

Operator (>) or Manager (#)

### Parameters

**<GROUP-IP>**

Specifies the IP address of the group. Format: A.B.C.D

**source <SOURCE-IP>**

Specifies the IP address of the source. Format: A.B.C.D

**all-vrfs**

Specify **all-vrfs** to show information for all VRFs.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing IGMP joined group details for group 239.1.1.10:

```
switch# show ip igmp group 239.1.1.10
```

IGMP group information for group 239.1.1.10

```
Interface Name : vlan2
VRF Name : default
```

```
Group Address : 239.1.1.10
Last Reporter : 100.1.1.10
```

| Vers | Mode | Uptime  | Expires | V1<br>Timer | V2<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|---------|---------|-------------|-------------|----------------------|--------------------|
| 3    | EXC  | 16m 34s | 2m 27s  |             |             |                      |                    |

Showing IGMP joined group details for group 239.1.1.10 and source 10.1.1.10:

```
switch# show ip igmp group 239.1.1.10 source 10.1.1.10
```

```
Interface Name : vlan2
VRF Name : default
Group Address : 239.1.1.10
Source Address : 10.1.1.10
```

| Mode | Uptime | Expire |
|------|--------|--------|
| ---- | -----  | -----  |
|      | 0m 13s | 4m 7s  |

Showing IGMP joined group details for group 239.1.1.10 for all VRFs:

```
switch# show ip igmp group 239.1.1.10 all-vrfs
```

IGMP group information for group 239.1.1.10

```
Interface Name : vlan10
VRF Name : default

Group Address : 239.1.1.10
Last Reporter : 100.1.1.10
```

| Vers | Mode | Uptime | Expires | V1<br>Timer | V2<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|-------------|----------------------|--------------------|
| ---- | ---- | -----  | -----   | -----       | -----       | -----                | -----              |
| 3    | EXC  | 17m 5s | 4m 2s   |             |             |                      |                    |

Showing IGMP joined group details for group 239.1.1.10 source 10.1.1.10 for all VRFs:

```
switch# show ip igmp group 239.1.1.10 source 10.1.1.10 all-vrfs
```

```
Interface Name : vlan10
VRF Name : default
Group Address : 239.1.1.10
Source Address : 10.1.1.10
```

| Mode | Uptime | Expire |
|------|--------|--------|
| ---- | -----  | -----  |
|      | 0m 39s | 3m 41s |

Showing IGMP joined group details group 239.1.1.10 for the default VRF:

```
switch# show ip igmp group 239.1.1.10 vrf default
```

IGMP group information for group 239.1.1.10

```
Interface Name : vlan2
VRF Name : default

Group Address : 239.1.1.10
Last Reporter : 100.1.1.10
```

| Vers | Mode | Uptime  | Expires | V1<br>Timer | V2<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|---------|---------|-------------|-------------|----------------------|--------------------|
| ---- | ---- | -----   | -----   | -----       | -----       | -----                | -----              |
| 3    | EXC  | 17m 35s | 3m 32s  |             |             |                      |                    |

Showing IGMP joined group details group 239.1.1.10 source 10.1.1.10 for the default VRF:

```
switch# show ip igmp group 239.1.1.10 source 10.1.1.10 vrf default
```

```
Interface Name : vlan10
VRF Name : default
Group Address : 239.1.1.10
Source Address : 10.1.1.10
```

| Mode | Uptime | Expire |
|------|--------|--------|
| ---- | -----  | -----  |
|      | 0m 59s | 3m 21s |

## show ip igmp groups

### Syntax

```
show ip igmp groups [all-vrfs]
```

### Description

Shows IGMP group information, or you can display group information by VRF.

### Command context

Operator (>) or Manager (#)

### Parameters

#### all-vrfs

Specify `all-vrfs` to show information for all VRFs.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing IGMP group information:

```
switch# show ip igmp groups
```

IGMP group information for group 239.1.1.10

```
Interface Name : vlan2
VRF Name : default
```

```
Group Address : 239.1.1.10
Last Reporter : 100.1.1.10
```

| Vers | Mode | Uptime | Expires | V1<br>Timer | V2<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|-------------|----------------------|--------------------|
| ---- | ---- | -----  | -----   | -----       | -----       | -----                | -----              |
| 3    | EXC  | 0m 36s | 3m 44s  |             |             |                      |                    |

IGMP group information for group 239.1.1.11

```
Interface Name : vlan2
VRF Name : default
```

```
Group Address : 239.1.1.11
Last Reporter : 100.1.1.10
```

| Vers | Mode | Uptime | Expires | V1<br>Timer | V2<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|-------------|----------------------|--------------------|
| ---- | ---- | -----  | -----   | -----       | -----       | -----                | -----              |

```

3 EXC 0m 36s 3m 44s
```

Showing IGMP groups for all VRFs:

```
switch# show ip igmp groups all-vrfs
```

IGMP group information for group 239.1.1.1

Interface Name : vlan20  
VRF Name : default

Group Address : 239.1.1.1  
Last Reporter : 200.1.1.10

| Vers | Mode | Uptime | Expires | V1<br>Timer | V2<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|-------------|----------------------|--------------------|
| 3    | EXC  | 0m 13s | 4m 7s   |             |             |                      |                    |

IGMP group information for group 239.1.1.2

Interface Name : vlan20  
VRF Name : default

Group Address : 239.1.1.2  
Last Reporter : 200.1.1.10

| Vers | Mode | Uptime | Expires | V1<br>Timer | V2<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|-------------|----------------------|--------------------|
| 3    | EXC  | 0m 13s | 4m 7s   |             |             |                      |                    |

Showing IGMP groups for the default VRF:

```
switch# show ip igmp groups vrf default
```

IGMP group information for group 239.1.1.10

Interface Name : vlan2  
VRF Name : default

Group Address : 239.1.1.10  
Last Reporter : 100.1.1.10

| Vers | Mode | Uptime | Expires | V1<br>Timer | V2<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|-------------|----------------------|--------------------|
| 3    | EXC  | 9m 23s | 3m 20s  |             |             |                      |                    |

IGMP group information for group 239.1.1.11

Interface Name : vlan2  
VRF Name : default

Group Address : 239.1.1.11  
Last Reporter : 100.1.1.10

| Vers | Mode | Uptime | Expires | V1<br>Timer | V2<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|-------------|----------------------|--------------------|
| 3    | EXC  | 9m 23s | 3m 20s  |             |             |                      |                    |

## show ip igmp interface

### Syntax

```
show ip igmp interface {vlan <VLAN-ID>}
```

### Description

Shows IGMP configuration information for a specific interface (VLAN).

### Command context

Operator (>) or Manager (#)

### Parameters

**vlan** <VLAN-ID>

Specifies a VLAN. Values: 1-4094.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing IGMP configuration information for interface VLAN 2:

```
switch# show ip igmp interface vlan 2
```

```
IGMP Configured Version : 3
IGMP Operating Version : 3
Querier State : Querier
Querier IP [this switch] : 20.1.1.1
Querier Uptime : 1m 46s
Querier Expiration Time : 0m 1s
Snoop Enabled on VLAN : True
```

## show ip igmp interface counters

### Syntax

```
show ip igmp interface {vlan <VLAN-ID>} counters
```

### Description

Shows IGMP counter details for a specific interface or VLAN interface.

### Command context

Operator (>) or Manager (#)

### Parameters

**vlan** <VLAN-ID>

Specifies a VLAN. Values: 1-4094.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing IGMP counters for interface VLAN 2:

```
switch# show ip igmp interface vlan 2 counters
```

IGMP Counters

Interface Name : vlan2  
VRF Name : default  
Membership Timeout : 0

|                                   | Rx    | Tx    |
|-----------------------------------|-------|-------|
|                                   | ----- | ----- |
| V1 All Hosts Queries              | 0     | 0     |
| V2 All Hosts Queries              | 0     | 0     |
| V3 All Hosts Queries              | 0     | 29    |
| V2 Group Specific Queries         | 0     | 0     |
| V3 Group Specific Queries         | 0     | 2     |
| Group And Source Specific Queries | 0     | 2     |
| V3 Member Reports                 | 0     | N/A   |
| V2 Member Reports                 | 0     | N/A   |
| V1 Member Reports                 | 0     | N/A   |
| V2 Member Leaves                  | 0     | N/A   |
| Packets dropped by ACL            | 0     | N/A   |

## show ip igmp interface group

### Syntax

```
show ip igmp interface {vlan <VLAN-ID>} group <GROUP-ID> [source <SOURCE-IP>]
```

### Description

Shows IGMP joined group information for a specific interface or VLAN interface, or specify a source IP.

### Command context

Operator (>) or Manager (#)

### Parameters

**vlan <VLAN-ID>**

Specifies a VLAN. Values: 1-4094.

**<GROUP-ID>**

Specifies the IP address of the group. Format: A.B.C.D

**source <SOURCE-IP>**

Specifies the IP address of the source. Format: A.B.C.D

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing IGMP joined group details for group 239.1.1.1 for interface VLAN 10:

```
switch# show ip igmp interface vlan 10 group 239.1.1.1
```

IGMP group information for group 239.1.1.1

Interface Name : vlan10  
VRF Name : default

Group Address : 239.1.1.1  
Last Reporter : 100.1.1.10

| Vers | Mode | Uptime | Expires | V1<br>Timer | V2<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|-------------|----------------------|--------------------|
| 3    | INC  | 8m 10s | 2m 21s  |             |             | 1                    |                    |

Group Address : 239.1.1.1  
Source Address : 10.1.1.1

| Mode | Uptime | Expire |
|------|--------|--------|
| INC  | 8m 10s | 2m 21s |

Showing IGMP joined group details for group 239.1.1.1 for interface VLAN 10 with source details for 10.1.1.1:

```
switch# show ip igmp interface vlan 10 group 239.1.1.1 source 10.1.1.1
```

Interface Name : vlan10  
VRF Name : default  
Group Address : 239.1.1.1  
Source Address : 10.1.1.1

| Mode | Uptime | Expire |
|------|--------|--------|
| INC  | 8m 52s | 3m 51s |

## show ip igmp interface groups

### Syntax

```
show ip igmp interface {vlan <VLAN-ID>} groups
```

### Description

Shows IGMP group information for a specific interface or VLAN interface.

### Command context

Operator (>) or Manager (#)

### Parameters

**vlan <VLAN-ID>**

Specifies a VLAN. Values: 1-4094.

### Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing IGMP groups for interface VLAN 2:

```
switch# show ip igmp interface vlan 2 groups
```

IGMP group information for group 239.1.1.1

Interface Name : vlan2  
VRF Name : default  
  
Group Address : 239.1.1.1  
Last Reporter : 100.1.1.10

| Vers | Mode | Uptime | Expires | V1<br>Timer | V2<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|-------------|----------------------|--------------------|
| 3    | INC  | 4m 40s | 3m 51s  |             |             | 1                    |                    |

Group Address : 239.1.1.1  
Source Address : 10.1.1.1

| Mode | Uptime | Expire |
|------|--------|--------|
| INC  | 4m 40s | 3m 51s |

IGMP group information for group 239.1.1.2

Interface Name : vlan2  
VRF Name : default  
  
Group Address : 239.1.1.2  
Last Reporter : 100.1.1.10

| Vers | Mode | Uptime | Expires | V1<br>Timer | V2<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|-------------|----------------------|--------------------|
| 3    | INC  | 4m 40s | 3m 51s  |             |             | 1                    |                    |

Group Address : 239.1.1.2  
Source Address : 10.1.1.1

| Mode | Uptime | Expire |
|------|--------|--------|
| INC  | 4m 40s | 3m 51s |

## show ip igmp interface statistics

### Syntax

```
show ip igmp interface {vlan <VLAN-ID>} statistics
```

### Description

Shows IGMP statistics for a specific interface or VLAN interface, including groups joined.

### Command context

Operator (>) or Manager (#)



## Parameters

**vlan** <VLAN-ID>

Specifies a VLAN. Values: 1-4094.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing IGMP statistics for interface VLAN 2:

```
switch# show ip igmp interface vlan 2 statistics

IGMP statistics

Interface Name : vlan2
VRF Name : default

Number of Include Groups : 2
Number of Exclude Groups : 0
Number of Static Groups : 0
Total Multicast Groups Joined : 2
```

# show ip igmp static-groups

## Syntax

```
show ip igmp static-groups [all-vrfs]
```

## Description

Shows IGMP static groups, or shows information by VRF.

## Command context

Operator (>) or Manager (#)

## Parameters

**all-vrfs**

Specify `all-vrfs` to show information for all VRFs.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing IGMP static-group information:

```
switch# show ip igmp static-groups

IGMP Static Group Address Information

VRF Name default
Interface Name Group Address
```

```

vlan10 238.1.1.1
```

Showing IGMP statics-group information for all VRFs:

```
switch# show ip igmp static-groups all-vrfs
```

IGMP Static Group Address Information

```
VRF Name :default
Interface Name Group Address

```

```
vlan10 238.1.1.1
```

## show ip igmp statistics

### Syntax

```
show ip igmp statistics [all-vrfs]
```

### Description

Shows IGMP statistics, including groups joined, or shows statistics by VRF.

### Command context

Operator (>) or Manager (#)

### Parameters

#### all-vrfs

Specify `all-vrfs` to show information for all VRFs.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing IGMP statistics:

```
switch# show ip igmp statistics
IGMP statistics
```

```
VRF Name : default
```

```
Number of Include Groups : 1
Number of Exclude Groups : 0
Number of Static Groups : 0
Total Multicast Groups Joined : 1
```

Showing IGMP statistics for all VRFs:

```
switch# show ip igmp statistics all-vrfs
IGMP statistics
```

```
VRF Name : default
```

```
Number of Include Groups : 1
Number of Exclude Groups : 0
```

```
Number of Static Groups : 0
Total Multicast Groups Joined : 1
```

## ip igmp snooping {enable|disable}

### Syntax

```
ip igmp snooping {enable | disable}
```

### Description

Enables or disables IGMP snooping on the VLAN. By default, IGMP snooping is disabled.

### Command context

```
config-vlan
```

### Parameters

**{enable | disable}**

Specifies enabling or disabling IGMP snooping on the VLAN. Default: disable.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enable IGMP snooping on a VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ip igmp snooping enable
```

Disable IGMP snooping on a VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ip igmp snooping disable
```

## ip igmp snooping apply access-list

### Syntax

```
ip igmp snooping apply access-list <ACL-NAME>
```

```
no ip igmp snooping apply access-list <ACL-NAME>
```

### Description

Configures the ACL on a particular interface to filter the IGMP join or leave packets based on rules set in the particular ACL name.

The **no** form of this command unconfigures the rules set for the ACL.



---

**NOTE:** This configuration will override the ACL associated with IGMP snooping on the corresponding L2 VLAN.

---

## Command context

config-vlan

## Parameters

### access-list

Associates an ACL with the IGMP.

<ACL-NAME>

Specifies the name of the ACL.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

Existing classifier commands are used to configure the ACL. In case an IGMPv3 packet with multiple group addresses is received, it will only process the permitted group addresses based on the ACL rule set, and any existing joins will time out. If there is no match or if there is a deny rule match, the packet is dropped.



**NOTE:** If the access list is configured for both L2 VLAN and L3 VLAN, the L3 VLAN configuration will be applied.

---

## Examples

Configuring the ACL to filter IGMP packets based on rules set in access list `mygroup`:

```
switch(config)# access-list ip mygroup
switch(config-acl-ip)# permit igmp any 239.1.1.1
switch(config-acl-ip)# exit
switch(config)# interface vlan 2
switch(config-vlan)# ip igmp snooping apply access-list mygroup
```

Configuring the ACL to remove the rules set in access list `mygroup`:

```
switch(config-vlan)# no ip igmp snooping apply access-list mygroup
```

# ip igmp snooping auto

## Syntax

ip igmp snooping auto <PORT-LIST>

no ip igmp snooping auto <PORT-LIST>

## Description

Configures the specified ports in auto mode. In auto mode traffic flow is controlled by the IGMP joins/leaves. Auto mode is the default.

The `no` form of this command removes auto mode ports for the VLAN.

## Command context

config-vlan

## Parameters

**auto** <PORT-LIST>

Specifies a list of ports to be configured as auto ports. You can specify a single port, a comma-separated list of ports, or a range of ports such as 1/1/1-1/1/3.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Configure auto ports for the VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ip igmp snooping auto 1/1/1
switch(config-vlan)# ip igmp snooping auto 1/1/1,1/1/2-1/1/3
```

# ip igmp snooping blocked

## Syntax

ip igmp snooping blocked <PORT-LIST>

no ip igmp snooping blocked <PORT-LIST>

## Description

Configures the specified ports in blocked mode. In blocked mode, joins and traffic are always blocked on this port.

The **no** form of this command disables blocked ports.

## Command context

config-vlan

## Parameters

**blocked** <PORT-LIST>

Specifies a list of ports to be configured in blocked mode. You can specify a single port, a comma-separated list of ports or a range of ports such as 1/1/1-1/1/3.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring blocked ports for the VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ip igmp snooping blocked 1/1/1
switch(config-vlan)# ip igmp snooping blocked 1/1/1-1/1/2
switch(config-vlan)# ip igmp snooping blocked 1/1/3,1/1/4-1/1/6
```

```
switch(config-vlan)# no ip igmp snooping blocked 1/1/1
switch(config-vlan)# no ip igmp snooping blocked 1/1/1-1/1/2
```

## ip igmp snooping drop-unknown

### Syntax

```
ip igmp snooping drop-unknown {vlan-shared | vlan-exclusive}
```

```
no ip igmp snooping drop-unknown
```

### Description

Configures drop-unknown mode. While IGMP snooping is enabled, the traffic will be forwarded only to ports that made an IGMP request for the multicast. Drop unknown filters ensure that packets are not forwarded to ports that did not make a request for the traffic stream. This could either be a filter across all VLANs (vlan-shared) or per VLAN (vlan-exclusive). The default is vlan-shared.

The `no` form of this command disables drop unknown on the switch.

### Command context

```
config
```

### Parameters

#### **vlan-shared**

Enables shared VLAN filter on the switch. Default: `vlan-shared`.

#### **vlan-exclusive**

Enables exclusive drop unknown filter per VLAN.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring shared VLAN filter on the switch:

```
switch(config)# ip igmp snooping drop-unknown vlan-shared
```

Configuring exclusive drop unknown filter per VLAN:

```
switch(config)# ip igmp snooping drop-unknown vlan-exclusive
```

Disabling drop unknown on the switch:

```
switch(config)# no ip igmp snooping drop-unknown
```

## ip igmp snooping fastlearn

### Syntax

```
ip igmp snooping fastlearn <PORT-LIST>
```

```
no ip igmp snooping fastlearn <PORT-LIST>
```

## Description

Enables the port to learn group information when receiving a topology change notification. By default, fast learn is not enabled on ports.

The `no` form of this command disables fast learn on the specified ports.

## Command context

`config`

## Parameters

**fastlearn** *<PORT-LIST>*

Specifies a list of one or more ports to be configured as fast learn ports. You can specify a single port, a comma-separated list of ports or a range of ports such as 1/1/1-1/1/3. You may also enter an L2 LAG (1-128).

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring fast learn ports:

```
switch(config)# ip igmp snooping fastlearn 1/1/3
switch(config)# ip igmp snooping fastlearn 1/1/1-1/1/2
switch(config)# ip igmp snooping fastlearn 1/1/5,1/1/6
```

# ip igmp snooping fastleave

## Syntax

`ip igmp snooping fastleave` *<PORT-LIST>*

`no ip igmp snooping fastleave` *<PORT-LIST>*

## Description

Enables the switch to immediately remove the IGMP client from its IGMP table and cease transmitting multicast traffic to the client.

The `no` form of this command disables fastleave on the specified ports.

## Command context

`config-vlan`

## Parameters

**fastleave** *<PORT-LIST>*

Specifies a list of one or more ports to be configured as fastleave ports. You can specify a single port, a comma-separated list of ports or a range of ports such as 1/1/1-1/1/3.

## Authority

Administrators or local user group members with execution rights for this command.



## Usage

IGMP fastleave is configured for ports on a per-VLAN basis. Upon receiving a Leave Group message, the querier sends an IGMP Group-Specific Query message out of the interface to ensure that no other receivers are connected to the interface. If receivers are directly attached to the switch, it is inefficient to send the membership query as the receiver wanting to leave is the only connected host.

When a fastleave enabled switch port is connected to a single host and receives a leave, the switch does not wait for the querier status update interval, but instead immediately removes the IGMP client from its IGMP table and ceases transmitting multicast traffic to the client. (If the switch detects multiple end nodes on the port, Fastleave does not activate regardless of whether one or more of these end nodes are IGMP clients.) This processing speeds up the overall leave process and also eliminates the CPU overhead of having to generate an IGMP Group-Specific Query message.

## Examples

Configuring fastleave ports for the VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ip igmp snooping fastleave 1/1/1
switch(config-vlan)# ip igmp snooping fastleave 1/1/1-1/1/2
switch(config-vlan)# ip igmp snooping fastleave 1/1/1,1/1/2-1/1/3
switch(config-vlan)# no ip igmp snooping fastleave 1/1/1
switch(config-vlan)# no ip igmp snooping fastleave 1/1/1-1/1/2
```

## ip igmp snooping forced fastleave

### Syntax

```
ip igmp snooping forced-fastleave <PORT-LIST>
```

```
no ip igmp snooping forced-fastleave <PORT-LIST>
```

### Description

Configures the specified ports in forced fastleave mode.

The **no** form of this command disables forced fastleave on the specified ports.

### Command context

config-vlan

### Parameters

**fastleave** <PORT-LIST>

Specifies a list of one or more ports to be configured as forced fastleave ports. You can specify a single port, a comma-separated list of ports or a range of ports such as 1/1/1-1/1/3.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

With forced fastleave enabled, IGMP speeds up the process of blocking unnecessary multicast traffic to a switch port that is connected to multiple end nodes. When a port having multiple end nodes receives a leave group request from one end node for a given multicast group, forced fastleave activates and waits for a second to receive a join request from any other member of the same group on that port. If the port does not receive a join request for that group within the forced fastleave interval, the switch then blocks any further traffic to that group on that port.

## Examples

Configuring forced-fastleave ports for the VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ip igmp snooping forced-fastleave 1/1/1
switch(config-vlan)# ip igmp snooping forced-fastleave 1/1/1-1/1/2
switch(config-vlan)# ip igmp snooping forced-fastleave 1/1/1,1/1/2-1/1/3
switch(config-vlan)# no ip igmp snooping forced-fastleave 1/1/1
switch(config-vlan)# no ip igmp snooping forced-fastleave 1/1/1-1/1/2
```

## ip igmp snooping forward

### Syntax

```
ip igmp snooping forward <PORT-LIST>
```

```
no ip igmp snooping forward <PORT-LIST>
```

### Description

Configures the specified ports in forward mode. In forward mode, traffic is always forwarded on this port, irrespective of joins.

The **no** form of this command disables forward ports.

### Command context

config-vlan

### Parameters

**forward** <PORT-LIST>

Specifies a list of ports to be configured in forward mode. You can specify a single port, a comma-separated list of ports or a range of ports such as 1/1/1-1/1/3.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring forward ports for the VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ip igmp snooping forward 1/1/1
switch(config-vlan)# ip igmp snooping forward 1/1/1-1/1/2
switch(config-vlan)# ip igmp snooping forward 1/1/1,1/1/2-1/1/3
switch(config-vlan)# no ip igmp snooping forward 1/1/1
switch(config-vlan)# no ip igmp snooping forward 1/1/1-1/1/2
```

## ip igmp snooping static-group

### Syntax

```
ip igmp snooping static-group <MULTICAST-IP-ADDRESS>
```

```
no ip igmp snooping static-group <MULTICAST-IP-ADDRESS>
```

## Description

Configures an IGMP snooping static multicast group. You can configure a maximum of 32 IGMP snooping static groups.

The `no` form of this command disables static multicast group.

## Command context

`config-vlan`

## Parameters

**<MULTICAST-IP-ADDRESS>**

Specifies the IGMP static multicast group IP address. Format: A.B.C.D

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring IGMP snooping static group:

```
switch(config)# vlan 2
switch(config-vlan)# ip igmp snooping static-group 239.1.1.1
switch(config-vlan)# no ip igmp snooping static-group 239.1.1.1
```

# ip igmp snooping version

## Syntax

`ip igmp snooping version <VERSION>`

## Description

Configures the IGMP snooping version on the VLAN.

## Command context

`config-vlan`

## Parameters

**<VERSION>**

Specifies the IGMP snooping version. Select 2 for IGMPv2 (RFC2236). Select 3 for IGMPv3 (RFC3376).  
Values: 2 or 3.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring IGMP snooping version on the VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ip igmp snooping version 2
```

## no ip igmp snooping

### Syntax

```
no ip igmp snooping
```

### Description

Disables all IGMP snooping configurations on the VLAN.

### Command context

```
config-vlan
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Disabling all IGMP snooping configurations on the VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# no ip igmp snooping
```

## show ip igmp snooping

### Syntax

```
show ip igmp snooping
```

### Description

Shows IGMP snooping configuration information and status for all VLANs.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing IGMP snooping configuration and status:

```
switch# show ip igmp snooping

IGMP Snooping Protocol Info

Total VLANs with IGMP enabled : 1
IGMP Drop Unknown Multicast : Global

VLAN ID : 1
VLAN Name : DEFAULT_VLAN_1
IGMP Snooping is not enabled

VLAN ID : 2
```

```
VLAN Name : VLAN2
IGMP Configured Version : 3
IGMP Operating Version : 3
Querier Address [this switch] : 20.1.1.1
Querier Port :
Querier UpTime :0m 21s
Querier Expiration Time :0m 2s
```

## show ip igmp snooping counters

### Syntax

```
show ip igmp snooping counters
```

### Description

Shows IGMP snooping query packet Tx, Rx, and Error packet counter details.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

### Examples

Showing IGMP snooping packet counters:

```
switch# show ip igmp snooping counters
IGMP Snooping VLAN Counters

Rx Counters :

V1 All Hosts Queries 0
V2 All Hosts Queries 0
V3 All Hosts Queries 3
V2 Group Specific Queries 0
V3 Group Specific Queries 0
Group And Source Specific Queries 0
V1 Member Reports 0
V2 Member Reports 0
V3 Member Reports 2
V2 Member Leaves 0

Tx Counters :

Flood on vlan 44
V2 Group Specific Queries 0
V3 Group Specific Queries 0

Errors:

Unknown Message Type 0
Malformed Packets 0
Bad Checksum 0
Packet received on IGMP-disabled Interface 0
Interface Wrong Version Queries 0
Packets dropped by ACL 0
```

Port Counters:

Membership Timeout 0

## show ip igmp snooping groups

### Syntax

```
show ip igmp snooping groups
```

### Description

Shows IGMP snooping group information.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing IGMP snooping groups:

```
switch# show ip igmp snooping groups
IGMP Group Address Information
```

| VLAN | ID | Group Address | Expires | UpTime | Last Reporter | Type   |
|------|----|---------------|---------|--------|---------------|--------|
| 2    |    | 239.1.1.3     | 0m 4s   | 0m 10s | 10.1.1.1      | Filter |

## show ip igmp snooping static-groups

### Syntax

```
show ip igmp snooping static-groups
```

### Description

Shows IGMP snooping static group details.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing IGMP snooping static group details:

```
switch# show ip igmp snooping static-groups
```

## IGMP Static Group Address Information

| VLAN | ID | Group      | Address |
|------|----|------------|---------|
| 10   |    | 239.1.1.10 |         |
| 10   |    | 239.1.1.11 |         |

## show ip igmp snooping statistics

### Syntax

```
show ip igmp snooping statistics
```

### Description

Shows IGMP snooping protocol information and the joined group statistics.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing IGMP snooping statistics:

```
switch# show ip igmp snooping statistics
IGMP Snooping Protocol Info

Total VLANs with IGMP enabled : 1
IGMP Drop Unknown Multicast : Global

IGMP Snooping Joined Groups Statistics

VLAN ID VLAN Name Total Static INCLUDE EXCLUDE
----- -
1 DEFAULT_VLAN_1 0 0 0 0
2 VLAN10 2 2 0 0
```

## show ip igmp snooping vlan

### Syntax

```
show ip igmp snooping vlan <VLAN-ID> [group [<group-ip>]]
 [source <source-ip>]]
```

### Description

Shows IGMP snooping protocol information for the specified VLAN. You can also specify a group and source to show group and source information.

### Command context

Operator (>) or Manager (#)

## Parameters

### <VLAN-ID>

Specifies a VLAN. Range: 1-4094.

### group <group-ip>

Specifies a group to display port and group information. Format: A.B.C.D

### source <source-ip>

Specifies a source to display source information for the group. Format: A.B.C.D

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing IGMP snooping protocol information for VLAN 2:

```
switch# show ip igmp snooping vlan 2
```

IGMP Snooping Protocol Info

Total VLANs with IGMP enabled : 1  
IGMP Drop Unknown Multicast : Global

VLAN ID : 2  
VLAN Name : VLAN2  
IGMP Configured Version : 3  
IGMP Operating Version : 3  
Querier Address : 20.1.1.1  
Querier Port : 1/1/1  
Querier UpTime :  
Querier Expiration Time :

| Active Group Address | Tracking | Vers | Mode | Uptime | Expires |
|----------------------|----------|------|------|--------|---------|
| 239.1.1.2            | Filter   | 3    | INC  | 0m 27s | 0m 13s  |

Showing IGMP snooping group information for group 239.1.1.2 on VLAN 2:

```
switch# show ip igmp snooping vlan 2 group 239.1.1.2
```

IGMP ports and group information for group 239.1.1.2

VLAN ID : 2  
VLAN Name : VLAN2

Group Address : 239.1.1.2  
Last Reporter : 10.1.1.1  
Group Type : Filter

| Port  | Vers | Mode | Uptime | Expires | V1<br>Timer | V2<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|-------|------|------|--------|---------|-------------|-------------|----------------------|--------------------|
| 1/1/6 | 3    | INC  | 0m 41s | 3m 39s  |             |             | 3                    | 0                  |

Group Address : 239.1.1.2  
Source Address : 30.1.1.1  
Source Type : Filter

| Port | Mode | Uptime | Expires | Configured Mode |
|------|------|--------|---------|-----------------|
|------|------|--------|---------|-----------------|



```

1/1/6 INC 0m 41s 3m 39s Auto

Group Address : 239.1.1.2
Source Address : 30.1.1.2
Source Type : Filter

Port Mode Uptime Expires Configured Mode

1/1/6 INC 0m 41s 3m 39s Auto

Group Address : 239.1.1.2
Source Address : 30.1.1.3
Source Type : Filter

Port Mode Uptime Expires Configured Mode

1/1/6 INC 0m 41s 3m 39s Auto

```

Showing IGMP snooping group information for group 239.1.1.2 on VLAN 2 and source 30.1.1.1:

```

switch# show ip igmp snooping vlan 2 group 239.1.1.2 source 30.1.1.1

VLAN ID : 2
VLAN Name : VLAN2
Group Address : 239.1.1.2
Source Address : 30.1.1.1
Source Type : Filter

Port Mode Uptime Expires Configured Mode

1/1/6 INC 0m 41s 3m 39s Auto

```

## show ip igmp snooping vlan counters

### Syntax

```
show ip igmp snooping vlan <VLAN-ID> counters
```

### Description

Shows IGMP snooping query packet Tx, Rx, Error packet counters for the specified VLAN.

### Command context

Operator (>) or Manager (#)

### Parameters

**<VLAN-ID>**

Specifies a VLAN. Range: 1-4094.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing IGMP snooping counters for VLAN 2:

```

Switch# show ip igmp snooping vlan 2 counters
IGMP Snooping VLAN Counters

VLAN ID : 2
VLAN Name : VLAN2

Rx Counters :

V1 All Hosts Queries 0
V2 All Hosts Queries 0
V3 All Hosts Queries 3
V2 Group Specific Queries 0
V3 Group Specific Queries 0
Group And Source Specific Queries 0
V1 Member Reports 0
V2 Member Reports 0
V3 Member Reports 2
V2 Member Leaves 0

Tx Counters :

Flood on vlan 71
V2 Group Specific Queries 0
V3 Group Specific Queries 0

Errors:

Unknown Message Type 0
Malformed Packets 0
Bad Checksum 0
Packet received on IGMP-disabled Interface 0
Interface Wrong Version Queries 0
Packet dropped by ACL 0

Port Counters:

Membership Timeout 0
Switch#

```

## show ip igmp snooping vlan group port

### Syntax

```

show ip igmp snooping vlan <VLAN-ID> group port
<PORT-ID>

```

### Description

Shows IGMP snooping group details for the specified VLAN. It shows information about all IGMP snooping groups or sources learned on a particular port.

### Command context

Operator (>) or Manager (#)

### Parameters

**<VLAN-ID>**

Specifies a VLAN. Range: 1-4094.

### <PORT-ID>

Specifies a port of a VLAN to display information about all IGMPv3 snooping groups/sources learn on a particular port.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing IGMP snooping group details for VLAN 2 port 1/1/6:

```
switch# show ip igmp snooping vlan 2 group port 1/1/6
```

```
VLAN ID : 2
VLAN Name : VLAN2
```

```
Group Address : 239.1.1.1
Last Reporter : 10.1.1.1
Group Type : Filter
```

| Port  | Vers | Mode | Uptime | Expires | V1<br>Timer | V2<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|-------|------|------|--------|---------|-------------|-------------|----------------------|--------------------|
| 1/1/6 | 2    | EXC  | 0m 21s | 1m 12s  |             | 2m 48s      | 0                    | 0                  |

```
VLAN ID : 2
VLAN Name : VLAN2
```

```
Group Address : 239.1.1.2
Last Reporter : 10.1.1.1
Group Type : Filter
```

| Port  | Vers | Mode | Uptime | Expires | V1<br>Timer | V2<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|-------|------|------|--------|---------|-------------|-------------|----------------------|--------------------|
| 1/1/6 | 2    | EXC  | 0m 21s | 1m 32s  |             | 2m 48s      | 0                    | 0                  |

## show ip igmp snooping vlan statistics

### Syntax

```
show ip igmp snooping vlan <VLAN-ID> statistics
```

### Description

Shows IGMP snooping statistics details for the specified VLAN. It also shows information on the different groups joined in the VLAN.

### Command context

Operator (>) or Manager (#)

### Parameters

#### <VLAN-ID>

Specifies a VLAN. Range: 1-4094.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing IGMP snooping statistics for VLAN 2:

```
switch# show ip igmp snooping vlan 2 statistics
IGMP Snooping statistics
```

```
VLAN ID : 2
VLAN Name : VLAN2
```

```
Number of Include Groups : 1
Number of Exclude Groups : 0
Number of Static Groups : 1
Total Multicast Groups Joined : 2
```

## clear update-log

### Syntax

```
clear update-log
```

### Description

Clears stored log files of any In-System Programming updates on the system.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

## show needed-updates

### Syntax

```
show needed-updates [next-boot [primary|secondary]]
```

### Description

Displays whether any programmable devices are in need of an update.

Without the `next-boot` parameter, this command displays needed updates relative to the currently running AOS-CX image.

With the `next-boot` parameter, this command displays needed updates relative to an AOS-CX image file in the persistent storage of the switch, which might be different from the currently running image. If either the `primary` or `secondary` parameter is specified, this command queries that specific AOS-CX image file. Otherwise, it queries the default AOS-CX image file as set by the most recent `boot system` or `boot set-default` command.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

## allow-unsupported-transceiver

### Syntax

```
allow-unsupported-transceiver [confirm | log-interval {none | <INTERVAL>}]

no allow-unsupported-transceiver
```

### Description

Allows unsupported transceivers to be enabled or establish connections. Only 1G and 10G transceivers are enabled by this command and unsupported transceivers of other speeds will remain disabled.

The `no` form of this command disallows using unsupported transceivers. This is the default.

### Command context

config

### Parameters

#### **confirm**

Specifies that unsupported transceiver warnings are to be automatically confirmed.

#### **log-interval none**

Disables unsupported transceiver logging.

#### **log-interval <INTERVAL>**

Sets the unsupported transceiver logging interval in minutes. Default: 1440 minutes. Range: 1440 to 10080 minutes.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

When none of the parameters are specified it will display a warning message to accept the warranty terms. With `confirm` option the warning message is displayed but the user is not prompted to (y/n) answering. Warranty terms must be agreed to as part of enablement and the support is on best effort basis.

### Examples

Allowing unsupported transceivers with follow-up confirmation:

```
switch(config)# allow-unsupported-transceiver
Warning: The use of unsupported transceivers, DACs, and AOCs is at your
own risk and may void support and warranty. Please see HPE Warranty terms
and conditions.
```

```
Do you agree and do you want to continue (y/n)? y
```

Allowing unsupported transceivers with confirmation in command syntax:

```
switch(config)# allow-unsupported-transceiver confirm
```

Warning: The use of unsupported transceivers, DACs, and AOCs is at your own risk and may void support and warranty. Please see HPE Warranty terms and conditions.

Configuring unsupported transceiver logging with an interval of every 48 hours:

```
switch(config)# allow-unsupported-transceiver log-interval 2880
```

Disabling unsupported transceiver logging:

```
switch(config)# allow-unsupported-transceiver log-interval none
```

Disallowing unsupported transceivers with follow-up confirmation:

```
switch(config)# no allow-unsupported-transceiver
```

Warning: Unsupported transceivers, DACs, and AOCs will be disabled, which could impact network connectivity. Use 'show allow-unsupported-transceiver' to identify unsupported transceivers, DACs, and AOCs.

Continue (y/n)? **y**

Disallowing unsupported transceivers with confirmation in command syntax:

```
switch(config)# no allow-unsupported-transceiver confirm
```

Warning: Unsupported transceivers, DACs, and AOCs will be disabled, which could impact network connectivity. Use 'show allow unsupported-transceiver' to identify unsupported transceivers, DACs, and AOCs.

```
switch(config)#
```

## default interface

### Syntax

```
default interface <INTERFACE-ID>
```

### Description

Sets an interface (or a range of interfaces) to factory default values.

### Command context

config

### Parameters

<INTERFACE-ID>

Specifies the ID of a single interface or range of interfaces. Format: member/slot/port or member/slot/port-member/slot/port to specify a range.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Resetting an interface:

```
switch(config)# default default interface 1/1/1
```

Resetting an range of interfaces:

```
switch(config)# default default interface 1/1/1-1/1/10
```

## description

### Syntax

```
description <DESCRIPTION>
```

```
no description
```

### Description

Associates descriptive information with an interface to help administrators and operators identify the purpose or role of an interface.

The `no` form of this command removes a description from an interface.

### Command context

```
config-if
```

### Parameters

**<DESCRIPTION>**

Specify a description for the interface. Range: 1 to 64 ASCII characters (including space, excluding question mark).

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the description for an interface to **DataLink 01**:

```
switch(config-if)# description DataLink 01
```

Removing the description for an interface.

```
switch(config-if)# no description
```

## energy-efficient-ethernet

### Syntax

```
energy-efficient-ethernet
```

### Description

Enables auto-negotiation of Energy-Efficient Ethernet (EEE) on an interface. EEE Negotiation is established only on auto-link negotiation with supported link partners.

### Command context

```
config
```



## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# energy-efficient-ethernet
```

Disabling Energy Efficient Ethernet on an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no energy-efficient-ethernet
```

# flow-control

## Syntax

## Description

The no form disables flow control support on the current interface.

## Command context

config-if

## Authority

Administrators or local user group members with execution rights for this command.

## Parameters

## Examples

# interface

## Syntax

```
interface <PORT-NUM>
```

## Description

Switches to the `config-if` context for a physical port. This is where you define the configuration settings for the logical interface associated with the physical port.

## Command context

config

## Parameters

<PORT-NUM>

Specifies a physical port number. Format: member/slot/port.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring an interface:

```
switch(config)# interface 1/1/1
switch(config-if)#
```

## interface vlan

### Syntax

```
interface vlan <VLAN-ID>
```

```
no interface vlan <VLAN-ID>
```

### Description

Creates an interface VLAN also know as an SVI (switched virtual interface) and changes to the `config-if-vlan` context. The specified VLAN must already be defined on the switch.

The `no` form of this command deletes an interface VLAN.

### Command context

`config`

### Parameters

**<VLAN-ID>**

Specifies the interface ID. Range: 2 to 4094

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch# config
switch(config)# vlan 10
switch(config-vlan-10)# exit
switch(config)# interface vlan 10
switch(config-if-vlan)#
```

## ip address

### Syntax

```
ip address <IPv4-ADDR>/<MASK> [secondary]
```

```
no ip address <IPv4-ADDR>/<MASK> [secondary]
```

### Description

Sets an IP/IPv6 address on the interface VLAN.

The `no` form of this command removes the IP/IPv6 address from the interface.

### Command context

config-if-vlan

## Parameters

### <IPV4-ADDR>

Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. You can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100.

### <MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

### secondary

Specifies a secondary IP address.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Assigning the IP address 192.168.199.1 with a mask of 24 bits to interface VLAN 10:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# ip address 192.168.199.1/24
```

Removing the IP address 192.168.199.1 with a mask of 24 bits from interface VLAN 10:

```
switch(config)# interface vlan 10
switch(config-if-vlan)# no ip address 192.168.199.1/24
```

## ip mtu

## Syntax

ip mtu <VALUE>

ip no mtu

## Description

Sets the IP MTU (maximum transmission unit) for an interface. This defines the largest IP packet that can be sent or received by the interface.

The no form of this command sets the IP MTU to the default value 1500.

## Command context

config-if-vlan

## Parameters

### <VALUE>

Specifies the IP MTU in bytes. Range: 68 to 9198. Default: 1500.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the IP MTU to 576 bytes:

```
switch(config-if-vlan)# ip mtu 576
```

## ip source-interface

### Syntax

```
ip source-interface {sflow | tftp | radius | tacacs | ntp | syslog | dhcp-relay
| simplivity | dns | all} {interface <IFNAME> | <IPV4-ADDR>} [vrf <VRF-NAME>]
```

```
no ip source-interface {sflow | tftp | radius | tacacs | ntp | syslog | dhcp-relay
| simplivity | dns | all} [interface <IFNAME> | <IPV4-ADDR>] [vrf <VRF-NAME>]
```

### Description

Sets a single source IP address for a feature on the switch. This ensures that all traffic sent the feature has the same source IP address regardless of how it egresses the switch. You can define a single global address that applies to all supported features, or an individual address for each feature.

This command provides two ways to set the source IP addresses: either by specifying a static IP address, or by using the address assigned to a switch interface. If you define both options, then the static IP address takes precedence.

The **no** form of this command deletes the single source IP address for all supported services, or a specific service.

### Command contexts

config

### Parameters

**sflow | tftp | radius | tacacs | ntp | syslog | dhcp-relay | simplivity | dns | all**

Sets a single source IP address for a specific service. The **all** option sets a global address that applies to all protocols that do not have an address set. For DHCP relay, the address is used as both the source IP and GIADDR.

**interface <IFNAME>**

Specifies the name of the interface from which the specified service obtains its source IP address. The interface must have a valid IP address assigned to it. If the interface has both a primary and secondary IP address, the primary IP address is used.

**<IPV4-ADDR>**

Specifies the source IP address to use for the specified service. The IP address must be defined on the switch, and it must exist on the specified VRF (which is the **default** VRF, if the **vrf** option is not used). Specify the address in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the IPv4 address 10.10.10.5 as the global single source address:

```
switch# config
switch(config)# ip source-interface all 10.10.10.5
```

Clearing the global single source IP address 10.10.10.5:

```
switch(config)# no ip source-interface all 10.10.10.5
```

## ipv6 address

### Syntax

```
ipv6 address <IPV6-ADDR>/<MASK>{eui64 | [tag <ID>]}
```

```
no ipv6 address <IPV6-ADDR>/<MASK>
```

### Description

Sets an IPv6 address on the interface.

The `no` form of this command removes the IPv6 address on the interface.



**NOTE:** This command automatically creates an IPv6 link-local address on the interface. However, it does not add the `ipv6 address link-local` command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the `ipv6 address link-local` command.

### Command context

config-if

### Parameters

#### <IPV6-ADDR>

Specifies the IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55.

#### <MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

#### eui64

Configure the IPv6 address in the EUI-64 bit format.

#### tag <ID>

Configure route tag for connected routes. Range: 0 to 4294967295. Default: 0.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the IPv6 address 2001:0db8:85a3::8a2e:0370:7334 with a mask of 24 bits:

```
switch(config-if)# ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

Removing the IP address 2001:0db8:85a3::8a2e:0370:7334 with mask of 24 bits:

```
switch(config-if)# no ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

## ipv6 source-interface

### Syntax

```
ipv6 source-interface {sflow | tftp | radius | tacacs | ntp | syslog | dhcp-relay | simplivity | dns | all} [interface <IFNAME> | <IPV6-ADDR>] [vrf]
```

```
no ipv6 source-interface {sflow | tftp | radius | tacacs | ntp | syslog | dhcp-relay | simplivity | dns | all} [interface <IFNAME> | <IPV6-ADDR>] [vrf]
```

### Description

Sets a single source IP address for a feature on the switch. This ensures that all traffic sent the feature has the same source IP address regardless of how it egresses the switch. You can define a single global address that applies to all supported features, or an individual address for each feature.

This command provides two ways to set the source IP addresses: either by specifying a static IP address, or by using the address assigned to a switch interface. If you define both options, then the static IP address takes precedence.

The **no** form of this command deletes the single source IP address for all supported protocols, or a specific protocol.

### Command context

config

### Parameters

**sflow | tftp | radius | tacacs | ntp | syslog | dhcp-relay | simplivity | dns | all**

Sets a single source IP address for a specific protocol. The **all** option sets a global address that applies to all protocols that do not have an address set.

**interface <IFNAME>**

Specifies the name of the interface from which the specified protocol obtains its source IP address.

**<IPV6-ADDR>**

Specifies the source IP address to use for the specified protocol. The IP address must be defined on the switch, and it must exist on the specified VRF (which is the **default** VRF). Specify the IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring the IPv6 address 2001:DB8::1 as the global single source address:

```
switch# config
switch(config)# ip source-interface all 2001:DB8::1/32
```

Stop the source IP address from using the IP address on interface 1/1/1 on VRF default.

```
switch(config)# no ip source-interface all interface 1/1/1 vrf default
```

Clear the source IP address 2001:DB8::1.

```
switch(config)# no ip source-interface all 2001:DB8::1
```

## mtu

### Syntax

```
mtu <VALUE>
```

```
no mtu
```

### Description

Sets the MTU (maximum transmission unit) for an interface. This defines the maximum size of a layer 2 (Ethernet) frame. Frames larger than the MTU (1500 bytes by default) are dropped.

To support jumbo frames (frames larger than 1522 bytes), increase the MTU as required by your network. A frame size of up to 9198 bytes is supported.

The largest possible layer 1 frame will be 18 bytes larger than the MTU value to allow for link layer headers and trailers.

The `no` form of this command sets the MTU to the default value 1500.

### Command context

```
config-if
```

### Parameters

**<VALUE>**

Specifies the MTU in bytes. Range: 46 to 9198. Default: 1500.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the MTU on interface `1/1/1` to 1000 bytes:

Setting the MTU on interface `1/1/1` to the default value:

## show allow-unsupported-transceiver

### Syntax

```
show allow-unsupported-transceiver
```

### Description

Displays configuration and status of unsupported transceivers.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing unallowed unsupported transceivers:

```
switch(config)# show allow-unsupported-transceiver
```

```
Allow unsupported transceivers : no
Logging interval : 1440 minutes
```

| Port   | Type       | Status      |
|--------|------------|-------------|
| 1/1/31 | SFP-SX     | unsupported |
| 1/1/32 | SFP-1G-BXD | unsupported |
| 1/1/2  | SFP28DAC3  | unsupported |

Showing allowed unsupported transceivers:

```
switch# show allow-unsupported-transceiver
```

```
Allow unsupported transceivers : yes
Logging interval : 1440 minutes
```

| Port   | Type       | Status              |
|--------|------------|---------------------|
| 1/1/31 | SFP-SX     | unsupported-allowed |
| 1/1/32 | SFP-1G-BXD | unsupported-allowed |
| 1/1/2  | SFP28DAC3  | unsupported         |

## show interface

### Syntax

```
show interface [<IFNAME>|<IFRANGE>] [brief | physical | extended [non-zero]]
```

```
show interface [lag | vlan] [<ID>] [brief | physical]
```

```
show interface [lag | | vlan] [<ID>] [extended | non-zero]
```

### Description

Displays active configurations and operational status information for interfaces.

### Command context

config

### Parameters

**<IFNAME>**

Specifies a interface name.

**<IFRANGE>**

Specifies the port identifier range.

**brief**

Displays brief info in tabular format.

**physical**

Display the physical connection info in tabular format.



### **extended**

Displays additional statistics.

### **non-zero**

Displays only non zero statistics.

### **LAG**

Displays LAG interface information.

### **VLAN**

Displays VLAN interface information.

### **<LAG-ID>**

Specifies the LAG number. Range: 1-256

### **<VLAN-ID>**

Specifies the VLAN ID. Range: 1-4094

## **Authority**

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## **Examples**

The following example shows when interface 1/1/1 is configured:

```
switch# show interface 1/1/1
Interface 1/1/1 is up
Admin state is up
Link state: up for 1 minute (since Thu Nov 26 10:26:34 UTC 2020)
Link transitions: 3
Description:
Hardware: Ethernet, MAC Address: 88:3a:30:47:d1:df
MTU 1500
Type 1GbT
Full-duplex
qos trust cos
Speed 1000 Mb/s
Auto-negotiation is on
Energy-Efficient Ethernet is disabled
Flow-control: off
Error-control: off
MDI mode: MDIX
VLAN Mode: native-untagged
Native VLAN: 1
Allowed VLAN List: all
```

| Statistics | Rx    | Tx    |
|------------|-------|-------|
| -----      | ----- | ----- |
| Packets    | 53    | 196   |
| Unicast    | 20    | 10    |
| Multicast  | 25    | 174   |
| Broadcast  | 8     | 12    |
| Errors     | 0     | 0     |
| Dropped    | 0     | 0     |
| CRC/FCS    | 0     | n/a   |
| Pause      | 0     | 0     |
| Collision  | n/a   | 0     |
| Bytes      | 9995  | 25010 |

When the interface is currently linked at a downshifted speed:

```
switch(config-if)# show interface 1/1/1

Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active
```

When the interface is currently linked with energy-efficient-ethernet negotiated:

```
Interface 1/1/1 is up
...
Energy-Efficient Ethernet is enabled
```

## show interface dom

### Syntax

```
show interface [<INTERFACE-ID>] dom [detail]
```

### Description

Shows diagnostics information and alarm/warning flags for the optical transceivers (SFP, SFP+). This information is known as DOM (Digital Optical Monitoring). DOM information also consists of vendor determined thresholds which trigger high/low alarms and warning flags.

### Command context

Manager (#)

### Parameters

**<INTERFACE-ID>**

Specifies an interface. Format: member/slot/port.

**detail**

Show detailed information.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# show interface dom
```

| Port  | Type    | Channel | Temperature<br>(Celsius) | Voltage<br>(Volts) | Tx Bias<br>(mA) | Rx Power<br>(mW/dBm) | Tx Power<br>(mW/dBm) |
|-------|---------|---------|--------------------------|--------------------|-----------------|----------------------|----------------------|
| 1/1/1 | SFP+SR  |         | 47.65                    | 3.31               | 8.40            | 0.08, -10.96         | 0.63, -2.49          |
| 1/1/2 | SFP+SR  |         | n/a                      | n/a                | n/a             | n/a                  | n/a                  |
| 1/1/3 | SFP+DA3 |         | 42.10                    | 3.24               | n/a             | n/a                  | n/a                  |

# show interface energy-efficient ethernet

## Syntax

```
show interface [<IFNAME>|<IFRANGE>] energy-efficient-ethernet
```

## Description

Displays Energy-Efficient Ethernet information for the interface.

## Command context

config

## Parameters

### <IFNAME>

Specifies the name of an interface on the switch. Use the format `member/slot/port` (for example, `1/1/1`).

### <IFRANGE>

Specifies the port identifier range of an interface on the switch. Use the format `member/slot/port` (for example, `1/1/1`).

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

The following example shows when the interfaces are Energy-Efficient Ethernet capable:

```
switch# show interface energy-efficient-ethernet
```

| Port  | Enabled | Negotiated | Speed<br>(MB/s) | TX Wake<br>Time (us) | RX Wake<br>Time (us) |
|-------|---------|------------|-----------------|----------------------|----------------------|
| 1/1/1 | no      | no         | --              | --                   | --                   |
| 1/1/2 | yes     | yes        | 100             | 36                   | 36                   |
| 1/1/3 | yes     | yes        | 1000            | 17                   | 17                   |
| 1/1/4 | no      | no         | --              | --                   | --                   |
| 1/1/5 | yes     | no         | 1000            | --                   | --                   |

```
switch# show interface 1/1/1 energy-efficient-ethernet
```

| Port  | Enabled | Negotiated | Speed<br>(Mb/s) | TX Wake<br>Time (us) | RX Wake<br>Time (us) |
|-------|---------|------------|-----------------|----------------------|----------------------|
| 1/1/1 | no      | no         | 1000            | --                   | --                   |

```
switch#
```

# show interface transceiver

## Syntax

```
show interface [<INTERFACE-ID>] transceiver [detail | threshold-violations]
```

## Description

Displays information about transceivers present in the switch. The information shown varies for different transceiver types and manufacturers. Only basic information is shown for unsupported HPE and third-party transceivers installed in the switch and they are also identified with an asterisk in the output.

## Command context

Manager (#)

## Parameters

**<INTERFACE-ID>**

Specifies the name or range of an interface on the switch. Use the format `member/slot/port` (for example, 1/3/1).

### detail

Show detailed information for the interfaces.

### threshold-violations

Show threshold violations for transceivers.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Showing summary transceiver information with identification of unsupported transceivers:

```
switch(config)# show interface transceiver
```

| Port   | Type   | Product<br>Number | Serial<br>Number | Part<br>Number |
|--------|--------|-------------------|------------------|----------------|
| 1/1/15 | SFP+SR | J9150D            | CN92KJW05P       | 1990-4634      |
| 1/1/16 | SFP+SR | J9150D            | CN92KJW1RG       | 1990-4634      |

Showing detailed transceiver information:

```
switch(config)# show interface transceiver detail
```

```
Transceiver in 1/1/15
```

```
Interface Name : 1/1/15
Type : SFP+SR
Connector Type : LC
Wavelength : 850nm
Transfer Distance : 0.00km (SMF), 20m (OM1), 80m (OM2), 300m (OM3)
Diagnostic Support : DOM
Product Number : J9150D
Serial Number : CN92KJW05P
Part Number : 1990-4634
```

#### Status

```
Temperature : 30.38C
Voltage : 3.26V
Tx Bias : 5.54mA
Rx Power : 0.56mW, -2.52dBm
Tx Power : 0.62mW, -2.08dBm
```

```
Recent Alarms:
```

Recent Errors:

Transceiver in 1/1/16

Interface Name : 1/1/16  
Type : SFP+SR  
Connector Type : LC  
Wavelength : 850nm  
Transfer Distance : 0.00km (SMF), 20m (OM1), 80m (OM2), 300m (OM3)  
Diagnostic Support : DOM  
Product Number : J9150D  
Serial Number : CN92KJW1RG  
Part Number : 1990-4634

Status

Temperature : 30.62C  
Voltage : 3.28V  
Tx Bias : 5.64mA  
Rx Power : 0.61mW, -2.15dBm  
Tx Power : 0.59mW, -2.29dBm

Recent Alarms:

Recent Errors:

Showing detailed transceiver information with identification of unsupported transceivers:

switch# **show interface transceiver detail**

Transceiver in 1/1/2

Interface Name : 1/1/2  
Type : SFP+ER (unsupported)  
Connector Type : LC  
Wavelength : 3590nm  
Transfer Distance : 80m (SMF), 0m (OM1), 0m (OM2), 0m (OM3)  
Diagnostic Support : DOM  
Vendor Name : INNOLIGHT  
Vendor Part Number : TR-PX15Z-NHP  
Vendor Part Revision: 1A  
Vendor Serial number: MYxxxxxxx

Status

Temperature : 28.88C  
Voltage : 3.30V  
Tx Bias : 65.53mA  
Rx Power : 0.00mW, -inf  
Tx Power : 1.47mW, 1.67dBm

Recent Alarms:

Rx Power low alarm  
Rx Power low warning

Recent Errors:

Rx loss of signal

Showing transceiver threshold-violations:

switch(config)# **show interface transceiver threshold-violations**

| Port   | Type   | Channel# | Recent Threshold Violations |
|--------|--------|----------|-----------------------------|
| 1/1/15 | SFP+SR |          | none                        |
| 1/1/16 | SFP+SR |          | none                        |

switch#

## show ip interface

### Syntax

```
show ip interface <INTERFACE-ID>
```

### Description

Shows status and configuration information for an IPv4 interface.

### Command context

Manager (#)

### Parameters

<INTERFACE-ID>

Specifies the name of an interface. Format: member/slot/port.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# show ip interface vlan1

Interface vlan1 is up
Admin state is up
Hardware: Ethernet, MAC Address: f8:60:f0:c9:11:60
IP MTU 1500
IPv4 address 10.120.3.8/26
```

## show ip source-interface

### Syntax

```
show ip source-interface {sflow | tftp | radius | tacacs | all} [vrf <VRF-NAME>]
```

### Description

Shows single source IP address configuration settings.

### Command context

Manager (#)

### Parameters

**sflow | tftp | radius | tacacs | all**

Shows single source IP address configuration settings for a specific protocol. The **all** option shows the global setting that applies to all protocols that do not have an address set.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing single source IP address configuration settings for sFlow:

```
switch# show ip source-interface sflow

Source-interface Configuration Information

Protocol Source Interface

sflow 10.10.10.1
```

Showing single source IP address configuration settings for all protocols:

```
switch# show ip source-interface all

Source-interface Configuration Information

Protocol Src-Interface Src-IP VRF

all vlan2 2.2.2.2 default

switch#
```

## show ipv6 interface

### Syntax

```
show ipv6 interface <INTERFACE-ID>
```

### Description

Shows status and configuration information for an IPv6 interface.

### Command context

Manager (#)

### Parameters

<INTERFACE-ID>

Specifies an interface ID. Format: member/slot/port.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

```
switch# show ipv6 interface vlan2

Interface vlan2 is up
Admin state is up
IPv6 address:
 2001::1/64 [VALID]
IPv6 link-local address: fe80::883a:3080:247:c1c0/64 [VALID]
IPv6 Forwarding feature: enabled
IPv6 multicast groups locally joined:
 ff02::1 ff02::1:ff00:1 ff02::1:ff47:c1c0 ff02::1:ff00:0
```

```
ff02::2
IPv6 MTU 1500
IPv6 unicast reverse path forwarding: none
IPv6 load sharing: none
switch#
```

## show ipv6 source-interface

### Syntax

```
show ipv6 source-interface {sflow | tftp | radius | tacacs | all} [vrf <VRF-NAME>]
```

### Description

Shows single source IP address configuration settings.

### Command context

Manager (#)

### Parameters

**sflow | tftp | radius | tacacs | all**

Shows single source IP address configuration settings for a specific protocol. The **all** option shows the global setting that applies to all protocols that do not have an address set.

**vrf <VRF-NAME>**

Specifies the name of a VRF.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Showing single source IP address configuration settings for sFlow:

```
switch# show ipv6 source-interface sflow

Source-interface Configuration Information

Protocol Source Interface

sflow 2001:DB8::1
```

Showing single source IP address configuration settings for all protocols:

```
switch# show ipv6 source-interface all

Source-interface Configuration Information

Protocol Src-Interface Src-IP VRF

all vlan2 2001::1 default
switch#
```



# shutdown

## Syntax

```
shutdown
```

```
no shutdown
```

## Description

Disables an interface. Interfaces are disabled by default when created.  
The `no` form of this command enables an interface.

## Command context

```
config-if
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Disabling an interface:

```
switch(config-if) # shutdown
```

Enabling an interface:

```
switch(config-if) # no shutdown
```

## client track ip

### Syntax

```
client track ip
```

### Description

Enables client IP address tracking on the switch. The default is disabled on global and VLAN levels.

Admin users can enable client IP address tracking at the VLAN level.



**NOTE:** Tracking enabling will take effect only if the client IP address tracking is enabled at system and VLAN level.

The `no` form of the command disables client IP address tracking. If tracking is disabled at switch level, it will be stopped even if it is enabled at VLAN or port level.

### Command context

```
config
```

### Parameters

None

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Enable client IP address tracking at switch level:

```
switch(config)# client track ip
```

Enable client IP address tracking on VLAN 100:

```
switch(config)# vlan 100
```

```
switch(config-vlan-100)# client track ip
```

Enable client IP address tracking on VLANs 10 to 100:

```
switch(config)# vlan 10-100
```

```
switch(config-vlan-<10-100>)# client track ip
```

## client track ip { enable | disable | auto }

### Syntax

```
client track ip { enable | disable | auto }
```

## Description

Enables client IP address tracking on the specified set of interfaces. Tracking will take effect only if client IP address tracking is enabled at both the system level and for the VLAN to which the port belongs. Default: auto.

The `no` form of the command disables client IP address tracking on the specified set of interfaces.

## Command context

`config-if`

## Parameters

### ***enable***

Specifies that all client IP addresses will be tracked in the port.

### ***disable***

Specifies that client IP addresses will not be tracked in the port.

### ***auto***

Specifies the following:

- For LLDP devices: Only the specified client IP address will be tracked in the port and other client IP addresses will not be tracked.
- For non-LLDP devices: All client IP addresses will be tracked in the port.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Enable client IP address tracking on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# client track ip enable
```

Enable client IP address tracking on interfaces 1/1/1 to 1/1/5:

```
switch(config)# interface 1/1/1-1/1/5
switch(config-if-<1/1/1-1/1/5>)# client track ip enable
```

# client track ip client-limit

## Syntax

```
client track ip client-limit <CLIENT-LIMIT>
```

## Description

Configures the maximum number of clients to be tracked on the specified set of interfaces.

For the 6100 Switch Series the `no` form of the command resets the client limit to the default value of 128.

## Command context

`config-if`

## Parameters

### *CLIENT-LIMIT*

Specifies the maximum number of clients tracked on a port. Required. For the 6100 Switch Series, Range: 1-128. Default: 128..

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Configure the maximum number of clients to be tracked on interface 1/1/5:

```
switch(config)# interface 1/1/5
switch(config-if)# client track ip client-limit 32
```

## client track ip update-interval

## Syntax

```
client track ip update-interval <INTERVAL>
```

## Description

Configures how often client IP addresses are updated.

The `no` form of the command resets the update interval to the default of 1800 seconds.

## Command context

```
config-if
```

## Parameters

### *INTERVAL*

Specifies the update interval in seconds. Required. Range: 60-28000. Default: 1800.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Configure the update interval for an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# client track ip update-interval 600
```

## client track ip update-method probe

## Syntax

```
client track ip update-method probe
```

## Description

Enables probing the client to update the IP address.

The probe is sent to all clients on the tracking list that have an IP address in the following scenarios:

1. IP packets are not received from the clients during the IP address update cycle.
2. There is no IP packet from a learned IP address. In this case, a probe will be sent for the IP address to confirm if it is still owned by that client.

The `no` form of the command disables probing.

### Command context

`config`

### Parameters

None

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Disable probing to update the client IP address:

```
switch(config)# no client track ip update-method probe
```

## show capacities

### Syntax

```
show capacities
```

### Description

Shows the capacities configured on the switch.

### Command context

Operator (>)

### Parameters

None

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```
switch# show capacities
```

```
System Capacities:
Capacities Name Value

Maximum number of Access Control Entries configurable in a system 14336
Maximum number of Access Control Lists configurable in a system 1024
Maximum number of class entries configurable in a system 1024
Maximum number of classes configurable in a system 512
Maximum number of entries in an Access Control List 1024
Maximum number of entries in a class 1024
```

|                                                                               |      |
|-------------------------------------------------------------------------------|------|
| Maximum number of entries in a policy                                         | 1024 |
| Maximum number of classifier policies configurable in a system                | 512  |
| Maximum number of policy entries configurable in a system                     | 1024 |
| Maximum number of clients supported for tracking the IP address in the system | 128  |

```
switch# show capacities client-track-ip-client-limit
```

| System Capacities: Filter Client Track IP Client Limit                        |       |
|-------------------------------------------------------------------------------|-------|
| Capacities Name                                                               | Value |
| -----                                                                         |       |
| Maximum number of clients supported for tracking the IP address in the system |       |

## show client ip { count | port | vlan }

### Syntax

```
show client ip { count | port | vlan }
```

### Description

Shows number of client IP addresses or information about client IP addresses tracked on ports and VLANs.

### Command context

Operator (>)

### Parameters

#### *count*

Displays number of clients tracked.

#### *port*

Displays client IP addresses tracked on the ports.

#### *vlan*

Displays client IP addresses tracked on the VLANs.

### Authority

Administrators or local user group members with execution rights for this command.

## ip-sla responder

### Syntax

```
ip-sla responder <SLA-NAME> {udp-echo | tcp-connect | udp-jitter-voip} <PORT-NUM>
 [source {<SOURCE-IPV4-ADDR> | <IFNAME>}] [vrf <VRF-NAME>]
no ip-sla responder <SLA-NAME>
```

### Description

Selects the IP-SLA responder. The responder can be configured for udp-echo, tcp-connect, udp-jitter-voip type. It requires the SLA name, SLA type, and port number as arguments. Source IP/interface ID is a must for type udp-jitter-voip and optional for other types.

The **no** form of this command removes the IP-SLA responder.

### Command context

config

### Parameters

#### <SLA-NAME>

Specifies the SLA name.

#### udp-echo

Enables responder for udp-echo probes.

#### tcp-connect

Selects TCP connect as the IP-SLA test mechanism.

#### vrf <VRF-NAME>

Specifies the name of the VRF to use.

#### udp-jitter-voip

Selects VOIP jitter as the IP-SLA test mechanism.

#### <PORT-NUM>

Specifies the port number to listen for IP-SLA probes. Range: 1 to 65535.

#### [source {<SOURCE-IPV4-ADDR> | <IFNAME>}]

Selects the source IPv4 address for SLA probes or the source interface to use for sending IP-SLA probes.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config)# ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
switch(config)# ip-sla responder SLA1 udp-echo 8000 source 1/1/1
```

```
switch(config)# no ip-sla responder <SLA-NAME>
```

## show ip-sla responder

### Syntax

```
show ip-sla responder <SLA-NAME>
```

### Description

Shows the given IP-SLA responder configuration and operation status.

### Command context

config

### Parameters

**<SLA-NAME>**

Specifies the SLA name.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config)# show ip-sla responder SLA3
```

```
SLA Name : SLA3
IP-SLA Type : Udp-echo
VRF : Default
Responder Port : 8000
Responder IP : 2.2.2.3
Responder Interface : 1/1/1
Responder Status : Running
```

## show ip-sla responder results

### Syntax

```
show ip-sla responder <SLA-NAME> <SOURCE-IPV4-ADDR> <PORT-NUM> results
```

### Description

Shows the given ip-sla responder statistics for a given source IP and port. This command is only applicable for the sources where source IP and port are configured.

### Command context

config



## Parameters

### <SLA-NAME>

Specifies the SLA name.

### <SOURCE-IPV4-ADDR>

Specifies the source IPV4 address.

### <PORT-NUM>

Specifies the port number. Range: 1 to 65535.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch# show ip-sla responder SLA1 2.2.2.1 8000 results
```

```
IP-SLA Type : Udp-echo
VRF Name : Default
Source IP : 2.2.2.1
Source Port : 8000
Responder Port : 8888
Responder IP : 2.2.2.3
Responder Interface :
Responder Status : Running
Packets Received : 2
Packets Sent : 2
```

## description

### Syntax

```
description <DESC>
```

```
no description
```

### Description

Associates a text description with an IP tunnel for identification purposes.

The `no` form of this command removes the description from an IP tunnel.

### Command context

```
config-ip-if
```

### Parameters

**<DESC>**

Specifies the descriptive text to associate with the IP tunnel. Range: 1 to 64 printable ASCII characters.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Defines a description for IPv6 in IPv4 tunnel 27.

```
switch(config)# interface tunnel 27 mode ip 6in4
switch(config-ip-if)# description Network 3 Tunnel 27
```

Removes the description for IPv6 in IPv4 tunnel 27.

```
switch(config)# interface tunnel 27
switch(config-ip-if)# no description
```

Defines a description for IPv6 in IPv6 tunnel 8.

```
switch(config)# interface tunnel 8 mode ip 6in6
switch(config-ip-if)# description Network 4 Tunnel 8
```

Removes the description for IPv6 in IPv6 tunnel 8.

```
switch(config)# interface tunnel 8
switch(config-ip-if)# no description
```

## destination ip

### Syntax

```
destination ip <IPV4-ADDR>
```

```
no destination ip <IPV4-ADDR>
```

### Description

Sets the destination IP address for an IP tunnel. Specify the address of the interface on the remote device to which the tunnel will be established.

The `no` form of this command deletes the destination IP address from an IP tunnel.

### Command context

```
config-gre-if
```

```
config-ip-if
```

### Parameters

**<IPV4-ADDR>**

Specifies the destination IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Defines the destination IP address to be `10.10.20.1` for IPv6 in IPv4 tunnel 27.

```
switch(config)# interface tunnel 27 mode ip 6in4
switch(config-ip-if)# destination ip 10.10.20.1
```

Deletes the destination IP address `10.10.20.1` from IPv6 in IPv4 tunnel 27.

```
switch(config)# interface tunnel 27
switch(config-ip-if)# no destination ip 10.10.20.1
```

## destination ipv6

### Syntax

```
destination ipv6 <IPVv6-ADDR>
```

```
no destination ipv6 <IPV6-ADDR>
```

### Description

Sets the destination IPv6 address for an IP tunnel. Specify the address of the interface on the remote device to which the tunnel will be established.

The `no` form of this command deletes the destination IPv6 address from an IP tunnel.

### Command context

```
config-ip-if
```

## Parameters

### <IPv6-ADDR>

Specifies the tunnel IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Defines the destination IPv6 address to be 2001:DB8::1 for IPv6 in IPv6 tunnel .

```
switch(config)# interface tunnel 8 mode ip 6in6
switch(config-ip-if)# destination ipv6 2001:DB8::1
```

Deletes the destination IPv6 address 2001:DB8::1 from IPv6 in IPv6 tunnel 8.

```
switch(config)# interface tunnel 8
switch(config-ip-if)# no destination ipv6 2001:DB8::1
```

# interface tunnel

## Syntax

```
interface tunnel <TUNNEL-NUMBER> mode {gre ipv4 | ip 6in4 | ip 6in6}
```

```
interface tunnel <EXISTING-TUNNEL-NUMBER>
```

```
no interface tunnel <EXISTING-TUNNEL-NUMBER>
```

## Description

Creates or updates an IP tunnel. After you enter the command, the firmware switches to the configuration context for the tunnel.

If the specified tunnel exists, this command switches to the context for the tunnel.

By default, all tunnels are automatically assigned to the default VRF when they are created.

The **no** form of this command deletes an existing IP tunnel.

## Command context

```
config
```

## Parameters

```
mode {gre ipv4 | ip 6in4 | ip 6in6}
```

Creates an IP tunnel. Choose one of the following options:

- **ip 6in4**: Creates an IPv4 tunnel for IPv6 traffic.
- **ip 6in6**: Creates an IPv6 tunnel for IPv6 traffic.

### <TUNNEL-NUMBER>

Specifies the number for a new tunnel. Range: 1 to 127. Numbering is shared between all tunnels, so the same tunnel number cannot be used for an IPv6 in IPv4 tunnel and a GRE tunnel.

### **<EXISTING-TUNNEL-NUMBER>**

Specifies the number for an existing IP tunnel. Range: 1 to 127.

### **Command context**

config-ip-if

### **Examples**

Defines a new IPv6 in IPv4 tunnel with number 27.

```
switch(config)# interface tunnel 27 mode ip 6in4
switch(config-ip-if)#
```

Switches to the config-ip-if context for existing tunnel 27.

```
switch(config)# interface tunnel 27
switch(config-ip-if)#
```

Deletes IPv6 in IPv4 tunnel 27.

```
switch(config)# no interface tunnel 27
```

Defines a new IPv6 in IPv6 tunnel with number 8.

```
switch(config)# interface tunnel 8 mode ip 6in6
switch(config-ip-if)#
```

## **ip address**

### **Syntax**

```
ip address <IPV4-ADDR>/<MASK>
```

```
no ip address <IPV4-ADDR>/<MASK>
```

### **Description**

Sets the local IP address of a GRE tunnel. This address identifies the tunnel interface for routing. It must be on the same subnet as the tunnel address assigned on the remote device.

The `no` form of this command deletes the local IP address assigned to a GRE tunnel.

### **Command context**

config-gre-if

### **Parameters**

#### **<IPV4-ADDR>**

Specifies the tunnel IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

You can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100.

#### **<MASK>**

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 32.

### **Authority**

Administrators or local user group members with execution rights for this command.

## Examples

Defines the local IP address 10.10.10.1 for GRE tunnel 33.

```
switch(config)# interface tunnel 33 mode gre ipv4
switch(config-gre-if)# ip address 10.10.10.1/24
```

Deletes the local IP address 10.10.10.1 for GRE tunnel 33.

```
switch(config)# interface tunnel 33
switch(config-gre-if)# no ip address 10.10.10.1/24
```

## ipv6 address

### Syntax

```
ipv6 address <IPV6-ADDR>/<MASK>
```

```
no ipv6 address <IPV6-ADDR>/<MASK>
```

### Description

Sets the local IP address of an IPv6 to IPv4 tunnel or of an IPv6 to IPv6 tunnel. This address identifies the tunnel interface for routing. It must be on the same subnet as the tunnel address assigned on the remote device.

The `no` form of this command deletes the local IP address assigned to an IPv6 to IPv4 tunnel.

### Command context

```
config-ip-if
```

```
config-if
```

### Parameters

**<IPV6-ADDR>**

Specifies the tunnel IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

**<MASK>**

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 32.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Defines the local IP address 2001:DB8:5::1/64 for tunnel 8 for an IPv6 to IPv6 tunnel.

```
switch(config)# interface tunnel 8 mode ip 6in6
switch(config-ip-if)# ipv6 address 2001:DB8:5::1/64
```

Deletes the local IP address 2001:DB8::1/32 for tunnel 8.

```
switch(config)# interface tunnel 8
switch(config-ip-if)# no ipv6 address 2001:DB8:5::1/64
```

# ip mtu

## Syntax

```
ip mtu <VALUE>
```

## Description

Sets the MTU (maximum transmission unit) for an IP interface. The default value is 1500 bytes.

The `no` form of this command sets the MTU to the default value of 1500 bytes.

## Command context

```
config-ip-if
```

## Parameters

<VALUE>

Specifies the MTU in bytes. Range: 1,280 bytes to 9,192 bytes.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

The IP MTU is the largest IP packet that can be sent or received by the interface. For a tunnel, the IP MTU is the maximum size of the IP payload. To enable jumbo packet forwarding through the tunnel, set the IP MTU of the tunnel to a value greater than 1500. Also set the MTU and the IP MTU values for the underlying physical interface that the tunnel is using to a value greater than 1,500 bytes. The IP MTU of the tunnel must also be greater than or equal to the MTU of the ingress interface on the switch. The IP MTU value of the tunnel must also be less than or equal to the IP MT of the underlying interface that the tunnel is using.

## Examples

Sets the MTU on IPv6 in IPv4 tunnel 27 to 1000 bytes.

```
switch(config)# interface tunnel 27 mode ip 6in4
switch(config-ip-if)# mtu 1000
```

Sets the MTU on IPv6 in IPv4 tunnel 27 to the default value.

```
switch(config)# interface tunnel 27 mode ip 6in4
switch(config-ip-if)# ip mtu
```

Sets the MTU on IPv6 in IPv6 tunnel 8 to 900 bytes.

```
switch(config)# interface tunnel 8 mode ip 6in6
switch(config-ip-if)# ip mtu 9000
```

Sets the MTU on IPv6 in IPv6 tunnel 8 to the default value.

```
switch(config)# interface tunnel 8 mode ip 6in6
switch(config-ip-if)# ip mtu
```

## show interface tunnel

### Syntax

```
show interface tunnel [<TUNNEL-NUMBER>] [vsx-peer]
```

### Description

Shows configuration settings for all IP tunnels, or a specific tunnel.

### Command context

Manager (#)

### Parameters

**<TUNNEL-NUMBER>**

Specifies the number of an IP tunnel. Range: 1 to 127.

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Shows configuration settings for tunnel 12, which is an IPv6 in IPv6 tunnel in the following example.

```
switch# show interface tunnel12

Interface tunnel12 is up
Admin state is up
tunnel type IPv6 in IPv6
tunnel interface IPv6 address 4::1/64
tunnel source IPv6 address 2::1
tunnel destination IPv6 address 2::2
tunnel ttl 60
Description: Network2 Tunnel
RX
 0 input packets 0 bytes
 0 dropped
TX
 0 output packets 0 bytes
 0 dropped
```

## show running-config interface tunnel

### Syntax

```
show running-config interface tunnel<TUNNEL-NUMBER> [vsx-peer]
```

### Description

Shows the commands used to configure an IP tunnel.



## Command context

Manager (#)

## Parameters

**<TUNNEL-NUMBER>**

Specifies the number of an IP tunnel. Range: 1 to 127.

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Shows the configuration for IPv6 in IPv4 tunnel.

```
switch# show running-config interface tunnel5
interface tunnel5 mode ip 6in4
source ip 10.10.10.12
destination ip 22.20.20.20
ip6 address 2001:DB8:5::1/64
ttl 60
no shutdown
description Network10
```

Shows the configuration for IPv6 in IPv6 tunnel.

```
switch# show running-config interface tunnel1
interface tunnel 1 mode ip 6in6
description Network2 Tunnel
source ipv6 2::1
destination ipv6 2::2
ipv6 address 4::1/64
ttl 60
```

# shutdown

## Syntax

shutdown  
no shutdown

## Description

This command disables an IP interface. IP interfaces are disabled by default when created.

The **no** form of this command enables an IP interface.

## Command context

config-ip-if

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enables IPv6 in IPv4 interface 27.

```
switch(config)# interface tunnel 27 mode ip 6in4
switch(config-ip-if)# no shutdown
```

Disables IPv6 in IPv4 interface 27.

```
switch(config)# interface tunnel 27 mode ip 6in4
switch(config-ip-if)# shutdown
```

# source ip

## Syntax

```
source ip <IPv4-ADDR>
```

```
no source ip <IPv4-ADDR>
```

## Description

Sets the source IP address for an IP tunnel. Specify the IP address of a layer 3 interface on the switch. Tunnels can have the same source IP address and different destination IP addresses.

The `no` form of this command deletes the source IP address for an IP tunnel.

## Command context

```
config-ip-if
```

## Parameters

**<IPv4-ADDR>**

Specifies the source IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Defines the source IP address to be 10.10.10.1 for IPv6 in IPv4 tunnel 27.

```
switch(config)# interface tunnel 27 mode ip 6in4
switch(config-ip-if)# source ip 10.10.10.1
```

Deletes the source IP address 10.1.10.1 from IPv6 in IPv4 tunnel 27.

```
switch(config)# interface tunnel 27
switch(config-ip-if)# no source ip 10.10.10.1
```

## source ipv6

### Syntax

```
source ipv6 <IPV6-ADDR>
```

```
no source ipv6
```

### Description

Sets the source IPv6 address to be used for the encapsulation.

The `no` form of this command deletes the source IPv6 address for an IP tunnel.

### Command context

```
config-ip-if
```

### Parameters

**<IPV6-ADDR>**

Specifies the tunnel IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Defines the source IPv6 address to be 2001:DB8::1 for IPv6 in IPv6 tunnel 8.

```
switch(config)# interface tunnel 8 mode ip 6in6
switch(config-ip-if)# source ipv6 2001:DB8::1
```

Deletes the source IP address 2001:DB8::1 from IPv6 in IPv6 tunnel 8.

```
switch(config)# interface tunnel 8
switch(config-ip-if)# no source ipv6 2001:DB8::1
```

## ttl

### Syntax

```
ttl <COUNT>
```

```
no ttl
```

### Description

Sets the TTL (time-to-live), also known as the hop count, for tunneled packets. If not configured, the default value of 64 is used for the tunnel. (The hop count of the original packets is not changed.) A maximum of four different TTL values can be used at the same time by all tunnels on the switch. For example, if tunnel-1 has TTL 10, tunnel-2 has TTL 20, tunnel-3 has TTL 30, and tunnel-4 has TTL 40, then tunnel-5 cannot have a unique TTL value, it must reuse one of the values assigned to the other tunnels (10, 20, 30, 40).

The `no` form of this command sets TTL to the default value of 64.

## Command context

config-ip-if

## Parameters

<COUNT>

Specifies the hop count. Range: 1 to 255. Default: 64.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Defines a TTL of 55 for IPv6 in IPv4 tunnel 27.

```
switch(config)# interface tunnel 27 mode ip 6in4
switch(config-ip-if)# ttl 55
```

Sets the TTL for IPv6 in IPv4 tunnel 27 to the default value of 64.

```
switch(config)# interface tunnel 27
switch(config-ip-if)# no ttl
```

## ipv4 source-binding

### Syntax

```
ipv4 source-binding <VLAN-ID> <IPV4-ADDR> <MAC-ADDR> <IFNAME>
```

```
no ipv4 source-binding <VLAN-ID> <IPV4-ADDR> <MAC-ADDR> <IFNAME>
```

### Description

Adds static IPv4 client source binding information to the switch IP binding database. Although DHCPv4 snooping is often used to dynamically populate the binding database, this command is available for manually adding entries to the switch IP binding database.



**NOTE:** Statically configured IP binding information supersedes any dynamically collected binding information for the same client.

The no form of this command removes the specified binding that was statically configured with the `ipv4 source-binding` command. The no form has no effect on bindings that were dynamically configured with DHCPv4 snooping.

### Command context

```
config
```

### Parameters

**<VLAN-ID>**

Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.

**<IPV4-ADDR>**

Specifies the client IPv4 unicast address.

**<MAC-ADDR>**

Specifies the client MAC address.

**<IFNAME>**

Specifies the interface on which the client is connected.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Adding a static IPv4 binding:

```
switch(config)# ipv4 source-binding 1 10.2.1.4 00:50:56:96:e4:cf 1/1/1
```

Removing a IPv4 binding:

```
switch(config)# no ipv4 source-binding 1 10.2.1.4 00:50:56:96:e4:cf 1/1/1
```

## ipv4 source-lockdown

### Syntax

```
ipv4 source-lockdown
```

```
no ipv4 source-lockdown
```

### Description

Enables IPv4 source lockdown for all VLANs on the selected interface (port).

The no form of this command disables IPv4 source lockdown for the selected interface (port).

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling IPv4 source lockdown on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv4 source-lockdown
```

Enabling IPv4 source lockdown on interface lag112:

```
switch(config)# interface lag112
switch(config-if)# ipv4 source-lockdown
```

Disabling IPv4 source lockdown on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no ipv4 source-lockdown
```

## ipv4 source-lockdown hardware retry

### Syntax

```
ipv4 source-lockdown hardware retry <VLAN-ID> <IPV4-ADDR>
```

### Description

Retries the IPv4 source lockdown hardware programming for a client identified by VLAN and IPv4 address.

### Command context

```
config
```

### Parameters

**<VLAN-ID>**

Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.

**<IPV4-ADDR>**

Specifies the client IPv4 unicast address.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Configure IPv4 source lockdown hardware retry for the client on VLAN 10.

```
switch(config)# ipv4 source-lockdown hardware retry 10 1.1.2.1
```

## show ipv4 source-binding

### Syntax

```
show ipv4 source-binding [vsx-peer]
```

### Description

Shows all IPv4 static source binding information irrespective of source lockdown configuration..

### Command context

Operator (>) or Manager (#)

### Parameters

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing all IPv4 source binding information:

```
switch# show ipv4 source-binding
```

| PORT  | VLAN | MAC-ADDRESS       | HW-STATUS | FROM   | IPv4-ADDRESS |
|-------|------|-------------------|-----------|--------|--------------|
| 1/1/1 | 2    | aa:bb:cc:dd:ee:ff | Yes       | static | 1.2.3.4      |
| 1/1/2 | 12   | aa:ab:cc:dd:ee:ff | Yes       | static | 10.20.30.40  |

## show ipv4 source-lockdown

### Syntax

```
show ipv4 source-lockdown [binding [interface <IFNAME> | ip <IPV4-ADDR> |
 mac <MAC-ADDR> | vlan <VLAN-ID>] |
 interface <IFNAME>] [vsx-peer]
```

## Description

Shows summary or detailed IPv4 source lockdown information. When entered without parameters, summary status information for all interfaces (ports) in the binding database is shown.

## Command context

Operator (>) or Manager (#)

## Parameters

### **binding**

Specifies that detailed lockdown binding record information is to be displayed. The binding database record can be identified by any one of `interface` (port), `ip`, `mac`, or `vlan`.

### **interface <IFNAME>**

Specifies the client interface (port). When entered without the `binding` parameter, the summary status information is displayed for the specified interface.

### **ip <IPV4-ADDR>**

Specifies the client IPv4 unicast address.

### **mac <MAC-ADDR>**

Specifies the client MAC address.

### **vlan <VLAN-ID>**

Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.

### **[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing the summary status information for all interfaces in the binding database:

```
switch# show ipv4 source-lockdown
```

| INTERFACE | LOCKDOWN | HW-STATUS |
|-----------|----------|-----------|
| 1/1/1     | Yes      | Yes       |
| 1/1/2     | Yes      | No        |
| lag112    | Yes      | Yes       |

Showing the summary status information for the specified interface in the binding database:

```
switch# show ipv4 source-lockdown interface 1/1/2
```

| INTERFACE | LOCKDOWN | HW-STATUS |
|-----------|----------|-----------|
| 1/1/2     | Yes      | No        |

Showing the detailed binding record and related information for all interfaces in the binding database:



```
switch# show ipv4 source-lockdown binding
```

```
Interface Name : 1/1/1
VLAN Id : 2000
MAC Address : 00:50:56:96:e4:cf
IP Address : 192.168.142.113
Time Remaining : static
Lockdown Status : Yes
Hardware Status : Yes
Hardware Error Reason : --

Interface Name : 1/1/2
VLAN Id : 100
MAC Address : 00:50:56:96:04:4d
IP Address : 120.168.43.52
Time Remaining : 115 seconds
Lockdown Status : Yes
Hardware Status : No
Hardware Error Reason : Resource unavailable

Interface Name : lag112
VLAN Id : 12
MAC Address : 00:50:56:96:d8:3d
IP Address : 120.168.76.182
Time Remaining : static
Lockdown Status : Yes
Hardware Status : Yes
Hardware Error Reason : --
```

Showing the detailed binding record and related information for interface 1/1/2:

```
switch# show ipv4 source-lockdown binding interface 1/1/2
```

```
Interface Name : 1/1/2
VLAN Id : 100
MAC Address : 00:50:56:96:04:4d
IP Address : 120.168.43.52
Time Remaining : 115 seconds
Lockdown Status : Yes
Hardware Status : No
Hardware Error Reason : Resource unavailable
```

Showing the detailed binding record and related information for interface lag112 (identified in this example command by the IP address):

```
switch# show ipv4 source-lockdown binding ip 120.168.76.182
```

```
Interface Name : lag112
VLAN Id : 12
MAC Address : 00:50:56:96:d8:3d
IP Address : 120.168.76.182
Time Remaining : static
Lockdown Status : Yes
Hardware Status : Yes
Hardware Error Reason : --
```

Showing the detailed binding record and related information for interface 1/1/1 (identified in this example command by the MAC address):

```
switch# show ipv4 source-lockdown binding mac 00:50:56:96:e4:cf
```

```
Interface Name : 1/1/1
VLAN Id : 2000
MAC Address : 00:50:56:96:e4:cf
```

```
IP Address : 192.168.142.113
Time Remaining : static
Lockdown Status : Yes
Hardware Status : Yes
Hardware Error Reason : --
```

Showing the detailed binding record and related information for interface 1/1/2 (identified in this example command by the VLAN):

```
switch# show ipv4 source-lockdown binding vlan 100
```

```
Interface Name : 1/1/2
VLAN Id : 100
MAC Address : 00:50:56:96:04:4d
IP Address : 120.168.43.52
Time Remaining : 115 seconds
Lockdown Status : Yes
Hardware Status : No
Hardware Error Reason : Resource unavailable
```

## ipv6 source-binding

### Syntax

```
ipv6 source-binding <VLAN-ID> <IPv6-ADDR> <MAC-ADDR> <IFNAME>
```

```
no ipv6 source-binding <VLAN-ID> <IPv6-ADDR> <MAC-ADDR> <IFNAME>
```

### Description

Adds static IPv6 client source binding information to the switch IPv6 binding database. Although DHCPv6 snooping is often used to dynamically populate the binding database, this command is available for manually adding entries to the switch IPv6 binding database.



**NOTE:** Statically configured IPv6 binding information supersedes any dynamically collected binding information for the same client.

The no form of this command removes the specified binding that was statically configured with the `ipv6 source-binding` command. The no form has no effect on bindings that were dynamically configured with DHCPv6 snooping.

### Command context

```
config
```

### Parameters

**<VLAN-ID>**

Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.

**<IPv6-ADDR>**

Specifies the client IPv6 address.

**<MAC-ADDR>**

Specifies the client MAC address.

**<IFNAME>**

Specifies the interface on which the client is connected.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Adding a static IPv6 binding:

```
switch(config)# ipv6 source-binding 2 2000::2 00:12:11:44:55:12 1/1/28
```

Removing a IPv6 binding:

```
switch(config)# no ipv6 source-binding 2 2000::2 00:12:11:44:55:12 1/1/28
```

## ipv6 source-lockdown

### Syntax

```
ipv6 source-lockdown
```

```
no ipv6 source-lockdown
```

### Description

Enables IPv6 source lockdown for all VLANs on the selected interface (port).

The no form of this command disables IPv6 source lockdown for the selected interface (port).

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling IPv6 source lockdown on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 source-lockdown
```

Enabling IPv6 source lockdown on interface lag112:

```
switch(config)# interface lag112
switch(config-if)# ipv6 source-lockdown
```

Disabling IPv6 source lockdown on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no ipv6 source-lockdown
```

## ipv6 source-lockdown hardware retry

### Syntax

```
ipv6 source-lockdown hardware retry <VLAN-ID> <IPv6-ADDR>
```

### Description

Retries the IPV6 source lockdown hardware programming for a client identified by VLAN and IPv6 address.

### Command context

```
config
```

### Parameters

**<VLAN-ID>**

Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.

**<IPV6-ADDR>**

Specifies the client IPv6 address.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Configure IPv6 source lockdown hardware retry for the client on VLAN 1.

```
switch(config)# ipv6 source-lockdown hardware retry 1 2000::2
```

## show ipv6 source-binding

### Syntax

```
show ipv6 source-binding [vsx-peer]
```

### Description

Shows all IPv6 static source binding information irrespective of source lockdown configuration.

### Command context

Operator (>) or Manager (#)

### Parameters

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing all IPv6 source binding information:

```
switch# show ipv6 source-binding
```

| PORT   | VLAN | MAC-ADDRESS       | HW-STATUS | FROM   | IPv6-ADDRESS |
|--------|------|-------------------|-----------|--------|--------------|
| 1/1/1  | 1234 | 00:50:56:96:e4:cf | --        | static | 3000::1      |
| 1/1/1  | 1    | 00:50:56:96:04:4d | --        | static | 3000::2      |
| 1/1/24 | 1    | 00:01:01:00:00:01 | Yes       | static | 1001::1      |

## show ipv6 source-lockdown

### Syntax

```
show ipv6 source-lockdown [binding [interface <IFNAME> | ip <IPV6-ADDR> |
 mac <MAC-ADDR> | vlan <VLAN-ID>] |
 interface <IFNAME>] [vsx-peer]
```

## Description

Shows summary or detailed IPv6 source lockdown information. When entered without parameters, summary status information for all interfaces (ports) in the binding database is shown.

## Command context

Operator (>) or Manager (#)

## Parameters

### **binding**

Specifies that detailed lockdown binding record information is to be displayed. The binding database record can be identified by any one of `interface` (port), `ip`, `mac`, or `vlan`.

### **interface <IFNAME>**

Specifies the client interface (port). When entered without the `binding` parameter, the summary status information is displayed for the specified interface.

### **ip <IPv6-ADDR>**

Specifies the client IPv6 address.

### **mac <MAC-ADDR>**

Specifies the client MAC address.

### **vlan <VLAN-ID>**

Specifies the ID of an existing VLAN on which the client is connected. Range: 1 to 4094.

### **[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing the summary status information for all interfaces in the binding database:

```
switch# show ipv6 source-lockdown
```

| INTERFACE | LOCKDOWN | HW-STATUS |
|-----------|----------|-----------|
| 1/1/1     | Yes      | Yes       |
| 1/1/2     | Yes      | Yes       |
| lag112    | Yes      | --        |

Showing the summary status information for the specified interface in the binding database:

```
switch# show ipv6 source-lockdown interface 1/1/2
```

| INTERFACE | LOCKDOWN | HW-STATUS |
|-----------|----------|-----------|
| 1/1/2     | Yes      | No        |

Showing the detailed binding record and related information for all interfaces in the binding database:

```
switch# show ipv6 source-lockdown binding
```

```
Interface Name : 1/1/1
VLAN Id : 1234
MAC Address : 00:50:56:96:e4:cf
IP Address : aaaa:bbbb:cccc:dddd:eeee:1234
Time Remaining : static
Lockdown Status : Yes
Hardware Status : Yes
Hardware Error Reason : --
```

```
Interface Name : 1/1/2
VLAN Id : 1234
MAC Address : 00:50:56:96:04:4d
IP Address : 4000::1
Time Remaining : 3290 seconds
Lockdown Status : Yes
Hardware Status : No
Hardware Error Reason : Resource unavailable
```

```
Interface Name : lag112
VLAN Id : 151
MAC Address : 00:50:56:96:d8:3d
IP Address : 1001::5
Time Remaining : 1200 seconds
Lockdown Status : No
Hardware Status : --
Hardware Error Reason : --
```

Showing the detailed binding record and related information for interface 1/1/2:

```
switch# show ipv6 source-lockdown binding interface 1/1/2
```

```
Interface Name : 1/1/2
VLAN Id : 1234
MAC Address : 00:50:56:96:04:4d
IP Address : 4000::1
Time Remaining : 3290 seconds
Lockdown Status : Yes
Hardware Status : No
Hardware Error Reason : Resource unavailable
```

Showing the detailed binding record and related information for interface 1/1/2 (identified in this example command by the IP address):

```
switch# show ipv6 source-lockdown binding ip 4000::1
```

```
Interface Name : 1/1/2
VLAN Id : 1234
MAC Address : 00:50:56:96:04:4d
IP Address : 4000::1
Time Remaining : 515 seconds
Lockdown Status : No
Hardware Status : --
Hardware Error Reason : --
```

Showing the detailed binding record and related information for interface 1/1/1 (identified in this example command by the MAC address):

```
switch# show ipv6 source-lockdown binding mac 00:50:56:96:e4:cf
```

```
Interface Name : 1/1/1
VLAN Id : 1234
MAC Address : 00:50:56:96:e4:cf
```

```
IP Address : aaaa:bbbb:cccc:dddd:eeee:1234
Time Remaining : static
Lockdown Status : Yes
Hardware Status : Yes
Hardware Error Reason : --
```

Showing the detailed binding record and related information for interface lag112 (identified in this example command by the VLAN):

```
switch# show ipv6 source-lockdown binding vlan 151
```

```
Interface Name : lag112
VLAN Id : 151
MAC Address : 00:50:56:96:d8:3d
IP Address : 1001::5
Time Remaining : 1200 seconds
Lockdown Status : No
Hardware Status : --
Hardware Error Reason : --
```



## ipv6 address <global-unicast-address>

### Syntax

```
ipv6 address <global-unicast-address>
no ipv6 address <global-unicast-address>
```

### Description

Sets a global unicast address on the interface.

The `no` form of this command removes the global unicast address on the interface.



**NOTE:** This command automatically creates an IPv6 link-local address on the interface. However, it does not add the command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the command.

### Command context

```
config-if
```

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Enabling a global unicast address:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address 3731:54:65fe:2::a7
```

Disabling a global unicast address:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ipv6 address 3731:54:65fe:2::a7
```

## ipv6 address autoconfig

### Syntax

```
ipv6 address autoconfig
no ipv6 address autoconfig
```

### Description

Enables the interface to automatically obtain an IPv6 address using router advertisement information and the EUI-64 identifier.

The `no` form of this command disables address auto-configuration.



**NOTE:**

- A maximum of 15 autoconfigured addresses are supported.
- This command automatically creates an IPv6 link-local address on the interface. However, it does not add the command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the command.

---

**Command context**

`config-if`

**Authority**

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

**Usage**

The IPv6 SLAAC feature lets the router obtain the IPv6 address for the interface it is configured through the SLAAC method. This feature is not available on the `mgmt` VRF.

**Example**

Enabling unicast autoconfiguring:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address autoconfig
```

Disabling unicast autoconfiguring:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# no ipv6 address autoconfig
```

## ipv6 address link-local

**Syntax**

```
ipv6 address link-local [<IPv6-ADDR>/<MASK>]
```

**Description**

Enables IPv6 on the current interface. If no address is specified, an IPv6 link-local address is auto-generated for the interface. If an address is specified, auto-configuration is disabled and the specified address/mask is assigned to the interface.

To disable IPv6 link-local on the interface, remove `ipv6 address link-local`, `ipv6 address <global-ipv6-address>`, and `ipv6 address autoconfig` from the interface.



---

**NOTE:** This feature is not available on the management VRF.

---

**Command context**

`config-if`

## Parameters

### <IPV6-ADDR>

Specifies the IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55.

### <MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Enabling IPv6 link-local on the interface:

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 address link-local
```

# ipv6 nd cache-limit

## Syntax

```
ipv6 nd cache-limit <CACHELIMIT>
```

```
no ipv6 nd cache-limit [<CACHELIMIT>]
```

## Description

Configures the limit on the number of neighbor entries in the ND cache. The no form of this command sets the cache limit to the default value.

## Command context

```
config
```

## Parameters

### <CACHELIMIT>

Specifies the neighbor cache entries limit. Range: 1-131072. Default: 131072.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the cache limit to 20.

```
switch(config)# ipv6 nd cache-limit 20
```

## ipv6 nd dad attempts

### Syntax

```
ipv6 nd dad attempts <NUM-ATTEMPTS>
```

```
no ipv6 nd dad attempts [<NUM-ATTEMPTS>]
```

### Description

Configures the number of neighbor solicitations to be sent when performing duplicate address detection (DAD) for a unicast address configured on an interface.

The `no` form of this command sets the number of attempts to the default value.

### Command context

```
config-if
```

### Parameters

**dad attempts** <NUM-ATTEMPTS>

Specifies the number of neighbor solicitations to send. Range: 0-15. Default: 1.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd dad attempts 5
```

## ipv6 nd hop-limit

### Syntax

```
ipv6 nd hop-limit <HOPLIMIT>
```

```
no ipv6 nd hop-limit [<HOPLIMIT>]
```

### Description

Configures the hop limit to be sent in RAs.

The `no` form of this command resets the hop limit to 0. This reset eliminates the hop limit from the RAs that originate on the interface, so the host determines the hop limit.

### Command context

```
config-if
```

### Parameters

**hop-limit** <HOPLIMIT>

Specifies the hop limit. Range: 0-255. Default: 64.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd hop-limit 64
```

## ipv6 nd mtu

### Syntax

```
ipv6 nd mtu <MTU-VALUE>

no ipv6 nd mtu [<MTU-VALUE>]
```

### Description

Configures the MTU size to be sent in the RA messages.

The `no` form of this command sets hop limit to the default value.

### Command context

config-if

### Parameters

**<MTU-VALUE>**

Specifies the MTU size. Range: 1280-65535 bytes. Default: 1500 bytes.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd mtu 1300
```

## ipv6 nd ns-interval

### Syntax

```
ipv6 nd ns-interval <TIME>

no ipv6 nd ns-interval [<TIME>]
```

### Description

Configures the ND time between DAD neighbor solicitations sent for an unresolved destination, or between duplicate address detection neighbor solicitation requests. Increase this setting when neighbor solicitation retries or failures are occurring, or in a slow (WAN) network.

The `no` form of this command sets the ns-interval to the default value.

### Command context

config-if

## Parameters

### <TIME>

Specifies the neighbor solicitation interval. Range: 1000-3600000 milliseconds. Default: 1000 milliseconds.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ns-interval 1200
```

# ipv6 nd prefix

## Syntax

```
ipv6 nd prefix <IPV6-ADDR>/<PREFIX-LEN>
 [no-advertise | [valid <LIFETIME-VALUE> preferred
 <LIFETIME-VALUE>] | no-autoconfig | no-onlink]

no ipv6 nd prefix <IPV6-ADDR>/<PREFIX-LEN> [no-advertise
 | [valid <LIFETIME-VALUE> preferred <LIFETIME-VALUE>
] | no-autoconfig | no-onlink]

ipv6 nd prefix default [no-advertise | [valid <LIFETIME-VALUE>
 preferred <LIFETIME-VALUE>] | no-autoconfig | no-onlink]}

no ipv6 nd prefix default [no-advertise | [valid <LIFETIME-VALUE>
 preferred <LIFETIME-VALUE>] | no-autoconfig | no-onlink]}
```

## Description

Specifies prefixes for the routing switch to include in RAs transmitted on the interface. IPv6 hosts use the prefixes in RAs to autoconfigure themselves with global unicast addresses. The autoconfigured address of a host is composed of the advertised prefix and the interface identifier in the current link-local address of the host.

By default, advertise, autoconfig, and onlink are set.

The **no** form of this command removes the configuration on the interface.

## Command context

config-if

## Parameters

### <IPV6-ADDR>/<PREFIX-LEN>

Specifies the IPv6 prefix to advertise in RA. Format: X:X::X:X/M

### default

Specifies apply configuration to all on-link prefixes that are not individually set by the **ipv6 ra prefix <IPV6-ADDR>/<PREFIX-LEN>** command. It applies the same valid and preferred lifetimes, link state, autoconfiguration state, and advertise options to the advertisements sent for all on-link prefixes that are

not individually configured with a unique lifetime. This also applies to the prefixes for any global unicast addresses configured later on the same interface.

Using default once, and then using it again with any new parameter values results in the new values replacing the former values in advertisements. If default is used without the `no-advertise`, `no-autoconfig`, or `no-onlink` parameter, the advertisement setting for the absent parameter is returned to its default setting.

#### **no-advertise**

Specifies do not advertise prefix in RA.

#### **valid <LIFETIME-VALUE>**

Specifies the total time, in seconds, the prefix remains available before becoming unusable. After preferred-lifetime expiration, any autoconfigured address is deprecated and used only for transactions only before preferred-lifetime expires. If the valid lifetime expires, the address becomes invalid.

You can enter a value in seconds or enter `valid infinite` which sets infinite lifetime. Default: 2,592,000 seconds which is 30 days. Range: 0–4294967294 seconds.

#### **preferred <LIFETIME-VALUE>**

Specifies the span of time during which the address can be freely used as a source and destination for traffic. This setting must be less than or equal to the corresponding valid-lifetime setting.

You can enter a value in seconds or enter `preferred infinite` which sets infinite lifetime. Default: 604,800 seconds which is seven days. Range: 0–4294967294 seconds.

#### **no-autoconfig**

Specifies do not use prefix for autoconfiguration.

#### **no-onlink**

Specifies do not use prefix for onlink determination.

### **Authority**

Administrators or local user group members with execution rights for this command.

### **Examples**

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd prefix 4001::1/64 valid 30 preferred 10 no-autoconfig no-onlink
```

## **ipv6 nd ra dns search-list**

### **Syntax**

```
ipv6 nd ra dns search-list <DOMAIN-NAME> [lifetime <TIME>]
```

```
no ipv6 nd ra dns search-list <DOMAIN-NAME>
```

### **Description**

Configures the DNS Search List (DNSSL) to include in Router Advertisements (RAs) transmitted on the interface.

The `no` form of this command removes the DNS Search List from the RAs transmitted on the interface.

### **Command context**

```
config-if
```

## Parameters

**<DOMAIN-NAME>**

Specifies the domain names for DNS queries.

**lifetime <TIME>**

Specifies lifetime in seconds. Range: 4-4294967295 seconds. Default: 1800 seconds.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

- DNSSL contains the domain names of DNS suffixes or IPv6 hosts to append to short, unqualified domain names for DNS queries.
- Multiple DNS domain names can be added to the DNSSL by using the command repeatedly.
- A maximum of eight server addresses are allowed.

## Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns search-list test.com lifetime 500
```

# ipv6 nd ra dns server

## Syntax

ipv6 nd ra dns server <IPv6-ADDR> [lifetime <TIME>]

no ipv6 nd ra dns server <IPv6-ADDR>

## Description

Configures the IPv6 address of a preferred Recursive DNS Server (RDNSS) to be included in Router Advertisements (RAs) transmitted on the interface.

The **no** form of this command removes the configured DNS server from the RAs transmitted on the interface.

## Command context

config-if

## Parameters

**<IPv6-ADDR>**

Specifies the RDNSS address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55.

**lifetime <TIME>**

Specifies IPv6 DNS server lifetime in seconds. Range: 4-4294967295 seconds. Default: 1800 seconds.



## Authority

Administrators or local user group members with execution rights for this command.

## Usage

- Including RDNSS information in RAs provides DNS server configuration for connected IPv6 hosts without requiring DHCPv6.
- Multiple servers can be configured on the interface by using the command repeatedly.
- A maximum of eight server addresses are allowed.

## Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns server 2001::1 lifetime 400
```

# ipv6 nd ra lifetime

## Syntax

```
ipv6 nd ra lifetime <TIME>
```

```
no ipv6 nd ra lifetime [<TIME>]
```

## Description

Configures the lifetime, in seconds, for the routing switch to be used as a default router by hosts on the current interface.

The `no` form of this command sets lifetime to the default of 1800 seconds.

## Command context

```
config-if
```

## Parameters

**<TIME>**

Specifies lifetime in seconds of a default router. A setting of 0 for default router lifetime in an RA indicates that the routing switch is not a default router on the interface. Range: 0-9000 seconds. Default: 1800 seconds.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

- A given host on an interface refreshes the default router lifetime for a specific router each time the host receives an RA from that router.
- A specific router ceases to be a default router candidate for a given host if the default router lifetime expires before the host is updated with a new RA from the router.

## Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra lifetime 1200
```

## ipv6 nd ra managed-config-flag

### Syntax

```
ipv6 nd ra managed-config-flag

no ipv6 nd ra managed-config-flag
```

### Description

Controls the M flag setting in RAs the router transmits on the current interface. Enable the M flag to indicate that hosts can obtain IP address through DHCPv6. The M flag is disabled by default.

The `no` form of this command turns off (disables) the M flag.

### Command context

config-if

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

- Enabling the M flag directs hosts to acquire their IPv6 addressing for the current interface from a DHCPv6 server.
- When the M-bit is enabled, receiving hosts ignore the O flag setting, which is configured using the command `ipv6 nd ra other-config-flag`.
- When the M-bit is disabled (the default), receiving hosts expect to receive their IPv6 addresses from RA.

| M flag | O flag | Description                                                                                                                                                                          |
|--------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | 0      | Indicates that no information is available via DHCPv6.                                                                                                                               |
| 0      | 1      | Indicates that other configuration information is available via DHCPv6. Examples of such information are DNS-related information or information on other servers within the network. |
| 1      | 0      | Indicates that addresses are available via Dynamic Host Configuration Protocol (DHCPv6).                                                                                             |
| 1      | 1      | If the M flag is set, the O flag is redundant and can be ignored because DHCPv6 will return all available configuration information.                                                 |

## Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra managed-config-flag
```

## ipv6 nd ra max-interval

### Syntax

```
ipv6 nd ra max-interval <TIME>
```

```
no ipv6 nd ra max-interval [<TIME>]
```

### Description

Configures the maximum interval between transmissions of IPv6 RAs on the interface. The interval between RA transmissions on an interface is a random value that changes every time an RA is sent. The interval is calculated to be a value between the current max-interval and min-interval settings.

The `no` form of this command returns the setting to its default, provided the default value is less than the default lifetime value.

### Command context

```
config-if
```

### Parameters

**<TIME>**

Specifies the maximum advertisement time in seconds. Range: 4-1800. Default: 600 seconds.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

- This value has one setting per interface. The setting does not apply to RAs sent in response to a router solicitation received from another device.
- Attempting to set max-interval to a value that is not sufficiently larger than the current min-interval also results in an error message.

### Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra max-interval 30
```

## ipv6 nd ra min-interval

### Syntax

```
ipv6 nd ra min-interval <TIME>
```

```
no ipv6 nd ra min-interval [<TIME>]
```

### Description

Configures the minimum interval between transmissions of IPv6 RAs on the interface. The interval between RA transmissions on an interface is a random value that changes every time an RA is sent. The interval is calculated to be a value between the current max-interval and min-interval settings.

The `no` form of this command returns the setting to its default, provided the default value is less than the current max-interval setting.

## Command context

config-if

## Parameters

<TIME>

Specifies a minimum advertisement time in seconds. Range: 3-1350. Default: 200 seconds.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

- This value has one setting per interface and does not apply to RAs sent in response to a router solicitation received from another device.
- The min-interval must be less than the max-interval. Attempting to set min-interval to a higher value results in an error message.

## Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra min-interval 25
```

# ipv6 nd ra other-config-flag

## Syntax

```
ipv6 nd ra other-config-flag
```

```
no ipv6 nd ra other-config-flag
```

## Description

Controls the O-bit in RAs the router transmits on the current interface; but is ignored unless the M-bit is disabled in RAs. Configure to set the O-bit in RA messages for host to obtain network parameters through DHCPv6. The other-config-flag is disabled by default.

For more information on configuring the M-bit, see `ipv6 nd ra managed-config-flag`.

The `no` form of this command turns off (disables) the setting for this command in RAs.

## Command context

config-if

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

Enabling the O-bit while the M-bit is disabled directs hosts on the interface to acquire their other configuration information from DHCPv6. Examples of such information are DNS-related information or information on other servers within the network.

## Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra other-config-flag
```

## ipv6 nd ra reachable-time

### Syntax

```
ipv6 nd ra reachable-time <TIME>
```

```
no ipv6 nd ra reachable-time [<TIME>]
```

### Description

Sets the amount of time that the interface considers a device to be reachable after receiving a reachability confirmation from the device.

The `no` form of this command sets the reachable time to the default value of 0. (no limit).

### Command context

```
config-if
```

### Parameters

**<TIME>**

Specifies the reachable time in milliseconds. Range: 1000-3600000. Default: 0 (no limit).

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra reachable-time 2000
```

## ipv6 nd ra retrans-timer

### Syntax

```
ipv6 nd ra retrans-timer <TIME>
```

```
no ipv6 nd ra retrans-timer [<TIME>]
```

### Description

Configures the period (retransmit timer) between ND solicitations sent by a host for an unresolved destination, or between DAD neighbor solicitation requests. By default, hosts on the interface use their own locally configured NS-interval settings instead of using the value received in the RAs.

Increase this timer when neighbor solicitation retries or failures are occur, or in a "slow" (WAN) network. The `no` form of this command sets the value to the default of 0.

### Command context

`config-if`

### Parameters

**<TIME>**

Specifies the retransmit timer value in milliseconds. Range: 0 - 4294967295 milliseconds. Default: 0 (Use locally configured NS-interval).

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra retrans-timer 400
```

## ipv6 nd router-preference

### Syntax

`ipv6 nd router-preference {high | medium | low}`

`no ipv6 nd router-preference [high | medium | low]`

### Description

Specifies the value that is set in the Default Router Preference (DRP) field of Router Advertisements (RAs) that the switch sends from an interface. An interface with a DRP value of high will be preferred by other devices on the network over interfaces with an RA value of medium or low.

The `no` form of this command set the value to the default of medium.

### Command context

`config-if`

### Parameters

**high**

Sets DRP to high.

**medium**

Sets DRP to medium. Default.

**low**

Sets DRP to low.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd router-preference high
```

## ipv6 nd suppress-ra

### Syntax

```
ipv6 nd suppress-ra [<SUPPRESS-OPTION>]

no ipv6 nd ra supress-ra [<SUPPRESS-OPTION>]
```

### Description

Configures suppression of IPv6 Router Advertisement transmissions on an interface. The **no** form of this command restores transmission of IPv6 Router Advertisement and options.

### Command context

config-if

### Parameters

**suppress-ra** [<SUPPRESS-OPTION>]

Specifies suppressing RA transmissions. Entering **suppress-ra** without any options, suppresses all RA messages (default). Or you can enter one of the following options.

#### **dnssl**

Specifies suppressing DNSSL options in RA messages.

#### **mtu**

Specifies suppressing MTU options in RA messages.

#### **rdnss**

Specifies suppressing RDNSS options in RA messages.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd suppress-ra mtu dnssl rdnss
switch(config-if-vlan)# no ipv6 nd suppress-ra mtu dnssl rdnss
```

## show ipv6 nd global traffic

### Syntax

```
show ipv6 nd global traffic [vsx-peer]
```

### Description

Displays IPV6 Neighbor Discovery traffic details on a device.

## Command context

Operator (>) or Manager (#)

## Parameters

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

```
switch# show ipv6 nd global traffic
ICMPv6 packet Statistics (sent/received)
 Total Messages : 18/0
 Error Messages : 0/0
 Destination Unreachables : 0/0
 Time Exceeded : 0/0
 Parameter Problems : 0/0
 Echo Request : 0/0
 Echo Replies : 0/0
 Redirects : 0/0
 Packet Too Big : 0/0
 Router Advertisements : 4/0
 Router Solicitations : 0/0
 Neighbor Advertisements : 0/0
 Neighbor Solicitations : 3/0
 Duplicate router RA received : 0/0
ICMPv6 MLD Statistics (sent/received)
 V1 Queries : 0/0
 V2 Queries : 0/0
 V1 Reports : 0/0
 V2 Reports : 11/0
 V1 Leaves : 0/0
```

# show ipv6 nd interface

## Syntax

```
show ipv6 nd interface [<IF-NAME> | all-vrfs | vrf <VRF-NAME>]
 [vsx-peer]
```

## Description

Displays neighbor discovery information for an interface. If no options are specified, displays information for the default VRF.

## Command context

Operator (>) or Manager (#)



## Parameters

### <IF-NAME>

Displays information about the specified IPv6 enabled interface.

### all-vrfs

Displays information about interfaces in all VRFs.

### vrf <VRF-NAME>

Displays information about interfaces in a particular VRF. Or, if <VRF-NAME> is not specified, information for the default VRF is displayed.

### [vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing information for all VRFs:

```
switch# show ipv6 nd interface all-vrfs
```

```
List of IPv6 Interfaces for VRF default
Interface vlan2 is up
 Admin state is up
 IPv6 address:
 IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
 ICMPv6 active timers:
 Last Router-Advertisement sent:
 Next Router-Advertisement sent in:
 Router-Advertisement parameters:
 Periodic interval: 200 to 600 secs
 Router Preference: medium
 Send "Managed Address Configuration" flag: false
 Send "Other Stateful Configuration" flag: false
 Send "Current Hop Limit" field: 64
 Send "MTU" option value: 1500
 Send "Router Lifetime" field: 1800
 Send "Reachable Time" field: 0
 Send "Retrans Timer" field: 0
 Suppress RA: true
 Suppress MTU in RA: true
 ICMPv6 error message parameters:
 Send redirects: false
 ICMPv6 DAD parameters:
 Current DAD attempt: 1
```

```
List of IPv6 Interfaces for VRF red
Interface vlan3 is up
 Admin state is up
 IPv6 address:
 2001::1/64 [VALID]
 IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
 ICMPv6 active timers:
 Last Router-Advertisement sent:
```

```
Next Router-Advertisement sent in:
Router-Advertisement parameters:
 Periodic interval: 200 to 600 secs
 Router Preference: medium
 Send "Managed Address Configuration" flag: false
 Send "Other Stateful Configuration" flag: false
 Send "Current Hop Limit" field: 64
 Send "MTU" option value: 1500
 Send "Router Lifetime" field: 1800
 Send "Reachable Time" field: 0
 Send "Retrans Timer" field: 0
 Suppress RA: true
 Suppress MTU in RA: true
ICMPv6 error message parameters:
 Send redirects: false
ICMPv6 DAD parameters:
 Current DAD attempt: 1
```

Showing information for interface vlan 2:

```
switch# show ipv6 nd interface vlan 2
Interface vlan2 is up
 Admin state is up
 IPv6 address:
 IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
 ICMPv6 active timers:
 Last Router-Advertisement sent:
 Next Router-Advertisement sent in:
 Router-Advertisement parameters:
 Periodic interval: 200 to 600 secs
 Router Preference: high
 Send "Managed Address Configuration" flag: false
 Send "Other Stateful Configuration" flag: false
 Send "Current Hop Limit" field: 64
 Send "MTU" option value: 1500
 Send "Router Lifetime" field: 1800
 Send "Reachable Time" field: 0
 Send "Retrans Timer" field: 0
 Suppress RA: true
 Suppress MTU in RA: true
 ICMPv6 error message parameters:
 Send redirects: false
 ICMPv6 DAD parameters:
 Current DAD attempt: 1
```

Showing information for the default VRF:

```
switch# show ipv6 nd interface

List of IPv6 Interfaces for VRF default
Interface vlan2 is up
 Admin state is up
 IPv6 address:
 2001::1/64 [VALID]
 IPv6 link-local address: fe80::7272:cfff:fee7:a8b9/64 [VALID]
 ICMPv6 active timers:
 Last Router-Advertisement sent: 6 Secs
 Next Router-Advertisement sent in: 7 Secs
 Router-Advertisement parameters:
 Periodic interval: 3 to 13 secs
 Router Preference: medium
 Send "Managed Address Configuration" flag: false
 Send "Other Stateful Configuration" flag: false
 Send "Current Hop Limit" field: 64
```

```
Send "MTU" option value: 1500
Send "Router Lifetime" field: 1900
Send "Reachable Time" field: 0
Send "Retrans Timer" field: 0
Suppress RA: true
Suppress MTU in RA: true
ICMPv6 error message parameters:
Send redirects: false
ICMPv6 DAD parameters:
Current DAD attempt: 1
```

## show ipv6 nd interface prefix

### Syntax

```
show ipv6 nd interface prefix [all-vrfs | vrf <VRF-NAME>] [vsx-peer]
```

### Description

Shows IPv6 prefix information for all VRFs or a specific VRF. If no options are specified, shows information for the default VRF.

### Command context

Operator (>) or Manager (#)

### Parameters

#### **all-vrfs**

Shows prefix information for all VRFs.

#### **vrf <VRF-NAME>**

Name of a VRF.

#### **[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing prefix information for the default VRF:

```
switch# show ipv6 nd interface prefix

List of IPv6 Interfaces for VRF default
List of IPv6 Prefix advertised on vlan2
 Prefix : 4545::/65
 Enabled : Yes
 Validlife time : 2592000
 Preferred lifetime : 604800
 On-link : Yes
 Autonomous : Yes
```

Showing information for VRF red:

```
switch# show ipv6 nd interface prefix vrf red
```

```
List of IPv6 Interfaces for VRF red
List of IPv6 Prefix advertised on vlan3
 Prefix : 2001::/64
 Enabled : Yes
 Validlife time : 2592000
 Preferred lifetime : 604800
 On-link : Yes
 Autonomous : Yes
```

## show ipv6 nd ra dns search-list

### Syntax

```
show ipv6 nd ra dns search-list [vsx-peer]
```

### Description

Displays domain name information on all interfaces.

### Command context

Operator (>) or Manager (#)

### Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns search-list test.com
switch# show ipv6 nd ra dns search-list
Recursive DNS Search List on: 1
 Suppress DNS Search List: Yes
 DNS Search 1: test.com lifetime 1800
```

## show ipv6 nd ra dns server

### Syntax

```
show ipv6 nd ra dns server [vsx-peer]
```

### Description

Displays DNS server information on all interfaces.

### Command context

Operator (>) or Manager (#)

## Parameters

### [vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 nd ra dns server 2001::1
switch# show ipv6 nd ra dns server
Recursive DNS Server List on: 1
 Suppress DNS Server List: Yes
 DNS Server 1: 2001::1 lifetime 1800
```

## diag-dump irdp basic

### Syntax

```
diag-dump irdp basic
```

### Description

Displays diagnostic information for IRDP.

### Command context

Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# diag-dump irdp basic
=====
[Start] Feature irdp Time : Thu Jun 8 09:50:28 2017
=====

[Start] Daemon hpe-rdiscd

Interface: 1/1/1 (state : Up)
rdis ipv4 (enabled: 0, max:600, min:450, hold:1800, pref:0, isBcast:0)
Router IPs - 192.168.1.2,
Interface: 1/1/2 (state : Up)
rdis ipv4 (enabled: 0, max:600, min:450, hold:1800, pref:0, isBcast:0)
Router IPs - 192.168.2.2,

[End] Daemon hpe-rdiscd

[End] Feature irdp
=====
Diagnostic dump captured for feature irdp
```

## ip irdp

### Syntax

```
ip irdp [broadcast | multicast]
```

```
no ip irdp
```

## Description

Enables IRDP on an interface and specifies the packet type that is used to send advertisements. By default, the packet type is set to `multicast`. IRDP is only supported on layer 3 interfaces.

The `no` form of this command disables IRDP on an interface.

## Command context

`config-if`

## Parameters

### **broadcast**

Advertisements are sent as broadcast packets to IP address 255.255.255.255.

### **multicast**

Advertisements are sent as multicast packets to the multicast group with IP address 24.0.0.1. Default.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling IRDP on interface `1/1/1` with packet type set to the default value (multicast).

```
switch(config)# interface 1/1/1
switch(config-if)# ip irdp
```

Enabling IRDP on interface `1/1/1` with packet type set to **broadcast**.

```
switch(config)# interface 1/1/1
switch(config-if)# ip irdp broadcast
```

Disabling IRDP.

```
switch(config)# interface 1/1/1
switch(config-if)# no ip irdp
```

## ip irdp holdtime

## Syntax

```
ip irdp holdtime <TIME>
```

## Description

Specifies the maximum amount of time the host will consider an advertisement to be valid until a newer advertisement arrives. When a new advertisement arrives, hold time is reset. Hold time must be greater than or equal to the maximum advertisement interval. Therefore, if the hold time for an advertisement expires, the host can reasonably conclude that the router interface that sent the advertisement is no longer available. The default hold time is three times the maximum advertisement interval.

## Command context

`config-if`

## Parameters

**<TIME>**

Specifies the lifetime of router advertisements sent from this interface. Range: 4 to 9000 seconds.  
Default: 1800 seconds.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Setting the hold time for interface 1/1/1 to 5000 seconds:

```
switch(config)# interface 1/1/1
switch(config-if)# ip irdp holdtime 5000
```

# ip irdp maxadvertinterval

## Syntax

```
ip irdp maxadvertinterval <TIME>
```

## Description

Specifies the maximum router advertisement interval.

## Command context

config-if

## Parameters

**<TIME>**

Specifies the maximum time allowed between the sending of unsolicited router advertisements. Range: 4 to 1800 seconds. Default: 600 seconds.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Setting the advertisement interval for interface 1/1/1 to 30 seconds:

```
switch(config)# interface 1/1/1
switch(config-if)# ip irdp maxadvertinterval 30
```

# ip irdp minadvertinterval

## Syntax

```
ip irdp minadvertinterval <TIME>
```

## Description

Specifies the minimum amount of time the switch waits between sending router advertisements. By default, this value is automatically set by the switch to be 75% of the value configured for maximum router advertisement interval. Use this command to override the automatically configured value.



## Command context

config-if

## Parameters

**<TIME>**

Specifies the minimum time allowed between the sending of unsolicited router advertisements. Range: 3 to 1800 seconds. Default: 450 seconds (75% of the default value for maximum router advertisement interval).

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Setting the minimum advertisement interval for interface 1/1/1 to 25 seconds:

```
switch(config)# interface 1/1/1
switch(config-if)# ip irdp minadvertinterval 25
```

# ip irdp preference

## Syntax

ip irdp preference **<LEVEL>**

## Description

Specifies the IRDP preference level. If a host receives multiple router advertisement messages from different routers, the host selects the router that sent the message with the highest preference as the default gateway.

## Command context

config-if

## Parameters

**<LEVEL>**

Specifies the IRDP preference level. Range: -2147483648 to 2147483647. Default: 0.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Setting the IRDP preference level for interface 1/1/1 to 25.

```
switch(config)# interface 1/1/1
switch(config-if)# ip irdp preference 25
```

# show ip irdp

## Syntax

```
show ip irdp [vsx-peer]
```

## Description

Displays IRDP configuration settings.

## Command context

Manager (#)

## Parameters

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch# show ip irdp
```

ICMP Router Discovery Protocol

| Interface | Status   | Advertising<br>Address | Minimum<br>Interval | Maximum<br>Interval | Holdtime | Preference |
|-----------|----------|------------------------|---------------------|---------------------|----------|------------|
| 1/1/1     | Enabled  | multicast              | 6                   | 8                   | 10       | 10         |
| 1/1/2     | Disabled | multicast              | 450                 | 600                 | 1800     | 0          |
| 1/1/3     | Enabled  | broadcast              | 450                 | 600                 | 1800     | 115        |

## accept-lifetime

### Syntax

```
accept-lifetime [start-time <time> <month>/<day>/<year>] {duration {<seconds> | infinite} | end-time <time> <month>/<day>/<year>}
```

### Description

Configures the duration for which the key is valid for receiving packets.

The `no` form of this command configures the key packet receiving duration to the default value of an infinite time.

### Command context

```
config-keychain-key
```

### Parameters

#### start-time

Time at which the key chain lifetime starts. Required. Format: HH:MM:SS

#### end-time

Time at which the key chain lifetime expires. Required. Format: HH:MM:SS

#### day

Day of the month. Required. Range: 1-31.

#### month

Month of the year. Required.

#### year

Year. Required. Range: 2020-2050

#### duration

Time in seconds. Optional. Range: 1-2147483646.

#### infinite

Specifies infinite time for the key. Optional.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Configuring the duration for which the key is valid for receiving packets:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020 duration 1000
```

```
switch(config-keychain-key)# accept-lifetime start-time 10:10:10 10/25/2020 duration infinite
switch(config-keychain-key)# accept-lifetime end-time 10:10:10 11/25/2020
switch(config-keychain-key)# accept-lifetime duration 1000
switch(config-keychain-key)# accept-lifetime duration infinite
```

Configuring the key packet receiving duration to the default value of an infinite time:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no accept-lifetime
```

## key

### Syntax

```
key <key-id>
```

### Description

Creates the key for a key chain and enters the key chain key context. A maximum of 64 keys can be configured per key chain.

The `no` form of this command deletes the key from the key chain.

### Command context

```
config-keychain
```

### Parameters

#### **key-id**

ID of the key. Required. Range: 1-255.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Creating a key for a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
```

Deleting a key from a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# no key 1
```

## keychain

### Syntax

```
keychain <keychain-name>
```

## Description

Creates the key chain and enters the key chain context. A maximum of 64 key chains can be configured in the system.

The `no` form of this command removes the key chain if it is not used by any subscribers.

## Command context

`config`

## Parameters

### **keychain-name**

Name of the key chain. Required.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Creating a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
```

Removing a key chain:

```
switch# configure terminal
switch(config)# keychain ospf_keys
```

# key-string

## Syntax

`key-string {ciphertext | plaintext} <key>`

## Description

Sets the password used for the key. The password is always stored in encrypted format in the OVSDb. The key will be considered invalid if the password is not specified for the key.

The `no` form of this command deletes the password used for the key.

## Command context

`config-keychain-key`

## Parameters

### **ciphertext**

Password in encrypted format. Required.

### **plaintext**

Password in simple-text format. Required.

### **key**

Password for key authentication. Required.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Setting the password for the key:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# key-string plaintext key
<cr>
switch(config-keychain-key)# key-string ciphertext ASDFkjdkj
<cr>
```

Deleting the password from the key:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no key-string
<cr>
```

# send-lifetime

## Syntax

```
send-lifetime [start-time <time> <month>/<day>/<year>] {duration {<seconds> | infinite} | end-time <time> <month>/<day>/<year>}
```

## Description

Configures the duration for which the key is valid for sending packets.

The `no` form of this command configures the key packet sending duration to the default value of an infinite time.

## Command context

config-keychain-key

## Parameters

### start-time

Time at which the key chain lifetime starts. Required. Format: HH:MM:SS

### end-time

Time at which the key chain lifetime expires. Required. Format: HH:MM:SS

### day

Day of the month. Required. Range: 1-31.

### month

Month of the year. Required.

### year

Year. Required. Range: 2020-2050

### **duration**

Time in seconds. Optional. Range: 1-2147483646.

### **infinite**

Specifies infinite time for the key. Optional.

### **Authority**

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### **Examples**

Configuring the duration for which the key is valid for sending packets:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020 duration 1000
switch(config-keychain-key)# send-lifetime start-time 10:10:10 10/25/2020 duration infinite
switch(config-keychain-key)# send-lifetime end-time 10:10:10 11/25/2020
switch(config-keychain-key)# send-lifetime duration 1000
switch(config-keychain-key)# send-lifetime duration infinite
```

Configuring the key packet sending duration to the default value of an infinite time:

```
switch# configure terminal
switch(config)# keychain ospf_keys
switch(config-keychain)# key 1
switch(config-keychain-key)# no send-lifetime
```

## **show capacities keychain**

### **Syntax**

```
show capacities keychain
```

### **Description**

Shows the maximum number of key chains and keys configurable in a key chain.

### **Command context**

Operator (>)

### **Parameters**

None

### **Authority**

Administrators or local user group members with execution rights for this command.

### **Example**

```
switch# show capacities keychain
```

```
System Capacities: Filter Keychain
Capacities Name
```

Value

## show keychain

### Syntax

```
show keychain [<keychain-name>]
```

### Description

Shows information about configured and active keys of a named key chain or (if `keychain-name` is not specified) all configured key chains.

### Command context

Operator (>)

### Parameters

#### **keychain-name**

Name of the key chain. Optional.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```
switch# show keychain
Keychain Name : ospf_keys
 Number of Keys : 2
 Active Send Key ID : 7
 Active Recv Key IDs : 7, 200

 Key ID : 7
 Key string : AQBapZ10HiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
 Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
 Recv Key Validity : 00:00:01 10/1/2020 to infinite
 Key ID : 200
 Key string : AQBapZ10HiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
 Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
 Recv Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021

Keychain Name : bgp_keys
 Number of Keys : 2
 Active Send Key ID : 7
 Active Recv Key IDs : 7

 Key ID : 7
 Key string : AQBapZ10HiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
 Send Key Validity : 00:00:01 10/26/2020 to 23:59:01 10/1/2021
 Recv Key Validity : 00:00:01 10/22/2020 to infinite
 Key ID : 8
 Key string : AQBapZ10HiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
 Send Key Validity : 00:00:01 10/1/2021 to 23:59:01 10/1/2021
 Recv Key Validity : 00:00:01 10/1/2021 to 23:59:01 10/1/2021
```



```

Keychain Name : ospf_keys
 Number of Keys : 2
 Active Send Key ID : 7
 Active Recv Key IDs : 7, 200

 Key ID : 7
 Key string : AQBapZ10HiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
 Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
 Recv Key Validity : 00:00:01 10/1/2020 to infinite

 Key ID : 200
 Key string : AQBapZ10HiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
 Send Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021
 Recv Key Validity : 00:00:01 10/1/2020 to 23:59:01 10/1/2021

```

## show running-config keychain

### Syntax

```
show runnning-config keychain
```

### Description

Shows the configurations for key chain protocol.

### Command context

Operator (>)

### Parameters

None

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```

switch# show running-config keychain
keychain ospf_keys
 key 1
 key-string ciphertext AQBapZ10HiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
 accept-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
 send-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
 key 45
 key-string ciphertext AQBapZ10HiO9W3JwRqnjtLfbV73BPLS1S6TGVg+Lzl7N4e5eBAAAAPWaPBE=
 accept-lifetime start-time 10:10:10 10/25/2020 end-time 10:10:10 11/25/2020
 key 33
switch#

```

## downshift enable

### Syntax

```
downshift-enable
```

```
no downshift-enable
```

### Description

Enables/disables automatic speed downshift on an interface that supports downshift, generally 1GBASE-T ports. When enabled, downshift allows an interface to link at a lower advertised speed when unable to establish a stable link at the maximum speed. Downshifting only applies to physical interfaces that are not members of a LAG and is only available when auto-negotiation is enabled. When only one speed is advertised, downshift will not be triggered.

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch(config-if)# interface 1/1/1
switch(config-if)# downshift-enable
```

Warning: this is a non-standard mode for use only when standards-based auto-negotiation is not able to establish a stable link. Enabling this may cause the port to link at a lower than expected speed and should not be used on ports that are members of a LAG. Support calls may require this feature to be disabled

Continue (y/n)?

```
switch(config-if)#
```

When automatic downshift is enabled:

```
switch(config-if)# show running-config interface
interface 1/1/1
 downshift-enable
```

Disabling automatic speed downshift:

```
switch(config-if)# interface 1/1/1
switch(config-if)# no downshift-enable
```

# show interface

## Syntax

```
show interface [<IFNNAME>|<IFRANGE>] [brief | physical | extended [non-zero]]
```

```
show interface [lag | loopback | tunnel | vlan] [<ID>] [brief | physical]
```

```
show interface [lag | loopback | tunnel | vlan] [<ID>] [extended | non-zero]
```

## Description

Displays active configurations and operational status information for interfaces.

## Command context

config

## Parameters

**<IFNAME>**

Specifies a interface name.

**<IFRANGE>**

Specifies the port identifier range.

**brief**

Displays brief info in tabular format.

**extended**

Displays the physical connection info in tabular format.

**non-zero**

Displays only non zero statistics.

**LAG**

Displays LAG interface information

**LOOPBACK**

Displays loopback interface information.

**TUNNEL**

Displays tunnel interface information.

**VLAN**

Displays VLAN interface information.

**<LAG-ID>**

Specifies the LAG number. Range: 1-256

**<LOOPBACK-ID>**

Specifies the LOOPBACK number. Range: 0-255

**<TUNNEL-ID>**

Specifies the tunnel ID. Range: 1-255

**<VLAN-ID>**

Specifies the VLAN ID. Range: 1-4094

## VXLAN

Displays the VXLAN interface information.

### <VXLAN-ID>

Specifies the VXLAN interface identifier. Default: 1

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing interfaces configured for L2 forwarding:

```
switch(config-if)# show interface
Interface 1/1/1 is down (Administratively down)
Admin state is down
State information: Administratively down
Link transitions: 2
Description: Backup data center link
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 9198
Type SFP-SX
Full-duplex
qos trust none
Speed 0 Mb/s
Auto-negotiation is on
Flow-control: off
Error-control: off
VLAN Mode: access
Access VLAN: 1
Rx
 1386055 total packets 586397526 total bytes
 1374287 unicast packets
 11764 multicast packets
 4 broadcast packets
 0 errors 0 dropped
 0 CRC/FCS
Tx
 2475612 total packets 1003506711 total bytes
 2311806 unicast packets
 100369 multicast packets
 63437 broadcast packets
 0 errors 2462773 dropped
 0 collision
Interface 1/2/1 is down (Administratively down)
Admin state is down
State information: Administratively down
Link transitions: 0
Description:
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 9198
Type QSFP+SR4
Full-duplex
qos trust none
Speed 0 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: access
```

```

Access VLAN: 1
Rx
 0 total packets 0 total bytes
 0 unicast packets
 0 multicast packets
 0 broadcast packets
 0 errors 0 dropped
 0 CRC/FCS
Tx
 0 total packets 0 total bytes
 0 unicast packets
 0 multicast packets
 0 broadcast packets
 0 errors 0 dropped
 0 collision

```

When the interface is currently linked at a downshifted speed:

```

switch(config-if)# show interface 1/1/1

Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active

```

## show interface downshift-enable

### Syntax

```
show interface [<IFNAME>|<IFRANGE>] downshift-enable
```

### Description

Displays speed downshift information, including the interface speed status and configuration.

### Command context

config

### Parameters

**<IFNAME>**

Specifies a interface name.

**<IFRANGE>**

Specifies the port identifier range.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing automatic downshift information:

```

switch(config-if)# show interface downshift-enable

Port Downshift Speed
 Enabled | Active Status | Config

1/1/1 yes yes 100M-FDx auto

```

|       |     |    |          |          |
|-------|-----|----|----------|----------|
| 1/1/2 | yes | no | 1G       | auto     |
| 1/1/3 | yes | no | 100M-FDx | 100M-FDx |
| 1/1/4 | no  | no | --       | auto     |

Showing automatic downshift information on per interface:

```
switch(config-if)# show interface 1/1/2 downshift-enable
```

| Port  | Downshift |        | Speed  |        |
|-------|-----------|--------|--------|--------|
|       | Enabled   | Active | Status | Config |
| 1/1/2 | yes       | no     | 1G     | auto   |

## show running-config interface

### Syntax

```
show running-config interface [<IFNAME>|<IFRANGE>]
```

```
show running-config interface [lag | loopback | tunnel | vlan] [<ID>]
```

### Description

Displays active configurations of various switch interfaces.

### Command context

config

### Parameters

#### <IFNAME>

Specifies a interface name.

#### <IFRANGE>

Specifies the port identifier range.

#### LAG

Specifies LAG interface information

#### LOOPBACK

Specifies loopback interface information.

#### TUNNEL

Specifies tunnel interface information.

#### VLAN

Specifies VLAN interface information.

#### <LAG-ID>

Specifies the LAG number. Range: 1-256.

#### <LOOPBACK-ID>

Specifies the LOOPBACK number. Range: 0-255.

#### <TUNNEL-ID>

Specifies the tunnel ID. Range: 1-255.

**<VLAN-ID>**

Specifies the VLAN ID. Range: 1-4094.

**VXLAN**

Specifies the VXLAN interface information.

**<VXLAN-ID>**

Specifies the VXLAN interface identifier. Default: 1.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing 1/1/2 interface configuration:

```
switch(config-if)# show running-config interface 1/1/2

interface 1/1/2
 no shutdown
 description DC-23
 exit
```

Showing loopback interfaces configured:

```
switch(config-if)# show running-config interface loopback

interface loopback 1
 description lb interface 1
 exit
interface loopback 2
 description lb interface 2
 exit
```

Showing loopback interfaces not configured:

```
switch(config-if)# show running-config interface loopback

No loopback interfaces configured.
```

## description

### Syntax

```
description <TEXT>
```

```
no description <TEXT>
```

### Description

Provides a brief description of the LAG interface. The description text is saved in the configuration of the LAG. It is available even after a reboot.

The `no` form of this command removes the description of the LAG interface from the configuration.

### Command context

```
config-lag-if
```

### Parameters

**<TEXT>**

Specifies the description of the LAG interface.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```
switch(config)# interface lag 10
switch(config-lag-if)# description This LAG is used for an example.
switch(config-lag-if)# show running-config
...
vlan 1
interface lag 10
 description This LAG is used for an example.
interface lag 60
switch(config-lag-if)#
```

## hash

### Syntax

```
hash [l2-src-dst | l3-src-dst | l4-src-dst]
```

### Description

This command controls the selection of an interface in a group of aggregate interfaces. The hash type value helps transmit a frame. This configuration must be done at the LAG interface level.



## Command context

config-lag-if

## Parameters

### 12-src-dst

Specifies the load-balancing calculation to include only Layer 2 items, such as source and destination MAC addresses.

### 13-src-dst

Specifies the load-balancing calculation to include only Layer 3 items, such as source and destination IP addresses. Default setting.

### 14-src-dst

Specifies the load-balancing calculation to include only Layer 4 items, such as source and destination UDP/TCP ports.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config-lag-if)# hash 12-src-dst
```

# interface lag

## Syntax

```
interface lag <ID>
```

```
no interface lag <ID>
```

## Description

Creates a Link Aggregation Group (LAG) interface represented by an ID.

The `no` form of this command deletes a LAG interface represented by an ID.

## Command context

config

## Parameters

<ID>

Specifies a LAG interface ID.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

Keep in mind the following requirements when adding interfaces to a LAG:

- To determine the maximum number of LAG interfaces for your type of switch, look at the output from the `show capacities lag` command; however, the number of LAGs that can be created depends on

the availability of the physical interface since each LAG interface needs at least one physical interface as a member link.

- After the maximum limit of members is reached in a LAG, an additional port cannot be added to the aggregation group. If a port belongs to a card type with a different speed than the other aggregation members, the port can still be added to the aggregation group. If dynamic LAG is enabled, any port member with a speed different than other aggregation members is blocked or ineligible from the same aggregation group. Any operational keys/attributes or configuration changes might affect the aggregation states of the member ports.
- The nondefaults configuration on an interface is removed automatically when the interface is added to a link aggregation. For example: Assume that you remove a member interface from an existing LAG and add it to another LAG. The software removes the nondefault configurations on the interface when it is added to the new LAG.

## Examples

Creating a Link Aggregation Group (LAG) interface represented by an ID of 100:

```
switch(config)# interface lag 100
```

Deleting a Link Aggregation Group (LAG) interface represented by an ID of 100:

```
switch(config)# no interface lag 100
```

## lacp hash

*Supported only on the 6100 and 8400 Switch Series.*

### Syntax

```
lacp hash [l2-src-dst | l3-src-dst | l4-src-dst]
```

### Description

This command controls the selection of an interface in a group of aggregate interfaces. The hash type value helps transmit a frame. This configuration must be done at the global level.

### Command context

```
config
```

### Parameters

#### **l2-src-dst**

Specifies the load-balancing calculation to include only Layer 2 items, such as source and destination MAC addresses.

#### **l3-src-dst**

Specifies the load-balancing calculation to include only Layer 3 items, such as source and destination IP addresses.

#### **l4-src-dst**

Specifies the load-balancing calculation to include only Layer 4 items, such as source and destination UDP/TCP ports.

### Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# lacp hash 12-src-dst
```

## lacp mode

### Syntax

```
lacp mode {active | passive}
no lacp mode {active | passive}
```

### Description

Sets an LACP mode to active or passive.

The `no` form of this command sets the LACP mode to `off`, returning the LAG to a static mode aggregation.

### Command context

config-lag-if

### Parameters

#### active

Specifies that the local switch will transmit LACP Data Units (LACPDUs) to attempt to negotiate with the remote device.

#### passive

Specifies that the local switch will listen for LACPDUs from the remote device for LACP negotiation.



**NOTE:** A momentary traffic drop occurs because LACP partners reconverge when changing the mode from active to passive or passive to active.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the LACP mode to `active`:

```
switch(config)# interface lag 1
switch(config-lag-if)# lacp mode active
```

Setting the LACP mode to `off`:

```
switch(config)# interface lag 1
switch(config-lag-if)# no lacp mode active
```

## lacp port-id

### Syntax

```
lacp port-id <PORT-ID>

no lacp port-id
```

## Description

Sets the LACP port ID value of the member interface of the LAG.

The `no` form of this command removes the LACP port ID value from the interface.

## Command context

`config-if`

## Parameters

**<PORT-ID>**

Specifies a port ID value. Range: 1 to 65535.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting an LACP port ID to a value of 10:

```
switch(config-if) # lacp port-id 10
```

Removing the LACP port ID value:

```
switch(config-if) # no lacp port-id
```

# lacp port-priority

## Syntax

`lacp port-priority <PORT-PRIORITY>`

`no lacp port-priority`

## Description

Sets an LACP port priority value for the member interface of the LAG.

The `no` form of this command reverts the LACP port priority to the default, which is 1.

## Command context

`config-if`

## Parameters

**<PORT-PRIORITY>**

Specifies a port priority value. Range: 1 to 65535.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting a LACP port priority value of 10:

```
switch(config-if) # lacp port-priority 10
```

Reverting the LACP port ID to the default:

```
switch(config-if)# no lacp port-priority
```

## lacp rate

### Syntax

```
lacp rate {fast | slow}
no lacp rate
```

### Description

Sets an LACP heartbeat request time to fast or slow.

The `no` form of the command sets an LACP rate to `slow`.

### Command context

```
config-lag-if
```

### Parameters

#### **fast**

Specifies the heartbeat request to every second, and the timeout period is a three-consecutive heartbeat loss that is 3 seconds.

#### **slow**

Specifies the heartbeat request to every 30 seconds. The timeout period is three-consecutive heartbeat loss that is 90 seconds. Default setting.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the LACP heartbeat request time to `fast`:

```
switch(config)# interface lag 1
switch(config-lag-if)# lacp rate fast
```

Resetting the LACP heartbeat request time to the default, which is `slow`:

```
switch(config)# interface lag 1
switch(config-lag-if)# no lacp rate
```

Another way to set the LACP heartbeat request time to the default, which is `slow`:

```
switch(config)# interface lag 1
switch(config-lag-if)# lacp rate slow
```

## lacp system-priority

### Syntax

```
lacp system-priority <SYSTEM-PRIORITY-VALUE>
no lacp system-priority
```

## Description

Sets a Link Aggregation Control Protocol (LACP) system priority.

The `no` form of this command sets an LACP system priority to the default, which is 65534.

## Command context

`config`

## Parameters

**<SYSTEM-PRIORITY-VALUE>**

Specifies a system priority value. Range: 0 to 65535.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting a Link Aggregation Control Protocol (LACP) system priority to 100:

```
switch(config)# lacp system-priority 100
```

Setting an LACP system priority to the default (65534):

```
switch(config)# no lacp system-priority
```

A momentary traffic drop can be seen in case the LACP state machine must renegotiate.

# lag

## Syntax

`lag <ID>`

`no lag <ID>`

## Description

Adds an interface to a specified LAG interface ID.

The `no` form of this command removes an interface from a specified LAG interface ID. The member loses its LACP configuration when removed from the LAG. The member also reaches the default state with an administrative shutdown. For 6300 and 6400 series switches, the administrative state is enabled. Configurations, such as MTU and UDLD, are retained.

## Command context

`config-if`

## Parameters

**<ID>**

Specifies a LAG interface ID. Range: 1 to 256.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

- All members of the LAG must have the same speed. If a member comes up late with a different speed, it will not participate in the LAG/LACP. The hardware restriction is applied before adding an interface to LAG. The member belongs to the card type that has the same maximum speed as the reference port card type.
- To move an interface from LagA to LagB, first remove the interface from LagA and then add it to LagB. When a member is attached to a LAG, the nondefault configurations on the member are removed silently.
- After removing a physical interface from a LAG, the interface associated with the LAG becomes L3 ports with default L3 configurations and administrative down. For example, suppose interface 1/1/1 was part of LAG 3 and you had administratively enabled the interface. If you later remove interface 1/1/1 from LAG 3, the administrative status automatically changes to down. If you want to use the interface again, you must administratively enable it again.

## Examples

Adding an interface to a Link Aggregation Group (LAG) represented by an ID of 100:

```
switch(config)# interface 1/1/1
switch(config-if)# lag 100
```

Deleting an interface from a Link Aggregation Group (LAG) represented by an ID of 100:

```
switch(config)# interface 1/1/1
switch(config-if)# no lag 100
```

## show interface

### Syntax

```
show interfaces <LAG-NAME> [vsx-peer]
```

### Description

Displays information about a specific LAG.

### Command context

Operator (>) or Manager (#)

### Parameters

<LAG-NAME>

Specifies a LAG name.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Displaying information about LAG 100:

```
switch# show interface lag100
Aggregate lag100 is up
Admin state is up
Description :
MAC Address : 48:0f:cf:af:43:9c
Aggregated-interfaces : 1/1/2
Aggregation-key : 100
Aggregate mode : active
Speed : 2000 Mb/s
L3 Counters: Rx Disabled, Tx Disabled
qos trust none
VLAN Mode: access
Access VLAN: 1
Rx
 20 input packets 2584 bytes
 0 input error 0 dropped
 0 CRC/FCS
Tx
 45 output packets 5658 bytes
 0 input error 4 dropped
 0 collision
```

## show lacp aggregates

### Syntax

```
show lacp aggregates [<LAG-NAME>] [vsx-peer]
```

### Description

Displays all LACP aggregate information configured for all LAGs, or for a specific LAG.

### Command context

Operator (>) or Manager (#)

### Parameters

**<LAG-NAME>**

Optional: Specifies a lag name.

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Displaying LACP aggregate information configured for lag10:

```
switch# show lacp aggregates lag10
```



```
Aggregate-name : lag10
Aggregated-interfaces : 1/1/1 1/1/2
Heartbeat rate : slow
Hash : 13-src-dst
Aggregate mode : active
```

Displaying LACP aggregates:

```
switch# show lacp aggregates
```

```
Aggregate-name : lag1
Aggregated-interfaces : 1/1/27 1/1/28 1/1/29
Heartbeat rate : slow
Hash : 13-src-dst
Aggregate mode : active
```

```
Aggregate-name : lag2
Aggregated-interfaces : 1/1/48
Heartbeat rate : slow
Hash : 12-src-dst
Aggregate mode : passive
```

## show lacp configuration

### Syntax

```
show lacp configuration [vsx-peer]
```

### Description

Displays global LACP configuration.

### Command context

Operator (>) or Manager (#)

### Parameters

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

## show lacp interfaces

### Syntax

```
show lacp interfaces [<IFNAME>] [vsx-peer]
```

## Description

Displays an LACP configuration of the physical interfaces, including VSXs. If an interface name is passed as argument, it only displays an LACP configuration of a specified interface.

## Command context

Operator (>) or Manager (#)

## Parameters

<IFNAME>

Optional: Specifies an interface name.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

This example displays an LACP configuration of the physical interfaces. One of the interfaces has the lacp-block forwarding state. If a VSX switch has loop protect enabled on an interface and a loop occurs, VSX blocks the interface to stop the loop. The forwarding state of the blocked interface is set to lacp-block.

```
switch# show lacp interfaces
```

State abbreviations :

A - Active                      P - Passive                      F - Aggregable    I - Individual  
S - Short-timeout    L - Long-timeout    N - InSync              O - OutofSync  
C - Collecting              D - Distributing  
X - State m/c expired                      E - Default neighbor state

Actor details of all interfaces:

| Intf  | Aggr name | Port id | Port Pri | State  | System-id         | System Pri | Aggr Key | Forwarding State |
|-------|-----------|---------|----------|--------|-------------------|------------|----------|------------------|
| 1/1/1 | lag10     | 17      | 1        | ALFOE  | 70:72:cf:37:a3:5c | 20         | 10       | lacp-block       |
| 1/1/2 | lag128    | 69      | 1        | ALFNCD | 70:72:cf:37:a3:5c | 20         | 128      | up               |
| 1/1/3 | lag128    | 14      | 1        | ALFNCD | 70:72:cf:37:a3:5c | 20         | 128      | up               |
| 1/1/4 | lag128    |         |          |        |                   |            |          | down             |
| 1/1/5 | lag20     |         |          |        |                   |            |          | up               |

Partner details of all interfaces:

| Intf  | Aggr name | Partner Port-id | Port Pri | State  | System-id         | System Priority | Aggr Key |
|-------|-----------|-----------------|----------|--------|-------------------|-----------------|----------|
| 1/1/1 | lag10     | 0               | 65534    | PLFOEX | 00:00:00:00:00:00 | 65534           | 0        |
| 1/1/2 | lag128    | 69              | 1        | PLFNCD | 70:72:cf:8c:60:a7 | 65534           | 128      |
| 1/1/3 | lag128    | 14              | 1        | PLFNCD | 70:72:cf:8c:60:a7 | 65534           | 128      |
| 1/1/4 | lag128    |                 |          |        |                   |                 |          |
| 1/1/5 | lag20     |                 |          |        |                   |                 |          |

Displaying static LAG:

```
switch# show lacp interfaces
```

```
State abbreviations :
```

```
A - Active P - Passive F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync O - OutofSync
C - Collecting D - Distributing
X - State m/c expired E - Default neighbor state
```

```
Actor details of all interfaces:
```

| Intf  | Aggr<br>Name | Port<br>Id | Port<br>Pri | State | System-id | System<br>Pri | Aggr<br>Key | Forwarding<br>State |
|-------|--------------|------------|-------------|-------|-----------|---------------|-------------|---------------------|
| 1/1/1 | lag10        |            |             |       |           |               |             | up                  |
| 1/1/2 | lag10        |            |             |       |           |               |             | up                  |

```
Partner details of all interfaces:
```

| Intf  | Aggr<br>Name | Port<br>Id | Port<br>Pri | State | System-id | System<br>Pri | Aggr<br>Key |
|-------|--------------|------------|-------------|-------|-----------|---------------|-------------|
| 1/1/1 | lag10        |            |             |       |           |               |             |
| 1/1/2 | lag10        |            |             |       |           |               |             |

Displaying an LACP configuration of the 1/1/1 interface:

```
switch# show lacp interfaces 1/1/1
```

```
State abbreviations :
```

```
A - Active P - Passive F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync O - OutofSync
C - Collecting D - Distributing
X - State m/c expired E - Default neighbor state
```

```
Aggregate-name : lag1
```

|                 | Actor             | Partner           |
|-----------------|-------------------|-------------------|
| Port-id         | 28                | 31                |
| Port-priority   | 1                 | 1                 |
| Key             | 1                 | 1                 |
| State           | ALFNCD            | ALFNCD            |
| System-id       | 98:f2:b3:68:40:a0 | 98:f2:b3:68:60:a6 |
| System-priority | 65534             | 65534             |

Displaying an LACP configuration after loop-protect is enabled on the primary VSX switch:

```
switch# show lacp interfaces
```

```
State abbreviations :
```

```
A - Active P - Passive F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync O - OutofSync
C - Collecting D - Distributing
X - State m/c expired E - Default neighbor state
```

```
Actor details of all interfaces:
```

| Intf   | Aggr<br>Name | Port<br>Id | Port<br>Pri | State  | System-ID         | System<br>Pri | Aggr<br>Key | Forwarding<br>State |
|--------|--------------|------------|-------------|--------|-------------------|---------------|-------------|---------------------|
| 1/4/14 | lag1 (mc)    | 206        | 1           | ALFNCD | f8:60:f0:06:49:00 | 65534         | 1           | up                  |
| 1/5/15 | lag2 (mc)    |            |             |        |                   |               |             | down                |

Partner details of all interfaces:

| Intf   | Aggr<br>Name | Port<br>Id | Port<br>Pri | State  | System-ID         | System<br>Pri | Aggr<br>Key |
|--------|--------------|------------|-------------|--------|-------------------|---------------|-------------|
| 1/4/14 | lag1 (mc)    | 130        | 1           | ALFNCD | f8:60:f0:06:87:00 | 65534         | 1           |
| 1/5/15 | lag2 (mc)    |            |             |        |                   |               |             |

Displaying an LACP configuration after loop-protect is enabled on the secondary VSX switch:

```
switch# show lacp interfaces
```

State abbreviations :

A - Active                      P - Passive                      F - Aggregable I - Individual  
S - Short-timeout              L - Long-timeout              N - InSync                      O - OutofSync  
C - Collecting                D - Distributing  
X - State m/c expired                      E - Default neighbor state

Actor details of all interfaces:

| Intf  | Aggr<br>Name | Port<br>Id | Port<br>Pri | State  | System-ID         | System<br>Pri | Aggr<br>Key | Forwarding<br>State |
|-------|--------------|------------|-------------|--------|-------------------|---------------|-------------|---------------------|
| 1/3/2 | lag1 (mc)    | 1130       | 1           | ALFNCD | f8:60:f0:06:49:00 | 65534         | 1           | up                  |
| 1/9/3 | lag2 (mc)    |            |             |        |                   |               |             | down                |

Partner details of all interfaces:

| Intf  | Aggr<br>Name | Port<br>Id | Port<br>Pri | State  | System-ID         | System<br>Pri | Aggr<br>Key |
|-------|--------------|------------|-------------|--------|-------------------|---------------|-------------|
| 1/3/2 | lag1 (mc)    | 131        | 1           | ALFNCD | f8:60:f0:06:87:00 | 65534         | 1           |
| 1/9/3 | lag2 (mc)    |            |             |        |                   |               |             |

## shutdown

### Syntax

shutdown

no shutdown

### Description

Sets every interface in the LAG operationally down.

The **no** form of this command sets every interface operationally up.

### Command context

config-lag-if

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting every interface in the LAG to shutdown:

```
switch(config)# interface lag 1
switch(config-lag-if)# shutdown
```

Resetting every interface in the LAG to the default (up):

```
switch(config)# interface lag 1
switch(config-lag-if)# no shutdown
```

## vlan trunk native

### Syntax

```
vlan trunk native <VLAN-ID>
```

```
no vlan trunk native [<VLAN-ID>]
```

### Description

Assigns a native VLAN ID to a LAG interface.

The `no` form of this command removes a native VLAN from a LAG interface and assigns VLAN ID 1 as its native VLAN.

### Command context

```
config-lag-if
```

### Command context

```
config-if
```

### Parameters

**<VLAN-ID>**

Specifies the number of the VLAN ID to assign. The VLAN ID must exist.

Maximum number of VLANs supported: 512 (6100).

VLAN ID range: 2 to 4094.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

By default, VLAN ID 1 is assigned as the LAG VLAN ID for all LAG interfaces. VLANs can only be assigned to a nonrouted (layer 2) interface or LAG interface.

Only one VLAN ID can be assigned as the native VLAN. For the interface to forward the native VLAN traffic, the interface has to be allowed explicitly by entering `vlan trunk allowed <ID>` where the ID is the native VLAN ID. This setting is also applicable to the physical interface.

### Examples

Configuring a Layer 2 dynamic aggregation group with native VLAN ID 1 assigned to LAG 1:

For 6100 and 6200 switch series:

```
switch(config)# interface lag 1
switch(config-lag-if)# no shutdown
switch(config-lag-if)# lacp mode active
switch(config-lag-if)# vlan trunk native 1
switch(config-lag-if)# vlan trunk allowed 1
```

Configuring a Layer 2 dynamic aggregation group with native VLAN ID 20 assigned to LAG 1:

For 6100 and 6200 switch series:

```
switch(config)# interface lag 1
switch(config-lag-if)# no shutdown
switch(config-lag-if)# lacp mode active
switch(config-lag-if)# vlan trunk native 20
switch(config-lag-if)# vlan trunk allowed 20
```

Removing a native VLAN from LAG 1:

```
switch(config)# interface lag 1
switch(config-lag-if)# no vlan trunk native
```

## clear lldp neighbors

### Syntax

```
clear lldp neighbors
```

### Description

Clears all LLDP neighbor details.

### Command context

Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Clearing all LLDP neighbor details:

```
switch# clear lldp neighbors
```

## clear lldp statistics

### Syntax

```
clear lldp statistics
```

### Description

Clears all LLDP neighbor statistics.

### Command context

Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Clearing all LLDP neighbor statistics:

```
switch# clear lldp statistics
```

## lldp

### Syntax

```
lldp
```

```
no lldp
```

### Description

Enables LLDP support globally on all active interfaces. By default, LLDP is enabled.

The `no` form of this command disables LLDP support globally on all active interfaces. It does not remove any LLDP configuration settings.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling LLDP:

```
switch(config)# lldp
```

Disabling LLDP:

```
switch(config)# no lldp
```

## lldp dot3

### Syntax

```
lldp dot3 {poe | macphy}
```

```
no lldp dot3 {poe | macphy}
```

### Description

Sets the 802.3 TLVs to be advertised. By default, advertisement of both POE and MAC/PHY TLVs is enabled.

The `no` form of this command disables advertisement of 802.3 TLVs.

### Command context

```
config-if
```

### Parameters

**poe**

Specifies advertisement of power over Ethernet data link classification.

**macphy**

Specifies advertisement of media access control and physical layer information.



## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling advertisement of the POE TLV:

```
switch(config-if)# lldp dot3 poe
```

Disabling advertisement of the POE TLV:

```
switch(config-if)# no lldp dot3 poe
```

# lldp holdtime

## Syntax

```
lldp holdtime <TIME>
```

```
no lldp holdtime
```

## Description

Sets the holdtime that is used to calculate the LLDP Time-to-Live value. Time-to-Live defines the length of time that neighbors consider LLDP information sent by this agent as valid. When Time-to-Live expires, the information is deleted by the neighbor. Time-to-live is calculated by multiplying holdtime by the value of lldp timer.

The `no` form of this command sets the holdtime to its default value of 4.

## Command context

```
config
```

## Parameters

**<TIME>**

Specifies the holdtime in seconds. Range: 2 to 10. Default: 4.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the holdtime to 8 seconds:

```
switch(config)# lldp holdtime 8
```

Setting the holdtime to the default value of 4 seconds:

```
switch(config)# no lldp holdtime
```

## lldp management-ipv4-address

### Syntax

```
lldp management-ipv4-address <IPV4-ADDR>
```

```
no lldp management-ipv4-address
```

### Description

Defines the IPv4 management address of the switch which is sent in the management address TLV. One IPv4 and one IPv6 management address can be configured.

If you do not define an LLDP management address, then LLDP uses one of the following (in order):

- IP address of SVI [interface VLAN <vid>]
- Base MAC address of the switch

The `no` form of this command removes the IPv4 management address of the switch.

### Command context

```
config
```

### Parameters

**<IPV4-ADDR>**

Specifies the management address of the switch as an IPv4 format (x.x.x.x), where x is a decimal value from 0 to 255.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the management address to 10.10.10.2:

```
switch(config)# lldp management-ipv4-address 10.10.10.2
```

Removing the management address:

```
switch(config)# no lldp management-ipv4-address
```

## lldp management-ipv6-address

### Syntax

```
lldp management-ipv6-address <IPV6-ADDR>
```

```
no lldp management-ipv6-address
```

### Description

Defines the IPv6 management address of the switch. The management address is encapsulated in the management address TLV.

If you do not define an LLDP management address, then LLDP uses one of the following (in order):

- IP address of SVI [interface VLAN <vid>]
- Base MAC address of the switch

The `no` form of this command removes the IPv4 management address of the switch.

### Command context

`config`

### Parameters

**<IPV6-ADDR>**

Specifies an IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the management address to 2001:db8:85a3::8a2e:370:7334:

```
switch(config)# lldp management-ipv6-address 2001:0db8:85a3::8a2e:0370:7334
```

Removing the management address:

```
switch(config)# no lldp management-ipv6-address
```

## lldp med

### Syntax

```
lldp med [poe [priority-override] | capability | network-policy]
```

```
no med [poe [priority-override] | capability | network-policy]
```

### Description

Configures support for the LLDP-MED TLV. LLDP-MED (media endpoint devices) is an extension to LLDP developed by TIA to support interoperability between VoIP endpoint devices and other networking end-devices. The switch only sends the LLDP MED TLV after receiving a MED TLV from and connected endpoint device.

The `no` form of this command disables support for the LLDP MED TLV.

### Command context

`config-if`

### Parameters

**poe [priority-override]**

Specifies advertisement of power over Ethernet data link classification. The `priority-override` option overrides user-configured port priority for Power over Ethernet. When both `lldp dot3 poe` and `lldp med poe` are enabled, the `lldp dot3 poe3` setting takes precedence. Default: enabled.

### **capability**

Specifies advertisement of supported LLDP MED TLVs. The capability TLV is always sent with other MED TLVs, therefore it cannot be disabled when other MED TLVs are enabled. Default: enabled.

### **network-policy**

Network policy discovery lets endpoints and network devices advertise their VLAN IDs, and IEEE 802.1p (PCP and DSCP) values for voice applications. This TLV is only sent when a voice VLAN policy is present. Default: enabled.

### **Authority**

Administrators or local user group members with execution rights for this command.

### **Examples**

Enabling advertisement of the network policy TLV:

```
switch(config-if) # lldp med network-policy
```

Disabling advertisement of the network policy TLV:

```
switch(config-if) # no lldp med network-policy
```

## **lldp med-location**

### **Syntax**

```
lldp med-location {civic-addr | elin-addr }
```

```
no med-location {civic-addr | elin-addr }
```

### **Description**

Configures support for the LLDP-MED TLV. Supports only civic address and emergency location information number (ELIN). Coordinate-based location is not supported.

The **no** form of this command disables support for the LLDP MED TLV.

### **Command context**

config-if

### **Parameters**

#### **civic-addr**

Configures the LLDP MED civic location TLV.

#### **elin-addr**

Configures support for the LLDP MED emergency location TLV.

### **Authority**

Administrators or local user group members with execution rights for this command.

### **Examples**

Enabling support for the LLDP MED emergency location TLV:

```
switch(config-if) # lldp med-location elin-addr gher
```

Disabling support for the LLDP MED emergency location TLV:

```
switch(config-if)# no lldp med-location elin-addr gher
```

Enabling support for the LLDP MED civic address TLV:

```
switch(config-if)# lldp med-location civic-addr US 1 4 ret 6 tyu 7 tiyuo
```

Disabling support for the LLDP MED civic address TLV:

```
switch(config-if)# no lldp med-location civic-addr US 1 4 ret 6 tyu 7 tiyuo
```

## lldp receive

### Syntax

```
lldp receive
```

```
no lldp receive
```

### Description

Enables reception of LLDP information on an interface. By default, LLDP reception is enabled on all active interfaces.

The `no` form of this command disables reception of LLDP information on an interface.

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling LLDP reception on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp receive
```

Disabling LLDP reception on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp receive
```

## lldp reinit

### Syntax

```
lldp reinit <TIME>
```

```
no lldp reinit
```

### Description

Sets the amount of time (in seconds) to wait before performing LLDP initialization on an interface.

The `no` form of this command sets the reinitialization time to its default value of 2 seconds.

### Command context

config

### Parameters

**<TIME>**

Specifies the reinitialization time in seconds. Range: 1 to 10. Default: 2 seconds.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the reinitialization time to 5 seconds:

```
switch(config)# lldp reinit 5
```

Setting the reinitialization time to the default value of 2 seconds:

```
switch(config)# no lldp reinit
```

## lldp select-tlv

### Syntax

```
lldp select-tlv <TLV-NAME>
```

```
no lldp select-tlv <TLV-NAME>
```

### Description

Selects a TLV that the LLDP agent will send and receive. By default, all supported TLVs are sent and received. The `no` form of this command stops the LLDP agent from sending and receiving a specific TLV.

### Command context

config

### Parameters

**select-tlv <TLV-NAME>**

Specifies the TLV name to send. The following TLV names are supported:

- `management-address`: Selected as follows:
  1. IPv4 or IPV6 management address.
  2. If layer 2, the IP address of the SVI.
  3. Base MAC address of the switch.
- `port-description`: A description of the port.
- `port-vlan-id`: VLAN ID assigned to the port.

- **system-capabilities:** Identifies the primary switch functions that are enabled, such as routing.
- **system-description:** Description of the system, comprised of the following information: hardware serial number, hardware revision number, and firmware version.
- **system-name:** Host name assigned to the switch.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Stopping the LLDP agent from sending the **port-description** TLV:

```
switch(config)# no lldp select-tlv port-description
```

Enabling the LLDP agent to send the **port-description** TLV:

```
switch(config)# lldp select-tlv port-description
```

## lldp timer

### Syntax

```
lldp timer <TIME>
```

```
no lldp timer
```

### Description

Sets the interval (in seconds) at which local LLDP information is updated and TLVs are sent to neighboring network devices by the LLDP agent. The minimum setting for this timer must be four times the value of `lldp txdelay`.

For example, this is a valid configuration:

- `lldp timer = 16`
- `lldp txdelay = 4`

And, this is an invalid configuration:

- `lldp timer = 5`
- `lldp txdelay = 2`



**NOTE:** When copying a saved configuration to the running configuration, the value for `lldp timer` is applied before the value of `lldp txdelay`. This can result in a configuration error if the saved configuration has a value of `lldp timer` that is not four times the value of `lldp txdelay` in the running configuration.

For example, if the saved configuration has the settings:

- `lldp timer = 16`
- `lldp txdelay = 4`

And the running configuration has the settings:

- `lldp timer = 30`
- `lldp txdelay = 7`

Then you will see an error indicating that certain configuration settings could not be applied, and you will have to manually adjust the value of `lldp txdelay` in the running configuration.

The `no` form of this command sets the update interval to its default value of 30 seconds.

### Command context

`config`

### Parameters

**<TIME>**

Specifies the update interval (in seconds). Range: 5 to 32768. Default: 30.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the update interval to 7 seconds:

```
switch(config)# lldp timer 7
```

Setting the update interval to the default value of 30 seconds:

```
switch(config)# no lldp timer
```

## lldp transmit

### Syntax

```
lldp transmit
```

```
no lldp transmit
```

### Description

Enables transmission of LLDP information on specific interface. By default, LLDP transmission is enabled on all active interfaces.

The `no` form of this command disables transmission of LLDP information on an interface.



## Command context

config-if

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling LLDP transmission on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp transmit
```

Disabling LLDP transmission on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp transmit
```

# lldp txdelay

## Syntax

```
lldp txdelay <TIME>
```

```
no lldp txdelay
```

## Description

Sets the amount of time (in seconds) to wait before sending LLDP information from any interface. The maximum value for `txdelay` is 25% of the value of `lldp tx timer`.

The `no` form of this command sets the delay time to its default value of 2 seconds.

## Command context

config

## Parameters

**<TIME>**

Specifies the delay time in seconds. Range: 0 to 10. Default: 2.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the delay time to 8 seconds:

```
switch(config)# lldp txdelay 8
```

Setting the delay time to the default value of 2 seconds:

```
switch(config)# no lldp txdelay
```

# show lldp configuration

## Syntax

show lldp configuration [<INTERFACE-ID>]

## Description

Shows LLDP configuration settings for all interfaces or a specific interface.

## Command context

Manager (#)

## Parameters

<INTERFACE-ID>

Specifies an interface. Format: member/slot/port.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Showing configuration settings for all interfaces:

```
switch# show lldp configuration
```

```
LLDP Global Configuration
```

```
=====
```

```
LLDP Enabled : Yes
LLDP Transmit Interval : 30
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Time Interval : 2
LLDP Trap Enabled : No
```

```
TLVs Advertised
```

```
=====
```

```
Management Address
Port Description
Port VLAN-ID
System Capabilities
System Description
System Name
OUI
```

```
LLDP Port Configuration
```

```
=====
```

| PORT  | TX-ENABLED | RX-ENABLED |
|-------|------------|------------|
| 1/1/1 | Yes        | Yes        |
| 1/1/2 | Yes        | Yes        |
| 1/1/3 | Yes        | Yes        |
| 1/1/4 | Yes        | Yes        |
| 1/1/5 | Yes        | Yes        |
| 1/1/6 | Yes        | Yes        |

|        |     |     |
|--------|-----|-----|
| 1/1/7  | Yes | Yes |
| 1/1/8  | Yes | Yes |
| 1/1/9  | Yes | Yes |
| 1/1/10 | Yes | Yes |
| 1/1/11 | Yes | Yes |
| 1/1/12 | Yes | Yes |
| 1/1/13 | Yes | Yes |
| 1/1/14 | Yes | Yes |
| 1/1/15 | Yes | Yes |
| 1/1/16 | Yes | Yes |

This example shows configuration settings for interface 1/1/1.

```
switch# show lldp configuration 1/1/1
```

```
LLDP Global Configuration
=====
```

```
LLDP Enabled : Yes
LLDP Transmit Interval : 8
LLDP Hold Time Multiplier : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Time Interval : 2
```

```
LLDP Port Configuration
=====
```

| PORT  | TX-ENABLED | RX-ENABLED |
|-------|------------|------------|
| 1/1/1 | Yes        | Yes        |

## show lldp local-device

### Syntax

```
show lldp local-device
```

### Description

Shows global LLDP information advertised by the switch, as well as port-based data. If VLANs are configured on any active interfaces, the VLAN ID is only shown for trunk native or untagged VLAN IDs on access interfaces.

### Command context

Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Showing global LLDP information only:

```
switch# show lldp local-device
```

```
Global Data
=====
```

```
Chassis-ID : 88:3a:30:47:c1:c0
System Name : 6100
```

```
System Description : Aruba JL679A PL.10.06.0001-346-g56a12a8f4cf15
Management Address : 88:3a:30:47:c1:c0
Capabilities Available : Bridge, Router
Capabilities Enabled : Bridge, Router
TTL : 120
```

Showing all ports except 1/1/11 as administratively down:

```
switch# show lldp local-device
```

Global Data

=====

```
Chassis-ID : 88:3a:30:47:c1:c0
System Name : 6100
System Description : Aruba JL679A PL.10.06.0001-346-g56a12a8f4cf15
Management Address : 11.11.11.11
Capabilities Available : Bridge, Router
Capabilities Enabled : Bridge, Router
TTL : 120
```

Port Based Data

=====

```
Port-ID : 1/1/11
Port-Desc : "1/1/11"
Port Mgmt-Address : 11.11.11.11
Port VLAN ID : 1
Parent Interface : interface 1/1/11
```

```
switch#
```

In this example, all the ports except 1/1/11 are administratively down, and VLAN ID 100 is configured on this access interface.

```
switch# show lldp local-device
```

Global Data

=====

```
Chassis-ID : 1c:98:ec:e3:45:00
System Name : switch
System Description : Aruba
Management Address : 192.168.10.1
Capabilities Available : Bridge, Router
Capabilities Enabled : Bridge, Router
TTL : 120
```

Port Based Data

=====

```
Port-ID : 1/1/11
Port-Desc : "1/1/11"
Port VLAN ID : 100
Parent Interface : interface 1/1/11
```

## show lldp neighbor-info

### Syntax

```
show lldp neighbor-info [<INTERFACE-NAME>]
```

## Description

Displays information about neighboring devices for all interfaces or for a specific interface. The information displayed varies depending on the type of neighbor connected and the type of TLVs sent by the neighbor.

## Command context

Manager (#)

## Parameters

**<INTERFACE-NAME>**

Specifies the interface for which to show information for neighboring devices. Use the format `member/slot/port` (for example, 1/3/1).

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing LLDP information for all interfaces:

```
switch# show lldp neighbor-info
```

```
LLDP Neighbor Information
=====
```

```
Total Neighbor Entries : 1
Total Neighbor Entries Deleted : 5
Total Neighbor Entries Dropped : 0
Total Neighbor Entries Aged-Out : 3
```

| LOCAL-PORT | CHASSIS-ID        | PORT-ID | PORT-DESC | TTL | SYS-NAME |
|------------|-------------------|---------|-----------|-----|----------|
| 1/1/2      | 38:21:c7:5c:df:40 | 1/1/2   | 1/1/2     | 120 | switch   |
| 1/1/3      | f8:60:f0:c9:e0:a0 | 1/1/3   | 1/1/3     | 120 | switch   |

Showing information for interface 1/1/3 when it has only one switch connected as a neighbor:

```
Aruba-6100-Switch2# show lldp neighbor-info 1/1/3
```

```
Port : 1/1/3
Neighbor Entries : 1
Neighbor Entries Deleted : 1
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 1
Neighbor Chassis-Name : 6100
Neighbor Chassis-Description : Aruba JL679A PL.10.06.0001
Neighbor Chassis-ID : 88:3a:30:47:d1:c0
Neighbor Management-Address : 88:3a:30:47:d1:c0
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge, Router
Neighbor Port-ID : 1/1/3
Neighbor Port-Desc : 1/1/3
Neighbor Port VLAN ID : 1
TTL : 120
Neighbor Mac-Phy details
Neighbor Auto-neg Supported : True
Neighbor Auto-neg Enabled : True
```

```
Neighbor Auto-ned Advertised : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU Type : 1000 BASETFD
```

## show lldp neighbor-info detail

### Syntax

```
show lldp neighbor-info detail
```

### Description

Shows detailed LLDP neighbor information for all LLDP neighbor connected interfaces.

### Command context

Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

### Examples

Showing detailed LLDP information for all interfaces:

```
switch# show lldp neighbor-info detail
```

```
LLDP Neighbor Information
```

```
=====
```

```
Total Neighbor Entries : 6
Total Neighbor Entries Deleted : 2
Total Neighbor Entries Dropped : 0
Total Neighbor Entries Aged-Out : 2
```

```

```

```
Port : 1/1/1
Neighbor Entries : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : 6300
Neighbor Chassis-Description : Aruba ...
Neighbor Chassis-ID : 38:11:17:1a:d5:00
Neighbor Management-Address : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge, Router
Neighbor Port-ID : 1/1/4
Neighbor Port-Desc : 1/1/4
Neighbor Port VLAN ID : 1
TTL : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported : true
Neighbor Auto-Neg Enabled : true
Neighbor Auto-Neg Advertised : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type : 1000 BASETFD
```

```

```

```

Port : 1/1/2
Neighbor Entries : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : 6300
Neighbor Chassis-Description : Aruba ...
Neighbor Chassis-ID : 38:11:17:1a:d5:00
Neighbor Management-Address : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge, Router
Neighbor Port-ID : 1/1/5
Neighbor Port-Desc : 1/1/5
Neighbor Port VLAN ID : 1
TTL : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported : true
Neighbor Auto-Neg Enabled : true
Neighbor Auto-Neg Advertised : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type : 1000 BASETFD

```

```

Port : 1/1/3
Neighbor Entries : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : 6300
Neighbor Chassis-Description : Aruba ...
Neighbor Chassis-ID : 38:11:17:1a:d5:00
Neighbor Management-Address : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge, Router
Neighbor Port-ID : 1/1/6
Neighbor Port-Desc : 1/1/6
Neighbor Port VLAN ID : 1
TTL : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported : true
Neighbor Auto-Neg Enabled : true
Neighbor Auto-Neg Advertised : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type : 1000 BASETFD

```

```

Port : 1/1/46
Neighbor Entries : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : 6300
Neighbor Chassis-Description : Aruba ...
Neighbor Chassis-ID : 38:11:17:1a:d5:00
Neighbor Management-Address : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled : Bridge, Router
Neighbor Port-ID : 1/1/19
Neighbor Port-Desc : 1/1/19
Neighbor Port VLAN ID : 1
TTL : 120

```

```
Neighbor Mac-Phy details
Neighbor Auto-neg Supported : true
Neighbor Auto-Neg Enabled : true
Neighbor Auto-Neg Advertised : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type : 1000 BASETFD
```

---

```
Port : 1/1/47
Neighbor Entries : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : 6300
Neighbor Chassis-Description : Aruba ...
Neighbor Chassis-ID : 38:11:17:1a:d5:00
Neighbor Management-Address : 38:11:17:1a:d5:00
Chassis Cap
```

## show lldp statistics

### Syntax

```
show lldp statistics [<INTERFACE-ID>]
```

### Description

Shows global LLDP statistics or statistics for a specific interface.

### Command context

Manager (#)

### Parameters

**<INTERFACE-ID>**

Specifies an interface. Format: member/slot/port.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Showing global statistics for all interfaces:

```
switch# show copp-policy statistics
switch# show lldp statistics
LLDP Global Statistics
=====

Total Packets Transmitted : 25
Total Packets Received : 20
Total Packets Received And Discarded : 0
Total TLVs Unrecognized : 0

LLDP Port Statistics
=====
```

| PORT-ID | TX-PACKETS | RX-PACKETS | RX-DISCARDED | TLVS-UNKNOWN |
|---------|------------|------------|--------------|--------------|
|---------|------------|------------|--------------|--------------|



|        |    |    |   |   |
|--------|----|----|---|---|
| 1/1/1  | 25 | 20 | 0 | 0 |
| 1/1/2  | 0  | 0  | 0 | 0 |
| 1/1/3  | 0  | 0  | 0 | 0 |
| 1/1/4  | 0  | 0  | 0 | 0 |
| 1/1/5  | 0  | 0  | 0 | 0 |
| 1/1/6  | 0  | 0  | 0 | 0 |
| 1/1/7  | 0  | 0  | 0 | 0 |
| 1/1/8  | 0  | 0  | 0 | 0 |
| 1/1/9  | 0  | 0  | 0 | 0 |
| 1/1/10 | 0  | 0  | 0 | 0 |
| 1/1/11 | 0  | 0  | 0 | 0 |
| 1/1/12 | 0  | 0  | 0 | 0 |
| 1/1/13 | 0  | 0  | 0 | 0 |
| 1/1/14 | 0  | 0  | 0 | 0 |
| 1/1/15 | 0  | 0  | 0 | 0 |
| 1/1/16 | 0  | 0  | 0 | 0 |
| 1/1/17 | 0  | 0  | 0 | 0 |
| 1/1/18 | 0  | 0  | 0 | 0 |
| 1/1/19 | 0  | 0  | 0 | 0 |
| 1/1/20 | 0  | 0  | 0 | 0 |
| 1/1/21 | 0  | 0  | 0 | 0 |
| 1/1/22 | 0  | 0  | 0 | 0 |
| 1/1/23 | 0  | 0  | 0 | 0 |
| 1/1/24 | 0  | 0  | 0 | 0 |
| 1/1/25 | 0  | 0  | 0 | 0 |
| 1/1/26 | 0  | 0  | 0 | 0 |
| 1/1/27 | 0  | 0  | 0 | 0 |
| 1/1/28 | 0  | 0  | 0 | 0 |

Showing statistics for interface 1/1/1:

```
switch# show lldp statistics 1/1/1
```

```
LLDP Statistics
=====
```

```
Port Name : 1/1/1
Packets Transmitted : 159
Packets Received : 163
Packets Received And Discarded : 0
Packets Received And Unrecognized : 0
```

## show lldp tlv

### Syntax

```
show lldp tlv
```

### Description

Shows the LLDP TLVs that are configured for send and receive.

### Command context

Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch# show lldp tlv
```

```
TLVs Advertised
=====
```

```
Management Address
Port Description
Port VLAN-ID
System Capabilities
System Description
System Name
```

## aaa accounting all-mgmt

### Syntax

```
aaa accounting all-mgmt <CONNECTION-TYPE> start-stop {local | group <GROUP-LIST>}
```

```
no aaa accounting all-mgmt <CONNECTION-TYPE>
```

### Description

Defines accounting as being local (with the name `local`) (the default). Or defines a sequence of remote AAA server groups to be accessed for accounting purposes.

For remote accounting, the information is sent to the first reachable remote server that was configured with this command for remote accounting. If no remote server is reachable, local accounting remains available. Each available connection type (channel) can be configured individually as either local or using remote AAA server groups. All server groups named in your command, must exist. This command can be issued multiple times, once for each connection type. Local is always available for any connection type not configured for remote accounting.



**NOTE:** The system accounting log is not associated with any connection type (channel) and is therefore sent to the accounting method configured on the default connection type (channel) only.

The `no` form of this command removes for the specified connection type, any defined remote AAA server group accounting sequence. Local accounting is available for connection types without a configured remote AAA server group list (whether default or for the specific connection type).

### Command context

`config`

### Parameters

**<CONNECTION-TYPE>**

One of these connection types (channels):

#### **default**

Defines a list of accounting server groups to be used for the `default` connection type. This configuration applies to all other connection types (`console`, `https-server`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa accounting all-mgmt console...` to define the console accounting list, then this default configuration is used for console.

#### **console**

Defines a list of accounting server groups to be used for the `console` connection type.

#### **https-server**

Defines a list of accounting server groups to be used for the `https-server` (REST, Web UI) connection type.

## ssh

Defines a list of accounting server groups to be used for the `ssh` connection type.

## start-stop

Selects accounting information capture at both the beginning and end of a process.

## local

Selects local-only accounting when used without the `group` parameter.

## group <GROUP-LIST>

Specifies the list of remote AAA server group names. Each name can be specified one time. Predefined remote AAA group names `tacacs` and `radius` are available. Although not a group name, predefined name `local` is available. User-defined TACACS+ and RADIUS server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa group server` and servers are added to a server group with the command `server`.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

Local accounting is always active. It cannot be turned off.

## Examples

Setting local accounting for the default connection type:

```
switch(config)# aaa accounting all-mgmt default start-stop local
```

Setting local accounting for the console connection type:

```
switch(config)# aaa accounting all-mgmt console start-stop local
```

# aaa authentication limit-login-attempts

## Syntax

```
aaa authentication limit-login-attempts <ATTEMPTS> lockout-time <LOCKOUT-TIME>
```

```
no aaa authentication limit-login-attempts
```

## Description

Enables local login attempt limiting. If the number of failed local login attempts equals the configured threshold, the user is locked out for the configured duration.

The `no` form of this command disables local login attempt limits.



**NOTE:** This local login attempt limiting feature is only available when not using remote authentication through AAA servers (TACACS+ or RADIUS).

## Command context

config

## Parameters

### <ATTEMPTS>

Specifies the threshold of failed local login attempts that triggers user logout. Range: 1 to 10. For example, if <ATTEMPTS> is set to 1, a single failed login attempt triggers immediate user logout.

### <LOCKOUT-TIME>

Specifies the amount of time a user is locked out. Range: 1 to 3600 seconds.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling local login attempt failure limiting with a 20 second logout being triggered upon the fourth consecutive login attempt failure.

```
switch(config)# aaa authentication limit-login-attempts 4 lockout-time 20
```

Disabling login attempt failure limiting:

```
switch(config)# no aaa authentication limit-login-attempts
```

# aaa authentication login

## Syntax

```
aaa authentication login <CONNECTION-TYPE> {local | group <GROUP-LIST>}
```

```
no aaa authentication login <CONNECTION-TYPE>
```

## Description

Defines authentication as being local (with the name `local`) (the default). Or defines a sequence of remote AAA server groups to be accessed for authentication purposes. Each available connection type (channel) can be configured individually as either local or using remote AAA server groups. All server groups named in your command, must exist. This command can be issued multiple times, once for each connection type. Local is always available for any connection type not configured for remote AAA authentication.

The `no` form of this command removes for the specified connection type, any defined remote AAA server group authentication sequence. Local authentication is available for connection types without a configured remote AAA server group list (whether default or for the specific connection type).

## Command context

```
config
```

## Parameters

### <CONNECTION-TYPE>

One of these connection types (channels):

#### default

Defines a list of authentication server groups to be used for the `default` connection type. This configuration applies to all other connection types (`console`, `https-server`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa authentication login console...` to define the console authentication list, then this default configuration is used for console.

### **console**

Defines a list of authentication server groups to be used for the `console` connection type.

### **https-server**

Defines a list of authentication server groups to be used for the `https-server` (REST, Web UI) connection type.

### **ssh**

Defines a list of authentication server groups to be used for the `ssh` connection type.

### **local**

Selects local-only authentication when used without the `group` parameter.

### **group <GROUP-LIST>**

Specifies the list of remote AAA server group names. Each name can be specified one time. Predefined remote AAA group names `tacacs` and `radius` are available. Although not a group name, predefined name `local` is available. User-defined TACACS+ and RADIUS server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa group server` and servers are added to a server group with the command `server`.

If no AAA server is reachable, local authentication is attempted.

### **Authority**

Administrators or local user group members with execution rights for this command.

### **Examples**

Setting local authentication for the default connection type:

```
switch(config)# aaa authentication login default local
```

Setting local authentication for the console connection type:

```
switch(config)# aaa authentication login console local
```

## **aaa authentication minimum-password-length**

### **Syntax**

```
aaa authentication minimum-password-length <LENGTH>
```

```
no aaa authentication minimum-password-length
```

### **Description**

Enables minimum password length checking. Existing passwords shorter than the minimum length are unaffected. Length checking does not apply to ciphertext passwords. Length checking applies both to local and remote authentication.

The `no` form of this command disables minimum password length checking.

### **Command context**

```
config
```

## Parameters

### <LENGTH>

Specifies the minimum password length. Range: 1 to 32.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling password length checking, with a minimum length of 12.

```
switch(config)# aaa authentication minimum-password-length 12
```

Disabling minimum password length checking:

```
switch(config)# no aaa authentication minimum-password-length
```

# aaa authorization commands

## Syntax

```
aaa authorization commands <CONNECTION-TYPE> {local | none}
aaa authorization commands <CONNECTION-TYPE> group <GROUP-LIST>
```

```
no aaa authorization commands <CONNECTION-TYPE>
```

## Description

Defines authorization as being basic local RBAC (specified as `none`), or as full-fledged local RBAC specified as `local` (the default), or as remote TACACS+ (specified with `group <GROUP-LIST>`). Each available connection type (channel) can be configured individually. All server groups named in the command, must exist. This command can be issued multiple times, once for each connection type.

The `no` form of this command unconfigures authorization for the specified connection type, reverting to the default of `local`.



**NOTE:** Although only TACACS+ servers are supported for remote authorization, local authorization (basic or full-fledged) can be used with remote RADIUS authentication.

## Command context

`config`

## Parameters

### <CONNECTION-TYPE>

One of these connection types (channels):

#### default

Selects the `default` connection type for configuration. This configuration applies to all other connection types (`console`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa authorization commands console...` to define the console authorization list, then this default configuration is used for console.

#### console

Selects the `console` connection type for configuration.

## ssh

Selects the `ssh` connection type for configuration.

### local (the default)

When used alone without `group <GROUP-LIST>`, selects local authorization which can be used to provide authorization for a purely local setup without any remote AAA servers and also for when RADIUS is used for remote Authentication and Accounting but Authorization is local.

When used after `group`, provides for fallback (to full-fledged local authorization) when every server in every specified TACACS+ server group cannot be reached.



**NOTE:** If any TACACS+ server in the specified groups is reachable, but the command fails to be authorized by that server, the command is rejected and local authorization is never attempted. Local authorization is only attempted if every TACACS+ server cannot be reached.

### none

When used alone without `group <GROUP-LIST>`, selects basic local RBAC authorization, for use with the built-in user groups (administrators, operators, auditors).

When used after `group`, provides for fallback (to basic local RBAC authorization) when every server in every specified TACACS+ server group cannot be reached.



**NOTE:** With `none`, for users belonging to user-defined user groups, all commands can be executed regardless of what authorization rules are defined in such groups. For per-command local authorization, use `local` instead.

### group <GROUP-LIST>

Specifies the list of remote AAA server group names. Predefined remote AAA group name `tacacs` is available. User-defined TACACS+ server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa server group` and servers are added to a server group using command `server`.

It is recommended to always include either the special name `local` or `none` as the last name in the group list. If both `local` and `none` are omitted, and no remote AAA server is reachable (or the first reachable server cannot authorize the command), command execution for the current user will not be possible.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the authorization for default to `local`:

```
switch(config)# aaa authorization commands default local
```

Setting the authorization for the SSH interface to `none`:

```
switch(config)# aaa authorization commands ssh none
```



## show aaa accounting

### Syntax

```
show aaa accounting
```

### Description

Shows the accounting configuration per connection type (channel).

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Configuring and then showing local accounting for the default and console connection types:

```
switch(config)# aaa accounting all default start-stop local
switch(config)# aaa accounting all console start-stop local
switch(config)# exit
switch# show aaa accounting
AAA Accounting:
 Accounting Type : all
 Accounting Mode : start-stop
```

Accounting for default channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| local      | 0              |

Accounting for console channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| local      | 0              |

## show aaa authentication

### Syntax

```
show aaa authentication
```

### Description

Shows the authentication configuration per connection type (channel).

### Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

Configuring and then showing local authentication for the default and console connection types (channels):

```
switch(config)# aaa authentication login default local
switch(config)# aaa authentication login console local
switch(config)# exit
switch# show aaa authentication
```

```
AAA Authentication:
 Fail-through : Disabled
 Limit Login Attempts : Not set
 Lockout Time : 300
 Minimum Password Length : Not set
```

Authentication for default channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| local      | 0              |

Authentication for console channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| local      | 0              |

## show aaa authorization

### Syntax

```
show aaa authorization
```

### Description

Shows the authorization configuration per connection type (channel).

### Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

Configuring and then showing full-fledged local RBAC authorization for the default and console connection types (channels):

```
switch(config)# aaa authorization commands default none
switch(config)#
switch(config)# aaa authorization commands console none
switch(config)# exit
```

```
switch#
switch# show aaa authorization
```

Authorization for default channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| none       | 0              |

Authorization for console channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| none       | 0              |

## show ssh authentication-method

### Syntax

```
show ssh authentication-method
```

### Description

Shows the status of the SSH public key method and the local password-based (through SSH client) authentication method.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Showing the authentication methods.

```
switch# show ssh authentication-method
SSH publickey authentication : Enabled
SSH password authentication : Enabled
```

## show user

### Syntax

```
show user <USERNAME> authorized-key
```

### Description

Shows the SSH client public key list for a specified user.

### Command context

Operator (>) or Manager (#)

## Parameters

### <USERNAME>

Specifies the username for which you want to show the SSH client public key list.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Usage

Any user can show their own public key list; however, administrators can also show a public key list of other users.

## Examples

Showing a client public key:

```
switch# show user admin authorized-key

1. Key Type : RSA Key size : 2048
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDMtyMBmmAaF6r1zxf3DZNHSYVHBjhlbBlyAIqQ8DSHK
...
U+aE14UW/ifiukmK67sIHwK+FhhRYwPztQc5pjyOPk128a4pgKQaHCcOF169Z admin@switch
```

Showing two client public keys:

```
switch# show user admin authorized-key

1. Key Type : ECDSA Curve : nistp256
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBEqEFevZ0
...
l76V+D0svdCJ9Wo32zqI9OeAdTJw/eZYp5qknhNgS81HjAI6J/4/kAqdZAJbqQUiCAk= admin@switch

2. Key Type : RSA Key size : 2048
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDXQHrqV7+/GcMdOhr//IRjJkX7TQKupW89j80bL7xq8
...
j8qKuHWSN0/h/HxjzQJuYDVmZN5vG3DhpXbBZU1ZNnchVod13QLCesqA3VLKN admin@switch
```

# ssh password-authentication

## Syntax

```
ssh password-authentication
```

```
no ssh password-authentication
```

## Description

Enables the password-based authentication method for use with SSH clients.

The `no` form of this command disables the password-based authentication method for use with SSH clients.

## Command context

```
config
```

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

The switch ships with password-based authentication (for SSH clients) enabled. The maximum number of password retries is three.

## Examples

Enabling password authentication for use with SSH clients:

```
switch(config)# ssh password-authentication
```

Disabling password authentication for use with SSH clients:

```
switch(config)# no ssh password-authentication
```

## ssh public-key-authentication

### Syntax

```
ssh public-key-authentication
```

```
no ssh public-key-authentication
```

### Description

Enables the SSH public key authentication method. The switch ships with SSH public key authentication enabled.

The `no` form of this command disables the SSH public key authentication method.



**NOTE:** Although SSH public key authentication is enabled by default, it cannot be used until SSH public keys are added with the `user authorized-key` command.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling SSH public key authentication:

```
switch(config)# ssh public-key-authentication
```

Disabling SSH public key authentication:

```
switch(config)# no ssh public-key-authentication
```

## user authorized-key

### Syntax

```
user <USERNAME> authorized-key <PUBKEY>
```

```
no user <USERNAME> authorized-key [<KEYNUM>]
```

## Description

Copies an SSH client public key into the key list. If the key list and the public key do not exist, it creates a list with the public key. If the SSH client public key exists, the command appends the new key to the existing list. The client public key list holds a maximum of 32 client keys.

The `no` form of the command removes either one or all SSH public keys from the key list.

## Command context

`config`

## Parameters

**<USERNAME>**

Specifies the name of the user.

**<PUBKEY>**

Specifies the SSH client public key to be copied into the key list.

**<KEYNUM>**

Specifies the key number. The range is 1 to 32. Use the `show user <USERNAME> authorized-key` command to find the key number associated with the key.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Usage

Each key on the key list has a key identifier. The `show user <USERNAME> authorized-key` command displays the key identifier associated with the key.

Administrators can add and remove the public keys of themselves and other users. Operators can add and remove only their own public keys. If the public key authentication method is enabled, the client public key present is used by the SSH server to authenticate the client. The authentication method reverts to the password authentication method and prompts for a client password when one of the following occurs:

- The client public keys are not present.
- The server does not have the keys enabled.
- The public key method is disabled.

You can either remove all keys or a specific key. Each key on the key list has a key identifier. If you provide the key identifier in this command, the command removes the corresponding key from the list. If you provide no key identifier, the command removes all keys from the key list.

## Examples

Adding a public key:

```
switch(config)#user admin authorized-key ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBEqEFevZ0176V+D0svdCJ9Wo32zqI9OeAIdTJwT/eZYp50qkA
nhZNgS81HBjAI6QJ/4/kAyqdZ9oAjbiqQUiCAk= root@switch
```

Removing all SSH public keys from the list:

```
switch(config)# no user admin authorized-key
```

Removing the specified SSH public key from the list:

```
switch(config)# no user admin authorized-key 2
```

## logrotate maxsize

### Syntax

```
logrotate maxsize <MAX-SIZE>
```

```
no logrotate maxsize
```

### Description

Specifies the maximum allowed log file size.

The `no` form of this command resets the size of the log file to the default (100 MB).

### Command context

```
config
```

### Parameters

**<MAX-SIZE>**

Specifies the allowed size the log file can reach before it is compressed and stored locally or transferred to a remote host. The size is a value in the range of 10- 200 MB, and it cannot exceed 32 MB for transferred files.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

A log file that exceeds either the configured `<MAX-SIZE>` value or the `logrotate period`, triggers rotation for that log file. Log rotation occurs during the next hourly maintenance cycle.

Logs are stored locally (event logs in the `/var/log/event.log` file, and authentication logs in the `/var/log/auth.log` file) or transferred to the configured remote destination target using TFTP.

### Examples

```
switch(config)# logrotate maxsize 32
```

```
switch(config)# no logrotate maxsize
```

## logrotate period

### Syntax

```
logrotate period {daily | hourly | monthly | weekly}
```

```
no logrotate period
```



## Description

Sets the rotate period for the event logs, stored in the `/var/log/event.log` file, and authentication logs, stored in the `/var/log/auth.log` file. Defaults to `daily`.

A log file that exceeds either the `logrotate <MAX-SIZE>` value or the `logrotate period` (whichever happens first), triggers rotation for that log file.

The `no` form of this command resets the log rotation period to the default.

## Command context

`config`

## Parameters

### **daily**

Rotates log files on a daily basis (default) at 1:00 am local time.

### **hourly**

Rotates log files every hour at the first second of the hour.

### **monthly**

Rotates log files monthly on the first day of the month.

### **weekly**

Rotates log files once a week on Sunday.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config)# logrotate period weekly
```

# logrotate target

## Syntax

```
logrotate target {tftp://<IPV4_ADDR> | tftp://<IPV6_ADDR>}
```

```
no logrotate target
```

## Description

Specifies the target remote host Universal Resource Identifier (URI) using TFTP protocol to allow transfer of rotated and compressed files to a remote target. Rotated log files are stored locally (event logs in the `/var/log/event.log` file, and authentication logs in the `/var/log/auth.log` file) or transferred to the configured remote destination target.

The `no` form of this command resets the target to the default, which stores the rotated and compressed log files locally in `/var/log/`.

## Command context

`config`

## Parameters

**<IPV4\_ADDR>**

Specifies an IPv4 IP Address location for log file storage.

**<IPV6\_ADDR>**

Specifies an IPv6 IP Address location for log file storage.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

To transfer rotated log files remotely, use the TFTP protocol only, and make sure that the rotated log file is less than 32 MB in size. Use the Linux TFTP command to transfer the file. The rotated log file is removed from the local path `/var/log/` when the log file is moved to a TFTP server.

## Examples

Setting an IPv4 target:

```
switch(config)# logrotate target tftp://192.168.1.132
```

Setting an IPv6 target:

```
switch(config)# logrotate target tftp://2001:db8:0:1::128
```

Removing a logrotate target :

```
switch(config)# logrotate target tftp://1.1.1.1
switch(config)# do show logrotate
Logrotate configurations :
Period : daily
Maxsize : 10MB
Target : tftp://1.1.1.1

switch(config)# no logrotate target
switch(config)# do show logrotate
Logrotate configurations :
Period : daily
Maxsize : 10MB

switch(config)#
```

# show logrotate

## Syntax

```
show logrotate
```

## Description

Displays logrotate configuration details.

## Command context

Manager (#)

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch# show logrotate
Logrotate configurations :
Period : daily
Maxsize : 100MB
Target : local
switch#
```

## loop-protect

### Syntax

```
loop-protect
```

```
no loop-protect
```

### Description

Enables loop protection on a layer 2 interface or LAG. Loop protection packets are sent/received on the LAG and not the interface which are members of the LAG. Loop protection only works on layer 2 interfaces. If a layer 2 interface is changed to a layer 3 interface, all loop protection configuration settings are lost for that interface.

The `no` form of this command disables loop protection on a layer 2 interface or LAG.

### Command context

```
config-if
```

```
config-lag-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling loop protection on interface `1/1/1`:

```
switch# config
switch(config)# interface 1/1/1
switch(config-if)# loop-protect
```

Enabling loop protection on LAG 25:

```
switch# config
switch(config)# interface lag 25
switch(config-lag-if)# loop-protect
```

## loop-protect action

### Syntax

```
loop-protect action {do-not-disable | tx-disable | tx-rx-disable}
```

```
no loop-protect action {do-not-disable | tx-disable | tx-rx-disable}
```

### Description

Sets the action to be taken when a loop protection packet is received on a port.

If an action is configured after a loop is detected, then the new action only takes effect after the re-enable timer expires. To have the action take effect immediately, disable and then re-enable loop protect.

The `no` form of this command resets the action to the default (`tx-disable`).

### Command context

`config-if`

### Parameters

#### **do-not-disable**

No ports are disabled. On every transmit interval, the loop will be detected and the detection will be reported via an SNMP trap and an event log message.

#### **tx-disable**

The port that transmitted the loop detection packet is disabled. When this setting is enabled, environments with N loops, must have loop protection be configured on at least N-1 ports to have a loop free topology. Default.

#### **tx-rx-disable**

The ports that transmitted and received the loop detection packet are disabled.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```
switch(config-if) # loop-protect action do-not-disable
switch(config-if) # no loop-protect action do-not-disable
```

## loop-protect re-enable-timer

### Syntax

```
loop-protect re-enable-timer <TIME>
```

```
no loop-protect re-enable-timer <TIME>
```

### Description

Configures the time interval after which an interface disabled by loop protection is re-enabled. The loop protection timer is disabled by default.

The `no` form of this command disables the loop protect timer.

### Command context

`config`

### Parameters

**<TIME>**

Specify the number of seconds after which a disabled interface is re-enabled. Range: 15 to 604800.

### Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch# config
switch(config)# loop-protect re-enable-timer 60
```

## loop-protect transmit-interval

### Syntax

```
loop-protect transmit-interval <TIME>

no loop-protect transmit-interval [<TIME>]
```

### Description

Configures the time interval between successive loop protect packets sent on an interface. The `no` form of this command sets the time interval to the default value of 5 seconds.

### Command context

config

### Parameters

<TIME>

Configures the transmit interval in seconds. Range: 5 to 10. Default: 5.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch(config)# loop-protect transmit-interval 10
switch(config)# no loop-protect transmit-interval
```

## loop-protect trap loop-detected

### Syntax

```
loop-protect trap loop-detected

no loop-protect trap loop-detected
```

### Description

Enables sending SNMP traps for loop-protect related events. The `no` form of this command disables sending SNMP traps for loop-protect related events.

### Command context

config

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling the sending of SNMP traps:

```
switch# loop-protect trap loop-detected
```

Disabling the sending of SNMP traps:

```
switch# no loop-protect trap loop-detected
```

## loop-protect vlan

### Syntax

```
loop-protect vlan <VLAN-LIST>
```

```
no loop-protect vlan
```

### Description

Specifies the trunk allowed VLANs on which loop protection packets are sent. By default, loop protection packets are only sent on access VLANs and native VLANs on a port. To send loop protection packets on trunk allowed VLANs, the VLANs must be explicitly added using this command.

Loop protection can be configured on a maximum of VLANs across all interfaces.

The `no` form of this command removes loop protection from all VLANs on the interface.

### Command context

```
config-if
```

### Parameters

**<VLAN-LIST>**

Specifies the number of a single VLAN, or a series of numbers for a range of VLANs, separated by commas (1, 2, 3, 4), dashes (1-4), or both (1-4, 6).

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```
switch(config-if)# loop-protect vlan 2-6,10,15-20
```

## show loop-protect

### Syntax

```
show loop-protect [<INTERFACE-NAME>]
```

### Description

Shows loop protection information for all interfaces or for a specific interface.

### Command context

Operator (>) or Manager (#)

## Parameters

### <INTERFACE-NAME>

Specifies the name of a logical interface on the switch. This can be one of the following:

- An Ethernet interface associated with a physical port. Format: member/slot/port.
- A LAG (link aggregation group). Specify the ID of LAG . For example: lag100.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

```
switch# show loop-protect
```

```
Transmit Interval (sec) : 5
Port Re-enable Timer (sec) : Disabled
Loop Detected Trap : Enabled
```

```
Interface 1/1/1
 Loop-protect enabled : Yes
 Loop-Protect enabled VLANs :
 Action on loop detection : TX disable
 Loop detected count : 0
 Loop detected : No
 Interface status : up
```

```
Interface 1/1/2
 Loop-protect enabled : Yes
 Loop-Protect enabled VLANs :
 Action on loop detection : TX disable
 Loop detected count : 0
 Loop detected : No
 Interface status : up
```

```
switch# show loop-protect 1/1/3
```

```
Status and Counters - Loop Protection Information
```

```
Transmit Interval (sec) : 5
Port Re-enable Timer (sec) : 0
Loop Detected Trap : Disabled
```

```
Interface 1
 Loop-protect enabled : Yes
 Loop-Protect enabled VLANs :
 Action on loop detection : TX disable
 Loop detected count : 0
 Loop detected : No
 Interface status : up
```



## interface loopback

### Syntax

```
interface loopback <INSTANCE>
```

```
no interface loopback <INSTANCE>
```

### Description

Creates a loopback interface and enters loopback configuration mode.

The `no` form of this command deletes a loopback interface.

### Command context

```
config
```

### Parameters

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

```
switch(config)# interface loopback 1
switch(config-loopback-if)#
```

## ip address

### Syntax

```
ip address <IPV4-ADDR/MASK> [secondary]
```

```
no ip address <IPV4-ADDR/MASK> [secondary]
```

### Description

Sets the IPv4 address for a loopback interface.

The `no` form of this command reverses the set of the IPv4 address for a loopback interface.

### Command context

```
config
```

## Parameters

### <IPV4-ADDR>

Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

### <MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

### *secondary*

Indicates that the IPv4 address is a secondary address.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

```
switch(config)# interface loopback 1
switch(config-loopback-if)# ip address 16.93.50.2/24
switch(config-loopback-if)# ip address 20.1.1.1/24 secondary
```

## ipv6 address

## Syntax

ipv6 address <IPV6-ADDR/MASK>

## Description

Sets the IPv6 address for a loopback interface.

## Command context

config

## Parameters

### <IPV6-ADDR>

Specifies an IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

### <MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

```
switch(config)# interface loopback 1
switch(config-loopback-if)# ipv6 address fd00:5708::f02d:4df6/64
```

# show interface loopback

## Syntax

```
show interface loopback [brief | instance <ID>]
```

## Description

This command displays the configuration and status of loopback interfaces.

## Command context

Operator (>) or Manager (#)

## Parameters

### brief

Displays brief information about all configured loopback interfaces.

### instance <ID>

Displays the configuration and status of a loopback interface ID. Range: 1-255

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

```
switch# show interface loopback
```

```
Interface loopback1 is up
IPv4 address 192.168.1.1/24
```

```
Interface loopback2 is up
IPv4 address 182.168.1.1/24
```

```
switch# show interface loopback brief
```

```

Loopback IP Address Status
Interface

```

```
loopback1 10.1.1.1/24 up
loopback1 1111:2222:3333:4444::6666/128 up
```

```
switch# show interface loopback 1
```

```
Interface loopback1 is up
IPv4 address 192.168.1.1/24
```

## clear mac-address

### Syntax

```
clear mac-address {port <PORT-NUM> [vlan <VLAN-ID>] | vlan <VLAN-ID> [port <PORT-NUM>]}
```

### Description

Clears the dynamic learned MAC addresses on the specified port, VLAN, or combination of port and VLAN.

### Command context

Manager (#)

### Parameters

#### <PORT-NUM>

Specifies a physical port on the switch. Format: member/slot/port.

#### <VLAN-ID>

Specifies the number of a VLAN.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Clearing the learned MAC addresses on a port:

```
switch# clear mac-address port 1/1/1
```

Clearing the learned MAC addresses on a combination of a VLAN and a port:

```
switch# clear mac-address port 1/1/1 vlan 20
```

```
switch# clear mac-address vlan 2 port 1/1/3
```

## mac-address-table age-time

### Syntax

```
mac-address-table age-time <SECONDS>
```

```
no mac-address-table age-time [<SECONDS>]
```

### Description

Sets the maximum amount of time a MAC address remains in the MAC address table. When this time expires, the MAC address is removed.

The `no` form of this command resets the MAC aging timer to the default value (300 seconds).

### Command context

`config`

### Parameters

**age-time** *<SECONDS>*

Specifies the MAC address aging time in seconds. Range: 60 to 3600. Default: 300.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```
switch(config)# mac-address-table age-time 120
```

## mac-lockout

### Syntax

`mac-lockout` *<MAC-ADDR>*

`no mac-lockout` *<MAC-ADDR>*

### Description

Locks a MAC address globally on the switch and all VLANs. The switch drops all data packets addressed to or from the given address.

The `no` form of this command unlocks the MAC address globally on the switch and all VLANs.

### Command context

`config`

### Parameters

**<MAC-ADDR>**

Specifies the MAC address.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

MAC lockout is implemented on each switch individually. MAC lockout overrides MAC lockdown, port security (secure MAC), and 802.1X authentication. The MAC lockout feature is not intended to lock broadcast/multicast MAC addresses and switch agent MACs.

A maximum of MAC lockouts can be configured on a switch.

### Example

Enabling MAC lockout:

```
switch(config)# mac-lockout 00:00:00:00:00:01
```

Disabling MAC lockout:

```
switch(config)# no mac-lockout 00:00:00:00:00:01
```

## show mac-address-table

### Syntax

```
show mac-address-table
```

### Description

Shows MAC address table information.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing output when table entries exist:

```
switch# show mac-address-table
MAC age-time : 300 seconds
Number of MAC addresses : 5
```

| MAC Address       | VLAN | Type    | Port  |
|-------------------|------|---------|-------|
| 00:00:00:00:00:05 | 1    | dynamic | 1/1/2 |
| 00:00:00:00:00:06 | 2    | dynamic | 1/1/1 |

Showing output when there are no MAC table entries:

```
switch# show mac-address-table
No MAC entries found.
```

## show mac-address-table address

### Syntax

```
show mac-address-table address <MAC-ADDR>
```

### Description

Shows MAC address table information for a specific MAC address.

### Command context

Operator (>) or Manager (#)

### Parameters

<MAC-ADDR>

Specifies the MAC address.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch# show mac-address-table address 00:00:00:00:00:01
MAC age-time : 300 seconds
Number of MAC addresses : 2
```

| MAC Address       | VLAN | Type    | Port  |
|-------------------|------|---------|-------|
| 00:00:00:00:00:01 | 2    | dynamic | 1/1/1 |
| 00:00:00:00:00:01 | 1    | dynamic | 1/1/1 |

## show mac-address-table count

### Syntax

```
show mac-address-table count
 [dynamic | port <PORT-NUM> | vlan <VLAN-ID>]
```

### Description

Displays the number of MAC addresses.

### Command context

Operator (>) or Manager (#)

### Parameters

#### dynamic

Show the count of dynamically learned MAC addresses.

#### <PORT-NUM>

Specifies a physical port on the switch. Format: member/slot/port.

#### vlan <VLAN-ID>

Specifies the number of a VLAN.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing the number of MAC addresses:

```
switch# show mac-address-table count
Number of MAC addresses : 8
```

Showing the number of dynamically learned MAC addresses:

```
switch# show mac-address-table count dynamic
Number of MAC addresses : 8
```

Showing the number of MAC addresses per physical port on the switch:

```
switch# show mac-address-table count port 1/1/1
Number of MAC addresses : 2
```

Showing the number of MAC addresses per VLAN:

```
switch# show mac-address-table count vlan 100
Number of MAC addresses : 5
```

## show mac-address-table dynamic

### Syntax

```
show mac-address-table dynamic [port <PORT-NUM> | vlan <VLAN-ID>]
```

### Description

Shows MAC address table information about dynamically learned MAC addresses.

### Command context

Operator (>) or Manager (#)

### Parameters

**<PORT-NUM>**

Specifies a physical port on the switch. Format: member/slot/port.

**<VLAN-ID>**

Specifies the number of a VLAN.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing all dynamic MAC address table entries:

```
switch# show mac-address-table dynamic
MAC age-time : 300 seconds
Number of MAC addresses : 2
```

| MAC Address       | VLAN | Type    | Port  |
|-------------------|------|---------|-------|
| 00:00:00:00:00:05 | 1    | dynamic | 1/1/2 |
| 00:00:00:00:00:06 | 2    | dynamic | 1/1/1 |

Showing dynamic MAC address table entries for VLAN 1:

```
switch# show mac-address-table dynamic vlan 1
MAC age-time : 300 seconds
Number of MAC addresses : 1
```

| MAC Address       | VLAN | Type    | Port  |
|-------------------|------|---------|-------|
| 00:00:00:00:00:05 | 1    | dynamic | 1/1/2 |

Showing dynamic MAC address table entries for port 1/1/1:



```
switch# show mac-address-table dynamic port 1/1/1
MAC age-time : 300 seconds
Number of MAC addresses : 1
```

| MAC Address       | VLAN | Type    | Port  |
|-------------------|------|---------|-------|
| 00:00:00:00:00:06 | 2    | dynamic | 1/1/1 |

## show mac-address-table lockout

### Syntax

```
show mac-address-table lockout
```

### Description

Shows MAC lockout table information.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

```
switch# show mac-address-table lockout
Number of MAC lockout addresses :
```

| MAC Address       | Type   |
|-------------------|--------|
| 00:00:00:00:01:10 | static |
| 00:00:00:00:10:03 | static |

## show mac-address-table port

### Syntax

```
show mac-address-table port <PORT-NUM>
```

### Description

Shows the MAC address table entries for the specified port.

### Command context

Operator (>) or Manager (#)

### Parameters

<PORT-NUM>

Specifies a physical port on the switch. Format: member/slot/port.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing the MAC address table entries for port 1/1/1:

```
switch# show mac-address-table port 1/1/1
MAC age-time : 300 seconds
Number of MAC addresses : 1
```

| MAC Address       | VLAN | Type    | Port  |
|-------------------|------|---------|-------|
| 00:00:00:00:00:01 | 2    | dynamic | 1/1/1 |

## show mac-address-table static

### Syntax

```
show mac-address-table static
```

### Description

Shows all statically configured MAC addresses.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

```
switch# show mac-address-table static
Number of MAC addresses : 2
```

| MAC Address       | VLAN | Port  |
|-------------------|------|-------|
| 00:00:00:00:10:02 | 1    | 1/1/1 |
| 00:00:00:00:10:03 | 1    | 1/1/1 |

## show mac-address-table vlan

### Syntax

```
show mac-address-table vlan <VLAN-ID>
```

### Description

Shows MAC addresses learned by or configured on the specified VLAN.

### Command context

Operator (>) or Manager (#)

## Parameters

**vlan** *<VLAN-ID>*

Specifies the VLAN ID.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

```
switch# show mac-address-table vlan 1
MAC age-time : 300 seconds
Number of MAC addresses : 1
```

| MAC Address       | VLAN | Type    | Port  |
|-------------------|------|---------|-------|
| 00:00:00:00:00:01 | 1    | dynamic | 1/1/1 |

# static-mac

## Syntax

**static-mac** *<MAC-ADDR>* **vlan** *<VLAN-ID>* **port** *<PORT-NUM>*

**no static-mac** *<MAC-ADDR>* **vlan** *<VLAN-ID>* **port** *<PORT-NUM>*

## Description

Adds a static MAC address to the MAC address table and associates it with a port or existing VLAN. Static MAC addresses can only be assigned to layer 2 (non-routed) interfaces. Static MAC addresses are not affected by the MAC address aging time.

The **no** form of this command deletes a static MAC address.

## Command context

config

## Parameters

**<MAC-ADDR>**

Specifies a MAC address (xx:xx:xx:xx:xx:xx), where x is a hexadecimal number from 0 to F.

**vlan** *<VLAN-ID>*

Specifies number of an existing VLAN.

**port** *<PORT-NUM>*

Specifies a physical port on the switch. Format: member/slot/port.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config)# static-mac 00:00:00:00:00:01 vlan 1 port 1/1/1
switch(config)# no static-mac 00:00:00:00:00:01 vlan 1 port 1/1/1
```

```
switch(config)# static-mac 00:00:00:00:00:01 vlan 1 port 1/1/2
1/1/2 is not an L2 port

switch(config)# static-mac 00:00:00:00:00:01 vlan 2 port 1/1/1
VLAN 2 not found
```

## default-gateway

### Syntax

```
default-gateway <IP-ADDR>
```

```
no default-gateway <IP-ADDR>
```

### Description

Assigns an IPv4 or IPv6 default gateway to the management interface. An IPv4 default gateway can only be configured if a static IPv4 address was assigned to the management interface. An IPv6 default gateway can only be configured if a static IPv6 address was assigned to the management interface. The default gateway should be on the same network segment.

The `no` form of this command removes the default gateway from the management interface.

### Command context

```
config-if-mgmt
```

### Parameters

**<IP-ADDR>**

Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting a default gateway with the IPv4 address of `198.168.5.1`:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# default-gateway 198.168.5.1
```

Setting an IPv6 address of `2001:DB8::1`:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# default-gateway 2001:DB8::1
```

## ip static

### Syntax

```
ip static <IP-ADDR>/<MASK>
```

```
no ip static <IP-ADDR>/<MASK>
```

## Description

Assigns an IPv4 or IPv6 address to the management interface.

The `no` form of this command removes the IP address from the management interface and sets the interface to operate as a DHCP client.

## Command context

`config-if-mgmt`

## Parameters

### <IP-ADDR>

Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

### <MASK>

Specifies the number of bits in an IPv4 or IPv6 address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 32 for IPv4, and 0 to 128 for IPv6.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting an IPv4 address of `198.51.100.1` with a mask of `24` bits:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# ip static 198.51.100.1/24
```

Setting an IPv6 address of `2001:DB8::1` with a mask of `32` bits:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# ip static 2001:DB8::1/32
```

# nameserver

## Syntax

```
nameserver <PRIMARY-IP-ADDR> [<SECONDARY-IP-ADDR>]
```

```
no nameserver <PRIMARY-IP-ADDR> [<SECONDARY-IP-ADDR>]
```

## Description

Assigns a primary or secondary IPv4 or IPv6 DNS server to the management interface. IPv4 DNS servers can only be configured if a static IPv4 address was assigned to the management interface. IPv6 DNS servers can only be configured if a static IPv6 address was assigned to the management interface. The default gateway should be on the same network segment.

The `no` form of this command removes the DNS servers from the management interface.

## Command context

`config-if-mgmt`

## Parameters

### <PRIMARY-IP-ADDR>

Specifies the IP address of the primary DNS server. Specify the address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

### <SECONDARY-IP-ADDR>

Specifies the IP address of the secondary DNS server. Specify the address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting primary and secondary DNS servers with the IPv4 addresses of 198.168.5.1 and 198.168.5.2 :

```
switch(config)# interface mgmt
switch(config-if-mgmt)# nameserver 198.168.5.1 198.168.5.2
```

Setting primary and secondary DNS servers with the IPv6 addresses of 2001:DB8::1 and 2001:DB8::2:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# nameserver 2001:DB8::1 2001:DB8::2
```

## show interface mgmt

## Syntax

```
show interface mgmt [vsx-peer]
```

## Description

Shows status and configuration information for the management interface.

## Command context

Manager (#)

## Parameters

### [vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

```
switch# show interface mgmt

Address Mode : static
Admin State : up
```

```
Mac Address : 02:42:ac:11:00:02
IPv4 address/subnet-mask : 192.168.1.10/16
Default gateway IPv4 : 192.168.1.1
IPv6 address/prefix : 2001:db8:0:1::129/64
IPv6 link local address/prefix: fe80::7272:cfff:fe80:e485/64
Default gateway IPv6 : 2001:db8:0:1::1
Primary Nameserver : 2001::1
Secondary Nameserver : 2001::2
```



## clear mirror

### Syntax

```
clear mirror [all | <SESSION-ID>]
```

### Description

Clears the mirror statistics for all configured mirror sessions or a specified session

### Command context

Manager (#)

### Parameters

**all**

Specifies all configured sessions.

**<SESSION-ID>**

Specifies a numeric identifier for the session. Range: 1 to 4

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Clearing mirror statistics for all configured mirror sessions:

```
switch# clear mirror all
```

Clearing mirror statistics for mirror session 1:

```
switch# clear mirror 1
```

## comment

### Syntax

```
comment <COMMENT>
```

```
no comment
```

### Description

Specifies a descriptive comment for the mirroring session.

The `no` form of this command removes the comment.

## Command context

config-mirror-<SESSION-ID>

## Parameters

<COMMENT>

A comment string of up to 64 characters composed of letters, numbers, underscores, dashes, spaces, and periods.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

Comments are optional and can be added or removed at any time without affecting the state of the mirroring session.

Adding a comment to a session that already has a comment replaces the existing comment.

## Examples

Adding a comment:

```
switch(config-mirror-3) # comment This Mirror will be removed during next maintenance window
```

Removing the comment for mirroring session 3:

```
switch(config-mirror-3) # no comment
```

# copy tshark-pcap

## Syntax

```
copy tshark-pcap <REMOTE-URL> [vrf <VRF-NAME>]
```

## Description

Copies the tshark capture data to a file on a TFTP or SFTP server.

## Command context

Manager (#)

## Parameters

<REMOTE-URL>

Specifies the capture file on a remote TFTP or SFTP server. The URL syntax is:

```
{tftp:// | sftp://<USER>@} {<IP>|<HOST>} [:<PORT>] [;blocksize=<SIZE>]/<FILE>
```

**vrf <VRF-NAME>**

Specifies the name of a VRF. Default: default.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Copying the capture data to a file on SFTP server 10.0.0.2:

```
switch# copy tshark-pcap sftp://root@10.0.0.2/file.pcap

root@10.0.0.2's password:
Connected to 10.0.0.2.
sftp> put packets.pcap file.pcap
Uploading packets.pcap to /root/file.pcap
packets.pcap 100% 156 219.8KB/s 00:00
Copied successfully.
```

## destination cpu

### Syntax

```
destination cpu
```

```
no destination cpu
```

### Description

The command causes the mirror session to transmit mirrored packets to the switch CPU. This destination may be configured for multiple sessions, however only one such configured session may be active at a given time.

The diagnostic utility Tshark may be used to view and capture packets transmitted to the CPU through this route. Ctrl+C must be entered to terminate a Tshark capture session. More details can be found in the *Supportability Guide*.

The `no` form of this command will immediately stops mirroring traffic to the CPU, but will not remove any sources from the mirror configuration.

### Command context

```
config-mirror-<SESSION-ID>
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring a mirror session with CPU as the destination.

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination cpu
```

Removing the destination entirely.

```
switch(config-mirror-1)# no destination cpu
```

## destination interface

### Syntax

```
destination interface {<INTERFACE-ID>|<LAG-NAME>}
```

```
no destination interface
```

## Description

Configures the specified interface as the destination of the mirrored traffic.

The `no` form of this command immediately disables the mirroring session and removes the specified destination interface from the configuration.

## Command context

`config-mirror-<SESSION-ID>`

## Parameters

**<INTERFACE-ID>**

Specifies a interface. Format: member/slot/port.

**<LAG-NAME>**

Specifies a LAG (link aggregation group) identifier.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

Only one destination is allowed per session.

Configuring a different destination interface in an enabled mirroring session causes all mirrored traffic to use the new destination interface. This action might cause a temporary suspension of mirrored source traffic during the reconfiguration.

## Examples

Configuring a mirroring session and adding an interface as a destination:

```
switch(config)# mirror session 1
switch(config-mirror-1)# destination interface 1/1/1
```

Replacing the existing destination with different interface:

```
switch(config-mirror-1)# destination interface 1/1/12
```

Removing a destination:

```
switch(config-mirror-1)# no destination interface
```

# destination tunnel

## Syntax

## Description

Specifies the tunnel where all mirrored traffic for the session is transmitted. Only one tunnel destination is allowed per session.

You may configure multiple mirror sessions with the same source/destination IP address pair, however, only one of those sessions sharing the same source/destination IP address pair can be enabled at a given time.

ERSPAN is not supported leaving the switch by the OOB port. If VRF management is configured for an ERSPAN session, the session will be in "mirror\_err\_tunnel\_oob\_port\_not\_supported" operation status.

ERSPAN is not supported leaving the switch encapsulated within another tunnel (e.g. GRE IPv4). When the path to the destination IP address will leave via a tunnel, the session will be in "tunnel\_route\_resolution\_not\_populated" operation status.



**NOTE:** The interface/LAG used to transmit ERSPAN packets should not be a source in the same mirror session.

The `no` form of this command will cease the use of the tunnel and disable the session.

### Command context

`config-mirror-<SESSION-ID>`

### Authority

Administrators or local user group members with execution rights for this command.

### Parameters

**<TUNNEL-IPV4-ADDR>**

Specifies the tunnel address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

**<SOURCE-IPv4-ADDR>**

Specifies the source address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

**<DSCP-VALUE>**

Specifies the DSCP value to be carried within the DS field of ERSPAN packet header. Range: 0 to 63.  
Default: 0.

**<VRF-NAME>**

Specifies a VRF name. Default: default.

### Examples

Creating a Mirror Session and adding tunnel destination, source, dscp, and VRF:

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination tunnel 1.1.1.1 source 2.2.2.2 dscp 10 vrf default
```

Replacing the existing tunnel destination:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp 10 vrf default
```

Replacing the existing destination with a different DSCP value:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp 2 vrf default
```

Replacing the existing destination with a different VRF:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp 2 vrf newvrf
```

Removing the destination:

```
switch(config-mirror-1)# no destination tunnel
```

# diagnostic

## Syntax

`diagnostic`

```
diag utilities tshark [file]
diag utilities tshark [delete-file]
```

## Description

Captures packets from a mirror-to-cpu session, and save the most recent 32MB to pcap file which can then be copied and analyzed. When capturing a mirror-to-cpu session to a file, packets will not be dumped to the console.



**NOTE:** The `diagnostic` command must be entered prior to the `diag utilities tshark` command.

Use the `delete-file` form of this command to delete the most recent capture file.

Since `file` and `delete-file` are optional, the behavior of the base command `diag utilities tshark` does **not** save anything to a file, and instead dumps the tshark session to the console until **CTRL + c** is entered.

## Command context

Manager (#)

## Parameters

### **file**

Saves captured packets to a temporary file.

### **delete-file**

Deletes the most recent captured file.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Performing diagnostic:

```
switch# diagnostic

switch# diagnostic utilities tshark file
Inspecting traffic mirrored to the CPU until Ctrl-C is entered
^CEnding traffic inspection.
```

# disable

## Syntax

`disable`

## Description

Disables the mirroring session specified by the current command context.

## Command context

config-mirror-<SESSION-ID>

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

By default, mirroring sessions are disabled.

When a mirroring session is disabled, the `show mirror` command for that session ID shows an `Admin Status of disable` and an `Operation Status of disabled`.

## Example

Disabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# disable
```

# enable

## Syntax

enable

## Description

Enables the mirroring session for the current command context.

## Command context

config-mirror-<SESSION-ID>

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

By default, mirroring sessions are disabled.

When a mirroring session is enabled, the `show mirror` command for that session ID shows an `Admin Status of enable` and an `Operation Status of enabled`.

If sFlow is enabled on an interface and a mirroring session specifies the same interface as the source of received traffic (the source is configured with a direction of `rx` or `both`):

The attempt to enable the mirroring session fails and an error is returned.



**NOTE:** When adding, removing, or changing the configuration of a source interface in an enabled mirroring session, packets from other mirror sources using the same destination interface might be interrupted.

## Example

Configuring and enabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# source interface 1/1/2 rx
switch(config-mirror-3)# destination interface 1/1/3
switch(config-mirror-3)# comment Monitor router port ingress-only traffic
switch(config-mirror-3)# enable
```

## mirror session

### Syntax

```
mirror session <SESSION-ID>
```

```
no mirror session <SESSION-ID>
```

### Description

Creates a mirroring session configuration context or enters an existing mirroring session configuration context.

From this context, you can enter commands to configure and enable or disable the mirroring session.

The `no` form of this command removes an existing mirroring session from the configuration.

### Command context

```
config
```

### Parameters

**<SESSION-ID>**

Specifies the session identifier. Range: 1 to 4

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch(config)# mirror session 1
switch(config-mirror-1)#

switch(config)# mirror session 3
switch(config-mirror-3)#

switch(config)# no mirror session 1
switch(config)#
```

## show mirror

### Syntax

```
show mirror [<SESSION-ID>]
```

### Description

Shows information about mirroring sessions. If `<SESSION-ID>` is not specified, then the command shows a summary of all configured mirroring sessions. If `<SESSION-ID>` is specified, then the command shows detailed information about the specified mirroring session.



## Command context

Operator (>) or Manager (#)

## Parameters

**<SESSION-ID>**

Specifies the session identifier. Range: 1 to 4

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Usage

Admin Status indicates the configured status. Admin Status is one of the following values:

### **enable**

The mirroring session is enabled.

### **disable**

The mirroring session has been configured but not yet enabled, or has been disabled.

Operation Status indicates the status of the mirroring session. Operation Status is one of the following values:

### **dest\_doesnt\_exist**

The configured destination interface is not found in the system. The mirroring session cannot be enabled.

### **destination\_shutdown**

The mirroring session is enabled, but the destination interface is shut down. No traffic can be monitored.

### **disabled**

The mirroring session is disabled and is not in an error condition.

### **enabled**

The mirroring session is enabled.

### **external/driver\_error**

An internal ASIC hardware error occurred.

### **hit\_active\_sessions\_capacity**

The mirroring session could not be enabled because the maximum number of supported mirroring sessions are already enabled.

### **internal\_error**

An invalid parameter was passed to the ASIC software layer.

### **no\_dest\_configured**

The mirroring session does not have a destination interface configured.

### **no\_name\_configured**

A software error occurred. The mirroring session does not have a session ID in its configuration.

#### **null\_mirror**

A software error occurred. The session object reference is invalid.

#### **out\_of\_memory**

The system is out of memory, reboot recommended.

#### **tunnel\_route\_resolution\_not\_populated**

If the destination tunnel IP address is not reachable.

#### **unknown\_error**

An unexpected error occurred.

### **Examples**

Showing summary information about all configured mirroring sessions:

```
switch# show mirror
ID Admin Status Operation Status
--- -
1 enable enabled
2 disable disabled
3 disable disabled
4 enable internal_error
```

Showing detailed information about a single mirroring session:

```
switch# show mirror 3
Mirror Session: 3
Admin Status: disable
Operation Status: disabled
Comment: Monitor router port ingress-only traffic
Source: interface 1/1/2 rx
Destination: interface 1/1/3
Output Packets: 0
Output Bytes: 0
switch#
```

## **source interface**

### **Syntax**

```
source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
```

```
no source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
```

### **Description**

Configures the specified interface (either an Ethernet port or a LAG) as a source of traffic to be mirrored.

The **no** form of this command ceases mirroring traffic from the specified source interface and removes the source interface from the mirroring session configuration.

### **Command context**

```
config-mirror-<SESSION-ID>
```

## Parameters

### <PORT-NUM>

Specifies a physical port on the switch. Use the format `member/slot/port` (for example, 1/3/1).

### <LAG-NAME>

Specifies the identifier for the LAG (link aggregation group).

### <DIRECTION>

Selects the direction of traffic to be mirrored from this source interface. There is no default for this parameter. Valid values are the following:

#### **both**

Mirror both transmitted and received packets.

#### **rx**

Mirror only received packets.

#### **tx**

Mirror only transmitted packets.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

There is a limit of four source interfaces in each direction of a given mirror session. However, there is a practical limit to the amount of traffic that a mirror destination can transmit. For example, mirroring session with multiple 10G sources can overwhelm a single 10G destination.



**NOTE:** When adding, removing, or changing the configuration of a source port in an enabled mirroring session, packets from other mirror sources using the same destination port might be interrupted.

## Examples

Configuring a mirrored traffic source interface:

```
switch(config-mirror-1)# source interface
LAG-NAME Enter a LAG name. For example, lag10
PORT-NUM Enter a port number
```

Creating a mirroring session and configuring a source interface to mirror both transmitted and received packets:

```
switch(config)# mirror session 1
switch(config-mirror-1)# source interface 1/1/1 both
```

Creating a second mirroring session and configuring two source interfaces. One port mirroring only transmitted packets and the other mirroring both transmitted and received packets:

```
switch(config)# mirror session 2
switch(config-mirror-2)# source interface 1/1/3 tx
switch(config-mirror-2)# source interface 1/2/1 both
```

Removing the first source interface:

```
switch(config-mirror-2)# no source interface 1/2/3
```

Configuring a source interface to mirror received packets only:

```
switch(config-mirror-3)# source interface 1/1/2 rx
```

Configuring a source interface to mirror both transmitted and received packets:

```
switch(config-mirror-1)# source interface 1/1/1 both
```

Configuring a LAG as source interface to mirror both transmitted and received packets:

```
switch(config-mirror-4)# source interface lag1 both
```

Stopping the mirroring of received packets from a configured source interface:

```
switch(config-mirror-4)# no source interface lag1 rx
```

## MLD snooping functionality

Multicast Listener Discovery (MLD) snooping optimizes multicast traffic across the network to prevent traffic from flooding ports on a VLAN.

- For example, one of the features of MLD snooping lets you configure the network so that traffic is forwarded only to ports that initiate an MLD request for multicast.
- Another feature of MLD lets you enable a setting so that packets that do not match the configured version are dropped.
- You can also block ports from traffic.

## MLD snooping global configuration commands

```
[no] ipv6 mld snooping [drop-unknown [vlan-shared | vlan-exclusive]]
```

### Syntax

```
[no] ipv6 mld snooping [drop-unknown [vlan-shared | vlan-exclusive]]
```

### Description

This command configures the drop unknown mode. While MLD snooping is enabled, the traffic will be forwarded only to ports that initiate an MLD request for multicast. Drop unknown mode can be a filter across all VLANs (vlan-shared) or per VLAN (exclusive-vlan). The default configuration is vlan-shared.

The `no` form of this command disables drop unknown mode on the switch.

### Command context

config

### Parameters

#### **vlan-shared**

Required: Enable shared VLAN filter on the switch.

#### **vlan-exclusive**

Required: Enable exclusive drop unknown filter per VLAN.

### Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# ipv6 mld snooping drop-unknown vlan-shared
switch(config)# ipv6 mld snooping drop-unknown vlan-exclusive
switch(config)# no ipv6 mld snooping drop-unknown
```

# MLD snooping VLAN configuration commands

## ipv6 mld snooping

### Syntax

```
ipv6 mld snooping {enable | disable}
```

### Description

This command enables or disables MLD snooping on the VLAN.

The **no** form of this command disables all MLD snooping configurations on the VLAN.

### Command context

```
config-vlan-<VLAN-ID>
```

### Parameters

#### **enable**

Required: Enable MLD snooping on the VLAN.

#### **disable**

Required: Disable MLD snooping on the VLAN.

### Authority

Administrators or local user group members with execution rights for this command.

## Example

Enable MLD snooping on VLAN 2:

```
switch(config)# vlan 2
switch(config-vlan)# ipv6 mld snooping enable
switch(config-vlan)# ipv6 mld snooping disable
```

Remove all MLD snooping configurations on VLAN 2:

```
switch(config)# vlan 2
switch(config-vlan)# no ipv6 mld snooping
```

## ipv6 mld snooping fastlearn

### Syntax

```
ipv6 mld snooping fastlearn <port-list>
```

### Description

This command enables the port to learn group information on receiving topology change notification.

The **no** form of this command disables fastlearn on the ports.

## Command context

config

## Parameters

### port-list

Required: 1/1/1-1/1/2, ports to be configured as fastlearn ports.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# ipv6 mld snooping fastlearn 1/1/3
switch(config)# ipv6 mld snooping fastlearn 1/1/1-1/1/2
switch(config)# ipv6 mld snooping fastlearn 1/1/5,1/1/6
```

## ipv6 mld snooping fastleave

## Syntax

ipv6 mld snooping fastleave <port-list>

## Description

Enables the switch to immediately remove an interface from the bridge table upon receiving the leave group message.

The `no` form of this command disables fastleave configuration on the ports.

## Command context

config-vlan

## Parameters

### fastleave <PORT-LIST>

Specifies a list of one or more ports to be configured as fastleave ports. You can specify a single port, a comma-separated list of ports or a range of ports such as 1/1/1-1/1/3.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

MLD fastleave is configured for ports on a per-VLAN basis. By default, the querier sends a MLD Group-Specific Query message out of the interface, upon which the leave group message is received to ensure that no other receivers are connected to the interface. If receivers are directly attached to the switch, it is inefficient to send the membership query as the receiver wanting to leave is the only connected host. Fastleave processing eliminates the MLD Group-Specific Query message. Thus, it allows the switch to immediately remove an interface from the bridge table upon receiving the leave Group message. This processing speeds up the overall leave process and also eliminates the CPU overhead of having to generate an MLD Group-Specific Query message.

## Example

Configure fastleave ports for the VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ipv6 mld snooping fastleave 1/1/1
switch(config-vlan)# ipv6 mld snooping fastleave 1/1/1-1/1/2
switch(config-vlan)# ipv6 mld snooping fastleave 1/1/1,1/1/2-1/1/3
switch(config-vlan)# no ipv6 mld snooping fastleave 1/1/1
switch(config-vlan)# no ipv6 mld snooping fastleave 1/1/1-1/1/2
```

## ipv6 mld snooping forced fastleave

### Syntax

```
ipv6 mld snooping forced fastleave <port-list>
```

### Description

Configures the given ports in forced fastleave mode.

The **no** form of this command disables forced fastleave configuration on the ports.

### Command context

config-vlan

### Parameters

**fastleave** <PORT-LIST>

Specifies a list of one or more ports to be configured as forced fastleave ports. You can specify a single port, a comma-separated list of ports or a range of ports such as 1/1/1-1/1/3.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

With forced fastleave enabled, MLD speeds up the process of blocking unnecessary multicast traffic to a switch port that is connected to multiple end nodes. When a port having multiple end nodes receives a leave group request from one end node for a given multicast group, forced fastleave activates and waits a small amount of time to receive a join request from any other member of the same group on that port. If the port does not receive a join request for that group within the forced fastleave interval, the switch then blocks any further traffic to that group on that port.

### Example

Configure forced-fastleave ports for the VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ipv6 mld snooping forced-fastleave 1/1/1
switch(config-vlan)# ipv6 mld snooping forced-fastleave 1/1/1-1/1/2
switch(config-vlan)# ipv6 mld snooping forced-fastleave 1/1/1,1/1/2-1/1/3
switch(config-vlan)# no ipv6 mld snooping forced-fastleave 1/1/1
switch(config-vlan)# no ipv6 mld snooping forced-fastleave 1/1/1-1/1/2
```

## ipv6 mld snooping [version <ver>]

### Syntax

```
ipv6 mld snooping [version <ver>]
```

### Description

This command configures the MLD snooping version on the VLAN. MLD version 2 is the default.



## Command context

config-vlan-<VLAN-ID>

## Parameters

**ver**

Required: 1-2, MLD snooping version.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# vlan 2
switch(config-vlan)# ipv6 mld snooping version 2
```

## ipv6 mld snooping apply access-list

## Syntax

ipv6 mld snooping apply access-list <ACL-NAME>

no ipv6 mld snooping apply access-list <ACL-NAME>

## Description

Configures the ACL on a particular interface to filter the MLD join or leave packets based on rules set in the particular ACL name.

The **no** form of this command disables the rules set for the ACL.



**NOTE:** This configuration will override the ACL associated with IGMP snooping on the corresponding L2 VLAN.

## Command context

config-vlan

## Parameters

**access-list**

Associates an ACL with the IGMP.

**<ACL-NAME>**

Specifies the name of the ACL.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

Existing classifier commands are used to configure the ACL. In case an MLDv2 packet with multiple group addresses is received, it will only process the permitted group addresses based on the ACL rule set, and any existing joins will time out. If there is no match or if there is a deny rule match, the packet is dropped.



**NOTE:** If the access list is configured for both L2 VLAN and L3 VLAN, the L3 VLAN configuration will be applied.

## Examples

Configuring the ACL to filter MLD packets based on rules set in access list `mygroup`:

```
switch(config)# access-list ipv6 mygroup
switch(config-acl-ip)# permit icmpv6 any ff55::1
switch(config-acl-ip)# exit
switch(config)# interface vlan 2
switch(config-vlan)# ipv6 mld snooping apply access-list mygroup
```

Configuring the ACL to remove the rules set in access list `mygroup`:

```
switch(config-vlan)# no ipv6 mld snooping apply access-list mygroup
```

## ipv6 mld snooping [auto <port-list>]

### Syntax

```
ipv6 mld snooping [auto <port-list>]
```

### Description

This command configures the given ports in auto mode, which is the default port mode.

### Command context

config-vlan-<VLAN-ID>

### Parameters

#### port-list

Required: 1/1/1-1/1/2, ports to be configured as auto ports.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Configure auto ports for the VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ipv6 mld snooping auto 1/1/1
switch(config-vlan)# ipv6 mld snooping auto 1/1/1,1/1/2-1/1/3
```

## ipv6 mld snooping [blocked <port-list>]

### Syntax

```
ipv6 mld snooping [blocked <port-list>]
```

### Description

By default ports are configured in auto mode. This command configures the given ports in blocked mode. The `no` form of this command removes blocked ports.

### Command context

config-vlan-<VLAN-ID>

## Parameters

### port-list

Required: 1/1/1-1/1/2, ports to be configured as blocked ports.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Configure blocked ports for the VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ipv6 mld snooping blocked 1/1/1
switch(config-vlan)# ipv6 mld snooping blocked 1/1/1-1/1/2
switch(config-vlan)# ipv6 mld snooping blocked 1/1/3,1/1/4-1/1/6
switch(config-vlan)# no ipv6 mld snooping blocked 1/1/1
switch(config-vlan)# no ipv6 mld snooping blocked 1/1/1-1/1/2
```

## ipv6 mld snooping [forward <port-list>]

## Syntax

```
ipv6 mld snooping [forward <port-list>]
```

## Description

By default ports are configured in auto mode. This command configures the given ports in forward mode. The `no` form of this command disables forward ports.

## Command context

config-vlan-<VLAN-ID>

## Parameters

### port-list

Required: 1/1/1-1/1/2, ports to be configured as forward ports.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Configure forward ports for the VLAN:

```
switch(config)# vlan 2
switch(config-vlan)# ipv6 mld snooping forward 1/1/1
switch(config-vlan)# ipv6 mld snooping forward 1/1/1-1/1/2
switch(config-vlan)# ipv6 mld snooping forward 1/1/1,1/1/2-1/1/3
switch(config-vlan)# no ipv6 mld snooping forward 1/1/1
switch(config-vlan)# no ipv6 mld snooping forward 1/1/1-1/1/2
```

## ipv6 mld snooping [static-group <X:X::X:X>]

## Syntax

```
ipv6 mld snooping [static-group <X:X::X:X>]
```

## Description

This command configures static multicast group.

The `no` form of this command disables static multicast group.

## Command context

`config-vlan-<VLAN-ID>`

## Parameters

### `static-group`

Required: `<x:x::x:x>`, MLD static multicast group.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# vlan 2
switch(config-vlan)# ipv6 mld snooping static-group ff12::c
switch(config-vlan)# no ipv6 mld snooping static-group ff12::c
```

# MLD snooping show commands

## show ipv6 mld snooping

## Syntax

`show ipv6 mld snooping`

## Description

This command shows MLD snooping configuration details for all VLANs.

## Command context

Manager (#)

## Parameters

None

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ipv6 mld snooping
```

MLD Snooping Protocol Info

```
Total VLANs with MLD enabled : 1
Current count of multicast groups joined : 0
```

```
MLD Drop Unknown Multicast : Global
```

```

VLAN ID : 1
VLAN Name : DEFAULT_VLAN_1
MLD Snooping is not enabled

VLAN ID : 2
VLAN Name : VLAN2
MLD Configured Version : 2
MLD Operating Version : 2
Querier Address [this switch] : fe80::218:71ff:fec4:2f00
Querier Port :
Querier UpTime :0m 21s
Querier Expiration Time :0m 2s

```

## show ipv6 mld snooping [counters]

### Syntax

```
show ipv6 mld snooping [counters]
```

### Description

This command shows MLD snooping query packet Tx, Rx, and Error packet counter details.

### Command context

Manager (#)

### Parameters

#### counters

Optional, show MLD snooping counters.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# show ipv6 mld snooping counters
MLD Snooping VLAN Counters
```

Rx Counters :

```

V1 All Hosts Queries 0
V2 All Hosts Queries 0
V2 Group Specific Queries 0
Group And Source Specific Queries 0
V1 Member Reports 0
V2 Member Reports 0
V1 Member Leaves 0

```

Tx Counters :

```

Flood on vlan 44
V1 Group Specific Queries 0

```

V2 Group Specific Queries 0

#### Errors:

|                                           |   |
|-------------------------------------------|---|
| Unknown Message Type                      | 0 |
| Malformed Packets                         | 0 |
| Bad Checksum                              | 0 |
| Packet received on MLD-disabled Interface | 0 |
| Interface Wrong Version Queries           | 0 |
| Packets dropped by ACL                    | 0 |

#### Port Counters:

|                    |   |
|--------------------|---|
| Membership Timeout | 0 |
|--------------------|---|

switch#

## show ipv6 mld snooping [groups]

### Syntax

```
show ipv6 mld snooping [groups]
```

### Description

This command shows MLD snooping group details for the specified VLAN.

### Command context

Manager (#)

### Parameters

#### groups

Optional, show MLD snooping groups information.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# show ipv6 mld snooping groups
```

MLD Group Address Information

| VLAN | ID | Group Address | Expires | UpTime | Last Reporter | Type   |
|------|----|---------------|---------|--------|---------------|--------|
| 10   |    | ff12::c       | 3m 54s  | 0m 26s | 2001::1       | Filter |
| 10   |    | ff12::d       | 4m 17s  | 0m 3s  | 2001::1       |        |

## show ipv6 mld snooping [statistics]

### Syntax

```
show ipv6 mld snooping [statistics]
```

### Description

This command shows MLD snooping statistics information.

## Command context

Manager (#)

## Parameters

### statistics

Optional, show MLD snooping statistics.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ipv6 mld snooping statistics
MLD Snooping Protocol Info
```

```
Total VLANs with MLD enabled : 1
Current count of multicast groups joined : 2
```

```
MLD Drop Unknown Multicast : Global
```

```
MLD Snooping Joined Groups Statistics
```

| VLAN ID | VLAN Name      | Total | Static | INCLUDE | EXCLUDE |
|---------|----------------|-------|--------|---------|---------|
| 1       | DEFAULT_VLAN_1 | 0     | 0      | 0       | 0       |
| 2       | VLAN2          | 2     | 2      | 0       | 0       |

**show ipv6 mld snooping [vlan <vlan-id> [counters]]**

## Syntax

```
show ipv6 mld snooping [vlan <vlan-id> [counters]]
```

## Description

This command shows MLD snooping protocol information and number of different groups joined for the VLAN.

## Command context

Manager (#)

## Parameters

### vlan-id

Required, 1-4094, shows MLD snooping information.

### counters

Optional, shows MLD query packet Tx, Rx, Error packet counters on a specified VLAN.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ipv6 mld snooping vlan 2 counters
MLD Snooping VLAN Counters

VLAN ID : 2
VLAN Name : VLAN2

Rx Counters :

V1 All Hosts Queries 0
V2 All Hosts Queries 0
V1 Group Specific Queries 0
V2 Group Specific Queries 0
Group And Source Specific Queries 0
V1 Member Reports 0
V2 Member Reports 0
V1 Member Leaves 0

Tx Counters :

Flood on vlan 71
V1 Group Specific Queries 0
V2 Group Specific Queries 0

Errors:

Unknown Message Type 0
Malformed Packets 0
Bad Checksum 0
Packet received on MLD-disabled Interface 0
Interface Wrong Version Queries 0
Packets dropped by ACL 0

Port Counters:

Membership Timeout 0
switch#
```

**show ipv6 mld snooping [vlan <vlan-id> [statistics]]**

## Syntax

```
show ipv6 mld snooping [vlan <vlan-id> [statistics]]
```

## Description

This command shows MLD snooping statistics details for the specified VLAN, including the number of different groups joined for the VLAN.

## Command context

Manager (#)

## Parameters

**vlan-id**

Required, 1-4094, shows MLD query packet Tx, Rx, error packet counters on VLAN.



## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ipv6 mld snooping vlan 2 statistics
MLD Snooping statistics
```

```
VLAN ID : 2
VLAN Name : VLAN2
```

```
Number of Include Groups : 1
Number of Exclude Groups : 0
Number of Static Groups : 1
Total Multicast Groups Joined : 2
```

```
show ipv6 mld snooping [vlan <vlan-id> [group [<group-ip>]
[source <source-ip>]]]
```

## Syntax

```
show ipv6 mld snooping [vlan <vlan-id> [group [<group-ip>] [source <source-ip>]]]
```

## Description

This command shows MLD snooping details for the specified VLAN, including the number of different groups joined for the VLAN.

## Command context

Manager (#)

## Parameters

### **vlan-id**

Required: 1-4094, shows MLD protocol information for the specified VLAN.

### **group-ip**

Optional: X:X::X:X, MLD source information for the specified group.

### **source-ip**

Optional: X:X::X:X, MLD source information for the specified group.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ipv6 mld snooping vlan 2
```

```
MLD Snooping Protocol Info
```

```
Total VLANs with MLD enabled : 2
Current count of multicast groups joined : 0
```

```

MLD Drop Unknown Multicast : Global

VLAN ID : 2
VLAN Name : VLAN2
MLD Configured Version : 2
MLD Operating Version : 2
Querier Address [this switch] : fe80::218:71ff:fec4:2f00
Querier Port :
Querier UpTime :0m 21s
Querier Expiration Time :0m 2s

```

| Active Group Address | Tracking | Vers  | Mode  | Uptime | Expires |
|----------------------|----------|-------|-------|--------|---------|
| -----                | -----    | ----- | ----- | -----  | -----   |
| ff05::2:1            | Filter   | 2     | EXC   | 0m 17s | 4m 3s   |

```
switch# show ipv6 mld snooping vlan 2 group
```

```
MLD ports and group information for group ff05::2:1
```

```

VLAN ID : 2
VLAN Name : VLAN2

Group Address : ff05::2:1
Last Reporter : 2001::1
Group Type : Filter

```

| Port  | Vers  | Mode  | Uptime | Expires | V1<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|-------|-------|-------|--------|---------|-------------|----------------------|--------------------|
| ----- | ----- | ----- | -----  | -----   | -----       | -----                | -----              |
| 1/1/1 | 2     | EXC   | 0m 5s  | 4m 15s  | 4m 15s      | 0                    | 0                  |

```
switch# show ipv6 mld snooping vlan 2 group ff05::2:1
```

```
MLD ports and group information for group ff05::2:1
```

```

VLAN ID : 2
VLAN Name : VLAN2

Group Address : ff05::2:1
Last Reporter : 2001::1
Group Type : Filter

```

| Port  | Vers  | Mode  | Uptime | Expires | V1<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|-------|-------|-------|--------|---------|-------------|----------------------|--------------------|
| ----- | ----- | ----- | -----  | -----   | -----       | -----                | -----              |
| 1/1/1 | 2     | EXC   | 0m 5s  | 4m 15s  | 4m 15s      | 0                    | 0                  |

```
switch# show ipv mld snooping vlan 2 group ff05::2:1 source 3000::3
```

```

VLAN ID : 2
VLAN Name : VLAN2
Group Address : ff05::2:1
Source Address : 3000::3
Source Type : Filter

```

| Port  | Mode | Uptime | Expires | Configured Mode |
|-------|------|--------|---------|-----------------|
| 1/1/1 | INC  | 0m 27s | 3m 53s  | Auto            |

```
show ipv6 mld snooping [vlan <vlan-id> [group [port
<port_id>]]]
```

### Syntax

```
show ipv6 mld snooping [vlan <vlan-id> [group [port <port_id>]]]
```

### Description

This command shows MLD snooping details for the specified VLAN, including the number of different groups joined for the VLAN.

### Command context

Manager (#)

### Parameters

#### port-id

Required: <PORT>, shows MLD protocol information for the specified port of a VLAN.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# show ipv6 mld snooping vlan 2 group port 1/1/1
```

```
VLAN ID : 2
VLAN Name : VLAN2
```

```
Group Address : ff05::2:1
Last Reporter : fe80::1
Group Type : Filter
```

| Port  | Vers | Mode | Uptime | Expires | V1<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|-------|------|------|--------|---------|-------------|----------------------|--------------------|
| 1/1/1 | 2    | INC  | 1m 46s | 2m 34s  |             | 3                    | 0                  |

```
Group Address : ff05::2:1
Source Address : 3000::1
Source Type : Filter
```

| Port  | Mode | Uptime | Expires | Configured Mode |
|-------|------|--------|---------|-----------------|
| 1/1/1 | INC  | 1m 46s | 2m 34s  | Auto            |

```
Group Address : ff05::2:1
Source Address : 3000::2
```

Source Type : Filter

| Port  | Mode | Uptime | Expires | Configured Mode |
|-------|------|--------|---------|-----------------|
| 1/1/1 | INC  | 1m 46s | 2m 34s  | Auto            |

Group Address : ff05::2:1  
Source Address : 3000::3  
Source Type : Filter

| Port  | Mode | Uptime | Expires | Configured Mode |
|-------|------|--------|---------|-----------------|
| 1/1/1 | INC  | 1m 46s | 2m 34s  | Auto            |

## show ipv6 mld snooping [static-groups]

### Syntax

```
show ipv6 mld snooping [static-groups]
```

### Description

This command shows MLD snooping static group details, including the number of static groups joined.

### Command context

Manager (#)

### Parameters

None

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# show ipv6 mld snooping static-groups
```

MLD Static Group Address Information

| VLAN ID | Group Address |
|---------|---------------|
| 10      | ff12::1       |
| 10      | ff12::2       |

## MLD configuration commands for interface VLAN

### ipv6 mld {enable | disable}

### Syntax

```
ipv6 mld {enable | disable}
```

## Description

This command enables or disables MLD on the interface VLAN.

## Command context

config-if-vlan

## Parameters

### enable

Required: Enable MLD on the interface VLAN.

### disable

Required: Disable MLD on the interface VLAN.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld enable
switch(config-if-vlan)# ipv6 mld disable
```

## ipv6 mld apply access-list

## Syntax

```
ipv6 mld apply access-list <ACL-NAME>
```

```
no ipv6 mld apply access-list <ACL-NAME>
```

## Description

Configures the ACL on a particular interface to filter the MLD join or leave packets based on rules set in the particular ACL name.

The `no` form of this command disables the rules set for the ACL.

## Command context

config-vlan

## Parameters

### access-list

Associates an ACL with the IGMP.

### <ACL-NAME>

Specifies the name of the ACL.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

Existing classifier commands are used to configure the ACL. In case an MLDv2 packet with multiple group addresses is received, it will only process the permitted group addresses based on the ACL rule set, and any existing joins will time out. If there is no match or if there is a deny rule match, the packet is dropped.

## Examples

Configuring the ACL to filter MLD packets based on rules set in access list `mygroup`:

```
switch(config)# access-list ipv6 mygroup
switch(config-acl-ip)# permit icmpv6 any ff55::1
switch(config-acl-ip)# exit
switch(config)# interface vlan 2
switch(config-vlan)# ipv6 mld apply access-list mygroup
```

Configuring the ACL to remove the rules set in access list `mygroup`:

```
switch(config-vlan)# no ipv6 mld apply access-list mygroup
```

## no ipv6 mld

### Syntax

```
no ipv6 mld
```

### Description

This command removes all MLD configurations on the interface.

### Authority

Administrators or local user group members with execution rights for this command.

### Parameters

None

### Command context

```
config-if
```

### Example

```
switch(config)# interface vlan 1
switch(config-if)# no ipv6 mld
```

## ipv6 mld querier

### Syntax

```
ipv6 mld querier
```

### Description

This command configures MLD querier.

The `no` form of this command disables MLD querier.

### Command context

```
config-if-vlan
```

### Parameters

None

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld querier
switch(config-if-vlan)# no ipv6 mld querier
```

**ipv6 mld querier [interval <interval-value>]**

## Syntax

ipv6 mld querier [interval <interval-value>]

## Description

This command configures MLD querier interval. The default interval-value is 125.

## Command context

config-if-vlan

## Parameters

**interval-value**

Required: 5-300, configures MLD querier interval.



**NOTE:** Default interval-value is 125. Use the `no ipv6 mld querier interval` command to set interval-value to the default.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld querier interval 100
switch(config-if-vlan)# no ipv6 mld querier interval
```

**ipv6 mld last-member-query-interval <interval-value>**

## Syntax

ipv6 mld last-member-query-interval <interval-value>

## Description

This command configures MLD last member query interval value in seconds. The default interval-value is 1 second.

## Command context

config-if-vlan

## Parameters

### interval-value

Required: 1-2, configures MLD last-member-query-interval.



**NOTE:** Default interval-value is 1 second. Use the `no ipv6 mld last-member-query-interval` command to set interval-value to the default.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld last-member-query-interval 2
switch(config-if-vlan)# no ipv6 mld last-member-query-interval
```

**ipv6 mld querier query-max-response-time <response-time>**

## Syntax

ipv6 mld querier query-max-response-time <response-time>

## Description

This command configures MLD max response time value in seconds. The default max-response-time-value is 10 seconds.

## Command context

config-if-vlan

## Parameters

### max-response-time-value

Required: 10-128, configures MLD querier max-response-time.



**NOTE:** Default max-response-time-value is 10 seconds. Use the `no ipv6 mld querier query-max-response-time` command to set max-response-time-value to the default.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld query-max-response-time 50
switch(config-if-vlan)# no ipv6 mld query-max-response-time
```

**ipv6 mld robustness**

## Syntax

ipv6 mld robustness <value>



## Description

This command configures MLD robustness. The robustness value represents the number of times the querier retries queries on the connected subnets. The default robustness-value is 2 seconds.

## Command context

config-if-vlan

## Parameters

**robustness-value**

Required: 1-7, configures MLD robustness.



**NOTE:** Default robustness-value is 2 seconds. Use the `no ipv6 mld robustness` command to set robustness-value to the default.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld robustness 5
switch(config-if-vlan)# no ipv6 mld robustness
```

**ipv6 mld static-group <multicast-group-ip>**

## Syntax

ipv6 mld static-group <multicast-group-ip>

## Description

This command configures MLD static group.

## Command context

config-if-vlan

## Parameters

**multicast-group-ip**

Required: X:X::X:X, configures MLD static group.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld static-group ff12::c
switch(config-if-vlan)# no ipv6 mld static-group ff12::c
```

## ipv6 mld version <version>

### Syntax

```
ipv6 mld version <version>
```

### Description

This command configures MLD version.

### Command context

```
config-if-vlan
```

### Parameters

#### version

Required: 1-2, configures MLD version.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld version 2
```

## ipv6 mld version <version> [strict]

### Syntax

```
ipv6 mld version <version> [strict]
```

### Description

This command configures MLD strict version. Packets that do not match the configured version will be dropped. By default, strict option is not enabled.

### Command context

```
config-if-vlan
```

### Parameters

#### version

Required: 1-2, configures MLD version.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```
switch(config)# interface vlan 2
switch(config-if-vlan)# ipv6 mld version 2 strict
switch(config-if-vlan)# no ipv6 mld version 2 strict
```

# MLD show commands for interface VLAN

## show ipv6 mld

### Syntax

```
show ipv6 mld
```

### Description

This command shows MLD configuration on VLAN.

### Command context

Manager (#)

### Parameters

None

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# show ipv6 mld
```

```
VRF Name : default
Interface : vlan10
MLD Configured Version : 2
MLD Operating Version : 2
Querier State : Querier
Querier IP [this switch] : fe80::7272:cfff:fe96:d3ec
Querier Uptime : 39m 44s
Querier Expiration Time : 0m 31s
MLD Snoop Enabled on VLAN : True
```

## show ipv6 mld [interface <intf\_id> | vlan <vlan-id>]

### Syntax

```
show ipv6 mld [interface <intf_id> | vlan <vlan-id>]
```

### Description

This command shows MLD configuration on a specific VLAN.

### Command context

Manager (#)

### Parameters

**vlan-id**

Required: 1-4094, shows MLD configuration on a specified VLAN.

## intf-id

Required: IFNAME, shows MLD configuration on a specified interface.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ipv6 mld interface vlan 10
```

```
MLD Configured Version : 2
MLD Operating Version : 2
Querier State : Querier
Querier IP [this switch] : fe80::7272:cfff:fe96:d3ec
Querier Uptime : 40m 42s
Querier Expiration Time : 1m 39s
MLD Snoop Enabled on VLAN : True
```

```
switch# show ipv6 mld interface 1/1/2
```

```
MLD Configured Version : 2
MLD Operating Version : 2
Querier State : Querier
Querier IP [this switch] : fe80::7272:cfff:fe96:d3ec
Querier Uptime : 40m 42s
Querier Expiration Time : 1m 39s
MLD Snoop Enabled on VLAN : True
```

## show ipv6 mld [all-vrfs ]

## Syntax

```
show ipv6 mld [all-vrfs]
```

## Description

This command shows MLD information for the specified VRF.

## Command context

Manager (#)

## Parameters

### all-vrfs

Optional: shows MLD information status for all VRFs.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch(config)# show ipv6 mld all-vrfs
VRF Name : default
Interface : vlan2
```

```

MLD Configured Version : 2
MLD Operating Version : 2
Querier State : Querier
Querier IP [this switch] : fe80::a00:9ff:fe06:67cd
Querier Uptime : 23m 53s
Querier Expiration Time : 0m 17s
MLD Snoop Enabled on VLAN : True

```

| Active Group Address | Vers | Mode | Uptime | Expires |
|----------------------|------|------|--------|---------|
| ff05::2:1            | 2    | INC  | 3m 56s | 1m 47s  |

**show ipv6 mld [interface <intf-id> | vlan <vlan-id>]  
[counters]]**

### Syntax

```
show ipv6 mld [interface <intf-id> | vlan <vlan-id>] [counters]]
```

### Description

This command shows MLD query packet Tx and Rx on a specific VLAN.

### Command context

Manager (#)

### Parameters

#### vlan-id

Required: 1-4094, shows MLD configuration on a specified VLAN.

#### intf-id

Required: IFNAME, shows MLD configuration on a specified interface.

#### counters

Optional: Shows MLD query packet counter Tx-Rx on a specified VLAN.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# show ipv6 mld interface vlan 2 counters
```

MLD Counters

```

Interface Name : vlan2
VRF Name : default
Membership Timeout : 0

```

|                           | Rx | Tx |
|---------------------------|----|----|
| V1 All Hosts Queries      | 0  | 0  |
| V2 All Hosts Queries      | 0  | 0  |
| V1 Group Specific Queries | 0  | 0  |
| V2 Group Specific Queries | 0  | 2  |

|                                   |   |     |
|-----------------------------------|---|-----|
| Group And Source Specific Queries | 0 | 2   |
| V2 Member Reports                 | 0 | N/A |
| V1 Member Reports                 | 0 | N/A |
| V1 Member Leaves                  | 0 | N/A |
| Packets dropped by ACL            | 0 | N/A |

```
switch# show ipv6 mld interface 1/1/1 counters
```

#### MLD Counters

```
Interface Name : 1/1/1
VRF Name : default
Membership Timeout : 0
```

|                                   | Rx    | Tx    |
|-----------------------------------|-------|-------|
|                                   | ----- | ----- |
| V1 All Hosts Queries              | 0     | 0     |
| V2 All Hosts Queries              | 0     | 0     |
| V1 Group Specific Queries         | 0     | 0     |
| V2 Group Specific Queries         | 0     | 0     |
| Group And Source Specific Queries | 0     | 0     |
| V2 Member Reports                 | 0     | N/A   |
| V1 Member Reports                 | 0     | N/A   |
| V1 Member Leaves                  | 0     | N/A   |
| Packets dropped by ACL            | 0     | N/A   |

```
show ipv6 mld [interface <intf-id> | vlan <vlan-id>]
[groups]]
```

### Syntax

```
show ipv6 mld [interface <intf-id> | vlan <vlan-id>] [groups]]
```

### Description

This command shows MLD groups joined details.

### Command context

Manager (#)

### Parameters

#### vlan-id

Required: 1-4094, shows MLD information on a specified VLAN.

#### intf-id

Required: IFNAME, shows MLD information on a specified interface.

#### groups

Optional: Shows MLD groups information on a specified interface.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ipv mld interface vlan 2 groups
```

```
MLD group information for group ff05::2:1
```

```
Interface Name : vlan2
VRF Name : default
```

```
Group Address : ff05::2:1
Last Reporter : fe80::1
```

| Vers | Mode | Uptime | Expires | V1<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|----------------------|--------------------|
| 2    | INC  | 6m 2s  | 0m 4s   |             | 1                    |                    |

```
Group Address : ff05::2:1
Source Address : 3000::1
```

| Mode | Uptime | Expire |
|------|--------|--------|
| INC  | 6m 2s  | 0m 4s  |

```
show ipv6 mld [interface (<intf-id> | vlan <vlan-id>)
[group <group_ip>] [source <source_ip>]]]
```

## Syntax

```
show ipv6 mld [interface (<intf-id> | vlan <vlan-id>) [group <group_ip>] [source <source_ip>]]]
```

## Description

This command shows MLD joined group details on a specified interface.

## Command context

Manager (#)

## Parameters

### group\_ip

Required: X:X::X:X, shows MLD joined group details.

### source\_ip

Required: X:X::X:X, shows MLD joined group details for a specified source.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ipv mld interface vlan 2 group ff55::5
```

```
MLD group information for group ff55::5
```

```
Interface Name : vlan2
```

```
VRF Name : default
Group Address : ff55::5
Last Reporter : fe80::1
```

| Vers | Mode | Uptime | Expires | V1<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|----------------------|--------------------|
| 2    | INC  | 6m 2s  | 0m 4s   |             | 1                    |                    |

```
Group Address : ff55::5
Source Address : 3000::1
```

| Mode | Uptime | Expire |
|------|--------|--------|
| INC  | 6m 2s  | 0m 4s  |

```
switch# show ipv6 mld interface vlan 2 group ff55::5 source 3000::1
```

```
Interface Name : vlan2
VRF Name : default
Group Address : ff55::5
Source Address : 3000::1
```

| Mode | Uptime | Expire |
|------|--------|--------|
| INC  | 9m 37s | 2m 0s  |

## show ipv6 mld [groups]

### Syntax

```
show ipv6 mld [groups]
```

### Description

This command shows MLD groups joined details.

### Command context

Manager (#)

### Parameters

#### groups

Options: shows MLD groups information.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# show ipv6 mld groups
```

```
MLD group information for group ff05::2:11
```

```
Interface Name : vlan2
```



VRF Name : default

Group Address : ff05::2:11

Last Reporter : 2001::1

| Vers | Mode | Uptime | Expires | V1<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|----------------------|--------------------|
| 1    |      | 2m 27s | 1m 53s  | 1m 53s      |                      |                    |

MLD group information for group ff05::2:12

Interface Name : vlan2

VRF Name : default

Group Address : ff05::2:12

Last Reporter : 2001::1

| Vers | Mode | Uptime | Expires | V1<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|----------------------|--------------------|
| 1    |      | 0m 3s  | 4m 18s  | 4m 18s      |                      |                    |

## show ipv6 mld groups [all-vrfs]

### Syntax

```
show ipv6 mld groups [all-vrfs]
```

### Description

This command shows MLD groups joined details on VRFs.

### Command context

Manager (#)

### Parameters

**all-vrfs**

Optional: shows MLD groups joined in all VRFs.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# show ipv6 mld groups all-vrfs
```

MLD group information for group ff05::2:11

Interface Name : vlan1

VRF Name : default

Group Address : ff05::2:11

Last Reporter : 2001::1

| V1 | Sources | Sources |
|----|---------|---------|
|----|---------|---------|

```

Vers Mode Uptime Expires Timer Forwarded Blocked

1 4m 4s 2m 38s 2m 38s

switch# show ipv6 mld groups vrf default

MLD group information for group ff05::2:11

Interface Name : vlan2
VRF Name : default

Group Address : ff05::2:11
Last Reporter : 2001::1

V1 Sources Sources
Vers Mode Uptime Expires Timer Forwarded Blocked

1 5m 25s 1m 17s 1m 17s

```

## show ipv6 mld [interface <intf-id> [counters]]

### Syntax

```
show ipv6 mld [interface <intf-id> [counters]]
```

### Description

This command shows MLD query packet Tx and Rx on a specific interface.

### Command context

Manager (#)

### Parameters

#### intf-id

Required: shows MLD configuration on a specified interface.

#### counters

Optional: shows MLD query packet counter Tx-Rx on a specified interface.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# show ipv6 mld interface 1/1/1 counters
```

MLD Counters

```

Interface Name : 1/1/1
VRF Name : default
Membership Timeout : 0

```

|                           | Rx    | Tx    |
|---------------------------|-------|-------|
|                           | ----- | ----- |
| V1 All Hosts Queries      | 0     | 0     |
| V2 All Hosts Queries      | 0     | 9     |
| V1 Group Specific Queries | 0     | 0     |
| V2 Group Specific Queries | 0     | 0     |

|                                   |   |     |
|-----------------------------------|---|-----|
| Group And Source Specific Queries | 0 | 0   |
| V2 Member Reports                 | 0 | N/A |
| V1 Member Reports                 | 0 | N/A |
| V1 Member Leaves                  | 0 | N/A |
| Packets dropped by ACL            | 0 | N/A |

**show ipv6 mld [interface <intf-id> [statistics]]**

### Syntax

```
show ipv6 mld [interface <intf-id> [statistics]]
```

### Description

This command shows MLD statistics on a specific interface.

### Command context

Manager (#)

### Parameters

#### intf-id

Required: shows MLD statistics on a specified interface.

#### statistics

Optional: shows MLD statistics on a specified interface.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# show ipv6 mld interface 1/1/1 statistics
```

MLD statistics

```
Interface Name : 1/1/1
VRF Name : default
```

```
Number of Include Groups : 2
Number of Exclude Groups : 0
Number of Static Groups : 0
Total Multicast Groups Joined : 2
```

**show ipv6 mld [interface <intf-id> [groups]]**

### Syntax

```
show ipv6 mld [interface <intf-id> [groups]]
```

### Description

This command shows MLD groups joined details.

### Command context

Manager (#)

## Parameters

### **intf-id**

Required: shows MLD configuration on a specified interface.

### **groups**

Optional: shows MLD groups information.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ipv6 mld interface 1/1/1 groups
```

MLD group information for group ff55::1

Interface Name : 1/1/1  
VRF Name : default

Group Address : ff55::1  
Last Reporter : fe80::a00:9ff:fe77:1062

| Vers | Mode | Uptime | Expires | V1<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|----------------------|--------------------|
| 2    | EXC  | 0m 14s | 4m 6s   |             |                      |                    |

```
show ipv6 mld [interface (<intf-id> | vlan <vlan-id>)
[group <group_ip>] [source <source_ip>]]]
```

## Syntax

```
show ipv6 mld [interface (<intf-id> | vlan <vlan-id>) [group <group_ip>] [source <source_ip>]]]
```

## Description

This command shows MLD joined group details on a specified interface.

## Command context

Manager (#)

## Parameters

### **group\_ip**

Required: X:X::X:X, shows MLD joined group details.

### **source\_ip**

Required: X:X::X:X, shows MLD joined group details for a specified source.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ipv mld interface vlan 2 group ff55::5
```

MLD group information for group ff55::5

```
Interface Name : vlan2
VRF Name : default
```

```
Group Address : ff55::5
Last Reporter : fe80::1
```

| Vers | Mode | Uptime | Expires | V1<br>Timer | Sources<br>Forwarded | Sources<br>Blocked |
|------|------|--------|---------|-------------|----------------------|--------------------|
| 2    | INC  | 6m 2s  | 0m 4s   |             | 1                    |                    |

```
Group Address : ff55::5
Source Address : 3000::1
```

| Mode | Uptime | Expire |
|------|--------|--------|
| INC  | 6m 2s  | 0m 4s  |

```
switch# show ipv mld interface vlan 2 group ff55::5 source 3000::1
```

```
Interface Name : vlan2
VRF Name : default
Group Address : ff55::5
Source Address : 3000::1
```

| Mode | Uptime | Expire |
|------|--------|--------|
| INC  | 9m 37s | 2m 0s  |

**show ipv6 mld [group <group\_ip> [all-vrfs]]**

## Syntax

```
show ipv6 mld [group <group_ip> [all-vrfs]]
```

## Description

This command shows MLD joined group details on VRF.

## Command context

Manager (#)

## Parameters

### group\_ip

Required: X:X::X:X, shows MLD joined group details.

### all-vrfs

Optional: shows MLD groups joined in all VRFs.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ipv6 mld group ff55::1

MLD group information for group ff55::1

Interface Name : 1/1/1
VRF Name : default

Group Address : ff55::1
Last Reporter : fe80::a00:9ff:fe77:1062

Vers Mode Uptime Expires V1 Sources Sources
----- -
2 EXC 3m 12s 3m 46s Timer Forwarded Blocked

switch# show ipv6 mld group ff05::2:11 all-vrfs

MLD group information for group ff05::2:11

Interface Name : vlan2
VRF Name : default

Group Address : ff05::2:11
Last Reporter : 2001::1

Vers Mode Uptime Expires V1 Sources Sources
----- -
1 1m 16s 3m 4s 3m 4s Timer Forwarded Blocked
```

```
show ipv6 mld [group <group_ip> [source <source_ip> [all-
vrfs]]]
```

## Syntax

```
show ipv6 mld [group <group_ip> [source <source_ip> [all-vrfs]]]
```

## Description

This command shows MLD joined group details for a source on VRF.

## Command context

Manager (#)

## Parameters

### group\_ip

Required: X:X::X:X, shows MLD joined group details.

### source\_ip

Required: X:X::X:X, shows MLD joined group details for a source.

## **all-vrfs**

Optional: shows MLD groups joined in all VRFs.

## **Authority**

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## **Example**

```
switch# show ipv6 mld group ff05::2:1 source 3000::1

Interface Name : vlan2
VRF Name : default
Group Address : ff05::2:1
Source Address : 3000::1

Mode Uptime Expire

INC 0m 53s 3m 27s

switch# show ipv6 mld group ff05::2:1 source 3000::1 all-vrfs

Interface Name : vlan2
VRF Name : default
Group Address : ff05::2:1
Source Address : 3000::1

Mode Uptime Expire

INC 1m 38s 4m 5s
```

## **show ipv6 mld [interface vlan <vlan-id> [statistics]]**

## **Syntax**

```
show ipv6 mld [interface vlan <vlan-id> [statistics]]
```

## **Description**

This command shows MLD statistics on a specific interface VLAN.

## **Command context**

Manager (#)

## **Parameters**

### **vlan-id**

Required: 1-4094, shows MLD information on a specified VLAN.

### **statistics**

Optional: shows MLD query packet Tx, Rx, Error packet counters on a specified VLAN.

## **Authority**

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ipv6 mld interface vlan 2 statistics
```

MLD statistics

```
Interface Name : vlan2
VRF Name : default
```

```
Number of Include Groups : 2
Number of Exclude Groups : 0
Number of Static Groups : 0
Total Multicast Groups Joined : 2
```

## show ipv6 mld [static-groups [all-vrfs]]

### Syntax

```
show ipv6 mld [static-groups [all-vrfs]]
```

### Description

This command shows MLD static groups.

### Command context

Manager (#)

### Parameters

**all-vrfs**

Optional: shows MLD groups joined in all VRFs.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ipv6 mld static-groups
```

MLD Static Group Address Information

```
VRF Name :default
```

```
Interface Name Group Address
```

```

vlan2 ff12::c
vlan2 ff12::d
```

```
switch# show ipv6 mld static-groups all-vrfs
```

MLD Static Group Address Information

```
VRF Name :default
```

```
Interface Name Group Address
```

```

vlan2 ff12::c
vlan2 ff12::d
```



## show ipv6 mld counters

### Syntax

```
show ipv6 mld [counters]
```

### Description

This command shows MLD counters.

### Command context

Manager (#)

### Parameters

**vrf**

Optional: shows MLD counter status in a specific VRF.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# show ipv6 mld counters
```

MLD Counters

```
Interface Name : vlan2
VRF Name : default
Membership Timeout : 0
```

|                                   | Rx    | Tx    |
|-----------------------------------|-------|-------|
|                                   | ----- | ----- |
| V1 All Hosts Queries              | 0     | 0     |
| V2 All Hosts Queries              | 0     | 12    |
| V1 Group Specific Queries         | 0     | 0     |
| V2 Group Specific Queries         | 0     | 0     |
| Group And Source Specific Queries | 0     | 0     |
| V2 Member Reports                 | 0     | N/A   |
| V1 Member Reports                 | 0     | N/A   |
| V1 Member Leaves                  | 0     | N/A   |
| Packets dropped by ACL            | 0     | N/A   |

```
switch# show ipv6 mld counters vrf default
```

MLD Counters

```
Interface Name : vlan2
VRF Name : default
Membership Timeout : 0
```

|                                   | Rx    | Tx    |
|-----------------------------------|-------|-------|
|                                   | ----- | ----- |
| V1 All Hosts Queries              | 0     | 0     |
| V2 All Hosts Queries              | 0     | 12    |
| V1 Group Specific Queries         | 0     | 0     |
| V2 Group Specific Queries         | 0     | 0     |
| Group And Source Specific Queries | 0     | 0     |
| V2 Member Reports                 | 0     | N/A   |
| V1 Member Reports                 | 0     | N/A   |

|                        |   |     |
|------------------------|---|-----|
| V1 Member Leaves       | 0 | N/A |
| Packets dropped by ACL | 0 | N/A |

## MLD configuration commands for interface

### ipv6 mld {enable | disable}

#### Syntax

```
ipv6 mld {enable | disable}
```

#### Description

This command enables or disables MLD on the interface.

#### Command context

```
config-if
```

#### Parameters

##### enable

Required: Enable MLD on the interface.

##### disable

Required: Disable MLD on the interface.

x

#### Authority

Administrators or local user group members with execution rights for this command.

#### Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld enable
switch(config-if)# ipv6 mld disable
```

### ipv6 mld apply access-list

#### Syntax

```
ipv6 mld apply access-list <ACL-NAME>
```

```
no ipv6 mld apply access-list <ACL-NAME>
```

#### Description

Configures the ACL on a particular interface to filter the MLD join or leave packets based on rules set in the particular ACL name.

The `no` form of this command removes the rules set for the ACL.

#### Command context

```
config-vlan
```

## Parameters

### access-list

Associates an ACL with the IGMP.

### <ACL-NAME>

Specifies the name of the ACL.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

Existing classifier commands are used to configure the ACL. In case an MLDv2 packet with multiple group addresses is received, it will only process the permitted group addresses based on the ACL rule set, and any existing joins will time out. If there is no match or if there is a deny rule match, the packet is dropped.

## Examples

Configuring the ACL to filter MLD packets based on rules set in access list `mygroup`:

```
switch(config)# access-list ipv6 mygroup
switch(config-acl-ip)# permit icmpv6 any ff55::1
switch(config-acl-ip)# exit
switch(config)# interface vlan 1
switch(config-vlan)# ipv6 mld apply access-list mygroup
```

Configuring the ACL to remove the rules set in access list `mygroup`:

```
switch(config-vlan)# no ipv6 mld apply access-list mygroup
```

## no ipv6 mld

## Syntax

```
no ipv6 mld
```

## Description

This command removes all MLD configurations on the interface.

## Authority

Administrators or local user group members with execution rights for this command.

## Parameters

None

## Command context

```
config-if
```

## Example

```
switch(config)# interface vlan 1
switch(config-if)# no ipv6 mld
```

## ipv6 mld querier

### Syntax

```
ipv6 mld querier
```

### Description

This command configures MLD querier. This functionality will allow the interface to join in the querier-election process.

### Command context

```
config-if
```

### Parameters

None

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld querier
switch(config-if)# no ipv6 mld querier
```

## ipv6 mld querier [interval <interval-value>]

### Syntax

```
ipv6 mld querier [interval <interval-value>]
```

### Description

This command configures MLD querier interval. The default interval-value is 125.

### Command context

```
config-if
```

### Parameters

#### interval-value

Required: 5-300, configures MLD querier interval.



**NOTE:** Default interval-value is 125. Use the `no ipv6 mld querier interval` command to set interval-value to the default.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld querier interval 100
switch(config-if)# no ipv6 mld querier interval
```

## ipv6 mld last-member-query-interval

### Syntax

```
ipv6 mld last-member-query-interval <interval-value>
```

### Description

This command configures MLD last member query interval value in seconds. The default interval-value is 1 second.

### Command context

config-if

### Parameters

**interval-value**

Required: 1-2, configures MLD last-member-query-interval.



**NOTE:** Default interval-value is 1 second. Use the `no ipv6 mld last-member-query-interval` command to set interval-value to the default.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld last-member-query-interval 2
switch(config-if)# no ipv6 mld last-member-query-interval
```

## ipv6 mld querier query-max-response-time

### Syntax

```
ipv6 mld querier query-max-response-time <response-time>
```

### Description

This command configures MLD max response time value in seconds. The default max-response-time-value is 10 seconds.

### Command context

config-if

### Parameters

**max-response-time-value**

Required: 10-128, configures MLD querier max-response-time.



**NOTE:** Default max-response-time-value is 10 seconds. Use the `no ipv6 mld querier query-max-response-time` command to set max-response-time-value to the default.

### Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld query-max-response-time 50
switch(config-if)# no ipv6 mld query-max-response-time
```

## ipv6 mld robustness

### Syntax

```
ipv6 mld robustness <value>
```

### Description

This command configures MLD robustness. The robustness value represents the number of times the querier retries queries on the connected subnets. The default robustness-value is 2 seconds.

### Command context

```
config-if
```

### Parameters

**robustness-value**

Required: 1-7, configures MLD robustness.



**NOTE:** Default robustness-value is 2 seconds. Use the `no ipv6 mld robustness` command to set robustness-value to the default.

### Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld robustness 5
switch(config-if)# no ipv6 mld robustness
```

## ipv6 mld static-group

### Syntax

```
ipv6 mld static-group <multicast-group-ip>
```

### Description

This command configures MLD static group.

### Command context

```
config-if
```

### Parameters

**multicast-group-ip**

Required: X:X::X:X, configures MLD static group.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld static-group ff12::c
switch(config-if)# no ipv6 mld static-group ff12::c
```

## ipv6 mld version

### Syntax

```
ipv6 mld version <version>
```

### Description

This command configures MLD version.

### Command context

config-if

### Parameters

#### version

Required: 1-2, configures MLD version.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld version 2
```

## ipv6 mld version [strict]

### Syntax

```
ipv6 mld version <version> [strict]
```

### Description

This command configures MLD strict version. Packets that do not match the configured version will be dropped. By default, strict option is not enabled.

### Command context

config-if

### Parameters

#### version

Required: 1-2, configures MLD version.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

```
switch(config)# interface vlan 1
switch(config-if)# ipv6 mld version 2 strict
switch(config-if)# no ipv6 mld version 2 strict
```



## show spanning-tree

### Syntax

```
show spanning-tree
```

### Description

Shows priority, address, Hello-time, Max-age, and Forward-delay for bridge and root node.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

### Example

Showing spanning tree standard information:

```
switch# show spanning-tree
Spanning tree status : Enabled Protocol: MSTP
```

MST0

```
Root ID
 Priority : 32768, Root
 MAC-Address : 48:0F:CF:AF:04:76
 Hello time : 2 seconds
 Max Age : 20 seconds
 Forward Delay : 15 seconds
```

```
Bridge ID
 Priority : 32768
 MAC-Address : 48:0F:CF:AF:04:76
 Hello time : 2 seconds
 Max Age : 20 seconds
 Forward Delay : 15 seconds
```

| PORT  | ROLE       | STATE      | COST  | PRIORITY | TYPE     | BPDU-Tx | BPDU-Rx | TCN-Tx | TCN-Rx |
|-------|------------|------------|-------|----------|----------|---------|---------|--------|--------|
| 1/1/1 | Designated | Forwarding | 20000 | 128      | P2P Edge | 100     | 60      | 20     | 10     |
| 1/1/2 | Designated | Forwarding | 20000 | 128      | P2P      | 100     | 60      | 20     | 10     |
| 1/1/3 | Designated | Forwarding | 20000 | 128      | Shr      | 100     | 60      | 20     | 10     |
| 1/1/4 | Designated | Forwarding | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/5 | Alternate  | Loop-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/6 | Alternate  | Root-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |

```
Number of topology changes : 4
Last topology change occurred : 516 seconds ago
```

## show spanning-tree detail

### Syntax

```
show spanning-tree detail
```

## Description

Shows spanning tree detail including CIST and corresponding port information.

## Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

Showing spanning tree detailed information:

```
switch# show spanning-tree detail
Spanning tree status : Enabled Protocol: MSTP

MST0
 Root ID Priority : 32768
 MAC-Address: 48:0f:cf:af:04:76
 This bridge is the root
 Hello time(in seconds):2 Max Age(in seconds):20
 Forward Delay(in seconds):15

 Bridge ID Priority : 32768
 MAC-Address: 48:0f:cf:af:04:76
 Hello time(in seconds):2 Max Age(in seconds):20
 Forward Delay(in seconds):15

PORT ROLE STATE COST PRIORITY TYPE BPDU-Tx BPDU-Rx TCN-Tx TCN-Rx

1/1/1 Designated Forwarding 20000 128 P2P Edge 100 60 20 10
1/1/2 Designated Forwarding 20000 128 P2P 100 60 20 10
1/1/3 Designated Forwarding 20000 128 Shr 100 60 20 10
1/1/4 Designated Forwarding 20000 128 Shr Edge 100 60 20 10
1/1/5 Alternate Loop-Inc 20000 128 Shr Edge 100 60 20 10
1/1/6 Alternate Root-Inc 20000 128 Shr Edge 100 60 20 10

Topology change flag : True
Number of topology changes : 4
Last topology change occurred : 516 seconds ago
Timers: Hello expiry 1 , Forward delay expiry 18

Port 1/1/1
Designated root has priority :32768 Address: 48:0f:cf:af:04:76
Designated bridge has priority :32768 Address: 48:0f:cf:af:04:76
Designated port :1/1/1
Number of transitions to forwarding state : 3
Bpdus sent 347, received 9
TCN_Tx: 20, TCN_Rx: 10

Port 1/1/2
Designated root has priority :32768 Address: 48:0f:cf:af:04:76
Designated bridge has priority :32768 Address: 48:0f:cf:af:04:76
Designated port :1/1/2
Number of transitions to forwarding state : 3
Bpdus sent 350, received 11
TCN_Tx: 20, TCN_Rx: 10

Port lag1 ID 321
Designated root has priority : 32768 Address: 48:0F:CF:AF:04:76
Designated bridge has priority : 32768 Address: 48:0F:CF:AF:04:76
Designated port id : 321
Multi-Chassis role : active
Number of transitions to forwarding state : 3
BPDUs sent : 340
BPDUs received : 5
TCN_Tx: 20, TCN_Rx: 10
```

## show spanning-tree inconsistent-ports

### Syntax

```
show spanning-tree inconsistent-ports [instance <INSTANCE-ID>]
```

### Description

Shows ports blocked by STP protection functions such as Root guard, Loop guard, BPDU guard, and RPVST guard in addition to MSTI information.

### Command context

Operator (>) or Manager (#)

### Parameters

<INSTANCE-ID>

Specifies the MSTP instance ID. Range: 0 to 64.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

*On the 6400 Switch Series, interface identification differs.*

Showing spanning tree inconsistent ports:

```
switch# show spanning-tree inconsistent-ports
Instance ID Blocked Port Reason

0 1/1/13 BPDU Guard
```

Showing inconsistent port information for instances 1-4:

```
switch# show spanning-tree inconsistent-ports instance 1-4
Instance ID Blocked Port Reason

1 1/1/3 Root Guard
2 1/1/7 BPDU Guard
3 1/1/9 Loop Guard
4 1/1/37 RPVST Guard
```

## show spanning-tree mst

### Syntax

```
show spanning-tree mst
```

### Description

Shows MSTP configuration and status information for each instance.

### Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing MSTP configuration and status information:

```
switch# show spanning-tree mst
MST0
Vlans mapped : 2,4-4094
Bridge Address : 48:0F:CF:AF:04:76
Priority : 32768
Root
Regional Root
Operational Hello time : 2 seconds Forward delay: 15 seconds
 Max-age : 20 seconds TxHoldCount : 6 pps
Configured Hello time : 2 seconds Forward delay: 15 seconds
 Max-age : 20 seconds Max-Hops : 20
Root Address : 48:0F:CF:AF:04:76 Priority : 32768
 Port : 0 Path cost : 0
Regional Root Address : 48:0F:CF:AF:04:76 Priority : 32768
 Internal cost: 0 Rem Hops : 20
```

| PORT  | ROLE       | STATE      | COST  | PRIORITY | TYPE     | BPDU-Tx | BPDU-Rx | TCN-Tx | TCN-Rx |
|-------|------------|------------|-------|----------|----------|---------|---------|--------|--------|
| 1/1/1 | Designated | Forwarding | 20000 | 128      | P2P Edge | 100     | 60      | 20     | 10     |
| 1/1/2 | Designated | Forwarding | 20000 | 128      | P2P      | 100     | 60      | 20     | 10     |
| 1/1/3 | Designated | Forwarding | 20000 | 128      | Shr      | 100     | 60      | 20     | 10     |
| 1/1/4 | Designated | Forwarding | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/5 | Alternate  | Loop-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/6 | Alternate  | Root-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |

```
Topology change flag : True
Number of topology changes : 4
Last topology change occurred : 516 seconds ago
```

```
MST1
Vlans mapped: 1
Bridge Address : 48:0F:CF:AF:04:76 Priority: 32768
Root Address : 48:0F:CF:AF:04:76 Priority: 32768
 Port : 0 Cost : 0
 Rem Hops : 20
```

| PORT  | ROLE       | STATE      | COST  | PRIORITY | TYPE     | BPDU-Tx | BPDU-Rx | TCN-Tx | TCN-Rx |
|-------|------------|------------|-------|----------|----------|---------|---------|--------|--------|
| 1/1/1 | Designated | Forwarding | 20000 | 128      | P2P Edge | 100     | 60      | 20     | 10     |
| 1/1/2 | Designated | Forwarding | 20000 | 128      | P2P      | 100     | 60      | 20     | 10     |
| 1/1/3 | Designated | Forwarding | 20000 | 128      | Shr      | 100     | 60      | 20     | 10     |
| 1/1/4 | Designated | Forwarding | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/5 | Alternate  | Loop-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/6 | Alternate  | Root-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |

```
Topology change flag : True
Number of topology changes : 4
Last topology change occurred : 516 seconds ago
```

```
MST2
Vlans mapped: 3
Bridge Address : 48:0F:CF:AF:04:76 Priority: 32768
Root Address : 48:0F:CF:AF:04:76 Priority: 32768
 Port : 0 Cost : 0
 Rem Hops : 20
```

| PORT  | ROLE       | STATE      | COST  | PRIORITY | TYPE     | BPDU-Tx | BPDU-Rx | TCN-Tx | TCN-Rx |
|-------|------------|------------|-------|----------|----------|---------|---------|--------|--------|
| 1/1/1 | Designated | Forwarding | 20000 | 128      | P2P Edge | 100     | 60      | 20     | 10     |
| 1/1/2 | Designated | Forwarding | 20000 | 128      | P2P      | 100     | 60      | 20     | 10     |
| 1/1/3 | Designated | Forwarding | 20000 | 128      | Shr      | 100     | 60      | 20     | 10     |
| 1/1/4 | Designated | Forwarding | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/5 | Alternate  | Loop-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/6 | Alternate  | Root-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |

```
Topology change flag : True
Number of topology changes : 4
Last topology change occurred : 516 seconds ago
```

## show spanning-tree mst-config

### Syntax

```
show spanning-tree mst-config
```

### Description

Shows MSTP instance and corresponding VLAN information.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing configuration information for MST instances and corresponding VLANs:

```
switch# show spanning-tree mst-config
MST configuration information
 MST config ID : reg
 MST config revision : 1
 MST config digest : 2D2BC9A32097B463C48EE1817673FA2D
 Number of instances : 2

Instance ID Member VLANs

0 2,4-4094
1 1
2 3
```

## show spanning-tree mst detail

### Syntax

```
show spanning-tree mst detail
```

### Description

Shows detailed information for all MST instances.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Showing detailed information for all MST instances:

```
switch# show spanning-tree mst detail
MST0
```

```

Vlans mapped: 2,4-4094
Bridge Address: 48:0F:CF:AF:04:76 Priority: 32768
Root
Regional Root
Operational Hello time : 2 seconds Forward delay: 15 seconds
Max-age : 20 seconds TxHoldCount : 6 pps
Configured Hello time : 2 seconds Forward delay: 15 seconds
Max-age : 20 seconds Max-Hops : 20
Root Address : 48:0F:CF:AF:04:76 Priority : 32768
Port : 0 Path cost : 0
Regional Root Address : 48:0F:CF:AF:04:76 Priority : 32768
Internal cost: 0 Rem Hops : 20

```

| PORT  | ROLE       | STATE      | COST  | PRIORITY | TYPE     | BPDU-Tx | BPDU-Rx | TCN-Tx | TCN-Rx |
|-------|------------|------------|-------|----------|----------|---------|---------|--------|--------|
| 1/1/1 | Designated | Forwarding | 20000 | 128      | P2P Edge | 100     | 60      | 20     | 10     |
| 1/1/2 | Designated | Forwarding | 20000 | 128      | P2P      | 100     | 60      | 20     | 10     |
| 1/1/3 | Designated | Forwarding | 20000 | 128      | Shr      | 100     | 60      | 20     | 10     |
| 1/1/4 | Designated | Forwarding | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/5 | Alternate  | Loop-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/6 | Alternate  | Root-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |

```

Topology change flag : True
Number of topology changes : 4
Last topology change occurred : 516 seconds ago

```

```

Port 1/1/1
Designated root address : 48:0F:CF:AF:04:76
Designated regional root address : 48:0F:CF:AF:04:76
Designated bridge address : 48:0F:CF:AF:04:76
Priority : 32768
BPDUs sent : 638
BPDUs received : 9
Message expiry : 1 second
Forward delay expiry : 18 seconds
Forward transitions : 3
TCN_Tx: 10, TCN_Rx: 10

```

```

Port 1/1/2
Designated root address : 48:0F:CF:AF:04:76
Designated regional root address : 48:0F:CF:AF:04:76
Designated bridge address : 48:0F:CF:AF:04:76
Priority : 32768
BPDUs sent : 641
BPDUs received : 11
Message expiry : 1 second
Forward delay expiry : 18 seconds
Forward transitions : 3
TCN_Tx: 10, TCN_Rx: 10

```

```

MST1
Vlans mapped: 1
Bridge Address : 48:0F:CF:AF:04:76 Priority: 32768
Root Address : 48:0F:CF:AF:04:76 Priority: 32768
Port : 0 Cost : 0
Rem Hops: 20

```

| PORT  | ROLE       | STATE      | COST  | PRIORITY | TYPE     | BPDU-Tx | BPDU-Rx | TCN-Tx | TCN-Rx |
|-------|------------|------------|-------|----------|----------|---------|---------|--------|--------|
| 1/1/1 | Designated | Forwarding | 20000 | 128      | P2P Edge | 100     | 60      | 20     | 10     |
| 1/1/2 | Designated | Forwarding | 20000 | 128      | P2P      | 100     | 60      | 20     | 10     |
| 1/1/3 | Designated | Forwarding | 20000 | 128      | Shr      | 100     | 60      | 20     | 10     |
| 1/1/4 | Designated | Forwarding | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/5 | Alternate  | Loop-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/6 | Alternate  | Root-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |

```

Topology change flag : True
Number of topology changes : 4
Last topology change occurred : 516 seconds ago

```

```

Port 1/1/1
Designated root address : 48:0F:CF:AF:04:76
Designated bridge address : 48:0F:CF:AF:04:76
Priority : 32768
BPDUs sent : 638
BPDUs received : 9
Message expiry : 1 second
Forward delay expiry : 18 seconds
Forward transitions : 4
TCN_Tx: 10, TCN_Rx: 10

```

```

Port 1/1/2
Designated root address : 48:0F:CF:AF:04:76
Designated bridge address : 48:0F:CF:AF:04:76
Priority : 32768
BPDUs sent : 641
BPDUs received : 11
Message expiry : 1 second
Forward delay expiry : 18 seconds
Forward transitions : 4
TCN_Tx: 10, TCN_Rx: 10

MST2
Vlans mapped: 3
Bridge Address : 48:0F:CF:AF:04:76 Priority: 32768
Root Address : 48:0F:CF:AF:04:76 Priority: 32768
 Port : 0 Cost : 0
 Rem Hops: 20

```

| PORT  | ROLE       | STATE      | COST  | PRIORITY | TYPE     | BPDU-Tx | BPDU-Rx | TCN-Tx | TCN-Rx |
|-------|------------|------------|-------|----------|----------|---------|---------|--------|--------|
| 1/1/1 | Designated | Forwarding | 20000 | 128      | P2P Edge | 100     | 60      | 20     | 10     |
| 1/1/2 | Designated | Forwarding | 20000 | 128      | P2P      | 100     | 60      | 20     | 10     |
| 1/1/3 | Designated | Forwarding | 20000 | 128      | Shr      | 100     | 60      | 20     | 10     |
| 1/1/4 | Designated | Forwarding | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/5 | Alternate  | Loop-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/6 | Alternate  | Root-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |

```

Topology change flag : True
Number of topology changes : 4
Last topology change occurred : 516 seconds ago

```

```

Port 1/1/1
Designated root address : 48:0F:CF:AF:04:76
Designated bridge address : 48:0F:CF:AF:04:76
Priority : 32768
BPDUs sent : 638
BPDUs received : 9
Message expiry : 1 second
Forward delay expiry : 18 seconds
Forward transitions : 3
TCN_Tx: 10, TCN_Rx: 10

```

```

Port 1/1/2
Designated root address : 48:0F:CF:AF:04:76
Designated bridge address : 48:0F:CF:AF:04:76
Priority : 32768
BPDUs sent : 641
BPDUs received : 11
Message expiry : 1 second
Forward delay expiry : 18 seconds
Forward transitions : 3
TCN_Tx: 10, TCN_Rx: 10

```

## show spanning-tree mst <INSTANCE-ID>

### Syntax

```
show spanning-tree mst <INSTANCE-ID>
```

### Description

Displays MSTP configurations for the given instance ID.

### Command context

Operator (>) or Manager (#)

### Parameters

<INSTANCE-ID>

Specifies the MSTP instance number. Range: 0 to 64.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch# show spanning-tree mst 1
```

```
MST1
Vlans mapped: 1
Bridge Address : 48:0F:CF:AF:04:76 Priority: 32768
Root Address : 48:0F:CF:AF:04:76 Priority: 32768
 Port : 0 Cost : 0
 Rem Hops: 20
```

| PORT  | ROLE       | STATE      | COST  | PRIORITY | TYPE     | BPDU-Tx | BPDU-Rx | TCN-Tx | TCN-Rx |
|-------|------------|------------|-------|----------|----------|---------|---------|--------|--------|
| 1/1/1 | Designated | Forwarding | 20000 | 128      | P2P Edge | 100     | 60      | 20     | 10     |
| 1/1/2 | Designated | Forwarding | 20000 | 128      | P2P      | 100     | 60      | 20     | 10     |
| 1/1/3 | Designated | Forwarding | 20000 | 128      | Shr      | 100     | 60      | 20     | 10     |
| 1/1/4 | Designated | Forwarding | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/5 | Alternate  | Loop-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/6 | Alternate  | Root-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |

```
Topology change flag : True
Number of topology changes : 4
Last topology change occurred : 516 seconds ago
```

## show spanning-tree mst <INSTANCE-ID> detail

### Syntax

```
show spanning-tree mst <INSTANCE-ID> detail
```

### Description

Displays MSTP configurations for the given instance ID with corresponding port details.

### Command context

Operator (>) or Manager (#)

### Parameters

<INSTANCE-ID>

Specifies the MSTP instance number. Range: 0 to 64.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch# show spanning-tree mst 1 detail
```

```
MST1
Vlans mapped: 1
Bridge Address : 48:0F:CF:AF:04:76 Priority: 32768
Root Address : 48:0F:CF:AF:04:76 Priority: 32768
 Port : 0 Cost : 0
 Rem Hops: 20
```

| PORT  | ROLE       | STATE      | COST  | PRIORITY | TYPE     | BPDU-Tx | BPDU-Rx | TCN-Tx | TCN-Rx |
|-------|------------|------------|-------|----------|----------|---------|---------|--------|--------|
| 1/1/1 | Designated | Forwarding | 20000 | 128      | P2P Edge | 100     | 60      | 20     | 10     |
| 1/1/2 | Designated | Forwarding | 20000 | 128      | P2P      | 100     | 60      | 20     | 10     |



|       |            |            |       |     |          |     |    |    |    |
|-------|------------|------------|-------|-----|----------|-----|----|----|----|
| 1/1/3 | Designated | Forwarding | 20000 | 128 | Shr      | 100 | 60 | 20 | 10 |
| 1/1/4 | Designated | Forwarding | 20000 | 128 | Shr Edge | 100 | 60 | 20 | 10 |
| 1/1/5 | Alternate  | Loop-Inc   | 20000 | 128 | Shr Edge | 100 | 60 | 20 | 10 |
| 1/1/6 | Alternate  | Root-Inc   | 20000 | 128 | Shr Edge | 100 | 60 | 20 | 10 |

Topology change flag : True  
Number of topology changes : 4  
Last topology change occurred : 516 seconds ago

Port 1/1/1  
Designated root address : 48:0F:CF:AF:04:76  
Designated bridge address : 48:0F:CF:AF:04:76  
Priority : 32768  
BPDUs sent : 667  
BPDUs received : 9  
Message expiry : 0 second  
Forward delay expiry : 18 seconds  
Forward transitions : 4  
TCN\_Tx: 10, TCN\_Rx: 10

Port 1/1/2  
Designated root address : 48:0F:CF:AF:04:76  
Designated bridge address : 48:0F:CF:AF:04:76  
Priority : 32768  
BPDUs sent : 670  
BPDUs received : 11  
Message expiry : 0 second  
Forward delay expiry : 18 seconds  
Forward transitions : 4  
TCN\_Tx: 10, TCN\_Rx: 10

## show spanning-tree mst interface

### Syntax

```
show spanning-tree mst <INSTANCE-ID> interface <IFNAME>
```

### Description

Shows MSTP configurations for the given instance ID with corresponding port details.

### Command context

Operator (>) or Manager (#)

### Parameters

**<INSTANCE-ID>**

Specifies the MSTP instance number. Range: 0 to 64.

**<IFNAME>**

Specifies an interface.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing MST configuration and port details:

```
switch# show spanning-tree mst 1 interface 1/1/1
Port 1/1/1
```

| Instance | Role | State | Cost | Priority | Vlans mapped |
|----------|------|-------|------|----------|--------------|
|----------|------|-------|------|----------|--------------|

|   |            |            |       |     |   |
|---|------------|------------|-------|-----|---|
| 1 | Designated | Forwarding | 20000 | 128 | 1 |
|---|------------|------------|-------|-----|---|

## show spanning-tree summary port

### Syntax

```
show spanning-tree summary port
```

### Description

Shows spanning tree port summary information.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

### Example

*On the 6400 Switch Series, interface identification differs.*

Showing summary of spanning tree ports:

```
switch# show spanning-tree summary port

STP status : Enabled
Protocol : MSTP
BPDU guard timeout value : None
BPDU guard enabled interfaces : 1/1/1-1/1/9,1/1/11,1/1/13,1/1/15,1/1/17,1/1/19,
 1/1/21,lag1,lag2
BPDU filter enabled interfaces : None
Root guard enabled interfaces : 1/1/3
Loop guard enabled interfaces : 1/1/2
TCN guard enabled interfaces : 1/1/1-1/1/3

Interface count by state

Instance ID Blocking Listening Learning Forwarding

0 2 0 0 15
1 2 0 0 15
2 2 0 0 15

Total = 3 6 0 0 45
```

## show spanning-tree summary root

### Syntax

```
show spanning-tree summary root
```

### Description

Shows spanning tree root summary information.

## Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

*On the 6400 Switch Series, interface identification differs.*

Showing spanning tree root summary:

```
switch# show spanning-tree summary root
```

```
STP status : Enabled
Protocol : MSTP
System ID : 70:72:cf:32:50:f5
```

Root bridge for STP Instance : 0,1,2

| Instance ID | Priority | Root ID           | Root cost | Hello Time | Max Age | Fwd Dly | Root Port |
|-------------|----------|-------------------|-----------|------------|---------|---------|-----------|
| 0           | 32768    | 70:72:cf:32:50:f5 | 0         | 2          | 20      | 15      | n/a       |
| 1           | 32768    | 70:72:cf:32:50:f5 | 0         | 2          | 20      | 15      | n/a       |
| 2           | 32768    | 70:72:cf:32:50:f5 | 200       | 2          | 20      | 15      | 1/1/1     |

# spanning-tree

## Syntax

```
spanning-tree
```

```
no spanning-tree
```

## Description

Enables the spanning tree protocol on the switch.

The `no` form of this command disables the spanning tree protocol on the switch.

## Command context

```
config
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling spanning tree:

```
switch(config)# spanning-tree
```

Disabling spanning tree:

```
switch(config)# no spanning-tree
```

# spanning-tree bdpu-filter

## Syntax

```
spanning-tree bdpu-filter
```

```
no spanning-tree bdpu-filter
```

## Description

Enables the BPDU filter for the interface.

The BPDU filter feature allows control of spanning tree participation on a per-port basis. It can be used to exclude specific ports from becoming part of spanning tree operations. A port with the BPDU filter enabled will ignore incoming BPDU packets and stay locked in the spanning tree forwarding state. All other ports maintain their role. Typical uses for this parameter include:

- To have MSTP operations running on selected ports of the switch rather than every port of the switch at a time.
- To prevent the spread of errant BPDU frames.
- To eliminate the need for a topology change when a port's link status changes. For example, ports that connect to servers and workstations can be configured to remain outside of spanning tree operations.
- To protect the network from denial of service attacks that use spoofing BPDUs by dropping incoming BPDU frames. For this scenario, BPDU protection offers a more secure alternative, implementing port shut down and a detection alert when errant BPDU frames are received.



**NOTE:** Ports configured with the BPDU filter mode remain active (learning and forward frames). However, spanning tree cannot receive or transmit BPDUs on the port. The port remains in a forwarding state, permitting all broadcast traffic. This can create a network storm if there are any loops (that is, redundant links) using these ports. If you suddenly have a high load, disconnect the link and disable the BPDU filter (using the no command.)

The `no` form of the command sets the BPDU filter status to the default of disabled on the interface.

## Command context

```
config-if
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling the BPDU filter on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree bdpu-filter
```

Disabling BPDU filter on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree bdpu-filter
```

## spanning-tree bpdu-guard

### Syntax

```
spanning-tree bpdu-guard
```

```
no spanning-tree bpdu-guard
```

### Description

Enables the BPDU guard on the selected switch interface. When BPDU guard is enabled, interfaces receiving MSTP BPDUs become disabled.

BPDU protection is a security feature designed to protect the active MSTP topology by preventing spoofed BPDU packets from entering the MSTP domain. In a typical implementation, BPDU protection would be applied to edge ports connected to end user devices that do not run MSTP. If MSTP BPDU packets are received on a protected port, this feature disables that port and alerts the network manager using an SNMP trap.

Occasionally a hardware or software failure can cause MSTP to fail, creating forwarding loops that can cause network failures where unidirectional links are used. The non-designated port transitions in a faulty manner because the port is no longer receiving MSTP BPDUs.

The `no` form of the command disables BPDU guard on the selected interface.

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling the BPDU guard on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree bpdu-guard
```

Disabling BPDU guard on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree bpdu-guard
```

## spanning-tree bpdu-guard timeout

### Syntax

```
spanning-tree bpdu-guard timeout <INTERVAL>
```

```
no spanning-tree bpdu-guard timeout
```

### Description

Enables and configures the auto re-enable timeout in seconds for all interfaces with BPDU guard enabled. When an interface is disabled after receiving an unauthorized BPDU it will automatically be re-enabled after the timeout expires. The default is for the interface to stay disabled until manually re-enabled.

The `no` form of the command disables BPDU guard timeout on the interface. This is the default.

## Command context

config

## Parameters

**<Interval>**

Specifies the re-enable timeout in seconds. Range: 1 to 65535.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Enabling the BPDU guard timeout on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree bpdu-guard timeout 10
```

Disabling BPDU guard timeout on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree bpdu-guard
```

# spanning-tree config-name

## Syntax

```
spanning-tree config-name <CONFIG-NAME>
```

```
no spanning-tree config-name [<CONFIG-NAME>]
```

## Description

Sets the configuration name for the MST region in which the switch resides.

All switches within an MST region must have identical configuration names. For more than one MSTP switch in the same MST region, the identical region name must be configured on all such switches. If the default configuration name is retained on a switch, it cannot exist in the same MST region with another switch.

The **no** form of this command overwrites the currently configured name with the default name. The default name is a text string using the hexadecimal representation of the system MAC address.

## Command context

config

## Parameters

**<CONFIG-NAME>**

Specifies the configuration name for the MST region in which the switch resides. Default: text string using the hexadecimal representation of the MAC address of the switch. Range: 1 - 32 nonblank characters (case-sensitive).

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the configuration name to MST0:

```
switch(config)# spanning-tree config-name MST0
```

Setting the configuration name to the default value:

```
switch(config)# no spanning-tree config-name
```

## spanning-tree config-revision

### Syntax

```
spanning-tree config-revision <REVISION-NUMBER>
```

```
no spanning-tree config-revision [<REVISION-NUMBER>]
```

### Description

Configures the revision number for the MST region in which the switch resides. All switches within an MST region must have identical revision numbers. Use this setting to differentiate between region configurations. For example, when changing configuration settings within a region where you want to track the configuration versions you use, or when creating a new region from a subset of switches in a current region and you want to maintain the same region name.

The `no` form of this command overwrites the currently configured revision number of the MST region and sets it to the default value of 0.

### Command context

config

### Parameters

**<REVISION-NUMBER>**

Specifies the revision number for the MST region in which the switch resides. Range: 0 - 65535. Default: 0.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the revision to 40:

```
switch(config)# spanning-tree config-revision 40
```

Setting the revision to the default value:

```
switch(config)# no spanning-tree config-revision
```

## spanning-tree cost

### Syntax

```
spanning-tree cost <PORT-COST>
```

```
no spanning-tree cost [<PORT-COST>]
```

## Description

Sets individual port cost for MSTI 0.

For a given port, the path cost setting can be different for different MSTIs to which the port may belong.

The switch uses the path cost to determine which ports are the forwarding ports in the MSTI; that is, which links to use for the active topology of the MSTI and which ports to block.

The `no` form of the command sets the port cost for MSTI 0 instance to the default value.

## Command context

`config-if`

## Parameters

**<PORT-COST>**

Specifies the cost of the port for MSTI 0. Range: 1-200,000,000. Default is calculated from the port link speed:

- 10 Mbps link speed equals a path cost of 2,000,000.
- 100 Mbps link speed equals a path cost of 200,000.
- 1 Gbps link speed equals a path cost of 20,000.
- 2 Gbps link speed equals a path cost of 10,000.
- 10 Gbps link speed equals a path cost of 2,000.
- 100 Gbps link speed equals a path cost of 200.
- 1 Tbps link speed equals a path cost of 20.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the cost to 2000 on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree cost 2000
```

Setting the cost to the default on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree cost
```

# spanning-tree forward-delay

## Syntax

`spanning-tree forward-delay <DELAY-IN-SECS>`

`no spanning-tree forward-delay [<DELAY-IN-SECS>]`



## Description

Configures the time the switch waits between transitions from listening to learning and from learning to forwarding states.

The `no` form of this command sets forward delay time for the bridge to the default of 15 seconds.

## Command context

`config`

## Parameters

**<DELAY-IN-SECS>**

Specifies the forward delay time in seconds. Default: 15 seconds. Range: 4-30.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting forward delay to 6 seconds:

```
switch(config)# spanning-tree forward-delay 6
```

Setting forward delay to the default of 15 seconds:

```
switch(config)# no spanning-tree forward-delay
```

# spanning-tree hello-time

## Syntax

```
spanning-tree hello-time <HELLO-IN-SECS>
```

```
no spanning-tree hello-time [<HELLO-IN-SECS>]
```

## Description

Configures the transmission interval between consecutive Bridge Protocol Data Units (BPDU) that the switch sends as a root bridge. The hello time interval is inserted in outbound BPDUs.

The `no` form of this command sets hello time to the default of 2 seconds.

## Command context

`config`

## Parameters

**<HELLO-IN-SECS>**

Specifies the hello time interval in seconds. Default: 2 seconds. Range: 2-10.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the hello time interval to 6 seconds:

```
switch(config)# spanning-tree hello-time 6
```

Setting the hello time interval to the default of 2 seconds:

```
switch(config)# no spanning-tree hello-time
```

## spanning-tree instance cost

### Syntax

```
spanning-tree instance <INSTANCE-ID> cost <PORT-COST>
```

```
no spanning-tree instance <INSTANCE-ID> cost [<PORT-COST>]
```

### Description

Sets the individual port cost for an MSTI. The switch uses the path cost to determine which links to use for the active topology of the MSTI (forwarding ports) and which ports to block. The path cost setting for a port can be different on each MSTI to which the port belongs.

The **no** form of this command sets the port cost for an MSTI to the default value.

### Command context

config-if

### Parameters

**<INSTANCE-ID>**

Specifies the MSTI number. Range: 1-64.

**<PORT-COST>**

Specifies the cost of the port for the MSTI. Range: 1-200000000. Default value is calculated from the port link speed:

- 10 Mbps link speed equals a path cost of 2000000.
- 100 Mbps link speed equals a path cost of 200000.
- 1 Gbps link speed equals a path cost of 20000.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the port 1/1/1 cost for MSTI 1 to 2000:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree instance 1 cost 2000
```

Setting the port 1/1/1 cost for MSTI 1 to the default:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree instance 1 cost
```

## spanning-tree instance port-priority

### Syntax

```
spanning-tree instance <INSTANCE-ID> port-priority <PRIORITY-MULTIPLIER>
```

```
no spanning-tree instance <INSTANCE-ID> port-priority [<PRIORITY-MULTIPLIER>]
```

### Description

Configures the priority as a priority multiplier for the specified ports in the specified MST instance.

For a given port, the priority setting can be different for different MST instances to which the port may belong.

The `no` form of this command sets the port priority to the default value of 8 for the MST instance. The default priority value is derived by multiplying 8 by 16.

### Command context

config-if

### Parameters

**<INSTANCE-ID>**

Specifies the MSTP instance number. Range: 1-64.

**<PRIORITY-MULTIPLIER>**

Specifies the priority as a multiplier. Default: 8. Range: 0 to 15.

The priority range for a port in a given MST instance is 0 to 255. However, this command specifies the priority as a multiplier (0 to 15) of 16. When you specify a priority multiplier of 0 to 15, the actual priority assigned to the switch is: (priority-multiplier) x 16.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the port 1/1/1 priority for instance 1 to 8:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree instance 1 port-priority 8
```

Setting the port 1/1/1 priority for instance 1 to the default:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree instance 1 port-priority
```

## spanning-tree instance priority

### Syntax

```
spanning-tree instance <INSTANCE-ID> priority <PRIORITY-MULTIPLIER>
```

```
no spanning-tree instance <INSTANCE-ID> priority [<PRIORITY-MULTIPLIER>]
```

## Description

Sets the switch priority for the specified MST instance.

The `no` form of this command sets the priority for the specified instance to the default of 8.

## Command context

`config`

## Parameters

**<INSTANCE-ID>**

Specifies the MSTP instance number. Range: 1 to 64.

**<PRIORITY-MULTIPLIER>**

Specifies the priority as a multiplier. Default: 8. Range: 0 to 15.

The priority range for an MSTP switch is 0-61440. However, this command specifies the priority as a multiplier (0 - 15) of 4096. That is, when you specify a priority multiplier value of 0 - 15, the actual priority assigned to the switch is: (priority-multiplier) x 4096. For example, with 2 as the priority-multiplier on a given MSTP switch, the switch priority setting is 8,192.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the priority multiplier for instance 1 to 5:

```
switch(config)# spanning-tree instance 1 priority 5
```

Setting the priority multiplier for instance 1 to the default of 8:

```
switch(config)# no spanning-tree instance 1 priority
```

# spanning-tree instance vlan

## Syntax

```
spanning-tree instance <INSTANCE-ID> vlan <VLAN-ID>
```

```
no spanning-tree instance <INSTANCE-ID> vlan <VLAN-ID>
```

## Description

Creates a new instance with VLANs mapped or maps VLANs to an existing instance.

Each instance must have at least one VLAN mapped to it. When VLANs are mapped to an instance, they are automatically unmapped from the instance they were mapped to before. Any MSTP instance can have all the VLANs configured on the switch.

The `no` form of this command removes the specified VLAN from the MSTP instance.

## Command context

`config`

## Parameters

**<INSTANCE-ID>**

Specifies the MSTP instance number. Range: 1 to 64.

**<VLAN-ID>**

Specifies a VLAN ID number.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Mapping VLAN 1 to instance 1:

```
switch(config)# spanning-tree instance 1 vlan 1
```

Removing VLAN 1 from instance 1:

```
switch(config)# no spanning-tree instance 1 vlan 1
```

# spanning-tree link-type

## Syntax

```
spanning-tree link-type {point-to-point|shared}
```

## Description

Specifies the link type of the interface, which is normally derived from the duplex setting of the port. The default setting depends on the duplex mode of the port: full-duplex ports are point-to-point, half-duplex ports are shared.

## Command context

```
config-if
```

## Parameters

**point-to-point**

Specifies the link type as point-to-point.

**shared**

Specifies the link type as shared.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the link type to point-to-point on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree link-type point-to-point
```

Setting the link type to shared on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree link-type shared
```

## spanning-tree loop-guard

### Syntax

```
spanning-tree loop-guard
```

```
no spanning-tree loop-guard
```

### Description

Enables the loop guard on the interface. STP loop guard is best applied on blocking or forwarding ports. The `no` form of the command sets the loop guard status to the default of disabled on the interface.

### Command context

```
config-if
```

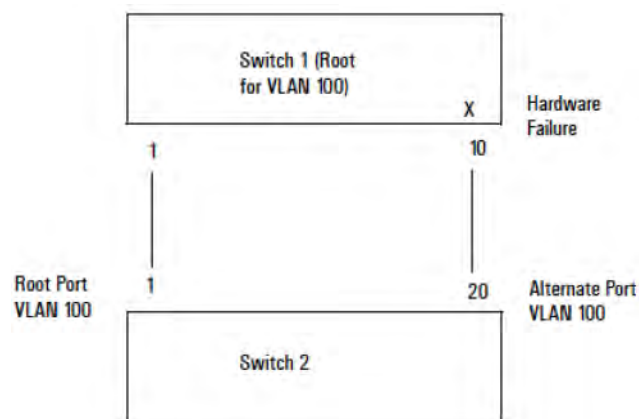
### Authority

Administrators or local user group members with execution rights for this command.

### Usage

Occasionally a hardware or software failure can cause MSTP to fail, creating forwarding loops that can cause network failures where unidirectional links are used. The non-designated port transitions in a faulty manner because the port is no longer receiving MSTP BPDUs.

Loop guard causes the non-designated port to go into the MSTP loop inconsistent state instead of the forwarding state. In the loop inconsistent state the port prevents data traffic and BPDU transmission through the link, therefore avoiding the loop creation. When BPDUs again are received on the inconsistent port, it resumes normal MSTP operation automatically.



In this example, the transmission from switch 1 port 10 to switch 2 port 20 is blocked due to a hardware failure. Switch 2 port 20 does not receive BPDUs and goes into a forwarding state, creating a loop.

When loop guard is configured for switch 2 port 20, this port goes from a forwarding state to an inconsistent state, and does not forward the traffic through the link, thus avoiding loop creation.

### Examples

Enabling the loop guard on interface `1/1/1`:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree loop-guard
```

Disabling loop guard on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree loop-guard
```

## spanning-tree max-age

### Syntax

```
spanning-tree max-age <AGE-IN-SECS>
```

```
no spanning-tree max-age [<AGE-IN-SECS>]
```

### Description

Sets the maximum age timer, which specifies the maximum age value that the switch inserts in outbound BPDU packets it sends as a root bridge. Max-age is the interval, specified in the BPDU, that BPDU data remains valid after its reception.

The bridge recomputes the spanning tree topology if it does not receive a new BPDU before max-age expiry.

The `no` form of this command sets the max-age value to the default of 20 seconds.

### Command context

```
config
```

### Parameters

**<AGE-IN-SECS>**

Specifies the max-age in seconds. Range: 6 to 40. Default: 20.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the max-age to 10 seconds:

```
switch(config)# spanning-tree max-age 10
```

Setting the max-age to the default of 20 seconds:

```
switch(config)# no spanning-tree max-age
```

## spanning-tree max-hops

### Syntax

```
spanning-tree max-hops <HOP-COUNT>
```

```
no spanning-tree max-hops [<HOP-COUNT>]
```

## Description

Configures the max hop setting that the switch inserts into BPDUs that it sends out as the root bridge. The max hop setting determines the number of bridges in an MST region that a BPDU can traverse before it is discarded.

The `no` form of this command sets the maximum number of hops to the default of 20.

## Command context

`config`

## Parameters

**<HOP-COUNT>**

Specifies the maximum number of hops. Range: 1 to 40. Default: 20.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the hop count to 10:

```
switch(config)# spanning-tree max-hops 10
```

Setting the max-age to the default of 20:

```
switch(config)# no spanning-tree max-hops
```

# spanning-tree mode

## Syntax

```
spanning-tree mode {mstp|rpvst}
```

## Description

Sets the spanning tree mode to either MSTP mode (Multiple-instance Spanning Tree Protocol) or RPVST mode (Rapid Per VLAN Spanning Tree).

The `no` form of this command sets the spanning tree mode to the default value `mstp`.

## Command context

`config`

## Parameters

**mstp**

Sets the mode to MSTP (Multiple-instance Spanning Tree Protocol), which applies the STP (spanning tree protocol) separately for each set of VLANs (called an MSTI - multiple spanning tree instance).

**rpvst**

Sets the mode to RPVST (Rapid Per VLAN Spanning Tree).

## Authority

Administrators or local user group members with execution rights for this command.



## Examples

Enabling MSTP mode:

```
switch(config)# spanning-tree mode mstp
```

Enabling RPVST mode:

```
switch(config)# spanning-tree mode rpvt
```

## spanning-tree port-priority

### Syntax

```
spanning-tree port-priority <PRIORITY-MULTIPLIER>
```

```
no spanning-tree port-priority [<PRIORITY-MULTIPLIER>]
```

### Description

Configures the port priority. The priority of a port can be different for each MST instance to which it belongs.

The `no` form of the command sets the port priority for MST instance 0 to the default of 8. The default priority value is derived by multiplying 8 by 8. For LAG interfaces the default is 4.

### Command context

config-if

### Parameters

**<PRIORITY-MULTIPLIER>**

Specifies the port priority as a multiplier. Default: 8, except for LAG interfaces where the default is 4. Range: 0 to 15.

The priority range for a port in a given MSTI is 0 to 255. However, this command specifies the priority as a multiplier (0 to 15) of 16. When you specify a priority multiplier of 0 to 15, the actual priority assigned to the switch is: (priority-multiplier) x 16.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the port priority to 8 on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree port-priority 8
```

Setting the port priority to the default on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree port-priority
```

# spanning-tree port-type

## Syntax

```
spanning-tree port-type {admin-edge|admin-network}

no spanning-tree port-type [admin-edge|admin-network]
```

## Description

Sets the STP port type for the interface.

Port types include: admin-edge and admin-network.

The `no` form of the command sets the port type to the default of admin-network.

## Command context

config-if

## Parameters

### admin-edge

Specifies the port type as administrative edge. During spanning tree establishment, ports with admin-edge enabled transition immediately to the forwarding state.

### admin-network

Specifies the port type as administrative network. When this option is selected, the port looks for BPDUs for the first 3 seconds. If there are none, the port is classified as an edge port and immediately starts forwarding packets. If BPDUs are seen on the port, the port is classified as a non-edge port and normal STP operation commences on that port.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the port type to admin-edge on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree port-type admin-edge
```

Setting the port type to admin-network on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree port-type admin-network
```

Setting the port type to the default of admin-network on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree port-type
```

# spanning-tree priority

## Syntax

```
spanning-tree priority <PRIORITY-MULTIPLIER>
```

```
no spanning-tree priority [<PRIORITY-MULTIPLIER>]
```

## Description

Configures the switch (bridge) priority for the designated region in which the switch resides.

The switch compares this priority with the priorities of other switches in the same region to determine the root switch for the region. The lower the priority value, the higher the priority.

The `no` form of this command sets the bridge priority to the default of 8. The default priority value is derived by multiplying 8 by 4096.

## Command context

```
config
```

## Parameters

**<PRIORITY-MULTIPLIER>**

Specifies the priority as a multiplier. Range: 0 to 15. Default: 8.

The priority range for an MSTP switch is 0-61440. However, this command specifies the priority as a multiplier (0 to 15) of 4096. That is, when you specify a priority multiplier value of 0 to 15, the actual priority assigned to the switch is: (priority-multiplier) x 4096. For example, with 2 as the priority-multiplier on a given MSTP switch, the switch priority setting is 8,192.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

Every switch running an instance of MSTP has a Bridge Identifier, which is a unique identifier that helps distinguish this switch from all others. The switch with the lowest Bridge Identifier is elected as the root for the tree. The Bridge Identifier is composed of a configurable priority component (2 bytes) and the bridge's MAC address (6 bytes). You can change the priority component provides flexibility in determining which switch will be the root for the tree, regardless of its MAC address.

## Examples

Setting the priority multiplier to 12:

```
switch(config)# spanning-tree priority 12
```

Setting the priority multiplier to the default of 8:

```
switch(config)# no spanning-tree priority
```

## spanning-tree root-guard

### Syntax

```
spanning-tree root-guard
```

```
no spanning-tree root-guard
```

### Description

Enables the root guard on the interface.

When a port is enabled as root-guard, it cannot be selected as the root port even if it receives superior STP BPDUs. The port is assigned an "alternate" port role and enters a blocking state if it receives superior MSTP BPDUs.

A superior BPDU contains both "better" information on the root bridge and path cost to the root bridge, which would normally replace the current root bridge selection.

The `no` form of the command sets the root guard status to the default of disabled on the interface.

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling the root guard on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree root-guard
```

Disabling root guard on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree root-guard
```

## spanning-tree rpvst-filter

### Syntax

```
spanning-tree rpvst-filter
```

```
no spanning-tree rpvst-filter
```

### Description

Enables the RPVST filter for the interface.

When the RPVST filter is enabled, the ingressing RPVST proprietary BPDUs are dropped after copying to CPU whereas the standard IEEE RPVST BPDUs are still allowed. This helps in preventing the flooding of RPVST proprietary BPDUs under an MSTP-RPVST interop environment.



**NOTE:** If the neighboring switch is running RPVST then this pair of switches will not converge as RPVST BPDUs will not reach them.

If enabling RPVST filter causes a high traffic load, shutdown the port and reconfigure the BPDU filter with the CLI command: `no spanning tree rpvst-filter`.

RPVST filter is disabled by default.

### Command context

`config-if`

### Authority

Administrators or local user group members with execution rights for this command.

### Example

*On the 6400 Switch Series, interface identification differs.*

Enabling the RPVST filter on interface 1/1/1:

```
switch# configure terminal
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree rpvst-filter
```

Disabling RPVST filter on interface 1/1/1:

```
switch# configure terminal
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree rpvst-filter
```

## spanning-tree rpvst-guard

### Syntax

`spanning-tree rpvst-guard`

`no spanning-tree rpvst-guard`

### Description

Enables RPVST guard on the switch interface. When RPVST guard is enabled on an interface, it will disable that interface if RPVST BPDUs are received on it.

The `no` form of the command sets the RPVST guard status to the default of disabled on the interface.

### Command context

`config-if`

### Authority

Administrators or local user group members with execution rights for this command.

### Example

*On the 6400 Switch Series, interface identification differs.*

Enabling RPVST guard on interface 1/1/1:

```
switch# configure terminal
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree rpvst-guard
```

Disabling RPVST guard on interface 1/1/1:

```
switch# configure terminal
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree rpvst-guard
```

## spanning-tree rpvst-mstp interconnect vlan

### Syntax

```
spanning-tree rpvst-mstp-interconnect-vlan <VLAN-ID>
no spanning-tree rpvst-mstp-interconnect-vlan
```

### Description

Configures the VLAN that has to be used to interconnect RPVST and MSTP domains. VLAN 1 is used by default.

The `no` form of this command removes the VLAN configuration.

- It is required to create the interconnect VLAN and then configure RPVST spanning tree on it.
- The same interconnect VLAN must be kept on all the switches in the network.
- Adding or deleting the interconnect VLAN triggers a re-convergence in the network.
- Deleting a VLAN that is configured as the interconnect VLAN does not reset the value to the default.

### Command context

config

### Parameters

<VLAN-ID>

Specifies the number of a VLAN.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

This example configures VLAN 10 to used to interconnect RPVST and MSTP domains.

```
switch#(config)# spanning-tree rpvst-mstp-interconnect-vlan 10
```

## spanning-tree tcn-guard

### Syntax

```
spanning-tree tcn-guard
no spanning-tree tcn-guard
```

### Description

Enables the TCN (Topology Change Notification) guard in the interface. When enabled for a port, the port stops propagating received topology change notifications and topology changes to other ports.

The `no` form of the command sets the TCN guard status to the default of disabled on the interface.

## Command context

config-if

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling TCN guard on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree tcn-guard
```

Disabling TCN guard on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree tcn-guard
```

# spanning-tree transmit-hold-count

## Syntax

```
spanning-tree transmit-hold-count <COUNT>
```

```
no spanning-tree transmit-hold-count [<COUNT>]
```

## Description

Sets the maximum number of BPDUs per second that the switch can send from an interface.

The `no` form of this command sets the transmit-hold-count to the default of 6.

## Command context

config

## Parameters

<COUNT>

Specifies the number of BPDUs that can be sent per second. Range: 1 to 10. Default: 6.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the transmit-hold-count to 5:

```
switch(config)# spanning-tree transmit-hold-count 5
```

Setting the transmit-hold-count to the default of 6:

```
switch(config)# no spanning-tree transmit-hold-count
```

# spanning-tree trap

## Syntax

```
spanning-tree trap {new-root|topology-change [instance <0-64>] | errant-bpdu |
 root-guard-inconsistency | loop-guard-inconsistency}

no spanning-tree trap {new-root|topology-change [instance <0-64>] | errant-bpdu |
 root-guard-inconsistency | loop-guard-inconsistency}
```

## Description

This command enables SNMP traps for new root, topology change event, errant-bpdu received event, root-guard inconsistency, and loop-guard inconsistency notifications. It is disabled by default.

The **no** form of this command disables SNMP traps for new root and topology change event notifications.

## Command context

config

## Parameters

### **new-root**

Enabling SNMP notification when a new root is elected on any MST instance on the switch.

### **topology-change**

Enabling SNMP notification when a topology change event occurs in the specified MST instance on the switch.

### **errant-bpdu**

Enabling SNMP notification when an errant bpdu is received by any MST instance on the switch.

### **root-guard-inconsistency**

Enabling SNMP notification when the root-guard finds the port inconsistent for any MST instance on the switch.

### **loop-guard-inconsistency**

Enabling SNMP notification when the loop-guard finds the port inconsistent for any MST instance on the switch.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling SNMP notification when a new root is elected:

```
switch(config)# spanning-tree trap new-root
```

Enabling SNMP notification when a topology change event occurs:

```
switch(config)# spanning-tree trap topology-change
```

Enabling SNMP notification when a topology change event occurs on instance 1:

```
switch(config)# spanning-tree trap topology-change instance 1
```

Disabling SNMP traps when a topology change event notification occurs on instance 1:



```
switch(config)# no spanning-tree trap topology-change instance 1
```

## clear mvrp statistics

### Syntax

```
clear mvrp statistics [<PORT-NUM> | <PORT-LIST> | LAG <LAG-NUM>]
```

### Command context

Manager (#)

### Description

Resets the MVRP statistic counters globally or for the specified ports or LAG.

### Parameters

**<PORT-NUM>**

Specifies a port number.

**<PORT-LIST>**

Specifies a list of ports.

**LAG <LAG-NUM>**

Specifies a Link Aggregation number. Range: 1 to 128.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch# clear mvrp statistics 1/1/1
```

## mvrp

### Syntax

```
mvrp
```

```
no mvrp
```

### Description

Enables the MVRP feature globally or on a specific interface. By default, MVRP is disabled.

The `no` form of this command disables MVRP.



**NOTE:** MVRP and VLAN translation cannot be enabled on the same interface.

## Command context

```
config
config-if
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling MVRP globally:

```
switch(config)# mvrp
```

Enabling MVRP on an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# mvrp
```

# mvrp registration

## Syntax

```
mvrp registration {normal | fixed | forbidden [<VLAN-LIST>]}

no mvrp registration forbidden {<VLAN-LIST>}
```

## Description

Configures the MVRP registrar state which determines how an MVRP participant responds to MRP messages. The default registration mode is normal.

The `no` command removes the specified VLANs from the forbidden list.

## Command context

```
config-if
```

## Parameters

### normal

Enables dynamic registration and deregistration of VLANs on the interface, and propagates VLAN information to other switches on the network. Default.

### fixed

Disables dynamic deregistration of VLANs and drops received MVRP frames. The interface does not deregister dynamic VLANs or register new dynamic VLANs.

### forbidden

Disables dynamic registration of VLANs and drops received MVRP frames. The MVRP participant does not register new dynamic VLANs or re-register a deregistered dynamic VLAN.

### <VLAN-LIST>

Disables dynamic registration of VLANs and drops received MVRP frames for specific VLANs only. Normal behavior applies to all other VLANs. Specify the number of a single VLAN, or a series of numbers for a range of VLANs, separated by commas (1, 2, 3, 4), dashes (1-4), or both (1-4,6).

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config)# switch(config-if)# mvrp registration forbidden 10
```

```
switch(config-if)# mvrp registration fixed
```

```
switch(config-if)# mvrp registration forbidden 1,2,10-20
```

## mvrp timer

### Syntax

```
mvrp timer {join | leave | leaveall | periodic} <TIME>
```

```
no mvrp timer {join | leave | leaveall | periodic}
```

### Description

Sets an MVRP timer.

The `no` form of this command sets the specified timer to its default value.

### Command context

config-if

### Parameters

**join** <TIME>

Sets the join timer. You can use the timer to space MVRP join messages. To ensure that join messages are transmitted to other participants, an MRP participant waits for the specified period of the join timer before sending a join message. The Join timer must be less than half of the Leave Timer. Range: 20 to 100 in centiseconds. Default: 20.

**leave** <TIME>

Sets the leave timer for the port, specifying the time that the registrar state machine waits in the LV state before transiting to the MT state. The leave timer must be at least twice the join timer and must be less than the leave all timer. Range: 40 - 1000000 centiseconds. Default: 300 centiseconds.

**leaveall** <TIME>

Sets the leave all timer for the port, specifying the frequency with which the leave all state machine generates leave all PDUs. Range: 500 to 1000000 centiseconds. Default: 1000.

**periodic** <TIME>

Sets the periodic timer for the port, specifying the frequency with which the periodic transmission state machine generates periodic events. The periodic timer is set to 1 second when it is started. Range: 100 to 1000000 centiseconds. Default: 100.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config-if)# mvrp timer join 22
```

## show mvrp config

### Syntax

```
show mvrp config [<PORT-NUM> | <PORT-LIST> | LAG <LAG-NUM>]
```

### Description

Displays the MVRP configuration for all L2 ports or optionally for the ports specified.

### Command context

Operator (>) or Manager (#)

### Parameters

**<PORT-NUM>**

Specifies displaying information for a particular port number.

**<PORT-LIST>**

Specifies displaying information for a list of ports.

**LAG <LAG-NUM>**

Specifies displaying information by LAG. Range: 1 to 128.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

```
switch# show mvrp config
Configuration and Status - MVRP
Global MVRP status : Disabled
Port Status Registration Join Leave LeaveAll Periodic
 Status Type Timer Timer Timer Timer

1/1/1 Disabled Normal 20 300 1000 100
1/1/2 Disabled Normal 20 300 1000 100
1/1/3 Disabled Normal 20 300 1000 100
```

## show mvrp state

### Syntax

```
show mvrp state [<VLAN-ID> | <VLAN-ID> <PORT-NUM>]
```

### Description

Displays the MVRP Registrar and Applicant state machine information for all ports on which MVRP is enabled, or for specific ports.

## Command context

Operator (>) or Manager (#)

## Parameters

<VLAN-ID>

<PORT-NUM>

Specifies a physical port on the switch. Format: member/slot/port.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

```
switch# show mvrp state 1
Configuration and Status - MVRP state for VLAN 1
Port VLAN Registrar Applicant
 State State

1/1/1 1 MT QA
```

```
switch# show mvrp state 10 1/1/1
Configuration and Status - MVRP state for VLAN 10
Port VLAN Registrar Applicant Forbid
 State State Mode

1/1/1 10 MT LO Yes
switch#
```

# show mvrp statistics

## Syntax

```
show mvrp statistics [<PORT-LIST>]
```

## Description

Displays MVRP statistics for all ports or on the ports specified in the list.

## Command context

Operator (>) or Manager (#)

## Parameters

<PORT-LIST>

Specifies a list of ports. When specifying a list of ports, the ports for which there are no statistics will be listed in the output.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

```

switch# show mvrp statistics
Status and Counters - MVRP
MVRP statistics for port : 1/1/1

Failed registration : 0
Last PDU origin : 48:0f:cf:af:b1:76
Total PDU Transmitted : 13127
Total PDU Received : 327
Frames Discarded : 0
Message type Transmitted Received

New 0 0
Empty 50029394 1264
In 0 4
Join Empty 1425 48
Join In 563 555
Leave 0 0
Leaveall 12218 25

```

```

switch# show mvrp statistics 1/1/1

Status and Counters - MVRP
MVRP statistics for port : 1/1/1

Failed registration : 0
Last PDU origin : 48:0f:cf:af:b1:76
Total PDU Transmitted : 14874
Total PDU Received : 327
Frames Discarded : 0
Message type Transmitted Received

New 0 0
Empty 57181612 1264
In 0 4
Join Empty 1425 48
Join In 563 555
Leave 0 0
Leaveall 13965 25

```

## clear nd-snooping binding

### Syntax

```
clear nd-snooping binding {all | ipv6 <IPV6-ADDR> vlan <VLAN-ID> |
 port <PORT-NUM> | vlan <VLAN-ID>}
```

### Description

Clears ND snooping binding entries.

### Command context

Operator (>) or Manager (#)

### Parameters

#### all

Specifies that all ND binding information is to be cleared.

#### ip <IPV6-ADDR> vlan <VLAN-ID>

Specifies the IPv6 address and VLAN for which all ND binding information is to be cleared.

#### port <PORT-NUM>

Specifies the port for which all ND binding information is to be cleared.

#### vlan <VLAN-ID>

Specifies the VLAN for which all ND binding information is to be cleared. Range: 1 to 4094.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Clearing all ND binding information for 5000::1:

```
switch(config)# clear nd-snooping binding ipv6 5000::1
```

Clearing all ND binding information for 5000::1 vlan 1:

```
switch(config)# clear nd-snooping binding ipv6 5000::1 vlan 1
```

Clearing all ND binding information for port 1/1/10:

```
switch(config)# clear nd-snooping binding port 1/1/10
```

Clearing all ND binding information for VLAN 10:

```
switch(config)# clear nd-snooping binding vlan 10
```



Clearing all ND binding information:

```
switch(config)# clear nd-snooping binding all
```

## clear nd-snooping statistics

### Syntax

```
clear nd-snooping statistics
```

### Description

Clears all ND snooping statistics.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Clear all ND snooping statistics:

```
switch# clear nd-snooping statistics
```

## nd-snooping

### Syntax

```
nd-snooping {enable|disable}
```

```
no nd-snooping {enable|disable}
```

### Description

Enables or disables ND snooping. ND snooping is disabled by default. ND snooping is not supported on the management interface.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling ND snooping:

```
switch(config)# nd-snooping enable
```

Disabling ND snooping:

```
switch(config)# nd-snooping disable
```

## nd-snooping (in config-vlan context)

### Syntax

```
nd-snooping
```

```
no nd-snooping
```

### Description

Enables ND snooping in the `config-vlan` context. ND snooping is disabled by default for all VLANs.

The `no` form of the command disables ND snooping on the specified VLAN, flushing all the IPv6 bindings learned for this VLAN since ND snooping was enabled for this VLAN.

### Command context

```
config-vlan
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling ND snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# nd-snooping
switch(config-vlan-100)# exit
switch(config)#
```

Disabling ND snooping on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no nd-snooping
switch(config-vlan-100)# exit
switch(config)#
```

## nd-snooping mac-check

### Syntax

```
nd-snooping mac-check
```

```
no nd-snooping mac-check
```

### Description

This command enables verification of the hardware address field in ND snooping packets. When enabled, the ICMPv6 target link layer address field and the source MAC address must be the same for packets received on untrusted ports or else the packets are dropped. This ND snooping MAC verification is enabled by default.

The `no` form of the command disables ND snooping MAC verification.

### Command context

```
config
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling ND snooping MAC verification:

```
switch(config)# nd-snooping mac-check
```

Disabling ND snooping MAC verification:

```
switch(config)# no nd-snooping mac-check
```

# nd-snooping prefix-list

## Syntax

```
nd-snooping prefix-list <IPV6-ADDR>
```

```
no nd-snooping prefix-list <IPV6-ADDR>
```

## Description

Configures the ND snooping prefix list for the selected VLAN and the specified IPv6 address prefix. ND snooping must be enabled both globally and on this VLAN before this prefix list configuration takes effect. The no form of this command removes the prefix list configuration for the selected VLAN and IPv6 address.

## Command context

```
config-vlan
```

## Parameters

<IPV6-ADDR>

Specifies the IPv6 address.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Configuring ND snooping prefix-list on VLAN 1:

```
switch(config)# vlan 1
switch(config-vlan-1)# nd-snooping prefix-list 2001::1/64
switch(config-vlan-1)# exit
switch(config)#
```

Remove configuration of ND snooping prefix-list on VLAN 100:

```
switch(config)# vlan 1
switch(config-vlan-1)# no nd-snooping prefix-list 2001::1/64
switch(config-vlan-1)# exit
switch(config)#
```

## nd-snooping max-bindings

### Syntax

```
nd-snooping max-bindings <MAX-BINDINGS>
```

```
no nd-snooping max-bindings
```

### Description

Sets the maximum number of ND bindings allowed on the selected interface. For all interfaces on which this command is not run, the default max bindings applies.

The no form of the command reverts max bindings for the selected interface to its default.

### Command context

```
config-if
```

### Parameters

**<MAX-BINDINGS>**

Specifies the maximum number of ND bindings. You can use the `show capacities` command to see the maximum available for your switch model.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Set the ND max bindings to 768 on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# nd-snooping max-bindings 768
switch(config-if)# exit
switch(config)#
```

Revert ND max bindings to its default on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no nd-snooping max-bindings
switch(config-if)# exit
switch(config)#
```

## nd-snooping nd-guard

### Syntax

```
nd-snooping nd-guard
```

```
no nd-snooping nd-guard
```

### Description

This command enables ND guard on the selected VLAN.

The no form of the command disables ND guard and deletes all the IPv6 bindings learned on the VLAN.



---

**NOTE:** ND snooping must be enabled in both the global context and the config-vlan context before this command can be used.

---

### Command context

config-vlan

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling ND snooping ND guard on VLAN 100:

```
switch(config)# nd-snooping enable
switch(config)# vlan 100
switch(config-vlan-100)# nd-snooping nd-guard
switch(config-vlan-100)# exit
switch(config)#
```

Disabling ND snooping ND guard on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no nd-snooping nd-guard
switch(config-vlan-100)# exit
switch(config)#
```

## nd-snooping ra-guard

### Syntax

nd-snooping ra-guard [log]

no nd-snooping ra-guard

### Description

This command enables Routing Advertisement (RA) guard on the selected VLAN. When enabled, ingress Routing Advertisement (RA) and Routing Redirect (RR) packets on the selected VLAN are blocked on untrusted ports. The packets are forwarded when received on trusted ports.

The no form of the command disables RA guard on the VLAN.



---

**NOTE:** ND snooping must be enabled in both the global context and the config-vlan context before this command can be used.

---

### Command context

config-vlan

### Parameters

[log]

Logs messages along with drop functionality.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling ND snooping RA guard on VLAN 100:

```
switch(config)# nd-snooping enable
switch(config)# vlan 100
switch(config-vlan-100)# nd-snooping ra-guard
switch(config-vlan-100)# exit
switch(config)#
```

Enabling ND snooping RA guard on VLAN 100 with event logging on dropped packets:

```
switch(config)# nd-snooping enable
switch(config)# vlan 100
switch(config-vlan-100)# nd-snooping ra-guard log
switch(config-vlan-100)# exit
switch(config)#
```

Disabling ND snooping RA guard on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no nd-snooping ra-guard
switch(config-vlan-100)# exit
switch(config)#
```

## nd-snooping ra-drop

### Syntax

```
nd-snooping ra-drop
```

```
no nd-snooping ra-drop
```

### Description

This command enables Routing Advertisement (RA) drop on the selected VLAN. When enabled, ingress RA packets on the selected VLAN are blocked on both trusted and untrusted ports. When disabled, RA packets are forwarded on the selected VLAN with ND snooping trusted port validation. RA drop is disabled by default.



**NOTE:** ND snooping must be enabled in both the config context and the config-vlan context before this command can be used.

The no form of the command disables ND snooping RA drop on the selected VLAN.

### Command context

```
config-vlan
```

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling ND snooping RA drop on VLAN 100:

```
switch(config)# nd-snooping enable
vlan 100
switch(config-vlan-100)# nd-snooping ra-drop
```

```
switch(config-vlan-100)# exit
switch(config)#
```

Disabling ND snooping RA drop on VLAN 100:

```
switch(config)# vlan 100
switch(config-vlan-100)# no nd-snooping ra-drop
switch(config-vlan-100)# exit
switch(config)#
```

## nd-snooping trust

### Syntax

```
nd-snooping trust
```

```
no nd-snooping trust
```

### Description

Enables ND snooping trust on the selected interface (port). Only server packets received on trusted ports are forwarded. All the ports are untrusted by default.

The no form of the command disables ND snooping trust on the selected port.

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling ND snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# nd-snooping trust
switch(config-if)# exit
switch(config)#
```

Disabling ND snooping trust on interface 2/2/1:

```
switch(config)# interface 2/2/1
switch(config-if)# no nd-snooping trust
switch(config-if)# exit
switch(config)#
```

## show nd-snooping

### Syntax

```
show nd-snooping [vlan <VLAN-ID>] [vsx-peer]
```

### Description

Shows either all ND snooping configuration or the configuration for the specified VLAN.

## Command context

Operator (>) or Manager (#)

## Parameters

**vlan** <VLAN-ID>

Specifies the VLAN for which the ND configuration is to be shown. Range: 1 to 4094.

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing all ND snooping configuration:

```
switch(config)# show nd-snooping
```

```
ND Snooping Information
=====
```

```
ND Snooping : Enabled
ND Snooping Enabled VLANs : 1
RA Drop Enabled VLANs : 2-3
MAC Address Check : Disabled
```

| PORT  | TRUST | MAX-BINDINGS | CURRENT-BINDINGS |
|-------|-------|--------------|------------------|
| 1/1/1 | Yes   | -            | -                |
| 1/1/2 | Yes   | -            | -                |
| 1/1/3 | No    | 100          | 10               |
| 1/1/4 | No    | 200          | 10               |
| 1/1/5 | No    | 300          | 10               |

Showing ND snooping configuration for VLAN 2:

```
switch(config)# show nd-snooping vlan 2
```

```
ND Snooping Information
=====
```

```
ND Snooping : Enabled
MAC Address Check : Disabled
RA Drop : Disabled
```

| PORT  | TRUST | MAX-BINDINGS | CURRENT-BINDINGS |
|-------|-------|--------------|------------------|
| 1/1/1 | Yes   | -            | -                |
| 1/1/2 | Yes   | -            | -                |
| 1/1/3 | No    | 100          | 10               |



## show nd-snooping binding

### Syntax

```
show nd-snooping binding [vsx-peer]
```

### Description

Shows the ND snooping binding configuration.

### Command context

Operator (>) or Manager (#)

### Parameters

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing the ND snooping binding configuration:

```
switch# show nd-snooping binding
```

| PORT  | IPV6-ADDRESS                            | MAC-ADDRESS       | VLAN | TIME-LEFT | STATE     |
|-------|-----------------------------------------|-------------------|------|-----------|-----------|
| 1/1/1 | 2001::1                                 | 00:00:0A:01:02:03 | 1    | 600       | Valid     |
| 1/1/2 | fe80::250:56ff:fe9a:143c                | 00:00:0B:01:02:03 | 2    | -         | Tentative |
| 1/1/3 | 2001:1111:2222:3333:4444:5555:6666:7777 | 00:00:0C:01:02:03 | 4094 | -         | Testing   |

## show nd-snooping prefix-list

### Syntax

```
show nd-snooping prefix-list [vsx-peer]
```

### Description

Shows the ND snooping prefix list information.

### Command context

Operator (>) or Manager (#)

### Parameters

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing the ND snooping prefix list information:

```
switch# show nd-snooping prefix-list
```

| VLAN | IPV6-ADDRESS-PREFIX | SOURCE  |
|------|---------------------|---------|
| 1    | 2001::/64           | Static  |
| 4094 | 3001::/64           | Dynamic |

## show nd-snooping statistics

### Syntax

```
show nd-snooping statistics [vsx-peer]
```

### Description

Shows the global ND snooping statistics.

### Command context

Operator (>) or Manager (#)

### Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing the global ND snooping statistics:

```
switch(config)# show nd-snooping statistics
```

| PACKET-TYPE | ACTION  | REASON                                          | COUNT |
|-------------|---------|-------------------------------------------------|-------|
| RA          | forward | RA packets received on trusted port             | 20    |
| RA          | drop    | RA packets received on untrusted port           | 45    |
| NS          | forward | NS packets received on trusted port             | 52    |
| NS          | forward | NS packets received on untrusted port           | 95    |
| NS          | drop    | NS packets failed MAC check                     | 14    |
| NS          | drop    | NS packets failed Prefix check                  | 12    |
| NS          | drop    | NS packets failed on max-binding limit          | 0     |
| NS          | drop    | NS packets failed ND snooping validation checks | 20    |
| NA          | forward | NA packets received on trusted port             | 17    |
| NA          | forward | NA packets received on untrusted port           | 30    |
| NA          | drop    | NA packets failed Prefix check                  | 15    |

|    |      |                                                 |   |
|----|------|-------------------------------------------------|---|
| NA | drop | NA packets failed on max-binding limit          | 2 |
| NA | drop | NA packets failed ND snooping validation checks | 5 |

## clear nae-data

### Syntax

```
clear nae-data
```

### Description

Clears all data collected by the Aruba Network Analytics Engine.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Example

```
switch# clear nae-data
This action will delete NAE data monitored so far.
Do you want to continue (y/n)?
```

## show nae-agent

### Syntax

```
show nae-agent [vsx-peer]
```

### Description

Shows information about the Aruba Network Analytics Engine agents that are available on the switch.

### Command context

Manager (#)

### Parameters

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Administrators or local user group members with execution rights for this command.

## Usage

This command shows the following information about the Aruba Network Analytics Engine agents that are available on the switch:

### Agent Name

The name of the agent. Length: 3 through 80 characters.

### Script Name

The name of the script. Length: 3 through 80 characters.

Example: `system_resource_monitor_mm1.default`

### Version

The version number of the script.

### Origin

The origin of the script:

**system**

Indicates that the script is provided as part of the system software.

**user**

Indicates that a user loaded the script.

### Disabled

Indicates whether the agent is disabled or active on the switch:

**true**

Indicates that the agent is disabled.

**false**

Indicates that the agent is active on the switch.

### Status

The current state of the agent. Status values are the following:

#### CRITICAL

The agent has encountered a critical error during execution. For information about the error, see the Analytics Dashboard of the Web UI.

#### MAJOR

The agent has encountered a major error during execution. For information about the error, see the Analytics Dashboard of the Web UI.

#### MINOR

The agent has encountered a minor error during execution. For information about the error, see the Analytics Dashboard of the Web UI.

#### NORMAL

Indicates that the agent is actively monitoring network conditions and handling events.

## Example

```
switch# show nae-agent
```

| Agent Name                      | Script Name               | Version | Origin | Disabled | Status  | Error |
|---------------------------------|---------------------------|---------|--------|----------|---------|-------|
| com.arubanetworks.monitor.agent | com.arubanetworks.monitor | 1.0     | user   | true     | UNKNOWN | NONE  |

|                                       |                                 |     |        |       |         |       |
|---------------------------------------|---------------------------------|-----|--------|-------|---------|-------|
| com.arubanetworks.wildcard.vlan.agent | com.arubanetworks.wildcard.vlan | 1.0 | user   | false | UNKNOWN | ERROR |
| system_resource_monitor.default       | system_resource_monitor         | 1.0 | system | false | NORMAL  | NONE  |

## show nae-script

### Syntax

```
show nae-script [vsx-peer]
```

### Description

Shows information about the Aruba Network Analytics Engine scripts that are available on the switch.

### Command context

Manager (#)

### Parameters

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

This command shows the following information about the Aruba Network Analytics Engine scripts that are available on the switch:

#### Script Name

The name of the script. Length: 3 through 80 characters.

Example: `system_resource_monitor_mm1.default`

#### Version

The version number of the script.

#### Origin

The origin of the script:

**system**

Indicates that the script is provided as part of the system software.

**user**

Indicates that a user loaded the script.

#### Status

The current state of the script. Status values are the following:

**CREATED**

The script has been uploaded to the switch, but script validation has not begun.

## ERROR

The script validation process detected an error that would result in execution errors if an agent runs the script. Resolve the error by modifying the script. For information about the error, see the **Analytics Dashboard** of the Web UI.

## VALIDATING

The script syntax and components (manifest, parameters, monitor, condition, and action) are in the process of being validated.

## VALIDATED

The script syntax and components (manifest, parameters, monitor, condition, and action) have been validated and no errors have been found.

## Example

```
switch# show nae-script
```

| Script Name                         | Version | Origin | Status    |
|-------------------------------------|---------|--------|-----------|
| fan_monitor                         | 1.0     | system | VALIDATED |
| interface_link_flap_monitor         | 1.0     | system | VALIDATED |
| interface_link_state_monitor        | 1.0     | system | VALIDATED |
| interface_tx_rx_stats_monitor       | 1.0     | system | VALIDATED |
| lag_imbalance_monitor               | 1.0     | system | VALIDATED |
| lag_status_monitor                  | 1.0     | system | VALIDATED |
| power_supply_monitor                | 1.0     | system | VALIDATED |
| stp_bpdu_tcn_rate_monitor           | 1.0     | system | VALIDATED |
| system_resource_monitor_mm1.default | 1.0     | system | VALIDATED |
| system_resource_monitor_mm2.default | 1.0     | system | VALIDATED |
| temp_sensor_monitor                 | 1.0     | system | VALIDATED |

## ntp authentication

### Syntax

```
ntp authentication
```

```
no ntp authentication
```

### Description

Enables support for authentication when communicating with an NTP server.

The `no` form of this command disables authentication support.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling authentication support:

```
switch(config)# ntp authentication
```

Disabling authentication support:

```
switch(config)# no ntp authentication
```

## ntp authentication-key

### Syntax

```
ntp authentication-key <KEY-ID> {md5 | sha1} <PASSWORD> [trusted]
```

```
no ntp authentication-key <KEY-ID>
```

### Description

Defines an authentication key that is used to secure the exchange with an NTP time server. This command provides protection against accidentally synchronizing to a time source that is not trusted.

The `no` form of this command removes an authentication key.

### Command context

```
config
```



## Parameters

**authentication-key** <KEY-ID>

Specifies an identification number to uniquely identify the key. Range: 1 to 65534.

**md5**

Specifies md5 encryption to protect the key.

**sh1**

Specifies sh1 encryption to protect the key

<PASSWORD>

Specifies the md5 or sh1 password associated with the key. Range: 8 to 16 characters.

**trusted**

Specifies that this is a trusted key. When NTP authentication is enabled, the switch only synchronizes with time servers that transmit packets containing a trusted key.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Defining key 10 with the **md5** password **myPassword** as a trusted key:

```
switch(config)# ntp authentication-key 10 md5 myPassword trusted
```

Removing key 10:

```
switch(config)# no ntp authentication-key 10
```

# ntp disable

## Syntax

```
ntp disable
```

## Description

Disables the NTP client on the switch. The NTP client is disabled by default.

## Command context

```
config
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Disabling the NTP client.

```
switch(config)# ntp disable
```

## ntp enable

### Syntax

```
ntp enable
```

```
no ntp enable
```

### Description

Enables the NTP client on the switch to automatically adjust the local time and date on the switch. The NTP client is disabled by default.

The `no` form of this command disables the NTP client.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling the NTP client.

```
switch(config)# ntp enable
```

Disabling the NTP client.

```
switch(config)# no ntp enable
```

## ntp server

### Syntax

```
ntp server <IP-ADDR> [key <KEY-NUM>] [minpoll <MIN-NUM>] [maxpoll <MAX-NUM>] [burst | iburst] [prefer] [version <VER-NUM>]
```

```
no ntp server <IP-ADDR>
```

```
no ntp server <IP-ADDR> [burst] [iburst] [prefer] [key-id <KEY-NUM>]
```

### Description

Defines an NTP server to use for time synchronization, or updates the settings of an existing server with new values. Up to eight servers can be defined.

The `no` form of this command removes a configured NTP server.

### Command context

```
config
```

### Parameters

**server <IP-ADDR>**

Specifies the address of an NTP server as a DNS name, an IPv4 address (`x.x.x.x`), where `x` is a decimal number from 0 to 255, or an IPv6 address (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

When specifying an IPv4 address, you can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100.

When specifying an IPv6 address, you can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address 2222:0000:3333:0000:0000:0000:4444:0055 becomes 2222:0:3333::4444:55.

**key <KEY-NUM>**

Specifies the key to use when communicating with the server. A trusted key must be defined with the command `ntp authentication-key` and authentication must be enabled with the command `ntp authentication`. Range: 1 to 65534.

**minpoll <MIN-NUM>**

Specifies the minimum polling interval in seconds, as a power of 2. Range: 4 to 17. Default: 6 (64 seconds).

**maxpoll <MAX-NUM>**

Specifies the maximum polling interval in seconds, as a power of 2. Range: 4 to 17. Default: 10 (1024 seconds).

**burst**

Send a burst of packets instead of just one when connected to the server. Useful for reducing phase noise when the polling interval is long.

**iburst**

Send a burst of six packets when not connected to the server. Useful for reducing synchronization time at startup. Range: 1 to 4094.

**prefer**

Make this the preferred server.

**version <VER-NUM>**

Specifies the version number to use for all outgoing NTP packets. Range: 3 or 4.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Defining the ntp server pool.ntp.org, using iburst, and NTP version 4.

```
switch(config)# ntp server pool.ntp.org iburst version 4
```

Removing the ntp server pool.ntp.org.

```
switch(config)# no ntp server pool.ntp.org
```

Defining the ntp server my-ntp.mydomain.com and makes it the preferred server.

```
switch(config)# ntp server my-ntp.mydomain.com prefer
```

## ntp trusted-key

### Syntax

```
ntp trusted-key <KEY-ID>
```

```
no ntp trusted-key <KEY-ID>
```

### Description

Sets a key as trusted. When NTP authentication is enabled, the switch only synchronizes with time servers that transmit packets containing a trusted key.

The `no` form of this command removes the trusted designation from a key.

### Command context

```
config
```

### Parameters

**<KEY-ID>**

Specifies the identification number of the key to set as trusted. Range: 1 to 65534.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Defining key 10 as a trusted key.

```
switch(config)# ntp trusted-key 10
```

Removing trusted designation from key 10:

```
switch(config)# no ntp trusted-key 10
```

## ntp vrf

### Syntax

```
ntp vrf <VRF-NAME>
```

### Description



**NOTE:** 6100 only supports default VRF.

Specifies the VRF on which the NTP client communicates with an NTP server.

### Command context

```
config
```

### Parameters

**<VRF-NAME>**

Specifies the name of a VRF.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Setting the switch to use the default VRF for NTP client traffic.

```
switch(config)# ntp vrf default
```

# show ntp associations

## Syntax

```
show ntp associations
```

## Description

Shows the status of the connection to each NTP server. The following information is displayed for each server:

- Tally code : The first character is the Tally code:
  - (blank): No state information available (e.g. non-responding server)
  - x : Out of tolerance (discarded by intersection algorithm)
  - . : Discarded by table overflow (not used)
  - - : Out of tolerance (discarded by the cluster algorithm)
  - + : Good and a preferred remote peer or server (included by the combine algorithm)
  - # : Good remote peer or server, but not utilized (ready as a backup source)
  - \* : Remote peer or server presently used as a primary reference
  - o : PPS peer (when the prefer peer is valid)
- ID: Server number.
- NAME: NTP server FQDN/IP address (Only the first 24 characters of the name are displayed).
- REMOTE: Remote server IP address.
- REF\_ID: Reference ID for the remote server (Can be an IP address).
- ST: (Stratum) Number of hops between the NTP client and the reference clock.
- LAST: Time since the last packet was received in seconds unless another unit is indicated.
- POLL: Interval (in seconds) between NTP poll packets. Maximum (1024) reached as server and client sync.
- REACH: 8-bit octal number that displays status of the last eight NTP messages (377 = all messages received).

## Command context

Manager (#)

## Parameters

There are no parameters for show ntp associations.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ntp associations
```

| ID  | NAME           | REMOTE       | REF-ID | ST | LAST | POLL | REACH |
|-----|----------------|--------------|--------|----|------|------|-------|
| 1   | 192.0.1.1      | 192.0.1.1    | .INIT. | 16 | -    | 64   | 0     |
| * 2 | time.apple.com | 17.253.2.253 | .GPSs. | 2  | 70   | 128  | 377   |

## show ntp authentication-keys

### Syntax

```
show ntp authentication-keys
```

### Description

Shows the currently defined authentication keys.

### Command context

Manager (#)

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch# show ntp authentication-keys
```

| Auth key | Trusted | MD5 password |
|----------|---------|--------------|
| 10       | No      | *****        |
| 20       | Yes     | *****        |

## show ntp servers

### Syntax

```
show ntp servers
```

### Description

Shows all configured NTP servers.

### Command context

Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch# show ntp servers
```

| NTP SERVER | KEYID | MINPOLL | MAXPOLL | OPTION | VER      |
|------------|-------|---------|---------|--------|----------|
| 192.0.1.18 | -     | 5       | 10      | iburst | 3        |
| 192.0.1.19 | -     | 6       | 10      | none   | 4        |
| 192.0.1.20 | -     | 6       | 8       | burst  | 3 prefer |

## show ntp statistics

### Syntax

```
show ntp statistics
```

### Description

Shows global NTP statistics. The following information is displayed:

- Rx-pkts: Total NTP packets received.
- Current Version Rx-pkts: Number of NTP packets that match the current NTP version.
- Old Version Rx-pkts: Number of NTP packets that match the previous NTP version.
- Error pkts: Packets dropped due to all other error reasons.
- Auth-failed pkts: Packets dropped due to authentication failure.
- Declined pkts: Packets denied access for any reason.
- Restricted pkts: Packets dropped due to NTP access control.
- Rate-limited pkts: Number of packets discarded due to rate limitation.
- KOD pkts: Number of Kiss of Death packets sent.

### Command context

```
config
```

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

```
switch(config)# show ntp statistics
Rx-pkts 100
Current Version Rx-pkts 80
Old Version Rx-pkts 20
```

```
Err-pkts 2
Auth-failed-pkts 1
Declined-pkts 0
Restricted-pkts 0
Rate-limited-pkts 0
KoD-pkts 0
```

## show ntp status

### Syntax

```
show ntp status
```

### Description

Shows the status of NTP on the switch.

### Command context

Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Displaying the status information when the switch is not synced to an NTP server:

```
switch# show ntp status
NTP is enabled.
NTP authentication is enabled.
NTP is using the default VRF for NTP server connections.

Wed Nov 23 23:29:10 PDT 2016
NTP uptime: 187 days, 1 hours, 37 minutes, 48 seconds

Not synchronized with an NTP server.
```

Displaying the status information when the switch is synced to an NTP server:

```
switch# show ntp status
NTP is enabled.
NTP authentication is enabled.
NTP is using the default VRF for NTP server connections.

Wed Nov 23 23:29:10 PDT 2016
NTP uptime: 187 days, 1 hours, 37 minutes, 48 seconds

Synchronized to NTP Server 17.253.2.253 at stratum 2.
Poll interval = 1024 seconds.
Time accuracy is within 0.994 seconds
Reference time: Thu Jan 28 2016 0:57:06.647 (UTC)
```



## area

### Syntax

```
area <AREA-ID>
```

```
no area <AREA-ID>
```

### Description

Creates a normal area, with <AREA-ID> set if not present. If the area is already present and it is not a normal area, then this command changes the area type to normal.

The `no` form of this command deletes the area with the <AREA-ID> specified. Area can be of any type (nssa, nssa no-summary, stub, stub no-summary, and default normal area).

### Command context

```
config-ospf-<PROCESS-ID>
```

### Parameters

**<AREA-ID>**

Specifies the area ID in one of the following formats.

- OSPF area identifier in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
- OSPF area identifier in decimal format. Range: 0 to 4294967295.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating a normal area:

Deleting an area:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no area 1
```

## area default-metric

### Syntax

```
area <AREA-ID> default-metric <COST>
```

```
no area <AREA-ID> default-metric
```

## Description

Sets the cost of the default route announced to NSSA or stub areas.

The `no` form of this command resets the cost of the default route announced to NSSA or stub areas, to the default value of 1.

## Command context

```
config-ospf-<PROCESS-ID>
```

## Parameters

**<AREA-ID>**

Specifies area ID in one of the following formats.

- OSPF area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.
- OSPF area identifier in decimal format. Range: 0 to 4294967295.

**default-metric <COST>**

Sets the cost of default-summary LSAs announced to NSSA or stub areas, to the specified value. Default cost: 1. Range: 0 to 16777215.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting cost for default LSA summary:

Setting cost for default LSA summary to default:

# area nssa

## Syntax

```
area <AREA-ID> nssa [no-summary]
```

```
no area <AREA-ID> nssa [no-summary]
```

## Description

Creates the NSSA area (Not So Stubby Area) with `<AREA-ID>` if not present. If area is present and not NSSA area, this command changes the area type to NSSA area. If `no-summary` is used, area type will be NSSA No-Summary.

The `no` form of this command unsets the area type as NSSA. That is, the configured area will be changed to default normal area. The `no area <AREA-ID> nssa no-summary` command enables sending inter-area routes into NSSA, but will not unset the area as NSSA.

## Command context

```
config-ospf-<PROCESS-ID>
```

## Parameters

**<AREA-ID>**

Specifies the area ID in one of the following formats.

- OSPF area identifier in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
- OSPF area identifier in decimal format. Range: 0 to 4294967295.

#### **nssa [no-summary]**

Specifies Not So Stubby Area (NSSA) area type. If area is present and not NSSA area, parameter changes the area type to NSSA area. If `no-summary` is specified, area type will be NSSA No-Summary, which means do not inject inter-area routes into NSSA.

#### **Authority**

Administrators or local user group members with execution rights for this command.

#### **Examples**

Creating an NSSA area:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 1 nssa
switch(config-ospf-1)# area 1 nssa no-summary
```

Unsetting the area as NSSA

```
switch(config)# router ospf 1
switch(config-ospf-1)# no area 1 nssa
switch(config-ospf-1)# no area 1 nssa no-summary
```

## **clear ip ospf neighbors**

#### **Syntax**

```
clear ip ospf [<PROCESS-ID>] neighbor [<NEIGHBOR>] [interface [<INTERFACE-NAME>]]
[all-vrfs | vrf <VRF-NAME>]
```

#### **Description**

Resets the neighbor and clears the OSPF neighbor information.

#### **Command context**

Operator (>) or Manager (#)

#### **Parameters**

**<PROCESS-ID>**

Specifies the OSPFv2 process ID to clear the statistics for the particular OSPFv2 process. Range: 1 to 63.

**<NEIGHBOR>**

Specifies the router ID of a neighbor.

**<INTERFACE-NAME>**

Specifies the OSPFv2 statistics to clear for the specified interface.

**all-vrfs**

Select to clear the OSPFv2 statistics for all VRFs.

**vrf** <VRF-NAME>

Specifies the name of a VRF.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Clearing the OSPFv2 neighbor information:

## clear ip ospf statistics

### Syntax

```
clear ip ospf [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Clear the OSPF event statistics.

### Command context

Operator (>) or Manager (#)

### Parameters

<PROCESS-ID>

OSPF process ID. Clear the statistics for the particular OSPF process. Range: 1 to 63.

<INTERFACE-NAME>

Clear the OSPF statistics for the specified interface.

**all-vrfs**

Optionally select to clear the OSPF statistics for all VRFs.

**vrf** <VRF-NAME>

Optionally select to clear the OSPF statistics for a particular VRF. If the VRF is not specified, information for the default VRF is cleared.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Clearing the OSPF event statistics:

```
switch# clear ip ospf statistics
switch# clear ip ospf statistics interface 1/1/1
```

## default-information originate

### Syntax

```
default-information originate
```

```
no default-information originate
```

### Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors if it is present in the routing table. The `no` form of this command disables advertisement of the default route.

### Command context

```
config-ospf-<PROCESS-ID>
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate
```

## default-information originate always

### Syntax

```
default-information originate always
```

```
no default-information originate always
```

### Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors, regardless if it is present in the routing table or not.

The `no` form of this command sets the OSPF administrative distance to the default of 110.

### Command context

```
config-ospf-<PROCESS-ID>
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate always
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate always
```

## default-metric

### Syntax

```
default-metric <METRIC-VALUE>
```

```
no default-metric
```

### Description

Sets the default metric for redistributed routes in the OSPF.

The `no` form of this command sets the default metric to be used for redistributed routes into OSPF to the default of 25.

### Command context

```
config-ospf-<PROCESS-ID>
```

### Parameters

<METRIC-VALUE>

Specifies the default metric value to use for redistributed routes. Default: 25. Range: 0-1677214.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting default metric for redistributed routes:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-metric 37
```

Setting default metric for redistributed routes to the default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-metric
```

## disable

### Syntax

```
disable
```

### Description

Disables the OSPF process.



---

**IMPORTANT:** This command does not remove the OSPF configurations.

---

### Command context

config-ospf-<PROCESS-ID>

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Disabling OSPF process:

```
switch(config)# router ospf 1
switch(config-ospf-1)# disable
```

## distance

### Syntax

distance <DISTANCE>

no distance

### Description

Defines an administrative distance for OSPF. Administrative distance is used as a criteria to select the best route when multiple routes are present from different routing protocols.

The `no` form of this command sets the OSPF administrative distance to the default of 110.

### Command context

config-ospf-<PROCESS-ID>

### Parameters

<DISTANCE>

Specifies OSPF administrative distance. Range: 1-255. Default: 110.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting OSPF administrative distance:

```
switch(config)# router ospf 1
switch(config-ospf-1)# distance 100
```

Setting OSPF administrative distance to the default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no distance
```

# enable

## Syntax

enable

## Description

Enables the OSPF process, if disabled. By default the OSPF process is enabled.

## Command context

config-ospf-<PROCESS-ID>

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling OSPF process:

```
switch(config)# router ospf 1
switch(config-ospf-1)# enable
```

# graceful-restart

## Syntax

graceful-restart {restart-interval <SECONDS> | helper}

no graceful-restart {restart-interval | helper}

## Description

Configures graceful restart parameters for OSPF.

The **no** form of this command sets the restart interval to the default interval of 120 seconds or disables the helper mode, depending on the parameter specified.

## Command context

config-ospf-<PROCESS-ID>

## Parameters

Choose one of the following parameters:

**restart-interval** <SECONDS>

Specifies the time another router should wait for this router to gracefully restart and selects the maximum interval (in seconds) that another router should wait. Default: 120 seconds. Range: 5-1800.

**helper**

Specifies that the router will participate in the graceful restart of a neighbor router.

## Authority

Administrators or local user group members with execution rights for this command.



## Examples

Enabling OSPF nonstop forwarding:

```
switch(config)# router ospf 1
switch(config-ospf-1)# graceful-restart restart-interval 50
switch(config-ospf-1)# graceful-restart helper
```

Setting restart-interval to default, disable helper mode:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no graceful-restart restart-interval
switch(config-ospf-1)# no graceful-restart helper
```

## ip ospf area

### Syntax

```
ip ospf <PROCESS-ID> area <AREA-ID>
```

```
no ip ospf <PROCESS-ID> area <AREA-ID>
```

### Description

Runs the OSPF protocol on the interface with the configured IPv4 address for the area specified. The interfaces which have an IP address configured in this network or in a subset of this network, will participate in the OSPF protocol.

To move an interface to a new area, unmap the existing area and then associate a new area with the interface.

The `no` form of this command disables OSPF on the interface and removes the interface from the area. Interfaces which have an IP address configured on the network or in a subset of the network, stop participating in the OSPF protocol.

### Command context

config-if-vlan

### Parameters

**<PROCESS-ID>**

Specifies the OSPF process Id. Range: 1 to 63.

**<AREA-ID>**

Specifies the OSPF area ID in one of the following formats.

- Area identifier in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
- Area identifier in decimal format. Range: 0 to 4294967295.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting OSPF network for the area:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf 1 area 1
switch(config-if-vlan)# ip ospf 1 area 0.0.0.1
```

Disabling OSPF network for the area:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf 1 area 1
switch(config-if-vlan)# no ip ospf 1 area 0.0.0.1
```

## ip ospf authentication

### Syntax

```
ip ospf authentication {message-digest | simple-text | null | keychain}
```

```
no ip ospf authentication
```

### Description

Sets the authentication type that will be used for authentication with the neighbor router.

The **no** form of this command deletes the authentication type used for a particular authentication with the neighbor router and sets to null authentication.

### Command context

config-if-vlan

### Parameters

Choose one of the following authentication types.

#### message-digest

Sets authentication type as message-digest.

#### simple-text

Sets authentication type as simple-text.

#### null

Sets authentication type as null.

#### keychain

Sets the authentication type to use the key chain.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting OSPF authentication type on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication simple-text
```

Deleting OSPF authentication type on the interface and sets it to null:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf authentication
```

## ip ospf authentication-key

### Syntax

```
ip ospf authentication-key {ciphertext | plaintext} <PASSWORD>
```

```
no ip ospf authentication-key
```

### Description

Sets the authentication password used for simple-text authentication. If the password is given in ciphertext it will be decrypted and applied to the protocol.

The `no` form of this command deletes the authentication password used for simple-text authentication.

### Command context

```
config-if-vlan
```

### Parameters

#### ciphertext

Specifies providing the password in encrypted format.

#### plaintext

Specifies providing the password in simple-text format.

#### <PASSWORD>

Specifies the password for simple-text authentication.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting OSPF simple-text authentication password on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf authentication-key ciphertext AQBapdAVzEZb205RmH33+4cEctMayhvmuOVeYmXs8Y7ANu2eBgAAABO5xecopg==
switch(config-if-vlan)# ip ospf authentication-key plaintext secure
```

Deleting OSPF simple-text authentication password on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf authentication-key
```

## ip ospf cost

### Syntax

```
ip ospf cost <INTERFACE-COST>
```

```
no ip ospf cost
```

### Description

Sets the cost (metric) associated with a particular interface. The interface cost is used as a parameter to calculate the best routes.

The `no` form of this command sets the cost (metric) associated with a particular interface to the default cost 1.

### Command context

config-if-vlan

### Parameters

**<INTERFACE-COST>**

Specifies the interface cost value. Range: 1 to 65535. Default: 1.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting OSPF interface cost

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf cost 100
```

Setting the OSPF interface cost to default

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf cost
```

## ip ospf dead-interval

### Syntax

ip ospf dead-interval **<INTERVAL>**

no ip ospf dead-interval

### Description

Sets the interval after which a neighbor is declared dead if no hello packet is received on the OSPF interface.

The `no` form of this command sets the interval after which a neighbor is declared dead, to the default for the OSPF interface. The default value is 40 seconds (generally 4 times the hello packet interval).

### Command context

config-if-vlan

### Parameters

**<INTERVAL>**

Specifies the time interval for the dead interval, in seconds. Range: 1 to 65535. Default: 40.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting OSPF dead interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf dead-interval 30
```

Setting OSPF dead interval to default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf dead-interval
```

## ip ospf hello-interval

### Syntax

```
ip ospf hello-interval <INTERVAL>
```

```
no ip ospf hello-interval
```

### Description

Sets the time interval between OSPF hello packets for the OSPF interface.

The `no` form of this command sets the time interval OSPF hello packets to the default of 10 seconds for the OSPF interface.

### Command context

```
config-if-vlan
```

### Parameters

<INTERVAL>

Specifies the time interval for the hello interval, in seconds. Range: 1 to 65535. Default: 10.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting OSPF hello interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf hello-interval 30
```

Setting OSPF hello interval to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf hello-interval
```

## ip ospf keychain

### Syntax

```
ip ospf keychain <KEYCHAIN-NAME>
```

```
no ip ospf keychain
```

### Description

Sets the key chain for md5 authentication. A key chain configures rotating keys for packet authenticating, reducing the risk of keys being compromised.

The **no** form of this command deletes the key chain used for md5 authentication.

### Command context

config-if

### Parameters

**<KEYCHAIN-NAME>**

Name of key chain to be used for md5 authentication.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting OSPFv2 key chain authentication:

```
switch(config)# interface 1/1/1
switch(config-if)# ip ospf keychain ospf_keys
```

Deleting OSPFv2 key chain authentication:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip ospf keychain
```

## ip ospf message-digest-key md5

### Syntax

ip ospf authentication message-digest

ip ospf message-digest-key <KEY-ID> md5 {ciphertext | plaintext} <KEY>

no ip ospf message-digest-key <KEY-ID>

### Description

Sets the message digest password used for message digest authentication. If the md5 key is given in ciphertext, it will be decrypted and applied to the protocol.

The **no** form of this command deletes the message digest password used for message digest authentication.

### Command context

config-if-vlan

### Parameters

**ciphertext**

Specifies providing the md5 key in encrypted format.

**plaintext**

Specifies providing the md5 key in simple-text format.

**<KEY-ID>**

Specifies the password for md5 (message-digest) authentication. Range: 1 to 255.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting OSPF message-digest authentication password on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf message-digest-key 1 md5 ciphertext AQBapdAVzEZb205RmH33+4cEctMayhvmuOVeYmXs8Y7ANu2eBgAABO5xecopg==
switch(config-if-vlan)# ip ospf message-digest-key 1 md5 plaintext secure
```

Deleting OSPF message-digest authentication password on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf message-digest-key 1
```

# ip ospf network

## Syntax

```
ip ospf network {broadcast | point-to-point}
```

```
no ip ospf network
```

## Description

Configures the OSPF network type for the interface.

The **no** form of this command sets the network type for the interface to the system default which is broadcast network.

## Command context

config-if-vlan

## Parameters

Choose one of the following parameters as the interface network type.

### **broadcast**

Specifies the OSPF network type as a broadcast multi-access network.

### **point-to-point**

Specifies the OSPF network type as a point-to-point network.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting OSPF network type for the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf network broadcast
switch(config-if-vlan)# ip ospf network point-to-point
```

Disabling OSPF network type for the interface to system default of broadcast network:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf network
```

## ip ospf passive

### Syntax

```
ip ospf passive

no ip ospf passive
```

### Description

Configures the interface as an OSPF passive interface. With this setting, the interface participates in OSPF but does not send or receive packets on that interface.

The `no` form of this command resets the interface as active. With this setting, the interface starts sending and receiving OSPF packets.

### Command context

```
config-if-vlan
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the interface as OSPF passive interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf passive
```

Setting the interface as OSPF active interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf passive
```

## ip ospf priority

### Syntax

```
ip ospf priority <PRIORITY-VALUE>

no ip ospf priority
```

### Description

Sets the OSPF priority for the interface. The larger the numeric value of the priority, the higher the chances for it to become the designated router. Setting a priority of zero makes the router ineligible to become a designated router or back up designated router.

The `no` form of this command sets the OSPF priority for the interface to the default of 1.

### Command context

```
config-if-vlan
```

### Parameters

**<PRIORITY-VALUE>**

Specifies the OSPF priority value. Range: 0 to 255. Default: 1.



## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting OSPF priority for the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf priority 50
```

Disabling OSPF priority for the interface to default:

```
switch(config)# interface vlan1
switch(config-if-vlan)# no ip ospf priority
```

# ip ospf retransmit-interval

## Syntax

```
ip ospf retransmit-interval <INTERVAL>
```

```
no ip ospf retransmit-interval
```

## Description

Sets the time between retransmitting lost link state advertisements for the OSPF interface.

The `no` form of this command sets the time between retransmitting lost link state advertisements to the default of 5 seconds for the OSPF interface.

## Command context

```
config-if-vlan
```

## Parameters

**<INTERVAL>**

Specifies the retransmit interval, in seconds. Range: 1 to 3600. Default: 5.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting OSPF retransmit interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf retransmit-interval 30
```

Setting OSPF retransmit interval to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf retransmit-interval
```

## ip ospf shutdown

### Syntax

```
ip ospf shutdown
```

```
no ip ospf shutdown
```

### Description

Disables OSPF on the interface. The interface state changes to Down. It does not remove the interface from the OSPF area. To remove the interface, use the command `no ip ospf area`.

The `no` form of this command re-enables OSPF on the interface

### Command context

```
config-if-vlan
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Disabling OSPF on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf shutdown
```

Re-enabling OSPF on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf shutdown
```

## ip ospf transit-delay

### Syntax

```
ip ospf transit-delay <DELAY>
```

```
no ip ospf transit-delay
```

### Description

Sets the time delay in link state transmission for the OSPF interface.

The `no` form of this command sets the delay in link state transmission to the default of 1 second for the OSPF interface.

### Command context

```
config-if-vlan
```

### Parameters

<DELAY>

Specifies the transit delay in seconds. Range: 1 to 3600. Default: 1.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting OSPF transit delay on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ip ospf transit-delay 30
```

Setting OSPF transit delay to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ip ospf transit-delay
```

# keychain

## Syntax

```
keychain <KEYCHAIN-NAME>
```

```
no keychain
```

## Description

Sets the key chain for md5 authentication. A key chain configures rotating keys for packet authenticating, reducing the risk of keys being compromised.

The `no` form of this command deletes the key chain used for md5 authentication.

## Command context

```
config-router-vlink
```

## Parameters

<KEYCHAIN-NAME>

Name of key chain to be used for md5 authentication.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting OSPF virtual link key chain authentication:

```
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# keychain ospf_keys
```

Deleting OSPF virtual link key chain authentication:

```
switch# configure terminal
switch(config)# router ospf 1
switch(config-ospf-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink)# no keychain
```

## max-metric router-lsa

### Syntax

```
max-metric router-lsa [on-startup [<ADVERT-TIME>]]
```

```
no max-metric router-lsa [on-startup]
```

### Description

Sets the protocol to advertise a maximum metric so that other routers do not prefer this router as an intermediate hop in their shortest path first (SPF) calculations. If the on-startup parameter is used, the router is configured to advertise a maximum metric at startup for the time mentioned in seconds or for a default value of 600 seconds.

The `no` form of this command advertises the normal cost metrics instead of advertising the maximized cost metric. This setting causes the router to be considered in traffic forwarding.

### Command context

```
config-ospf-<PROCESS-ID>
```

### Parameters

**on-startup** **<ADVERT-TIME>**

Specifies the time in seconds to advertise self as stub-router on startup. If no time is specified, the default time of 600 seconds is used. Range: 5 to 86400. Default: 600.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting to maximize the cost metrics for Router LSA:

```
switch(config)# router ospf 1
switch(config-ospf-1)# max-metric router-lsa
switch(config-ospf-1)# max-metric router-lsa on-startup
switch(config-ospf-1)# max-metric router-lsa on-startup 3000
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no max-metric router-lsa
switch(config-ospf-1)# no max-metric router-lsa on-startup
```

## passive-interface default

### Syntax

```
passive-interface default
```

```
no passive-interface
```

### Description

Configures all OSPF interfaces as passive.

The `no` form of this command sets all OSPF interfaces as active.

## Command context

config-ospf-<PROCESS-ID>

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting OSPF-enabled interfaces as passive:

```
switch(config)# router ospf 1
switch(config-ospf-1) # passive-interface default
```

Setting OSPF-enabled interfaces as active:

```
switch(config)# router ospf 1
switch(config-ospf-1) # no passive-interface
```

# redistribute

## Syntax

```
redistribute {bgp | connected | static | rip | ospf <PROCESS-ID>} [route-map <ROUTE-MAP-NAME>]
no redistribute {bgp | connected | static | rip | ospf <PROCESS-ID>} [route-map <ROUTE-MAP-NAME>]
```

## Description

Redistributes routes originating from other protocols, or from another OSPF process, to the current OSPF process.

The **no** form of this command disables redistribution of routes to the current OSPF process.

## Command context

config-ospf-<PROCESS-ID>

## Parameters

### bgp

Specifies redistributing BGP (Border Gateway Protocol) routes.

### connected

Specifies redistributing connected (directly attached subnet or host).

### static

Specifies redistributing static routes.

### rip

Specifies redistributing RIP routes.

### ospf <PROCESS-ID>

Specifies redistributing routes from the specified OSPFv2 process ID.

### route-map <ROUTE-MAP-NAME>

Specifies the name of a route map. To create a route map, use the **route-map** command.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Redistributing routes to OSPF:

Disabling redistributing routes to OSPF:

# reference-bandwidth

## Syntax

```
reference-bandwidth <BANDWIDTH>
```

```
no reference-bandwidth
```

## Description

Sets the reference bandwidth for OSPFv2. If the OSPFv2 interface cost is not explicitly set, then the cost of all the OSPFv2 interfaces is recalculated based on the reference bandwidth and link speed of the interface.

For VLAN interfaces the link speed value is taken as 1 Gbps (if the OSPFv2 interface cost is not explicitly set).

The `no` form of this command sets the reference bandwidth for OSPF to the default of 100000 Mbps.

## Command context

```
config-ospf-<PROCESS-ID>
```

## Parameters

**<BANDWIDTH>**

Specifies the reference bandwidth used to calculate the cost of an interface in Mbps. Range: 1 to 4000000. Default: 100000.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the reference bandwidth:

```
switch(config)# router ospf 1
switch(config-ospf-1)# reference-bandwidth 40000
```

Setting the reference bandwidth to the default value:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no reference-bandwidth
```

# rfc1583-compatibility

## Syntax

```
rfc1583-compatibility
```

```
no rfc1583-compatibility
```

## Description

Enables OSPF compatibility with RFC1583 (backward compatibility). If RFC1583 compatibility is enabled, then the route cost calculation follows a different method.

The `no` form of this command disables OSPF compatibility with RFC1583 (backward compatibility). By default the RFC1583 compatibility is disabled.

## Command context

```
config-ospf-<PROCESS-ID>
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling OSPF RFC1583 compatibility:

```
switch(config)# router ospf 1
switch(config-ospf-1)# rfc1583-compatibility
```

Disabling OSPF RFC1583 compatibility:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no rfc1583-compatibility
```

# router ospf

## Syntax

```
router ospf <PROCESS-ID> [vrf <VRF-NAME>]
```

```
no router ospf <PROCESS-ID> [vrf <VRF-NAME>]
```

## Description

Creates an OSPF process (if not created already) on a VRF, and switches to the OSPF router instance context. Up to eight OSPF processes are supported per VRF.

The `no` form of this command removes the OSPF instance.

## Command context

```
config
```

## Parameters

**<PROCESS-ID>**

Specifies an OSPF process ID. Range: 1 to 63.

**vrf <VRF-NAME>**

Specifies a VRF name for the OSPF process. Default: default.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config)# router ospf 1
switch(config-ospf-1)#
```

```
switch(config)# no router ospf 1
```

## router-id

### Syntax

```
router-id <ROUTER-ADDR>
```

```
no router-id
```

### Description

Sets an ID for the router in an IPv4 address format.

The `no` form of this command unconfigures the router-id for the instance and sets the router-id to the default as follows: the router-id is selected dynamically as equal to the highest loopback address on the router, or the highest active interface if there are no loopback addresses. If no IP address is configured on any interfaces on the router, OSPF will not form an adjacency.

### Command context

```
config-ospf-<PROCESS-ID>
```

### Parameters

<ROUTER-ADDR>

Specifies the Router ID in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting router-id in the OSPF context:

```
switch(config)# router ospf 1
switch(config-ospf-1)# router-id 1.1.1.1
```

Unconfiguring router-id:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no router-id
```

## show ip ospf

### Syntax

```
show ip ospf [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays general OSPF, area, state, and configuration information.



## Command context

Operator (>) or Manager (#)

## Parameters

**<PROCESS-ID>**

Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 63.

**all-vrfs**

Optionally select to display general OSPF information for all VRFs.

**vrf <VRF-NAME>**

Specifies the name of a VRF. Default: default.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing general OSPF configurations:

```
switch# show ip ospf 1
Routing Process 1 with ID : 1.1.1.1 VRF default

OSPFv2 Protocol is enabled
Graceful-restart is configured
Restart Interval: 120, State: inactive
Last Graceful Restart Exit Status: none
Area Border: false
AS Border: false
SPF: Start Time: 200ms, Hold Time: 1000ms, Max Wait Time: 5000ms
LSA: Start Time: 5000ms, Hold Time: 0ms, Max Wait Time: 0ms
LSA Arrival: 1000ms
Maximum Paths to Destination: 4
Number of external LSAs 0, checksum sum 0
Summary address:
 prefix 20.1.1.0/24 advertise tag 10
Number of areas is 2, 2 normal, 0 stub, 0 NSSA
Number of active areas is 1, 1 normal, 0 stub, 0 NSSA
BFD is disabled
Reference Bandwidth: 100000 Mbps
Area (0.0.0.0) (Active)
 Interfaces in this Area: 1 Active Interfaces: 1
 Passive Interfaces: 0 Loopback Interfaces: 0
 SPF calculation has run 7 times
 Area ranges:
 Number of LSAs: 3, checksum sum 75093
Area (0.0.0.1) (Inactive)
 Interfaces in this Area: 0 Active Interfaces: 0
 Passive Interfaces: 0 Loopback Interfaces: 0
 SPF calculation has run 2 times
 Area ranges:
 ip-prefix 20.1.1.1/24, inter-area, advertise
 Number of LSAs: 0, checksum sum 0
```

## show ip ospf border-routers

### Syntax

```
show ip ospf [<PROCESS-ID>]
 border-routers [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays the OSPF routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR).

### Command context

Operator (>) or Manager (#)

### Parameters

**<PROCESS-ID>**

Enter an OSPF process ID to display the OSPF routing table entries for ABR and ASBR for the particular OSPF process. Range: 1 to 63.

**all-vrfs**

Select to display OSPF border router information for all VRFs.

**vrf <VRF-NAME>**

Specify the name of a VRF. Default: default.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing OSPF border routers information:

```
switch# show ip ospf border-routers
OSPF Process ID 1 VRF default Internal Routing Table

Codes: i - Intra-area route, I - Inter-area route

i 2.2.2.2 [10], ABR, Area 0.0.0.0, SPF 15 via
 10.1.1.2, Interface 1/1/1
i 2.2.2.2 [10], ABR, Area 0.0.0.1, SPF 15 via
 11.1.1.12, Interface 1/1/2
```

## show ip ospf interface

### Syntax

```
show ip ospf [<PROCESS-ID>] interface [<interface-name>] [brief]
[all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays general OSPF, area, state, and configuration information.

## Command context

Operator (>) or Manager (#)

## Parameters

**<PROCESS-ID>**

Enter an OSPF process ID to display general OSPF information for a particular OSPF process. Range: 1 to 63.

**<interface-name>]**

Specify the name of an OSPF interface.

**brief**

Display a brief overview of OSPF configuration information.

**all-vrfs**

Select to display general OSPF information for all VRFs.

**vrf <VRF-NAME>**

Specify the name of a VRF. Default: default.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing general OSPF configuration settings for the default VRF:

```
switch# show ip ospf 1
Routing Process 1 with ID : 1.1.1.1 VRF default

OSPFv2 Protocol is enabled
Graceful-restart is configured
Restart Interval: 120, State: inactive
Last Graceful Restart Exit Status: none
Area Border: false
AS Border: false
SPF: Start Time: 200ms, Hold Time: 1000ms, Max Wait Time: 5000ms
LSA: Start Time: 5000ms, Hold Time: 0ms, Max Wait Time: 0ms
LSA Arrival: 1000ms
Maximum Paths to Destination: 4
Number of external LSAs 0, checksum sum 0
Summary address:
 prefix 20.1.1.0/24 advertise tag 10
Number of areas is 2, 2 normal, 0 stub, 0 NSSA
Number of active areas is 1, 1 normal, 0 stub, 0 NSSA
BFD is disabled
Reference Bandwidth: 100000 Mbps
Area (0.0.0.0) (Active)
 Interfaces in this Area: 1 Active Interfaces: 1
 Passive Interfaces: 0 Loopback Interfaces: 0
 SPF calculation has run 7 times
 Area ranges:
 Number of LSAs: 3, checksum sum 75093
Area (0.0.0.1) (Inactive)
 Interfaces in this Area: 0 Active Interfaces: 0
 Passive Interfaces: 0 Loopback Interfaces: 0
 SPF calculation has run 2 times
```

```
Area ranges:
 ip-prefix 20.1.1.1/24, inter-area, advertise
Number of LSAs: 0, checksum sum 0
```

Showing OSPF configuration settings for all interfaces:

```
switch# show ip ospf interface
Interface 1/1/1 is up, line protocol is up

IP address 10.1.1.1/24, Process ID 1 VRF default, area 0.0.0.0
 State Backup-dr, Status up, Network type Broadcast
 Link Speed: 10000 Mbps
 Cost Configured NA, Calculated 10
 Transit delay 1 sec, Router priority 1
 Designated Router IP: 10.1.1.2
 Backup Designated Router IP: 10.1.1.1
 Timer Intervals: Hello 10, Dead 40, Retransmit 5
 No authentication
 Number of Link LSAs: 0, checksum sum 0
 BFD is disabled

Interface 1/1/2 is up, line protocol is up

IP address 11.1.1.1/24, Process ID 1 VRF default, area 0.0.0.1
 State Dr, Status up, Network type Broadcast
 Link Speed: 10000 Mbps
 Cost Configured NA, Calculated 10
 Transit delay 1 sec, Router priority 1
 Designated Router IP: 11.1.1.1
 No backup designated router on this network
 Timer Intervals: Hello 10, Dead 40, Retransmit 5
 No authentication
 Number of Link LSAs: 0, checksum sum 0
 BFD is disabled
```

## show ip ospf lsdb

### Syntax

```
show ip ospf [<PROCESS-ID>] lsdb [all-vrfs | vrf <VRF-NAME>]
 [area <AREA-ID>] [lsid <LINK-STATE-ID>]
 [adv-router {<ROUTER-ID> | self}]

show ip ospf [<PROCESS-ID>] lsdb [all-vrfs | vrf <VRF-NAME>]
 asbr-summary [area <AREA-ID>] [lsid <LINK-STATE-ID>]
 [adv-router {<router-id> | self}]

show ip ospf [<PROCESS-ID>] lsdb [all-vrfs | vrf <VRF-NAME>]
 external [lsid <LINK-STATE-ID>] [adv-router {<ROUTER-ID> | self}]

show ip ospf [<PROCESS-ID>] lsdb [all-vrfs | vrf <VRF-NAME>]
 router [area <AREA-ID>] [lsid <LINK-STATE-ID>]
 [adv-router {<ROUTER-ID> | self}]

show ip ospf [<PROCESS-ID>] lsdb [all-vrfs | vrf <VRF-NAME>]
 network [area <AREA-ID>] [lsid <LINK-STATE-ID>]
 [adv-router {<ROUTER-ID> | self}]
```

```
show ip ospf [<PROCESS-ID>] lsdbs [all-vrfs | vrf <VRF-NAME>]
summary [area <AREA-ID>] [lsid <LINK-STATE-ID>]
[adv-router {<ROUTER-ID> | self}]

show ip ospf [<PROCESS-ID>] lsdbs [all-vrfs | vrf <VRF-NAME>]
nssa-external [area <AREA-ID>] [lsid <LINK-STATE-ID>]
[adv-router {<ROUTER-ID> | self}]

show ip ospf [<PROCESS-ID>] lsdbs [all-vrfs | vrf <VRF-NAME>]
database-summary
```

## Description

Shows the OSPF link state database summary for different OSPF LSAs (Link State Advertisement). Use the parameters to get information for a particular LSA.

## Command context

Operator (>) or Manager (#)

## Parameters

### <PROCESS-ID>

Enter an OSPF process ID to display the OSPF link state database information for the specified OSPF process. Range: 1 to 63.

### all-vrfs

Select to display OSPF link state database information for all VRFs.

### vrf <VRF-NAME>

Specify the name of a VRF. Default: default.

**Optionally select one of the following parameters to filter the link state database information.**

- `asbr-summary`. Displays ASBR summary link states (LSA type 4).
- `external`. Displays external link states (LSA type 5). The external parameter does not take the area <AREA-ID> parameter.
- `router`. Displays router LSAs (LSA type 1).
- `network`. Displays network LSAs (LSA type 2).
- `summary`. Displays network-summary link states (LSA type 3).
- `nssa-external`. Displays NSSA external link states (LSA type 7).

### database-summary

Select to display the count of each type of LSA and each area in the database.

### area <AREA-ID>

Select to display information filtered for the specified area in one of the following formats.

- OSPF area identifier in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
- OSPF area identifier in decimal format. Value: 0 to 4294967295.

**lsid** <LINK-STATE-ID>

Select to display information filtered by link state identifier specified in IPv4 address format (A.B.C.D).

**adv-router** {<ROUTER-ID> | **self**}

Select to display link states for a particular advertising router. Specify either a Router ID of the advertising router or specify **self** to show self-originated link states.

### Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

### Examples

Showing OSPF link state database (LSDB) general information:

```
switch# show ip ospf lsdb
OSPF Router with ID (50.50.50.50) (Process ID 1 VRF default)
=====

Router Link State Advertisements (Area 0.0.0.0)

```

| LSID        | ADV Router  | Age | Seq#       | Checksum | Link Count |
|-------------|-------------|-----|------------|----------|------------|
| 40.40.40.40 | 40.40.40.40 | 930 | 0x80000004 | 0x2ea1   | 3          |
| 50.50.50.50 | 50.50.50.50 | 935 | 0x80000002 | 0x8b52   | 1          |
| 60.60.60.60 | 60.60.60.60 | 943 | 0x800003c5 | 0x9854   | 2          |

```


Network Link State Advertisements (Area 0.0.0.0)

```

| LSID          | ADV Router  | Age | Seq#       | Checksum |
|---------------|-------------|-----|------------|----------|
| 209.165.201.3 | 60.60.60.60 | 944 | 0x80000001 | 0x7179   |
| 192.0.2.1     | 50.50.50.50 | 935 | 0x80000001 | 0x516a   |

```


Inter-area Summary Link State Advertisements (Area 0.0.0.0)

```

| LSID          | ADV Router  | Age  | Seq#       | Checksum |
|---------------|-------------|------|------------|----------|
| 209.165.201.1 | 40.40.40.40 | 929  | 0x80000001 | 0x2498   |
| 209.165.201.1 | 50.50.50.50 | 928  | 0x80000001 | 0x5b2f   |
| 209.165.201.1 | 60.60.60.60 | 1265 | 0x800003c3 | 0xf49b   |
| 192.0.2.0     | 40.40.40.40 | 943  | 0x80000001 | 0x53f3   |
| 192.0.2.0     | 50.50.50.50 | 935  | 0x80000001 | 0x26f8   |
| 192.0.2.0     | 60.60.60.60 | 930  | 0x80000001 | 0x7b51   |

Showing ASBR summary link states:

```
switch# show ip ospf lsdb asbr-summary
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====

ASBR Summary Link State Advertisements (Area 0.0.0.0)

```

| LSID          | ADV Router  | Age | Seq#       | Checksum |
|---------------|-------------|-----|------------|----------|
| 209.165.201.3 | 60.60.60.60 | 944 | 0x80000001 | 0x7179   |
| 192.0.2.1     | 50.50.50.50 | 935 | 0x80000001 | 0x516a   |

Showing external link states:

```
switch# show ip ospf lsdb external
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====
```

AS External Link State Advertisements

| LSID          | ADV Router  | Age | Seq#       | Checksum |
|---------------|-------------|-----|------------|----------|
| 209.165.201.3 | 60.60.60.60 | 944 | 0x80000001 | 0x7179   |
| 192.0.2.1     | 50.50.50.50 | 935 | 0x80000001 | 0x516a   |

Showing database summary:

```
switch# show ip ospf lsdb database-summary
OSPF Router with ID (10.1.1.1) (Process ID 1 VRF default)
=====
```

Area 0.0.0.0 database summary

| LSA Type           | Count |
|--------------------|-------|
| Router             | 2     |
| Network            | 1     |
| Inter-area Summary | 1     |
| ASBR Summary       | 0     |
| NSSA External      | 0     |
| Subtotal           | 4     |

Process 1 database summary

| LSA Type           | Count |
|--------------------|-------|
| Router             | 2     |
| Network            | 1     |
| Inter-area Summary | 1     |
| ASBR Summary       | 0     |
| NSSA External      | 0     |
| AS External        | 0     |
| Total              | 4     |

Showing router LSAs:

```
switch# show ip ospf lsdb router
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====
```

Router Link State Advertisements (Area 0.0.0.0)

| LSID    | ADV Router | Age | Seq#       | Checksum | Link Count |
|---------|------------|-----|------------|----------|------------|
| 1.1.1.2 | 1.1.1.2    | 15  | 0x80000004 | 0xf526   | 1          |
| 2.2.2.1 | 2.2.2.1    | 14  | 0x80000005 | 0x6c5e   | 2          |
| 2.2.2.2 | 2.2.2.2    | 104 | 0x80000004 | 0xf51a   | 1          |

Showing network LSAs:

```
switch# show ip ospf lsdb network
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
```

```
=====
Network Link State Advertisements (Area 0.0.0.0)

```

| LSID    | ADV Router | Age | Seq#       | Checksum |
|---------|------------|-----|------------|----------|
| 1.1.1.2 | 1.1.1.2    | 141 | 0x80000001 | 0xc55e   |
| 2.2.2.2 | 2.2.2.2    | 230 | 0x80000001 | 0xa179   |

Showing network-summary link states:

```
sswitch# show ip ospf lsd summary
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====
```

```
Inter-area Summary Link State Advertisements (Area 0.0.0.0)

```

| LSID    | ADV Router | Age | Seq#       | Checksum |
|---------|------------|-----|------------|----------|
| 1.1.1.0 | 2.2.2.1    | 133 | 0x80000002 | 0xa089   |

```
Inter-area Summary Link State Advertisements (Area 0.0.0.1)

```

| LSID    | ADV Router | Age | Seq#       | Checksum |
|---------|------------|-----|------------|----------|
| 2.2.2.0 | 2.2.2.1    | 133 | 0x80000002 | 0x7caa   |

Showing NSSA external link states:

```
switch(config-ospf-1)# show ip ospf lsd nssa-external
OSPF Router with ID (2.2.2.1) (Process ID 1 VRF default)
=====
```

```
NSSA External Link State Advertisements (Area 0.0.0.1)

```

| LSID    | ADV Router | Age | Seq#       | Checksum |
|---------|------------|-----|------------|----------|
| 8.8.8.0 | 1.1.1.2    | 162 | 0x80000003 | 0xc7b2   |

## show ip ospf neighbors

### Syntax

```
show ip ospf [<PROCESS-ID>] neighbors [<NEIGHBOR-ID>]
 [interface <INTERFACE-NAME>] [detail | summary]
 [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays information about OSPF neighbors.

### Command context

Operator (>) or Manager (#)



## Parameters

### <PROCESS-ID>

Enter an OSPF process ID to display OSPF neighbor information for the particular OSPF process. Range: 1 to 63.

### neighbors <NEIGHBOR-ID>

Select to display information about a particular neighbor, specified in IPv4 format (A.B.C.D).

### interface <INTERFACE-NAME>

Select to display neighbor information only for the specified interface.

### detail

Select to display detailed information for all the neighbors.

### summary

Select to display summary information for the neighbors.

### all-vrfs

Select to display neighbor information for all VRFs.

### vrf <VRF-NAME>

Specify the name of a VRF. Default: default.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing OSPF neighbors information for the default VRF:

```
switch# show ip ospf neighbors
OSPF Process ID 1 VRF default
=====
```

Total Number of Neighbors: 1

| Neighbor ID | Priority | State   | Nbr Address | Interface |
|-------------|----------|---------|-------------|-----------|
| 2.2.2.2     | 1        | FULL/DR | 10.1.1.2    | 1/1/1     |

Showing OSPF neighbors information for VRF red:

```
switch# show ip ospf neighbors vrf red
OSPF Process ID 1 VRF red
=====
```

Total Number of Neighbors: 1

| Neighbor ID | Priority | State    | Nbr Address | Interface |
|-------------|----------|----------|-------------|-----------|
| 1.1.1.1     | 1        | FULL/BDR | 10.1.1.1    | 1/1/1     |

Showing OSPF neighbors information for a specific neighbor:

```
switch# show ip ospf neighbors 2.2.2.2
Neighbor 2.2.2.2, interface address 10.1.1.2

Process ID 1 VRF default, in area 0.0.0.0 via interface 1/1/1
Neighbor priority is 1, State is FULL
Options is 0x42
Dead timer due in 00:00:32
Time since last state change 00h:19m:17s
```

Showing OSPF neighbors information for a specific neighbor and interface:

```
switch# show ip ospf neighbors 2.2.2.2 interface 1/1/1
Neighbor 2.2.2.2, interface address 10.1.1.2

Process ID 1 VRF default, in area 0.0.0.0 via interface 1/1/1
Neighbor priority is 1, State is FULL
DR is 10.1.1.2, BDR is 10.1.1.1
Options is 0x42
Dead timer due in 00:00:38
Retransmission queue length 0
Time since last state change 00h:22m:21s
```

Showing detail information for OSPF neighbors:

```
switch# show ip ospf neighbors detail
Neighbor 2.2.2.2, interface address 10.1.1.2

Process ID 1 VRF default, in area 0.0.0.0 via interface 1/1/1
Neighbor priority is 1, State is FULL
DR is 10.1.1.2, BDR is 10.1.1.1
Options is 0x42
Dead timer due in 00:00:38
Retransmission queue length 0
Time since last state change 00h:22m:21s
```

Showing summary information for OSPF neighbors in default VRF:

```
switch# show ip ospf neighbors summary
OSPF Process ID 1 VRF default, Neighbor Summary
=====
```

| Interface | Down | Attempt | Init | TwoWay | ExStart | Exchange | Loading | Full | Total |
|-----------|------|---------|------|--------|---------|----------|---------|------|-------|
| 1/1/1     | 0    | 0       | 0    | 0      | 0       | 0        | 0       | 1    | 1     |
| 1/1/2     | 0    | 0       | 0    | 0      | 0       | 0        | 0       | 0    | 0     |
| Total     | 0    | 0       | 0    | 0      | 0       | 0        | 0       | 1    | 1     |

## show ip ospf routes

### Syntax

```
show ip ospf [<PROCESS-ID>] routes
 [<IPV4-ADDR>/<MASK>] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays OSPF routing table information.

### Command context

Operator (>) or Manager (#)

## Parameters

### <PROCESS-ID>

Enter an OSPF process ID to display information on the OSPF routing table for the particular OSPF process. Range: 1 to 63.

### <IPV4-ADDR>

Specify an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. .

### <MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 32.

### all-vrfs

Select to display OSPF routing table information for all VRFs.

### vrf <VRF-NAME>

Specify the name of a VRF. Default: default.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing OSPF routing table information:

```
switch# show ip ospf routes
Codes: i - Intra-area route, I - Inter-area route
 E1 - External type-1, E2 - External type-2

OSPF Process ID 1 VRF default, Routing Table

Total Number of Routes : 2

10.1.1.0/24 (i) area: 0.0.0.0
 directly attached to interface 1/1/1, cost 1 distance 110
20.1.1.0/24 (I)
 via 10.1.1.2 interface 1/1/1, cost 2 distance 110
```

Showing OSPF routing table information for a specific subnet:

```
switch# show ip ospf routes 10.1.1.0/2
Codes: i - Intra-area route, I - Inter-area route
 E1 - External type-1, E2 - External type-2

OSPF Process ID 1 VRF default, Routing Table for prefixes 10.1.1.0/24

Total Number of Routes : 1

10.1.1.0/24 (i) area: 0.0.0.0
 directly attached to interface 1/1/1, cost 1 distance 110
```

## show ip ospf statistics

### Syntax

```
show ip ospf [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays OSPF statistics.

### Command context

Operator (>) or Manager (#)

### Parameters

**<PROCESS-ID>**

Enter an OSPF Process ID to display information on the OSPF statistics for the particular OSPF process.  
Range: 1 to 63.

**all-vrfs**

Select to display OSPF statistics information for all VRFs.

**vrf <VRF-NAME>**

Specify the name of a VRF. Default: default.

### Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

### Examples

Showing OSPF statistics:

```
switch# show ip ospf statistics
OSPF Process ID 1 VRF default, Statistics (cleared 1h 16m 24s ago)

Unknown Interface Drops : 0
Unknown Virtual Interface Drops : 0
Bad Instance ID Drops : 0
Bad IP Header Length Drops : 0
Wrong OSPF Version Drops : 0
Bad Source IP Drops : 0
Resource Failure Drops : 0
Bad Header Length Drops : 0
Total Drops : 0
```

## show ip ospf statistics interface

### Syntax

```
show ip ospf [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>]
[all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays OSPF statistics for the OSPF-enabled interfaces.

## Command context

Operator (>) or Manager (#)

## Parameters

**<PROCESS-ID>**

Enter an OSPF process ID to display OSPF-enabled interface statistics information on the specified OSPF process. Range: 1 to 63.

**<INTERFACE-NAME>**

Select to display information only for the specified interface.

**all-vrfs**

Select to display OSPF-enabled interface statistics information for all VRFs.

**vrf <VRF-NAME>**

Specify the name of a VRF. Default: default.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing OSPF-enabled interfaces information:

```
switch# show ip ospf statistics interface 1/1/1
OSPF Process ID 1 VRF default, interface 1/1/1 statistics (cleared 0h 30m 28s ago)
=====
```

|                       |       |                        |      |
|-----------------------|-------|------------------------|------|
| Tx Hello Packets      | : 101 | Rx Hello Packets       | : 99 |
| Tx Hello Bytes        | : 101 | Rx Hello Bytes         | : 99 |
| Tx DD Packets         | : 101 | Rx DD Packets          | : 99 |
| Tx DD Bytes           | : 101 | Rx DD Bytes            | : 99 |
| Tx LS Request Packets | : 101 | Rx LS Requests Packets | : 99 |
| Tx LS Request Bytes   | : 101 | Rx LS Request Bytes    | : 99 |
| Tx LS Update Packets  | : 101 | Rx LS Update Packets   | : 99 |
| Tx LS Update Bytes    | : 101 | Rx LS Update Bytes     | : 99 |
| Tx LS Ack Packets     | : 101 | Rx LS Ack Packets      | : 99 |
| Tx LS Ack Bytes       | : 101 | Rx LS Ack Bytes        | : 99 |

|                               |        |
|-------------------------------|--------|
| Total Number of State Changes | : 8    |
| Number of LSAs                | : 29   |
| LSA Checksum Sum              | : 2345 |
| Total Transmit Failures       | : 29   |
| Total OSPF Packets Discarded  | : 999  |

| Reason                      | Packets Dropped |
|-----------------------------|-----------------|
| Invalid type                | 19              |
| Invalid length              | 9               |
| Invalid checksum            | 0               |
| Invalid version             | 23              |
| Bad or unknown source       | 67              |
| Area mismatch               | 1               |
| Self-originated             | 19              |
| Duplicate router ID         | 9               |
| Interface standby           | 0               |
| Total Hello packets dropped | 60              |
| Network Mask mismatch       | 10              |

|                          |      |
|--------------------------|------|
| Hello interval mismatch  | 10   |
| Dead interval mismatch   | 10   |
| Options mismatch         | 10   |
| MTU mismatch             | 10   |
| Neighbor ignored         | 10   |
| Authentication errors    | 12   |
| Type mismatch            | 6    |
| Authentication failures  | 6    |
| Wrong protocol           | 0    |
| Resource failures        | 0    |
| Bad LSA length           | 0    |
| Others                   | 0    |
| Total LSAs Ignored : 176 |      |
| Bad Type                 | : 10 |
| Bad Length               | : 56 |
| Invalid Data             | : 55 |
| Invalid Checksum         | : 55 |

## summary-address

### Syntax

```
summary-address <IPV4-ADDR>/<MASK> [no-advertise | tag <TAG-VALUE>]
```

```
no summary-address <prefix/length> [no-advertise | tag <tag-value>]
```

### Description

Summarizes the external routes with the matching address and mask. When advertising this route, its metric is set to the lowest cost path from among the routes that were summarized.

The `no` form of this command disables route summarization.



**NOTE:** This command only works for an ASBR (Autonomous System Boundary Router).

### Command context

```
config-ospf-<PROCESS-ID>
```

### Parameters

#### <IPV4-ADDR>

Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.

#### <MASK>

Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 32.

#### no-advertise

Do not advertise the aggregate route. Suppress routes that match the specified prefix/mask pair.

#### tag <TAG-VALUE>

Specify the tag for the aggregate route. The summary prefix will be advertised along with the tag value in External LSAs. Range: 0 to 4294967295

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting OSPF route summarization:

```
switch(config)# router ospf 1
switch(config-ospf-1)# summary-address 10.1.0.0/16
```

Disabling route summarization:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no summary-address 10.1.0.0/16
```

# timers lsa-arrival

## Syntax

```
timers lsa-arrival <DELAY>
```

```
no timers lsa-arrival
```

## Description

Configures the minimum delay between receiving the same LSA from a peer. The same LSA is an LSA that contains the same LSA ID number, LSA type, and advertising router ID. If an instance of the same LSA arrives sooner before the delay expires, the LSA is dropped. Generally, the LSA arrival timer should be set to a value less than or equal to the start-time value for the command `timers throttle lsa start` on the neighbor.

The `no` form of this command sets the LSA timers to default values.

## Command context

```
config-ospf-<PROCESS-ID>
```

## Parameters

**<DELAY>**

Specifies the delay in milliseconds. Range: 0 to 600000. Default: 1000.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the LSA arrival timer:

```
switch(config)# router ospf 1
switch(config-ospf-1)# timers lsa-arrival 10
```

Setting the LSA arrival timer to default:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no timers lsa-arrival
```

## timers throttle lsa

### Syntax

```
timers throttle lsa start-time <START-TIME> hold-time <HOLD-TIME> max-wait-time <WAIT-TIME>
no timers throttle lsa
```

### Description

Configures the timers for LSA generation.

The `no` form of this command sets the LSA timers to default values.

### Command context

```
config-ospf-<PROCESS-ID>
```

### Parameters

**start-time <START-TIME>**

Specifies the initial wait time in milliseconds after which LSAs are generated. When set to 0, the LSAs are generated without any delay. Range: 0 to 600000. Default: 5000.

**hold-time <HOLD-TIME>**

Specifies the amount of time, in milliseconds, between regeneration of an LSA. The hold time doubles each time the same LSA must be regenerated, until `max-wait-time` is reached. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0.

**max-wait-time <WAIT-TIME>**

Specifies the maximum wait time, in milliseconds, for regeneration of the same LSA. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the LSA timers:

```
switch(config)# router ospf 1
switch(config-ospf-1)# timers throttle lsa start-time 100 hold-time 1000 max-wait-time 10000
```

Setting LSA timers to default values:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no timers throttle lsa
```

## timers throttle spf

### Syntax

```
timers throttle spf start-time <START-TIME> hold-time <HOLD-TIME>
max-wait-time <WAIT-TIME>
no timers throttle spf
```



## Description

Configures timers for SPF calculation. There are three timers:

- `start-time` Is the initial delay before an SPF calculation is started. Default is 200 milliseconds.
- `hold-time` Is the progressive backoff time to wait before next scheduled SPF calculation. Default is 1000 milliseconds. If a route change event occurs during this period, the value doubles until it reaches the *max-wait-time*.
- `max-wait-time` Is the maximum time to wait before the next scheduled SPF calculation. Default is 5000 milliseconds. This is used to limit the SPF hold timer and also defines the time to be considered for which the OSPF LSDB has to be stable, after which the SPF throttle mechanism is reset.

The `no` form of this command sets all the configured non-default timers to default value.

## Command context

```
config-ospf-<PROCESS-ID>
```

## Parameters

### <START-TIME>

Time in milliseconds to set timer for initial SPF delay. Default: 200.

### <HOLD-TIME>

Time in milliseconds to set the minimum hold time between two consecutive SPF calculations. Default: 1000.

### <WAIT-TIME>

Time in milliseconds to set the maximum wait time between two consecutive SPF calculations. Default: 5000.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting non-default timer values for SPF throttling:

Setting default timer values for SPF throttling after configuring non-default values:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no timers throttling spf
```

# trap-enable

## Syntax

```
trap-enable
```

```
no trap-enable
```

## Description

Enables the notification of the events to be sent as traps to the SNMP management stations for OSPF.

The `no` form of this command disables the notification of the events to be sent as traps to the SNMP management stations for OSPF.

## Command context

config-ospf-<PROCESS-ID>

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling sending notification of events as traps:

```
switch(config)# router ospf 1
switch(config-ospf-1)# trap-enable
```

Disabling sending notification of events as traps:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no trap-enable
```

## area

### Syntax

```
area <AREA-ID>
```

```
no area <AREA-ID>
```

### Description

Creates a normal area with <AREA-ID> set if not present. If area is present and is not the normal area, this command changes the area type to normal area.

The `no` form of this command deletes the area with the <AREA-ID> specified. The area can be of any type (stub, stub no-summary, and default normal area).

### Command context

```
config-ospfv3-<PROCESS-ID>
```

### Parameters

**<AREA-ID>**

Specifies the area ID in one of the following formats.

- Area identifier in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
- Area identifier in decimal format. Range: 0 to 4294967295.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating a normal area:

Deleting an area:

## area authentication ipsec

### Syntax

```
area <AREA-ID> authentication ipsec spi <SPI-INDEX> <AUTH-TYPE> <KEY-TYPE> <AUTH-KEY>
```

```
no area <AREA-ID> authentication
```

### Description

Configures IPsec AH authentication for the specified area. OSPFv3 interfaces which have IPsec configured at the interface context will not use area level IPsec.

The `no` form of this command removes IPsec AH authentication for the specified area.



**NOTE:** IPsec is not supported for 6in6 tunnel interfaces.

### Command context

`config-ospfv3-<PROCESS-ID>`

### Parameters

#### **<AREA-ID>**

Specifies the area ID is one of the following formats.

- OSPF area identifier in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
- OSPF area identifier in decimal format. Range: 0 to 4294967295.

#### **spi <SPI-INDEX>**

Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256-4294967295 characters.

#### **<AUTH-TYPE>**

Specifies the algorithm to use for authentication: `md5` or `sha1`.

#### **<KEY-TYPE>**

Specifies the key type to use: `plaintext` (unencrypted), `hex-string` (encrypted) or `ciphertext` (encrypted).

#### **<AUTH-KEY>**

Specifies the key.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting area 1 to use IPsec authentication(AH):

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 1 authentication ipsec spi 256 sha1 plaintext abcd
```

Removing IPsec authentication (AH) on area 1:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no area 1 authentication
```

## area encryption ipsec

### Syntax

```
area <AREA-ID> encryption ipsec spi <SPI-INDEX> <AUTH-TYPE> <KEY-TYPE> <AUTH-KEY>
({<ENCR-TYPE> <KEY-TYPE> <ENCR-KEY>} | NULL)
```

```
no area <AREA-ID> encryption
```

## Description

Configures IPsec ESP with authentication/encryption algorithm type and key for the specified area. OSPFv3 interfaces with IPsec configured at interface context will not use area level IPsec ESP configuration.

The `no` form of this command removes IPsec ESP from the specified area.



**NOTE:** IPsec is not supported for 6in6 tunnel interfaces.

## Command context

`config-ospfv3-<PROCESS-ID>`

## Parameters

### <AREA-ID>

Specifies the area ID is one of the following formats.

- OSPF area identifier in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.
- OSPF area identifier in decimal format. Range: 0 to 4294967295.

### `spi` <SPI-INDEX>

Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295 characters.

### <AUTH-TYPE>

Specifies the algorithm to use for authentication: `md5` or `sha1`.

### <ENCR-TYPE>

Specifies the algorithm to use for encryption: `3des`, `aes`, `des` or `null`.



**NOTE:** `aes` will be considered AES128, AES192 or AES256 based on key length.

### <KEY-TYPE>

Specifies the key type to use: `plaintext` (unencrypted) `hex-string` (encrypted) or `ciphertext` (encrypted).

### <AUTH-KEY>

Specifies authentication key.

### <ENCR-KEY>

Specifies encryption key.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting area 0 to use IPsec ESP:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 0 encryption ipsec spi 256 md5 plaintext abcd des plaintext abcdefab
```

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 0 encryption ipsec spi 256 md5 plaintext abcd null
```

Removing IPsec ESP on area 0:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no area 0 encryption
```

## clear ipv6 ospfv3 neighbors

### Syntax

```
clear ipv6 ospfv3 [<PROCESS-ID>] neighbor [<NEIGHBOR>] [interface [<INTERFACE-NAME>]]
[all-vrfs | vrf <VRF-NAME>]
```

### Description

Resets the neighbor and clears the OSPF neighbor information.

### Command context

Operator (>) or Manager (#)

### Parameters

**<PROCESS-ID>**

Specifies the OSPFv3 process ID to clear the statistics for the particular OSPFv3 process. Range: 1 to 63.

**<NEIGHBOR>**

Specifies the router ID of a neighbor.

**<INTERFACE-NAME>**

Specifies the OSPFv3 statistics to clear for the specified interface.

**all-vrfs**

Select to clear the OSPFv3 statistics for all VRFs.

**vrf <VRF-NAME>**

Specifies the name of a VRF.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Clearing the OSPFv3 neighbor information:

## clear ipv6 ospfv3 statistics

### Syntax

```
clear ipv6 ospfv3 [<PROCESS-ID>] statistics [interface [<INTERFACE-NAME>]]
[all-vrfs | vrf <VRF-NAME>]
```

## Description

Clears the OSPFv3 event statistics.

## Command context

Operator (>) or Manager (#)

## Parameters

**<PROCESS-ID>**

Specifies the OSPFv3 process ID to clear the statistics for the particular OSPFv3 process. Range: 1 to 63.

**<INTERFACE-NAME>**

Specifies the OSPFv3 statistics to clear for the specified interface.

**all-vrfs**

Select to clear the OSPFv3 statistics for all VRFs.

**vrf <VRF-NAME>**

Specifies the name of a VRF.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Clearing the OSPFv3 event statistics:

# default-metric

## Syntax

```
default-metric <METRIC-VALUE>
```

```
no default-metric
```

## Description

Sets the default metric for redistributed routes in the OSPFv3.

The `no` form of this command sets the default metric to be used for redistributed routes into OSPFv3 to the default of 25.

## Command context

```
config-ospfv3-<PROCESS-ID>
```

## Parameters

**<METRIC-VALUE>**

Specifies the default metric value to use for redistributed routes. Range: 1 to 1677214. Default: 25.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting default metric for redistributed routes:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# default-metric 36
```

Setting default metric for redistributed routes to the default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no default-metric
```

## disable

### Syntax

disable

### Description

Disables the OSPFv3 process. By default OSPFv3 process is enabled.



---

**IMPORTANT:** This command does not remove the OSPFv3 configurations.

---

### Command context

config-ospfv3-<PROCESS-ID>

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Disabling OSPFv3 process:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# disable
```

## distance

### Syntax

distance <DISTANCE>

no distance

### Description

Defines an administrative distance for OSPFv3. Administrative distance is used as a criteria to select the best route when the same route is learned by multiple routing protocols.

The **no** form of this command sets the OSPFv3 administrative distance to the default of 110.

### Command context

config-ospfv3-<PROCESS-ID>



## Parameters

### <DISTANCE>

Specifies OSPFv3 administrative distance. Range: 1 to 255. Default: 110.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting OSPF administrative distance:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# distance 100
```

Setting OSPF administrative distance to the default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no distance
```

# enable

## Syntax

enable

## Description

Enables OSPFv3 process when disabled. By default OSPFv3 process is enabled.

## Command context

config-ospfv3-<PROCESS-ID>

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Enabling OSPFv3 process:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# enable
```

# default-information originate

## Syntax

default-information originate

no default-information originate

## Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors if it is present in the routing table.

The **no** form of this command disables advertisement of the default route.

## Command context

config-ospf-<PROCESS-ID>

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate
```

# default-information originate always

## Syntax

default-information originate always

no default-information originate always

## Description

Configures OSPF to advertise the default route (0.0.0.0/0) to its neighbors, regardless if it is present in the routing table or not.

The **no** form of this command sets the OSPF administrative distance to the default of 110.

## Command context

config-ospf-<PROCESS-ID>

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# default-information originate always
```

Disabling advertisement of the default route:

```
switch(config)# router ospf 1
switch(config-ospf-1)# no default-information originate always
```

# graceful-restart

## Syntax

```
graceful-restart {restart-interval <INTERVAL> | helper | strict-lsa-check}
```

```
no graceful-restart {restart-interval | helper | strict-lsa-check}
```

## Description

Configures graceful restart for OSPFv3. By default graceful restart is enabled on the OSPFv3 router.

The `no` form of this command sets the restart interval to the default of 120 seconds or disables helper mode depending on the specified parameters.

## Command context

```
config-ospfv3-<PROCESS-ID>
```

## Parameters

**restart-interval** <INTERVAL>

Specifies the time another router waits for this router to gracefully restart and selects the maximum time to wait in seconds. Range: 5 to 1800. Default: 120.

**helper**

Specifies that the router will participate in the graceful restart of a neighbor router.

**strict-lsa-check**

Enable strict LSA checking when acting as a restart helper for a restarting peer.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling OSPF graceful restart:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# graceful-restart restart-interval 40
switch(config-ospfv3-1)# graceful-restart helper
switch(config-ospfv3-1)# graceful-restart strict-lsa-check
```

Setting the restart interval to default, disabling helper mode, and strict LSA checking:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no graceful-restart restart-interval
switch(config-ospfv3-1)# no graceful-restart helper
switch(config-ospfv3-1)# no graceful-restart strict-lsa-check
```

# ipv6 ospfv3 area

## Syntax

```
ipv6 ospfv3 <PROCESS-ID> area <AREA-ID>
```

```
no ipv6 ospfv3 <PROCESS-ID> area <area-id>
```

## Description

Runs the OSPFv3 protocol on the interface for the area specified.

To move an interface to a new area, unmap the existing area and then associate a new area with the interface.

The `no` form of this command disables OSPF on the interface and removes the interface from the area. Interfaces which have an IP address configured on the network or in a subset of the network, stop participating in the OSPF protocol

## Command context

config-if-vlan

## Parameters

**<PROCESS-ID>**

Specifies the OSPFv3 process ID. Range: 1 to 63.

**<AREA-ID>**

Specifies the area ID in one of the following formats.

- Area ID in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
- Area ID as a decimal value. Range: 0-4294967295.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting OSPFv3 network for the area:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 1 area 1
switch(config-if-vlan)# ipv6 ospfv3 1 area 0.0.0.1
```

Disabling OSPFv3 network for the area:

```
switch(config)# interface 1/1/1
switch(config-if-vlan)# no ipv6 ospfv3 1 area 1
switch(config-if-vlan)# no ipv6 ospfv3 1 area 0.0.0.1
```

# ipv6 ospfv3 authentication null

## Syntax

```
ipv6 ospfv3 authentication null
```

## Description

Configures null authentication on an interface which disables IPsec authentication.

## Command context

config-if-vlan

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Disabling IPsec on interface VLAN 1:

```
switch(config)# interface van 1
switch(config-if-vlan)# ipv6 ospfv3 authentication null
```

# ipv6 ospfv3 authentication ipsec

## Syntax

```
ipv6 ospfv3 authentication ipsec spi <SPI-INDEX> <AUTH-TYPE> <KEY-TYPE> <AUTH-KEY>

no ipv6 ospfv3 authentication
```

## Description

Configures IPsec AH authentication. OSPFv3 interfaces that have IPsec configured at the interface context will not use area level IPsec.

The `no` form of this command removes IPsec AH authentication for the specified area.

## Command context

config-if-vlan

## Parameters

**spi** <SPI-INDEX>

Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec AH header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256-4294967295 characters.

<AUTH-TYPE>

Specifies the algorithm to use for authentication: `md5` or `sha1`.

<KEY-TYPE>

Specifies the key type to use: `plaintext` (not encrypted) or `hex-string` (encrypted) or `ciphertext` (encrypted).

<AUTH-KEY>

Specifies the key.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting interface VLAN 1 to use IPsec authentication (AH):

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 authentication ipsec spi 256 md5 plaintext abcd
```

Removing IPsec authentication (AH) on interface VLAN 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 authentication
```

## ipv6 ospfv3 cost

### Syntax

```
ipv6 ospfv3 cost <INTERFACE-COST>
```

```
no ipv6 ospfv3 cost
```

### Description

Sets the cost (metric) associated with a particular interface. The interface cost is used as a parameter to calculate the best routes.

The `no` form of this command sets the cost (metric) associated with a particular interface to the default of 1.

### Command context

```
config-if-vlan
```

### Parameters

**<INTERFACE-COST>**

Specifies the interface cost value. Range: 1 to 65535. Default: 1.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting OSPFv3 interface cost:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 cost 100
```

Setting the OSPFv3 interface cost to default:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 cost
```

## ipv6 ospfv3 dead-interval

### Syntax

```
ipv6 ospfv3 dead-interval <INTERVAL>
```

```
no ipv6 ospfv3 dead-interval
```

### Description

Sets the interval after a neighbor is declared dead when no hello packet is received on the OSPFv3 interface.

The `no` form of this command sets the interval after which a neighbor is declared dead, to the default for the OSPFv3 interface. The default value is 40 seconds (generally four times the hello packet interval).

## Command context

config-if-vlan

## Parameters

**<INTERVAL>**

Specifies the time interval for the dead interval, in seconds. Range: 1 to 65535. Default: 40.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting OSPFv3 dead interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 dead-interval 30
```

Setting OSPFv3 dead interval to default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 dead-interval
```

# ipv6 ospfv3 encryption ipsec

## Syntax

```
ipv6 ospfv3 encryption ipsec spi <SPI-INDEX> <AUTH-TYPE> <KEY-TYPE> <AUTH-KEY>
({<ENCR-TYPE> <KEY-TYPE> <ENCR-KEY>} | null)
```

```
no ipv6 ospfv3 encryption
```

## Description

Configures IPsec ESP. OSPFv3 interfaces that have IPsec configured at the interface context will not use area level IPsec ESP.

The **no** form of this command removes IPsec ESP for the specified area.

## Command context

config-if-vlan

## Parameters

**spi <SPI-INDEX>**

Specifies the Security Parameters Index (SPI) to use. The SPI is an identification tag carried in the IPsec ESP header. It enables the receiving OSPF process to select and use the Security Association (SA) from the SA table. The SPI must be unique on the switch. Range: 256 to 4294967295 characters.

**<AUTH-TYPE>**

Specifies the algorithm to use for authentication: **md5** or **sha1**.

**<ENCR-TYPE>**

Specifies the algorithm to use for encryption: **des**, **3des** or **aes**.

#### <KEY-TYPE>

Specifies the key type to use: `plaintext` (not encrypted), `hex-string` (encrypted) or `ciphertext` (encrypted).

#### <AUTH-KEY>

Specifies the authentication key.

#### <ENCR-KEY>

Specifies the encryption key.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting interface VLAN 1 to use IPsec ESP:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1 plaintext abcdef aes
plaintext abcdefabcdefabcd
```

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 encryption ipsec spi 256 sha1 plaintext abcdef null
```

Removing IPsec on interface VLAN 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 encryption
```

## ipv6 ospfv3 encryption null

### Syntax

```
ipv6 ospfv3 encryption null
```

### Description

Configures NULL ESP on an interface which disables IPsec ESP.

### Command context

```
config-if-vlan
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Disable IPsec ESP on interface VLAN 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 encryption null
```



## ipv6 ospfv3 hello-interval

### Syntax

```
ipv6 ospfv3 hello-interval <INTERVAL>
```

```
no ipv6 ospfv3 hello-interval
```

### Description

Sets the time interval between OSPFv3 hello packets for the OSPFv3 interface.

The `no` form of this command sets the time interval between OSPFv3 hello packets to the default for the OSPFv3 interface of 10 seconds.

### Command context

```
config-if-vlan
```

### Parameters

<INTERVAL>

Specifies the time interval between hello packets, in seconds. Range: 1 to 65535. Default: 10.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting OSPFv3 hello interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 hello-interval 30
```

Setting OSPFv3 hello interval to default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 hello-interval
```

## ipv6 ospfv3 network

### Syntax

```
ipv6 ospfv3 network {broadcast|point-to-point}
```

```
no ipv6 ospfv3 network
```

### Description

Configures the network type for the interface. By default the network type is broadcast network.

The `no` form of this command sets the network type for the interface to the system default of broadcast network.

### Command context

```
config-if-vlan
```

## Parameters

### **broadcast**

Specifies the OSPFv3 network type as a broadcast multiaccess network.

### **point-to-point**

Specifies the OSPFv3 network type as a point-to-point network.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting OSPFv3 network type for the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 network broadcast
switch(config-if-vlan)# ipv6 ospfv3 network point-to-point
```

Disabling OSPFv3 network type for the interface to system default of broadcast network:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 network
```

## ipv6 ospfv3 passive

### Syntax

```
ipv6 ospfv3 passive
```

```
no ipv6 ospfv3 passive
```

### Description

Configures the interface as an OSPFv3 passive interface. With this setting, the interface participates in the OSPF, but does not send or receive OSPF packets on that interface.

The **no** form of this command resets the interface as active. With this setting, the interface starts sending and receiving OSPF packets.

### Command context

```
config-if-vlan
```

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the interface as OSPFv3 passive interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 passive
```

Setting the interface as OSPFv3 active interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 passive
```

## ipv6 ospfv3 priority

### Syntax

```
ipv6 ospfv3 priority <number-value>
```

```
no ipv6 ospfv3 priority
```

### Description

Sets the OSPFv3 priority for the interface. The larger the numeric value of the priority, the higher the chance it will become the designated router. Setting a priority of 0 makes the router ineligible to become a designated router or back up designated router.

The `no` form of this command sets the OSPFv3 priority for the interface to the default of 1.

### Command context

```
config-if-vlan
```

### Parameters

**<number-value>**

Specifies the OSPFv3 priority value. Default: 1. Range: 0 to 255.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the OSPFv3 priority for the interface:

```
switch(config)# interface vlan /1
switch(config-if-vlan)# ipv6 ospfv3 priority 50
```

Setting the OSPFv3 priority for the interface to the default of 1:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 priority
```

## ipv6 ospfv3 retransmit-interval

### Syntax

```
ipv6 ospfv3 retransmit-interval <INTERVAL>
```

```
no ipv6 ospfv3 retransmit-interval
```

### Description

Sets the time between retransmitting lost link state advertisements for the OSPFv3 interface.

The `no` form of this command sets the time between retransmitting lost link state advertisements to the default 5 seconds.

### Command context

```
config-if-vlan
```

## Parameters

### <INTERVAL>

Specifies the time interval for the retransmit interval, in seconds. Range: 1 to 3600. Default: 5

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting OSPFv3 retransmit interval on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 retransmit-interval 30
```

Setting OSPFv3 retransmit interval to the default on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 retransmit-interval
```

## ipv6 ospfv3 shutdown

## Syntax

```
ipv6 ospfv3 shutdown
```

```
no ipv6 ospfv3 shutdown
```

## Description

Disables OSPFv3 on the interface. The interface state changes to Down. It does not remove the interface from the OSPF area. To remove the interface, use the command `no ip ospf area`.

The `no` form of this command re-enables OSPFv3 on the interface.

## Command context

```
config-if-vlan
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Disabling OSPFv3 on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 shutdown
```

Re-enabling OSPFv3 on the interface:

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 shutdown
```

## ipv6 ospfv3 transit-delay

### Syntax

```
ipv6 ospfv3 transit-delay <DELAY>
```

```
no ipv6 ospfv3 transit-delay
```

### Description

Sets the time delay in Link state transmission for the OSPFv3 interface.

The `no` form of this command sets the transit delay in Link state transmission to the default of 1 second.

### Command context

```
config-if-vlan
```

### Parameters

**<DELAY>**

Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting OSPFv3 transit delay on the interface

```
switch(config)# interface vlan 1
switch(config-if-vlan)# ipv6 ospfv3 transit-delay 30
```

Setting OSPFv3 transit delay to default on the interface

```
switch(config)# interface vlan 1
switch(config-if-vlan)# no ipv6 ospfv3 transit-delay
```

## maximum-paths

### Syntax

```
maximum-paths <MAXIMUM>
```

```
no maximum-paths
```

### Description

Sets the maximum number of ECMP routes that OSPFv3 can support.

The `no` form of this command sets the maximum number of ECMP routes that OSPFv3 can support to the default value of 4.

### Command context

```
config-ospfv3-<PROCESS-ID>
```

## Parameters

### <MAXIMUM>

Specifies the maximum number of ECMP routes. Range: 1 to 8. Default: 4.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting maximum number of parallel routes:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# maximum-paths 3
```

Setting maximum number of parallel paths to the default value of 4:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no maximum-paths
```

# max-metric router-lsa

## Syntax

```
max-metric router-lsa [on-startup <INTERVAL>]
```

```
no max-metric router-lsa [on-startup]
```

## Description

Sets the protocol to advertise a maximum metric so that other routers do not prefer the router as an intermediate hop in their shortest path first (SPF) calculations. If the on-startup parameter is used, the router is configured to advertise a maximum metric at startup. That is, for the time specified in seconds, or the default value of 600 seconds.

To disable advertisement of the maximum metric, use the `no` form of the command.

The `no` form of this command advertises the normal cost metrics instead of advertising the maximized cost metric. This setting causes the router to be considered in traffic forwarding.

## Command context

```
config-ospfv3-<PROCESS-ID>
```

## Parameters

**on-startup <INTERVAL>**

Automatically advertises the stub Router-LSA (or maximize the router-LSA cost metric) for a specified time interval upon OSPFv3 startup.

Specifies the time in seconds. Range: 5 to 86400. Default: 600.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting to maximize the cost metrics for Router LSA:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# max-metric router-lsa
switch(config-ospfv3-1)# max-metric router-lsa on-startup 3000
```

Setting to advertise the normal cost metrics instead of advertising the maximized cost metric:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no max-metric router-lsa
switch(config-ospfv3-1)# no max-metric router-lsa on-startup
```

## passive-interface default

### Syntax

```
passive-interface default
```

```
no passive-interface default
```

### Description

Sets all OSPFv3 interfaces as passive.

The `no` form of this command sets all the OSPFv3 interfaces as active.

### Command context

```
config-ospfv3-<PROCESS-ID>
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting OSPFv3-enabled interfaces as passive:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# passive-interface default
```

Setting OSPFv3-enabled interfaces as active:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no passive-interface default
```

## redistribute

### Syntax

```
redistribute {connected | static | ripng | ospfv3 <PROCESS-ID>} [route-map <ROUTE-MAP-NAME>]
```

```
no redistribute {connected | static} [route-map <ROUTE-MAP-NAME>]
```

### Description

Redistributes routes originating from other protocols, or from another OSPF process, to the current OSPF process.

The `no` form of this command disables redistribution of routes to the current OSPF process.

## Command context

config-ospfv3-<PROCESS-ID>

## Parameters

### connected

Specifies redistributing connected routes (directly attached subnet or host) to OSPFv3.

### static

Specifies redistributing static routes to OSPFv3.

### ripng

Specifies redistributing RIPv3 routes.

### ospfv3 <PROCESS-ID>

Specifies redistributing routes from the specified OSPFv3 process ID.

### route-map <ROUTE-MAP-NAME>

Specifies the name of a route map. To create a route map, use the `route-map` command.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Redistributing routes to OSPFv3:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)#
switch(config-ospfv3-1)# redistribute connected
switch(config-ospfv3-1)# redistribute static
switch(config-ospfv3-1)# redistribute ripng
switch(config-ospfv3-1)# redistribute ospfv3 1
switch(config-ospfv3-1)# redistribute route-map static_networks
```

Disabling redistributing routes to OSPFv3:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)#
switch(config-ospfv3-1)# no redistribute connected
switch(config-ospfv3-1)# no redistribute static
switch(config-ospfv3-1)# no redistribute ripng
switch(config-ospfv3-1)# no redistribute ospfv3 1
switch(config-ospfv3-1)# no redistribute route-map static_networks
```

## reference-bandwidth

### Syntax

reference-bandwidth <BANDWIDTH>

no reference-bandwidth

### Description

Sets the reference bandwidth for OSPFv3. If the OSPFv3 interface cost is not explicitly set, then the cost of all the OSPFv3 interfaces is recalculated based on the reference bandwidth and link speed of the interface.



For VLAN interfaces the calculated link speed value is 1 Gbps (if the OSPFv3 interface cost is not explicitly set).

The `no` form of this command sets the reference bandwidth for OSPF to the default of 100000 Mbps.

### Command context

```
config-ospfv3-<PROCESS-ID>
```

### Parameters

**<BANDWIDTH>**

Specifies the reference bandwidth used to calculate the cost of an interface in Mbps. Range: 1 to 4000000. Default: 100000.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the reference bandwidth:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# reference-bandwidth 40000
```

Setting the reference bandwidth to the default value:

```
switch(config)# routerv3 ospf 1
switch(config-ospfv3-1)# no reference-bandwidth
```

## retransmit-interval

### Syntax

```
retransmit-interval <INTERVAL>
```

```
no retransmit-interval
```

### Description

Sets the time between retransmitting lost link state advertisements for virtual links.

The `no` form of this command sets the time between retransmitting lost link state advertisements to the default of 5 seconds for virtual links.

### Command context

```
config-router-vlink
```

### Parameters

**<INTERVAL>**

Specifies the time interval for the retransmit interval, in seconds. Range: 1 to 3600. Default: 5.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting OSPFv3 virtual links retransmit interval:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# retransmit-interval 30
```

Setting OSPFv3 virtual links retransmit interval to default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no retransmit-interval
```

## router-id

### Syntax

```
router-id <ROUTER-ADDRESS>
```

```
no router-id
```

### Description

Sets an ID for the router in an IPv4 address format.

The **no** form of this command unconfigures the router-id for the instance and sets the router-id to the default. The router-id is changed to the dynamically selected router-id. The default router-id 0.0.0.0 updates to the routing stack that triggers to auto-elect a router-id based on the highest IP address of loopback interface, or the highest IP address of interfaces.

### Command context

```
config-ospfv3-<PROCESS-ID>
```

### Parameters

**<ROUTER-ADDRESS>**

Specifies the router address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting router-id in the OSPFv3 context:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1) # router-id 1.1.1.1
```

Unconfiguring router-id:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1) # no router-id
```

## router ospfv3

### Syntax

```
router ospfv3 <PROCESS-ID> [vrf <VRF-NAME>]
```

```
no router ospfv3 <PROCESS-ID> [vrf <VRF-NAME>]
```

### Description

Creates the OSPFv3 process (if not created already) and enters the router OSPFv3 instance context. Optionally if specified, you can specify a named VRF, or the default VRF if the <vrf-name> is not specified. Only one OSPFv3 process is allowed per VRF.

The `no` form of this command removes the OSPFv3 instance. If a VRF is specified, it removes the OSPF instance from the named VRF, or the default VRF if the <var-name> is not specified.

### Command context

config

### Parameters

**<PROCESS-ID>**

Specifies an OSPFv3 process ID. Length: 1 to 63.

**vrf <VRF-NAME>**

Specifies the name of a VRF. Default: default.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Entering the router OSPFv3 instance:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)#
```

Setting the router OSPFv3 VRF instance:

```
switch(config)# router ospfv3 1 vrf vrf_red
```

Removing the router OSPFv3 instance:

```
switch#(config)# no router ospfv3 1
```

## show ipv6 ospfv3

### Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows OSPFv3 information including area, state, and configuration information.

## Command context

Operator (>) or Manager (#)

## Parameters

**<PROCESS-ID>**

Specifies an OSPFv3 process ID optionally to show OSPFv3 information for a particular OSPFv3 process.  
Range: 1 to 63.

**all-vrfs**

Select to show OSPFv3 information for all VRFs.

**vrf <VRF-NAME>**

Specifies the name of a VRF. Default: default.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

Showing general OSPFv3 configurations:

```
switch# show ipv6 ospfv3 1
Routing Process 1 with ID: 1.1.1.1 VRF default

OSPFv3 Protocol is enabled
Graceful-restart is configured
Restart Interval: 120, State: inactive
Last Graceful Restart Exit Status: none
Area Border: false
AS Border: false
SPF: Start Time: 200ms, Hold Time: 1000ms, Max Wait Time: 5000ms
LSA: Start Time: 5000ms, Hold Time: 0ms, Max Wait Time: 0ms
LSA Arrival: 1000ms
Maximum Paths to Destination: 4
Number of External LSAs 0, checksum sum 0
Summary address:
 prefix fd00::0/64 advertise tag 10
Number of areas is: 2, 2 normal, 0 stub, 0 NSSA
Number of active areas is: 1, 1 normal, 0 stub, 0 NSSA
Reference Bandwidth: 100000 Mbps
Area (0.0.0.0) (Active)
 Interfaces in this Area: 1 Active Interfaces: 1
 Passive Interfaces: 0 Loopback Interfaces: 0
 SPF calculation has run 9 times
 Number of LSAs: 4, checksum sum 195745
Area (0.0.0.1) (Inactive)
 Interfaces in this Area: 0 Active Interfaces: 0
 Passive Interfaces: 0 Loopback Interfaces: 0
 SPF calculation has run 8 times
 Area ranges:
 fd00::1/64, inter-area, advertise
 Number of LSAs: 0, checksum sum 0
```

## show ipv6 ospfv3 border-routers

### Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] border-routers [all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows the OSPFv3 routing table entries for Area Border Router (ABR) and Autonomous System Border Router (ASBR).

### Command context

Operator (>) or Manager (#)

### Parameters

**<PROCESS-ID>**

Specifies an OSPFv3 process ID to show the OSPFv3 routing table entries for ABR and ASBR for the particular OSPFv3 process. Range: 1-63.

**all-vrfs**

Select to show OSPFv3 border router information for all VRFs.

**vrf <VRF-NAME>**

Specifies the name of a VRF. Default: default.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Showing OSPFv3 border routers information for default VRF:

```
switch#
OSPFv3 Process ID 1 VRF default, Internal Routing Table

Codes: i - Intra-area route, I - Inter-area route
i 2.2.2.2 [10], ABR, Area 0.0.0.0, SPF 17 via
 fe80::98f2:b301:2c68:1d48, Interface 1/1/1
i 2.2.2.2 [10], ABR, Area 0.0.0.1, SPF 17 via
 fe80::98f2:b301:3068:1d48, Interface 1/1/2
```

## show ipv6 ospfv3 interface

### Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] interface [<INTERFACE-NAME>] [brief]
[all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows information about OSPFv3 enabled interfaces.

### Command context

Operator (>) or Manager (#)

## Parameters

### <PROCESS-ID>

Specifies an OSPFv3 process ID optionally to show the OSPFv3 enabled interfaces for the particular OSPFv3 process. Range: 1-63.

### <INTERFACE-NAME>

Selects to show information only for the specified OSPFv3 enabled interface.

### brief

Select to show the overview information for the OSPFv3 enabled interface.

### all-vrfs

Select to show OSPF-enabled interface information for all VRFs.

### vrf <VRF-NAME>

Specifies the name of a VRF. Default: default.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing OSPFv3 information for all interfaces in default VRF:

```
switch# show ipv6 ospfv3 interface
Interface 1/1/1 is up, Line Protocol is up

IPv6 address fe80::98f2:b301:468:be02
Process ID 1 VRF default, Area 0.0.0.0
 State Dr, Status up, Network type Broadcast
 Link Speed: 10000 Mbps
 Cost Configured NA, Calculated 10
 Transit Delay 1 sec, Router Priority 1
 Designated Router ID: 1.1.1.1
 Backup Designated Router ID: 2.2.2.2
 Timer Intervals: Hello 10, Dead 40, Retransmit 5
 Number of Link LSAs: 2, checksum sum 67473

Interface 1/1/2 is up, Line Protocol is up

IPv6 address fe80::98f2:b301:868:be02
Process ID 1 VRF default, Area 0.0.0.1
 State Waiting, Status up, Network type Broadcast
 Link Speed: 10000 Mbps
 Cost Configured NA, Calculated 10
 Transit Delay 1 sec, Router Priority 1
 No designated router on this network
 No backup designated router on this network
 Timer Intervals: Hello 10, Dead 40, Retransmit 5
 Number of Link LSAs: 1, checksum sum 24592
```

# show ipv6 ospfv3 neighbors

## Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] neighbors [<NEIGHBOR-ID>]
 [interface <INTERFACE-NAME>] [detail | summary]
 [all-vrfs | vrf <VRF-NAME>]
```

## Description

Shows information about OSPFv3 neighbors.

## Command context

Operator (>) or Manager (#)

## Parameters

### <PROCESS-ID>

Specifies an OSPFv3 process ID to show OSPFv3 neighbor information for the particular OSPFv3 process.  
Range: 1-63.

### neighbors <NEIGHBOR-ID>

Shows information about a particular neighbor, specified in IPv4 format (A.B.C.D).

### interface <INTERFACE-NAME>

Shows neighbor information only for the specified interface.

### detail

Shows detailed information for the neighbors.

### summary

Shows summary information for the neighbors.

### all-vrfs

Shows neighbor information for all VRFs.

### vrf <VRF-NAME>

Specifies the name of a VRF. Default: default.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing OSPFv3 neighbors information:

```
switch# show ipv6 ospfv3 neighbors
OSPFv3 Process ID 1 VRF default
=====

Total Number of Neighbors: 1

Neighbor ID Priority State Interface

2.2.2.2 1 FULL/BDR 1/1/1
Neighbor address fe80::98f2:b301:2c68:1d48
```

Showing OSPFv3 neighbors information for a specific neighbor:

```
switch# show ipv6 ospfv3 neighbors 2.2.2.2
Neighbor 2.2.2.2, Interface address fe80::98f2:b301:2c68:1d48

Process ID 1 VRF default, in Area 0.0.0.0 via Interface 1/1/1
 Neighbor priority is 1, State is FULL
 Options is 0x13
 Dead Timer due in 00:00:38
 Time since last state change 00h:20m:26s
```

Showing detail OSPFv3 neighbors information for a specific neighbor:

```
switch# show ipv6 ospfv3 neighbors 2.2.2.2 detail
Neighbor 2.2.2.2, Interface address fe80::98f2:b301:2c68:1d48

Process ID 1 VRF default, in Area 0.0.0.0 via Interface 1/1/1
 Neighbor priority is 1, State is FULL
 DR is 1.1.1.1, BDR is 2.2.2.2
 Options is 0x13
 Dead Timer due in 00:00:36
 Retransmission Queue Length 0
 Time since last state change 00h:20m:38s
```

Showing OSPFv3 neighbors information for interface 1/1/1:

```
switch# show ipv6 ospfv3 neighbors 2.2.2.2 interface 1/1/1
Neighbor 2.2.2.2, Interface address fe80::98f2:b301:2c68:1d48

Process ID 1 VRF default, in Area 0.0.0.0 via Interface 1/1/1
 Neighbor priority is 1, State is FULL
 Options is 0x13
 Dead Timer due in 00:00:32
 Time since last state change 00h:20m:52s
```

## show ipv6 ospfv3 routes

### Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] routes [<PREFIX/LENGTH>]
 [all-vrfs | vrf <VRF-NAME>]
```

### Description

Shows the OSPFv3 routing table information.

### Command context

Operator (>) or Manager (#)

### Parameters

#### <PROCESS-ID>

Specifies an OSPFv3 process ID that shows information from the OSPFv3 routing table for the particular OSPFv3 process. Range: 1-63.

#### <PREFIX/LENGTH>

Specifies the IPv6 destination prefix showing information about a particular destination prefix. For example, 2010:bd9::/32.



## **all-vrfs**

Select to show OSPFv3 routing table information for all VRFs.

## **vrf <VRF-NAME>**

Specifies the name of a VRF. Default: default.

## **Authority**

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## **Examples**

Showing OSPFv3 routing table information:

```
switch# show ipv6 ospfv3 routes
Codes: i - Intra-area route, I - Inter-area route
 E1 - External type-1, E2 - External type-2

OSPFv3 Process ID 1 VRF default, Routing Table

Total Number of OSPFv3 Routes : 2

111::/64 (i) area:0.0.0.0
 directly attached to interface 1/1/1, cost 1 distance 110
fd00::/64 (i) area:0.0.0.1
 directly attached to interface vlan10, cost 1 distance 110
```

Showing OSPFv3 routing table information for destination prefix fd00::/64:

```
switch# show ipv6 ospfv3 1 routes fd00::/64
Codes: i - Intra-area route, I - Inter-area route
 E1 - External type-1, E2 - External type-2

OSPFv3 Process ID 1 VRF default, Routing Table for prefixes fd00::/64

Total Number of OSPFv3 Routes : 1

fd00::/64 (i) area:0.0.0.1
 directly attached to interface vlan10, cost 1 distance 110
```

# **show ipv6 ospfv3 statistics**

## **Syntax**

```
show ipv6 ospfv3 [<PROCESS-ID>] statistics
 [all-vrfs | vrf <VRF-NAME>]
```

## **Description**

Shows OSPFv3 statistics.

## **Command context**

Operator (>) or Manager (#)

## Parameters

### <PROCESS-ID>

Specifies an OSPFv3 process ID that shows information on the OSPFv3 SPF statistics for the particular OSPFv3 process. Range: 1-63.

### all-vrfs

Select to show OSPFv3 SPF statistics information for all VRFs.

### vrf <VRF-NAME>

Specifies the name of a VRF. Default: default.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing OSPFv3 statistics information:

```
switch# show ipv6 ospfv3 statistics
OSPFv3 Process ID 1 VRF default, Statistics (cleared 3h 2m 21s ago)

Unknown Interface Drops : 0
Unknown Virtual Interface Drops : 0
Bad IPv6 Header Length Drops : 0
Wrong OSPFv3 Version Drops : 0
Bad Source IPv6 Drops : 0
Resource Failure Drops : 0
Bad Header Length Drops : 0
Total Drops : 0
```

# show ipv6 ospfv3 statistics interface

## Syntax

```
show ipv6 ospfv3 [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>]
 [all-vrfs | vrf <VRF-NAME>]
```

## Description

Shows the OSPFv3 statistics for the OSPFv3-enabled interfaces.

## Command context

Operator (>) or Manager (#)

## Parameters

### <PROCESS-ID>

Specifies an OSPFv3 process ID to show OSPF-enabled interface statistics information on the specified OSPFv3 process. Range: 1 to 63.

### <INTERFACE-NAME>

Selects to show information only for the specified interface.

## all-vrfs

Select to show OSPF-enabled interface statistics information for all VRFs.

## vrf <VRF-NAME>

Specifies the name of a VRF. Default: default.

## Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

## Example

Showing OSPFv3-enabled interfaces information for interface 1/1/1:

```
switch# show ipv6 ospfv3 statistics interface all-vrfs
OSPFv3 Process ID 1 VRF default, Interface vlan2000 Statistics (cleared 0h 2m 52s ago)
=====

Tx Hello packets : 0 Rx Hello packets : 0
Tx Hello bytes : 0 Rx Hello bytes : 0
Tx DD packets : 0 Rx DD packets : 0
Tx DD bytes : 0 Rx DD bytes : 0
Tx LS request packets : 0 Rx LS request packets : 0
Tx LS request bytes : 0 Rx LS request bytes : 0
Tx LS update packets : 0 Rx LS update packets : 0
Tx LS update bytes : 0 Rx LS update bytes : 0
Tx LS ack packets : 0 Rx LS ack packets : 0
Tx LS ack bytes : 0 Rx LS ack bytes : 0

Total IPsec packets processed : 0
Total IPsec bytes processed : 0
Total Number of State Changes : 0
Number of LSAs : 0
LSA Checksum Sum : 0

Total OSPFv3 Packets Discarded: 0

Reason Packets Dropped

Invalid Type 0
Invalid Length 0
Invalid Version 0
Bad or Unknown Source 0
Area Mismatch 0
Self-originated 0
Duplicate Router ID 0
Interface Standby 0
Total Hello Packets Dropped 0
 Hello Interval Mismatch 0
 Dead Interval Mismatch 0
 Options Mismatch 0
 MTU Mismatch 0
 Neighbor Ignored 0
Resource Failures 0
Bad LSA Length 0
Others 0
IPsec Authentication Errors 0
IPsec ESP Errors 0

Total LSAs Ignored : 0

Bad Type : 0
```

```
Bad Length : 0
Invalid Data : 0
Invalid Checksum : 0
```

## summary-address

### Syntax

```
summary-address <IPV6-ADDR>/<MASK> [no-advertise | tag <TAG-VALUE>]
```

```
no summary-address <prefix/length> [no-advertise | tag <tag-value>]
```

### Description

Summarizes the external routes with the matching address and mask. When advertising this route, its metric is set to the lowest cost path from among the routes that were summarized.

The `no` form of this command disables route summarization.



**NOTE:** This command only works for an ASBR (Autonomous System Boundary Router).

### Command context

```
config-ospfv3-<PROCESS-ID>
```

### Parameters

#### <IPV6-ADDR>

Specifies an IP address in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

#### <MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

#### no-advertise

Do not advertise the aggregate route. Suppress routes that match the specified prefix/mask pair.

#### tag <TAG-VALUE>

Specify the tag for the aggregate route. The summary prefix will be advertised along with the tag value in External LSAs. Range: 0 to 4294967295

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting OSPF route summarization:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# summary-address 2001:DB8::1/32
```

Disabling route summarization:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no summary-address 2001:DB8::1/32
```

## timers lsa-arrival

### Syntax

```
timers lsa-arrival <DELAY>
```

```
no timers lsa-arrival
```

### Description

Configures the minimum delay between receiving the same LSA from a peer. The same LSA is an LSA that contains the same LSA ID number, LSA type, and advertising router ID. If an instance of the same LSA arrives sooner before the delay expires, the LSA is dropped. Generally, the LSA arrival timer should be set to a value less than or equal to the start-time value for the command `timers throttle lsa start` on the neighbor.

The `no` form of this command sets the LSA timers to default values.

### Command context

```
config-ospfv3-<PROCESS-ID>
```

### Parameters

**<DELAY>**

Specifies the delay in milliseconds. Range: 0 to 600000. Default: 1000.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the LSA arrival timer:

```
switch(config)# router ospf 1
switch(config-ospfv3-1)# timers lsa-arrival 10
```

Setting the LSA arrival timer to default:

```
switch(config)# router ospf 1
switch(config-ospfv3-1)# no timers lsa-arrival
```

## timers throttle lsa

### Syntax

```
timers throttle lsa start-time <START-TIME> hold-time <HOLD-TIME> max-wait-time <WAIT-TIME>
```

```
no timers throttle lsa
```

### Description

Configures the timers for LSA generation.

The `no` form of this command sets the LSA timers to default values.

### Command context

```
config-ospfv3-<PROCESS-ID>
```

## Parameters

**start-time** <START-TIME>

Specifies the initial wait time in milliseconds after which LSAs are generated. When set to 0, the LSAs are generated without any delay. Range: 0 to 600000. Default: 5000.

**hold-time** <HOLD-TIME>

Specifies the amount of time, in milliseconds, between regeneration of an LSA. The hold time doubles each time the same LSA must be regenerated, until `max-wait-time` is reached. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0.

**max-wait-time** <WAIT-TIME>

Specifies the maximum wait time, in milliseconds, for regeneration of the same LSA. When set to 0, LSA regeneration time is not increased. Range: 0 to 600000. Default: 0.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the LSA timers:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# timers throttle lsa start-time 100 hold-time 1000 max-wait-time 10000
```

Setting LSA timers to default values:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no timers throttle lsa
```

# timers throttle spf

## Syntax

```
timers throttle spf start-time <START-TIME> hold-time <HOLD-TIME>
max-wait-time <WAIT-TIME>
```

```
no timers throttle spf
```

## Description

Configures timers for SPF calculation. There are three timers:

- `start-time` Is the initial delay before an SPF calculation is started. Default is 200 milliseconds.
- `hold-time` Is the progressive backoff time to wait before next scheduled SPF calculation. Default is 1000 milliseconds. If a route change event occurs during this period, the value doubles until it reaches the `max-wait-time`.
- `max-wait-time` Is the maximum time to wait before the next scheduled SPF calculation. Default is 5000 milliseconds. This is used to limit the SPF hold timer and also defines the time to be considered for which the OSPF LSDB has to be stable, after which the SPF throttle mechanism is reset.

The `no` form of this command sets all the configured non-default timers to default value.

## Command context

```
config-ospf-<PROCESS-ID>
```

## Parameters

### <START-TIME>

Time in milliseconds to set timer for initial SPF delay. Default: 200.

### <HOLD-TIME>

Time in milliseconds to set the minimum hold time between two consecutive SPF calculations. Default: 1000.

### <WAIT-TIME>

Time in milliseconds to set the maximum wait time between two consecutive SPF calculations. Default: 5000.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting non-default timer values for SPF throttling:

Setting default timer values for SPF throttling after configuring non-default values:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no timers throttling spf
```

# transit-delay

## Syntax

```
transit-delay <DELAY>
```

```
no transit-delay
```

## Description

Sets the time delay in Link state transmission for virtual links.

The `no` form of this command sets the delay in Link state transmission to the default of 1 second for virtual links.

## Command context

```
config-router-vlink
```

## Parameters

### <DELAY>

Specifies the time delay for the transit delay, in seconds. Range: 1 to 3600. Default: 1.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting OSPFv3 virtual links transit delay:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# transit-delay 30
```

Setting OSPFv3 virtual links transit delay to default:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# area 100 virtual-link 100.0.1.1
switch(config-router-vlink6)# no transit-delay
```

## trap-enable

### Syntax

```
trap-enable
```

```
no trap-enable
```

### Description

Enables the notification of the events to be sent as traps to the SNMP management stations for OSPFv3.

The `no` form of this command disables the notification of the events to be sent as traps to the SNMP management stations for OSPFv3.

### Command context

```
config-ospfv3
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling sending notification of events as traps:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# trap-enable
```

Disabling sending notification of events as traps:

```
switch(config)# router ospfv3 1
switch(config-ospfv3-1)# no trap-enable
```



## ping

### Syntax

```
ping <IPv4-address | hostname> [data-fill <pattern> | datagram-size <size> |
interval <time> | repetitions <number> | timeout <time> | tos <number> | ip-option
(include-timestamp | include-timestamp-and-address | record-route) | vrf <vrfname>]
```

### Description

Pings the specified IPv4 address or hostname with or without optional parameters.

### Command context

Operator (>) or Manager (#)

### Parameters

#### <IPv4-ADDR>

Selects the IPv4 address to ping.

#### <HOSTNAME>

Selects the hostname to ping. Range: 1-256 characters

#### data-fill <PATTERN>

Specifies the data pattern in hexadecimal digits to send. A maximum of 16 "pad" bytes can be specified to fill out the ICMP packet. Default: AB

#### datagram-size <SIZE>

Specifies the ping datagram size. Range: 0-65399, default: 100.

#### interval <TIME>

Specifies the interval between successive ping requests in seconds. Range: 1-60 seconds, default: 1 second.

#### repetitions <NUMBER>

Specifies the number of packets to send. Range: 1-10000 packets, default: Five packets.

#### timeout <TIME>

Specifies the ping timeout in seconds. Range: 1-60 seconds, default: 2 seconds.

#### tos <NUMBER>

Specifies the IP Type of Service to be used in Ping request. Range: 0-255

#### ip-option [include-timestamp | include-timestamp-and-address | record-route]

Specifies an IP option (record-route or timestamp option).

#### include-timestamp

Specifies the intermediate router time stamp.

**include-timestamp-and-address**

Specifies the intermediate router time stamp and IP address.

**record-route**

Specifies the intermediate router addresses.

**vrf <VRF-NAME>**

Specifies the virtual routing and forwarding (VRF) to use. When VRF option is not given, the default VRF is used.

**Authority**

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

**Examples**

Pinging an IPv4 address:

```
switch# ping 10.0.0.0
PING 10.0.0.0 (10.0.0.0) 100(128) bytes of data.
108 bytes from 10.0.0.0: icmp_seq=1 ttl=64 time=0.035 ms
108 bytes from 10.0.0.0: icmp_seq=2 ttl=64 time=0.034 ms
108 bytes from 10.0.0.0: icmp_seq=3 ttl=64 time=0.034 ms
108 bytes from 10.0.0.0: icmp_seq=4 ttl=64 time=0.034 ms
108 bytes from 10.0.0.0: icmp_seq=5 ttl=64 time=0.033 ms

--- 10.0.0.0 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.033/0.034/0.035/0.000 ms
```

Pinging the localhost:

```
switch# ping localhost
PING localhost (127.0.0.1) 100(128) bytes of data.
108 bytes from localhost (127.0.0.1): icmp_seq=1 ttl=64 time=0.060 ms
108 bytes from localhost (127.0.0.1): icmp_seq=2 ttl=64 time=0.035 ms
108 bytes from localhost (127.0.0.1): icmp_seq=3 ttl=64 time=0.043 ms
108 bytes from localhost (127.0.0.1): icmp_seq=4 ttl=64 time=0.041 ms
108 bytes from localhost (127.0.0.1): icmp_seq=5 ttl=64 time=0.034 ms

--- localhost ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3998ms
rtt min/avg/max/mdev = 0.034/0.042/0.060/0.011 ms
```

Pinging a server with a data pattern:

```
switch# ping 10.0.0.2 data-fill 1234123412341234acde123456789012
PATTERN: 0x1234123412341234acde123456789012
PING 10.0.0.2 (10.0.0.2) 100(128) bytes of data.
108 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.207 ms
108 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.187 ms
108 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.225 ms
108 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.197 ms
108 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.210 ms

--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.187/0.205/0.225/0.015 ms
```

Pinging a server with a datagram size:

```
switch# ping 10.0.0.0 datagram-size 200
PING 10.0.0.0 (10.0.0.0) 200(228) bytes of data.
208 bytes from 10.0.0.0: icmp_seq=1 ttl=64 time=0.202 ms
208 bytes from 10.0.0.0: icmp_seq=2 ttl=64 time=0.194 ms
208 bytes from 10.0.0.0: icmp_seq=3 ttl=64 time=0.201 ms
208 bytes from 10.0.0.0: icmp_seq=4 ttl=64 time=0.200 ms
208 bytes from 10.0.0.0: icmp_seq=5 ttl=64 time=0.186 ms

--- 10.0.0.0 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4000ms
rtt min/avg/max/mdev = 0.186/0.196/0.202/0.016 ms
```

Pinging a server with an interval specified:

```
switch# ping 9.0.0.2 interval 2
PING 9.0.0.2 (9.0.0.2) 100(128) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.199 ms
108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.192 ms
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.208 ms
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.182 ms
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.194 ms

--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 7999ms
rtt min/avg/max/mdev = 0.182/0.195/0.208/0.008 ms
```

Pinging a server with a specified number of packets to send:

```
switch# ping 9.0.0.2 repetitions 10
PING 9.0.0.2 (9.0.0.2) 100(128) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.213 ms
108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.204 ms
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.201 ms
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.184 ms
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.202 ms
108 bytes from 9.0.0.2: icmp_seq=6 ttl=64 time=0.184 ms
108 bytes from 9.0.0.2: icmp_seq=7 ttl=64 time=0.193 ms
108 bytes from 9.0.0.2: icmp_seq=8 ttl=64 time=0.196 ms
108 bytes from 9.0.0.2: icmp_seq=9 ttl=64 time=0.193 ms
108 bytes from 9.0.0.2: icmp_seq=10 ttl=64 time=0.200 ms

--- 9.0.0.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 8999ms
rtt min/avg/max/mdev = 0.184/0.197/0.213/0.008 ms
```

Pinging a server with a specified timeout:

```
switch# ping 9.0.0.2 timeout 3
PING 9.0.0.2 (9.0.0.2) 100(128) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.175 ms
108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.192 ms
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.190 ms
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.181 ms
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.197 ms

--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4000ms
rtt min/avg/max/mdev = 0.175/0.187/0.197/0.007 ms
```

Pinging a server with the specified IP Type of Service:

```
switch# ping 9.0.0.2 tos 2
PING 9.0.0.2 (9.0.0.2) 100(128) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.033 ms
```

```
108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.034 ms
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.031 ms
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.034 ms
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.031 ms

--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.031/0.032/0.034/0.006 ms
```

Pinging a server with the intermediate router time stamp:

```
switch# ping 9.0.0.2 ip-option include-timestamp
PING 9.0.0.2 (9.0.0.2) 100(168) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.031 ms
TS: 59909005 absolute
 0
 0
 0

108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.034 ms
TS: 59910005 absolute
 0
 0
 0

108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.038 ms
TS: 59911005 absolute
 0
 0
 0

108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.035 ms
TS: 59912005 absolute
 0
 0
 0

108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.037 ms
TS: 59913005 absolute
 0
 0
 0

--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.031/0.035/0.038/0.002 ms
```

Pinging a server with the intermediate router time stamp and address:

```
switch# ping 9.0.0.2 ip-option include-timestamp-and-address
PING 9.0.0.2 (9.0.0.2) 100(168) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.030 ms
TS: 9.0.0.2 60007355 absolute
 9.0.0.2 0
 9.0.0.2 0
 9.0.0.2 0

108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.037 ms
TS: 9.0.0.2 60008355 absolute
 9.0.0.2 0
 9.0.0.2 0
 9.0.0.2 0
```

```

108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.037 ms
TS: 9.0.0.2 60009355 absolute
 9.0.0.2 0
 9.0.0.2 0
 9.0.0.2 0

108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.038 ms
TS: 9.0.0.2 60010355 absolute
 9.0.0.2 0
 9.0.0.2 0
 9.0.0.2 0

108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.039 ms
TS: 9.0.0.2 60011355 absolute
 9.0.0.2 0
 9.0.0.2 0
 9.0.0.2 0

--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.030/0.036/0.039/0.005 ms

```

Pinging a server with the intermediate router address:

```

switch# ping 9.0.0.2 ip-option record-route
PING 9.0.0.2 (9.0.0.2) 100(168) bytes of data.
108 bytes from 9.0.0.2: icmp_seq=1 ttl=64 time=0.034 ms
RR: 9.0.0.2
 9.0.0.2
 9.0.0.2
 9.0.0.2

108 bytes from 9.0.0.2: icmp_seq=2 ttl=64 time=0.038 ms (same route)
108 bytes from 9.0.0.2: icmp_seq=3 ttl=64 time=0.036 ms (same route)
108 bytes from 9.0.0.2: icmp_seq=4 ttl=64 time=0.037 ms (same route)
108 bytes from 9.0.0.2: icmp_seq=5 ttl=64 time=0.035 ms (same route)

--- 9.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.034/0.036/0.038/0.001 ms

```

## ping6

### Syntax

```
ping6 {<IPv6-ADDR> | <HOSTNAME>} [data-fill <PATTERN> | datagram-size <SIZE> |
interval <TIME> | repetitions <NUMBER> | timeout <TIME> | vrf <VRF-NAME>]
```

### Description

Pings the specified IPv6 address or hostname with or without optional parameters.

### Command context

Operator (>) or Manager (#)

### Parameters

#### IPv6-ADDR

Selects the IPv6 address to ping.

## HOSTNAME

Selects the hostname to ping. Range: 1-256 characters

## data-fill <PATTERN>

Specifies the data pattern in hexadecimal digits to send. A maximum of 16 "pad" bytes can be specified to fill out the ICMP packet. Default: AB

## datagram-size <SIZE>

Specifies the ping datagram size. Range: 0-65399, default: 100.

## interval <TIME>

Specifies the interval between successive ping requests in seconds. Range: 1-60 seconds, default: 1 second.

## repetitions <NUMBER>

Specifies the number of packets to send. Range: 1-10000 packets, default: Five packets.

## timeout <TIME>

Specifies the ping timeout in seconds. Range: 1-60 seconds, default: 2 seconds.

## vrf <VRF-NAME>

Specifies the virtual routing and forwarding (VRF) to use. When this option is not provided, the default VRF is used.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Pinging an IPv6 address:

```
switch# ping6 2020::2
PING 2020::2(2020::2) 100 data bytes
108 bytes from 2020::2: icmp_seq=1 ttl=64 time=0.386 ms
108 bytes from 2020::2: icmp_seq=2 ttl=64 time=0.235 ms
108 bytes from 2020::2: icmp_seq=3 ttl=64 time=0.249 ms
108 bytes from 2020::2: icmp_seq=4 ttl=64 time=0.240 ms
108 bytes from 2020::2: icmp_seq=5 ttl=64 time=0.252 ms

--- 2020::2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4000ms
rtt min/avg/max/mdev = 0.235/0.272/0.386/0.059 ms
```

Pinging the localhost:

```
switch# ping6 localhost
PING localhost(localhost) 100 data bytes
108 bytes from localhost: icmp_seq=1 ttl=64 time=0.093 ms
108 bytes from localhost: icmp_seq=2 ttl=64 time=0.051 ms
108 bytes from localhost: icmp_seq=3 ttl=64 time=0.055 ms
108 bytes from localhost: icmp_seq=4 ttl=64 time=0.046 ms
108 bytes from localhost: icmp_seq=5 ttl=64 time=0.048 ms

--- localhost ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3998ms
rtt min/avg/max/mdev = 0.046/0.058/0.093/0.019 ms
```

Pinging a server with a data pattern:

```
switch# ping6 2020::2 data-fill ab
PATTERN: 0xab
PING 2020::2(2020::2) 100 data bytes
108 bytes from 2020::2: icmp_seq=1 ttl=64 time=0.038 ms
108 bytes from 2020::2: icmp_seq=2 ttl=64 time=0.074 ms
108 bytes from 2020::2: icmp_seq=3 ttl=64 time=0.076 ms
108 bytes from 2020::2: icmp_seq=4 ttl=64 time=0.075 ms
108 bytes from 2020::2: icmp_seq=5 ttl=64 time=0.077 ms

--- 2020::2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.038/0.068/0.077/0.015 ms
```

Pinging a server with a datagram size:

```
switch# ping6 2020::2 datagram-size 200
PING 2020::2(2020::2) 200 data bytes
208 bytes from 2020::2: icmp_seq=1 ttl=64 time=0.037 ms
208 bytes from 2020::2: icmp_seq=2 ttl=64 time=0.076 ms
208 bytes from 2020::2: icmp_seq=3 ttl=64 time=0.076 ms
208 bytes from 2020::2: icmp_seq=4 ttl=64 time=0.077 ms
208 bytes from 2020::2: icmp_seq=5 ttl=64 time=0.066 ms

--- 2020::2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3999ms
rtt min/avg/max/mdev = 0.037/0.066/0.077/0.016 ms
```

Pinging a server with an interval specified:

```
switch# ping6 2020::2 interval 5
PING 2020::2(2020::2) 100 data bytes
108 bytes from 2020::2: icmp_seq=1 ttl=64 time=0.043 ms
108 bytes from 2020::2: icmp_seq=2 ttl=64 time=0.075 ms
108 bytes from 2020::2: icmp_seq=3 ttl=64 time=0.074 ms
108 bytes from 2020::2: icmp_seq=4 ttl=64 time=0.075 ms
108 bytes from 2020::2: icmp_seq=5 ttl=64 time=0.075 ms

--- 2020::2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 19999ms
rtt min/avg/max/mdev = 0.043/0.068/0.075/0.014 ms
```

Pinging a server with a specified number of packets to send:

```
switch# ping6 2020::2 repetitions 6
PING 2020::2(2020::2) 100 data bytes
108 bytes from 2020::2: icmp_seq=1 ttl=64 time=0.039 ms
108 bytes from 2020::2: icmp_seq=2 ttl=64 time=0.070 ms
108 bytes from 2020::2: icmp_seq=3 ttl=64 time=0.076 ms
108 bytes from 2020::2: icmp_seq=4 ttl=64 time=0.076 ms
108 bytes from 2020::2: icmp_seq=5 ttl=64 time=0.071 ms
108 bytes from 2020::2: icmp_seq=6 ttl=64 time=0.078 ms

--- 2020::2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 4999ms
rtt min/avg/max/mdev = 0.039/0.068/0.078/0.015 ms
```



**NOTE:** On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

## crypto pki application

### Syntax

```
crypto pki application <APP-NAME> certificate <CERT-NAME>
```

```
no crypto pki application <APP-NAME> certificate
```

### Description

Associates a leaf certificate with a feature (application) on the switch. By default, all features are associated with the default, self-signed certificate `local-cert`. This certificate is created by the switch the first time it starts.

The `no` form of this command associates the specified feature with the default certificate.

### Command context

config

### Parameters

#### <APP-NAME>

Specifies the name of a feature on the switch:

- `est-client`: EST client
- `hsc`: Hardware switch controller.
- `https-server`: HTTPS server.
- `syslog-client`: Syslog client.

#### <CERT-NAME>

Specifies the name of an installed leaf certificate.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Associating the EST client with leaf certificate `leaf-cert1`:

```
switch(config)# crypto pki application est-client certificate leaf-cert1
```

Associating the syslog client with leaf certificate `leaf-cert`:

```
switch(config)# crypto pki application syslog-client certificate leaf-cert
```





**NOTE:** `syslog-client` communicates with syslog server over TLS.

You can associate a certificate with the `syslog-client` application by enrolling the certificate manually or through EST.

Setting the syslog client to use the default certificate:

```
switch(config)# no crypto pki application syslog-client certificate
```

Associating the HTTPS server with leaf certificate `leaf-cert2`:

```
switch(config)# crypto pki application https-server certificate leaf-cert2
```

## crypto pki certificate

### Syntax

```
crypto pki certificate <CERT-NAME>
```

```
no crypto pki certificate <CERT-NAME>
```

### Description

Creates a leaf certificate and changes to its context `config-cert-<CERT-NAME>`. If the specified leaf certificate exists, this command changes to its context.

The first time the switch starts it creates a self-signed, default leaf certificate called `local-cert`. This certificate is used by any switch application that does not have an associated leaf certificate.

The `no` form of this command deletes the specified leaf certificate. The default leaf certificate `local-cert` cannot be deleted.

### Command context

```
config
```

### Parameters

**<CERT-NAME>**

Specifies the name of a leaf certificate. Range: 1 to 32 alphanumeric characters (excluding ").

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating leaf certificate `leaf-cert`:

```
switch(config)# crypto pki certificate leaf-cert
switch(config-cert-leaf-cert)#
```

Deleting leaf certificate `leaf-cert`:

```
switch(config)# no crypto pki certificate leaf-cert
The leaf certificate has associated applications. Deleting the certificate
will make the applications use the default certificate local-cert.
Continue (y/n)? y
switch(config)#
```

## crypto pki ta-profile

### Syntax

```
crypto pki ta-profile <TA-NAME>
```

```
no crypto pki ta-profile <TA-NAME>
```

### Description

Creates a trust anchor (TA) profile and changes to the `config-ta-<TA-NAME>` context for the profile. Each TA profile stores the certificate for a trusted CA. Up to 64 profiles can be defined.

If the specified TA profile exists, this command changes to the `config-ta-<TA-NAME>` context for the profile.

The `no` form of this command removes the specified TA profile.



**NOTE:** When creating a new profile, If you exit the `config-ta-<TA-NAME>` context without importing the TA certificate, the profile is discarded.

### Command context

```
config
```

### Parameters

**<TA-NAME>**

Specifies the TA profile name. Range: 1 to 48 alphanumeric characters excluding ".".



**NOTE:** The TA profile name cannot end with `est-ta<nn>` where `<nn>` is 00 to 99. For example, `company-trust-anchor-est-ta01` is not allowed. This TA profile name suffix is reserved for TA profiles that are created for CA certificates from EST servers.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating the TA profile `root-cert`:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)#
```

Removing TA profile `root-cert`:

```
switch(config)# no crypto pki ta-profile root-cert
```

## enroll self-signed

### Syntax

```
enroll self-signed
```

### Description

Generates a key pair and generates a self-signed certificate with it.

The subject fields and key type of the current leaf certificate must be defined before running this command. If not, you are prompted to fill in the subject fields, and the key type is set to RSA 2048.

### Command context

```
config-cert-<CERT-NAME>
```

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Enrolling the leaf certificate **leaf-cert**:

```
switch(config-cert-leaf-cert)# enroll self-signed
You are enrolling a certificate with the following attributes:
Subject: C=US, ST=CA, L=Rocklin, OU=Site, O=Comp,
 CN=Leaf01
Key Type: RSA (2048)

Continue (y/n)? y
Self-signed certificate is created and enrolled successfully.

switch(config-cert-leaf-cert)#
```

## enroll terminal

### Syntax

```
enroll terminal
```

### Description

Generates a key pair and certificate signing request (CSR) for the current leaf certificate. Use the CSR to obtain a signed certificate from a certificate authority (CA), and then import the certificate onto the switch with the command `import terminal`.

The key type, and the certificate common name in the subject fields of the current leaf certificate must be completed before running this command.

### Command context

```
config-cert-<CERT-NAME>
```

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Enrolling the leaf certificate **leaf-cert**:

```
switch(config-cert-leaf-cert)# enroll terminal
You are enrolling a certificate with the following attributes:
Subject: C=US, ST=CA, L=Rocklin, OU=Site, O=Comp,
 CN=Leaf01
Key Type: RSA (2048)

Continue (y/n)? y

-----BEGIN CERTIFICATE REQUEST-----
```

```

MIIBOzCCAQwCAQAwYzEVMBMGA1UEAxMMcG9kMDEtODQwMC0xMQ4wDAYDVQQLEwV
nViYTEMMAoGA1UEChMDSFBFMRiWEAYDVQQHEw1Sb3Nldm1sbGUxCzAJBgNVBAGT
NBMQswCQYDVQQGEwJVUzCBnzANBghkqhkig9w0BAQEFAAOBjQAwgYkCgYEAtKcLS
...
GBAJ4L3lFFfWBEL+KAKpOGjZcVmw1BMqSKFtOFNF9nzmUmONmU3SKy6dzQ+6ynR
7Au22mf3lWDxzrtCC/dj5RtWJeJekxp2LCIK/3eRXUwbYveQDKcxH7j9ZB+BAp2
ace+2tA68F2vlgRCQ/hcQH0YmNuaq4Ne3w0dhm7HlUrx
-----END CERTIFICATE REQUEST-----
switch(config-cert-leaf-cert)#

```

## import (CA-signed leaf certificate)

### Syntax

```

import terminal ta-profile <TA-NAME> [password <PW>]
import <REMOTE-URL> ta-profile <TA-NAME> [password <PW>] [vrf <VRF-NAME>]
import <STORAGE-URL> ta-profile <TA-NAME> [password <PW>]

```

### Description

Imports a CA-signed leaf certificate and then validates the certificate against the specified TA profile. If the imported data includes a private key, the private key must match the leaf certificate being imported. If the imported data does not include a private key, the certificate must match a CSR that was previously generated with the command `enroll terminal` and must be signed by the CA whose root certificate is installed in the specified TA profile. The TA profile must exist and have a TA certificate configured.

### Parameters

#### terminal

Import the certificate by pasting PEM-format data at the console. Upon execution, the `config-cert-import` context is entered for certificate pasting. To complete certificate data entry press Control-D in your terminal program. Alternatively, the pasted certificate data can include at its end the delimiter `END_OF_CERTIFICATE` (after the `-----END CERTIFICATE-----` line), making entry of Control-D unnecessary.

#### ta-profile <TA-NAME>

Specifies the TA profile name. Range: 1 to 48 alphanumeric characters excluding ".".

#### <REMOTE-URL>

Specifies a certificate data file on a remote TFTP or SFTP server. The URL syntax is:

```
{tftp:// | sftp://<USER>@} {<IP>|<HOST>} [:<PORT>] [;blocksize=<SIZE>]/<FILE>
```

#### <STORAGE-URL>

Available on switch families that provide USB device file import capability, specifies a certificate data file on a USB storage device inserted in the switch USB port. The URL syntax is:

```
usb:/<FILE>
```

#### password <PW>

Specifies the plaintext password used to decrypt the private key in the imported certificate data. When this parameter is omitted, the password is prompted for as required. Range: 1 to 32 alphanumeric characters.

#### vrf <VRF-NAME>

Specifies the name of the VRF to use for the remote URL file transfer. The default is `mgmt`.

### Command context

```
config-cert-<CERT-NAME>
```

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

- The imported data must include all the intermediate CA certificates in the certificate chain leading to the certificate imported into the specified TA profile.
- This command cannot be used with the default certificate `local-cert`.
- The PEM data format is supported for all import sources. The PKCS#12 data format is supported for `<REMOTE-URL>` and `<STORAGE-URL>`.
- The PEM data must be delimited with these lines for the certificate data:

```
-----BEGIN CERTIFICATE-----
-----END CERTIFICATE-----
```

And the PEM data must be delimited with either of these line pairs for the private key data:

```
-----BEGIN PRIVATE KEY-----
-----END PRIVATE KEY-----

-----BEGIN ENCRYPTED PRIVATE KEY-----
-----END ENCRYPTED PRIVATE KEY-----
```

## Examples

Importing a leaf certificate from the console:

```
switch(config)# crypto pki certificate leaf-cert
switch(config-cert-leaf-cert1)# import terminal ta-profile root-cert
Paste the certificate in PEM format below, then hit enter and ctrl-D:
switch(config-cert-import)# -----BEGIN CERTIFICATE-----
switch(config-cert-import)# MIIFRDCCAyYgAwIBAgQP8nS2Vp15u0xXMdkDJzANBgkqhkiG9w0Bv
switch(config-cert-import)# MQswCQYDVQGEWJVUEOMAwGA1UCgwFXJ1YmDAgNBAMM1Jvb3QgQ0Ew
switch(config-cert-import)# HhcNMTkNDEwMjIwNT1WhcjIwMT0MjwNE1WjzQswQDVQGEWJVUzEL
...
switch(config-cert-import)# 1fIYZYQYla0AwFuPTTxBXHYwRxTPbUYU5umJfRPmE4VY8S9DQgcr
switch(config-cert-import)# 1NGNm3NG03GqPScs/TF9bVyFA5BOS5lmmkfRYK8D/kMTfRreSdxis
switch(config-cert-import)# YQ1ulNqShps=
switch(config-cert-import)# -----END CERTIFICATE-----
switch(config-cert-import)# -----BEGIN ENCRYPTED PRIVATE KEY-----
switch(config-cert-import)# MIIFDjBABgkqhkiG9wBBQ0wMzAbBgqkw0QwwDQIPJMN7sVGwCAgga
switch(config-cert-import)# MBQGCCqGS1b3DQMHAit+2qadNAASCgLYJ4Am3EfhH5p51Ggr86VqS
switch(config-cert-import)# IJ6L/UhEtH523nUkdV6gvAgoYaD83PswToAGv5VS8OMFTPttrn5/K
...
switch(config-cert-import)# OgSecqZsG6arbx0ESaYBir1c/6rPspcjbx283iD1MWOpesS2aEmOX
switch(config-cert-import)# iKnXnUMpVPfLc74ty2S41DtH0X9gf6aaljStg+7cND9XfGtjaV2+/
switch(config-cert-import)# cb4=
switch(config-cert-import)# -----END ENCRYPTED PRIVATE KEY-----
switch(config-cert-import)#
Enter import password: *****
Leaf certificate is validated with root-cert and imported successfully.
switch(config-cert-leaf-cert)#
```

Importing a leaf certificate from a remote file:

```
switch(config)# crypto pki certificate leaf-cert2
switch(config-cert-leaf-cert2)# import tftp://1.1.1.2/c2.p12 ta-profile root-cert
% Total % Received % Xferd Average Speed Time Time Time Current
 Dload Upload Total Spent Left Speed
100 3722 100 3722 0 0 391k 0 --:--:-- --:--:-- --:--:-- 391k
```

```

100 3722 100 3722 0 0 376k 0 --:--:-- --:--:-- --:--:-- 376k
Enter import password: *****
Leaf certificate is validated with root-cert and imported successfully.
switch(config-cert-leaf-cert2)#

```

## import (self-signed leaf certificate)

### Syntax

```

import terminal self-signed [password <PW>]
import <REMOTE-URL> self-signed [password <PW>] [vrf <VRF-NAME>]
import <STORAGE-URL> self-signed [password <PW>]

```

### Description

Imports a self-signed leaf certificate including its matching private key.

### Parameters

#### terminal

Import the certificate by pasting PEM-format data at the console. Upon execution, the `config-cert-import` context is entered for certificate pasting. To complete certificate data entry press Control-D in your terminal program. Alternatively, the pasted certificate data can include at its end the delimiter `END_OF_CERTIFICATE` (after the `-----END CERTIFICATE-----` line), making entry of Control-D unnecessary.

#### <REMOTE-URL>

Specifies a certificate data file on a remote TFTP or SFTP server. The URL syntax is:

```
{tftp:// | sftp://<USER>@} {<IP>|<HOST>} [:<PORT>] [;blocksize=<SIZE>]/<FILE>
```

#### <STORAGE-URL>

Available on switch families that provide USB device file import capability, specifies a certificate data file on a USB storage device inserted in the switch USB port. The URL syntax is:

```
usb:/<FILE>
```

#### password <PW>

Specifies the plaintext password used to decrypt the private key in the imported certificate data. When this parameter is omitted, the password is prompted for as required. Range: 1 to 32 alphanumeric characters.

#### vrf <VRF-NAME>

Specifies the name of the VRF to use for the remote URL file transfer. The default is `mgmt`.

### Command context

```
config-cert-<CERT-NAME>
```

### Authority

Administrators or local user group members with execution rights for this command.

## Usage

- This command cannot be used with the default certificate `local-cert`.
- The PEM data format is supported for all import sources. The PKCS#12 data format is supported for `<REMOTE-URL>` and `<STORAGE-URL>`.
- The PEM data must be delimited with these lines for the certificate data:

```
-----BEGIN CERTIFICATE-----
-----END CERTIFICATE-----
```

And the PEM data must be delimited with either of these line pairs for the private key data:

```
-----BEGIN PRIVATE KEY-----
-----END PRIVATE KEY-----

-----BEGIN ENCRYPTED PRIVATE KEY-----
-----END ENCRYPTED PRIVATE KEY-----
```

## Example

Importing a self-signed leaf certificate from the console:

```
switch(config)# crypto pki certificate ss-leaf-cert
switch(config-cert-ss-leaf-cert)# import terminal self-signed
Paste the certificate in PEM format below, then hit enter and ctrl-D:
switch(config-cert-import)# -----BEGIN CERTIFICATE-----
switch(config-cert-import)# MIID2TCCAsGgAwIBAgIJAKcrqokm6p9GMA0GCSqGSIB3DQEBCwUAM
switch(config-cert-import)# tDCCA5ygAwIBAgICEAEwDQYJKoZIhvcNAQELBQAwYgxCzABAYTA1
switch(config-cert-import)# VQQGEwJVUzELMAkGA1UECAwCQ0ExDTALBgNVBACMBFJvc2UxDDAKB
...
switch(config-cert-import)# +fWQLxhp+jKJGZGOZz/FENt2uSfZHxlXiu8n3g+EgqExenY1pBRJr
switch(config-cert-import)# VuEEoNb/YfkPXHHva4Zfx223q+f694wlVsHkENSzqr2goHpa2fOzq
switch(config-cert-import)# alewwdmVqCES+x8bvhf3C/6IB6ePkEsnMlHNTeM=
switch(config-cert-import)# -----END CERTIFICATE-----
switch(config-cert-import)# -----BEGIN ENCRYPTED PRIVATE KEY-----
switch(config-cert-import)# MIIFDjBABgkqhkiG9w0BBQ0wMzAbBgkqhkiG9w0BBQwwDgQIt8Ni3
switch(config-cert-import)# MBQGCCqGSIB3DQMHBAiBHrejkcdpdASCBMjVxrrYYPNt3V1abr9k8
switch(config-cert-import)# 5GE0U99awh9ys4360WR95xOFGThvjkTyRWG511nGwVeLZs/7TPXWI
...
switch(config-cert-import)# hzc5ZT/w2F08icRI5mFbGoTAAw9IIWMOXGweaWQJDyKGrhg89GrnV
switch(config-cert-import)# M2UuP/tYuuO328QcenKZEJmZKCbx78oFRR+pgma4oeMaFTIyXE6Pr
switch(config-cert-import)# GAdCK8tkDiJ9DKbqdm5W0/nTJfqwUqlf127dNrBAodsHdwr3UR99H
switch(config-cert-import)# SPo=
switch(config-cert-import)# -----END ENCRYPTED PRIVATE KEY-----
switch(config-cert-import)#
Enter import password: *****
Leaf certificate is validated as self-signed certificate and imported successfully.
switch(config-cert-ss-leaf-cert)#
```

Importing a leaf certificate from a remote file:

```
switch(config)# crypto pki certificate ss-leaf-cert2
switch(config-cert-ss-leaf-cert2)# import tftp://1.1.1.2/ss2.p12 self-signed
% Total % Received % Xferd Average Speed Time Time Time Current
 Dload Upload Total Spent Left Speed
100 3230 100 3230 0 0 875k 0 --:--:-- --:--:-- --:--:-- 875k
100 3230 100 3230 0 0 831k 0 --:--:-- --:--:-- --:--:-- 831k
Enter import password: *****
Leaf certificate is validated as self-signed certificate and imported successfully.
switch(config-cert-ss-leaf-cert2)#
```

## key-type

### Syntax

```
key-type {rsa [key-size <K-SIZE>] | ecdsa [curve-size <C-SIZE>]}
```

### Description

Sets the key type and key size for the current leaf certificate. The key type of the default certificate `local-cert` cannot be changed.

### Command context

```
config-cert-<CERT-NAME>
```

### Parameters

#### rsa

Specifies the key type as RSA.

#### key-size <K-SIZE>

Specifies the RSA key size in bits. Supported values: 2048, 3072, 4096. Default: 2048

#### ecdsa

Specifies the key type as ECDSA.

#### curve-size <C-SIZE>

Specifies the ECDSA elliptic curve size in bits. Supported values: 256, 348, 521. Default: 256

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting RSA encryption on the leaf certificate `leaf-cert`:

```
switch(config)# crypto pki certificate leaf-cert
switch(config-cert-leaf-cert)# key-type rsa key-size 3072
```

Setting ECDSA encryption on the leaf certificate `leaf-cert`:

```
switch(config)# crypto pki certificate leaf-cert
switch(config-cert-leaf-cert)# key-type ecdsa curve-size 521
```

## ocsp disable-nonce

### Syntax

```
ocsp disable-nonce
```

```
no ocsp disable-nonce
```

### Description

Configures exclusion of the nonce from OCSP requests. A nonce is a unique identifier that an OCSP client inserts in an OCSP request and expects the OCSP responder to include it in the corresponding OCSP response. The nonce mechanism helps prevent replay attacks in which a malicious player attempts to masquerade as the OCSP responder. Although the nonce is included by default, it can be excluded. Some OCSP responders choose to not support the use of the nonce due to performance considerations.



The `no` form of this command re-enables nonce inclusion in OCSF requests.

### Command context

```
config-ta-<TA-NAME>
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Disable inclusion of the nonce in OCSF requests for TA profile `root-cert`:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)# ocsp disable-nonce
```

Enable inclusion of the nonce in OCSF requests for TA profile `root-cert`:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)# no ocsp disable-nonce
```

## ocsp enforcement-level

### Syntax

```
ocsp enforcement-level {strict | optional}
```

```
no enforcement-level
```

### Description

Sets either strict or reduced enforcement of the OCSF check of certificates. Strict enforcement is enabled by default.

The `no` form of this command resets enforcement to its default of `strict`.

### Command context

```
config-ta-<TA-NAME>
```

### Parameters

#### **strict**

Sets strict OCSF checking of certificates. The certificate is accepted only if all possible checking (including validation failures, software system errors, configuration errors, transactional errors) is successful.

#### **optional**

Sets reduced OCSF checking of certificates. The certificate is accepted unless one or more of these validation errors occur:

- Response signature invalid.
- Nonce in response mismatch.
- Certificate revoked, but only when revocation checking is possible. if revocation check is not possible, the certificate is still accepted if there are no other validation errors.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting reduced OCSP checking of certificates:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)# ocsp enforcement-level optional
```

Setting strict OCSP checking of certificates:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)# ocsp enforcement-level strict
```

## ocsp url

### Syntax

```
ocsp url {primary | secondary} <URL>
```

```
no ocsp url {primary | secondary}
```

### Description

Configures the OCSP responder URLs that the current TA profile uses to verify the revocation status of an X.509 digital certificate. These URLs override the OCSP responder URL contained within the peer certificate being verified (as well as URLs defined in any intermediate CAs in the chain of trust).

If no OCSP responder URLs are defined for a TA profile (default setting), then the OCSP responder URL in the peer certificate is used for revocation status checking. (The OCSP responder URL is contained in a certificate's Authority Information Access field, which is an X.509 v3 certificate extension.)

The **no** form of this command deletes the specified OCSP responder URL (primary or secondary) from the current TA profile.

### Command context

```
config-ta-<TA-NAME>
```

### Parameters

```
{primary | secondary} <URL>
```

Specify the HTTP URL of the primary or secondary OCSP responder using either a fully qualified domain name or IPv4 address.

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Defining the primary OCSP URL for the TA profile **root-cert**:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)# revocation-check ocsp
switch(config-ta-root-cert)# ocsp url primary http://ocsp-server.site.com
```

Removing the primary OCSP URL from the TA profile **root-cert**:

```
switch(config)# crypto pki ta-profile oot-cert
switch(config-ta-root-cert)# revocation-check ocsp
switch(config-ta-root-cert)# no ocsp url primary
```

## ocsp vrf

### Syntax

```
ocsp vrf <VRF-NAME>
```

```
no ocsp vrf
```

### Description

Sets the VRF that the switch uses to communicate with OCSF responders for OCSF checking. VRF `mgmt` is used by default.

The `no` form of this command resets the VRF to its default `mgmt`.

### Command context

```
config-ta-<TA-NAME>
```

### Parameters

**<VRF-NAME>**

Specifies the name of the VRF the switch uses to communicate with OCSF responders. Default: `mgmt`.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Reverting the OCSF responder VRF to its default:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)# no ocsp vrf
```

## revocation-check ocsp

### Syntax

```
revocation-check ocsp
```

```
no revocation-check
```

### Description

Enables certificate revocation checking for the current profile using the online certificate status protocol (OCSF).

The `no` form of this command disables certificate revocation checking for the current profile.

### Command context

```
config-ta-<TA-NAME>
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling revocation checking for the TA profile `root-cert`:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)# revocation-check ocs
```

Disabling revocation checking for the TA profile **root-cert**:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)# no revocation-check
```

## show crypto pki application

### Syntax

```
show crypto pki application
```

### Description

Shows certificate information for all features (applications) using leaf certificates that are managed by PKI.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Showing certificate information for all features (applications) using leaf certificates:

```
switch# show crypto pki application1
```

| Associated Applications | Certificate Name | Cert Status                      |
|-------------------------|------------------|----------------------------------|
| https-server            |                  | not configured, using local-cert |
| syslog-client           | local-cert       | valid                            |
| hsc                     | xhscert          | invalid, using local-cert        |

## show crypto pki certificate

### Syntax

```
show crypto pki certificate [<CERT-NAME> [plaintext | pem]]
```

### Description

Shows a list of all configured leaf certificates, or detailed information for a specific leaf certificate.

Possible values for Cert Status are: CSR pending, expired, expires soon, installed, malformed, not yet known.

Possible values for EST Status are: enroll failed, enroll pending, enroll retrying, enroll success, n/a (certificate is not EST-enrolled), reenroll failed, reenroll pending, reenroll retrying.

### Command context

Manager (#)

## Parameters

### <CERT-NAME>

Specifies the leaf certificate name. Range: 1 to 32 alphanumeric characters excluding ".

### plaintext

Shows certificate information in plain text.

### pem

Shows certificate information in PEM format.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing a list of all configured leaf certificates:

```
switch# show crypto pki certificate
```

| Certificate Name | Cert Status | EST Status      | Associated Applications  |
|------------------|-------------|-----------------|--------------------------|
| device-identity  | installed   | n/a             | none                     |
| pod01-99-1       | installed   | n/a             | https-server, est-client |
| syslog-1         | CSR pending | enroll retrying | syslog-client            |
| leaf-cert1       | installed   | enroll success  | none                     |
| leaf-cert2       | CSR pending | enroll failed   | none                     |

Showing detailed information (in plaintext format) for leaf certificate **pod01-99-1**:

```
switch# show crypto pki certificate pod01-99-1 plaintext
```

```
Certificate Name: pod01-99-1
Associated Applications:
 https-server, est-client
Certificate Status: installed
EST Status: n/a
Certificate Type: regular
Intermediates:
 Subject: C = US, ST = CA, O = Company, OU = Lab-IT, CN = DeviceCA
 Issuer: C = US, ST = CA, O = Company, OU = Lab-IT, CN = Lab-CA
 Serial Number: 0x02
 Subject: C = US, ST = CA, O = Company, OU = Lab-IT, CN = Lab-CA
 Issuer: C = US, ST = CA, O = Company, OU = Lab-IT, CN = Lab-Root
 Serial Number: 0x01
Certificate:
 Data:
 Version: 1 (0x0)
 Serial Number: 14529416756121781768 (0xc9a2db8f3e3f4608)
 Signature Algorithm: sha256WithRSAEncryption
 Issuer: C=US, ST=CA, OU=Lab-IT, O=Company, CN=DeviceCA
 Validity
 Not Before: Jan 12 23:36:57 2018 GMT
 Not After : Nov 1 23:36:57 2020 GMT
 Subject: C=US, ST=CA, OU=Lab-IT, O=Company, CN=pod01-99-1
 Subject Public Key Info:
 Public Key Algorithm: rsaEncryption
 Public-Key: (2048 bit)
 Modulus:
 00:a0:cd:ef:1b:f9:b8:bd:39:fc:7a:0e:00:17:ff:
 2b:72:d8:4e:d4:df:49:36:ca:3a:f9:05:05:d7:e3:
 d1:97:29:71:e6:33:b8:bb:8e:f0:ee:a6:e4:4a:f8:
```

```

...
fe:dd:d9:a0:af:59:47:25:b4:34:06:af:03:1d:33:
30:c3:85:fe:5c:e7:19:7f:ff:3a:b2:21:b8:e8:ed:
83:09
Exponent: 65537 (0x10001)
Signature Algorithm: sha256WithRSAEncryption
39:f6:03:86:03:d9:05:61:39:25:5f:0d:75:cc:05:ae:04:7e:
4c:a3:13:0b:f0:1e:af:68:0e:40:9f:ed:48:b6:5e:56:8c:53:
46:5b:c9:a4:e0:b0:bc:31:4b:a7:5d:0a:ed:7c:9c:f6:bf:1e:
...
39:f5:26:58:68:e2:13:ec:94:ac:60:8e:4b:b0:ba:45:cf:d6:
6a:4b:9f:7d:ae:3f:e5:2e:81:fe:ac:b3:65:44:35:47:a5:2f:
89:e7:58:a0

```

Showing detailed information (in PEM format) for leaf certificate **leaf-cert1** with a status of CSR pending:

```

switch# show crypto pki certificate leaf-cert1 pem

Certificate Name: leaf-cert1

Associated Applications:
 syslog-client
Certificate Status: CSR pending
EST Status: enroll retrying
Certificate Type: regular
-----BEGIN CERTIFICATE REQUEST-----

MIICtTCCAzoCAQAwcDEWMBQGA1UEAxMNc3lzbG9nLTg0MBYGA1UECjMPQ
XJlYmEtUm9zZXZpbGx1MQ4wDAYDVQQKEyTESMBAGA1EBxMJU9zZXZpbG
xlMQswCQYDVQQIEwJDQTELMAGA1UEBhMCVVMwggEiMSIb3DQEBAQUAA4I
...
cw2ytN6Idgh81k59x6DH7V/eORaKd5lq+oO7nkr6+QBf5L3f5Kb+TOFio
lei+EdCHMxxc07MK0n3dkziSW25HFUGsyEXVMK+BiD3zbKDoUe6XVhvqI
mamXyghigLYDcbsn6WVw==
-----END CERTIFICATE REQUEST-----

```

## show crypto pki ta-profile

### Syntax

```
show crypto pki ta-profile [<TA-NAME>]
```

### Description

Shows a list of all configured TA profiles, or detailed information for a specific profile.



**NOTE:** This command shows information for both directly-configured TA profiles and TA profiles that were dynamically downloaded from EST servers.

### Command context

Manager (#)

### Parameters

**<TA-NAME>**

Specifies the TA profile name. Range: 1 to 48 alphanumeric characters excluding ".".

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing a list of all configured TA profiles:

```
switch# show crypto pki ta-profile
```

| Profile Name | TA Certificate    | Revocation Check |
|--------------|-------------------|------------------|
| -----        | -----             | -----            |
| BASE_CA      | Installed,valid   | disabled         |
| BASE02_CA    | Installed,expired | disabled         |
| root-cert    | Installed,valid   | OCSP             |
| ROOT-A_CA    | Not Installed     | OCSP             |
| EST-Service1 | Installed,valid   | None             |
| EST-Service2 | Installed,valid   | None             |

Showing detailed information for TA profile **root-cert**:

```
switch# show crypto pki ta-profile root-cert
```

```
TA Profile Name : root-cert
Revocation Check : OCSP
 OSCP Primary URL : http://ocsp1.domain.com
 OSCP Secondary URL : Not Configured
 OSCP Disable-nonce : false
 OSCP Enforcement Level: strict
 OSCP VRF : mgmt
TA Certificate: Installed and valid
Version: 3 (0x2)
Serial Number:
 74:e6:6d:22:3f:52:cc:94:43:41:ab:66:a8:8d:47:b1
Signature Algorithm: sha1withRSAEncryption
Issuer: OU=DeviceTrust, OU=Operations, O=Site, C=US,
 CN=Site Trusted Computing Root CA 1.0
Validity
 Not Before: Sep 14 03:12:06 2007 GMT
 Not After : Sep 14 03:21:14 2032 GMT
Subject: OU=DeviceTrust, OU=Operations, O=Site, C=US,
 CN=Site Trusted Computing Root CA 1.0
Subject Public Key Info:
 Public Key Algorithm: rsaEncryption
 RSA Public Key: (2048 bit)
 Modulus (2048 bit):
 30:0d:06:09:2a:86:48:86:f7:0d:01:01:01:05:33:
 03:82:01:0f:00:30:82:01:3a:02:82:01:01:00:ac:
 3d:60:3a:2e:ca:a4:34:db:5c:3b:6b:07:df:73:62:
 ...
 20:c8:df:63:14:5a:e8:d3:ea:83:d8:47:a3:b5:2e:
 bb:64:51:f0:be:13:b6:91:e4:32:45:58:5e:1f:0d:
 02:03:01:00:01
 Exponent: 65537 (0x10001)
X509v3 extensions:
 X509v3 Key Usage:
 Digital Signature, Certificate Signing, CRL Signing
 X509v3 Basic Constraints:
 CA:TRUE, pathlen:4
 X509v3 Subject Key Identifier:
 eb:d7:ec:db:8a:cb:f2:51:d5:06:e1:42:7b:39:a7:d0:1e:31:6e:bf

Signature Algorithm: sha1withRSAEncryption
1c:90:f3:a4:f0:0d:e2:e3:e9:ae:01:e1:7d:a7:13:e2:cc:0b:
17:31:26:92:a2:5d:1d:19:60:54:03:13:9b:e1:73:6c:e4:b3:
01:4f:4e:ae:61:bd:ae:b6:12:d3:ab:08:ae:8c:47:92:d7:0d:
...
ca:cf:11:78:55:6d:06:49:fa:d4:8d:f3:ef:7f:79:38:35:5d:
```

```
16:5a:57:7f:a8:dc:b0:f8:a2:04:0d:17:0b:bb:58:32:30:e0:
2d:a8:37:a2
```

## ta-certificate

### Syntax

```
ta-certificate { [import [terminal]] | import {<REMOTE-URL> | <STORAGE-URL>} }
```

### Description

Imports a CA certificate for use in the current TA profile. The certificate must be in PEM format. The PEM data must be delimited with these lines:

```
-----BEGIN CERTIFICATE-----
-----END CERTIFICATE-----
```



**NOTE:** Only the first certificate in the PEM data is imported. Any additional certificates are ignored.

### Command context

```
config-ta-<TA-NAME>
```

### Parameters

**[import [terminal]]**

Import the certificate by pasting at the console (the default). This form of importing is selected whether `ta-certificate` is entered without parameters or if only `import` is entered or if `import terminal` is entered. Upon execution, the `config-ta-cert` context is entered for certificate pasting. To complete certificate data entry press Control-D in your terminal program. Alternatively, the pasted certificate data can include at its end the delimiter `END_OF_CERTIFICATE` (after the `-----END CERTIFICATE-----` line), making entry of Control-D unnecessary..

**import <REMOTE-URL>**

Import the certificate from a file on a remote TFTP or SFTP server. The URL syntax is:

```
{tftp:// | sftp://<USER>@} {<IP>|<HOST>} [:<PORT>] [;blocksize=<SIZE>]/<FILE>
```

**import <STORAGE-URL>**

Available on switch families that provide USB device file import capability, import the certificate from a file on a USB storage device inserted in the switch USB port. The URL syntax is:

```
usb:/<FILE>
```

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Importing a certificate into the TA profile `root-cert` by pasting PEM-format certificate data at the console:

```
switch(config)# crypto pki ta-profile root-cert
switch(config-ta-root-cert)# ta-certificate import terminal
Paste the certificate in PEM format below, then hit enter and ctrl-D:
switch(config-ta-cert)# -----BEGIN CERTIFICATE-----
switch(config-ta-cert)# MIIDuTCCAqECCQCuoxeJ2ZNYcjANBgkqhkiG9w0BAQsFADCBqzELMAEBh
switch(config-ta-cert)# VVMxEzARBgNVBAgMCkNhbGlmb3JuaWExEDAOBgNVBACMB1JvY2tsDAKBg
switch(config-ta-cert)# BAoMA0hQTjEVMBMGA1UECwwMSFBOUm9zZXZpbGx1MSowKAYDVQocG5zdz
...
```



```

switch(config-ta-cert)# x3Wff3dFZ8o9sd5LVAHneH/ztb9MP34z+le1V346r12L2kpxmTOVJVyTO
switch(config-ta-cert)# BIzD/ST/HaWI+0S+S80rm93PSscEbb9GWk7vshh5EnW/moehBKcE40lzy
switch(config-ta-cert)# 3LvMLZcSSSe5J2Ca2XIhfDme8UaNZ7syGYMsAW0nG7yYHWkEOQu9s
switch(config-ta-cert)# -----END CERTIFICATE-----
switch(config-ta-cert)#
The certificate you are importing has the following attributes:
Issuer: C=US, ST=CA, L=Rocklin, O=Company, OU=Site,
 CN=site.com/emailAddress=test.ca@site.com
Subject: C=US, ST=CA, L=Rocklin, O=Company, OU=Site,
 CN=9000/emailAddress=test.ca@site.com
Serial Number: 12121221634631568498 (0xaea51217d5945772)

TA certificate import is allowed only once for a TA profile
Do you want to accept this certificate (y/n)? y
TA certificate accepted.
switch(config-ta-root-cert)#

```

Importing a certificate into the TA profile **root-cert2** from file **rcert2-data** on the USB device:

```

switch(config)# crypto pki ta-profile root-cert2
switch(config-ta-root-cert2)# ta-certificate import usb:/rcert2-data
The certificate you are importing has the following attributes:
Issuer: C=US, ST=California, L=Rocklin, O=Company, OU=Site,
 CN=site.com/emailAddress=test.ca@site.com
Subject: C=US, ST=California, L=Rocklin, O=Company, OU=Site,
 CN=9000/emailAddress=test.ca@site.com
Serial Number: 12121221634631568498 (0xaea51217d5945772)

TA certificate import is allowed only once for a TA profile
Do you want to accept this certificate (y/n)? y
TA certificate accepted.
switch(config-ta-root-cert2)#

```

## subject

### Syntax

```

subject [common-name <COMMON-NAME>] [country <COUNTRY>] [locality <LOCALITY>]
 [org <ORG-NAME>] [org-unit <ORG-UNIT>] [state <STATE>]

```

### Description

Sets the subject fields for the current leaf certificate. If the `common-name` parameter is not specified, then you are prompted to define a value for each field. If a configured value exists for any field, it is presented as the default.

The subject fields of the default certificate `local-cert` cannot be changed.

### Command context

```

config-cert-<CERT-NAME>

```

### Parameters

**common-name** <COMMON-NAME>

Specifies the common name.

**country** <COUNTRY>

Specifies the region.

**locality** <LOCALITY>

Specifies the locality.

**org** <ORG-NAME>

Specifies the organization.

**org-unit** <ORG-UNIT>

Specifies the organizational unit.

**state** <STATE>

Specifies the state.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting subject fields for the leaf certificate **leaf-cert**:

```
switch(config-cert-leaf-cert)# subject common-name Leaf01 country US
locality CA org Company org-unit Site state CA
```

Setting subject fields for the leaf certificate **leaf-cert** interactively:

```
switch(config-cert-leaf-cert)# subject
Do you want to use the switch serial number as the common name (y/n)? n
Enter Common Name : Leaf01
Enter Org Unit : Site
Enter Org Name : Company
Enter Locality : Rocklin
Enter State : CA
Enter Country : US
switch(config-cert-leaf-cert)#
```



**NOTE:** On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

## arbitrary-label

### Syntax

```
arbitrary-label <LABEL>
```

```
no arbitrary-label
```

### Description

Within the EST profile context, configures the generic optional label (also known as arbitrary label) to be concatenated to the EST server URL that is configured with the `url` command. There is no arbitrary label configured by default. Any existing arbitrary label is replaced by this command. The use of arbitrary labels is optional.

RFC 7030 allows the use of arbitrary labels so that one EST server may serve multiple CAs with the same server URL that gets concatenated with different arbitrary labels. The same label is used for every request made under a particular EST profile.

Some EST schemes use arbitrary labels in a more sophisticated way, defining different labels for different types of requests under the same EST profile. For example, the CA certificate request could use the generic label (configured with this `arbitrary-label` command), the certificate enrollment request could use the enrollment label (configured with the `arbitrary-label-enrollment` command), and the re-enrollment request could use the re-enrollment label (configured with the `arbitrary-label-reenrollment` command). Note that only one label of each of the three available types can be configured in any EST profile.

The `no` form of this command removes the generic arbitrary label.

### Command context

```
config-est-<EST-NAME>
```

### Parameters

**<LABEL>**

Specifies the generic arbitrary label. Range: Up to 64 characters.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring the URL and generic arbitrary label. Note that with the URL and arbitrary label configured in this example, the final URL the switch uses to request CA certificates from the EST server is `https://est-service999.com/.well-known/est/rsa2048/cacerts`.

```
switch(config)# crypto pki est-profile EST-service1
switch(config)# url https://est-service999.com/.well-known/est
switch(config-est-EST-service1)# arbitrary-label rsa2048
```

Removing the generic arbitrary label:

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# no arbitrary-label
```

## arbitrary-label-enrollment

### Syntax

```
arbitrary-label-enrollment <LABEL>
```

```
no arbitrary-label-enrollment
```

### Description

Within the EST profile context, configures the arbitrary enrollment label to be concatenated to the EST server URL that is configured with the `url` command. This label is specific to the enrollment operation. There is no arbitrary enrollment label configured by default. Any existing arbitrary enrollment label is replaced by this command. The use of arbitrary enrollment labels is optional.

When the enrollment label is not configured, the generic arbitrary label (created with the `arbitrary-label` command) is used (if configured) for enrollment.

RFC 7030 allows the use of arbitrary labels so that one EST server may serve multiple CAs with the same server URL that gets concatenated with different arbitrary labels. The same label is used for every request made under a particular EST profile.

Some EST schemes use arbitrary labels in a more sophisticated way, defining different labels for different types of requests under the same EST profile. For example, the CA certificate request could use the generic label (configured with the `arbitrary-label` command), the certificate enrollment request could use the enrollment label (configured with this `arbitrary-label-enrollment` command), and the re-enrollment request could use the re-enrollment label (configured with the `arbitrary-label-reenrollment` command). Note that only one label of each of the three available types can be configured in any EST profile.

The `no` form of this command removes the arbitrary enrollment label.

### Command context

```
config-est-<EST-NAME>
```

### Parameters

<LABEL>

Specifies the arbitrary enrollment label. Range: Up to 64 characters.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring the arbitrary enrollment label:

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# arbitrary-label-enrollment ipsec-v7
```

Removing the arbitrary enrollment label :

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# no arbitrary-label-enrollment
```

## arbitrary-label-reenrollment

### Syntax

```
arbitrary-label-reenrollment <LABEL>
```

```
no arbitrary-label-reenrollment
```

### Description

Within the EST profile context, configures the arbitrary re-enrollment label to be concatenated to the EST server URL that is configured with the `url` command. This label is specific to the re-enrollment operation. There is no arbitrary re-enrollment label configured by default. Any existing arbitrary re-enrollment label is replaced by this command. The use of arbitrary re-enrollment labels is optional.

When the re-enrollment label is not configured, the generic arbitrary label (created with the `arbitrary-label` command) is used (if configured) for re-enrollment.

RFC 7030 allows the use of arbitrary labels so that one EST server may serve multiple CAs with the same server URL that gets concatenated with different arbitrary labels. The same label is used for every request made under a particular EST profile.

Some EST schemes use arbitrary labels in a more sophisticated way, defining different labels for different types of requests under the same EST profile. For example, the CA certificate request could use the generic label (configured with the `arbitrary-label` command), the certificate enrollment request could use the enrollment label (configured with the `arbitrary-label-enrollment` command), and the re-enrollment request could use the re-enrollment label (configured with this `arbitrary-label-reenrollment` command). Note that only one label of each of the three available types can be configured in any EST profile.

The `no` form of this command removes the arbitrary re-enrollment label.

### Command context

```
config-est-<EST-NAME>
```

### Parameters

<LABEL>

Specifies the arbitrary re-enrollment label. Range: Up to 64 characters.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring the arbitrary re-enrollment label:

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# arbitrary-label-reenrollment ipsec-v7
```

Removing the arbitrary re-enrollment label :

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# no arbitrary-label-reenrollment
```

## crypto pki est-profile

### Syntax

```
crypto pki est-profile <EST-NAME>
```

```
no crypto pki est-profile <EST-NAME>
```

### Description

Creates a certificate Enrollment over Secure Transport (EST) profile and changes to the `config-est-<EST-NAME>` context for the profile. Each EST profile stores information about the EST service, including EST server URL. Up to 16 profiles can be created.

If the specified EST profile exists, this command changes to the `config-est-<EST-NAME>` context for the profile.

The `no` form of this command deletes the specified EST profile. It also deletes the TA profiles whose CA certificates were downloaded from the corresponding EST server, and the leaf certificates that were enrolled using this EST profile.



**NOTE:** The deletion of the related TA profiles and enrolled certificates is permanent. If the EST profile is in the startup configuration and the EST profile is deleted but this deletion is not updated in the startup configuration before a switch reboot, the EST profile will still exist after the reboot but the related TA profiles and enrolled certificates will not exist.

### Command context

```
config
```

### Parameters

**<EST-NAME>**

Specifies the EST profile name. Range: Up to 32 alphanumeric characters (excluding ").

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating EST profile **EST-Service1**:

```
switch(config)# crypto pki est-profile EST-Service1
switch(config-est-service1)#
```

Removing EST profile **service1**:

```
switch(config)# no crypto pki est-profile EST-Service1
```

## enroll est-profile

### Syntax

```
enroll est-profile <EST-NAME>
```

### Description

Enrolls a leaf certificate through a remote EST (Enrollment over Secure Transport) server.

Per RFC 7030, EST enables clients to request certificate signing services over secure TLS connections. The switch generates a key pair and the corresponding CSR. The CSR is sent to the EST server to request signing, and the signed certificate is returned to the switch where it is validated. If the whole process succeeds, the certificate can be used as a leaf certificate on the switch. When the leaf certificate approaches its expiry date, it will be renewed automatically through the same EST server.

Each enrollment or re-enrollment attempt starts with a `/cacerts` request sent to the EST server to get the latest chain of CA certificates. After the enrollment or re-enrollment succeeds, this chain of CA certificates will be compared with those downloaded previously from the same EST server. Updates will be made as appropriate.

The subject fields of the current leaf certificate must be defined before running this command. If the common name subject field is not configured, this command is rejected.

This command cannot be used to enroll or renew the default certificate "local-cert."

## Command context

```
config-cert-<CERT-NAME>
```

## Parameters

**<EST-NAME>**

Specifies an existing EST profile name. Range: Up to 32 alphanumeric characters (excluding ").

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Enrolling leaf certificate **leaf-cert1** through the EST server identified in EST profile **EST-service1**:

```
switch(config-cert-leaf-cert1)# enroll est-profile EST-service1
You are enrolling a certificate with the following attributes:
 Subject: C=US, ST=CA, L=Roseville, OU=Aruba-Roseville, O=Aruba,
 CN=leaf-cert1
 Key Type: RSA (2048 bits)

Continue (y/n)? y
Certificate enrollment via EST-service1 has been initiated.
Please use `show crypto pki certificate leaf-cert1` to check its status.

switch(config-cert-leaf-cert1)#
```

# reenrollment-lead-time

## Syntax

```
reenrollment-lead-time <LEAD-TIME>
```

```
no reenrollment-lead-time
```

## Description

Within the EST profile context, sets the certificate re-enrollment lead time which is the number of days before certificate expiry date that certificate re-enrollment will be initiated.

The no form of this command resets the EST server re-enrollment lead time to its default of 2 days.

## Command context

config-est-<EST-NAME>

## Parameters

<LEAD-TIME>

Specifies the certificate re-enrollment lead time in days. Range: 0 to 30 days. Default: 2 days.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the certificate re-enrollment lead time to 15 days:

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# reenrollment-lead-time 15
```

Resetting the certificate re-enrollment lead time to its default of 2 days :

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# no reenrollment-lead-time
```

# retry-count

## Syntax

retry-count <RETRIES>

no retry-count

## Description

Within the EST profile context, sets the maximum number of retries to be attempted after the initial certificate enrollment request fails.

The no form of this command resets the maximum number of certificate enrollment request retries to its default of 3.

## Command context

config-est-<EST-NAME>

## Parameters

<RETRIES>

Specifies the maximum number of certificate enrollment request retries. Range: 0 to 32 retries. Default: 3 retries.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the retry count to 5 retries:

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# retry-count 5
```



Resetting the retry count to its default of 3 retries:

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# no retry-count
```

## retry-interval

### Syntax

```
retry-interval <INTERVAL>
```

```
no retry-interval
```

### Description

Within the EST profile context, sets the interval at which a failed certificate enrollment request is retried. The no form of this command resets the enrollment request retry interval to its default of 30 seconds.

### Command context

```
config-est-<EST-NAME>
```

### Parameters

<INTERVAL>

Specifies the enrollment request retry interval in seconds. Range: 30 to 600 seconds. Default: 30 seconds.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the certificate enrollment request retry interval to 45 seconds:

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# retry-interval 45
```

Resetting the retry interval to its default of 30 seconds:

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# no retry-interval
```

## show crypto pki est-profile

### Syntax

```
show crypto pki est-profile [<EST-NAME>]
```

### Description

Shows a list of all configured EST profiles, or detailed information for a specific profile.

### Command context

Manager (#)

## Parameters

### <EST-NAME>

Specifies the EST profile name. Range: Up to 32 alphanumeric characters excluding ".

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing a list of all configured EST profiles:

```
switch# show crypto pki est-profile
```

| Profile Name | Downloaded<br>TA Profiles | Enrolled<br>Certificates |
|--------------|---------------------------|--------------------------|
| EST-service1 | 2                         | 3                        |
| EST-service2 | 1                         | 2                        |
| EST-service3 | 2                         | 0                        |

Showing detailed information for EST profile **EST-service1**:

```
switch# show crypto pki est-profile EST-service1
Profile Name : EST-service1
Service VRF : mgmt
Service URL : https://est-service999.com
Arbitrary Label : not configured
Arbitrary Label Enrollment : /ipsec-VP7
Arbitrary Label Reenrollment : not configured
Authentication Username : est1
Authentication Password :
AQBApREALpWYm2z7L1LanOtR3vGkqhBN1hBUU2CuvQXUF/ggYgAAnAnGTnKq49P4c
dNQ6UqPbjHL4XzCO0T04djkhsUXPKGfnsWuFEONveh+JbEobqKImfwJjc3eWHiaUb
eNpPx2zN2Q1DdyxAAQi4rmKr8LITMTTmd7qr
Retry Interval : 45 seconds
Retry Count : 5 times
Reenrollment Lead Time : 2 days
Downloaded TA Profiles : 2
Enrolled Certificates :
leaf-cert1
leaf-cert2
leaf-cert3
```

## url

### Syntax

```
url <URL>
```

```
no url
```

### Description

Within the EST profile context, configures the URL of the certificate enrollment EST server. This is not configured by default. Any existing URL is replaced by this command.

The no form of this command removes the EST server URL within the selected EST profile. The removal of the URL does not affect the TA profiles and enrolled certificates from the EST server.

## Command context

config-est-<EST-NAME>

## Parameters

<URL>

Specifies the EST server URL. Range: Up to 192 characters.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

- The configuration and update of the EST profile URL triggers the sending of a `/cacerts` request to the EST server. A successful request will result in a chain of trusted CA certificates being downloaded from the EST server. Each CA certificate, either root CA certificates or intermediate CA certificates, will be saved as a TA profile, with TA profile name `<est-name>-est-taNN` with NN representing two numerical digits. This TA profile naming scheme with the `-est-taNN` suffix is reserved for TA profiles downloaded from EST servers.
- Upon connection with an EST server, the switch authenticates the server by validating the server certificate. For this validation to succeed, a TA profile needs to pre-exist in the switch with a CA certificate from the issuer chain of the server certificate. Once the server is authenticated, all CA certificates in its `/cacerts` response will be trusted, with no further validation occurring for them.
- The TA profiles with CA certificates downloaded from an EST server will have their revocation check set to OCSP, enforcement set to optional, and the OCSP VRF set to the same as that of the EST profile.

## Examples

Configuring the EST server URL :

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# url https://est-service999.com/.well-known/est
```

Removing the EST server URL :

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# no url
```

## username

### Syntax

```
username <USERNAME> password [ciphertext <CIPHERTEXT-PASSWORD> |
 plaintext <PLAINTEXT-PASSWORD>]
```

no username

### Description

Within the EST profile context, configures the user account information for the EST server that is used to authenticate the switch before accepting requests from the switch. This is not configured by default. Any existing username and password is replaced by this command.

When entered without either optional `ciphertext` or `plaintext` parameters, the plaintext password is prompted for twice, with the characters entered masked with "\*" symbols.

The no form of this command removes the user account information within the selected EST profile.

There are two ways the EST client on a CX switch can prove itself to an EST server: a certificate, and/or username and password. At least one of the two must be configured for the EST request to succeed. If both are configured, certificate authentication will be used. If a certificate is not configured or certificate authentication fails, and username and password is configured, the username and password will be sent to the EST server for authentication.

## Command context

`config-est-<EST-NAME>`

## Parameters

### <USERNAME>

Specifies the EST server account user name. The exact user name requirements are set by the chosen EST service. Range: Up to 32 alphanumeric characters.

### ciphertext <CIPHERTEXT-PASSWORD>

Specifies the EST server account password as Base64 ciphertext. No password prompts are provided and the ciphertext password is validated before the configuration is applied for the user.



**NOTE:** The ciphertext password must be gotten from the EST service.

### plaintext <PLAINTEXT-PASSWORD>

Specifies the password without prompting. The password is visible as cleartext when entered but is encrypted thereafter. The exact password requirements are set by the chosen EST service. Range: Up to 64 alphanumeric characters.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring an EST user with prompted cleartext password entry :

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# username est1 password
Enter password: *****
Confirm password: *****
switch(config-est-EST-service1)#
```

Configuring an EST user with direct cleartext password entry:

```
switch(config)# crypto pki est-profile EST-service2
switch(config-est-EST-service2)# username est1 password plaintext concept_leap739
```

Configuring an EST user with ciphertext password entry :

```
switch(config)# crypto pki est-profile EST-service3
switch(config-est-EST-service3)# username est1 password ciphertext
AQBpRALpWYm2z7L1LanOtr3vGkqhN1hBU2CuvQXUF/ggYgAAAHWaPqxU6nAnGTnKq49P4cdNQ6U
qPbjHL4Xz00T04djkUPKGfnsWuFEONveh+JbEobq63+1k80qBKImfwJjc3eWHiaUbeNpPx2zN2Q
1DdyxAAQi4rmKr8LITMTTmd7qr
```

Removing the EST user account information for EST profile EST-service2:

```
switch(config)# crypto pki est-profile EST-service2
switch(config-est-EST-service2)# no username
```

## vrf

### Syntax

```
vrf <VRF-NAME>
```

```
no vrf
```

### Description

Within the EST profile context, selects the VRF through which the EST server can be reached. Any existing VRF selection is replaced by this command. When this command is not used, VRF `mgmt` is used by default on switch families supporting the `mgmt` VRF, otherwise the default VRF named `default` is used.

The `no` form of this command selects the default VRF either `mgmt` or `default`.

### Command context

```
config-est-<EST-NAME>
```

### Parameters

**<VRF-NAME>**

Specifies the name of the VRF to use for EST server communication.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Selecting VRF `it-services` for EST server communications:

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# vrf it-services
```

Resetting the VRF to its default of `mgmt` for EST server communications:

```
switch(config)# crypto pki est-profile EST-service1
switch(config-est-EST-service1)# no vrf
```

All PoE configuration commands except `threshold` configuration and `always-on poe` configuration are entered at the `config-if` context. The PoE threshold command is used at the system level whereas the `always-on poe` command is set at the slot level. These commands can only be configured in the global configuration context.

## lldp dot3 poe

### Syntax

```
lldp dot3 poe
```

```
no lldp dot3 poe
```

### Description

Enables 802.3 TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. LLDP dot3 TLV is by default enabled for PoE.

The `no` form of this command disables 802.3 TLV list in LLDP.

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling 802.3 TLV list in LLDP:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dot3 poe
```

Disabling 802.3 TLV list in LLDP:

```
switch(config-if)# no lldp dot3 poe
```

## lldp med poe

### Syntax

```
lldp med poe [priority-override]
```

```
no lldp med poe [priority-override]
```

### Description

Enables MED TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. Also enables the lldp-MED TLV priority to override user configured port priority for Power over Ethernet. When

both dot3 and MED are enabled, dot 3 will take precedence. MED TLV is by default enabled for PoE. Priority over-ride is by default disabled.

The `no` form of this command disables MED TLV list in LLDP.

### Command context

`config-if`

### Parameters

**[priority-override]**

System defined name of the interface.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling and disabling LLDP MED PoE:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp med poe
switch(config-if)# no lldp med poe
```

Enabling and disabling LLDP MED PoE priority override:

```
switch(config-if)# lldp med poe priority-override
```

## power-over-ethernet

### Syntax

`power-over-ethernet`

`no power-over-ethernet`

### Description

Enables per-interface power distribution. Per-port power is enabled by default with priority low. PoE cannot be disabled for individual ports when Quick PoE is enabled for the entire switch or line module.

The `no` form of this command disables per-interface power distribution.

### Command context

`config-if`

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling per-interface power distribution:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet
```

Disabling per-interface power distribution:

```
switch(config-if)# no power-over-ethernet
```

Showing Quick PoE enabled:

```
switch(config-if)# power-over-ethernet quick-poe 1/1
switch(config-if)# interface 1/1/1
switch(config-if)# no power-over-ethernet
Interface PoE cannot be disabled when Quick PoE is enabled.
```

## power-over-ethernet allocate-by

### Syntax

```
power-over-ethernet allocate-by {usage | class}

no power-over-ethernet allocate-by {usage | class}
```

### Description

Configures the power allocation method. Power allocation method is initially based on usage. PSE Allocated power value will change to LLDP negotiated power if and when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocated Power Value will be the actual instantaneous power draw and reserve power based on actual consumption. In allocate-by class, power allocation is based on PD requested class and PSE allocated power value will be the LLDP negotiated power when LLDP exchange takes place between PSE and PD. . When there is no LLDP negotiation, PSE Allocate Power will be based on PD class. Reserve power is based on PD Class. By default, power allocation is by usage.

The **no** form of this command resets the action to default.

### Command context

config-if

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring the power allocation method:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet allocate-by usage
switch(config-if)# power-over-ethernet allocate-by class
```

Resetting power allocation method:

```
switch(config-if)# no power-over-ethernet allocate-by class
```

## power-over-ethernet always-on

### Syntax

```
power-over-ethernet always-on <MODULE-ID>

no power-over-ethernet always-on <MODULE-ID>
```





---

**NOTE:** This command is not available on the 6100 Switch Series.

---

## Description

Always-on PoE is a feature that provides the ability to the switch to continue to provide power across a soft reboot. It is applicable only to the interfaces which were connected and delivering before the soft reboot. Also, power will not be delivered if power to the switch is interrupted. This command enables or disables the always-on PoE feature at the switch or the slot level. By default, always-on PoE is enabled at the switch or the slot level.

The `no` form of this command disables power distribution on soft reboot.

## Command context

`config`

## Parameters

**<MODULE-ID>**

Module number to apply always-on PoE configuration.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling per-interface power distribution:

```
switch(config)# power-over-ethernet always-on 1/1
```

Disabling per-interface power distribution:

```
switch(config)# no power-over-ethernet always-on 1/1
```

# power-over-ethernet assigned-class

## Syntax

```
power-over-ethernet assigned-class {3 | 4 | 6}
```

```
no power-over-ethernet assigned-class
```

## Description

Limit PoE power based on the assigned class. When an user assigns a maximum class to an interface, the PSE will limit the maximum power delivered to the PD up to a total power draw not exceeding the PSE assigned-class power. Power demotion occurs when a PD requested class is higher than the PSE assigned class, permitting the PD to receive power and operate in a reduced power mode. PoE ports cannot set an assigned class when Quick PoE is enabled on the subsystem. The default assigned class is 4 for 2-pair capable PSE and 6 for 4-pair capable PSE.

The `no` form of this command resets the action to default.

## Command context

`config-if`

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting PoE assigned class:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet assigned-class 4
```

Resetting PoE assigned class to default:

```
switch(config-if)# no power-over-ethernet assigned-class 4
```

Showing Quick PoE enabled:

```
switch(config)# power-over-ethernet quick-poe 1/1
switch(config)# interface 1/1/1
switch(config)# power-over-ethernet assigned-class 4
Interface assigned class cannot be configured when Quick PoE is enabled.
```

# power-over-ethernet pre-std-detect

## Syntax

```
power-over-ethernet pre-std-detect
```

```
no power-over-ethernet pre-std-detect
```

## Description

Before IEEE 802.3 released the first Power over Ethernet standard (802.3af), vendors had shipped PoE capable switches and PD's. As we are backward compatible Aruba will support both IEEE standard and pre-standard 802.3af Power over Ethernet PD's concurrently. This CLI allows the user to enable or disable pre-802.3af-standard device detection and powering on the specific port. When pre-std-detect is enabled, power will be delivered on PairA only. Default is disabled.

The `no` form of this command resets the action to default.

## Command context

```
config-if
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling standard device detection:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet pre-std-detect
```

Disabling standard device detection:

```
switch(config-if)# no power-over-ethernet pre-std-detect
```

## power-over-ethernet priority

### Syntax

```
power-over-ethernet priority {critical | high | low}
```

```
no power-over-ethernet priority {critical | high | low}
```

### Description

Sets PoE priority for an interface. Specifying critical, high, or low indicates the priority of the interface in the event of power over-subscription. Within the same priority level, higher power-priority line-module ports have higher precedence. With same PoE priority and same line-module priority, lower numbered line-module ports have higher precedence. Per-interface PoE priority is low by default.

The `no` form of this command resets the priority to default PoE priority "low".

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring PoE priority:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet priority critical
switch(config-if)# power-over-ethernet priority high
```

Resetting the PoE priority to default:

```
switch(config-if)# no power-over-ethernet priority high
```

## power-over-ethernet quick-poe

### Syntax

```
power-over-ethernet quick-poe <MODULE-ID>
```

```
no power-over-ethernet
```



**NOTE:** This command is not available on the 6100 Switch Series.

### Description

Quick PoE is a feature that provides the ability for the switch to provide power to the connected powered device as soon as switch goes through cold reboot. When quick PoE is enabled on the subsystem PoE port disablement and PD demotion is not allowed. Also quick PoE enablement is not allowed if any of the port is disabled on the subsystem. User should not over-subscribe the PoE power when quick PoE is enabled. Quick PoE saved configuration will work irrespective of the configuration change at reboot.

Enables quick PoE feature on the switch or the subsystem level. By default, quick-PoE is disabled for the subsystem.

The `no` form of this command disables quick PoE.

## Command context

config-if

## Parameters

**<MODULE-ID>**

Specifies module number for quick PoE configuration .

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling and disabling quick PoE:

```
switch(config)# power-over-ethernet quick-poe 1/2
switch(config)# no power-over-ethernet quick-poe 1/1
```

```
switch(config-if)# power-over-ethernet quick-poe 1/1
PoE must be enabled on all interfaces before enabling Quick PoE
```

```
switch(config-if)# power-over-ethernet quick-poe 1/3
All interfaces must use the default assigned class before enabling Quick PoE
```

# power-over-ethernet threshold

## Syntax

power-over-ethernet threshold <PERCENTAGE>

no power-over-ethernet threshold <PERCENTAGE>

## Description

Sets the threshold at which the system will send an excess power consumption notification trap. Default value is 80 percentage.

The **no** form of this command resets the action to default.

## Command context

config

## Parameters

**<PERCENTAGE>**

Excess power consumption trap threshold. Range 1-99.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the power-over-ethernet threshold:

```
switch(config)# power-over-ethernet threshold 75
```

Resetting the power-over-ethernet threshold to default:

```
switch(config-if)# no power-over-ethernet threshold 75
```

## power-over-ethernet priority

### Syntax

```
power-over-ethernet trap
```

```
no power-over-ethernet trap
```

### Description

This command enables/disables the SNMP trap generation for PoE related events at system level. PoE trap generation is enabled by default.

The `no` form of this command resets the priority to default PoE priority "low".

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling SNMP trap generation for PoE:

```
switch(config)# power-over-ethernet trap
```

Disabling SNMP trap generation for PoE:

```
switch(config-if)# no power-over-ethernet trap
```

## show lldp local

### Syntax

```
show lldp local-device [<INTERFACE-ID>]
```

### Description

Displays information advertised by the switch if the LLDP feature is enabled by user.

### Command context

Operator (>) or Manager (#)

### Parameters

<INTERFACE-ID>

Specifies an interface. Format: member/slot/port

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing LLDP local device:

```
switch# show lldp local-device 1/1/10
Local Port Data
=====

Port-ID : 1/1/10
Port-Desc : "1/1/10"
Port VLAN ID : 0

PoE Plus Information

PoE Device Type : Type 2 PSE
Power Source : Primary
Power Priority : low
PSE Allocated Power: 25.0 W
PD Requested Power: 25.0 W
```

## show lldp neighbor

### Syntax

```
show lldp neighbor [<INTERFACE-ID>]
```

### Description

Displays detailed information about a particular neighbor connected to a particular interface.

### Command context

Operator (>) or Manager (#)

### Parameters

**<INTERFACE-ID>**

Specifies an interface. Format: member/slot/port

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing LLDP neighbor information when there is only one neighbor:

```
switch# show lldp neighbor-info 1/1/10

Port : 1/1/10
Neighbor Entries : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : 84:d4:7e:ce:5d:68
Neighbor Chassis-Description : ArubaOS (MODEL: 325), Version Aruba IAP
Neighbor Chassis-ID : 84:d4:7e:ce:5d:68
Neighbor Management-Address : 169.254.41.250
Chassis Capabilities Available : Bridge, WLAN
Chassis Capabilities Enabled :
```

```

Neighbor Port-ID : 84:d4:7e:ce:5d:68
Neighbor Port-Desc : eth0
TTL : 120
Neighbor Port VLAN ID :
Neighbor PoEplus information : DOT3
Neighbor Device Type : TYPE2 PD
Neighbor Power Priority : Unkown
Neighbor Power Source : Primary
Neighbor Power Requested : 25.0 W
Neighbor Power Allocated : 0.0 W
Neighbor Power Supported : No
Neighbor Power Enabled : No
Neighbor Power Class : 5
Neighbor Power Paircontrol : No
Neighbor Power Pairs : SIGNAL

```

## show power-over-ethernet

### Syntax

```
show power-over-ethernet [<IFRANGE> [brief]
```

### Description

Displays the status information of the full system. Displays the brief status of all port or given port if parameter brief is used. Displays the detailed status of given port.

### Command context

Operator (>) or Manager (#)

### Parameters

**<IFNAME>**

Display the detailed status of given port.

**brief**

Display the brief status of all ports or the given port.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

```
switch# show power-over-ethernet
```

```
System Power Status for member 1
```

```

Configured Power Status : No redundancy
Operational Power Status : No redundancy
Total Available Power : 740 W
Total Failover Pwr Avl : 0 W
Total Redundancy Power : 0 W
Total Power Drawn : 0 W +/- 6W
Total Power Reserved : 0 W
Total Remaining Power : 740 W
Trap Threshold : 80 %
Trap Enabled : Yes
Always-on PoE Enabled : 1/1

```

Quick PoE Enabled : None

#### Internal Power

| PS | Total Power<br>(Watts) | Status |
|----|------------------------|--------|
| 1  | 0                      | Absent |
| 2  | 740                    | Ok     |

#### System Power Status for member 2

Configured Power Status : No redundancy  
Operational Power Status : No redundancy  
Total Available Power : 600 W  
Total Failover Pwr Avl : 0 W  
Total Redundancy Power : 0 W  
Total Power Drawn : 0 W +/- 6W  
Total Power Reserved : 0 W  
Total Remaining Power : 600 W  
Trap Threshold : 80 %  
Trap Enabled : Yes  
Always-on PoE Enabled : None  
Quick PoE Enabled : None

#### Internal Power

| PS | Total Power<br>(Watts) | Status |
|----|------------------------|--------|
| 1  | 0                      | Absent |
| 2  | 600                    | Ok     |

switch# **show power-over-ethernet member 1**

#### System Power Status for member 1

Configured Power Status : No redundancy  
Operational Power Status : No redundancy  
Total Available Power : 740 W  
Total Failover Pwr Avl : 0 W  
Total Redundancy Power : 0 W  
Total Power Drawn : 0 W +/- 6W  
Total Power Reserved : 0 W  
Total Remaining Power : 740 W  
Trap Threshold : 80 %  
Trap Enabled : No  
Always-on PoE Enabled : 1/1  
Quick PoE Enabled : 1/1

#### Internal Power

| PS | Total Power<br>(Watts) | Status |
|----|------------------------|--------|
| 1  | 0                      | Absent |
| 2  | 740                    | Ok     |

switch# **show power-over-ethernet brief**

#### Status and Configuration Information for PoE

##### Member 1 Power Status

Available: 370 W Reserved: 55.60 W Remaining: 314.40 W  
Always-on PoE Enabled: 1/1



Quick PoE Enabled: None

| PoE Port | Pwr En | Power Priority | Pre-std Detect | Alloc Act | PSE Pwr Rsrvd | PD Pwr Draw | PoE Port Status | PD Sign | Cls | Type |
|----------|--------|----------------|----------------|-----------|---------------|-------------|-----------------|---------|-----|------|
| 1/1/1    | Yes    | Low            | Off            | Class     | 0.0 W         | 0.0 W       | Denied          | None    | 4   | 2    |
| 1/1/2    | Yes    | Critical       | Off            | Usage     | 1.6 W         | 1.5 W       | Delivering*     | Single  | 0   | 1    |
| 1/1/3    | Yes    | High           | Off            | Class     | 54.0 W        | 25.5 W      | Delivering*^    | Dual    | 1/3 | 3    |
| 1/1/4    | No     | Low            | On             | Usage     | 0.0 W         | 0.0 W       | Disabled        | None    | N/A | N/A  |

Member 2 Power Status

Available: 600 W Reserved: 0.00 W Remaining: 600 W

Always-on PoE Enabled: None

Quick PoE Enabled: None

| PoE Port | Pwr En | Power Priority | Pre-std Detect | Alloc Act | PSE Pwr Rsrvd | PD Pwr Draw | PoE Port Status | PD Sign | Cls | Type |
|----------|--------|----------------|----------------|-----------|---------------|-------------|-----------------|---------|-----|------|
| 2/1/1    | Yes    | Low            | Off            | Class     | 0.0 W         | 0.0 W       | Searching       | None    | N/A | N/A  |
| 2/1/2    | Yes    | Critical       | Off            | Usage     | 0.0 W         | 0.0 W       | Searching       | None    | N/A | N/A  |
| 2/1/3    | Yes    | High           | Off            | Class     | 0.0 W         | 0.0 W       | Searching       | None    | N/A | N/A  |
| 2/1/4    | No     | Low            | On             | Usage     | 0.0 W         | 0.0 W       | Disabled        | None    | N/A | N/A  |

\*This port may go down in the event of a PSU failure.

^This port is power demoted due to user config or power availability.

switch# **show power-over-ethernet brief**

Status and Configuration Information for PoE

Power Status

Available: 370 W Reserved: 55.60 W Remaining: 314.40 W

Always-on PoE Enabled: 1/1,1/3,1/4,1/7

Quick PoE Enabled: None

| PoE Port | Pwr En | Power Priority | Pre-std Detect | Alloc Act | PSE Pwr Rsrvd | PD Pwr Draw | PoE Port Status | PD Sign | Cls | Type |
|----------|--------|----------------|----------------|-----------|---------------|-------------|-----------------|---------|-----|------|
| 1/1/1    | Yes    | Low            | Off            | Class     | 0.0 W         | 0.0 W       | Denied          | None    | 4   | 2    |
| 1/1/2    | Yes    | Critical       | Off            | Usage     | 1.6 W         | 1.5 W       | Delivering*     | Single  | 0   | 1    |
| 1/1/3    | Yes    | High           | Off            | Class     | 54.0 W        | 25.5 W      | Delivering^     | Dual    | 1/3 | 3    |
| 1/1/4    | No     | Low            | On             | Usage     | 0.0 W         | 0.0 W       | Disabled        | None    | N/A | N/A  |

\*This port may go down in the event of a PSU failure.

^This port is power demoted due to user config or power availability.

Showing sample output for power-over-ethernet brief per-port:

switch# **show power-over-ethernet 1/1/1 brief**

Status and Configuration Information for port 1/1/1

Member 1Power Status

Available: 370 W Reserved: 55.60 W Remaining: 314.40 W

Always-on PoE Enabled: 1/1

| PoE Port | Pwr En | Power Priority | Pre-std Detect | Alloc Act | PSE Pwr Rsrvd | PD Pwr Draw | PoE Port Status | PD Sign | Cls | Type |
|----------|--------|----------------|----------------|-----------|---------------|-------------|-----------------|---------|-----|------|
| 1/1/1    | Yes    | Low            | Off            | Class     | 0.0 W         | 0.0 W       | Denied          | None    | 4   | 2    |

Showing sample output for power-over-ethernet brief for interface range:

switch# **show power-over-ethernet 1/1/1-1/1/2 brief**

#### Status and Configuration Information for port 1/1/1-1/1/2

##### Member 1Power Status

Available: 370 W Reserved: 55.60 W Remaining: 314.40 W

Always-on PoE Enabled: 1/1

| PoE Port | Pwr En | Power Priority | Pre-std Detect | Alloc Act | PSE Pwr Rsrvd | PD Pwr Draw | PoE Port Status | PD Sign | Cls | Type |
|----------|--------|----------------|----------------|-----------|---------------|-------------|-----------------|---------|-----|------|
| -----    | ---    | -----          | -----          | -----     | -----         | -----       | -----           | -----   | --- | ---- |
| 1/1/1    | Yes    | Low            | Off            | Class     | 0.0 W         | 0.0 W       | Denied          | None    | 4   | 2    |
| 1/1/2    | Yes    | Critical       | Off            | Usage     | 1.6 W         | 1.5 W       | Delivering*     | Single  | 0   | 1    |

Showing sample output for power-over-ethernet for a missing line card:

```
switch# show power-over-ethernet 1/3 brief
```

Module 1/3 is not physically present.

Showing sample output for power-over-ethernet brief for a missing member:

```
switch# show power-over-ethernet member 3 brief
```

Member 3 is not physically present.

Showing sample output for power-over-ethernet port when physical interface is not present:

```
switch# show power-over-ethernet 2/1/1
```

Interface 2/1/1 is not present.

## aaa authentication port-access auth-mode

### Syntax

```
aaa authentication port-access auth-mode {client-mode | device-mode}
```

### Description

Configures the authentication mode for the port. By default, client mode is enabled.

### Command context

config-if

### Parameters

#### client-mode

Selects client mode. In this mode, all clients connecting to the port are sent for authentication.

The maximum number of clients allowed to connect to the port is limited by the client limit value configured with the `aaa authentication port-access client-limit` command.

#### device-mode

Selects device mode. In this mode, only the first client connecting to the port is sent for authentication. Once this client is authenticated, the port is considered as open and all subsequent clients trying to connect on that port are not sent for authentication.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring device mode authentication for a port:

```
switch(config-if)# aaa authentication port-access auth-mode device-mode
```

## aaa authentication port-access auth-precedence

### Syntax

```
aaa authentication port-access auth-precedence [dot1x mac-auth | mac-auth dot1x]
```

```
no aaa authentication port-access auth-precedence [dot1x mac-auth | mac-auth dot1x]
no aaa authentication port-access auth-precedence
```

### Description

Configures the per port authentication precedence using the space separator.

By default, 802.1X authentication (`dot1x`) takes a higher precedence than MAC authentication (`mac-auth`).

The **no** form of the command resets the port access authentication precedence to the default, 802.1X authentication followed by MAC authentication.

### Command context

config-if

### Parameters

**dot1x mac-auth**

Specifies that the port access authentication precedence is 802.1X authentication followed by MAC authentication.

**mac-auth dot1x**

Specifies that the port access authentication precedence is MAC authentication followed by 802.1X authentication.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring MAC authentication precedence on a port:

```
switch(config-if)# aaa authentication port-access auth-precedence mac-auth dot1x
```

Resetting the authentication precedence to the default value:

```
switch(config-if)# no aaa authentication port-access auth-precedence mac-auth dot1x
```

## aaa authentication port-access auth-priority

### Syntax

```
aaa authentication port-access auth-priority [dot1x mac-auth | mac-auth dot1x]
```

```
no aaa authentication port-access auth-priority [dot1x mac-auth | mac-auth dot1x]
```

```
no aaa authentication port-access auth-priority
```

### Description

Configures the authentication priority using the space separator to specific interface.

Default **auth-priority** with concurrent onboarding is 802.1x followed by MAC authentication. With **authentication precedence**, the default **auth-priority** follows the **auth-precedence** order.

The **no** form of the command resets the port access authentication priority to the default, is same as the configured **auth-precedence** order.

The authentication priority is useful in deployments where clients such as wireless access points (APs), IT-compliant-laptops or phones, or laptops without pre-loaded supplicant software must download the supplicant software or firmware patches before attempting 802.1X authentication. In such cases, configure the MAC authentication as the primary authentication method followed by 802.1X for the authentication order. Meanwhile, configure 802.1x as the primary authentication priority and MAC authentication as secondary to enforce access based on 802.1X. Thus the client (or end access device) will initially be authenticated by MAC authentication with the access required to onboard and install the software or patches, and subsequently attempt the 802.1X authentication.

Reauthentication will be triggered for all high priority methods and not just the final successful authentication method.

### Command context

config-if

### Parameters

#### **dot1x mac-auth**

Specifies that the port access authentication priority is 802.1X authentication followed by MAC authentication.

#### **mac-auth dot1x**

Specifies that the port access authentication priority is MAC authentication followed by 802.1X authentication.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring MAC authentication priority on a port:

```
switch(config-if)# aaa authentication port-access auth-priority mac-auth dot1x
```

Resetting the authentication priority to the default value:

```
switch(config-if)# no aaa authentication port-access auth-priority mac-auth dot1x
switch(config-if)# no aaa authentication port-access auth-priority
```

### Sample configuration:

```
interface 1/1/1
 no shutdown
 no routing
 vlan access 1
 aaa authentication port-access auth-precedence mac-auth dot1x
 aaa authentication port-access auth-priority dot1x mac-auth
```

## aaa authentication port-access client-limit

### Syntax

```
aaa authentication port-access client-limit <CLIENTS>
no aaa authentication port-access client-limit
```

### Description

Configures the maximum number of clients that can simultaneously connect to a port.

The no form of this command resets the number of clients to the default.

### Command context

config-if

## Parameters

### <CLIENTS>

Specifies the maximum number of clients. Default: 1. Range: 1 to 32 (6100).

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring the client limit for a port:

```
switch(config-if)# aaa authentication port-access client-limit 25
```

# aaa authentication port-access (role)

## Syntax

```
aaa authentication port-access [critical-role|preauth-role|reject-role|auth-role] <ROLE-NAME>
```

```
no aaa authentication port-access [critical-role|preauth-role|reject-role|auth-role]
```

## Description

Configures the role to assign to the clients depending on the client authentication state.

The **no** form of the command disassociates the roles that you assign to clients based on the authentication state.

## Command context

config-if

## Parameters

### **critical-role**

Specifies the role that is applied when the RADIUS server is unreachable for authentication or when there is a request timeout.

### **preauth-role**

Specifies the role that is applied when authentication is still in progress.

### **reject-role**

Specifies the role that is applied when authentication has failed.

### **auth-role**

Specifies the role that is applied to authenticated clients when a specific role is not assigned in the RADIUS server.

### <ROLE-NAME>

Specifies the role name.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring critical role for clients:

```
switch(config-if)# aaa authentication port-access critical-role role1
```

## port-access allow-flood-traffic

### Syntax

```
port-access allow-flood-traffic {enable | disable}
```

### Description

Enables or disables transmission of flood traffic, such as broadcast, multicast, and unknown unicast messages through a security enabled port on which no client has been authenticated.

By default, transmission of flood traffic is disabled.

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

This command can be used to allow Wake-on-LAN packets on security enabled ports, before a client is authenticated.

### Examples

Enabling flood traffic on a port:

```
switch(config-if)# port-access allow-flood-traffic enable
```

## port-access fallback-role

### Syntax

```
port-access fallback-role <ROLE-NAME>
```

```
no port-access fallback-role <ROLE-NAME>
```

### Description

Configures the fallback role to assign to the clients onboarding on a port. This role is applied only when no derived role is applied to the clients.

The `no` form of the command resets the fallback role.

### Command context

```
config-if
```

### Parameters

**<ROLE-NAME>**

Specifies the fallback role name. The maximum number of characters supported is 64.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

Following are the conditions for the fallback role to be applied on onboarding devices:

- The device profile local MAC match feature with block-until-profile-applied mode is configured.
- Device profile along with AAA is configured but no match was found for the device profile client.
- AAA method with no reject or critical role is configured, and the connection to RADIUS server failed.
- 802.1X authentication is enabled on the port, but the supplicant of the device timed out to respond to the authentication request.

## Example

Configuring fallback role for a port:

```
switch(config)# interface 1/1/3
switch(config-if)# port-access fallback-role fallback01
```

# port-access log-off client

## Syntax

```
port-access log-off client mac <MAC-ADDRESS>
port-access log-off client interface <INTERFACE-NAME>
port-access log-off client role <ROLE-NAME>
```

## Description

Logs off the client connected to a port access-enabled interface.

## Command context

Manager (#)

## Parameters

**mac** <MAC-ADDRESS>

Specifies the client MAC address.

**interface** <INTERFACE-NAME>

Specifies the client interface.

**role** <ROLE-NAME>

Specifies the client role.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Logging a client off from the switch, specifying the MAC address:

```
switch# port-access log-off client mac 00:50:56:bd:04:2d
```



Logging a client off from the switch, specifying the interface:

```
switch# port-access log-off client interface 1/1/1
```

Logging a client off from the switch, specifying the role:

```
switch# port-access log-off client role r1
```

## port-access onboarding-method precedence

### Syntax

```
port-access onboarding-method precedence [aaa device-profile | device-profile aaa]
```

```
no port-access onboarding-method precedence [aaa device-profile | device-profile aaa]
```

### Description

Configures the precedence for the method to be used to authenticate onboarding devices for each interface.

The `no` form of the command resets the authentication method precedence to the default precedence of AAA followed by device profile.

AAA includes the 802.1X and MAC authentication methods whose precedence can be configured using the `aaa authentication port-access auth-precedence` command. Here, the default precedence is 802.1X authentication.

For example, if you configure AAA (both 802.1X and MAC) authentication methods and device profile on a port, by default, the authentication precedence would be 802.1X, then MAC, and lastly device profile.

### Command context

config-if

### Parameters



**NOTE:** aaa here refers to the authentication precedence configured using the `aaa authentication port-access auth-precedence` command.

#### aaa device-profile

Specifies that the precedence for per port onboarding authentication method is AAA followed by device profile.

#### device-profile aaa

Specifies that the precedence for per port onboarding authentication method is device profile followed by AAA.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring AAA method precedence on a port:

```
switch(config)# interface 1/1/1
switch(config-if)# port-access onboarding-method precedence device-profile aaa
```

Resetting the authentication method precedence:

```
switch(config)# interface 1/1/1
switch(config-if)# no port-access onboarding-method precedence device-profile aaa
```

## port-access onboarding-method concurrent

### Syntax

```
port-access onboarding-method concurrent <enable | disable>
```

### Description

Configures all methods to start concurrently for faster onboarding process. If authentication priority is not configured when enabling concurrent onboarding, the priority will be 802.1x followed by `mac-auth` and `device-profile`.

Default priority for concurrent onboarding is 802.1x followed by `mac-auth` and `device-profile`.

When enabling concurrent onboarding on the port, existing clients will be de-authenticated and freshly onboarded concurrently.

When concurrent onboarding is enabled, then `auth-precedence` will be ignored.

If concurrent onboarding is configured, the client will stay in pre-auth role till it gets succeeded by one authentication method or gets failed by all the authentication methods.

When the authentication method with the highest priority fails, the profile of the next successful authentication method is applied.

If all methods fail, the reject or critical role is applied based on the 802.1X authentication failure reason and continues to reauthenticate with the 802.1X method.

Reauthentication will be triggered for all high priority methods and not just the final successful authentication method.

Some RADIUS server may block the client when it receives two requests, `mac-auth` and 802.1X, from the same client at the same time. This is because the RADIUS server allows only one authentication request. In such cases, concurrent onboarding is not feasible. To prevent such scenarios, configure `auth-precedence` with `auth-priority`.

### Command context

```
config-if
```

### Parameters

#### **enable**

Enable clients to be onboarded concurrently.

#### **disable**

Disable clients to be onboarded concurrently.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling concurrent onboarding on a port:

```
switch(config)# interface 1/1/1
switch(config-if)# port-access onboarding-method concurrent enable
```

Disabling concurrent onboarding on a port:

```
switch(config)# interface 1/1/1
switch(config-if)# port-access onboarding-method concurrent disable
```

### Sample configuration:

```
interface 1/1/1
 no shutdown
 no routing
 vlan access 999
 !aaa authentication port-access auth-precedence mac-auth dot1x
 port-access onboarding-method concurrent enable
```

## port-access reauthenticate interface

### Syntax

```
port-access reauthenticate interface <INTERFACE-NAME>
```

### Description

Forcefully reauthenticates all clients connected to an interface.



**NOTE:** Clients that are in the `HELD` state are ignored.

### Command context

Manager (#)

### Parameters

<INTERFACE-NAME>

Specifies the interface name.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring reauthentication of all clients on a port (6200 and 6300 Switch Series):

```
switch# port-access reauthenticate interface 1/1/1
```

Configuring reauthentication of all clients on a port (6400 Switch Series):

```
switch# port-access reauthenticate interface 1/3/1
```

## port-access security violation action

### Syntax

```
port-access security violation action [notify | shutdown]
```

```
no port-access security violation action
```

## Description

Configures the action that the switch must take whenever a security violation occurs at a port, such as the number of clients exceeds the configured client limit.

The `no` form of the command resets the action to the default action, `notify`.

## Command context

`config-if`

## Parameters

### **notify**

Specifies that the switch notifies any security violation as an event or log in the syslog server. This action is the default.

### **shutdown**

Specifies that the switch shuts down the port where the client limit has exceeded.

A port that is shut down can be configured to auto-recover after a recovery period that can be configured with the `port-access security violation action shutdown auto-recovery` and `port-access security violation action shutdown recovery-timer` commands.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring the shutdown security violation action for a port:

```
sswitch(config-if) # port-access security violation action shutdown
```

Resetting the security violation action to the default value:

```
switch(config-if) # no port-access security violation action
```

## port-access security violation action shutdown auto-recovery

## Syntax

```
port-access security violation action shutdown auto-recovery [enable | disable]
```

```
no port-access security violation action shutdown auto-recovery
```

## Description

Configures auto-recovery of the port when the security violation action is configured as shutdown.

This configuration allows the port that is shut down when a security violation occurs to automatically enable after the recovery timer expires.

The `no` form of the command resets auto-recovery to the default, `disable`.

## Command context

`config-if`

## Parameters

### **enable**

Enables auto-recovery of port when the security violation action is configured as shutdown.

### **disable**

Disables auto-recovery of port when the security violation action is configured as shutdown.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling auto-recovery of port:

```
switch(config-if)# port-access security violation action shutdown auto-recovery enable
```

Disabling auto-recovery of port:

```
switch(config-if)# no port-access security violation action shutdown auto-recovery
```

# port-access security violation action shutdown recovery-timer

## Syntax

```
port-access security violation action shutdown recovery-timer <RECOVERY-TIME>
```

```
no port-access security violation action shutdown recovery-timer
```

## Description

Configures security violation recovery timer for the port when the security violation action is configured as shutdown.

The `no` form of the command resets the shutdown recovery timer to the default, 10.

## Command context

config-if

## Parameters

**<RECOVERY-TIME>**

Specifies the recovery timer (in seconds) after which the port, which is shut down because of security violation, is automatically enabled. Default: 10. Range: 10 to 600.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring the shutdown recovery-timer on a port:

```
switch(config-if)# port-access security violation action shutdown recovery-timer 60
```

Resetting the shutdown recovery-timer to the default value:

```
switch(config-if)# no port-access security violation action shutdown recovery-timer
```

## show aaa authentication port-access interface client-status

### Syntax

```
show aaa authentication port-access interface {all|<INTERFACE-NAME>}
 client-status [mac <MAC-ADDRESS>]
```

### Description

Shows information about active port access sessions. The output can be filtered by interface or MAC address.

### Command context

Operator (>) or Manager (#)

### Parameters

**all**

Specifies all interfaces.

**<INTERFACE-NAME>**

Specifies the interface name.

**<MAC-ADDRESS>**

Specifies the client MAC address.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing information for all ports.

```
switch# show aaa authentication port-access interface all client-status
```

```
Port Access Client Status Details
```

```
Client 00:50:56:96:93:d6, John Doe
```

```
=====
```

```
Session Details
```

```

```

```
Port : 1/1/13
Session Time : 30s
```

```
Authentication Details
```

```

```

```
Status : dot1x Authenticated
Auth Precedence : dot1x - Authenticated, mac-auth - Not attempted
```

```
Authorization Details
```

```

```

```
Role : Employee
Status : Applied
```

```

Client 00:50:56:96:50:28
=====
Session Details

Port : 1/1/14
Session Time : 10s

Authentication Details

Status : mac-auth Authenticated
Auth Precedence : dot1x - Unauthenticated, mac-auth - Authenticated

Authorization Details

Role : RADIUS_773420618
Status : Applied

```

## show port-access clients

### Syntax

```
show port-access clients [interface <INTERFACE-NAME>] [mac <MAC-ADDRESS>]
```

### Description

Shows summarized active port access client information. The output can be filtered by interface or MAC address.

### Command context

Operator (>) or Manager (#)

### Parameters

**<INTERFACE-NAME>**

Specifies the interface name.

**<MAC-ADDRESS>**

Specifies the client MAC address.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing information for all clients.

```
switch# show port-access clients
```

```
Port Access Clients
```

| Port  | MAC Address       | Onboarded Method | Status  | Role                   |
|-------|-------------------|------------------|---------|------------------------|
| 1/1/4 | 00:50:56:bd:04:c8 | port-security    | Success |                        |
| 1/1/5 | 00:50:56:bd:32:07 |                  | Success | reject-role, Reject    |
| 1/1/5 | 00:50:56:bd:32:08 |                  | Fail    | critical-..., Critical |

|         |                   |                |             |                         |
|---------|-------------------|----------------|-------------|-------------------------|
| 1/1/6   | 00:50:56:bd:50:43 | mac-auth       | Success     | auth-role, Auth         |
| 1/1/6   | 00:50:56:bd:50:45 | dot1x          | Success     | RADIUS_773420618        |
| 1/1/19  | 08:97:34:ad:e4:00 | device-profile | Success     | ap-role                 |
| 1/1/20  | 00:50:56:bd:32:08 |                | In-Progress | preauth-role, Preauth   |
| 1/1/20  | 00:50:56:bd:32:06 |                | In-Progress |                         |
| 1/1/20  | 00:50:56:bd:32:09 |                | Fail        |                         |
| 1/1/212 | 00:60:56:bd:50:43 | mac-auth       | Success     | fallback-role, Fallback |

Showing information for clients on a particular interface:

```
switch# show port-access clients interface 1/1/5
```

Port Access Clients

| Port  | MAC Address       | Onboarded Method | Status  | Role                   |
|-------|-------------------|------------------|---------|------------------------|
| 1/1/5 | 00:50:56:bd:32:07 |                  | Success | reject-role, Reject    |
| 1/1/5 | 00:50:56:bd:32:08 |                  | Fail    | critical-..., Critical |

Showing information for a particular client MAC address:

```
switch# show port-access clients mac 00:50:56:bd:50:43
```

Port Access Clients

| Port  | MAC Address       | Onboarded Method | Status  | Role            |
|-------|-------------------|------------------|---------|-----------------|
| 1/1/6 | 00:50:56:bd:50:43 | mac-auth         | Success | auth-role, Auth |

## show port-access clients detail

### Syntax

```
show port-access clients [interface <INTERFACE-NAME>] [mac <MAC-ADDRESS>] detail
```

### Description

Shows detailed active port access clients information. The output can be filtered by interface or MAC address.

### Command context

Operator (>) or Manager (#)

### Parameters

**<INTERFACE-NAME>**

Specifies the interface name.

**<MAC-ADDRESS>**

Specifies the client MAC address.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.



## Examples

Showing detailed information for clients on a particular interface (one client):

```
switch# show port-access clients interface 1/1/7 detail
```

```
Port Access Client Status Details:
```

```

```

```
Client 2c:41:38:7f:35:b9, John Doe
```

```
=====
```

```
Session Details
```

```

```

```
Port : 1/1/7
Session Time : 203s
IPv4 Address : 10.10.10.10
IPv6 Address :
```

```
Authentication Details
```

```

```

```
Status : mac-auth Authenticated
Auth Precedence : dot1x - Unauthenticated, mac-auth - Authenticated
```

```
Authorization Details
```

```

```

```
Role : RADIUS_21963402
Status : Applied
```

```
Role Information:
```

```

```

```
Name : RADIUS_21963402
```

```
Type : radius
```

```

```

```
Reauthentication Period : 333 secs
Authentication Mode :
Session Timeout :
Client Inactivity Timeout :
Description :
```

```
Access VLAN : 10
Native VLAN :
Allowed Trunk VLANs :
Access VLAN Name :
Native VLAN Name :
Allowed Trunk VLAN Names :
MTU :
QOS Trust Mode :
STP Administrative Edge Port :
PoE Priority : low
```

```
Policy : PERMIT-ALL_87364653
```

```
Access Policy Details:
```

```

```

```
Policy Name : PERMIT-ALL_87364653
```

```
Policy Type : Local
```

```
Policy Status : Applied
```

| SEQUENCE | CLASS | TYPE | ACTION |
|----------|-------|------|--------|
| 10       | dns   | ipv4 | permit |

```

20 dhcp ipv4 permit
30 clearpass-web ipv4 cir kbps 1024 exceed drop

```

#### Class Details:

```

```

```

class ip dns
 10 match tcp any any
class ip dhcp
 20 match any any any
class ip clearpass-web
 30 match any any any

```

Showing information for a particular client MAC address:

```
switch# show port-access clients mac 2c:41:38:7f:35:c8 detail
```

#### Port Access Client Status Detail

```

```

Client 2c:41:38:7f:35:c8, Jane Q Public

```
=====
```

#### Session Details

```

```

```

Port : 1/1/8
Session Time : 33s
IPv4 Address :
IPv6 Address :

```

#### Authentication Details

```

```

```

Status : mac-auth Authenticated
Auth Precedence : dot1x - Unauthenticated, mac-auth - Authenticated

```

#### Authorization Details

```

```

```

Role : RADIUS_21963402
Status : Applied

```

#### Role Information:

```

```

```

Name : RADIUS_21963402
Type : radius

```

```

```

```

Reauthentication Period : 333 secs
Authentication Mode :
Session Timeout :
Client Inactivity Timeout :
Description :

```

```

Access VLAN : 10
Native VLAN :
Allowed Trunk VLANs :
Access VLAN Name :
Native VLAN Name :
Allowed Trunk VLAN Names :
MTU :
QOS Trust Mode :
STP Administrative Edge Port :
PoE Priority : low

```

```
Policy : PERMIT-ALL_87364653
```

#### Access Policy Details:

```

Policy Name : PERMIT-ALL_87364653
Policy Type : Local
Policy Status : Applied
```

| SEQUENCE | CLASS | TYPE | ACTION |
|----------|-------|------|--------|
| 10       | dns   | ipv4 | permit |
| 20       | dhcp  | ipv4 | permit |

#### Class Details:

```

class ip dns
 10 match tcp any any
class ip dhcp
 20 match any any any
class ip clearpass-web
 30 match any any any
class ip web-traffic
 40 match any any any
```

## show port-access clients onboarding-method

### Syntax

```
show port-access clients onboarding-method <METHOD>
```

### Description

Shows active port access client information for the specified onboarding method.

### Command context

Operator (>) or Manager (#)

### Parameters

<METHOD>

Selects the onboarding method. Available methods: device-profile, dot1x, mac-auth, port-security.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing information for clients onboarded using MAC authentication.

```
switch# show port-access clients onboarding-method mac-auth
```

```
Port Access Clients
```

```
Status codes: device-mode

```

| Port    | MAC-Address       | Onboarding Method | Status  | Role                    |
|---------|-------------------|-------------------|---------|-------------------------|
| 1/1/6   | 00:50:56:bd:50:43 | mac-auth          | Success | auth-role, auth         |
| 1/1/212 | 00:60:56:bd:50:43 | mac-auth          | Success | fallback-role, fallback |

## show port-access security violation client-limit-exceeded interface

### Syntax

```
show port-access security violation client-limit-exceeded interface {all|<INTERFACE-NAME>}
```

### Description

Shows information related to information on the number of client-limit-exceeded security violations that have occurred. The output can be filtered by interface.

### Command context

Operator (>) or Manager (#)

### Parameters

**all**

Specifies all interfaces.

**<INTERFACE-NAME>**

Specifies the interface name.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing information for all ports.

```
switch# show port-access security violation client-limit-exceeded interface all
```

Client Limit Exceeded Violation Status Details

```
Port : 1/1/1
Violation Count : 10
Violation State : Violation Occurred
```

```
Port : 1/1/2
Violation Count : 10
Violation State : None
```

Showing information for a particular port.

```
switch# show port-access security violation client-limit-exceeded interface 1/1/1
```

Client Limit Exceeded Violation Status Details

```
Port : 1/1/1
Violation Count : 10
Violation State : None
```

## aaa authentication port-access dot1x authenticator

### Syntax

```
aaa authentication port-access dot1x authenticator {enable | disable}

no aaa authentication port-access dot1x authenticator
```

### Description

Enables or disables 802.1X authentication globally or at the port-level.

The `no` form of the command deletes global 802.1X configuration details and disables 802.1x authentication.

### Command context

```
config
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling 802.1X authentication globally:

```
switch(config)# aaa authentication port-access dot1x authenticator enable
```

Disabling 802.1X authentication globally:

```
switch(config)# aaa authentication port-access dot1x authenticator disable
```

Deleting and disabling global 802.1X authentication:

```
switch(config)# no aaa authentication port-access dot1x authenticator
```

Enabling 802.1X authentication on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator enable
```

Disabling 802.1X authentication on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator disable
```

Deleting and disabling 802.1X authentication configuration on a port:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator
```

## aaa authentication port-access dot1x authenticator auth-method

### Syntax

```
aaa authentication port-access dot1x authenticator auth-method eap-radius
no aaa authentication port-access dot1x authenticator auth-method eap-radius
```

### Description

Configures the authentication mechanism used to control access to the network. The configured authentication method will be used to authenticate 802.1X clients.

The `no` form of the command resets the authentication mechanism to the default, `eap-radius`.

### Command context

`config`

### Parameters

#### **eap-radius**

Specifies the EAP RADIUS as the 802.1X authentication method.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling the EAP RADIUS 802.1X authentication method on the switch:

```
switch(config)# aaa authentication port-access dot1x authenticator auth-method eap-radius
```

Resetting the EAP RADIUS 802.1X authentication method on the switch:

```
switch(config)# no aaa authentication port-access dot1x authenticator auth-method eap-radius
```

## aaa authentication port-access dot1x authenticator cached-reauth

### Syntax

```
aaa authentication port-access dot1x authenticator cached-reauth
no aaa authentication port-access dot1x authenticator cached-reauth
```

### Description

Enables cached reauthentication on a port. Cached reauthentication allows 802.1X reauthentications to succeed when the RADIUS server is unavailable. Users already authenticated retain their currently assigned RADIUS attributes.

The `no` form of the command disables the cached reauthentication on a port.

### Command context

`config-if`

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling cached reauthentication on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator cached-reauth
```

Disabling cached reauthentication on a port:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator cached-reauth
```

# aaa authentication port-access dot1x authenticator cached-reauth-period

## Syntax

```
aaa authentication port-access dot1x authenticator cached-reauth-period <PERIOD>
```

```
no aaa authentication port-access dot1x authenticator cached-reauth-period
```

## Description

Configures the period during which an authenticated client, which has failed to reauthenticate because the RADIUS server is unreachable, remains authenticated.

The `no` form of the command resets the cached reauthentication period to the default, 30 seconds.

## Command context

config-if

## Parameters

<PERIOD>

Specifies the cached reauthentication period (in seconds). Default: 30. Range: 30 to 86400.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring the cached reauthentication period on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator cached-reauth-period 300
```

Resetting the cached reauthentication period to the default value:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator cached-reauth-period
```

## aaa authentication port-access dot1x authenticator discovery-period

### Syntax

```
aaa authentication port-access dot1x authenticator discovery-period <PERIOD>

no aaa authentication port-access dot1x authenticator discovery-period
```

### Description

Configures the period the port waits to retransmit the next EAPOL request identity frame on an 802.1X enabled port that has no authenticated clients.

The `no` form of the command resets the discovery period to the default, 30 seconds.

### Command context

config-if

### Parameters

<PERIOD>

Specifies the discovery period (in seconds). Default: 30. Range: 1 to 65535.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring the discovery period on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator discovery-period 120
```

Resetting the discovery period to the default value:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator discovery-period
```

## aaa authentication port-access dot1x authenticator eapol-timeout

### Syntax

```
aaa authentication port-access dot1x authenticator eapol-timeout <EAPOL-TIMEOUT>

no aaa authentication port-access dot1x authenticator eapol-timeout
```

### Description

Configure the period the switch waits for a response from a client before retransmitting an EAPOL PDU. If the value is 0, the time period is calculated as per RFC 2988.



**NOTE:** As per RFC 2988 2.1: Before Round-Trip Time (RTT) measurement, set Retransmission Timeout (RTO) to 3 seconds for initial retransmission and then double the RTO to provide back off as per section 5.5. Limit the maximum RTO (RTOmax) to 20 seconds as per section 4.3 of RFC 3748.



The `no` form of the command resets the timeout period to the default.

### Command context

`config-if`

### Parameters

**<EAPOL-TIMEOUT>**

Specifies the EAPOL timeout period (in seconds). Range: 1 to 65535.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring EAPOL timeout on a port:

```
switch(config-if) # aaa authentication port-access dot1x authenticator eapol-timeout 120
```

Resetting the EAPOL timeout to the default value:

```
switch(config-if) # no aaa authentication port-access dot1x authenticator eapol-timeout
```

## aaa authentication port-access dot1x authenticator max-eapol-requests

### Syntax

```
aaa authentication port-access dot1x authenticator max-eapol-requests <MAX-EAPOL-REQUESTS>
```

```
no aaa authentication port-access dot1x authenticator max-eapol-requests
```

### Description

Configures the number of EAPOL requests to send to a supplicant that must time out before authentication fails and the authentication session ends.

The `no` form of the command resets the maximum number of EAPOL requests to the default, 5.

### Command context

`config-if`

### Parameters

**<MAX-EAPOL-REQUESTS>**

Specifies the maximum number of EAPOL requests. Default: 5. Range: 1 to 10.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring maximum EAPOL requests on a port:

```
switch(config-if) # aaa authentication port-access dot1x authenticator max-eapol-requests 3
```

Resetting the maximum EAPOL requests on a port to default:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator max-eapol-requests
```

## port-access dot1x authenticator max-retries

### Syntax

```
aaa authentication port-access dot1x authenticator max-retries <max-retries>
```

```
no aaa authentication port-access dot1x authenticator max-retries
```

### Description

Configures the maximum number of retries that the switch attempts to authenticate a client on a port before marking the client as unauthenticated.

The `no` form of the command resets the maximum number of retries to the default, 2.

### Command context

```
config-if
```

### Parameters

**<max-retries>**

Indicates the number of authentication attempts. Default: 2. Range: 1 to 10.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring maximum authentication attempts on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator max-retries 5
```

Resetting the maximum authentication attempts on a port to default:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator max-retries
```

## aaa authentication port-access dot1x authenticator quiet-period

### Syntax

```
aaa authentication port-access dot1x authenticator quiet-period <PERIOD>
```

```
no aaa authentication port-access dot1x authenticator quiet-period
```

### Description

Configures the period during which the port does not try to acquire a supplicant. This period begins after the last authentication attempt, authorized by the maximum retries parameter, fails.

You can configure the number of maximum retries with the `aaa authentication port-access dot1x authenticator max-retries` command.

The `no` form of the command resets the quiet period to the default, 60 seconds.

## Command context

config-if

## Parameters

<PERIOD>

Specifies the quiet period (in seconds). Default: 60. Range: 0 to 65535.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring quiet period on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator quiet-period 100
```

Resetting the quiet period on a port to default:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator quiet-period
```

# aaa authentication port-access dot1x authenticator radius server-group

## Syntax

```
aaa authentication port-access dot1x authenticator radius server-group <GROUP-NAME>
```

```
no aaa authentication port-access dot1x authenticator radius server-group
```

## Description

Configures the switch to use an existing RADIUS server group for 802.1X authentication.

The **no** form of the command resets the server group to the default, **radius**.

## Command context

config

## Parameters

<GROUP-NAME>

Specifies the name of the RADIUS server group.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring the switch to use RADIUS server group **employee**:

```
switch(config)# aaa authentication port-access dot1x authenticator radius server-group employee
```

Resetting RADIUS server group configuration to default:

```
switch(config)# no aaa authentication port-access dot1x authenticator radius server-group
```

## aaa authentication port-access dot1x authenticator reauth

### Syntax

```
aaa authentication port-access dot1x authenticator reauth
no aaa authentication port-access dot1x authenticator reauth
```

### Description

Enables periodic reauthentication of authenticated clients on the port.

The `no` form of the command disables periodic reauthentication.

### Command context

config-if

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling periodic reauthentication on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator reauth
```

Disabling periodic reauthentication on a port:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator reauth
```

## aaa authentication port-access dot1x authenticator reauth-period

### Syntax

```
aaa authentication port-access dot1x authenticator reauth-period <PERIOD>
no aaa authentication port-access dot1x authenticator reauth-period
```

### Description

Configures the period after which the authenticated clients are reauthenticated on the port. You must enable reauthentication on the port before configuring the reauthentication period.

The `no` form of the command resets the reauthentication period to the default, 3600 seconds.

### Command context

config-if

### Parameters

<PERIOD>

Specifies the reauthentication period (in seconds). Default: 3600. Range: 1 to 65535.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring reauthentication period on a port:

```
switch(config-if)# aaa authentication port-access dot1x authenticator reauth-period 100
```

Resetting the reauthentication period to the default value:

```
switch(config-if)# no aaa authentication port-access dot1x authenticator reauth-period
```

# clear dot1x authenticator statistics interface

## Syntax

```
clear dot1x authenticator statistics [interface <IF-NAME>]
```

## Description

Clears the 802.1X authentication statistics associated with the port and all the authenticator clients attached to this port.

If no interface is specified, the statistics is cleared for all 802.1X enabled ports.

## Command context

Manager (#)

## Parameters

**<IF-NAME>**

Specifies the interface name.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Clearing authentication statistics on a port (6200 and 6300 Switch Series):

```
switch# clear dot1x authenticator statistics interface 1/1/1
```

Clearing authentication statistics on a port (6400 Switch Series):

```
switch# clear dot1x authenticator statistics interface 1/3/1
```

Clearing authentication statistics on all ports:

```
switch# clear dot1x authenticator statistics
```

# show aaa authentication port-access dot1x authenticator interface client-status

## Syntax

```
show aaa authentication port-access dot1x authenticator interface {all|<IF-NAME>}
client-status [mac <MAC-ADDRESS>]
```

## Description

Shows information about active 802.1X authentication sessions. The output can be filtered by interface or MAC address.

## Command context

Operator (>) or Manager (#)

## Parameters

**all**

Specifies all interfaces.

**<IF-NAME>**

Specifies the interface name.

**<MAC-ADDRESS>**

Specifies the client MAC address.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing client status information for all ports.

```
switch# show aaa authentication port-access dot1x authenticator interface all client-status
Client FE:04:D7:50:89:37, johndoe, 1/1/1
=====

Authentication Details

 Status : Authenticated
 Type : Pass-Through
 EAP-Method : MD5
 Time Since Last State Change : 10s

Authentication Statistics

 Authentication : 0
 Authentication Timeout : 0
 EAP-Start While Authenticating : 0
 EAP-Logoff While Authenticating : 0
 Successful Authentication : 0
 Failed Authentication : 0
 Re-Authentication : 0
 Successful Re-Authentication : 0
 Failed Re-Authentication : 0
 EAP-Start When Authenticated : 0
 EAP-Logoff When Authenticated : 0
```

```

Re-Auths When Authenticated : 0
Cached Re-Authentication : 0

Client 9A:B4:59:97:D0:7E, janedoe, 1/1/1
=====

Authentication Details

Status : Authenticated
Type : Pass-Through
EAP-Method : TLS
Time Since Last State Change : 5s

Authentication Statistics

Authentication : 0
Authentication Timeout : 0
EAP-Start While Authenticating : 0
EAP-Logoff While Authenticating : 0
Successful Authentication : 0
Failed Authentication : 0
Re-Authentication : 0
Successful Re-Authentication : 0
Failed Re-Authentication : 0
EAP-Start When Authenticated : 0
EAP-Logoff When Authenticated : 0
Re-Auths When Authenticated : 0
Cached Re-Authentication : 0

```

## show aaa authentication port-access dot1x authenticator interface port-statistics

### Syntax

```
show aaa authentication port-access dot1x authenticator interface {all|<IF-NAME>} port-statistics
```

### Description

Shows information about 802.1X ports. The output can be filtered by interface.

### Command context

Operator (>) or Manager (#)

### Parameters

**all**

Specifies all interfaces.

**<IF-NAME>**

Specifies the interface name.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing information for all ports.

```
switch# show aaa authentication port-access dot1x authenticator interface all port-statistics
```

```
Port 1/1/1
```

```
=====
```

```
Client Details
```

```

```

```
Number of Clients : 1
Number of Authenticated Clients : 1
Number of Unauthenticated Clients : 0
```

```
Statistics
```

```

```

```
EAPOL Frames Received : 4
EAPOL Frames Transmitted : 3
EAPOL Start Frames Received : 1
EAPOL Logoff Frames Received : 0
EAPOL Response ID Frames Received : 2
EAPOL Response Frames Received : 1
EAPOL Request ID Frames Transmitted : 2
EAPOL Request Frames Transmitted : 1
EAPOL Invalid Frames Received : 0
EAPOL EAP Length Error Frames Received : 0
EAPOL Last Received Frame Version : 0
EAPOL Last Received Frame Client MAC : 0
```

```
Port 1/1/2
```

```
=====
```

```
Client Details
```

```

```

```
Number of Clients : 1
Number of Authenticated Clients : 1
Number of Unauthenticated Clients : 0
```

```
Statistics
```

```

```

```
EAPOL Frames Received : 4
EAPOL Frames Transmitted : 3
EAPOL Start Frames Received : 1
EAPOL Logoff Frames Received : 0
EAPOL Response ID Frames Received : 2
EAPOL Response Frames Received : 1
EAPOL Request ID Frames Transmitted : 2
EAPOL Request Frames Transmitted : 1
EAPOL Invalid Frames Received : 0
EAPOL EAP Length Error Frames Received : 0
EAPOL Last Received Frame Version : 0
EAPOL Last Received Frame Client MAC : 0
```



## aaa authentication port-access mac-auth

### Syntax

```
aaa authentication port-access mac-auth {enable | disable}
```

### Description

Enables or disables MAC authentication globally or at the port-level.

### Command context

config

config-if

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling MAC authentication on all interfaces:

```
switch(config)# aaa authentication port-access mac-auth
switch(config-macauth)# enable
```

Disabling MAC authentication on all interfaces:

```
switch(config)# aaa authentication port-access mac-auth
switch(config-macauth)# disable
```

Enabling MAC authentication on an interface:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# enable
```

Disabling MAC authentication on an interface:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# disable
```

## aaa authentication port-access mac-auth addr-format

### Syntax

```
aaa authentication port-access mac-auth addr-format {no-delimiter | single-dash |
multi-dash | multi-colon | no-delimiter-uppercase | single-dash-uppercase |
multi-dash-uppercase | multi-colon-uppercase}
```

```
no aaa authentication port-access mac-auth addr-format {no-delimiter | single-dash |
multi-dash |multi-colon | no-delimiter-uppercase | single-dash-uppercase |
multi-dash-uppercase | multi-colon-uppercase}
```

### Description

Configures the MAC address format that the switch must use in the RADIUS request message.

The `no` form of the command resets the MAC address format to the default, `no-delimiter`.

### Command context

config

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the MAC address format on the switch:

```
switch(config)# aaa authentication port-access mac-auth
switch(config-macauth)# addr-format single-dash
```

Resetting the MAC address format on the switch to its default:

```
switch(config)# aaa authentication port-access mac-auth
switch(config-macauth)# no addr-format
```

## aaa authentication port-access mac-auth auth-method

### Syntax

```
aaa authentication port-access mac-auth auth-method {chap | pap}
```

```
no aaa authentication port-access mac-auth auth-method
```

### Description

Configures the RADIUS authentication method for MAC authentication.

Following are the MAC authentication methods supported:

- CHAP
- PAP



---

**NOTE:** The PEAP-MSCHAPv2 method of authentication is not supported.

---

The `no` form of the command resets the authentication method to the default, `chap`.

### Command context

config

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring the RADIUS authentication method on the switch:

```
switch# config
switch(config)# aaa authentication port-access mac-auth
switch(config-macauth)# auth-method pap
```

Resetting the RADIUS authentication method on the switch:

```
switch(config)# no aaa authentication port-access mac-auth auth-method
```

# aaa authentication port-access mac-auth cached-reauth

## Syntax

```
aaa authentication port-access mac-auth cached-reauth
```

```
no aaa authentication port-access mac-auth cached-reauth
```

## Description

Enables cached reauthentication on a port. Cached reauthentication allows MAC reauthentications to succeed when the RADIUS server is unavailable. Users who are already authenticated, retain their currently assigned RADIUS attributes.

The `no` form of the command disables cached reauthentication.

## Command context

```
config-if
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling cached reauthentication on a port:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# cached-reauth
```

Disabling cached reauthentication on a port:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# no cached-reauth
```

## aaa authentication port-access mac-auth cached-reauth-period

### Syntax

```
aaa authentication port-access mac-auth cached-reauth-period <PERIOD>
no aaa authentication port-access mac-auth cached-reauth-period
```

### Description

Configures the period during which an authenticated client, which has failed to reauthenticate because the RADIUS server is unreachable, remains authenticated.

The `no` form of the command resets the cached reauthentication period to the default, 30 seconds.

### Command context

config-if

### Parameters

<PERIOD>

Specifies the cached reauthentication period (in seconds). Default: 30. Range: 30 to 86400.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring cached reauthentication period on a port:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# cached-reauth-period 300
```

Resetting the cached reauthentication period to the default value:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# no cached-reauth-period
```

## aaa authentication port-access mac-auth password

### Syntax

```
aaa authentication port-access mac-auth password {plaintext|ciphertext}<PASSWORD>
no aaa authentication port-access mac-auth password
```

### Description

Enables and configures the global password that the switch must use for MAC authentication. The password can be either in ciphertext or plaintext format.

The `no` form of the command disables the password for MAC authentication.

## Command context

config

## Parameters

**<PASSWORD>**

Specifies the global password to be used by all MAC authenticating devices.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the MAC authentication password:

```
switch(config)# aaa authentication port-access mac-auth
switch(config-macauth)# password {ciphertext|plaintext}
```

Disabling the MAC authentication password:

```
switch(config)# aaa authentication port-access mac-auth
switch(config-macauth)# no password
```

# aaa authentication port-access mac-auth quiet-period

## Syntax

```
aaa authentication port-access mac-auth quiet-period <PERIOD>
```

```
no aaa authentication port-access mac-auth quiet-period
```

## Description

Configures the period during which the switch does not try to authenticate a rejected client.

The **no** form of the command resets the quiet period to the default, 60 seconds.

## Command context

config-if

## Parameters

**<PERIOD>**

Specifies the quiet period (in seconds). Default: 60. Range: 0 to 65535.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring the quiet period on a port:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# quiet-period 65
```

Resetting the quiet period on a port to default:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# no quiet-period
```

## aaa authentication port-access mac-auth radius server-group

### Syntax

```
aaa authentication port-access mac-auth radius server-group <GROUP-NAME>

no aaa authentication port-access mac-auth radius server-group
```

### Description

Configures the MAC authentication server group.

The **no** form of the command resets the authentication server group to the default value, **radius**.

### Command context

config

### Parameters

<GROUP-NAME>

Specifies the MAC authentication server group name.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring the MAC authentication server group:

```
switch# config
switch(config)# aaa authentication port-access mac-auth
switch(config-macauth)# radius server-group group1
```

## aaa authentication port-access mac-auth reauth

### Syntax

```
aaa authentication port-access mac-auth reauth

no aaa authentication port-access mac-auth reauth
```

### Description

Enables periodic MAC reauthentication of authenticated clients on the port.

The **no** form of the command disables periodic MAC reauthentication on the port.

### Command context

config-if

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling reauthentication on a port:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# reauth
```

Disabling reauthentication on a port:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# no reauth
```

# aaa authentication port-access mac-auth reauth-period

## Syntax

```
aaa authentication port-access mac-auth reauth-period <PERIOD>
```

```
no aaa authentication port-access mac-auth reauth-period
```

## Description

Configures the period after which MAC authenticated clients must be reauthenticated on the port. You must first enable MAC reauthentication on the port before configuring the MAC reauthentication period.

The `no` form of the command resets the MAC reauthentication period to the default, 3600 seconds.

## Command context

```
config-if
```

## Parameters

<PERIOD>

Specifies the MAC reauthentication period (in seconds). Default: 3600. Range: 1 to 65535.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring the MAC reauthentication period on a port:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# reauth-period 60
```

Resetting the MAC reauthentication period to its default:

```
switch(config-if)# aaa authentication port-access mac-auth
switch(config-if-macauth)# no reauth-period
```

## clear mac-auth statistics

### Syntax

```
clear mac-auth statistics [interface <IF-NAME>]
```

### Description

Clears the MAC authentication statistics associated with the port and all the authenticator state machines associated to this port.

If no interface is specified, the statistics is cleared for all MAC authentication enabled ports.

### Command context

Operator (>) or Manager (#)

### Parameters

**<IF-NAME>**

Specifies the interface name.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Clearing MAC authentication statistics on a port (6200 and 6300 Switch Series):

```
switch# clear mac-auth statistics interface 1/1/1
```

Clearing MAC authentication statistics on a port (6400 Switch Series):

```
switch# clear mac-auth statistics interface 1/3/1
```

Clearing MAC authentication statistics on all ports:

```
switch# clear mac-auth statistics
```

## show aaa authentication port-access mac-auth interface client-status

### Syntax

```
show aaa authentication port-access mac-auth interface {all|<IF-NAME>}
client-status [mac <MAC-ADDRESS>]
```

### Description

Shows information about MAC authentication clients status. The output can be filtered by interface or MAC address.

### Command context

Operator (>) or Manager (#)



## Parameters

**all**

Specifies all interfaces.

**<IF-NAME>**

Specifies the interface name.

**<MAC-ADDRESS>**

Specifies the client MAC address.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing client status information for all ports:

```
switch# show aaa authentication port-access mac-auth interface all client-status
```

```
Port Access Client Status Details
```

```
Client AB:CD:DE:FF:AA:BB, 1/1/1
```

```
=====
```

```
Authentication Details
```

```

```

|                              |                 |
|------------------------------|-----------------|
| Status                       | : Authenticated |
| Type                         | : Pass-Through  |
| Auth-Method                  | : CHAP          |
| Time Since Last State Change | : 10 secs       |

```
Authentication Statistics
```

```

```

|                              |     |
|------------------------------|-----|
| Authentication               | : 1 |
| Authentication Timeout       | : 0 |
| Successful Authentication    | : 1 |
| Failed Authentication        | : 0 |
| Re-Authentication            | : 0 |
| Successful Re-Authentication | : 0 |
| Failed Re-Authentication     | : 0 |
| Re-Auths When Authenticated  | : 0 |
| Cached Re-Authentication     | : 0 |

```
Client DD:CD:AB:CS:EE:OI, 1/1/2
```

```
=====
```

```
Authentication Details
```

```

```

|                              |                                 |
|------------------------------|---------------------------------|
| Status                       | : Unauthenticated               |
| Type                         | : Pass-Through                  |
| Auth-Method                  | : CHAP                          |
| Auth Failure reason          | : Server reject/ Server timeout |
| Time Since Last State Change | : 15 secs                       |

```
Authentication Statistics
```

```

```

|                           |     |
|---------------------------|-----|
| Authentication            | : 1 |
| Authentication Timeout    | : 0 |
| Successful Authentication | : 0 |
| Failed Authentication     | : 1 |
| Re-Authentication         | : 0 |

```
Successful Re-Authentication : 0
Failed Re-Authentication : 0
Re-Auths When Authenticated : 0
Cached Re-Authentication : 0
```

Showing status information for a client:

```
switch# show aaa authentication port-access mac-auth interface 1/1/1 client-status mac ab:cd:de:ff:aa:bb

Port Access Client Status Details

Client AB:CD:DE:FF:AA:BB, 1/1/1
=====
Authentication Details

 Status : Authenticated
 Type : Pass-Through
 Auth-Method : CHAP
 Time Since Last State Change : 10 secs

Authentication Statistics

 Authentication : 1
 Authentication Timeout : 0
 Successful Authentication : 1
 Failed Authentication : 0
 Re-Authentication : 0
 Successful Re-Authentication : 0
 Failed Re-Authentication : 0
 Re-Auths When Authenticated : 0
 Cached Re-Authentication : 0
```

## show aaa authentication port-access mac-auth interface port-statistics

### Syntax

```
show aaa authentication port-access mac-auth interface {all|<IF-NAME>} port-statistics
```

### Description

Shows information about MAC authentication ports. The output can be filtered by interface.

### Command context

Operator (>) or Manager (#)

### Parameters

**all**

Specifies all interfaces.

**<IF-NAME>**

Specifies the interface name.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing information for all ports.

```
switch# show aaa authentication port-access mac-auth interface all port-statistics
```

```
Port 1/1/1
```

```
=====
```

```
Client Details
```

```

```

```
Number of Clients : 3
Number of authenticated clients : 2
Number of unauthenticated clients : 1
Number of authenticating clients : 0
```

```
Port 1/1/2
```

```
=====
```

```
Client Details
```

```

```

```
Number of Clients : 4
Number of authenticated clients : 2
Number of unauthenticated clients : 2
Number of authenticating clients : 0
```

## port-access policy

### Syntax

```
port-access policy <POLICY-NAME>

 [<SEQUENCE-NUMBER>]
 class {ip|ipv6} <CLASS-NAME>
 action {<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}

 [<SEQUENCE-NUMBER>]
 comment ...
```

### Description

Creates or modifies policy and policy entries. A policy is made up of one or more policy entries ordered and prioritized by sequence numbers. Each entry has an IPv4/IPv6 class and one or more policy actions associated with it.

A policy must be applied to a role using the `associate policy` command.

The `no` form of the command can be used to delete either a policy (use `no` with the policy command) or an individual policy entry (use `no` with the sequence number).

### Command context

```
config
```

The `policy` command takes you into the `config-pa-policy` context where you enter the policy entries.

### Parameters

**<POLICY-NAME>**

Specifies the name of the policy.

**<SEQUENCE-NUMBER>**

Specifies a sequence number for the policy entry. Optional. Range: 1 to 4294967295.

**comment**

Specifies the description of an entry in the policy. Optional.

**class {ip|ipv6} <CLASS-NAME>**

Specifies a type of class, `ip` for IPv4 and `ipv6` for IPv6 policy. And specifies a class name.

**<REMARK-ACTIONS>**

Remark actions can be any of the following options: {`ip-precedence <IP-PRECEDENCE-VALUE>` | `dscp <DSCP-VALUE>` | `local-priority <LOCAL-PRIORITY-VALUE>`} where:

**ip-precedence** <IP-PRECEDENCE-VALUE>

Specifies the numeric IP precedence value. Range: 0 to 7.

**dscp** <DSCP-VALUE>

Specifies a Differentiated Services Code Point (DSCP) value. Enter either a numeric value (0 to 63) or a keyword as follows:

| Keyword | Status   | Syntax  | Description                                                   |
|---------|----------|---------|---------------------------------------------------------------|
| AF11    | Optional | Keyword | DSCP 10 (Assured Forwarding Class 1, low drop probability)    |
| AF12    | Optional | Keyword | DSCP 12 (Assured Forwarding Class 1, medium drop probability) |
| AF13    | Optional | Keyword | DSCP 14 (Assured Forwarding Class 1, high drop probability)   |
| AF21    | Optional | Keyword | DSCP 18 (Assured Forwarding Class 2, low drop probability)    |
| AF22    | Optional | Keyword | DSCP 20 (Assured Forwarding Class 2, medium drop probability) |
| AF23    | Optional | Keyword | DSCP 22 (Assured Forwarding Class 2, high drop probability)   |
| AF31    | Optional | Keyword | DSCP 26 (Assured Forwarding Class 3, low drop probability)    |
| AF32    | Optional | Keyword | DSCP 28 (Assured Forwarding Class 3, medium drop probability) |
| AF33    | Optional | Keyword | DSCP 30 (Assured Forwarding Class 3, high drop probability)   |
| AF41    | Optional | Keyword | DSCP 34 (Assured Forwarding Class 4, low drop probability)    |
| AF42    | Optional | Keyword | DSCP 36 (Assured Forwarding Class 4, medium drop probability) |
| AF43    | Optional | Keyword | DSCP 38 (Assured Forwarding Class 4, high drop probability)   |
| CS0     | Optional | Keyword | DSCP 0 (Class Selector 0: Default)                            |
| CS1     | Optional | Keyword | DSCP 8 (Class Selector 1: Scavenger)                          |

*Table Continued*

| Keyword | Status   | Syntax  | Description                                 |
|---------|----------|---------|---------------------------------------------|
| CS2     | Optional | Keyword | DSCP 16 (Class Selector 2: OAM)             |
| CS3     | Optional | Keyword | DSCP 24 (Class Selector 3: Signaling)       |
| CS4     | Optional | Keyword | DSCP 32 (Class Selector 4: Real time)       |
| CS5     | Optional | Keyword | DSCP 40 (Class Selector 5: Broadcast video) |
| CS6     | Optional | Keyword | DSCP 48 (Class Selector 6: Network control) |
| CS7     | Optional | Keyword | DSCP 56 (Class Selector 7)                  |
| EF      | Optional | Keyword | DSCP 46 (Expedited Forwarding)              |

**local-priority** *<LOCAL-PRIORITY-VALUE>*

Specifies a local priority value. Range: 0 to 7.

#### **<POLICE-ACTIONS>**

Police actions can be the following {cir kbps *<RATE-BPS>* exceed} where:

**cir kbps** *<RATE-BPS>*

Specifies a Committed Information Rate (CIR) value in Kilobits per second. Range: 1 to 4294967295.

**exceed**

Specifies action to take on packets that exceed the rate limit.

#### **<OTHER-ACTIONS>**

Other actions can be the following:

**drop**

Specifies drop traffic.

### **Authority**

Administrators or local user group members with execution rights for this command.

### **Usage**

- An applied policy processes the packet sequentially against policy and class entries in the list, until either the last policy entry in the list has been evaluated or the packet matches an entry. If there is no match, the packet will be dropped by one of the implicit `deny all` IPv4 and IPv6 entries.
- Entering an existing *<POLICY-NAME>* value will cause the existing policy to be modified, with any new *<SEQUENCE-NUMBER>* value creating an additional policy entry, and any existing *<SEQUENCE-NUMBER>* value replacing the existing policy entry with the same sequence number.
- If no sequence number is specified, a new policy entry will be appended to the end of the entry list with a sequence number equal to the highest policy entry currently in the list plus 10. The sequence numbers may be reordered with the `port-access policy <POLICY-NAME> resequence <STARTING-SEQ-NUM> <INCREMENT>` command.
- If a policy is configured without any action, the default action, `permit`, is applied for that policy.

## Examples

Creating a policy with several class entries:

```
switch(config)# port-access policy POL1
switch(config-pa-policy)# 10 class ip dns
switch(config-pa-policy)# 20 class ip dhcp
switch(config-pa-policy)# 30 class ip test action cir kbps 1024 exceed drop
switch(config-pa-policy)# exit
switch(config)# show port-access policy POL1
```

Access Policy Details:  
=====

Policy Name : POL1  
Policy Type : Local  
Policy Status : Applied

| SEQUENCE | CLASS | TYPE | ACTION                    |
|----------|-------|------|---------------------------|
| 10       | dns   | ipv4 | permit                    |
| 20       | dhcp  | ipv4 | permit                    |
| 30       | test  | ipv4 | cir kbps 1024 exceed drop |

Adding a comment to an existing class entry:

```
switch(config)# port-access policy POL1
switch(config-pa-policy)# 20 comment DHCP-PERMIT
switch(config-pa-policy)# exit
switch(config)# show run port-access policy POL1
```

```
port-access policy POL1
 10 class ip dns
 20 class ip dhcp
 20 comment DHCP-PERMIT
 30 class ip test action cir kbps 1024 exceed drop
```

Removing a comment from an existing class entry:

```
switch(config)# port-access policy POL1
switch(config-pa-policy)# no 20 comment
switch(config-pa-policy)# exit
switch(config)# show run port-access policy POL1
```

```
port-access policy POL1
 10 class ip dns
 20 class ip dhcp
 30 class ip test action cir kbps 1024 exceed drop
```

Modifying a policy by replacing one class with another at the same sequence number:

```
switch(config)# port-access policy POL1
switch(config-pa-policy)# 10 class ip mds action dscp af21
switch(config-pa-policy)# exit
switch(config)# show port-access policy POL1
```

Access Policy Details:  
=====

Policy Name : POL1  
Policy Type : Local  
Policy Status : Applied

| SEQUENCE | CLASS | TYPE | ACTION |
|----------|-------|------|--------|
|----------|-------|------|--------|

```

10 mds ipv4 dscp AF21
20 dhcp ipv4 permit
30 test ipv4 cir kbps 1024 exceed drop

```

Removing a class:

```

switch(config)# port-access policy POL1
switch(config-pa-policy)# no 10
switch(config-pa-policy)# exit
switch(config)# show port-access policy POL1

```

Access Policy Details:

=====

```

Policy Name : POL1
Policy Type : Local
Policy Status : Applied

```

| SEQUENCE | CLASS         | TYPE | ACTION                    |
|----------|---------------|------|---------------------------|
| 20       | dhcp          | ipv4 | permit                    |
| 30       | clearpass-web | ipv4 | cir kbps 1024 exceed drop |

## port-access policy copy

### Syntax

```
port-access policy <POLICY-NAME> copy <DESTINATION-POLICY>
```

### Description

Copies details of an existing policy to a new policy. Copying a policy copies all its entries as well.

### Command context

config

### Parameters

**<POLICY-NAME>**

Specifies the existing policy name.

**<DESTINATION-POLICY>**

Specifies the destination policy name.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Copying a policy:

```

switch(config)# port-access policy POL1 copy POL1_copy
switch(config)# show port-access policy

```

Access Policy Details:

=====

```

Policy Name : POL1

```



```
Policy Type : Local
Policy Status : Applied
```

| SEQUENCE | CLASS | TYPE | ACTION                    |
|----------|-------|------|---------------------------|
| 20       | dhcp  | ipv4 | permit                    |
| 30       | test  | ipv4 | cir kbps 1024 exceed drop |

```
Policy Name : POL1_copy
Policy Type : Local
Policy Status : Applied
```

| SEQUENCE | CLASS | TYPE | ACTION                    |
|----------|-------|------|---------------------------|
| 20       | dhcp  | ipv4 | permit                    |
| 30       | test  | ipv4 | cir kbps 1024 exceed drop |

## port-access policy resequence

### Syntax

```
port-access policy <POLICY-NAME> resequence <STARTING-SEQ-NUM> <INCREMENT>
```

### Description

Resequences numbering in a policy.

### Command context

```
config
```

### Parameters

**<POLICY-NAME>**

Specifies the policy where you want to resequence policy entries.

**<STARTING-SEQ-NUM>**

Specifies the sequence number to start resequencing from. Range: 1-4294967295.

**<INCREMENT>**

Specifies how much to increment the sequence numbers by.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Resequencing a policy starting at 5 with an increment of 10:

```
switch(config)# port-access policy POL1 resequence 5 10
switch(config)# show port-access policy POL1
```

```
Access Policy Details:
=====
```

```
Policy Name : POL1
Policy Type : Local
Policy Status : Applied
```

| SEQUENCE | CLASS | TYPE | ACTION |
|----------|-------|------|--------|
|----------|-------|------|--------|

|    |      |      |                           |
|----|------|------|---------------------------|
| 5  | dhcp | ipv4 | permit                    |
| 15 | test | ipv4 | cir kbps 1024 exceed drop |

## port-access policy reset

### Syntax

```
port-access policy <POLICY-NAME> reset
```

### Description

Resets the policy configuration to match the current hardware configuration of the policy.

### Command context

```
config
```

### Parameters

**<POLICY-NAME>**

Specifies the name of the policy to be reset.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Resetting a policy:

```
switch(config)# port-access policy POL2
switch(config-pa-policy)# 20 class ip dhcp
switch(config-pa-policy)# 40 class test2 action cir kbps 1024 exceed drop
switch(config-pa-policy)# exit
switch(config)# show port-access policy POL1-V2
```

Access Policy Details:

=====

```
Policy Name : POL2
Policy Type : Local
Policy Status : Applied
```

| SEQUENCE | CLASS | TYPE | ACTION                    |
|----------|-------|------|---------------------------|
| 20       | dhcp  | ipv4 | permit                    |
| 40       | test2 | ipv4 | cir kbps 1024 exceed drop |

```
switch(config)# port-access policy POLV2
switch(config-pa-policy)# 50 class ip test3 action cir kbps 1024 exceed drop
switch(config-pa-policy)# no 20
switch(config-pa-policy)# exit
switch(config)# show port-access policy POL2
```

Access Policy Details:

=====

```
Policy Name : POL2
Policy Type : Local
Policy Status : Rejected
```

| SEQUENCE | CLASS | TYPE | ACTION                    |
|----------|-------|------|---------------------------|
| 40       | test2 | ipv4 | cir kbps 1024 exceed drop |
| 50       | test3 | ipv4 | cir kbps 1024 exceed drop |

```

switch(config)# port-access policy POL2 reset
Following policy entries will be removed:
class ip test3 action cir kbps 1024 exceed drop

Following policy entries will be added:
20 class ip dhcp

Do you want to continue (y/n)? y
switch(config)# show port-access policy POL2

Access Policy Details:
=====

Policy Name : POL1-V2
Policy Type : Local
Policy Status : Applied

```

| SEQUENCE | CLASS | TYPE | ACTION                    |
|----------|-------|------|---------------------------|
| 20       | dhcp  | ipv4 | permit                    |
| 40       | test2 | ipv4 | cir kbps 1024 exceed drop |

## clear port-access policy hitcounts

### Syntax

```
clear port-access policy <POLICY-NAME> hitcounts {port | client}
```

### Description

Clears statistics and conform rate of policy applied on a port or client.

### Command context

Operator (>) or Manager (#)

### Parameters

**<POLICY-NAME>**

Specifies the policy name.

**port**

Clears the policy statistics matching policy name in the port mode.

**client**

Clears the policy statistics matching policy name in the client mode.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Clearing policy hit counts:

```
switch# show port-access policy POL6 hitcounts port
```

```
Port Access Policy Hit-Counts Details:
```

```
=====
```

```
Policy Name : POL4
Policy Type : Local
Policy Status : Applied
```

| SEQUENCE | CLASS | TYPE ACTION                    | CUR-RATE (kbps) |
|----------|-------|--------------------------------|-----------------|
| 3        | test8 | ipv4 cir kbps 1024 exceed drop | 512             |

```
Class Name : dhcp
Class Type : ipv4
```

| SEQUENCE | CLASS-ENTRY              | HIT-COUNT |
|----------|--------------------------|-----------|
| 10       | match icmp any any count | 0         |

```
Class Name : clearpass-web
Class Type : ipv4
```

| SEQUENCE | CLASS-ENTRY             | HIT-COUNT |
|----------|-------------------------|-----------|
| 15       | match udp any any count | 15101830  |

```
Class Name : web-traffic
Class Type : ipv4
```

| SEQUENCE | CLASS-ENTRY                                 | HIT-COUNT |
|----------|---------------------------------------------|-----------|
| 10       | match any any any count                     | 241       |
| 20       | match any 10.1.1.1 10.1.1.2 dscp AF11 count | 50        |

```
Class Name : class6
Class Type : ipv6
```

| SEQUENCE | CLASS-ENTRY                                                  | HIT-COUNT |
|----------|--------------------------------------------------------------|-----------|
| 10       | match any any any count                                      | 173       |
| 20       | match icmpv6 2001:db8:a::123 2001:db8:a::125 dscp AF11 count | 32        |

```
switch#
```

```
switch# clear port-access policy POL6 hitcounts port
```

```
switch#
```

```
switch# show port-access policy POL6 hitcounts port
```

```
Port Access Policy Hit-Counts Details:
```

```
=====
```

```
Policy Name : POL4
Policy Type : Local
Policy Status : Applied
```

| SEQUENCE | CLASS | TYPE ACTION                    | CUR-RATE (kbps) |
|----------|-------|--------------------------------|-----------------|
| 3        | test8 | ipv4 cir kbps 1024 exceed drop | 512             |

```
Class Name : dhcp
Class Type : ipv4
```

| SEQUENCE | CLASS-ENTRY | HIT-COUNT |
|----------|-------------|-----------|
|----------|-------------|-----------|

```

10 match icmp any any count 0

Class Name : clearpass-web
Class Type : ipv4

SEQUENCE CLASS-ENTRY HIT-COUNT

15 match udp any any count 0

Class Name : web-traffic
Class Type : ipv4

SEQUENCE CLASS-ENTRY HIT-COUNT

10 match any any any count 0
20 match any 10.1.1.1 10.1.1.2 dscp AF11 count 0

Class Name : class6
Class Type : ipv6

SEQUENCE CLASS-ENTRY HIT-COUNT

10 match any any any count 0
20 match icmpv6 2001:db8:a::123 2001:db8:a::125 dscp AF11
 count 0

```

## show port-access policy

### Syntax

```
show port-access policy [<POLICY-NAME>]
```

### Description

Shows various aspects of policies and their current usage. Details of a policy including the content of a specific policy is shown.

Following are the possible values for policy types:

- **Local**—User configured policy
- **Downloaded**—Downloaded user policy
- **RADIUS**—Policy obtained from the RADIUS server

Following are the possible values for policy status:

- **Applied**—Policy is successfully applied in the hardware.
- **Rejected**—Policy is not supported in the hardware.
- **In-Progress**—Policy is being processed in the hardware.
- **Failed**—Displayed when the switch fails to apply the policy configuration because the TCAM resources are unavailable or full.



**NOTE:** If a policy is configured without any action, the `show` command will represent such an entry with the action as `permit`.

## Command context

Operator (>) or Manager (#)

## Parameters

**<POLICY-NAME>**

Specifies the policy name.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing information for all policies:

```
switch(config)# show port-access policy
```

```
Access Policy Details:
```

```
=====
```

```
Policy Name : POL1
Policy Type : Local
Policy Status : Applied
```

| SEQUENCE | CLASS | TYPE | ACTION                    |
|----------|-------|------|---------------------------|
| 20       | dhcp  | ipv4 | permit                    |
| 30       | test  | ipv4 | cir kbps 1024 exceed drop |

```
Policy Name : POL1_copy
Policy Type : Local
Policy Status : Applied
```

| SEQUENCE | CLASS | TYPE | ACTION                    |
|----------|-------|------|---------------------------|
| 20       | dhcp  | ipv4 | permit                    |
| 30       | test  | ipv4 | cir kbps 1024 exceed drop |

Showing information for a particular policy:

```
switch(config)# show port-access policy POL1
```

```
Access Policy Details:
```

```
=====
```

```
Policy Name : POL1
Policy Type : Local
Policy Status : Applied
```

| SEQUENCE | CLASS | TYPE | ACTION                    |
|----------|-------|------|---------------------------|
| 20       | dhcp  | ipv4 | permit                    |
| 30       | test  | ipv4 | cir kbps 1024 exceed drop |

# show port-access policy hitcounts

## Syntax

Syntax that shows statistical information in the form of hit counts:

```
show port-access policy <POLICY-NAME> hitcounts {port | client}
```

## Description

Shows the statistics and the current rate of policy applied on the port or the client.

## Command context

Operator (>) or Manager (#)

## Parameters

**<POLICY-NAME>**

Specifies the policy name.

**port**

Shows policy statistics matching the policy name in the port mode.

**client**

Shows policy statistics matching the policy name in the client mode.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing policy hit counts (statistics) with current rate:

```
switch# show port-access policy POL6 hitcounts port
```

Port Access Policy Hit-Counts Details:

=====

Policy Name : POL1  
Policy Type : Local  
Policy Status : Applied

| SEQUENCE | CLASS | TYPE ACTION                    | CUR-RATE (kbps) |
|----------|-------|--------------------------------|-----------------|
| 30       | test8 | ipv4 cir kbps 1024 exceed drop | 512             |

Class Name : dhcp  
Class Type : ipv4

| SEQUENCE | CLASS-ENTRY              | HIT-COUNT |
|----------|--------------------------|-----------|
| 10       | match icmp any any count | 982150    |

Class Name : clearpass-web  
Class Type : ipv4

| SEQUENCE | CLASS-ENTRY             | HIT-COUNT |
|----------|-------------------------|-----------|
| 70       | match udp any any count | 15101830  |

Class Name : web-traffic  
Class Type : ipv4

| SEQUENCE | CLASS-ENTRY                                 | HIT-COUNT |
|----------|---------------------------------------------|-----------|
| 4        | match any any any count                     | 3194      |
| 5        | match any 10.1.1.1 10.1.1.2 dscp AF11 count | 1716      |

Class Name : class6  
Class Type : ipv6

| SEQUENCE | CLASS-ENTRY                                                  | HIT-COUNT |
|----------|--------------------------------------------------------------|-----------|
| 10       | match any any any count                                      | 0         |
| 20       | match icmpv6 2001:db8:a::123 2001:db8:a::125 dscp AF11 count | 0         |



## associate policy

### Syntax

```
associate policy <POLICY-NAME>
```

```
no associate policy <POLICY-NAME>
```

### Description

Associates the policy with the current role.

The `no` form of this command dissociates the policy from the role.

### Command context

`config-pa-role`

The `port-access role` command takes you into the `config-pa-role` context.

### Parameters

**<POLICY-NAME>**

Specifies the policy name to associate with the current role. The maximum number of characters allowed is 64.



**NOTE:** Only those policies created by using the `port-access policy` command are allowed to be associated with a role.

Policies created using the `policy` command are not allowed to be associated with a role.

Policies that are of the downloaded type are not allowed to be associated with a role.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Associating a policy with a role:

```
switch(config)# port-access role role01
switch(config-pa-role)# associate policy policy01
```

Dissociating a policy from the role:

```
switch(config-pa-role)# no associate policy poilcy01
```

## auth-mode

### Syntax

```
auth-mode {client-mode | device-mode}
```

### Description

Configures the authentication mode for the clients that are associated with the current role.

### Command context

```
config
```

The `port-access role` command takes you into the `config-pa-role` context.

### Parameters

#### **client-mode**

Selects client mode. In this mode, all clients connecting to the port are sent for authentication.

#### **device-mode**

Selects device mode. In this mode, only the first client connecting to the port is sent for authentication. Once this client is authenticated, the port is considered as open and all subsequent clients trying to connect on that port are not sent for authentication.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring the client authentication mode:

```
switch(config-pa-role)# auth-mode client-mode
```

## cached-reauth-period

### Syntax

```
cached-reauth-period [<PERIOD>]
```

```
no cached-reauth-period
```

### Description

Enables cached reauthentication, setting the period after which clients that associated with the current role must be reauthenticated.

The `no` form of this command disables cached authentication.

### Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

## Parameters

### <PERIOD>

Specifies the cached reauthentication period (in seconds) for clients associated with the role. Default: None. Range: 30 to 4294967295 .

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling cached reauthentication and setting its period to 200 seconds:

```
switch(config-pa-role) # cached-reauth-period 200
```

Disabling cached reauthentication:

```
switch(config-pa-role) # no cached-reauth-period
```

# client-inactivity timeout

## Syntax

```
client-inactivity timeout {<CLIENT-INACTIVITY-PERIOD> | none}
```

```
no client-inactivity timeout
```

## Description

Configures the period that the switch waits for a response from a client after which it removes the client from the role.

The `no` form of the command resets the timeout period to the default.

## Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

## Parameters

### <CLIENT-INACTIVITY-PERIOD>

Specifies the client inactivity time (in seconds). Default: 300. Range: 300 to 4294967295.

### none

Specifies that the client must not be deleted because of inactivity.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring client inactivity timer for a role:

```
switch(config-pa-role) # client-inactivity timeout 3600
```

## description

### Syntax

```
description <ROLE-DESCRIPTION>
```

### Description

Configures the role description.

### Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

### Parameters

**<ROLE-DESCRIPTION>**

Specifies the role description. Up to 255 characters.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring the role description:

```
switch(config-pa-role)# description student role
```

## mtu

### Syntax

```
mtu <MTU-SIZE>
```

### Description

Configures the MTU size of a client for a role.

### Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

### Parameters

**<MTU-SIZE>**

Specifies the MTU size in bytes of a client for a role. Range: 68 to 9198.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring client MTU size:

```
switch(config-pa-role)# mtu 9198
```

## poe-priority

### Syntax

```
poe-priority {critical | high | low}
```

```
no poe-priority
```

### Description

Configures the power distribution priority for the port access roles. High power consumption can be prevented using `poe-priority` control mechanism.

The `no` form of this command restores the power distribution to its default priority.

### Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

### Parameters

#### **critical**

Specifies critical priority.

#### **high**

Specifies high priority.

#### **low**

Specifies low priority.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring PoE priority for a new role:

```
switch(config)# port-access role role01
switch(config-pa-role)# poe-priority critical
```

Resetting PoE priority for the role to its default:

```
switch(config)# port-access role role01
switch(config-pa-role)# no poe-priority
```

## port-access role

### Syntax

```
port-access role <ROLE-NAME>
```

```
no port-access role <ROLE-NAME>
```

## Description

Creates a new port access role or modifies an existing role. This command takes you into the `config-pa-role` context. A maximum of 32 port access roles can be created.

The `no` form of this command deletes a role.

## Command context

`config`

## Parameters

**<ROLE-NAME>**

Specifies the role name. The maximum number of characters allowed is 64.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Creating a new role:

```
switch(config)# port-access role basic01
```

# reauth-period

## Syntax

`reauth-period <PERIOD>`

`no reauth-period`

## Description

Configures the period after which clients that associated with the current role must be reauthenticated.



**NOTE:** The reauthentication period configured here takes precedence over the reauthentication period configured at the port level.

## Command context

`config-pa-role`

The `port-access role` command takes you into the `config-pa-role` context.

## Parameters

**<PERIOD>**

Specifies the reauthentication period (in seconds) for clients associated with the role. Default: None. Range: 1 to 86400.



**NOTE:** Reauthentication period of less than 60 seconds is not recommended.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring reauthentication period:

```
switch(config-pa-role)# reauth-period 3000
```

## session timeout

### Syntax

```
session-timeout <SESSION-TIMEOUT>
```

```
no session-timeout
```

### Description

Configures the session timeout for the role. After the timeout period, the session will be disconnected.

### Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

### Parameters

<SESSION-TIMEOUT>

Specifies the session timeout (in seconds). Range: 1 to 4294967295.



---

**NOTE:** A session timeout period of less than 60 seconds is not recommended.

---

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring session timeout for a role:

```
switch(config-pa-role)# session timeout 3600
```

## show aaa authentication port-access interface client-status

### Syntax

```
show aaa authentication port-access interface {all | <IF-NAME>}
client-status [mac <MAC-ADDRESS>]
```

### Description

Shows information about the status of the role applied on ports.

### Command context

Operator (>) or Manager (#)

## Parameters

**all**

Specifies all interfaces.

**<IF-NAME>**

Specifies the instance name.

**<MAC-ADDRESS>**

Specifies the client MAC address.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing information about a client:

```
switch# show aaa authentication port-access interface all client-status mac 00:00:00:00:00:01

Port Access Client Status Details

Client 00:00:00:00:00:01
=====
Session Details

Port : 1/7/24
Session Time : 151s

Authentication Details

Status : mac-auth Authenticated
Auth Precedence : mac-auth - Authenticated, dot1x - Not attempted

Authorization Details

Role : UserRole_1
Status : Applied
```

## show port-access role

### Syntax

```
show port-access role {local | radius | name <ROLE-NAME>}
```

### Description

Shows information about roles configured locally, or downloaded from RADIUS server.

### Command context

Operator (>) or Manager (#)

## Parameters

**local**

Shows information about locally configured roles.



## radius

Shows information about roles downloaded from the RADIUS server.

### <ROLE-NAME>

Specifies the role name.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing locally configured role information:

```
switch# show port-access role local
```

### Role Information

```
Name : local_role_01
Type : local

Reauthentication Period : 333 secs
Authentication Mode :
Session Timeout :
Client Inactivity Timeout :
Tunneled Node Server Zone :
Tunneled Node Server Secondary Role :
Access VLAN :
Native VLAN :
Allowed Trunk VLANs :
MTU :
QoS Trust Mode :
PoE Priority : low
Policy :
```

Showing information for roles downloaded from a RADIUS server:

```
switch# show port-access role radius
```

### Role Information

```
Name : RADIUS_21963402
Type : radius

Reauthentication Period : 333 secs
Authentication Mode :
Session Timeout :
Client Inactivity Timeout :
Tunneled Node Server Zone :
Tunneled Node Server Secondary Role :
Access VLAN : 10
Native VLAN :
Allowed Trunk VLANs :
MTU :
QoS Trust Mode :
PoE Priority : low
Policy : PERMIT-ALL_87364653
```

## stp-admin-edge-port

### Syntax

```
stp-admin-edge-port
```

```
no stp-admin-edge-port
```

### Description

Configures the port as a spanning tree administrative edge port for the role. This configuration removes the port participation from STP interactions when onboarding devices. This in turn helps in faster onboarding of devices.

The `no` form of the command disables STP edge port functionality.



**NOTE:** If the port receives STP BPDU on the STP administrative edge configured port, the port will move to the STP state. You must configure the port as an STP administrative edge port only if you are sure that the connected device will not participate in STP interactions.

### Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Configuring STP edge port for a role:

```
switch(config)# port-access role role01
switch(config-pa-role)# stp-admin-edge-port
```

## trust-mode

### Syntax

```
trust-mode [dscp | cos | none]
```

```
no trust-mode
```

### Description

Configures QoS trust mode for the role.

The `no` form of this command configures the default trust mode for the role.

### Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

## Parameters

### **dscp**

Specifies to trust DSCP and preserve the 802.1p priority.

### **cos**

Specifies to trust 802.1p priority and retain DSCP or IP-ToS.

### **none**

Specifies to not trust any priority fields.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring DSCP trust mode for a role:

```
switch(config)# port-access role role01
switch(config-pa-role)# trust-mode dscp
```

# vlan

## Syntax

```
vlan {access | trunk native | trunk allowed} <VLAN-ID>
```

```
no vlan {access | trunk native | trunk allowed} <VLAN-ID>
```

Or

```
vlan {access name | trunk native name | trunk allowed name} <VLAN-NAME>
```

```
no vlan {access name | trunk native name | trunk allowed name} [<VLAN-NAME>]
```

## Description

Configures VLAN IDs or VLAN names, and VLAN modes for a port access role. You can configure either VLAN IDs or VLAN names, or a combination of both for a role.

The `no` form of the command deletes the VLAN configuration from the role. For trunk allowed VLAN names, you can delete the VLAN names individually or all names at once.

## Command context

```
config-pa-role
```

The `port-access role` command takes you into the `config-pa-role` context.

## Parameters

### **access <VLAN-ID>**

Specifies the VLAN ID for the access VLAN. Supports a single VLAN ID in the range 1 to 4094.

### **trunk native <VLAN-ID>**

Specifies the native VLAN ID on the trunk interface. Supports a single VLAN ID in the range 1 to 4094.

**trunk allowed <VLAN-ID>**

Specifies the list of tagged or allowed VLANs on the trunk interface. Supports a list of VLAN IDs in the range 1 to 4094. The 6300, 6400 Switch Series support a maximum of 1024 trunk allowed VLAN IDs. The 6200 Switch Series supports a maximum of 256 trunk allowed VLAN IDs.

**access name <VLAN-NAME>**

Specifies the VLAN name for the access VLAN. Supports a single VLAN name with a maximum length of 32 characters.

**trunk native name <VLAN-NAME>**

Specifies the native VLAN name on the trunk interface. Supports a single VLAN name with a maximum length of 32 characters.

**trunk allowed name <VLAN-NAME>**

Specifies the tagged or allowed VLAN name on the trunk interface. Supports a single VLAN name in each CLI with a maximum length of 32 characters. The switch supports a maximum of 50 trunk allowed VLAN names.

**Authority**

Administrators or local user group members with execution rights for this command.

**Usage**

Note the following points when configuring the VLAN IDs and names for a role:

- For VLAN access and VLAN trunk native respectively, it is recommended to configure only one of either VLAN ID or name for a role. In case both VLAN ID and name are configured, then VLAN ID takes precedence and is applied with the role.
- For VLAN trunk allowed, you can collectively configure a maximum of 50 names and 1024 VLAN IDs on the 6300, 6400 Switch Series, or 256 VLAN IDs on the 6200 Switch Series. In case this limit is exceeded in the role, then that role is rejected when applying it to an onboarding device.

**Examples**

Configuring VLAN modes and VLAN IDs for a new role:

```
switch(config)# port-access role role01
switch(config-pa-role)# vlan trunk native 10
switch(config-pa-role)# vlan trunk allowed 11-15
switch(config-pa-role)# vlan access 50
```

Configuring VLAN modes and VLAN names for a new role:

```
switch(config)# port-access role role10
switch(config-pa-role)# vlan trunk native name hpe01
switch(config-pa-role)# vlan trunk allowed name data
switch(config-pa-role)# vlan trunk allowed name voice
switch(config-pa-role)# vlan trunk allowed name video
```

Deleting VLAN configuration from a role:

```
switch(config-pa-role)# no vlan trunk native 10
switch(config-pa-role)# no vlan trunk allowed 10-15
switch(config-pa-role)# no vlan access 50
```

Deleting trunk allowed VLAN names from a role individually:

```
switch(config-pa-role)# no vlan trunk native name hpe01
switch(config-pa-role)# no vlan trunk allowed name data
switch(config-pa-role)# no vlan trunk allowed name voice
switch(config-pa-role)# no vlan trunk allowed name video
```

Deleting trunk allowed VLAN names from a role all at once:

```
switch(config-pa-role)# no vlan trunk native name hpe01
switch(config-pa-role)# no vlan trunk allowed name
```

## port-access port-security

### Syntax

```
port-access port-security {enable | disable}
```

### Description

Enables or disables port security globally or at the port level.

### Command context

```
config
```

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling port security globally:

```
switch(config)# port-access port-security enable
```

Disabling port security globally:

```
switch(config)# port-access port-security disable
```

Enabling port security on a port:

```
switch(config-if)# port-access port-security enable
```

Disabling port security on a port:

```
switch(config-if)# port-access port-security disable
```

## port-access port-security client-limit

### Syntax

```
port-access port-security client-limit <CLIENTS>
```

```
no port-access port-security client-limit
```

### Description

Configures the maximum number of clients that are allowed on a port. After configuring the maximum clients limit, the MAC addresses of the clients can be learned by one of the following methods:

- User can manually configure all MAC addresses by using the `mac-address` command.
- User can allow the port to dynamically learn all MAC addresses.
- User can configure a fixed number of MAC addresses and allow the switch to learn the remaining addresses dynamically.

The `no` form of the command resets the number of clients to the default, 1.

### Command context

`config-if`

### Parameters

**<CLIENTS>**

Specifies the maximum number of clients. Default: 1. Range: 1 to 32 (6100).

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring client limit on a port:

```
switch(config-if)# port-access port-security enable
switch(config-if-port-security)# client-limit 64
```

## port-access port-security mac-address

### Syntax

`port-access port-security mac-address <MAC-ADDRESS>`

`no port-access port-security mac-address <MAC-ADDRESS>`

### Description

Configures the MAC addresses of static clients.

The `no` form of this command removes an authorized static client from the port.

### Command context

`config-if`

### Parameters

**<MAC-ADDRESS>**

Specifies the MAC addresses of the static clients.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring a static client on a port:

```
switch(config-if)# port-access port-security
switch(config-if-port-security)# mac-address aa:bb:cc:dd:ee:ff
```

Deleting a static client on a port:

```
switch(config-if)# port-access port-security
switch(config-if-port-security)# no mac-address aa:bb:cc:dd:ee:ff
```

## show port-access port-security interface client-status

### Syntax

```
show port-access port-security interface {all|<IF-NAME>}
client-status [mac <MAC-ADDRESS>]
```

### Description

Shows port security clients status information for the ports. The output can be filtered by interface or MAC address.

### Command context

Operator (>) or Manager (#)

### Parameters

**all**

Selects all interfaces.

**<IF-NAME>**

Specifies the interface name.

**<MAC-ADDRESS>**

Specifies the client MAC address.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing client status information for all ports:

```
switch# show port-access port-security interface all client-status
```

Port Security Client Status Details

| Authorized-Clients | Port  |
|--------------------|-------|
| AB:CD:DE:FF:AA:BB  | 1/1/1 |
| DD:CD:AB:CD:EE:01  | 1/1/2 |

Showing client status information for a client:

```
switch# show port-access port-security interface 1/1/1 client-status mac ab:cd:de:ff:aa:bb
```

Port Security Client Status Details



| Authorized-Clients | Port  |
|--------------------|-------|
| AB:CD:DE:FF:AA:BB  | 1/1/1 |

Showing client status information for a port:

```
switch# show port-access port-security interface 1/1/1 client-status
```

Port Security Client Status Details

| Authorized-Clients | Port  |
|--------------------|-------|
| 70:00:02:b8:50:11  | 1/1/1 |
| 70:00:02:b8:50:3d  | 1/1/1 |
| 70:00:02:b8:50:46  | 1/1/1 |
| 70:00:02:b8:50:3b  | 1/1/1 |
| 70:00:02:b8:50:36  | 1/1/1 |
| 70:00:02:b8:50:48  | 1/1/1 |
| 70:00:02:b8:50:1e  | 1/1/1 |
| 70:00:02:b8:50:1a  | 1/1/1 |
| 70:00:02:b8:50:13  | 1/1/1 |
| 70:00:02:b8:50:26  | 1/1/1 |
| 70:00:02:b8:50:2d  | 1/1/1 |

## show port-access port-security interface port-statistics

### Syntax

```
show port-access port-security interface {all|<IF-NAME>} port-statistics
```

### Description

Shows port security statistics for the ports in a switch. The output can be filtered by interface.

### Command context

Operator (>) or Manager (#)

### Parameters

**all**

Selects all interfaces.

**<IF-NAME>**

Specifies the interface name.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing information for all ports.

```
switch# show port-access port-security interface all port-statistics
```

```
Port 1/1/1
```

```
=====
```

```
Client Details
```

```

```

```
Number of authorized clients : 2
```

## portfilter

### Syntax

```
portfilter <INTERFACE-LIST>
```

```
no portfilter [<INTERFACE-LIST>]
```

### Description

Configures the specified ports so they do not egress any packets that were received on the source port specified in interface context.

The `no` form of this command removes the port filter setting from one or more ingress ports/LAGs.

### Command context

```
config-if
```

```
config-lag-if
```

### Parameters

**<INTERFACE-LIST>**

Specifies a list of ports/LAGs to be blocked for egressing. Specify a single interface or LAG, or a range as a comma-separated list, or both. For example: `1/1/1, 1/1/3-1/1/6, lag2, lag1-lag4`.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

When a port filter configuration is applied on the same ingress physical port/LAG, the configuration is updated with the new sets of egress ports/LAGs that are to be blocked for egressing and that are not a part of its previous configuration. Duplicate updates on an existing port filter configuration are ignored.

When egress ports/LAGs are removed from the existing port filter configuration of an ingress port/LAG, egressing is allowed again on those egress ports/LAGs for all packets originating from the ingress port/LAG.

The `no portfilter [<IF-NAME-LIST>]` command removes port filter configurations from the egress ports/LAGs listed in the `<IF-NAME-LIST>` parameter only. All other egress ports/LAGs in the port filter configuration of the ingress port/LAG remain intact.

If no physical ports or LAGs are provided for the `no portfilter` command, the command removes the entire port filter configuration for the ingress port/LAG.

### Examples

Creating a filter that prevents packets received on port `1/1/1` from forwarding to ports `1/1/3-1/1/6` and to LAGs `1` through `4`:

```
switch(config)# interface 1/1/1
switch(config-if)# portfilter 1/1/3-1/1/6, lag1-lag4
```

Creating a filter that prevents packets received on LAG 1 from forwarding to ports 1/1/6 and LAGs 2 and 4:

```
switch(config)# interface lag 1
switch(config-lag-if)# portfilter 1/1/6,lag2,lag4
```

Removing filters from an existing configuration that allows back packets received on port 1/1/1 to forward to ports 1/1/6 and LAGs 3 and 4:

```
switch(config)# interface 1/1/1
switch(config-if)# no portfilter 1/1/6,lag3,lag4
```

Removing all filters from an existing configuration that allows back packets received on LAG 1 to forward to all the ports and LAGs:

```
switch(config)# interface lag 1
switch(config-lag-if)# no portfilter
```

## show portfilter

### Syntax

```
show portfilter [<IFNAME>]
```

### Description

Displays filter settings for all interfaces or a specific interface.

### Command context

Operator (>) or Manager (#)

### Parameters

**<IFNAME>**

Specifies the ingress interface name.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Displaying all port filter settings on the switch:

```
switch# show portfilter
Incoming Blocked
Interface Outgoing Interfaces

1/1/1 1/1/3-1/1/6,lag1-lag2
1/1/3 1/1/1,1/1/5,1/1/7,1/1/9,1/1/11,1/1/13,1/1/15,1/1/17,1/1/19,1/1/21,
 1/1/23,1/1/25,1/1/27,1/1/29,1/1/31,1/1/33,1/1/35
lag2 1/1/1,1/1/3-1/1/6
```

Displaying the port filter settings for port 1/1/1:

```
switch# show portfilter 1/1/1
Incoming Blocked
Interface Outgoing Interfaces
```

---

|       |                        |
|-------|------------------------|
| 1/1/1 | 1/1/3-1/1/6, lag1-lag2 |
|-------|------------------------|

Displaying the port filter settings for **LAG2**:

```
switch# show portfilter lag2
Incoming Blocked
Interface Outgoing Interfaces

lag2 1/1/1, 1/1/3-1/1/6
```

## apply qos

### Syntax

```
apply qos [queue-profile <QUEUE-NAME>] schedule-profile <SCHEDULE-NAME>
```

```
no apply qos schedule-profile <SCHEDULE-NAME>
```

### Description

Applies a queue profile and schedule profile globally to all Ethernet and LAG interfaces on the switch, or applies a schedule profile to a specific interface. When applied globally, the specified schedule profile is configured only on Ethernet interfaces and LAGs that do not already have their own schedule profile.

The same profile can be applied both globally and locally to an interface. This guarantees that an interface always uses the specified profile, even if the global profile is changed.

The `no` form of this command removes the specified schedule profile from an interface and the interface uses the global schedule profile. This is the only way to remove a schedule profile override from the interface.



---

**NOTE:** Interfaces may shut down briefly during reconfiguration.

---

### Command context

`config`

`config-if`

`config-lag-if`

### Parameters

**queue-profile** <QUEUE-NAME>

Specifies the name of the queue profile to apply. Range: 1 to 64 alphanumeric characters, including period (.), underscore (\_), and hyphen (-). This parameter is not supported in the `config-if` context.

**schedule-profile** <SCHEDULE-NAME>

Specifies the name of the schedule profile to apply. Range: 1 to 64 alphanumeric characters, including period (.), underscore (\_), and hyphen (-).

### Authority

Administrators or local user group members with execution rights for this command.

## Usage

- For a schedule profile to be complete and ready to be applied, it must define all queues specified in the queue profile. All queues must use the same algorithm, except for the highest numbered queue, which can be **strict**.
- Both the queue profile and the schedule profile must specify the same number of queues.
- Schedule profiles can be modified while applied, but only in ways where a single command will not result in the profile becoming invalid. For example, queue 7 can have the algorithm changed, and weighted queues can have their weights changed.
- Queues must be consecutively defined starting at queue number zero. For example, a four-queue profile with priority values defined for queues 0, 1, 2, 3 is valid, but a four-queue profile which defines priority values for queues 1, 3, 5, and 7 is not.

## Examples

The following commands illustrate a valid configuration where every CoS value is assigned to a queue and all assigned queues are defined.

```
switch(config)# qos trust cos
switch(config)# qos queue-profile Q1
switch(config)# map queue 0 cos 0
switch(config)# map queue 1 cos 1
switch(config)# map queue 2 cos 2
switch(config)# map queue 3 cos 3
switch(config)# map queue 4 cos 4
switch(config)# map queue 5 cos 5
switch(config)# map queue 6 cos 6
switch(config)# map queue 7 cos 7

switch(config)# qos schedule-profile S1
switch(config)# min-bandwidth queue 0 percent 5
switch(config)# min-bandwidth queue 1 percent 5
switch(config)# min-bandwidth queue 2 percent 10
switch(config)# min-bandwidth queue 3 percent 10
switch(config)# min-bandwidth queue 4 percent 20
switch(config)# min-bandwidth queue 5 percent 20
switch(config)# min-bandwidth queue 6 percent 10
switch(config)# min-bandwidth queue 7 percent 20
switch(config)# apply qos queue-profile Q1 schedule-profile S1
```

The following commands illustrate an invalid configuration because CoS 2 is not assigned to a queue.

```
switch(config)# qos trust cos
switch(config)# qos queue-profile Q1
switch(config)# map queue 0 cos 0
switch(config)# map queue 1 cos 1
switch(config)# map queue 3 cos 3
switch(config)# map queue 4 cos 4
switch(config)# map queue 5 cos 5
switch(config)# map queue 6 cos 6
switch(config)# map queue 7 cos 7

switch(config)# qos schedule-profile S1
switch(config)# min-bandwidth queue 0 percent 5
switch(config)# min-bandwidth queue 1 percent 5
switch(config)# min-bandwidth queue 3 percent 10
switch(config)# min-bandwidth queue 4 percent 20
switch(config)# min-bandwidth queue 5 percent 20
switch(config)# min-bandwidth queue 6 percent 10
```

```
switch(config)# min-bandwidth queue 7 percent 20
switch(config)# apply qos queue-profile Q1 schedule-profile S1
```

Applying the QoS profile **myprofile** and the schedule profile **myschedule** to all interfaces that do not have an applied interface-specific schedule profile:

```
switch(config)# apply qos queue-profile myprofile schedule-profile myschedule
```

## min-bandwidth

### Syntax

```
min-bandwidth queue <QUEUE-NUMBER> percent <VALUE>
```

```
no min-bandwidth queue <QUEUE-NUMBER>
```

### Description

Assigns the Guaranteed Minimum Bandwidth (GMB) algorithm and a percentage of bandwidth to a queue. GMB allocates available bandwidth among all non-empty queues in relation to their configured minimum bandwidth. Non-empty queues are serviced first in strict order up to their minimum bandwidth. If there is any remaining bandwidth the scheduler will strictly service any remaining non-empty queues.

The **no** form of this command only clears the algorithm for a queue if GMB has been assigned.

### Command context

config-schedule

### Parameters

**<QUEUE-NUMBER>**

Specifies the queue number. Range: 0 to 7.

**<VALUE>**

Specifies bandwidth percentage used for GMB scheduling. Range: 0 to 100.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Assigning queue 0 of schedule profile S1 the GMB scheduling algorithm with minimum bandwidth of 5 percent:

```
switch(config)# qos schedule-profile S1
switch(config-schedule)# min-bandwidth queue 0 percent 5
```

Removing GMB from queue 0:

```
switch(config)# qos schedule-profile s1
switch(config-schedule)# no min-bandwidth queue 0
```



## name queue

### Syntax

```
name queue <QUEUE-NUMBER> <DESCRIPTION>
```

```
no name queue <QUEUE-NUMBER>
```

### Description

Assigns a description to a queue in a queue profile. This is for identification purposes and has no effect on configuration.

The `no` form of this command removes the description associated with a queue.

### Command context

```
config-queue
```

### Parameters

**<QUEUE-NUMBER>**

Specifies the queue number. Range: 0 to 7.

**<DESCRIPTION>**

Specifies a queue description for identification purposes. Range: 1 to 64 alphanumeric characters, including period (.), underscore (\_), and hyphen (-).

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Assigning the description `priority-traffic` to queue 7:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)# name queue 7 priority-traffic
```

Removing the description from queue 7:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)# no name queue 7
```

## qos cos

### Syntax

```
qos cos <CODE-POINT>
```

```
no qos cos
```

### Description

Configures a CoS PCP remark for an Ethernet or LAG interface. Packets that ingress on the interface are remarked at egress using the configured CoS PCP value.

The remark only occurs when QoS trust mode on the interface is set to `none`.

If QoS trust mode is not set to `none`, then the remark is ignored, and the following commands will show the CoS remark status as `ignored` (incompatible Port Access Trust configuration) or `not applied` (incompatible QoS global/port Trust configuration):

- `show running-configuration`
- `show interface <PORT-NUM>`
- `show interface <PORT-NUM> qos`

The `no` form of this command removes a CoS remark on an interface.

### Command context

`config-if`

### Parameters

**<CODE-POINT>**

Specifies an 802.1 VLAN priority CoS value. Range: 0 to 7.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring a CoS remark of 3 on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# qos trust none
switch(config-if)# qos cos 3
```

Deleting a CoS remark of 3 on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no qos cos
```

## qos dscp

### Syntax

`qos dscp <CODE-POINT>`

`no qos dscp`

### Description

Configures a differentiated services code point (DSCP) remark for an Ethernet or LAG interface. IPv4 and IPv6 packets that ingress on the interface are remarked at egress using the configured DSCP value.

The remark only occurs when QoS trust mode on the interface is set to `none`. If a DSCP remark is configured and then trust mode is subsequently set to `cos` or `dscp`, then the DSCP remark is ignored.

The following commands will show the remark status as `ignored` (incompatible Port Access Trust configuration) or `not applied` (incompatible QoS global/port Trust configuration):

- `show running-configuration`
- `show interface <PORT-NUM>`
- `show interface <PORT-NUM> qos`

The `no` form of this command removes a CoS remark on an interface.

## Command context

`config-if`

## Parameters

**<CODE-POINT>**

Specifies an IP differentiated services code point value. Range: 0 to 63.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring a DSCP remark of **43** on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# qos trust none
switch(config-if)# qos dscp 43
```

Deleting a DSCP remark of **43** on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no dscp 43
```

# qos dscp-map

## Syntax

`qos dscp-map <CODE-POINT> [cos COS-CODE-POINT] [name <DESCRIPTION>]`

`no qos dscp-map <CODE-POINT>`

## Description

Defines the CoS assigned to incoming packets for a specific IP differentiated services code point (DSCP) value. The DSCP map values are used to prioritize incoming packets when QoS trust mode is set to **dscp**.

To see the default DSCP map settings, use the following command:

```
switch# show qos dscp-map default
DSCP code_point cos name

000000 0 0 CS0
000001 1 0
000010 2 0
000011 3 0
000100 4 0
000101 5 0
000110 6 0
000111 7 0
```

|        |    |   |      |
|--------|----|---|------|
| 001000 | 8  | 1 | CS1  |
| 001001 | 9  | 1 |      |
| 001010 | 10 | 1 | AF11 |
| 001011 | 11 | 1 |      |
| 001100 | 12 | 1 | AF12 |
| 001101 | 13 | 1 |      |
| 001110 | 14 | 1 | AF13 |
| 001111 | 15 | 1 |      |
| 010000 | 16 | 2 | CS2  |
| 010001 | 17 | 2 |      |
| 010010 | 18 | 2 | AF21 |
| 010011 | 19 | 2 |      |
| 010100 | 20 | 2 | AF22 |
| 010101 | 21 | 2 |      |
| 010110 | 22 | 2 | AF23 |
| 010111 | 23 | 2 |      |
| 011000 | 24 | 3 | CS3  |
| 011001 | 25 | 3 |      |
| 011010 | 26 | 3 | AF31 |
| 011011 | 27 | 3 |      |
| 011100 | 28 | 3 | AF32 |
| 011101 | 29 | 3 |      |
| 011110 | 30 | 3 | AF33 |
| 011111 | 31 | 3 |      |
| 100000 | 32 | 4 | CS4  |
| 100001 | 33 | 4 |      |
| 100010 | 34 | 4 | AF41 |
| 100011 | 35 | 4 |      |
| 100100 | 36 | 4 | AF42 |
| 100101 | 37 | 4 |      |
| 100110 | 38 | 4 | AF43 |
| 100111 | 39 | 4 |      |
| 101000 | 40 | 5 | CS5  |
| 101001 | 41 | 5 |      |
| 101010 | 42 | 5 |      |
| 101011 | 43 | 5 |      |
| 101100 | 44 | 5 |      |
| 101101 | 45 | 5 |      |
| 101110 | 46 | 5 | EF   |
| 101111 | 47 | 5 |      |
| 110000 | 48 | 6 | CS6  |
| 110001 | 49 | 6 |      |
| 110010 | 50 | 6 |      |
| 110011 | 51 | 6 |      |
| 110100 | 52 | 6 |      |
| 110101 | 53 | 6 |      |
| 110110 | 54 | 6 |      |
| 110111 | 55 | 6 |      |
| 111000 | 56 | 7 | CS7  |
| 111001 | 57 | 7 |      |
| 111010 | 58 | 7 |      |
| 111011 | 59 | 7 |      |
| 111100 | 60 | 7 |      |
| 111101 | 61 | 7 |      |
| 111110 | 62 | 7 |      |
| 111111 | 63 | 7 |      |

The `no` form of this command restores the assignments for a code point to the default setting.

### Command context

`config`

## Parameters

### <CODE-POINT>

Specifies an IP differentiated services code point. Range: 0 to 63. Default: 0.

### cos <COS-CODE-POINT>

Specifies an 802.1 VLAN priority CoS remark value. Range: 0 to 7. Default 0

### name <DESCRIPTION>

Specifies a description for the DSCP setting. The name is used for identification only, and has no effect on queue configuration. Range: 1 to 64 alphanumeric characters, including period (.), underscore (\_), and hyphen (-).

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting code point **41** to a CoS of **6**:

```
switch(config)# qos dscp-map 41 cos 6
```

Setting code point **41** to the default value:

```
switch(config)# no qos dscp-map 41
```

# qos queue-profile

## Syntax

```
qos queue-profile <NAME>
```

```
no qos queue-profile <NAME>
```

## Description

Creates a new QoS queue profile and switches to the `config-queue` context for the profile. Or, if the specified QoS queue profile exists, this command switches to the `config-queue` context for the profile.

A queue profile maps queues to local-priority values.

A queue profile named **factory-default** is defined by default and is automatically applied to all interfaces. It cannot be edited or deleted. Use the command `show qos queue-profile factory-default` to view this profile. Do not use `show running-configuration`, as it will only display changes from the initial values.

The `no` form of this command removes the specified QoS queue profile. Only profiles that are not currently applied can be removed.

## Command context

```
config
```

## Parameters

### <NAME>

Specifies the name of the QoS queue profile to create or configure. Range: 1 to 64 alphanumeric characters, including period (.), underscore (\_), and hyphen (-).

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Creating the profile **myprofile**:

```
switch(config)# qos queue-profile myprofile
switch(config-queue)#
```

Deleting the profile **myprofile**:

```
switch(config)# no qos queue-profile myprofile
```

# qos schedule-profile

## Syntax

```
qos schedule-profile <NAME>
```

```
no qos schedule-profile <NAME>
```

## Description

Creates a QoS schedule profile and switches to the `config-schedule` context for the profile. If the specified schedule profile exists, this command switches to the `config-schedule` context for the profile. The schedule profile determines the order in which queues are selected to transmit a packet, and the amount of service defined for each queue.

Queues in a schedule profile are numbered consecutively starting from zero. Queue zero is the lowest priority queue. The larger the queue number, the higher priority the queue has in scheduling algorithms.

A profile named **factory-default** is defined by default and applied to all interfaces. It cannot be edited or deleted. To see its settings, use the command:

```
switch# show qos schedule-profile factory-default
queue_num algorithm percent max-bandwidth_kbps

0 min-bandwidth 2
1 min-bandwidth 3
2 min-bandwidth 30
3 min-bandwidth 10
4 min-bandwidth 10
5 min-bandwidth 10
6 min-bandwidth 15
7 min-bandwidth 20
```

A profile named **strict** is predefined and cannot be edited or deleted. The strict profile services all queues of the queue profile to which it is applied, using the strict priority algorithm.

A schedule profile must be defined on all interfaces at all times.

There are two permitted configurations for a schedule profile:

1. All queues use the same scheduling algorithm (for example, GMB).
2. The highest queue number uses strict priority, and all remaining (lower) queues use the same algorithm (for example, GMB). This supports priority scheduling behavior necessary for the IETF RFC 3246 Expedited Forwarding specification (<https://tools.ietf.org/html/rfc3246>).

Only limited changes can be made to an applied schedule profile:

- The percentage of a GMB queue.
- The bandwidth of a strict queue.
- The algorithm of the highest numbered queue can be swapped between GMB and strict, and vice versa.

Applicable to REST: Any other changes will result in an unusable schedule profile, and the switch will revert to the `factory-default` profile until the profile is corrected.

The `no` form of this command removes the specified QoS schedule profile when it is not applied. Only profiles that are not currently applied to an interface can be removed.

### Command context

`config`

### Parameters

**<NAME>**

Specifies the name of the QoS schedule profile to create or configure. Range: 1 to 64 alphanumeric characters, including period (.), underscore (\_), and hyphen (-).

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating the schedule profile **myschedule**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)#
```

Deleting the schedule profile **myschedule**:

```
switch(config)# no qos schedule-profile myschedule
```

## qos trust

### Syntax

```
qos trust {none | cos | dscp}
```

```
no qos trust
```

### Description

Configures one of three modes that are applied globally on all Ethernet interfaces and LAGs that have not applied their own trust mode. The QoS mode determines how CoS values are assigned to packets.

In the `config` context:

- This command sets the trust mode that is globally applied to all interfaces that do not have a trust mode configured.
- The `no` form of this command restores all interfaces that do not currently have a trust mode configured to the default setting.

In the `config-if` context:

- This command sets the trust mode override for a specific interface.
- The `no` form of this command clears a trust mode override. The interface then uses the global setting. This is the only way to remove a trust mode override.

### Command context

`config`

`config-if`

### Parameters

#### **none**

Ignore all packet headers. Ingress packets are assigned the priority value configured for CoS map value 0.

#### **cos**

For 802.1 VLAN-tagged packets, use the priority code point field value from the outermost VLAN header as the index into the CoS map. If the packet is untagged, use the priority configured for CoS map value 0. Default.

#### **dscp**

For IP packets, use the DSCP value as the index into the DSCP map. For non-IP packets, use the priority configured for CoS map value 0.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Setting the global trust mode to **dscp**, which is applied to all interfaces that do not already have an individual trust mode configured. An override is then applied to interface **2/2/2**, and LAG 100, setting trust mode to **cos**:

```
switch(config)# qos trust dscp
switch(config)# interface 2/2/2
switch(config-if)# qos trust cos
switch(config-if)# interface lag 100
switch(config-if)# qos trust cos
```

## rate-limit

### Syntax

```
rate-limit {broadcast | multicast | unknown-unicast | icmp {ip-all | ipv4 | ipv6}} <RATE> kbps
no rate-limit {broadcast | multicast | unknown-unicast | icmp}
```

### Description

Sets the amount of traffic of a specific type that can ingress on an Ethernet interface, or on each port of a LAG interface. Rate limits are enforced separately on each individual member of a LAG, not on the LAG as a whole.

The `no` form of this command removes the traffic limit for the specified traffic type.



## Command context

config-if

## Parameters

**{broadcast | multicast | unknown-unicast| icmp {ip-all | ipv4 | ipv6}}**

Specifies the type of ingress traffic to which the rate limit applies: broadcast, multicast, unknown-unicast, or ICMP.

The multicast rate limit affects multicast and broadcast traffic. The broadcast rate limit only affects broadcast traffic. When both types are applied to the same interface, broadcast packets are limited to the lower of the two rate values. Layer 2 BPDU packets, like spanning tree, are also included in the multicast rate limit.

Unknown-unicast packets may be intended for devices whose addresses have temporarily aged out of network forwarding caches. Configuring rate limits can help provide balance between necessary and flooded traffic.

The ICMP rate limit can be configured to apply to IPv4, IPv6, or all IP traffic. Only one ICMP rate-limit can be configured at a time. Applying a new ICMP rate-limit replaces any previous ICMP rate-limit.

### <RATE>

Specifies the rate limit. Range: 64 to 100000000 kbps (in steps of 64 kbps). The actual rate limit varies with steps approximately equal to the minimum value. Verify the actual rate limit using the command `show interface <INTERFACE-NAME>`.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Limiting broadcast traffic to **1024 kbps** on interface **1/1/3**:

```
switch(config)# interface 1/1/3
switch(config-if)# rate-limit broadcast 1024 kbps
```

Limiting all ICMP IPv4 traffic to **10000 kbps** on interface **1/1/3**:

```
switch(config)# interface 1/1/3
switch(config-if)# rate-limit icmp ipv4 1024 kbps
```

Viewing the results of the previous configuration settings.

```
switch# show interface 1/1/3 qos
Interface 1/1/3 is up
Admin state is up
qos trust cos (global)
qos queue-profile factory-default (global)
qos schedule-profile factory-default (global)
rate-limit unknown-unicast 1024 kbps (1024 actual)
rate-limit broadcast 1024 kbps (1100 actual)
rate-limit multicast 1024 kbps (1100 actual)
rate-limit icmp ip-all 1024 kbps (1024 actual)

switch# show interface 1/1/3
Interface 1/1/3 is up
Admin state is up
Link state: up for 3 minutes (since Thu Nov 26 17:56:14 UTC 2020)
Link transitions: 1
```

```

Description:
Hardware: Ethernet, MAC Address: f8:60:f0:c9:21:bc
MTU 1500
Type 1GbT
Full-duplex
qos trust cos
rate-limit unknown-unicast 1024 kbps (1024 actual)
rate-limit broadcast 1024 kbps (1100 actual)
rate-limit multicast 1024 kbps (1100 actual)
rate-limit icmp ip-all 1024 kbps (1024 actual)
Speed 1000 Mb/s
Auto-negotiation is on
Energy-Efficient Ethernet is disabled
Flow-control: off
Error-control: off
MDI mode: MDIX
VLAN Mode: access
Access VLAN: 1
Rx
 0 total packets 0 total bytes
 0 unicast packets
 0 multicast packets
 0 broadcast packets
 0 errors 0 dropped
 0 CRC/FCS 0 pause
Tx
1057962 total packets 366066962 total bytes
 0 unicast packets
 0 multicast packets
1058039 broadcast packets
 0 errors 0 dropped
 0 collision 0 pause

```

## show interface queues

### Syntax

```
show interface <INTERFACE-NAME> queues [vsx-peer]
```

### Description

Shows the traffic transmitted per queue and the number of packets dropped due to the queue being full.

### Command context

Operator (>) or Manager (#)

### Parameters

**<INTERFACE-NAME>**

Specifies the name of an interface on the switch. Format: member/slot/port.

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Usage

Statistics include:

- **Tx Bytes:** Total bytes transmitted (available on the 6200 and 6300 Switch Series). The byte count may include packet headers and internal metadata that are removed before the packet is transmitted. Packet headers added when the packet is transmitted may not be included.
- **Tx Packets:** Total packets transmitted.
- **Tx Drops:** The number of packets dropped by a queue before it was sent. When traffic cannot be forwarded out an egress interface, it backs up at ingress. The more servicing assigned to a queue by a schedule profile, the less likely traffic destined for that queue will back up and be dropped. Tx Drops shows the sum of packets that were dropped across all line modules (due to insufficient capacity) by the ingress Virtual Output Queues (VOQs) destined for the egress port.

## Examples

Showing settings for interface 1/1/5:

```
switch# show interface 1/1/5
Interface 1/1/5 is down (Administratively down)
Admin state is down
State information: admin_down
Hardware: Ethernet, MAC Address: aa:55:aa:55:00:29
MTU 1500
Full-duplex
qos trust cos
qos schedule-profile default
qos shape 400000
qos cos 4 (not applied)
qos dscp 46 (not applied)
rate-limit broadcast 500 kbps
rate-limit multicast 4000 pps
Speed 0 Mb/s
Auto-Negotiation is turned on
Flow-control: off
```

Showing queue statistics for interface 1/1/5:

(Tx Bytes is available on the 6100, 6200, and 6300 Switch Series.)

```
switch# show interface 1/1/5 queues
Interface 5 is down
Admin state is up
```

|    | Tx Bytes | Tx Packets | Tx Drops |
|----|----------|------------|----------|
| Q0 | 0        | 0          | 3        |
| Q1 | 15356    | 73         | 0        |
| Q2 | 0        | 0          | 0        |
| Q3 | 0        | 0          | 0        |
| Q4 | 0        | 0          | 0        |
| Q5 | 0        | 0          | 0        |
| Q6 | 0        | 0          | 0        |
| Q7 | 0        | 0          | 0        |

## show interface qos

### Syntax

```
show interface <INTERFACE-NAME> qos
```

### Description

Shows various QoS settings for a specific interface.

### Command context

Operator (>) or Manager (#)

### Parameters

**<INTERFACE-NAME>**

Specifies the name of an interface on the switch. Format: member/slot/port or lag number.

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing QoS settings for interface 1/1/5:

```
switch# show interface 1/1/5 qos
Interface 1/1/5 is up
Admin state is up
qos trust none (global)
qos queue-profile factory-default (global)
qos schedule-profile eqs (override)
qos cos 5
qos dscp 47
rate-limit broadcast 40000 kbps (40000 actual)
rate-limit icmp ip-all 10000 kbps (10000 actual)

 Max-Bandwidth Kbps
Q6 100000
Q7 150000
```

## show qos dscp-map

### Syntax

```
show qos dscp-map [default]
```

### Description

Displays the current or default global QoS dscp-map.

## Command context

Operator (>) or Manager (#)

## Parameters

### default

Shows the factory default DSCP code point settings.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing the current DSCP map:

```
switch# show qos dscp-map
DSCP code_point cos name

000000 0 0 CS0
000001 1 0
000010 2 0
000011 3 0
000100 4 0
000101 5 0
...
101101 45 5
101110 46 6 new
101111 47 5
110000 48 6 CS6
...
111101 61 7
111110 62 7
111111 63 7
```

Showing the default DSCP map:

```
switch# show qos dscp-map default
DSCP code_point cos name

000000 0 0 CS0
000001 1 0
000010 2 0
000011 3 0
000100 4 0
000101 5 0
...
101101 45 5
101110 46 5
101111 47 5
110000 48 6 CS6
...
111100 60 7
111101 61 7
111110 62 7
111111 63 7
```

## show qos queue-profile

### Syntax

```
show qos queue-profile [<NAME>]
```

### Description

Shows the status of all queue profiles, or a specific queue profile.

### Command context

Operator (>) or Manager (#)

### Parameters

**<NAME>**

Specifies the name of a queue profile. Range 1 to 64 alphanumeric characters, including period (.), underscore (\_), and hyphen (-).

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Showing the status of the queue profile **Test**:

```
switch# show qos schedule-profile Test
queue_num algorithm percent max-bandwidth_kbps

0 min-bandwidth 10
1 min-bandwidth 10
2 min-bandwidth 30
3 min-bandwidth 10
4 min-bandwidth 10
5 min-bandwidth 10
6 min-bandwidth 10
7 min-bandwidth 10
```

Showing the status of the default queue profile **factory-default**:

```
switch# show qos schedule-profile default
queue_num algorithm percent max-bandwidth_kbps

0 min-bandwidth 2
1 min-bandwidth 3
2 min-bandwidth 30
3 min-bandwidth 10
4 min-bandwidth 10
5 min-bandwidth 10
6 min-bandwidth 15
7 min-bandwidth 20
```

## show qos schedule-profile

### Syntax

```
show qos schedule-profile [<NAME>]
```

## Description

Shows the status of all schedule profiles, or a specific schedule profile.

## Command context

Operator (>) or Manager (#)

## Parameters

<NAME>

Specifies the name of a queue or schedule profile. Range: 1 to 64 alphanumeric characters, including period (.), underscore (\_), and hyphen (-).

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Showing the status of schedule profile **myschedule**:

Showing the status of the default schedule profile:

Showing the status of schedule profile **Test**:

```
switch# show qos schedule-profile Test
queue_num algorithm percent max-bandwidth_kbps

0 min-bandwidth 10
1 min-bandwidth 10
2 min-bandwidth 30
3 min-bandwidth 10
4 min-bandwidth 10
5 min-bandwidth 10
6 min-bandwidth 10
7 min-bandwidth 10
```

Showing the status of the default schedule profile:

```
switch# show qos schedule-profile default
queue_num algorithm percent max-bandwidth_kbps

0 min-bandwidth 2
1 min-bandwidth 3
2 min-bandwidth 30
3 min-bandwidth 10
4 min-bandwidth 10
5 min-bandwidth 10
6 min-bandwidth 15
7 min-bandwidth 20
```

## show qos trust

### Syntax

```
show qos trust [default]
```

### Description

Shows the global QoS trust settings, or the factory default settings.

## Command context

Operator (>) or Manager (#)

## Parameters

### default

Shows the factory default QoS trust settings.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing the current QoS trust settings:

Showing the default QoS trust settings:

```
switch# show qos trust default
qos trust cos
```

# strict queue

## Syntax

```
strict queue <QUEUE-NUMBER> [[max-bandwidth <BANDWIDTH>]]
```

```
no strict queue <QUEUE-NUMBER>
```

## Description

Assigns the strict priority algorithm to a queue. Strict priority services all packets waiting in a queue, before servicing the packets in lower priority queues.

The `no` form of this command removes the queue configuration from the schedule profile. To remove only egress queue shaping, re-enter the `strict queue` command without the `max-bandwidth` parameter.

## Command context

config-schedule

## Parameters

### <QUEUE-NUMBER>

Specifies the number of the queue. Range: 0 to 7.

### max-bandwidth <BANDWIDTH>

Specifies the maximum bandwidth allowed on the queue in Kbps. Range: 49 to 100000000.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Assigning strict priority to queue 7 in the schedule profile `myschedule`:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# strict queue 7
```



Deleting strict priority from queue 7 in the schedule profile **myschedule**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# no strict queue 7
```

Assigning strict priority to queue 7 in the schedule profile **myschedule** with a maximum bandwidth of **10000 Kbps**:

```
switch(config)# qos schedule-profile myschedule
switch(config-schedule)# strict queue 7 max-bandwidth 10000
```



**NOTE:** On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

## radius dyn-authorization enable

### Syntax

```
radius dyn-authorization enable
```

```
no radius dyn-authorization enable
```

### Description

Enables RADIUS dynamic authorization. This command must be issued before the configuration set with other `radius dyn-authorization` commands takes effect.

The `no` form of this command disables RADIUS dynamic authorization.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling RADIUS dynamic authorization:

```
switch(config)# radius dyn-authorization enable
```

## radius dyn-authorization client

### Syntax

```
radius dyn-authorization client {<IPV4> | <IPV6> | <HOSTNAME>}
 [secret-key {plaintext | ciphertext} <PASSKEY>]
 [time-window <WIDTH>] [replay-protection {enable|disable}]
 [vrf <VRF-NAME>]
```

```
no radius dyn-authorization client {<IPV4> | <IPV6> | <HOSTNAME>} [vrf <VRF-NAME>]
```

### Description

Configures RADIUS dynamic authorization for the specified client on the specified (or `default`) VRF.

The `no` form of this command unconfigures RADIUS dynamic authorization for the specified client on the specified (or `default`) VRF.

## Command context

config

## Parameters

**<IPV4> | <IPV6> | <HOSTNAME>**

Specifies the client IPv4 address, IPv6 address, or host name.

**secret-key {plaintext | ciphertext} <PASSKEY>**

Specifies the dynamic authorization server (RADIUS server) shared secret key required for client access. Provide either a plaintext or an encrypted shared-secret passkey. As per RFC 2865, the shared-secret can be a mix of alphanumeric and special characters. Plaintext passkeys are between 1 and 32 alphanumeric and special characters.

**time-window <WIDTH>**

Specifies the width of the synchronization window (in seconds) between the RADIUS dynamic authorization client and the RADIUS dynamic authorization server. Default 300. Range: 1 to 65535.

**replay-protection {enable|disable}**

Enables or disables RADIUS dynamic authorization replay protection for the specified client on the specified (or default) VRF.

**vrf <VRF-NAME>**

Specifies the VRF on which the identified client is connected. When omitted, VRF `default` is assumed.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring RADIUS dynamic authorization with replay protection for a client on the default VRF:

```
switch(config)# radius dyn-authorization client 1.1.2.5 replay-protection enable
```

Configuring RADIUS dynamic authorization with time window and shared secret for a client on the default VRF:

```
switch(config)# radius dyn-authorization client 1.1.2.7 time-window 8 secret-key sk
```

# radius dyn-authorization port

## Syntax

**radius dyn-authorization port <PORT-NUMBER>**

## Description

Sets the RADIUS dynamic authorization server UDP/TCP port.

## Command context

config

## Parameters

**<PORT-NUMBER>**

Specifies the UDP or TCP port. Default UDP: 3799 and TCP:2083

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the RADIUS dynamic authorization server UDP port back to its default 3799:

```
switch(config)# radius dyn-authorization port 3799
```

Setting the RADIUS dynamic authorization server TCP port back to its default 2083:

```
switch(config)# radius dyn-authorization port 2083
```

# show radius dyn-authorization

## Syntax

```
show radius dyn-authorization
```

## Description

Shows RADIUS dynamic authorization configuration and summarized statistics for all clients configured for dynamic authorization.

## Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Usage

Show command output item identification:

- Radius Dynamic Authorization: Enabled or Disabled status, system wide.
- Radius Dynamic Authorization Port: The UDP or TCP port used for dynamic authorization (default 3799).
- Invalid Client Address in CoA Requests: The number of CoA (change of authorization) requests received with an incorrect DAC (dynamic authorization client) address.
- Invalid Client Address in Disconnect Requests: The number of disconnect requests received with incorrect DAC address.
- Disconnect Requests: The number of disconnect requests received from the DAC.
- Disconnect ACKs: The number of Disconnect-ACKs sent to the DAC.
- Disconnect NAKs: The number of Disconnect-NAKs sent to the DAC.
- CoA Requests: The number of CoA-requests received from the DAC.
- CoA ACKs: The number of CoA-ACKs sent to the DAC.
- CoA NAKs: The number of CoA-NAKs sent to the DAC.

## Example

Showing RADIUS dynamic authorization summarized statistics for all clients configured for dynamic authorization:

```
switch# show radius dyn-authorization
Status and Counters - RADIUS Dynamic Authorization Information

 RADIUS Dynamic Authorization : Enabled
 RADIUS Dynamic Authorization UDP Port : 3799
 Invalid Client Addresses in CoA Requests : 0
 Invalid Client Addresses in Disconnect Requests: 0

Dynamic Authorization Client Information
=====

IP Address : 1.1.2.1
VRF : adm2
Replay Protection : Disabled
Time Window : 20
Disconnect Requests : 1
Disconnect ACKs : 1
Disconnect NAKs : 0
CoA Requests : 7
CoA ACKs : 2
CoA-NAKs : 5
Shared-Secret : AQBapb+HsdpqV1Q3CPCBMQTG8ekK1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=

IP Address : 1.1.2.5
VRF : default
Replay Protection : Enabled
Time Window : 20
Disconnect Requests : 6
Disconnect ACKs : 6
Disconnect NAKs : 0
CoA Requests : 9
CoA ACKs : 5
CoA-NAKs : 4
Shared-Secret : AQBapb+HsdpqV1Q3CPCBMQTG8ekK1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=

IP Address : 1.1.2.7
VRF : default
Replay Protection : Disabled
Time Window : 8
Disconnect Requests : 6
Disconnect ACKs : 6
Disconnect NAKs : 0
CoA Requests : 9
CoA ACKs : 5
CoA-NAKs : 4
Shared-Secret : AQBapb+HsdpqV1Q3CPCBMQTG8ekK1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=
```

## show radius dyn-authorization client

### Syntax

```
show radius dyn-authorization client <IP-ADDR> [vrf <VRF-NAME>]
```

### Description

Shows RADIUS dynamic authorization statistics for the specified client on the specified VRF.

## Command context

Operator (>) or Manager (#)

## Parameters

**client** <IP-ADDR>

Specifies the client IPv4 or IPv6 address.

**vrf** <VRF-NAME>

Specifies the VRF on which the identified client is connected. When omitted, VRF `default` is assumed.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Usage

Show command output item identification:

- **Total Requests:** The number of Disconnect and CoA (change of authorization) requests received from the DAC (dynamic authorization client).
- **Authorize Only Requests:** The number of Disconnect and CoA requests received from the DAC with an "Authorize only" Service-Type attribute.
- **Malformed Requests:** The number of malformed Disconnect and CoA requests received from the DAC.
- **Bad Authenticator Requests:** The number of Disconnect and CoA requests received from this DAC with an invalid authenticator field.
- **Dropped Requests:** The number of Disconnect and CoA requests from this DAC that have been silently discarded for reasons other than malformed, bad authenticators, or unknown type.
- **Total ACK Responses:** The number of Disconnect-ACKs sent to the DAC.
- **Total NAK Responses:** The number of Disconnect-NAKs sent to the DAC.
- **Session Not Found Responses:** The number of Disconnect-NAKs sent to the DAC because no session context could be found.
- **User Sessions Modified:** The number of user sessions for which authorization changed due to Disconnect and CoA requests received from the DAC.

## Example

Showing RADIUS dynamic authorization statistics for client 1.1.2.1 on VRF default:

```
switch# show radius dyn-authorization client 1.1.2.1 vrf default
Status and Counters - RADIUS Dynamic Authorization Client Information
```

|                            |   |            |            |
|----------------------------|---|------------|------------|
| VRF Name                   | : | default    |            |
| Authorization Client       | : | 1.1.2.1    |            |
| Unknown Packets            | : | 55         |            |
| Message-Type               |   | Disconnect | CoA        |
| -----                      |   |            |            |
| Total Requests             |   | 2147483647 | 10         |
| Authorize Only Requests    |   | 10         | 10         |
| Malformed Requests         |   | 10         | 10         |
| Bad Authenticator Requests |   | 2147483647 | 2147483647 |
| Dropped Requests           |   | 10         | 10         |
| Total ACK Responses        |   | 10         | 10         |

|                             |    |    |
|-----------------------------|----|----|
| Total NAK Responses         | 10 | 10 |
| Session Not Found Responses | 10 | 10 |
| User Sessions Modified      | 20 | 20 |



**NOTE:** On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

## aaa accounting all-mgmt

### Syntax

```
aaa accounting all-mgmt <CONNECTION-TYPE> start-stop {local | group <GROUP-LIST>}
```

```
no aaa accounting all-mgmt <CONNECTION-TYPE>
```

### Description

Defines accounting as being local (with the name `local`) (the default). Or defines a sequence of remote AAA server groups to be accessed for accounting purposes.

For remote accounting, the information is sent to the first reachable remote server that was configured with this command for remote accounting. If no remote server is reachable, local accounting remains available. Each available connection type (channel) can be configured individually as either local or using remote AAA server groups. All server groups named in your command, must exist. This command can be issued multiple times, once for each connection type. Local is always available for any connection type not configured for remote accounting.



**NOTE:** The system accounting log is not associated with any connection type (channel) and is therefore sent to the accounting method configured on the default connection type (channel) only.

The `no` form of this command removes for the specified connection type, any defined remote AAA server group accounting sequence. Local accounting is available for connection types without a configured remote AAA server group list (whether default or for the specific connection type).

### Command context

`config`

### Parameters

**<CONNECTION-TYPE>**

One of these connection types (channels):

#### **default**

Defines a list of accounting server groups to be used for the `default` connection type. This configuration applies to all other connection types (`console`, `https-server`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa accounting all-mgmt console...` to define the console accounting list, then this default configuration is used for console.

#### **console**

Defines a list of accounting server groups to be used for the `console` connection type.



### **https-server**

Defines a list of accounting server groups to be used for the `https-server` (REST, Web UI) connection type.

### **ssh**

Defines a list of accounting server groups to be used for the `ssh` connection type.

### **start-stop**

Selects accounting information capture at both the beginning and end of a process.

### **local**

Selects local-only accounting when used without the `group` parameter.

### **group <GROUP-LIST>**

Specifies the list of remote AAA server group names. Each name can be specified one time. Predefined remote AAA group names `tacacs` and `radius` are available. Although not a group name, predefined name `local` is available. User-defined TACACS+ and RADIUS server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa group server` and servers are added to a server group with the command `server`.

## **Authority**

Administrators or local user group members with execution rights for this command.

## **Usage**

Local accounting is always active. It cannot be turned off.

## **Examples**

Defining the default accounting sequence based on two user-defined TACACS+ server groups, then the default TACACS+ server group, and finally (if needed), local accounting.

```
switch(config)# aaa accounting all-mgmt default start-stop group tg1 tg2 tacacs local
```

Defining the console accounting sequence based on two user-defined TACACS+ server groups, then the default TACACS+ server group, and finally (if needed), local accounting.

```
switch(config)# aaa accounting all-mgmt console start-stop group tg2 tg3 tacacs local
```

Defining the ssh accounting sequence based on one user-defined TACACS+ server group and then the default TACACS+ server group.

```
switch(config)# aaa accounting all-mgmt ssh start-stop group tg2 tacacs
```

Defining the default accounting sequence based on two user-defined RADIUS server groups, then the default RADIUS server group, and finally (if needed), local accounting.

```
switch(config)# aaa accounting all-mgmt default start-stop group rg1 rg2 radius local
```

Defining the https-server accounting sequence based on one user-defined RADIUS server group and then the default RADIUS server group.

```
switch(config)# aaa accounting all-mgmt https-server start-stop group rg1 radius
```

Setting local accounting for the default connection type:

```
switch(config)# aaa accounting all-mgmt default start-stop local
```

## aaa authentication allow-fail-through

### Syntax

```
aaa authentication allow-fail-through
```

```
no aaa authentication allow-fail-through
```

### Description

Enables authentication fail-through. When this option is enabled, the next server/authentication method is tried after an authentication failure.

The `no` form of this command disables authentication fail-through. If the system fails to authenticate with a reachable TACACS+ or RADIUS server, the system does not attempt to authenticate with the next TACACS+/RADIUS server.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Enabling authentication fail-through:

```
switch(config)# aaa authentication allow-fail-through
```

## aaa authentication login

### Syntax

```
aaa authentication login <CONNECTION-TYPE> {local | group <GROUP-LIST>}
```

```
no aaa authentication login <CONNECTION-TYPE>
```

### Description

Defines authentication as being local (with the name `local`) (the default). Or defines a sequence of remote AAA server groups to be accessed for authentication purposes. Each available connection type (channel) can be configured individually as either local or using remote AAA server groups. All server groups named in your command, must exist. This command can be issued multiple times, once for each connection type. Local is always available for any connection type not configured for remote AAA authentication.



**NOTE:** If you do not want local authentication to occur in cases where all AAA servers contacted reject the user's credentials, do not enable authentication fail-through (command `aaa authentication allow-fail-through`).

The `no` form of this command removes for the specified connection type, any defined remote AAA server group authentication sequence. Local authentication is available for connection types without a configured remote AAA server group list (whether default or for the specific connection type).

## Command context

config

## Parameters

**<CONNECTION-TYPE>**

One of these connection types (channels):

### **default**

Defines a list of authentication server groups to be used for the `default` connection type. This configuration applies to all other connection types (`console`, `https-server`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa authentication login console...` to define the console authentication list, then this default configuration is used for console.

### **console**

Defines a list of authentication server groups to be used for the `console` connection type.

### **https-server**

Defines a list of authentication server groups to be used for the `https-server` (REST, Web UI) connection type.

### **ssh**

Defines a list of authentication server groups to be used for the `ssh` connection type.

### **local**

Selects local-only authentication when used without the `group` parameter.

**group <GROUP-LIST>**

Specifies the list of remote AAA server group names. Each name can be specified one time. Predefined remote AAA group names `tacacs` and `radius` are available. Although not a group name, predefined name `local` is available. User-defined TACACS+ and RADIUS server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa group server` and servers are added to a server group with the command `server`.

If no AAA server is reachable, local authentication is attempted.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Defining the default authentication sequence based on two user-defined TACACS+ server groups, then the default TACACS+ server group, and finally (if needed), local authentication.

```
switch(config)# aaa authentication login default group tg1 tg2 tacacs local
```

Defining the console authentication sequence based on two user-defined TACACS+ server groups, then the default TACACS+ server group, and finally (if needed), local authentication.

```
switch(config)# aaa authentication login console group tg2 tg3 tacacs local
```

Defining the ssh authentication sequence based on one user-defined TACACS+ server group and then the default TACACS+ server group.

```
switch(config)# aaa authentication login ssh group tg2 tacacs
```

Defining the default authentication sequence based on two user-defined RADIUS server groups, then the default RADIUS server group, and finally (if needed), local authentication.

```
switch(config)# aaa authentication login default group rg1 rg2 radius local
```

Defining the https-server authentication sequence based on one user-defined RADIUS server group and then the default RADIUS server group.

```
switch(config)# aaa authentication login https-server group rg1 radius
```

Setting local authentication for the default connection type:

```
switch(config)# aaa authentication login default local
```

## aaa authorization commands

### Syntax

```
aaa authorization commands <CONNECTION-TYPE> {local | none}
aaa authorization commands <CONNECTION-TYPE> group <GROUP-LIST>
```

```
no aaa authorization commands <CONNECTION-TYPE>
```

### Description

Defines authorization as being basic local RBAC (specified as `none`), or as full-fledged local RBAC specified as `local` (the default), or as remote TACACS+ (specified with `group <GROUP-LIST>`). Each available connection type (channel) can be configured individually. All server groups named in the command, must exist. This command can be issued multiple times, once for each connection type.

The `no` form of this command unconfigures authorization for the specified connection type, reverting to the default of `local`.



**NOTE:** Although only TACACS+ servers are supported for remote authorization, local authorization (basic or full-fledged) can be used with remote RADIUS authentication.

### Command context

config

### Parameters

**<CONNECTION-TYPE>**

One of these connection types (channels):

#### default

Selects the `default` connection type for configuration. This configuration applies to all other connection types (`console`, `ssh`) that are not explicitly configured with this command. For example, if you do not use `aaa authorization commands console...` to define the console authorization list, then this default configuration is used for console.

#### console

Selects the `console` connection type for configuration.

#### ssh

Selects the `ssh` connection type for configuration.

## local (the default)

When used alone without `group <GROUP-LIST>`, selects local authorization which can be used to provide authorization for a purely local setup without any remote AAA servers and also for when RADIUS is used for remote Authentication and Accounting but Authorization is local.

When used after `group`, provides for fallback (to full-fledged local authorization) when every server in every specified TACACS+ server group cannot be reached.



**NOTE:** If any TACACS+ server in the specified groups is reachable, but the command fails to be authorized by that server, the command is rejected and local authorization is never attempted. Local authorization is only attempted if every TACACS+ server cannot be reached.

## none

When used alone without `group <GROUP-LIST>`, selects basic local RBAC authorization, for use with the built-in user groups (administrators, operators, auditors).

When used after `group`, provides for fallback (to basic local RBAC authorization) when every server in every specified TACACS+ server group cannot be reached.



**NOTE:** With `none`, for users belonging to user-defined user groups, all commands can be executed regardless of what authorization rules are defined in such groups. For per-command local authorization, use `local` instead.

## group <GROUP-LIST>

Specifies the list of remote AAA server group names. Predefined remote AAA group name `tacacs` is available. User-defined TACACS+ server group names may also be used. The remote AAA server groups are accessed in the order that the group names are listed in this command. Within each group, the servers are accessed in the order in which the servers were added to the group. Server groups are defined using command `aaa server group` and servers are added to a server group using command `server`.

It is recommended to always include either the special name `local` or `none` as the last name in the group list. If both `local` and `none` are omitted, and no remote AAA server is reachable (or the first reachable server cannot authorize the command), command execution for the current user will not be possible.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

### TACACS+ server authorization considerations



**NOTE:** Use caution when configuring authorization, as it has no fail through. If the switch is not configured properly, the switch might get into an unusable state in which all command execution is prohibited.

To prevent authorization difficulties:

- Make sure that all listed TACACS+ servers can authorize users for command execution.
- Make sure that credential database changes are promptly synchronized across all TACACS+ servers.

- Make sure either `local` or `none` is included as the last name in the group list. If both `local` and `none` are omitted, and no remote TACACS+ server is reachable (or the first reachable server cannot authorize), authorization will not be possible.
- Although not recommended, if you choose to omit both `local` and `none` from the list, and are manipulating configuration files, special caution is necessary. If the source configuration includes TACACS+ authorization and you are copying configuration from an existing switch into the running configuration of a new switch, and you have not yet configured the interface or routing information to reach the TACACS+ server, the switch will enter an unusable state, requiring hard reboot.

To avoid getting into this situation that can occur when `local` and `none` have been omitted, do either of the following:

- In the configuration source, delete or comment-out the line configuring remote authorization. Then, after the configuration copy and paste, manually configure authorization.
- Move the line configuring the authorization to the end of the source configuration before copying and pasting.

## Examples

Defining the default authorization sequence based on a user-defined TACACS+ server group, then the default TACACS+ server group, and finally (as a precaution), `local` authorization:

```
switch(config)# aaa authorization commands default group tg1 tacacs local
All commands will fail if none of the servers in the group list are reachable.
Continue (y/n)? y
```

Defining the console authorization sequence based on two user-defined TACACS+ server groups, and finally (as a precaution), `local` authorization:

```
switch(config)# aaa authorization commands console group tg1 tg2 local
All commands will fail if none of the servers in the group list are reachable.
Continue (y/n)? y
```

Setting the authorization for default to `local`:

```
switch(config)# aaa authorization commands default local
```

Setting the authorization for the SSH interface to `none`:

```
switch(config)# aaa authorization commands ssh none
```

## aaa group server

### Syntax

```
aaa group server {tacacs | radius} <SERVER-GROUP-NAME>
```

```
no aaa group server {tacacs | radius} <SERVER-GROUP-NAME>
```

### Description

Creates an AAA server group that is either empty or contains preconfigured RADIUS/TACACS+ servers. You can create a maximum of 28 server groups.

The `no` form of this command deletes a server group. Only a preconfigured user-defined RADIUS/TACACS+ server group can be deleted. RADIUS or TACACS+ servers that were in a deleted server group remain a part

of their default server group. The default server group for TACACS+ servers is `tacacs`. The default server group for RADIUS servers is `radius`.

### Command context

`config`

### Parameters

**server {tacacs | radius}**

Select either `tacacs` or `radius` for the server type.

**<SERVER-GROUP-NAME>**

Specifies the name of the server group to be created. The name of the server group can have a maximum of 32 characters.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating TACACS+ server group `sg1`:

```
switch(config)# aaa group server tacacs sg1
```

Creating RADIUS server group `sg3`:

```
switch(config)# aaa group server radius sg3
```

Deleting TACACS+ server group `sg1`:

```
switch(config)# no aaa group server tacacs sg1
```

Deleting RADIUS server group `sg3`:

```
switch(config)# no aaa group server radius sg3
```

## radius-server auth-type

### Syntax

`radius-server auth-type {pap | chap}`

`no radius-server auth-type`

### Description

Enables the CHAP or PAP authentication protocol, which is used for communication with the RADIUS servers, at the global level. You can override this command with a fine-grained per server `auth-type` configuration.

The `no` form of this command resets the global authentication mechanism for RADIUS to PAP, which is the default authentication mechanism for RADIUS.

### Command context

`config`

## Parameters

**auth-type** {pap | chap}

Selects either the PAP or CHAP authentication protocol.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Authenticating CHAP:

```
switch(config)# radius-server auth-type chap
```

Authenticating PAP:

```
switch(config)# radius-server auth-type pap
```

# radius-server host

## Syntax

```
radius-server host {<FQDN> | <IPv4> | <IPv6>} [key {plaintext <PASSKEY> |
 ciphertext <PASSKEY>}] [timeout <TIMEOUT-SECONDS>] [port <PORT-NUMBER>]
 [auth-type {pap | chap}] [acct-port <ACCT-PORT>] [retries <RETRY-COUNT>]
 [tracking {enable | disable}] [tracking-mode {any | dead-only}] [vrf <VRF-NAME>]

no radius-server host {<FQDN> | <IPv4> | <IPv6>} [port <PORT-NUMBER>] [vrf <VRF-NAME>]
```

## Description

Adds a RADIUS server. By default, the RADIUS server is associated with the server group named `radius`.

The `no` form of this command removes a previously added RADIUS server.



**NOTE:** For enhanced security with IPsec, the alternative command `radius-server host secure ipsec` is available. The standard non-IPsec `radius-server host` command does not modify any existing IPsec configuration. If IPsec is already configured for the RADIUS server, then IPsec will remain enabled for the server.

## Command context

config

## Parameters

{<FQDN> | <IPv4> | <IPv6>}

Specifies the RADIUS server as:

- <FQDN>: a fully qualified domain name.
- <IPv4>: an IPv4 address.
- <IPv6>: an IPv6 address.



**key** {plaintext <PASSKEY> | ciphertext <PASSKEY>}

Specifies either a plaintext or an encrypted local shared-secret passkey for the server. As per RFC 2865, the shared-secret can be a mix of alphanumeric and special characters. Plaintext passkeys are between 1 and 32 alphanumeric and special characters.

**timeout** <TIMEOUT-SECONDS>

Specifies the timeout. The range is 1 to 60 seconds. If a timeout is not specified, the value from the global timeout for RADIUS is used.

**port** <PORT-NUMBER>

Specifies the authentication port number. Range: 1 to 65535. Default RADIUS: 1812.

**auth-type** {pap | chap}

Selects either PAP (default) or CHAP authentication type. If this parameter is not specified, the RADIUS global default is used.

**acct-port** <ACCT-PORT>

Specifies the UDP accounting port number. Range: 1 to 65535. Default: 1813.

**retries** <RETRY-COUNT>

Specifies the number of retry attempts for contacting the specified RADIUS server. Range is 0 to 5 attempts. If no retry value is provided, the default value of 1 is used.

**tracking** {enable | disable}

Enables or disables server tracking for the RADIUS server. Tracked servers are probed at the start of each server tracking interval to check if they are reachable.

Use command `radius-server tracking` to configure RADIUS server tracking globally.



**NOTE:** Server tracking uses authentication request and response packets to determine server reachability status. The server tracking user name and password are used to form the request packet which is sent to the server with tracking enabled. Upon receiving a response to the request packet, the server is considered to be reachable.

**tracking-mode** {any | dead-only}

Configures tracking mode for the RADIUS server that has tracking enabled with the server. The tracking mode is used to monitor the status of RADIUS server reachability. The default tracking mode is `any`.

Sets the tracking mode to:

- `any`: track the RADIUS server irrespective of its server reachability.
- `dead-only`: track the RADIUS server only when the server is marked as unreachable.

**vrf** <VRF-NAME>

Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

If no local passkey is provided in the command, the server will use the global passkey. This command requires either the global or local passkey to be set; otherwise, the server will not be contacted. To set the global passkey, see [radius-server key](#).

If the fully qualified domain name is provided for the RADIUS server, a DNS server must be configured and accessible through the same VRF which is configured for the RADIUS server. This configuration is required for the resolution of the RADIUS server hostname to its IP address. If a DNS server is not available for this VRF, the RADIUS servers reachable through this VRF must be configured by means of their IP addresses only.

## Examples

Adding a RADIUS server with an IPv4 address and a named VRF:

```
switch(config)# radius-server host 1.1.1.1 vrf mgmt
```

Adding a RADIUS server with an IPv4 address, a port, and a named VRF:

```
switch(config)# radius-server host 1.1.1.2 port 32 vrf mgmt
```

Adding a RADIUS server with an IPv6 address:

```
switch(config)# radius-server host 2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Adding a RADIUS server with tracking enabled and tracking mode is set to dead-only:

```
switch(config)# radius-server host 1.1.1.1 tracking enable tracking-mode dead-only
```

Adding a RADIUS server with tracking disabled:

```
switch(config)# radius-server host 1.1.1.1 tracking disable
```

Deleting a RADIUS server with an IPv4 address and specified VRF:

```
switch(config)# no radius-server host 1.1.1.1 vrf mgmt
```

## radius-server host secure ipsec

### Syntax

Syntax for a RADIUS server that uses IPsec for authentication:

```
radius-server host {<FQDN> | <IPv4> | <IPv6>} [key {plaintext <PASSKEY> |
 ciphertext <PASSKEY>}] [timeout <TIMEOUT-SECONDS>] [port <PORT-NUMBER>]
[auth-type {pap | chap}] [acct-port <ACCT-PORT>] [retries <RETRY-COUNT>]
[tracking {enable | disable}] [tracking-mode {any | dead-only}] [vrf <VRF-NAME>]
secure ipsec authentication spi <SPI-INDEX> <AUTH-TYPE> <AUTH_KEY-TYPE> <AUTH-KEY>
```

```
no radius-server host {<FQDN> | <IPv4> | <IPv6>} [port <PORT-NUMBER>] [vrf <VRF-NAME>]
secure ipsec authentication
```

Syntax for a RADIUS server that uses IPsec for both authentication and encryption:

```
radius-server host {<FQDN> | <IPv4> | <IPv6>} [key {plaintext <PASSKEY> |
 ciphertext <PASSKEY>}] [timeout <TIMEOUT-SECONDS>] [port <PORT-NUMBER>]
[auth-type {pap | chap}] [acct-port <ACCT-PORT>] [retries <RETRY-COUNT>]
[tracking {enable | disable}] [tracking-mode {any | dead-only}] [vrf <VRF-NAME>]
secure ipsec encryption spi <SPI-INDEX> <AUTH-TYPE> <AUTH_KEY-TYPE> <AUTH-KEY>
<ENCRYPT-TYPE> <ENCRYPT-KEY-TYPE> <ENCRYPT-KEY>
```

```
no radius-server host {<FQDN> | <IPv4> | <IPv6>} [port <PORT-NUMBER>] [vrf <VRF-NAME>]
secure ipsec encryption
```

## Description

Adds a RADIUS server that uses IPsec for enhanced security (authentication and possibly encryption). By default, the RADIUS server is associated with the server group named `radius`.

The `no` form of this command removes a previously added RADIUS (with IPsec) server.



**NOTE:** Unless enhanced security with IPsec is required, use the `radius-server host` command instead.

## Command context

`config`

## Parameters

**{<FQDN> | <IPV4> | <IPV6>}**

Specifies the RADIUS server as:

- `<FQDN>`: a fully qualified domain name.
- `<IPV4>`: an IPv4 address.
- `<IPV6>`: an IPv6 address.

**key {plaintext <PASSKEY> | ciphertext <PASSKEY>}**

Selects either a plaintext or an encrypted local shared-secret passkey for the server. As per RFC 2865, shared-secret can be a mix of alphanumeric and special characters. Plaintext passkeys are between 1 and 32 alphanumeric and special characters.

**timeout <TIMEOUT-SECONDS>**

Specifies the timeout. The range is 1 to 60 seconds. If a timeout is not specified, the value from the global timeout for RADIUS is used.

**port <PORT-NUMBER>**

Specifies the authentication port number. Range: 1 to 65535. Default: 1812.

**auth-type {pap | chap}**

Selects either the PAP (the default) or CHAP authentication types. If this parameter is not specified, the RADIUS global default is used.

**acct-port <ACCT-PORT>**

Specifies the UDP accounting port number. Range: 1 to 65535. Default: 1813.

**retries <RETRY-COUNT>**

Specifies the number of retry attempts for contacting the specified RADIUS server. Range is 0 to 5 attempts. If no retry value is provided, the default value of 1 is used.

**tracking {enable | disable}**

Enables or disables server tracking for the RADIUS server. Tracked servers are probed at the start of each server tracking interval to check if they are reachable.

Use command `radius-server tracking` to configure RADIUS server tracking globally.



**NOTE:** Server tracking uses authentication request and response packets to determine server reachability status. The server tracking user name and password are used to form the request packet which is sent to the server with tracking enabled. Upon receiving a response to the request packet, the server is considered to be reachable.

**tracking-mode {any | dead-only}**

Configures tracking mode for the RADIUS server that has tracking enabled with the server. The tracking mode is used to monitor the status of RADIUS server reachability. The default tracking mode is `any`.

Sets the tracking mode to:

- `any`: track the RADIUS server irrespective of its server reachability.
- `dead-only`: track the RADIUS server only when the server is marked as unreachable.

**vrf <VRF-NAME>**

Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

**spi <SPI-INDEX>**

Specifies the Security Parameters Index. The SPI is an identification tag carried in the IPsec AH header. The SPI must be unique on the switch. Range: 256 to 4294967295.

**<AUTH-TYPE>**

Specifies the authentication algorithm: `md5`, `sha1`, or `sha256`.

**<AUTH\_KEY-TYPE>**

Specifies the authentication key type: `plaintext`, `hex-string`, or `ciphertext`.

**<AUTH-KEY>**

Specifies the authentication key. For **<AUTH-TYPE>** of `ciphertext`, this is the ciphertext string.

For **<AUTH-TYPE>** of `plaintext` or `hex-string`:

- `md5` (`plaintext`): 1 to 16 characters, (`hex-string`): 2 to 32 hexadecimal digits.
- `sha1` (`plaintext`): 1 to 20 characters, (`hex-string`): 2 to 40 hexadecimal digits.
- `sha256` (`plaintext`): 1 to 32 characters, (`hex-string`): 2 to 64 hexadecimal digits.

**<ENCRYPT-TYPE>**

Specifies the encryption algorithm: `3des`, `aes`, `des`, or `null`.

**<ENCRYPT-KEY-TYPE>**

Specifies the encryption key type: `plaintext`, `hex-string`, or `ciphertext`.

**<ENCRYPT-KEY>**

Specifies the encryption key. For **<ENCRYPT-TYPE>** of `ciphertext`, this is the ciphertext string.

For **<ENCRYPT-TYPE>** of `plaintext` or `hex-string`:

- `3des` (`plaintext`): 24 characters, (`hex-string`): 48 hexadecimal digits.
- `aes` (`plaintext`): 16, 24, or 32 characters, (`hex-string`): 32, 48, or 64 hexadecimal digits.
- `des` (`plaintext`): 8 characters, (`hex-string`): 16 hexadecimal digits.

**Authority**

Administrators or local user group members with execution rights for this command.

## Usage

If no local passkey is provided in the command, the server will use the global passkey. This command requires either the global or local passkey to be set; otherwise, the server will not be contacted. To set the global passkey, see [radius-server key](#).

If the fully qualified domain name is provided for the RADIUS server host, a DNS server must be configured and accessible through the same VRF as mentioned for the server host. This configuration is required for the resolution of the RADIUS server hostname to its IP address. If a DNS server is not available for this VRF, the RADIUS servers reachable through this VRF must be configured by means of their IP addresses only.

## Examples

Adding a RADIUS server with an IPv4 address, a plaintext passkey, and IPsec authentication (md5 plaintext).

```
switch(config)# radius-server host 1.1.1.1 key plaintext 98ab vrf mgmt secure
ipsec authentication spi 261 md5 plaintext 1abc
```

Adding a RADIUS server with an IPv4 address, IPsec authentication (MD5 plaintext), and IPsec encryption (AES plaintext):

```
switch(config)# radius-server host 1.1.1.2 vrf mgmt secure
ipsec encryption spi 262 md5 plaintext 9xyz aes plaintext 1234567890abcdef
```

Adding a RADIUS server with an IPv4 address, tracking enabled, tracking mode, IPsec authentication (MD5 plaintext), IPsec encryption (AES plaintext) is set to dead-only:

```
switch(config)# radius-server host 1.1.1.1 tracking enable tracking-mode dead-only
vrf mgmt secure ipsec encryption spi 262 md5 plaintext 9xyz aes plaintext 1234567890abcdef
```

Removing a RADIUS server with an IPv4 address and IPsec authentication:

```
switch(config)# no radius-server host 1.1.1.1 vrf mgmt secure ipsec authentication
```

Removing a RADIUS server with an IPv4 address and IPsec authentication and IPsec encryption:

```
switch(config)# no radius-server host 1.1.1.2 vrf mgmt secure ipsec encryption
```

## radius-server key

### Syntax

```
radius-server key {plaintext <GLOBAL-PASSKEY> | ciphertext <GLOBAL-PASSKEY>}
```

```
no radius-server key
```

### Description

Creates or modifies a RADIUS global passkey. The RADIUS global passkey is used as a shared-secret for encrypting the communication between all RADIUS servers and the switch. The RADIUS global passkey is required for authentication unless local passkeys have been set. By default, the RADIUS global passkey is empty. If the administrator has not set this key, the switch will not be able to perform RADIUS authentication. The switch will instead rely on the authentication mechanism configured with [aaa authentication login](#).

The `no` form of the command removes the global passkey.

### Authority

Administrators or local user group members with execution rights for this command.

## Parameters

**plaintext** <GLOBAL-PASSKEY>

Specifies the RADIUS global passkey in plaintext format with a length of 1 to 31 characters. As per RFC 2865, a shared-secret can be a mix of alphanumeric and special characters.

**ciphertext** <GLOBAL-PASSKEY>

Specifies the RADIUS global passkey in encrypted format.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Adding the global passkey:

```
switch(config)# radius-server key plaintext mypasskey123
```

Removing the global passkey:

```
switch(config)# no radius-server key
```

# radius-server retries

## Syntax

```
radius-server retries <0-5>
```

```
no radius-server retries
```

## Description

Sets at the global level the number of retries the switch makes before concluding that the RADIUS server is unreachable.

You can override this setting with a fine-grained per RADIUS server retries configuration. See [radius-server host](#).

The **no** form of this command resets the RADIUS global retries to the default retries value of 1.

## Command context

```
config
```

## Parameters

**retries** <0-5>

Specifies the number of retry attempts for contacting RADIUS servers. Range is 0 to 5 retries.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

```
switch(config)# radius-server retries 3
```

## radius-server timeout

### Syntax

```
radius-server timeout [<1-60>]
```

```
no radius-server timeout
```

### Description

Specifies the number of seconds to wait for a response from the RADIUS server before trying the next RADIUS server. If a value is not specified, a default value of 5 seconds is used. You can override this value with a fine-grained per server timeout configured for individual servers.

The `no` form of this command resets the RADIUS global authentication timeout to the default of 5 seconds.

### Command context

```
config
```

### Parameters

**timeout** <1-60>

Specifies the timeout interval of 1 to 60 seconds. The default is 5 seconds.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the RADIUS server timeout:

```
switch(config)# radius-server timeout 10
```

Resetting the timeout for the RADIUS server to the default:

```
switch(config)# no radius-server timeout
```

## radius-server tracking

### Syntax

```
radius-server tracking {interval <INTERVAL> | retries <RETRIES> | user-name <NAME>
 [password {plaintext <PLAINTEXT-PASSWORD> | ciphertext <CIPHERTEXT-PASSWORD>}]}
```

```
no radius-server tracking {interval | retries | user-name}
```

### Description

Configures RADIUS server tracking settings globally for all configured RADIUS servers that have tracking enabled with the `radius-server host` command on individual servers.

The `no` form of the command removes the specified configuration, reverting it to its default. The `no` form with `user-name` also clears the password (resets it to empty).

### Authority

Administrators or local user group members with execution rights for this command.

## Parameters

**interval** <INTERVAL>

Specifies the time interval, in seconds, to wait before checking the server reachability status. Default: 300. Range 60 to 84600.

**retries** <RETRIES>

Specifies the number of server retries. Default: Global RADIUS retries. Range: 0 to 5.

**user-name** <NAME> [password {plaintext <PLAINTEXT-PASSWORD> | ciphertext <CIPHERTEXT-PASSWORD>} ]

Specifies the user name to be used for server checking. The default user name is `radius-tracking-user` with an empty password.



**NOTE:** The user does not have to be configured on the server. Server tracking can still be performed with a user which is not configured on the server because authentication failure on the server achieves confirmation that the server is reachable.

The password for the server tracking user name is optional and may be entered as `plaintext` or pasted in as `ciphertext`. The plaintext password is visible as cleartext when entered but is encrypted thereafter. Command history does show the password as cleartext.



**NOTE:** Server tracking uses authentication request and response packets to determine server reachability status. The server tracking user name and password are used to form the request packet which is sent to the server with tracking enabled. Upon receiving a response to the request packet, the server is considered to be reachable.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring a tracking interval of 120 seconds:

```
switch(config)# radius-server tracking interval 120
```

Reverting the tracking interval to its default of 300 seconds:

```
switch(config)# no radius-server tracking interval
```

Configuring three retries:

```
switch(config)# radius-server tracking retries 3
```

Configuring user `radius-tracker` with a plaintext password.

```
switch(config)# radius-server tracking user-name radius-tracker password plaintext track$1
```

Reverting the tracking user name to its default of `radius-tracking-user`:

```
switch(config)# no radius-server tracking user-name
```



# server

## Syntax

```
server {<FQDN> | <IPv4> | <IPv6>} [port <PORT-NUMBER>] [vrf <VRF-NAME>]
```

```
no server {<FQDN> | <IPv4> | <IPv6>} [port <PORT-NUMBER>] [vrf <VRF-NAME>]
```

## Description

Adds a TACACS+/RADIUS server to a server-group. Only the configured TACACS+/RADIUS servers are allowed to be added within the server group. If the same server name exists with multiple ports or multiple VRFs, specify the server name, port, and VRF when adding the server to the server-group.

The `no` form of this command removes a TACACS+/RADIUS server from a server-group.

## Command context

config-sg

## Parameters

{<FQDN> | <IPv4> | <IPv6>}

Specifies the server as:

- <FQDN>: a fully qualified domain name.
- <IPv4>: an IPv4 address.
- <IPv6>: an IPv6 address.

**port** <PORT-NUMBER>

Specifies the authentication port number. Range: 1 to 65535. Default TACACS+ (TCP): 49, RADIUS (UDP): 1812.

If a port number is not provided, the system searches the TACACS+/RADIUS server by host name and sets the default authentication port. Group server priority is assigned based on the sequence in which the servers are added.

**vrf** <VRF-NAME>

Specifies the VRF name.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Adding a server to TACACS+ server group sg1 by providing an IPv4 address, port number, and VRF name:

```
switch(config)# aaa group server tacacs sg1
switch(config-sg)# server 1.1.1.2 port 32 vrf default
```

Adding a server to TACACS+ server group sg2 by providing an IPv6 address and default VRF:

```
switch(config)# aaa group server tacacs sg2
switch(config-sg)# server 2001:0db8:85a3:0000:0000:8a2e:0370:7334 vrf default
```

Adding a server to RADIUS server group sg3 by providing an IPv4 address, port number, and VRF name:

```
switch(config)# aaa group server radius sg3
switch(config-sg)# server 1.1.1.5 port 12 vrf default
```

Adding a server to RADIUS server group sg4 by providing an IPv6 address and default VRF:

```
switch(config)# aaa group server radius sg4
switch(config-sg)# server 2001:0db8:85a3:0000:0000:8a2e:0371:7334 vrf default
```

Adding a server to RADIUS server group sg4 by providing an IPv4 address, port number, and VRF name:

```
switch(config)# aaa group server radius sg4
switch(config-sg)# server 1.1.1.6 port 32 vrf vrf_red
```

Specifying an IPv4 address when removing a TACACS+ server from server group sg1:

```
switch(config)# aaa group server tacacs sg1
switch(config-sg)# no server 1.1.1.2 port 12 vrf default
```

Specifying an IPv6 address when removing a TACACS+ server from server group sg2 with the default VRF:

```
switch(config)# aaa group server tacacs sg2
switch(config-sg)# no server 2001:0db8:85a3:0000:0000:8a2e:0370:7334 vrf default
```

## show aaa accounting

### Syntax

```
show aaa accounting
```

### Description

Shows the accounting configuration per connection type (channel).

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

### Example

Configuring and then showing the accounting sequence for TACACS+ groups and local:

```
switch(config)# aaa accounting all default start-stop group tg1 tg2 tacacs local
switch(config)# aaa accounting all ssh start-stop group tg1 tg2
switch(config)# aaa accounting all console start-stop group tg4 tacacs local
switch(config)# aaa accounting all https-server start-stop local group tacacs tg3
switch(config)# exit
switch# show aaa accounting
```

```
AAA Accounting:
 Accounting Type : all
 Accounting Mode : start-stop
```

Accounting for default channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| tg1        | 0              |

|        |   |
|--------|---|
| tg2    | 1 |
| tacacs | 2 |
| local  | 3 |

Accounting for ssh channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| tg1        | 0              |
| tg2        | 1              |

Accounting for console channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| tg4        | 0              |
| tacacs     | 1              |
| local      | 2              |

Accounting for https-server channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| local      | 0              |
| tacacs     | 1              |
| tg3        | 2              |

Configuring and then showing the accounting sequence for RADIUS groups and local:

```
switch(config)# aaa accounting all default start-stop group rg1 rg2 radius local
switch(config)# aaa accounting all console start-stop group rg4 radius local
switch(config)# exit
switch# show aaa accounting
```

```
AAA Accounting:
 Accounting Type : all
 Accounting Mode : start-stop
```

Accounting for default channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| rg1        | 0              |
| rg2        | 1              |
| radius     | 2              |
| local      | 3              |

Accounting for console channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| tg4        | 0              |
| radius     | 1              |
| local      | 2              |

Configuring and then showing only local accounting for default:

```

switch(config)# aaa accounting all default start-stop local
switch(config)# exit
switch# show aaa accounting
AAA Accounting:
 Accounting Type : all
 Accounting Mode : start-stop

```

Accounting for default channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| local      | 0              |

## show aaa authentication

### Syntax

```
show aaa authentication
```

### Description

Shows the authentication configuration per connection type (channel).

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

### Example

Configuring TACACS+ authentication sequences and then showing the configuration per connection type (channel):

```

switch(config)# aaa authentication login default group tg1 tg2 tg3 tg4 tacacs local
switch(config)# aaa authentication login ssh group tg1 tg2
switch(config)# aaa authentication login console group tg4 tacacs local
switch(config)# aaa authentication login https-server local group tacacs tg3
switch(config)# exit
switch# show aaa authentication

```

```

AAA Authentication:
 Fail-through : Enabled
 Limit Login Attempts : Not set
 Lockout Time : 300
 Minimum Password Length : Not set

```

Authentication for default channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| tg1        | 0              |
| tg2        | 1              |
| tg3        | 2              |
| tg4        | 3              |
| tacacs     | 4              |
| local      | 5              |

Authentication for ssh channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| tg1        | 0              |
| tg2        | 1              |

Authentication for console channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| tg4        | 0              |
| tacacs     | 1              |
| local      | 2              |

Authentication for https-server channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| local      | 0              |
| tacacs     | 1              |
| tg3        | 2              |

Configuring RADIUS authentication sequences and then showing the configuration per connection type (channel):

```
switch(config)# aaa authentication login default group rg1 rg2 rg3 rg4 radius local
switch(config)# aaa authentication login console group rg4 radius local
switch(config)# exit
switch# show aaa authentication
```

AAA Authentication:

|                         |           |
|-------------------------|-----------|
| Fail-through            | : Enabled |
| Limit Login Attempts    | : Not set |
| Lockout Time            | : 300     |
| Minimum Password Length | : Not set |

Authentication for default channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| rg1        | 0              |
| rg2        | 1              |
| rg3        | 2              |
| rg4        | 3              |
| radius     | 4              |
| local      | 5              |

Authentication for console channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| rg4        | 0              |
| radius     | 1              |
| local      | 2              |

Configuring only default authentication and then showing the default connection type (channel):

```
switch(config)# aaa authentication login default local
switch(config)# exit
switch# show aaa authentication
```

```
AAA Authentication:
 Fail-through : Disabled
 Limit Login Attempts : Not set
 Lockout Time : 300
 Minimum Password Length : Not set
```

Authentication for default channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| local      | 0              |

## show aaa authorization

### Syntax

```
show aaa authorization
```

### Description

Shows the authorization configuration per connection type (channel).

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Configuring and then showing the authorization sequence for default and console connection types (channels):

```
switch(config)# aaa authorization commands default group tg1 tacacs none
All commands will fail if none of the servers in the group list are reachable.
Continue (y/n)? y
switch(config)# aaa authorization commands console group tg1 tg2 tacacs none
All commands will fail if none of the servers in the group list are reachable.
Continue (y/n)? y
switch(config)# exit
switch# show aaa authorization
```

Authorization for default channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| tg1        | 0              |
| tacacs     | 1              |
| none       | 2              |

Authorization for console channel:

| GROUP NAME | GROUP PRIORITY |
|------------|----------------|
| tg1        | 0              |
| tg2        | 1              |
| tacacs     | 2              |
| none       | 3              |

## show aaa server-groups

### Syntax

```
show aaa server-groups [tacacs | radius]
```

### Description

Shows TACACS+ and RADIUS AAA server group information for all server types or for the specified server type.

### Command context

Operator (>) or Manager (#)

### Parameters

#### tacacs

Narrows the command output to only TACACS+ servers.

#### radius

Narrows the command output to only RADIUS servers.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Showing all AAA server group information:

```
switch# show aaa server-groups

***** AAA Mechanism TACACS+ *****
```

| GROUP NAME | SERVER NAME                             | PORT | VRF     | PRIORITY |
|------------|-----------------------------------------|------|---------|----------|
| sg2        | 2001:0db8:85a3:0000:0000:8a2e:0370:7334 | 49   | default | 1        |
| sg1        | 1.1.1.2                                 | 12   | mgmt    | 1        |

```

tacacs (default) | FQDN.com | 32 | mgmt | 1
tacacs (default) | 1.1.1.1 | 49 | mgmt | 2
tacacs (default) | 1.1.1.2 | 12 | mgmt | 3
tacacs (default) | abc.com | 32 | vrf_red | 4
tacacs (default) | 2001:0db8:85a3:0000:0000:8a2e:0370:7334 | 49 | default | 5
tacacs (default) | 1.1.1.3 | 32 | vrf_blue | 6
***** AAA Mechanism RADIUS *****
```

| GROUP NAME       | SERVER NAME                             | PORT | VRF      | PRIORITY |
|------------------|-----------------------------------------|------|----------|----------|
| sg4              | 2001:0db8:85a3:0000:0000:8a2e:0370:7334 | 1812 | default  | 1        |
| sg3              | 1.1.1.5                                 | 12   | mgmt     | 1        |
| radius (default) | 1.1.1.4                                 | 1812 | mgmt     | 1        |
| radius (default) | 1.1.1.5                                 | 12   | mgmt     | 2        |
| radius (default) | abc1.com                                | 32   | mgmt     | 3        |
| radius (default) | 2001:0db8:85a3:0000:0000:8a2e:0370:7334 | 1812 | default  | 4        |
| radius (default) | 1.1.1.6                                 | 32   | vrf_red  | 5        |
| radius (default) | 1.1.1.7                                 | 32   | vrf_blue | 6        |

Showing TACACS+ server group information:

```
switch# show aaa server-groups tacacs
```

\*\*\*\*\* AAA Mechanism TACACS+ \*\*\*\*\*

| GROUP NAME       | SERVER NAME                             | PORT | VRF      | PRIORITY |
|------------------|-----------------------------------------|------|----------|----------|
| sg2              | 2001:0db8:85a3:0000:0000:8a2e:0370:7334 | 49   | default  | 1        |
| sg1              | 1.1.1.2                                 | 12   | mgmt     | 1        |
| tacacs (default) | FQDN.com                                | 32   | mgmt     | 1        |
| tacacs (default) | 1.1.1.1                                 | 49   | mgmt     | 2        |
| tacacs (default) | 1.1.1.2                                 | 12   | mgmt     | 3        |
| tacacs (default) | abc.com                                 | 32   | vrf_red  | 4        |
| tacacs (default) | 2001:0db8:85a3:0000:0000:8a2e:0370:7334 | 49   | default  | 5        |
| tacacs (default) | 1.1.1.3                                 | 32   | vrf_blue | 6        |

Showing RADIUS server group information:

```
switch# show aaa server-groups radius
```

\*\*\*\*\* AAA Mechanism RADIUS \*\*\*\*\*

| GROUP NAME       | SERVER NAME                             | PORT | VRF      | PRIORITY |
|------------------|-----------------------------------------|------|----------|----------|
| sg4              | 2001:0db8:85a3:0000:0000:8a2e:0370:7334 | 1812 | default  | 1        |
| sg3              | 1.1.1.5                                 | 12   | mgmt     | 1        |
| radius (default) | 1.1.1.4                                 | 1812 | mgmt     | 1        |
| radius (default) | 1.1.1.5                                 | 12   | mgmt     | 2        |
| radius (default) | abc1.com                                | 32   | mgmt     | 3        |
| radius (default) | 2001:0db8:85a3:0000:0000:8a2e:0370:7334 | 1812 | default  | 4        |
| radius (default) | 1.1.1.6                                 | 32   | vrf_red  | 5        |
| radius (default) | 1.1.1.7                                 | 32   | vrf_blue | 6        |



# show accounting log

## Syntax

```
show accounting log [last <QTY-TO-SHOW> | all]
```

## Description

Entered without optional parameters, this command shows all accounting log records for the current boot. Sensitive information is masked from the log, by being represented as asterisks.



**NOTE:** This `show accounting log` command replaces the `show audit-log` command that is supported only in 10.00 releases.

## Command context

Manager (#) or Auditor (auditor>)

## Parameters

**last <QTY-TO-SHOW>**

Specifies how many most-recent accounting log records to show for the current boot. Range: 1 to 1000.

**all**

Selects for showing, all accounting records from the current boot and the previous boot.

## Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

## Usage

The log message starts with the record type, which is specific to AOS-CX. Values are the following:

### USER\_START

Record of a user login action.

### USER\_END

Record of a user logout action.

### USYS\_CONFIG

Record of a command executed by the user.

The three types of accounting log information are identified by the `msg=` element starting with the `rec=` item as follows:

- Exec is identified with: `msg='rec=ACCT_EXEC'`
- Command is identified with: `msg='rec=ACCT_CMD'`
- System is identified with: `msg='rec=ACCT_SYSTEM'`

The user group is indicated by `priv-lvl`, which is specific to AOS-CX. Values are the following:

| Privilege level | User group     |
|-----------------|----------------|
| 1               | operators      |
| 15              | administrators |
| 19              | auditors       |

The value of `service` indicates which user interface was used:

**service=shell**

Indicates that the log entry is a result of a CLI command.

**service=https-server**

Indicates that the log entry is a result of a REST API request or a Web UI action.

The string value of `data` identifies the CLI command or REST API request that was executed.

These elements are shown in context under *Examples*.

## Examples

Showing the accounting log for the previous and current boot. Line breaks have been added for readability.

```
switch# show accounting log all
```

```

Local accounting logs from previous boot

```

```

type=DAEMON_START msg=audit(Nov 05 2018 23:00:58.607:9057) :
auditd start, ver=2.4.3 format=raw kernel=4.9.119-yocto-standard res=success

type=USER_START msg=audit(Nov 05 2018 23:06:42.398:42) :
msg='rec=ACCT_EXEC op=start session=CONSOLE timezone=UTC user=user1 priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no
hostname=8xxx addr=0.0.0.0 res=success'

type=USYS_CONFIG msg=audit(Nov 05 2018 23:06:42.399:43) :
msg='rec=ACCT_CMD op=stop session=CONSOLE timezone=UTC user=user1 priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no
data="enable" hostname=8xxx addr=0.0.0.0 res=success'

type=USYS_CONFIG msg=audit(Nov 05 2018 23:08:24.693:51) :
msg='rec=ACCT_CMD op=stop session=CONSOLE timezone=UTC user=user1 priv-lvl=1
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no
data="configure terminal" hostname=8xxx addr=0.0.0.0 res=success'

type=USYS_CONFIG msg=audit(Nov 05 2018 23:08:39.108:52) :
msg='rec=ACCT_CMD op=stop session=CONSOLE timezone=UTC user=user1 priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=yes
data="https-server rest access-mode read-write"
hostname=8xxx addr=0.0.0.0 res=success'

type=USER_START msg=audit(Nov 05 2018 23:10:57.238:58) :
msg='rec=ACCT_EXEC op=start session=REST timezone=UTC user=admin priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=https-server
data="http-method=POST http-uri=/rest/v1/login"
hostname=8xxx addr=127.0.0.1 res=success'

type=USYS_CONFIG msg=audit(Nov 05 2018 23:15:11.958:75) :
```

```

msg='rec=ACCT_CMD op=stop session=CONSOLE timezone=UTC user=user1 priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=yes
data="tacacs-server host 2.2.2.2" hostname=8xxx addr=0.0.0.0 res=success'

type=USYS_CONFIG msg=audit(Nov 05 2018 23:15:37.090:76) :
msg='rec=ACCT_CMD op=stop session=REST timezone=UTC user=admin priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=https-server
data="http-method=GET http-uri=/rest/v1/system/vrfs/mgmt/tacacs_servers"
hostname=8xxx addr=127.0.0.1 res=success'

type=USER_END msg=audit(Nov 05 2018 23:26:59.207:90) :
msg='rec=ACCT_EXEC op=stop session=REST timezone=UTC user=admin priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=https-server
data="http-method=POST http-uri=/rest/v1/logout"
hostname=8xxx addr=127.0.0.1 res=success'

type=USER_END msg=audit(Nov 05 2018 23:27:49.164:93) :
msg='rec=ACCT_EXEC op=stop session=CONSOLE timezone=UTC user=user1 priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no
hostname=8xxx addr=0.0.0.0 res=success'

Local accounting logs from current boot

type=DAEMON_START msg=audit(Nov 05 2018 23:32:05.642:626) :
auditd start, ver=2.4.3 format=raw kernel=4.9.119-yocto-standard res=success

type=USER_START msg=audit(Nov 05 2018 23:35:52.915:11) :
msg='rec=ACCT_EXEC op=start session=CONSOLE timezone=UTC user=admin priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no
hostname=8xxx addr=0.0.0.0 res=success'

type=USYS_CONFIG msg=audit(Nov 05 2018 23:35:52.917:12) :
msg='rec=ACCT_CMD op=stop session=CONSOLE timezone=UTC user=admin priv-lvl=15
auth-method=LOCAL auth-type=LOCAL service=shell isconfig=no data="enable"
hostname=8xxx addr=0.0.0.0 res=success'

```

## show radius-server

### Syntax

```
show radius-server [detail]
```

### Description

Shows configured RADIUS servers information.

### Command context

Operator (>) or Manager (#)

### Parameters

#### detail

Selects additional RADIUS server details and global parameters for showing.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Usage

When the `show radius-server` command shows `None` for the shared-secret, the passkey is missing.

## Examples

Showing a summary of the global RADIUS configuration:

```
switch# show radius-server
***** Global RADIUS Configuration *****
```

```
Shared-Secret: None
Timeout: 60
Auth-Type: pap
Retries: 5
Tracking Time Interval (seconds): 60
Tracking Retries: 5
Tracking User-name: radius-tracking-user
Tracking Password: None
Number of Servers: 1
```

| SERVER NAME                             | PORT | VRF     |
|-----------------------------------------|------|---------|
| 20.1.1.129                              | 1812 | default |
| 1.1.1.4                                 | 1812 | default |
| 1.1.1.5                                 | 12   | default |
| abc1.com                                | 32   | default |
| 2001:0db8:85a3:0000:0000:8a2e:0371:7334 | 1812 | default |

Showing a summary of a RADIUS server when the per-server shared key or the global RADIUS shared key is not set:

```
switch# show radius-server
***** Global RADIUS Configuration *****
```

```
Shared-Secret: None
Timeout: 60
Auth-Type: pap
Retries: 5
Tracking Time Interval (seconds): 60
Tracking Retries: 5
Tracking User-name: radius-tracking-user
Tracking Password: None
Number of Servers: 1
```

| SERVER NAME | PORT | VRF     |
|-------------|------|---------|
| 20.1.1.129  | 1812 | default |

Showing details of a global RADIUS configuration:

```
switch# show radius-server detail
***** Global RADIUS Configuration *****
```

```
Shared-Secret: AQBapb+HsdpqV1QcA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout: 5
Auth-Type: pap
Retries: 5
Tracking Time Interval (seconds): 60
Tracking Retries: 5
```

```
Tracking User-name: radius-tracking-user
Tracking Password: None
Number of Servers: 1
```

```
***** RADIUS Server Information *****
Server-Name : 20.1.1.129
Auth-Port : 1812
Accounting-Port : 1813
VRF : default
Shared-Secret : None
Timeout : 60
Retries : 5
Auth-Type : pap
Server-Group : radius
Default-Priority : 4
Tracking : disabled
Tracking-Mode : any
Reachability-Status : N/A
ClearPass-Username :
ClearPass-Password : None
```

Showing details of a RADIUS server when the per-server shared key and the global RADIUS shared key are not set:

```
switch# show radius-server detail
***** Global RADIUS Configuration *****

Shared-Secret: None
Timeout: 5
Auth-Type: pap
Retries: 1
Number of Servers: 1

***** RADIUS Server Information *****
Server-Name : 1.1.1.1
Auth-Port : 2083
VRF : default
Shared-Secret (default) : None
Timeout (default) : 5
Retries (default) : 1
Auth-Type (default) : pap
Server-Group (default) : radius
Default-Priority : 1
```

## show radius-server secure ipsec

### Syntax

```
show radius-server secure ipsec { server-list | host {<FQDN> | <IPV4> | <IPV6>}
 [port <PORT-NUMBER>] [vrf <VRF-NAME>]}
```

### Description

Shows information for one or all RADIUS servers configured with IPsec.

### Command context

Operator (>) or Manager (#)

## Parameters

### **server-list**

Selects all servers for showing.

### **host {<FQDN> | <IPv4> | <IPv6>}**

Specifies the RADIUS server as:

- <FQDN>: a fully qualified domain name.
- <IPv4>: an IPv4 address.
- <IPv6>: an IPv6 address.

### **port <PORT-NUMBER>**

Specifies the authentication port number. Range: 1 to 65535. Default: 1812.

### **vrf <VRF-NAME>**

Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Usage

The IPsec key is shown in an exportable ciphertext format.

## Examples

Showing information for RADIUS server 1.1.1.1 secured with IPsec:

```
switch# show radius-server secure ipsec host 1.1.1.1
IPsec : enabled
Protocol : ESP
Authentication : MD5
Encryption : AES
SPI : 1234
```

Showing information for all RADIUS servers secured with IPsec:

```
switch# show radius-server secure ipsec server-list
Server : 1.1.1.1
IPsec : enabled
Protocol : ESP
Authentication : MD5
Encryption : AES
SPI : 1234

Server : 1.1.1.2
IPsec : enabled
Protocol : ESP
Authentication : MD5
Encryption : AES
SPI : 12341
```

# show tacacs-server

## Syntax

```
show tacacs-server [detail]
```

## Description

Shows the configured TACACS+ servers.

## Command context

Operator (>) or Manager (#)

## Parameters

### detail

Selects additional TACACS+ server details and global parameters for showing.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing a summary of a global TACACS+ configuration with a shared-secret:

```
switch# show tacacs-server
***** Global TACACS+ Configuration *****

Shared-Secret: AQBapb+HsdpqV1Q3CPCBMQTG8e1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout: 5
Auth-Type: pap
Number of Servers: 5

SERVER NAME | PORT | VRF

1.1.1.1 | 49 | mgmt
1.1.1.2 | 12 | mgmt
abc.com | 32 | vrf_blue
2001:0db8:85a3:0000:0000:8a2e:0370:7334 | 49 | default
1.1.1.3 | 32 | vrf_red

```

Showing details of a global TACACS+ configuration:

```
switch# show tacacs-server detail
***** Global TACACS+ Configuration *****

Shared-Secret: AQBapb+HsdpqV1Q3CPCBMQTG8e1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout: 5
Auth-Type: pap
Number of Servers: 5

***** TACACS+ Server Information *****
Server-Name : 1.1.1.2
Auth-Port : 12
VRF : mgmt
Shared-Secret (default) : AQBapb+HsdpqV1Q3CPCBMQTG8eeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout (default) : 5
```

```

Auth-Type (default) : pap
Server-Group : sg1
Group-Priority : 1

Server-Name : 2001:0db8:85a3:0000:0000:8a2e:0370:7334
Auth-Port : 49
VRF : default
Shared-Secret (default) : AQBapb+HsdpqV1Q3CPCBMQTG8eeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout (default) : 5
Auth-Type (default) : pap
Server-Group : sg2
Group-Priority : 1

Server-Name : 1.1.1.1
Auth-Port : 49
VRF : mgmt
Shared-Secret (default) : AQBapb+HsdpqV1Q3CPCBMQTG8eeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout (default) : 5
Auth-Type (default) : pap
Server-Group (default) : tacacs
Default-Priority : 1

Server-Name : abc.com
Auth-Port : 32
VRF : vrf_red
Shared-Secret (default) : AQBapb+HsdpqV1Q3CPCBMQTG8eeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout : 15
Auth-Type (default) : pap
Server-Group (default) : tacacs
Default-Priority : 3

Server-Name : 1.1.1.3
Auth-Port : 32
VRF : vrf_blue
Shared-Secret : AQBapfnqbSswqKC476tdUFZ+AncIRY92hDTYkQCAAAAFEaHn43vNC
Timeout : 15
Auth-Type : chap
Server-Group (default) : tacacs
Default-Priority : 5

```

Showing TACACS+ server when per-server shared key and global TACACS+ shared key is not set:

```

switch# show tacacs-server
***** Global TACACS+ Configuration *****

```

```

Shared-Secret: None
Timeout: 5
Auth-Type: pap
Number of Servers: 1

```

| SERVER NAME | PORT | VRF     |
|-------------|------|---------|
| 1.1.1.1     | 49   | default |

Showing TACACS+ server details when per-server shared key and global TACACS+ shared key is not set:

```

switch# show tacacs-server detail
***** Global TACACS+ Configuration *****

```

```

Shared-Secret: None
Timeout: 5
Auth-Type: pap

```



```
Number of Servers: 1
```

```
***** TACACS+ Server Information *****
Server-Name : 1.1.1.1
Auth-Port : 49
VRF : default
Shared-Secret (default) : None
Timeout (default) : 5
Auth-Type (default) : pap
Server-Group (default) : tacacs
Default-Priority : 1
```

## show tacacs-server statistics

### Syntax

```
show tacacs-server statistics
```

### Description

Shows authentication statistics for all configured TACACS+ servers.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing TACACS+ server authentication statistics:

```
switch# show tacacs-server statistics
Server Name : tac1
Auth-Port : 49
VRF : mgmt
Authentication Statistics

Round Trip Time : 1
Pending Requests : 0
Timeout : 0
Unknown Types : 0
Packet Dropped : 0
Auth Start : 8
Auth challenge : 0
Auth Accepts : 4
Auth Rejects : 4
Auth reply malformed : 0
Tracking Requests : 0
Tracking Responses : 0
```

## show tech aaa

### Syntax

```
show tech aaa
```

## Description

Shows the AAA configuration settings.

## Command context

Manager (#)

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Showing the AAA configuration settings:

```
switch# show tech aaa
=====
Show Tech executed on Tue Feb 14 02:19:11 2017
=====
[Begin] Feature aaa
=====

Command : show aaa authentication

AAA Authentication:
 Fail-through : Enabled
 Limit Login Attempts : Not set
 Lockout Time : 300
 Minimum Password Length : Not set

Authentication for ssh channel:

GROUP NAME | GROUP PRIORITY

local | 0

Authentication for https-server channel:

GROUP NAME | GROUP PRIORITY

local | 0

Authentication for console channel:

GROUP NAME | GROUP PRIORITY

local | 0

Authentication for default channel:

GROUP NAME | GROUP PRIORITY

tacacs | 0
local | 1

Command : show aaa accounting
```

```

AAA Accounting:
 Accounting Type : all
 Accounting Mode : start-stop

Accounting for default channel:

GROUP NAME | GROUP PRIORITY

local | 0

Accounting for ssh channel:

GROUP NAME | GROUP PRIORITY

tacacs | 0
local | 1

Accounting for https-server channel:

GROUP NAME | GROUP PRIORITY

tacacs | 0

Command : show aaa authorization

Authorization for default channel:

GROUP NAME | GROUP PRIORITY

none | 0

Authorization for console channel:

GROUP NAME | GROUP PRIORITY

none | 0

Authorization for ssh channel:

GROUP NAME | GROUP PRIORITY

tacacs | 0
none | 1

Command : show aaa server-groups

***** AAA Mechanism TACACS+ *****

GROUP NAME | SERVER NAME | PORT | PRIORITY | VRF

tacacs (default) | 1.1.1.1 | 49 | 1 | mgmt

```

```

***** AAA Mechanism RADIUS *****

GROUP NAME | SERVER NAME | PORT | PRIORITY | VRF

Command : show tacacs-server detail

***** Global TACACS+ Configuration *****

Shared-Secret: G8ekK1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout: 5
Auth-Type: pap
Tracking: disabled
Tracking Time Interval (seconds): 300
Tracking User-name: tacacs-tracking-user
Tracking Password: None
Number of Servers: 1

***** TACACS+ Server Information *****
Server-Name : 1.1.1.1
Auth-Port : 49
VRF : mgmt
Shared-Secret : KCdmOMxMD26T0fQoXfJbtj9j2AUxlGn6eCAAAAF2MkfMTtoj qX
Timeout (default) : 5
Auth-Type (default) : pap
Server-Group (default) : tacacs
Default-Priority : 1
Tracking : disabled
Reachability-Status : N/A

Command : show radius-server detail

***** Global RADIUS Configuration *****

Shared-Secret: CPCBMQTG8ekK1cA+CyD0RvfbeA8BEgikCgAAAJOWZSNzA2SWrLA=
Timeout: 5
Auth-Type: pap
Retries: 1
Number of Servers: 0

=====
[End] Feature aaa
=====

=====
Show Tech commands executed successfully
=====

```

## tacacs-server auth-type

### Syntax

```
tacacs-server auth-type {pap | chap}
```

```
no tacacs-server auth-type
```

## Description

Enables the CHAP or PAP authentication protocol, which is used for communication with the TACACS+ servers, at the global level. You can override this command with a fine-grained per server `auth-type` configuration.

The `no` form of this command resets the global authentication mechanism for TACACS+ to PAP, which is the default authentication mechanism for TACACS+.

## Command context

`config`

## Parameters

`auth-type {pap | chap}`

Selects either the PAP or CHAP authentication protocol.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling command for CHAP authentication:

```
switch(config)# tacacs-server auth-type chap
```

Enabling command for PAP authentication:

```
switch(config)# tacacs-server auth-type pap
```

# tacacs-server host

## Syntax

```
tacacs-server host {<FQDN> | <IPV4> | <IPV6>} [key {plaintext <PASSKEY> |
 ciphertext <PASSKEY>}] [timeout <TIMEOUT-SECONDS>] [port <PORT-NUMBER>]
 [auth-type {pap | chap}] [tracking {enable | disable}] [vrf <VRF-NAME>]
```

```
no tacacs-server host {<FQDN> | <IPV4> | <IPV6>} [port <PORT-NUMBER>] [vrf <VRF-NAME>]
```

## Description

Adds a TACACS+ server. By default, the TACACS+ server is associated with the server group named `tacacs`.

The `no` form of this command removes a previously added TACACS+ server.

## Command context

`config`

## Parameters

`{<FQDN> | <IPV4> | <IPV6>}`

Specifies the TACACS+ server as:

- `<FQDN>`: a fully qualified domain name.
- `<IPv4>`: an IPv4 address.
- `<IPv6>`: an IPv6 address.

**key** {plaintext `<PASSKEY>` | ciphertext `<PASSKEY>`}

Specifies either a plaintext or an encrypted local shared-secret passkey for the server. As per RFC 2865, shared-secret can be a mix of alphanumeric and special characters. The length of shared-secret in plaintext format is fewer than 32 characters.

**timeout** `<TIMEOUT-SECONDS>`

Specifies the timeout. The range is 1 to 60 seconds. The default timeout is 5 seconds.

**port** `<PORT-NUMBER>`

Specifies the TCP authentication port number. Range: 1 to 65535. Default: 49.

**auth-type** {`pap` | `chap`}

Selects either PAP (default) or CHAP authentication type. If this parameter is not specified, the TACACS+ global default is used.

**tracking** {`enable` | `disable`}

Enables or disables server tracking for the server. Tracked servers are probed at the start of each server tracking interval to check if they are reachable. Unreachable servers are skipped in favor of servers that are proven to be reachable. Use command `tacacs-server tracking` to configure TACACS+ server tracking.

**vrf** `<VRF-NAME>`

Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

If no local passkey is provided in the command, the server will use the global passkey. This command requires either the global or local passkey to be set; otherwise, the server will not be contacted. To set the global passkey, see `tacacs-server key`.

If the fully qualified domain name is provided for the TACACS+ server, a DNS server must be configured and accessible through the same VRF which is configured for the TACACS+ server. This configuration is required for the resolution of the TACACS+ server hostname to its IP address. If a DNS server is not available for this VRF, the TACACS+ servers reachable through this VRF must be configured by means of their IP addresses only.

## Examples

Adding a TACACS+ server with an IPv4 address and a named VRF:

```
switch(config)# tacacs-server host 1.1.1.1 vrf mgmt
```

Adding a TACACS+ server with an IPv4 address, a port, and a named VRF:

```
switch(config)# tacacs-server host 1.1.1.2 port 32 vrf mgmt
```

Adding a TACACS+ server with an FQDN, a timeout, port number, and a named VRF:

```
switch(config)# tacacs-server host abc.com timeout 15 port 32 vrf vrf_blue
```

Adding a TACACS+ server with an IPv6 address:

```
switch(config)# tacacs-server host 2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Adding a TACACS+ server with an IPv4 address, plaintext passkey, timeout, port, authentication type, and VRF name:

```
switch(config)# tacacs-server host 1.1.1.3 key plaintext test-123 timeout 15 port 32 auth-type chap vrf vrf_red
```

Deleting a TACACS+ server with an IPv4 address and specified VRF:

```
switch(config)# no tacacs-server host 1.1.1.1 vrf mgmt
```

Deleting a TACACS+ server with an FQDN, port, and specified VRF:

```
switch(config)# no tacacs-server host abc.com port 32 vrf vrf_blue
```

## tacacs-server key

### Syntax

```
tacacs-server key {plaintext <GLOBAL-PASSKEY> | ciphertext <GLOBAL-PASSKEY>}
```

```
no tacacs-server key
```

### Description

Creates or modifies a TACACS+ global passkey. The TACACS+ global passkey is used as a shared-secret for encrypting the communication between all TACACS+ servers and the switch. The TACACS+ global passkey is required for authentication unless local passkeys have been set. By default, the TACACS+ global passkey is empty. If the administrator has not set this key, the switch will not be able to perform TACACS+ authentication. The switch will instead rely on the authentication mechanism configured with aaa authentication login.

The `no` form of the command removes the global passkey.

### Command context

```
config
```

### Parameters

**plaintext <GLOBAL-PASSKEY>**

Specifies the TACACS+ global passkey in plaintext format with a length of 1 to 31 characters. As per RFC 2865, shared-secret can be a mix of alphanumeric and special characters.

**ciphertext <GLOBAL-PASSKEY>**

Specifies the TACACS+ global passkey in encrypted format.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Adding the global passkey:

```
switch(config)# tacacs-server key plaintext mypasskey123
```

Removing the global passkey:

```
switch(config)# no tacacs-server key
```

## tacacs-server timeout

### Syntax

```
tacacs-server timeout [<1-60>]
```

```
no tacacs-server timeout
```

### Description

Specifies the number of seconds to wait for a response from the TACACS+ server before trying the next TACACS+ server. If a value is not specified, a default value of 5 seconds is used. You can override this value with a fine-grained per server timeout configured for individual servers.

The **no** form of this command resets the TACACS+ global authentication timeout to the default of 5 seconds.

### Command context

```
config
```

### Parameters

**timeout** <1-60>

Specifies the timeout interval of 1 to 60 seconds. The default is 5 seconds.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Specifying the TACACS+ server timeout:

```
switch(config)# tacacs-server timeout 10
```

Resetting the timeout for the TACACS+ server to the default:

```
switch(config)# no tacacs-server timeout
```

## tacacs-server tracking

### Syntax

```
tacacs-server tracking {interval <INTERVAL> | user-name <NAME> [password
 {plaintext <PLAINTEXT-PASSWORD> | ciphertext <CIPHERTEXT-PASSWORD>}]}
```

```
no tacacs-server tracking {interval | user-name}
```

### Description

Configures TACACS+ server tracking settings globally for all configured TACACS+ servers that have tracking enabled with the **tacacs-server host** command.

The **no** form of the command removes the specified configuration, reverting it to its default. The **no** form with **user-name** also clears the password (resets it to empty).



## Authority

Administrators or local user group members with execution rights for this command.

## Parameters

**interval** <INTERVAL>

Specifies the time interval, in seconds, to wait before checking the server reachability status. Default: 300. Range 60 to 84600.

**user-name** <NAME> [password {plaintext <PLAINTEXT-PASSWORD> | ciphertext <CIPHERTEXT-PASSWORD>} ]

Specifies the user name to be used for server tracking requests. The default user name is `tacacs-tracking-user` with an empty password.



**NOTE:** The user does not have to be configured on the server. Server tracking can still be performed with a user which is not configured on the server because authentication failure on the server achieves confirmation that the server is reachable.

The password for the server tracking user name is optional and may be entered as `plaintext` or pasted in as `ciphertext`. The plaintext password is visible as cleartext when entered but is encrypted thereafter. Command history does show the password as cleartext.



**NOTE:** Server tracking uses authentication request and response packets to determine server reachability status. The server tracking user name and password are used to form the request packet which is sent to the server with tracking enabled. Upon receiving a response to the request packet, the server is considered to be reachable.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring a tracking interval of 120 seconds:

```
switch(config)# tacacs-server tracking interval 120
```

Reverting the tracking interval to its default of 300 seconds:

```
switch(config)# no tacacs-server tracking interval
```

Configuring user `tacacs-tracker` with a plaintext password.

```
switch(config)# tacacs-server tracking user-name tacacs-tracker password plaintext track$1
```

Reverting the tracking user name to its default of `tacacs-tracking-user`:

```
switch(config)# no tacacs-server tracking user-name
```

## logging

### Syntax

```
logging {<IPV4-ADDR> | <IPV6-ADDR> | <HOSTNAME>}
 [udp [<PORT-NUM>] | tcp [<PORT-NUM>] | tls [<PORT-NUM>]]
 [include-auditable-events] [severity <LEVEL>] [vrf <VRF-NAME>]

logging {<IPV4-ADDR> | <IPV6-ADDR> | <HOSTNAME>}
 [tls [<PORT-NUM>]] [auth-mode {certificate|subject-name}]
 [legacy-tls-renegotiation] [include-auditable-events] [severity <LEVEL>]
 [vrf <VRF-NAME>]

no logging {<IPV4-ADDR> | <IPV6-ADDR> | <HOSTNAME>}
```

### Description

Enables syslog forwarding to a remote syslog server.

The **no** form of this command disables syslog forwarding to a remote syslog server.

### Command context

config

### Parameters

**{<IPV4-ADDR> | <IPV6-ADDR> | <HOSTNAME>}**

Selects the IPv4 address, IPv6 address, or host name of the remote syslog server. Required.

**[udp [<PORT-NUM>] | tcp [<PORT-NUM>] | tls [<PORT-NUM>]]**

Specifies the UDP port, TCP port, or TLS port of the remote syslog server to receive the forwarded syslog messages.

**udp [<PORT-NUM>]**

Range: 1 to 65535. Default: 514

**tcp [<PORT-NUM>]**

Range: 1 to 65535. Default: 1470

**tls [<PORT-NUM>]**

Range: 1 to 65535. Default: 6514

**include-auditable-events**

Specifies that auditable messages are also logged to the remote syslog server.

**severity <LEVEL>**

Specifies the severity of the syslog messages:

- **alert**: Forwards syslog messages with the severity of **alert** (6) and **emergency** (7).
- **crit**: Forwards syslog messages with the severity of **critical** (5) and above.
- **debug**: Forwards syslog messages with the severity of **debug** (0) and above.
- **emerg**: Forwards syslog messages with the severity of **emergency** (7) only.
- **err**: Forwards syslog messages with the severity of **err** (4) and above
- **info**: Forwards syslog messages with the severity of **info** (1) and above. Default.
- **notice**: Forwards syslog messages with the severity of **notice** (2) and above.
- **warning**: Forwards syslog messages with the severity of **warning** (3) and above.

#### **auth-mode**

Specifies the TLS authentication mode used to validate the certificate.

- **certificate**: Validates the peer using trust anchor certificate based authentication. Default.
- **subject-name**: Validates the peer using trust anchor certificates as well as subject-name based authentication.

#### **legacy-tls-renegotiation**

Enables the TLS connection with a remote syslog server supporting legacy renegotiation.

#### **vrf <VRF-NAME>**

Specifies the VRF used to connect to the syslog server. Optional. Default: **default**

### **Authority**

Administrators or local user group members with execution rights for this command.

### **Examples**

Enabling the syslog forwarding to remote syslog server 10.0.10.2:

```
switch(config)# logging 10.0.10.2
```

Enabling the syslog forwarding of messages with a severity of **err** (4) and above to TCP port 4242 on remote syslog server 10.0.10.9 with VRF **lab\_vrf**:

```
switch(config)# logging 10.0.10.9 tcp 4242 severity err vrf lab_vrf
```

Disabling syslog forwarding to a remote syslog server:

```
switch(config)# no logging
```

Enabling syslog forwarding over TLS to a remote syslog server using subject-name authentication mode:

```
switch(config)#logging example.com tls auth-mode subject-name
```

# logging facility

## Syntax

```
logging facility {local0 | local1 | local2 | local3
 | local4 | local5 | local6 | local7}
no logging facility
```

## Description

Sets the logging facility to be used for remote syslog messages. Default: `local7`

The `no` form of this command disables the logging facility to be used for remote syslog messages.

## Command context

`config`

## Parameters

`{local0 | local1 | local2 | local3 | local4 | local5 | local6 | local7}`

Selects the logging facility to be used for remote syslog messages. Required.

Specifies the severity of the syslog messages:

- `local0`
- `local1`
- `local2`
- `local3`
- `local4`
- `local5`
- `local6`
- `local7`

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Sets the `local5` logging facility to be used for remote syslog messages:

```
switch(config)# logging facility local5
```

# logging persistent-storage

## Syntax

```
logging persistent-storage [severity {alert|crit|debug|emerg|err|info|notice|warning}]
no logging persistent-storage
```

## Description

Enables or disables storage of logs in storage. Only logs of the specified severity and above will be preserved in the storage.

The `no` form of this command disables storage of logs in storage.

## Command context

`config`

## Parameters

**severity** <LEVEL>

Specifies the severity of the syslog messages:

- `alert`: Preserves syslog messages with the severity of `alert` (6) and `emergency` (7).
- `crit`: Preserves syslog messages with the severity of `critical` (5) and above. Default.
- `debug`: Preserves syslog messages with the severity of `debug` (0) and above.
- `emerg`: Preserves syslog messages with the severity of `emergency` (7) only.
- `err`: Preserves syslog messages with the severity of `err` (4) and above
- `info`: Preserves syslog messages with the severity of `info` (1) and above.
- `notice`: Preserves syslog messages with the severity of `notice` (2) and above.
- `warning`: Preserves syslog messages with the severity of `warning` (3) and above.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

These logs can be copied out by using the `copy support-files all` or `copy support-files previous-boot`.

## Examples

Enabling storage of logs in storage with severity `info`:

```
switch(config)#logging persistent-storage severity info
Logs will be written to storage and made available across reboot.
Do you want to continue (y/n)?
```

Disabling storage of logs in storage:

```
switch(config)# no logging persistent-storage
```

## Configuration commands

### `router rip`

#### Syntax

```
router rip <PROCESS-ID> [vrf <VRF-NAME>]
```

```
no router rip <PROCESS-ID> [vrf <VRF-NAME>]
```

#### Description

Creates RIP process if not already created and enters the `router rip <PROCESS-ID>` context for the VRF mentioned. If no VRF is mentioned, a default is used. Only one RIP process is allowed per VRF.

The `no` form of this command deletes the RIP instance for the VRF. If no VRF is mentioned the default is deleted.

#### Command context

```
config
```

#### Parameters

**<PROCESS-ID>**

Specifies name of the RIP process ID. Range: 1-63.

**vrf <VRF-NAME>**

Specifies VRF name.

#### Authority

Administrators or local user group members with execution rights for this command.

#### Examples

Creating RIP process and naming the VRF:

```
switch(config)# router rip 2 vrf red
```

Deleting RIP process:

```
switch(config)# no router rip 2 vrf red
```

## Interface commands

## ip rip

### Syntax

```
ip rip <PROCESS-ID> {all-ip | ip-address}

no ip rip <PROCESS-ID> {all-ip | ip-address}
```

### Description

Enables RIP process on an interface.

The `no` form of this command deletes the RIP process from an interface.

### Command context

config-if

### Parameters

**ip rip <PROCESS-ID>**

Specifies RIP process ID. Range: 1-63.

**all-ip**

Specifies RIP for all IP addresses configured on the interface.

**ip-address**

Specifies IP address for RIP on the interface.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.
- If `ip rip 1 all-ip` is configured and a new IP address is added to the interface, RIP configurations will not be applicable for the newly added IP address.

### Examples

Configuring RIP for all IP addresses configured on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 all-ip
```

Deleting RIP for all IP addresses configured on the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip 1 all-ip
```

## ip rip all-ip enable

### Syntax

```
ip rip all-ip enable

no ip rip all-ip enable
```

## Description

Enables RIP process for all RIP enabled IP addresses configured on interface.

The `no` form of this command disables RIP process on the interface.

## Command context

`config-if`

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

## Examples

Enabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip enable
```

Disabling RIP process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip enable
```

## `ip rip all-ip disable`

## Syntax

```
ip rip all-ip disable
```

```
no ip rip all-ip disable
```

## Description

Disables RIP process for all RIP enabled IP addresses configured on the interface.

The `no` form of this command enables RIP process for all RIP enabled IP addresses configured on the interface.

## Command context

`config-if`

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.



## Examples

Disabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip enable
```

Enabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip enable
```

## ip rip all-ip send disable

### Syntax

```
ip rip all-ip send disable
```

```
no ip rip all-ip send disable
```

### Description

Disables interface from sending RIP packets for all RIP enabled IP addresses.

The `no` form of this command enables interface to send RIP packets for all RIP enabled IP addresses.

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

- Default settings allow an interface to send RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

## Examples

Disabling interface from sending RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip send disable
```

Enabling interface to send RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip send disable
```

## ip rip all-ip receive disable

### Syntax

```
ip rip all-ip receive disable
```

```
no ip rip all-ip receive disable
```

## Description

Disables interface from receiving RIP packets for all enabled IP addresses.

The `no` form of this command enables interface to receive RIP packets for all RIP enabled IP addresses.

## Command context

`config-if`

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

## Examples

Disabling interface from receiving RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip all-ip receive disable
```

Enabling interface to receive RIP packets for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip rip all-ip receive disable
```

# Routing commands

## enable

### Syntax

`enable`

`no enable`

### Description

Enables RIP process if disabled. By default RIP process is enabled.

The `no` form of this command disables the RIP process.

### Command context

`config`

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling RIP process when disabled:

```
switch(config)# router rip 1
switch(config-rip-1)# enable
```

Disabling RIP process when enabled:

```
switch(config)# router rip 1
switch(config-rip-1)# no enable
```

## disable

### Syntax

```
disable
```

```
no disable
```

### Description

Disables RIP process.

The `no` form of this command enables the RIP process.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Disabling RIP process:

```
switch(config)# router rip 1
switch(config-rip-1)# disable
```

Enabling RIP process:

```
switch(config)# router rip 1
switch(config-rip-1)# no disable
```

## distance

### Syntax

```
distance <DISTANCE>
```

```
no distance
```

### Description

Configures administrative distance for RIP. Administrative distance is used as criteria to select the best route when multiple protocols have the same route.

The `no` form of this command sets the RIP administrative distance to the default. Default: 120.

### Command context

```
config
```

## Parameters

### <DISTANCE>

Specifies RIP administrative distance. Range: 1 to 255.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring administrative distance for RIP:

```
switch(config)# router rip 1
switch(config-rip-1)# distance 100
```

Setting administrative distance for RIP to default values:

```
switch(config)# router rip 1
switch(config-rip-1)# no distance
```

## maximum-paths

## Syntax

```
maximum-paths <MAX-VALUE>
```

```
no maximum-paths
```

## Description

Sets the maximum number of ECMP routes that RIP can support.

The `no` form of this command sets the maximum number of ECMP routes to the default value of 4.

## Command context

```
config
```

## Parameters

### <MAX-VALUE>

Sets the number of RIP ECMP routes. Range: 1-8.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting maximum number of RIP ECMP routes:

```
switch(config)# router rip 1
switch (config-rip-1)# maximum-paths 8
```

Setting maximum number of RIP ECMP routes to default:

```
switch(config)# router rip 1
switch (config-rip-1)# no maximum-paths
```

## redistribute

### Syntax

```
redistribute {bgp | connected | ospf <PROCESS-ID> | static}

no redistribute {bgp | connected | ospf <PROCESS-ID> | static}
```

### Description

Redistributes routes originating from other protocols into RIP.

The `no` form of this command disables redistribution of routes originating from other protocols into RIP.

### Command context

config

### Parameters

#### **bgp**

Specifies BGP routes to redistribute into RIP.

#### **connected**

Specifies connected routes (directly attached subnet or host) to redistribute into RIP.

#### **ospf <PROCESS-ID>**

Specifies OSPF route to redistribute into RIP.

#### **static**

Specifies static route to redistribute into RIP.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Redistributing BGP routes into RIP:

```
switch(config)# router rip 1
switch(config-rip-1)# redistribute bgp
```

Disabling BGP routes that originate from other protocols and redistribute into RIP:

```
switch(config)# router rip 1
switch(config-rip-1)# no redistribute bgp
```

## timers update

### Syntax

```
timers update <INTERVAL> timeout <DURATION> garbage-collection <PERIOD>

no timers
```

### Description

Configures RIP timers with specific values.

The `no` form of this command sets all RIP timers to default values.

## Command context

config

## Parameters

**timers update** <INTERVAL>

Specifies frequency at which RIP sends updates to all of its peers. Range: 1 to 2147484. Default: 30.

**timeout** <DURATION>

Specifies timeout duration from the point of the last refresh after a route is received from a peer timeout and is marked as expired. Range: 1 to 255. Default: 180.

**garbage-collection** <PERIOD>

Specifies amount of time route remains in routing table after route expiration. Range: 1 to 255. Default: 120.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring RIP timers with specific values:

```
switch(config)# router rip 1
switch(config-rip-1)# timers update 40 timeout 200 garbage-collection 150
```

Configuring RIP timers with default values:

```
switch(config)# router rip 1
switch(config-rip-1)# no timers
```

# RIPv2 clear commands

## clear ip rip statistics

### Syntax

```
clear ip rip [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]
```

### Description

Clears RIP event statistics.

### Command context

Operator (>) or Manager (#)

### Parameters

<PROCESS-ID>

Specifies RIP process ID. Range: 1-63

**all-vrfs**

Clears statistics for all VRFs.

### **vrf**

Selects VRF to clear statistics for.

**<VRF-NAME>**

Specifies VRF name.

### **Authority**

Administrators or local user group members with execution rights for this command.

### **Examples**

Clearing RIP event statistics:

```
switch# clear ip rip statistics
```

## **RIPv2 interface commands**

### **enable**

#### **Syntax**

enable

no enable

#### **Description**

Enables RIP process for RIP enabled IP address configured on interface.

The **no** form of this command disables RIP process on interface.

#### **Command context**

config-if

#### **Authority**

Administrators or local user group members with execution rights for this command.

#### **Usage**

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

### **Examples**

Enabling RIP process for RIP enabled IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# enable
```

Disabling RIP process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no enable
```

## disable

### Syntax

```
disable
```

```
no disable
```

### Description

Disables RIP process for RIP enabled IP addresses configured on interface.

The `no` form of this command enables RIP process on interface.

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Disabling RIP process for RIP enabled IP addresses configured on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# disable
```

Enabling RIP process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no disable
```

## send disable

### Syntax

```
send disable
```

```
no send disable
```

### Description

Disables an interface from sending RIP packets for a specific IP address.

The `no` form of this command enables interface for sending RIP packets for a specific IP address.

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.



## Usage

- Default settings allow an interface to send and receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

## Examples

Disabling interface from sending RIP packets for a specific IP address :

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# send disable
```

Enabling interface to send RIP packets for a specific IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no send disable
```

## receive disable

### Syntax

receive disable

no receive disable

### Description

Disables interface from receiving RIP packets for a specific IP address.

The **no** form of this command enables interface for receiving RIP packets for a specific IP address.

### Command context

config-if

### Authority

Administrators or local user group members with execution rights for this command.

## Usage

- Default settings allow an interface to receive RIP packets.
- If an IP address is removed from an interface configured with RIP, all RIP configurations will be removed from the interface.

## Examples

Disabling interface from receiving RIP packets for a specific IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# receive disable
```

Enabling interface for receiving RIP packets for a specific IP address:

```
switch(config)# interface 1/1/1
switch(config-if)# ip rip 1 10.1.1.1
switch(config-if-rip)# no receive disable
```

## RIPv2 show commands

### show capacities rip

#### Syntax

```
show capacities rip
```

#### Description

Displays maximum number of RIP interfaces, routes and process.

#### Command context

Operator (>) or Manager (#)

#### Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

#### Examples

Displaying maximum number of RIP interfaces, routes and process:

```
switch# show capacities rip

System Capacities: Filter RIP
Capacities Name Value

Maximum number of RIP interfaces configurable in the system 32
Maximum number of RIP processes supported across each VRF 1
Maximum number of routes in RIP supported across all VRFs 2540
```

### show capacities-status rip

#### Syntax

```
show capacities-status rip
```

#### Description

Displays number of RIP interfaces, routes and process configured in the system.

#### Command context

Operator (>) or Manager (#)

#### Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

#### Examples

Displaying number of RIP interfaces, routes and process:

```
switch# show capacities-status rip
```

| System Capacities Status: Filter RIP              |       |         |
|---------------------------------------------------|-------|---------|
| Capacities Name                                   | Value | Maximum |
| Number of RIP interfaces configured in the system | 0     | 32      |

## show ip rip

### Syntax

```
show ip rip [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays general RIP configuration.

### Command context

Operator (>) or Manager (#)

### Parameters

**<PROCESS-ID>**

Specifies RIP process ID. Range: 1-63.

**all vrfs**

Displays general RIP information for all VRFs.

**vrf**

Selects VRF to display general RIP information for.

**<VRF-NAME>**

Specifies VRF name.

### Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

### Usage

- Parameters display general RIP information for a specific RIP process.
- Parameters display general RIP information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

### Examples

Displaying general RIP configuration for all VRFs:

```
switch# show ip rip 34 all-vrfs
VRF : Default Process-ID : 34

RIP Version : RIPv2 Protocol Status : Enabled
Update Time : 60 sec Timeout Time : 240 sec
Garbage Collection Time : 250 sec ECMP : 6
Distance : 100 Redistribution : static,
 ospf 1
VRF : vrf_1 Process-ID : 34

```

|                         |           |                 |           |
|-------------------------|-----------|-----------------|-----------|
| RIP Version             | : RIPv2   | Protocol Status | : Enabled |
| Update Time             | : 30 sec  | Timeout Time    | : 180 sec |
| Garbage Collection Time | : 120 sec | ECMP            | : 4       |
| Distance                | : 120     | Redistribution  | : None    |

## show ip rip interface

### Syntax

```
show ip rip [<PROCESS-ID>] interface [<INTERFACE-NAME>] [brief] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays information about RIP enabled interfaces.

### Command context

Operator (>) or Manager (#)

### Parameters

**<PROCESS-ID>**

Specifies RIP process ID. Range: 1-63.

**<INTERFACE-NAME>**

Specifies interface.

**brief**

Shows brief overview information for the RIP interface.

**all-vrfs**

Displays interface information for all VRFs.

**vrf**

Selects specific VRF.

**<VRF-NAME>**

Specifies VRF.

### Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

### Usage

- Parameters display general RIP information for a specific RIP process.
- If a VRF is not mentioned, information for the default VRF is displayed.

### Examples

```
switch# show ip rip interface
Interface 1/1/1 is up, IP Address is 10.10.10.1/24

VRF : Default Process-ID : 1
Status : Oper Up Mode : Send and Receive
MTU : 500 Version : RIPv2
Poison Reverse : Enabled
```

```
Interface 1/1/2 is up, IP Address is 20.10.10.1/24
```

```

VRF : Default Process-ID : 1
Status : Admin Down Mode : Receive
MTU : 500 Version : RIPv2
Poision Reverse : Enabled
```

```
Interface 1/1/3 is up, IP Address is 30.10.10.1/24
```

```

VRF : Default Process-ID : 1
Status : Admin Down Mode : Send
MTU : 500 Version : RIPv2
Poision Reverse : Enabled
```

```
switch# show ip rip interface brief
VRF : default Process-ID : 1

```

```
Total Number of Interfaces: 2
```

| Interface | IP-Address/Mask | Status | MTU |
|-----------|-----------------|--------|-----|
| 1/1/1     | 10.10.10.1/24   | up     | 500 |
| 1/1/2     | 20.10.10.1/24   | up     | 500 |
| 1/1/3     | 30.10.10.1/24   | up     | 500 |

## show ip rip neighbors

### Syntax

```
show ip rip [<PROCESS-ID>] neighbors [<IP-ADDRESS>] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays information about RIP neighbors.

### Command context

Operator (>) or Manager (#)

### Parameters

**<PROCESS-ID>**

Specifies RIP process ID. Range: 1-63.

**<IP-ADDRESS>**

Specifies IP address of a specific neighbor to display information on.

**all-vrfs**

Displays neighbor information for all VRFs.

**vrf**

Selects VRF to display neighbor information.

**<VRF-NAME>**

Specifies VRF name.

### Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

## Usage

- Parameters display RIP neighbor information for a specific RIP process.
- Parameters display RIP neighbor information for a specific neighbor.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying RIP neighbor information for all VRFs:

```
switch# show ip rip neighbors all-vrfs
VRF : default Process-ID : 1

Total Number of Neighbors: 1

Peer-Address Type Last-Update Rcvd-Bad-Pkts Rcvd-Bad-Routes

1.1.1.2 RIPv2 0:0:7 4 5
```

## show ip rip routes

### Syntax

```
show ip rip [<PROCESS-ID>] routes [<PREFIX/LENGTH>] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays RIP routing table for a specific RIP process.

### Command context

Operator (>) or Manager (#)

### Parameters

**<PROCESS-ID>**

Specifies RIP process ID to display information for a specific RIP process. Range: 1-63.

**<PREFIX/LENGTH>**

Specifies the network prefix.

**all-vrfs**

Displays RIP routing information for all VRFs.

**vrf**

Selects VRF to display RIP routing information.

**<VRF-NAME>**

Specifies VRF name.

### Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

## Usage

- `<PREFIX/LENGTH>` is an optional parameter that displays RIP routing table information for a specific subnet.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying RIP routing table for all VRFs:

```
switch# show ip rip routes all-vrfs
VRF : default Process-ID : 1

Total Number of Routes : 6

Prefix Metric Interface Nexthop

10.1.0.0/16 2 1/1/1 30.1.1.2
20.1.2.0/24 3 1/1/1 30.1.1.2
30.1.1.0/24 1 1/1/1

VRF : vrf_1 Process-ID : 34

Prefix Metric Interface Nexthop

20.1.0.0/16 10 1/1/2 50.1.1.2
40.1.2.0/24 14 1/1/2 50.1.1.2
50.1.1.0/24 1 1/1/2
```

## show ip rip statistics

### Syntax

```
show ip rip [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays RIP statistics.

### Command context

Operator (>) or Manager (#)

### Parameters

#### <PROCESS-ID>

Specifies RIP process ID. Range: 1-63.

#### all-vrfs

Displays statistics information for all VRFs.

#### vrf

Selects VRF to display RIP statistics information for.

#### <VRF-NAME>

Specifies VRF name.

## Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

## Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying RIP statistics for all VRFs:

```
switch# show ip rip statistics all-vrfs
VRF : default Process-ID : 1

Global Route Changes : 50
Global Queries : 2
Last Cleared : 0h 30m 28s ago

VRF : vrf_1 Process-ID : 34

Global Route Changes : 20
Global Queries : 0
Last Cleared : 0h 30m 28s ago
```

## show ip rip statistics interface

### Syntax

```
show ip rip [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays RIP statistics for RIP enabled interfaces.

### Command context

Operator (>) or Manager (#)

### Parameters

**<PROCESS-ID>**

Specifies RIP process ID. Range: 1-63.

**<INTERFACE-NAME>**

Specifies name of interface.

**all-vrfs**

Displays RIP interface statistics for all VRFs.

**vrf**

Selects VRF to display RIP interface statistics.

**<VRF-NAME>**

Specifies VRF name.



## Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

## Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying RIP statistics for a RIP enabled interface:

```
switch# show ip rip statistics interface 1/1/1
VRF : default Process-ID : 1 interface 1/1/1

IP-Address Trigger-Updates Rcvd-Bad-Packets Rcvd-Bad-Routes

10.1.1.1 15 3 4
Last Cleared : 0h 30m 28s ago
```

## show running-config

### Syntax

```
show running config
```

### Description

Displays all running configurations for all protocols including RIP.

### Command context

Operator (>) or Manager (#)

## Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

## Examples

Displaying all running configurations for all protocols including RIP:

```
switch# show running-config
Current configuration:
!
!Version Halon 0.1.0 (Build: genericx86-64-Halon-0.1.0-master-20170309054955-dev)
!Schema version 0.1.8
lldp enable
timezone set utc
vrf blue
vrf green
vrf red
led base-loc_fdc on
led base-loc_on
led base-hlth_fdc fast_blink
led base-pwr_fdc on
```

```

!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospf 1 vrf red
router rip 1
 maximum-paths 5
 distance 1
router rip 1 vrf red
 default-information originate always
 maximum-paths 7
 distance 5
 redistribute ospf 1
 timers update 40 timeout 200 garbage-collection 120
vlan 1
 no shutdown
interface lag 44
 no shutdown
 ip address 33.1.1.1/24
 ip rip 1 33.1.1.1
 send disable
interface 1/1/1
 no shutdown
 ip address 33.44.1.1/24
 ip address 44.44.1.1/24 secondary
 ip rip 1 33.44.1.1
 send disable
 ip rip 1 44.44.1.1
 send disable
interface 1/1/2
interface loopback 2
 ip address 55.55.55.55/32
 ip rip 1 55.55.55.55

```

## Configuration commands

### router ripng

#### Syntax

```
router ripng <PROCESS-ID> [vrf <VRF-NAME>]
```

```
no router ripng <PROCESS-ID> [vrf <VRF-NAME>]
```

#### Description

Creates RIPng process if not already created and enters the `router ripng <PROCESS-ID>` context for the VRF mentioned. If no VRF is mentioned, a default is used. Only one RIPng process is allowed per VRF.

The `no` form of this command deletes the RIPng instance for the VRF. If no VRF is mentioned the default is deleted.

#### Command context

```
config
```

#### Parameters

**<PROCESS-ID>**

Specifies name of the RIPng process ID. Range: 1-63.

**vrf**

Sets VRF name for RIPng process.

**<VRF-NAME>**

VRF name for VRF.

#### Authority

Administrators or local user group members with execution rights for this command.

#### Examples

Creating RIPng process and naming the VRF:

```
switch(config)# router ripng 2 vrf red
```

Deleting RIPng process:

```
switch(config)# no router ripng 2 vrf red
```

## Interface commands

## ipv6 ripng

### Syntax

```
ipv6 ripng <PROCESS-ID>
```

```
no ipv6 ripng <PROCESS-ID>
```

### Description

Enables RIPng process on interface and creates a new context.

The `no` form of this command deletes RIPng process on interface.

### Command context

```
config-if
```

### Parameters

**<PROCESS-ID>**

Specifies RIPng process ID. Range: 1-63.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling RIPng process on an interface:

```
switch(config)# interface 1/1/1
switch (config-if)# ipv6 ripng 1
switch (config-if-ripng)#
```

Deleting RIPng process on an interface:

```
switch(config)# interface 1/1/1
switch (config-if)# no ipv6 ripng 1
```

## Routing commands

## enable

### Syntax

```
enable
```

```
no enable
```

### Description

Enables RIPng process if disabled. By default RIPng process is enabled.

The `no` form of this command disables the RIPng process.

### Command context

```
config
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling RIPng process when disabled:

```
switch(config)# router ripng 1
switch(config-ripng-1)# enable
```

Disabling RIPng process when enabled:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no enable
```

## disable

### Syntax

disable

no disable

### Description

Disables RIPng process.

The `no` form of this command enables the RIPng process.

### Command context

config

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Disabling RIPng process:

```
switch(config)# router ripng 1
switch(config-ripng-1)# disable
```

Enabling RIPng process:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no disable
```

## distance

### Syntax

distance <DISTANCE>

no distance

### Description

Configures administrative distance for RIPng. Administrative distance is used as criteria to select the best route when multiple protocols have the same route.

The `no` form of this command sets the RIPng administrative distance to the default. Default: 120.

## Command context

config

## Parameters

**<DISTANCE>**

Specifies RIPng administrative distance. Range: 1 to 255.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring administrative distance for RIPng:

```
switch(config)# router ripng 1
switch(config-ripng-1)# distance 100
```

Setting administrative distance for RIPng to default values:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no distance
```

## maximum-paths

### Syntax

maximum-paths <MAX-VALUE>

no maximum-paths

### Description

Sets the maximum number of ECMP routes that RIPng can support.

The `no` form of this command sets the maximum number of ECMP routes to the default value of 4.

## Command context

config

## Parameters

**<MAX-VALUE>**

Sets the number of RIPng ECMP routes. Range: 1-8.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting maximum number of RIPng ECMP routes:

```
switch(config)# router ripng 1
switch (config-ripng-1)# maximum-paths 8
```

Setting maximum number of RIPng ECMP routes to default:

```
switch(config)# router ripng 1
switch (config-ripng-1)# no maximum-paths
```

## redistribute

### Syntax

```
redistribute {bgp | connected | ospfv3 <PROCESS-ID> | static}

no redistribute {bgp | connected | ospfv3 <PROCESS-ID> | static}
```

### Description

Redistributes routes originating from other protocols into RIPng.

The **no** form of this command disables redistribution of routes originating from other protocols into RIPng.

### Command context

config

### Parameters

#### bgp

Specifies BGP routes to redistribute into RIPng.

#### connected

Specifies connected routes (directly attached subnet or host) to redistribute into RIPng.

#### ospfv3 <PROCESS-ID>

Specifies OSPFv3 route to redistribute into RIPng.

#### static

Specifies static route to redistribute into RIPng.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Redistributing BGP routes into RIPng:

```
switch(config)# router ripng 1
switch(config-ripng-1)# redistribute bgp
```

Disabling BGP routes that originate from other protocols and redistribute into RIPng:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no redistribute bgp
```

## timers update

### Syntax

```
timers update <INTERVAL> timeout <DURATION> garbage-collection <PERIOD>

no timers
```

## Description

Configures RIPng timers with specific values.

The `no` form of this command sets all RIPng timers to default values.

## Command context

`config`

## Parameters

**`timers update <INTERVAL>`**

Specifies frequency at which RIPng sends updates to all of its peers. Range: 1 to 2147484. Default: 30.

**`timeout <DURATION>`**

Specifies timeout duration from the point of the last refresh after a route is received from a peer timeout and is marked as expired. Range: 1 to 255. Default: 180.

**`garbage-collection <PERIOD>`**

Specifies amount of time route remains in routing table after route expiration. Range: 1 to 255. Default: 120.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring RIPng timers with specific values:

```
switch(config)# router ripng 1
switch(config-ripng-1)# timers update 40 timeout 200 garbage-collection 150
```

Configuring RIPng timers with default values:

```
switch(config)# router ripng 1
switch(config-ripng-1)# no timers
```

# RIPng clear commands

## clear ipv6 ripng statistics

### Syntax

```
clear ipv6 ripng [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]
```

### Description

Clears RIPng event statistics.

### Command context

Operator (>) or Manager (#)

### Parameters

**`<PROCESS-ID>`**

Specifies RIPng process ID. Range: 1-63



### **all-vrfs**

Clears statistics for all VRFs.

### **vrf**

Selects VRF to clear statistics for.

### **<VRF-NAME>**

Specifies VRF name.

### **Authority**

Administrators or local user group members with execution rights for this command.

### **Examples**

Clearing RIPng event statistics:

```
switch# clear ipv6 ripng statistics
```

## **RIPng interface commands**

### **enable**

#### **Syntax**

enable

no enable

#### **Description**

Enables RIPng process on interface.

The **no** form of this command disables RIPng process on interface.

#### **Command context**

config-if

#### **Authority**

Administrators or local user group members with execution rights for this command.

#### **Examples**

Enabling RIPng process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# enable
```

Disabling RIPng process on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# no enable
```

## disable

### Syntax

disable

no disable

### Description

Disables RIPng process on interface.

The `no` form of this command enables RIPng process on interface.

### Command context

config-if

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Disabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# disable
```

Enabling RIP process for all RIP enabled IP addresses on interface:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch(config-if-ripng)# no disable
```

## send disable

### Syntax

send disable

no send disable

### Description

Disables interface from sending RIPng packets. An interface can send RIPng packets by default.

The `no` form of this command enables interface to send RIPng packets, if disabled.

### Command context

config-if

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Disabling interface from sending RIPng packets:

```
switch(config)# interface 1/1/1
switch (config-if)# ipv6 ripng 1
switch (config-if-ripng)# send disable
```

Enabling interface to send RIPng packets:

```
switch(config)# interface 1/1/1
switch (config-if)# ipv6 ripng 1
switch (config-if-ripng)# no send disable
```

## receive disable

### Syntax

receive disable

no receive disable

### Description

Disables interface from receiving RIPng packets for all enabled IP addresses. An interface can receive RIPng packets by default.

The `no` form of this command enables interface to receive RIPng packets, if disabled.

### Command context

config-if

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Disabling interface from receiving RIPng packets:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch (config-if-ripng)# receive disable
```

Enabling interface to receive RIPng packets when disabled:

```
switch(config)# interface 1/1/1
switch(config-if)# ipv6 ripng 1
switch (config-if-ripng)# no receive disable
```

## RIPng show commands

### show capacities ripng

#### Syntax

show capacities ripng

#### Description

Displays the maximum number of RIPng interfaces, routes and process.

## Command context

Operator (>) or Manager (#)

## Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

## Examples

Displaying maximum number of RIPng interfaces, routes and process:

```
switch# show capacities ripng

System Capacities: Filter RIPng
Capacities Name Value

Maximum number of RIPng interfaces configurable in the system 32
Maximum number of RIPng processes supported across each VRF 1
Maximum number of routes in RIPng supported across all VRFs 2540
```

## show capacities-status ripng

### Syntax

```
show capacities-status ripng
```

### Description

Displays number of RIPng interfaces, routes and process configured in the system.

## Command context

Operator (>) or Manager (#)

## Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

## Examples

Displaying number of RIPng interfaces, routes and process:

```
switch# show capacities-status ripng

System Capacities Status: Filter RIPng
Capacities Name Value Maximum

Number of RIPng interfaces configured in the system 0 32
```

## show ipv6 ripng

### Syntax

```
show ipv6 ripng [<PROCESS-ID>] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays general RIPng configuration.

## Command context

Operator (>) or Manager (#)

## Parameters

**<PROCESS-ID>**

Specifies RIPng process ID. Range: 1-63.

**all vrfs**

Displays general RIPng information for all VRFs.

**vrf**

Selects VRF to display general RIPng information for.

**<VRF-NAME>**

Specifies VRF name.

## Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

## Usage

- Parameters display general RIPng information for a specific RIPng process.
- Parameters display general RIPng information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

Displaying general RIPng configuration for all VRFs:

```
switch# show ipv6 ripng 34 all-vrfs
VRF : Default Process-ID : 34

Protocol Status : Enabled ECMP : 6
Update Time : 60 sec Timeout Time : 240 sec
Garbage Collection Time : 250 sec Distance : 100
Redistribution : static,
 ospfv3 1

VRF : vrf_1 Process-ID : 34

Protocol Status : Enabled ECMP : 4
Update Time : 30 sec Timeout Time : 180 sec
Garbage Collection Time : 120 sec Distance : 120
Redistribution : None
```

## show ipv6 ripng interface

### Syntax

```
show ipv6 ripng [<PROCESS-ID>] interface [<INTERFACE-NAME>] [brief] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays information about RIPng enabled interfaces.

### Command context

Operator (>) or Manager (#)

## Parameters

### <PROCESS-ID>

Specifies RIPng process ID. Range: 1-63.

### <INTERFACE-NAME>

Specifies interface.

### brief

Shows brief overview information for RIPng interface.

### all-vrfs

Displays interface information for all VRFs.

### vrf

Selects specific VRF.

### <VRF-NAME>

Specifies VRF.

## Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

## Usage

- Parameters display general RIPng information for a specific RIPng process.
- Parameters display general RIPng information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

## Examples

```
switch# show ipv6 ripng interface
Interface 1/1/1 is up, IPv6 Address is fe80::7272:cfff:fe70:67a

VRF : Default Process-ID : 1
Status : Oper Up Mode : Send and Receive
MTU : 500 Poision Reverse : Enabled

Interface 1/1/2 is up, IPv6 Address is fe80::7272:cfff:fe70:67a

VRF : Default Process-ID : 1
Status : Admin Down Mode : Receive
MTU : 500 Poision Reverse : Enabled

Interface 1/1/3 is up, IPv6 Address is fe80::7272:cfff:fe70:67a

VRF : Default Process-ID : 1
Status : Admin Down Mode : Send
MTU : 500 Poision Reverse : Enabled

switch# show ipv6 ripng interface brief
VRF : default Process-ID : 1

Total Number of Interfaces: 2

Interface IPv6-Address Status MTU
```

|       |                          |    |     |
|-------|--------------------------|----|-----|
| 1/1/1 | fe80::7272:cfff:fe70:67a | up | 500 |
| 1/1/2 | fe80::7272:cfff:fe71:67a | up | 500 |

## show ipv6 ripng neighbors

### Syntax

```
show ipv6 ripng [<PROCESS-ID>] neighbors [<LINK-LOCAL-ADDRESS>] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays information about RIPng neighbors.

### Command context

Operator (>) or Manager (#)

### Parameters

#### <PROCESS-ID>

Specifies RIPng process ID. Range: 1-63.

#### neighbors

Specifies neighbor IP address.

#### <LINK-LOCAL-ADDRESS>

Specifies link-local address.

#### all-vrfs

Displays neighbor information for all VRFs.

#### vrf

Selects VRF to display neighbor information.

#### <VRF-NAME>

Specifies VRF name.

### Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

### Usage

- Parameters display RIPng neighbor information for a specific RIPng process.
- Parameters display RIPng neighbor information for a specific neighbor.
- Parameters display general RIPng information for a specific or all VRFs.
- If a VRF is not mentioned, information for the default VRF is displayed.

### Examples

Displaying RIPng neighbor information for all VRFs:

```
switch# show ipv6 ripng neighbors all-vrfs
VRF : default Process-ID : 1

```

Total Number of Neighbors: 1

| Peer-Address              | Type  | Last-Update | Rcvd-Bad-Pkts | Rcvd-Bad-Routes |
|---------------------------|-------|-------------|---------------|-----------------|
| fe80::7272:cfff:fe70:86ae | RIPng | 0:0:7       | 4             | 5               |

## show ipv6 ripng routes

### Syntax

```
show ipv6 ripng [<PROCESS-ID>] routes [<PREFIX/LENGTH>] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays RIPng routing table for a specific RIPng process.

### Command context

Operator (>) or Manager (#)

### Parameters

#### <PROCESS-ID>

Specifies RIPng process ID to display information for a specific RIPng process. Range: 1-63.

#### <PREFIX/LENGTH>

Specifies the network prefix.

#### all-vrfs

Displays RIPng routing information for all VRFs.

#### vrf

Selects VRF to display RIPng routing information.

#### <VRF-NAME>

Specifies VRF name.

### Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

### Usage

- <PREFIX/LENGTH> is an optional parameter that displays RIPng routing table information for a specific subnet.
- If a VRF is not mentioned, information for the default VRF is displayed.

### Examples

Displaying RIPng routing table for all VRFs:

```
switch# show ipv6 ripng routes all-vrfs
VRF : default Process-ID : 1

Prefix Metric Interface Nexthop

2001:DB8:10::/64 2 1/1/1 FE80::2E0:E6FF:FE1B:8242
```



```
2002:DB8:10::/64 3 1/1/1 FE80::2E0:E6FF:FE1B:8242
2003:DB8:10::/64 1 1/1/1
```

```
VRF : vrf_1 Process-ID : 34

```

| Prefix           | Metric | Interface | Nexthop                  |
|------------------|--------|-----------|--------------------------|
| 3001:DB8:10::/64 | 10     | 1/1/2     | FE80::2E0:E6FF:FE1B:8232 |
| 3002:DB8:10::/64 | 14     | 1/1/2     | FE80::2E0:E6FF:FE1B:8232 |
| 3003:DB8:10::/64 | 1      | 1/1/2     |                          |

## show ipv6 ripng statistics

### Syntax

```
show ipv6 ripng [<PROCESS-ID>] statistics [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays RIPng statistics.

### Command context

Operator (>) or Manager (#)

### Parameters

#### <PROCESS-ID>

Specifies RIPng process ID. Range: 1-63.

#### all-vrfs

Displays statistics information for all VRFs.

#### vrf

Selects VRF and displays RIPng statistics for it.

#### <VRF-NAME>

Specifies VRF name.

### Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

### Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

### Examples

Displaying RIPng statistics for all VRFs:

```
switch# show ipv6 ripng statistics all-vrfs
VRF : default Process-ID : 1

```

```
Global Route Changes : 50
Global Queries : 2
```

```

Last Cleared : 0h 30m 28s ago

VRF : vrf_1 Process-ID : 34

Global Route Changes : 20
Global Queries : 0
Last Cleared : 0h 30m 28s ago

```

## show ipv6 ripng statistics interface

### Syntax

```
show ipv6 ripng [<PROCESS-ID>] statistics interface [<INTERFACE-NAME>] [all-vrfs | vrf <VRF-NAME>]
```

### Description

Displays RIPng statistics for RIPng enabled interfaces.

### Command context

Operator (>) or Manager (#)

### Parameters

#### <PROCESS-ID>

Specifies RIPng process ID. Range: 1-63.

#### <INTERFACE-NAME>

Specifies name of interface.

#### all-vrfs

Displays RIPng interface statistics for all VRFs.

#### vrf

Selects VRF to display RIPng interface statistics.

#### <VRF-NAME>

Specifies VRF name.

### Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

### Usage

- Parameters can display information for all VRFs or a specific VRF.
- If a VRF is not mentioned, information for the default VRF is displayed.

### Examples

Displaying RIPng statistics for a RIPng enabled interface:

```

switch# show ipv6 ripng statistics interface 1/1/1
VRF : default Process-ID : 1 interface 1/1/1

IPv6-Address Trigger-Updates Rcvd-Bad-Packets Rcvd-Bad-Routes

```

```
fe80::7272:cfff:fe70:86ae
```

```
15
```

```
3
```

```
4
```

```
Last Cleared: 0h 30m 28s ago
```

## show running-config

### Syntax

```
show running config
```

### Description

Displays all running configurations for all protocols including RIPng.

### Command context

Operator (>) or Manager (#)

### Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

### Examples

Displaying all running configurations for all protocols including RIPng:

```
switch# show running-config
Current configuration:
!
!Version Halon 0.1.0 (Build: genericx86-64-Halon-0.1.0-master-20170309054955-dev)
!Schema version 0.1.8
lldp enable
timezone set utc
vrf green
vrf red
led base-loc_fdc on
led base-loc on
led base-hlth_fdc fast_blink
led base-pwr_fdc on
!
!
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospfv3 1
router ripng 1
 maximum-paths 5
 distance 1
 redistribute ospfv3 1
 timers update 40 timeout 200 garbage-collection 150
router ripng 1 vrf red
 default-information originate always
 maximum-paths 7
 distance 5
```

```
vlan 1
 no shutdown
interface lag 44
 no shutdown
 ipv6 address link-local
 ipv6 ripng 1
 send disable
interface 1
 no shutdown
 ipv6 address link-local
 ipv6 ripng 1
 receive disable
```

## show spanning-tree

### Syntax

```
show spanning-tree [detail]
```

### Description

Shows the spanning tree mode and information on the RPVST instances and optionally shows details on the RPVST instances.

### Command context

Operator (>) or Manager (#)

### Parameters

#### detail

Specifies showing details on the RPVST instances.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing spanning tree mode and RPVST instance information:

```

switch# show spanning-tree
Spanning tree status : Enabled Protocol: RPVST
Extended System-id : Enabled
Ignore PVID Inconsistency : Enabled
Path cost method : Long
RPVST-MSTP Interconnect VLAN : 1
VLAN1
 Root ID Priority : 32768
 MAC-Address: 70:72:cf:31:c9:23
 This bridge is the root
 Hello time(in seconds):2 Max Age(in seconds):20
 Forward Delay(in seconds):15

 Bridge ID Priority : 32768
 MAC-Address: 70:72:cf:31:c9:23
 Hello time(in seconds):2 Max Age(in seconds):20
 Forward Delay(in seconds):15

```

| PORT  | ROLE       | STATE      | COST  | PRIORITY | TYPE     | BPDU-Tx | BPDU-Rx | TCN-Tx | TCN-Rx |
|-------|------------|------------|-------|----------|----------|---------|---------|--------|--------|
| 1/1/1 | Designated | Forwarding | 20000 | 128      | P2P Edge | 100     | 60      | 20     | 10     |
| 1/1/2 | Designated | Forwarding | 20000 | 128      | P2P      | 100     | 60      | 20     | 10     |
| 1/1/3 | Designated | Forwarding | 20000 | 128      | Shr      | 100     | 60      | 20     | 10     |
| 1/1/4 | Designated | Forwarding | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/5 | Alternate  | Loop-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/6 | Alternate  | Root-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |

```

Number of topology changes : 4
Last topology change occurred : 516 seconds ago

```

Showing spanning tree mode and detailed RPVST instance information:

```

switch# show spanning-tree detail
Spanning tree status : Enabled Protocol: RPVST
Extended System-id : Enabled
Ignore PVID Inconsistency : Enabled
Path cost method : Long
RPVST-MSTP Interconnect VLAN : 1
Current Virtual Ports Count : 0
Maximum Allowed Virtual Ports : 2048

VLAN1
 Root ID Priority : 32768
 MAC-Address: 70:72:cf:31:c9:23
 This bridge is the root
 Hello time(in seconds):2 Max Age(in seconds):20
 Forward Delay(in seconds):15

 Bridge ID Priority : 32768
 MAC-Address: 70:72:cf:31:c9:23
 Hello time(in seconds):2 Max Age(in seconds):20
 Forward Delay(in seconds):15

```

| PORT  | ROLE       | STATE      | COST  | PRIORITY | TYPE     | BPDU-Tx | BPDU-Rx | TCN-Tx | TCN-Rx |
|-------|------------|------------|-------|----------|----------|---------|---------|--------|--------|
| 1/1/1 | Designated | Forwarding | 20000 | 128      | P2P Edge | 100     | 60      | 20     | 10     |
| 1/1/2 | Designated | Forwarding | 20000 | 128      | P2P      | 100     | 60      | 20     | 10     |
| 1/1/3 | Designated | Forwarding | 20000 | 128      | Shr      | 100     | 60      | 20     | 10     |
| 1/1/4 | Designated | Forwarding | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/5 | Alternate  | Loop-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/6 | Alternate  | Root-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |

```

Topology change flag : False
Number of topology changes : 1
Last topology change occurred : 33293 seconds ago

```

```

Port 1/1/1
Designated Root Priority : 32768 Address: 48:0F:CF:AF:22:1D
Designated Bridge Priority : 32768 Address: 48:0F:CF:AF:22:1D
Designated Port : 1/1/1
Forwarding-State transitions : 0
BPDUs sent 1582, received 1506
TCN_Tx: 10, TCN_Rx: 10

```

```

Port lag1
Designated Root Priority : 32768 Address: 48:0F:CF:AF:22:1D
Designated Bridge Priority : 32768 Address: 48:0F:CF:AF:22:1D
Designated Port : lag1
Forwarding-State transitions : 0
BPDUs sent 1402, received 1316
TCN_Tx: 10, TCN_Rx: 10
Multi-chassis role : active

```

## show spanning-tree inconsistent-ports

### Syntax

```
show spanning-tree inconsistent-ports [vlan <VLAN-ID>]
```

### Description

Shows ports blocked by STP protection functions such as Root guard, Loop guard, BPDU guard, and RPVST guard.

### Command context

Operator (>) or Manager (#)

### Parameters

<VLAN-ID>

Specifies a VLAN ID number.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing inconsistent port information:

```
switch# show spanning-tree inconsistent-ports
VLAN ID Blocked Port Reason

1 1/1/1 BPDU Guard
2 1/1/1 BPDU Guard
3 1/1/1 BPDU Guard
4 1/1/1 BPDU Guard
5 1/1/1 BPDU Guard
6 1/1/1 BPDU Guard
7 1/1/1 BPDU Guard
8 1/1/1 BPDU Guard
9 1/1/1 BPDU Guard
10 1/1/1 BPDU Guard
```

Showing inconsistent port information for VLANs 1 to 4:

```
switch# show spanning-tree inconsistent-ports vlan 1-4
VLAN ID Blocked Port Reason

1 1/1/3 Root Guard
2 1/1/7 BPDU Guard
3 1/1/9 Loop Guard
4 1/1/37 RPVST Guard
```

# show spanning-tree summary port

## Syntax

```
show spanning-tree summary port
```

## Description

Shows a summary of port-related spanning-tree configuration and status.

## Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

Showing a summary of port-related spanning tree information:

```
switch# show spanning-tree summary port

STP status : Enabled
Protocol : RPVST
BPDU guard Timeout value : None
```

```

BPDU guard enabled interfaces : None
BPDU filter enabled interfaces : None
Root guard enabled interfaces : None
Loop guard enabled interfaces : None
TCN guard enabled interfaces : None
RPVST filter enabled interfaces : None
RPVST guard enabled interfaces : None

```

Interface count by state

| VLAN       | Blocking | Listening | Learning | Forwarding |
|------------|----------|-----------|----------|------------|
| VLAN1      | 28       | 0         | 0        | 0          |
| VLAN2      | 0        | 0         | 0        | 0          |
| VLAN3      | 0        | 0         | 0        | 0          |
| VLAN4      | 0        | 0         | 0        | 0          |
| VLAN5      | 0        | 0         | 0        | 0          |
| VLAN6      | 0        | 0         | 0        | 0          |
| VLAN7      | 0        | 0         | 0        | 0          |
| VLAN8      | 0        | 0         | 0        | 0          |
| VLAN9      | 0        | 0         | 0        | 0          |
| VLAN10     | 0        | 0         | 0        | 0          |
| Total = 10 | 28       | 0         | 0        | 0          |

## show spanning-tree summary root

### Syntax

```
show spanning-tree summary root
```

### Description

Shows the summary of spanning tree root and configurations for all VLANs.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

### Example

Showing summary of spanning tree configurations:

```
switch# show spanning-tree summary root
```

```

STP status : Enabled
Protocol : RPVST
System ID : f8:60:f0:c9:70:40

```

Root bridge for VLANs : 1-10

| VLAN  | Priority | Root ID           | Root cost | Hello Time | Max Age | Fwd Dly | Root Port |
|-------|----------|-------------------|-----------|------------|---------|---------|-----------|
| VLAN1 | 32768    | f8:60:f0:c9:70:40 | 0         | 2          | 20      | 15      | 0         |
| VLAN2 | 32768    | f8:60:f0:c9:70:40 | 0         | 2          | 20      | 15      | 0         |
| VLAN3 | 32768    | f8:60:f0:c9:70:40 | 0         | 2          | 20      | 15      | 0         |



|        |       |                   |   |   |    |    |   |
|--------|-------|-------------------|---|---|----|----|---|
| VLAN4  | 32768 | f8:60:f0:c9:70:40 | 0 | 2 | 20 | 15 | 0 |
| VLAN5  | 32768 | f8:60:f0:c9:70:40 | 0 | 2 | 20 | 15 | 0 |
| VLAN6  | 32768 | f8:60:f0:c9:70:40 | 0 | 2 | 20 | 15 | 0 |
| VLAN7  | 32768 | f8:60:f0:c9:70:40 | 0 | 2 | 20 | 15 | 0 |
| VLAN8  | 32768 | f8:60:f0:c9:70:40 | 0 | 2 | 20 | 15 | 0 |
| VLAN9  | 32768 | f8:60:f0:c9:70:40 | 0 | 2 | 20 | 15 | 0 |
| VLAN10 | 32768 | f8:60:f0:c9:70:40 | 0 | 2 | 20 | 15 | 0 |

## show spanning-tree vlan

### Syntax

```
show spanning-tree vlan <VLAN-ID> [detail]
```

### Description

Displays the spanning tree mode and information on the RPVST instance of the specified VLAN and optionally displays details on the RPVST instance for the VLAN.

### Command context

Operator (>) or Manager (#)

### Parameters

<VLAN-ID>

Specifies the number of a VLAN.

[detail]

Specifies displaying details on the RPVST instance for the VLAN.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing spanning tree mode and RPVST instance information for VLAN 2:

```
switch# show spanning-tree vlan 2
VLAN2
Spanning tree status: Enabled Protocol: RPVST
 Root ID Priority : 32768
 MAC-Address: 70:72:cf:76:43:2a
 This bridge is the root
 Hello time(in seconds):2 Max Age(in seconds):20
 Forward Delay(in seconds):15

 Bridge ID Priority : 32768
 MAC-Address: 70:72:cf:76:43:2a
 Hello time(in seconds):2 Max Age(in seconds):20
 Forward Delay(in seconds):15
```

| PORT  | ROLE       | STATE      | COST  | PRIORITY | TYPE     | BPDU-Tx | BPDU-Rx | TCN-Tx | TCN-Rx |
|-------|------------|------------|-------|----------|----------|---------|---------|--------|--------|
| 1/1/1 | Designated | Forwarding | 20000 | 128      | P2P Edge | 100     | 60      | 20     | 10     |
| 1/1/2 | Designated | Forwarding | 20000 | 128      | P2P      | 100     | 60      | 20     | 10     |
| 1/1/3 | Designated | Forwarding | 20000 | 128      | Shr      | 100     | 60      | 20     | 10     |
| 1/1/4 | Designated | Forwarding | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/5 | Alternate  | Loop-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/6 | Alternate  | Root-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |

```
Number of topology changes : 4
Last topology change occurred : 516 seconds ago
```

Showing spanning tree mode and detailed RPVST instance information for VLAN 2:

```
switch# show spanning-tree vlan 2 detail
VLAN2
```

Spanning tree status: Enabled Protocol: RPVST

Root ID      Priority    : 32768  
              MAC-Address: 70:72:cf:76:43:2a  
              This bridge is the root  
              Hello time(in seconds):2    Max Age(in seconds):20  
              Forward Delay(in seconds):15

Bridge ID    Priority    : 32768  
              MAC-Address: 70:72:cf:76:43:2a  
              Hello time(in seconds):2    Max Age(in seconds):20  
              Forward Delay(in seconds):15

| PORT  | ROLE       | STATE      | COST  | PRIORITY | TYPE     | BPDU-Tx | BPDU-Rx | TCN-Tx | TCN-Rx |
|-------|------------|------------|-------|----------|----------|---------|---------|--------|--------|
| 1/1/1 | Designated | Forwarding | 20000 | 128      | P2P Edge | 100     | 60      | 20     | 10     |
| 1/1/2 | Designated | Forwarding | 20000 | 128      | P2P      | 100     | 60      | 20     | 10     |
| 1/1/3 | Designated | Forwarding | 20000 | 128      | Shr      | 100     | 60      | 20     | 10     |
| 1/1/4 | Designated | Forwarding | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/5 | Alternate  | Loop-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |
| 1/1/6 | Alternate  | Root-Inc   | 20000 | 128      | Shr Edge | 100     | 60      | 20     | 10     |

Topology change flag : False

Number of topology changes : 1

Last topology change occurred : 33293 seconds ago

Port 1/1/1

Designated root has priority :32768 Address: 48:0f:cf:af:22:1d

Designated bridge has priority :32768 Address: 48:0f:cf:af:22:1d

Designated port :1

Number of transitions to forwarding state : 0

BPDU sent 1582, received 1506

TCN\_Tx: 10, TCN\_Rx: 10

## spanning-tree bpdu-guard timeout

### Syntax

```
spanning-tree bpdu-guard timeout <INTERVAL>
```

```
no spanning-tree bpdu-guard timeout
```

### Description

Enables and configures the auto re-enable timeout in seconds for all interfaces with BPDU guard enabled. When an interface is disabled after receiving an unauthorized BPDU it will automatically be re-enabled after the timeout expires. The default is for the interface to stay disabled until manually re-enabled.

The `no` form of the command disables BPDU guard timeout on the interface. This is the default.

### Command context

config

### Parameters

<Interval>

Specifies the re-enable timeout in seconds. Range: 1 to 65535.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Enabling the BPDU guard timeout on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree bpduguard timeout 10
```

Disabling BPDU guard timeout on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree bpduguard
```

## spanning-tree extend-system-id

### Syntax

```
spanning-tree extend-system-id {enable | disable}
```

```
no spanning-tree extend-system-id
```

### Description

Configures use of extended system ID. When enabled, the VLAN ID is included in spanning tree packets. When disabled, the VLAN ID is set to NULL in the spanning tree packets.

By default, extended system ID is enabled. If you disable extended system ID, the bridge identifier field in the spanning tree packet is filled with zeros.

The `no` form of this command disables extended system ID.

### Command context

```
config
```

### Parameters

#### **enable**

Specifies enabling use of extended system ID.

#### **disable**

Specifies disabling use of extended system ID.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling extended system ID:

```
switch# config
switch(config)# spanning-tree extend-system-id enable
```

Disabling extended system ID:

```
switch# config
switch(config)# spanning-tree extend-system-id disable
switch(config)# no spanning-tree extend-system-id
```

# spanning-tree ignore-pvid-inconsistency

## Syntax

```
spanning-tree ignore-pvid-inconsistency {enable | disable}
```

```
no spanning-tree ignore-pvid-inconsistency
```

## Description

Configures port behavior when per-VLAN ID inconsistencies are present. For example, when the ports on both ends of a point-to-point link are untagged members of different VLANs, enabling this option allows RPVST+ to process untagged RPVST+ packets belonging to the peer's untagged VLAN as if they were received on the current device's untagged VLAN. When this option is disabled, RPVST+ blocks the link, causing traffic on the mismatched VLANs to be dropped.

If this option is enabled on multiple switches connected by hubs, there could be more than two VLANs involved in PVID mismatches that will be ignored by RPVST+.

If port VLAN memberships is misconfigured on a switch in the network, then enabling this option prevents RPVST+ from detecting the problem, which may result in packet duplication in the network since RPVST+ would not converge correctly.

This command affects all ports on the switch belonging to VLANs on which RPVST+ is enabled.

By default ignore per-VLAN ID inconsistency is disabled.

The `no` form of this command sets the ignore per-VLAN ID inconsistencies to disabled.

## Command context

```
config
```

## Parameters

### **enable**

Specifies ignore per-VLAN ID inconsistencies and allow RPVST to run on mismatched links.

### **disable**

Disables the ignore per-VLAN ID inconsistencies functionality.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling ignore per-VLAN ID inconsistencies:

```
switch# config
switch(config)# spanning-tree ignore-pvid-inconsistency enable
```

Disabling ignore per-VLAN ID inconsistencies:

```
switch# config
switch(config)# spanning-tree ignore-pvid-inconsistency disable
switch(config)# no spanning-tree ignore-pvid-inconsistency
```

## spanning-tree link-type

### Syntax

```
spanning-tree link-type {point-to-point | shared}
```

```
no spanning-tree link-type
```

### Description

Configures the link type of a port.

The `no` form of this command sets the spanning tree link type to the default value of `point-to-point`.

### Command context

```
config-if
```

### Parameters

#### **point-to-point**

Sets the spanning tree link type as point-to-point. Use this for full-duplex ports that provide a point-to-point link to devices such as a switch, bridge, or end-node. Default.

#### **shared**

Sets the spanning tree link type as shared. Use this when the port is connected to a hub.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting spanning tree link type to shared:

```
switch(config)# interface 1/1/1
switch(config-if)# spanning-tree link-type shared
```

Setting spanning tree link type to point-to-point for a port:

```
switch(config)# interface 1/1/1
switch(config-if)# no spanning-tree link-type
```

## spanning-tree mode

### Syntax

```
spanning-tree mode {mstp|rpvst}
```

### Description

Sets the spanning tree mode to either MSTP mode (Multiple-instance Spanning Tree Protocol) or RPVST mode (Rapid Per VLAN Spanning Tree).

The `no` form of this command sets the spanning tree mode to the default value `mstp`.

### Command context

```
config
```

## Parameters

### **mstp**

Sets the mode to MSTP (Multiple-instance Spanning Tree Protocol), which applies the STP (spanning tree protocol) separately for each set of VLANs (called an MSTI - multiple spanning tree instance).

### **rpvst**

Sets the mode to RPVST (Rapid Per VLAN Spanning Tree).

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling MSTP mode:

```
switch(config)# spanning-tree mode mstp
```

Enabling RPVST mode:

```
switch(config)# spanning-tree mode rpvst
```

# spanning-tree pathcost-type

## Syntax

```
spanning-tree pathcost-type {long | short}
```

```
no spanning-tree pathcost-type
```

## Description

Configures the spanning tree path cost type. The long mode provides support for the wider range of link speeds required by high-speed interfaces. All switches in the network must use the same path cost type or errors can occur in the spanning tree.

The `no` form of this command sets the spanning tree path cost type to the default value of `long`.

## Command context

`config`

## Parameters

### **long**

Specifies the spanning tree path cost type as a 32-bit value, allowing port cost values to be set in the range 1-200,000,000. Default.

### **short**

Specifies the spanning tree path cost type as a 16-bit value, allowing port cost values to be set in the range 1-65535.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting spanning tree path cost type to short:

```
switch# config
switch(config)# spanning-tree pathcost-type short
```

Setting spanning tree path cost type to long:

```
switch# config
switch(config)# spanning-tree pathcost-type long
```

Setting spanning tree path cost to default of long:

```
switch# config
switch(config)# no spanning-tree pathcost-type
```

## spanning-tree rpvst-mstp interconnect vlan

### Syntax

```
spanning-tree rpvst-mstp-interconnect-vlan <VLAN-ID>
no spanning-tree rpvst-mstp-interconnect-vlan
```

### Description

Configures the VLAN that has to be used to interconnect RPVST and MSTP domains. VLAN 1 is used by default.

The `no` form of this command removes the VLAN configuration.

- It is required to create the interconnect VLAN and then configure RPVST spanning tree on it.
- The same interconnect VLAN must be kept on all the switches in the network.
- Adding or deleting the interconnect VLAN triggers a re-convergence in the network.
- Deleting a VLAN that is configured as the interconnect VLAN does not reset the value to the default.

### Command context

```
config
```

### Parameters

**<VLAN-ID>**

Specifies the number of a VLAN.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

This example configures VLAN 10 to used to interconnect RPVST and MSTP domains.

```
switch#(config)# spanning-tree rpvst-mstp-interconnect-vlan 10
```

## spanning-tree tcn-guard

### Syntax

```
spanning-tree tcn-guard
```

```
no spanning-tree tcn-guard
```

### Description

Disables propagation of topology change notifications (TCNs) to other STP ports. Use this when you do not want topology changes to be noticed by peer devices.

The `no` form of this command, enables propagation of topology changes. By default, propagation is disabled.

### Command context

```
config-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling `tcn-guard`, which disables propagation of topology changes:

```
switch(config-if)# spanning-tree tcn-guard
```

Disabling `tcn-guard`, which enables propagation of topology changes:

```
switch(config-if)# no spanning-tree tcn-guard
```

## spanning-tree vlan

### Syntax

```
spanning-tree vlan <VLAN-LIST> [{hello-time | foward-delay | max-age | priority} <VALUE>]
```

```
no spanning-tree vlan <VLAN-LIST> [hello-time | foward-delay | max-age | priority]
```

### Description

Creates an RPVST instance for the specified VLAN. This command also allows for configuration of RPVST instance-specific time parameters.

The `no` form of this command removes the RPVST instance associated with the specified VLAN, and configures default values for RPVST instance-specific parameters.

### Command context

```
config
```

### Parameters

**<VLAN-LIST>**

Specifies the number of a single VLAN, or a series of numbers for a range of VLANs, separated by commas (1, 2, 3, 4), dashes (1-4), or both (1-4,6).



**hello-time <VALUE>**

Specifies the hello-time in seconds for the RPVST instance. Range: 2-10 seconds. Default: 2 seconds.

**forward-delay <VALUE>**

Specifies the forward-delay time in seconds for the RPVST instance. Range: 4-30 seconds. Default: 15 seconds.

**max-age <VALUE>**

Specifies the maximum age time in seconds for the RPVST instance. Range: 6-40 seconds. Default: 20 seconds.

**priority <VALUE>**

Specifies the priority for the RPVST instance. Priority value is configured as a multiple of 4096. Range: 0-15. Default: 8 which is 32768.

**Authority**

Administrators or local user group members with execution rights for this command.

**Examples**

Creating an RPVST instance for a list of VLANs and configuring various time parameters:

```
switch# config
switch(config)# spanning-tree vlan 2-5
switch(config)# spanning-tree vlan 2-5 hello-time 5
switch(config)# spanning-tree vlan 5 max-age 10
switch(config)# spanning-tree vlan 2-5 forward-delay 25
switch(config)# spanning-tree vlan 2-5 priority 5
```

Removing an RPVST instance for a list of VLANs and setting various time parameters to the default:

```
switch# config
switch(config)# no spanning-tree vlan 2-5
switch(config)# no spanning-tree vlan 2-5 hello-time
switch(config)# no spanning-tree vlan 2-5 forward-time
switch(config)# no spanning-tree vlan 2-5 max-age
switch(config)# no spanning-tree vlan 2-5 priority
```

## spanning-tree vlan cost

**Syntax**

```
spanning-tree vlan <VLAN-LIST> cost <PORT-COST>
```

```
no spanning-tree vlan <VLAN-LIST> cost
```

**Description**

Configures the spanning tree cost for the VLAN. This is the cost to reach the root port.

The **no** form of this command sets the port cost to the default value.

**Command context**

```
config-if
```

## Parameters

### <VLAN-LIST>

Specifies the number of a single VLAN, or a series of numbers for a range of VLANs, separated by commas (1, 2, 3, 4), dashes (1-4), or both (1-4,6).

### <PORT-COST>

Specifies the spanning tree cost for the VLAN. Range: 1-200,000,000. Default is calculated from the port link speed:

- 10 Mbps link speed equals a path cost of 2,000,000.
- 100 Mbps link speed equals a path cost of 200,000.
- 1 Gbps link speed equals a path cost of 20,000.
- 2 Gbps link speed equals a path cost of 10,000.
- 10 Gbps link speed equals a path cost of 2,000.
- 100 Gbps link speed equals a path cost of 200.
- 1 Tbps link speed equals a path cost of 20.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting port cost:

```
switch(config-if)# spanning-tree vlan 5 cost 100000
```

Setting port cost to the default:

```
switch(config-if)# no spanning-tree vlan 5 cost
```

# spanning-tree vlan port-priority

## Syntax

```
spanning-tree vlan <VLAN-LIST> port-priority <PRIORITY>
```

```
no spanning-tree vlan <VLAN-LIST> port-priority
```

## Description

Configures port priority. A port with the lowest priority number has the highest priority for use in forwarding traffic.

The `no` form of this command, sets the port priority to the default of 8.

## Command context

config-if

## Parameters

### <VLAN-LIST>

Specifies the number of a single VLAN, or a series of numbers for a range of VLANs, separated by commas (1, 2, 3, 4), dashes (1-4), or both (1-4,6).

### <PRIORITY>

Specifies the port priority. The value, configured as a multiple of 16, helps in determining the designated port. The lower a priority value, the higher the priority. Range: 1 to 15. Default: 8.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting port priority:

```
switch(config-if)# spanning-tree vlan 5 port-priority 10
```

Setting port priority to the default of 8:

```
switch(config-if)# no spanning-tree vlan 5 port-priority
```

# spanning-tree trap

## Syntax

```
spanning-tree trap {new-root | topology-change [vlan <VLAN-ID>] | errant-bpdu |
root-guard-inconsistency | loop-guard-inconsistency}
```

```
no spanning-tree trap {new-root|topology-change [vlan <VLAN-ID>] | errant-bpdu |
root-guard-inconsistency | loop-guard-inconsistency}
```

## Description

This command enables SNMP traps for new root, topology change event, errant-bpdu received event, root-guard inconsistency, and loop-guard inconsistency notifications. It is disabled by default.

The **no** form of this command disables SNMP traps for new root and topology change event notifications.

## Command context

```
config
```

## Parameters

### **new-root**

Enables SNMP notification when a new root is elected on any PVST vlan on the switch.

### **topology-change**

Enables SNMP notification when a topology change event occurred in specified PVST vlan on the switch.

### **errant-bpdu**

Enables SNMP notification when an errant bpdu is received by any PVST vlan on the switch.

### **root-guard-inconsistency**

Enables SNMP notification when the root-guard finds the port inconsistent for any PVST vlan on the switch.

### **loop-guard-inconsistency**

Enables SNMP notification when the loop-guard finds the port inconsistent for any PVST vlan on the switch.

## **Authority**

Administrators or local user group members with execution rights for this command.

## **Examples**

Commands enabling spanning-tree trap features:

```
switch(config)# spanning-tree trap
 new-root Enable notifications which are sent when a new root is elected
 topology-change Enable notifications which are sent when a topology change occurs
 errant-bpdu Enable notifications which are sent when an errant bpdu is received
 root-guard-inconsistency Enable notifications which are sent when root guard inconsistency occurs
 loop-guard-inconsistency Enable notifications which are sent when root guard inconsistency occurs
switch(config)# spanning-tree trap new-root
<cr>
switch(config)# spanning-tree trap topology-change
 vlan Enable topology change notification for the specified PVST vlan id.
switch(config)# spanning-tree trap topology-change vlan
 <0-64> Enable topology change information on the specified vlan id.
switch(config)# spanning-tree trap topology-change vlan 1
<cr>
switch(config)# spanning-tree trap errant-bpdu
<cr>
switch(config)# spanning-tree trap root-guard-inconsistency
<cr>
switch(config)# spanning-tree trap loop-guard-inconsistency
<cr>
```

Commands disabling spanning-tree trap features:

```
switch(config)# no spanning-tree trap
 new-root Disable notifications which are sent when a new root is elected
 topology-change Disable notifications which are sent when a topology change occurs
 errant-bpdu Disable notifications which are sent when an errant bpdu is received
 root-guard-inconsistency Disable notifications which are sent when root guard inconsistency occurs
 loop-guard-inconsistency Disable notifications which are sent when root guard inconsistency occurs
switch(config)# no spanning-tree trap new-root
<cr>
switch(config)# no spanning-tree trap topology-change
 instance Disable topology change notification for the specified PVST vlan id.
switch(config)# no spanning-tree trap topology-change vlan
 <0-64> Disable topology change information on the specified PVST vlan id.
switch(config)# no spanning-tree trap topology-change vlan 0
<cr>
switch(config)# no spanning-tree trap errant-bpdu
<cr>
switch(config)# no spanning-tree trap root-guard-inconsistency
<cr>
switch(config)# no spanning-tree trap loop-guard-inconsistency
<cr>
```

## diagnostic monitor

### Syntax

```
diagnostic monitor {line-module | management-module} [<SLOT-ID>]
```

```
no diagnostic monitor {line-module | management-module} [<SLOT-ID>]
```

### Description

Enables runtime diagnostics for all modules or for a specified module. This feature is enabled by default for all modules.

The `no` form of this command disables runtime diagnostics for all modules or for a specified module.

### Command context

config

### Parameters

#### line-module

Specifies the enabling of diagnostic monitoring specific to a line module.

#### management-module

Specifies the enabling of diagnostic monitoring specific to a management module.

#### <SLOT-ID>

Specifies the slot ID of a module. Format: member/slot.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

When no parameters are used in the command (`diagnostic monitor` or `no diagnostic monitor`), the command applies to all modules. This command impacts the diagnostics that run periodically. It does not affect on-demand diagnostics.

### Example

Enabling runtime diagnostics for a specified module:

```
switch(config)# diagnostic monitor management-module 1/1
```

## diag on-demand

### Syntax

```
diag on-demand {line-module | management-module} [<SLOT-ID>]
```

## Description

Runs the diagnostic tests for all modules or for a specified module.

## Command context

Manager (#)

## Parameters

**[line-module | management-module]**

Selects the options for enabling or disabling runtime diagnostics for a specific module.

**line-module**

Specifies the enabling of diagnostic monitoring specific to a line module.

**management-module**

Specifies the enabling of diagnostic monitoring specific to a management module.

**<SLOT-ID>**

Specifies the member/slot for management modules (1/1) and line modules (1/1).

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

When no parameters are used in the command (`diag on-demand`), the command applies to all modules.

## Examples

Running diagnostic tests for all modules on a 6100 switch:

```
switch# diag on-demand
Fetching Test results. Please wait ...

Module ID Diagnostics Success
 Performed

LineModule 1/1 13 100%
ManagementModule 1/1 13 100%
```

Running diagnostic tests for a specific module on a 6100 switch:

```
switch# diag on-demand management-module 1/1
Performing diagnostic tests. Please wait ...
Fetching Test results. Please wait ...

Module ID Diagnostics Success
 Performed

ManagementModule 1/1 13 100%
```

# show diagnostic

## Syntax

```
show diagnostic {line-module | management-module} [<SLOT-ID>]
 {brief | detail}
```

## Description

Displays the diagnostic test results for all modules or for a specified module.

## Command context

Manager (#)

## Parameters

[**line-module** | **management-module**]

Selects the options for enabling or disabling runtime diagnostics for a specific module.

**line-module**

Specifies the enabling of diagnostic monitoring specific to a line module.

**management-module**

Specifies the enabling of diagnostic monitoring specific to a management module.

<**SLOT-ID**>

Specifies the member/slot for management modules (1/1) and line modules (1/1)

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

When no parameters are used in the command (`show diagnostic`), the command applies to all modules.

## Example

Showing diagnostic test results in brief format for all modules on a 6100 switch:

```
switch# show diagnostic brief
Module ID Diagnostics Success
 ID Performed

ManagementModule 1/1 13 100%
LineModule 1/1 13 100%
```

Showing diagnostic test results in brief format for a specified module on a 6100 switch:

```
switch# show diagnostic line-module brief
Module ID Diagnostics Success
 ID Performed

LineModule 1/1 13 100%
```

Showing diagnostic test results in detail format for all modules on a 6100 switch:

```
switch# show diagnostic detail
Module : ManagementModule 1/1

Diagnostic Status Error Code History Code Successive Total Failure Total Last Run Timestamp First Run Timestamp

ddr_ccount Pass 0x0 0x0 0 0 109 2019-07-31 16:43:38 2019-07-31 07:44:55
emmc Pass 0x0 0x0 0 0 4 2019-07-31 16:08:04 2019-07-31 07:44:55
fan_ctrlr Pass 0x0 0x0 0 0 4 2019-07-31 16:08:04 2019-07-31 07:44:55
fepld Pass 0x0 0x0 0 0 109 2019-07-31 16:43:38 2019-07-31 07:44:54
fru_eeprom Pass 0x0 0x0 0 0 4 2019-07-31 16:08:04 2019-07-31 07:44:54
fru_eeprom_ul Pass 0x0 0x0 0 0 4 2019-07-31 16:08:04 2019-07-31 07:44:54
mm_lcb Pass 0x0 0x0 0 0 109 2019-07-31 16:43:37 2019-07-31 07:44:54
pmc Pass 0x0 0x0 0 0 109 2019-07-31 16:43:37 2019-07-31 07:44:54
rdimm_spd Pass 0x0 0x0 0 0 4 2019-07-31 16:08:04 2019-07-31 07:44:55
```

|           |      |     |     |   |   |   |                     |                     |
|-----------|------|-----|-----|---|---|---|---------------------|---------------------|
| rdimm_tmp | Pass | 0x0 | 0x0 | 0 | 0 | 4 | 2019-07-31 16:08:04 | 2019-07-31 07:44:55 |
| rtc       | Pass | 0x0 | 0x0 | 0 | 0 | 4 | 2019-07-31 16:08:04 | 2019-07-31 07:44:55 |
| tmp1      | Pass | 0x0 | 0x0 | 0 | 0 | 4 | 2019-07-31 16:08:04 | 2019-07-31 07:44:55 |
| tmp2      | Pass | 0x0 | 0x0 | 0 | 0 | 4 | 2019-07-31 16:08:04 | 2019-07-31 07:44:55 |

Module : LineModule 1/1

| Diagnostic     | Status | Error Code | History Code | Successive Failure Count | Total Failure Count | Total Iteration | Last Run Timestamp  | First Run Timestamp |
|----------------|--------|------------|--------------|--------------------------|---------------------|-----------------|---------------------|---------------------|
| lc_asic        | Pass   | 0x0        | 0x0          | 0                        | 0                   | 108             | 2019-07-31 16:43:37 | 2019-07-31 07:46:03 |
| poe_ctrlr_1_q1 | Pass   | 0x0        | 0x0          | 0                        | 0                   | 4               | 2019-07-31 16:08:16 | 2019-07-31 07:46:03 |
| poe_ctrlr_1_q2 | Pass   | 0x0        | 0x0          | 0                        | 0                   | 4               | 2019-07-31 16:08:16 | 2019-07-31 07:46:04 |
| poe_ctrlr_1_q3 | Pass   | 0x0        | 0x0          | 0                        | 0                   | 4               | 2019-07-31 16:08:16 | 2019-07-31 07:46:04 |
| poe_ctrlr_2_q1 | Pass   | 0x0        | 0x0          | 0                        | 0                   | 4               | 2019-07-31 16:08:16 | 2019-07-31 07:46:05 |
| poe_ctrlr_2_q2 | Pass   | 0x0        | 0x0          | 0                        | 0                   | 4               | 2019-07-31 16:08:16 | 2019-07-31 07:46:05 |
| poe_ctrlr_2_q3 | Pass   | 0x0        | 0x0          | 0                        | 0                   | 4               | 2019-07-31 16:08:16 | 2019-07-31 07:46:05 |
| poe_ctrlr_3_q1 | Pass   | 0x0        | 0x0          | 0                        | 0                   | 4               | 2019-07-31 16:08:16 | 2019-07-31 07:46:06 |
| poe_ctrlr_3_q2 | Pass   | 0x0        | 0x0          | 0                        | 0                   | 4               | 2019-07-31 16:08:16 | 2019-07-31 07:46:06 |
| poe_ctrlr_3_q3 | Pass   | 0x0        | 0x0          | 0                        | 0                   | 4               | 2019-07-31 16:08:17 | 2019-07-31 07:46:06 |
| poe_ctrlr_4_q1 | Pass   | 0x0        | 0x0          | 0                        | 0                   | 4               | 2019-07-31 16:08:17 | 2019-07-31 07:46:07 |
| poe_ctrlr_4_q2 | Pass   | 0x0        | 0x0          | 0                        | 0                   | 4               | 2019-07-31 16:08:17 | 2019-07-31 07:46:07 |
| poe_ctrlr_4_q3 | Pass   | 0x0        | 0x0          | 0                        | 0                   | 4               | 2019-07-31 16:08:17 | 2019-07-31 07:46:08 |

## show diagnostic events

### Syntax

```
show diagnostic events
```

### Description

Displays the diagnostic related event logs.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Showing diagnostic related event logs:

```
switch# show diagnostic events
2019-08-07:17:19:21.214532|hhmd|106001|ERR|
Diagnostic mm_mcbe failed with error code 0x380 on management module 1/1
2019-08-07:17:19:21.214554|hhmd|106001|ERR|
Diagnostic pmc failed with error code 0x4 on management module 1/1
2019-08-07:17:19:21.215532|hhmd|106001|ERR|
Diagnostic ledpld failed with error code 0x4 on management module 1/1
2019-08-07:17:19:21.353221|hhmd|106001|ERR|
Diagnostic mm_mcbe failed with error code 0x380 on management module 1/1
2019-08-07:17:19:21.354421|hhmd|106001|ERR|
Diagnostic pmc failed with error code 0x4 on management module 1/1
2019-08-07:17:19:21.453221|hhmd|106001|ERR|
Diagnostic ledpld failed with error code 0x4 on management module 1/1
```



# fastboot

## Syntax

`fastboot`

`no fastboot`

## Description

Enables fastboot for the system.

The `no` form of this command disables fastboot for the system.

## Command context

`config`

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

When fastboot is enabled, most tests under a Power On Self Test (POST) are skipped. By default, fastboot is enabled.

After disabling fastboot, save switch configurations and then reboot for POST to run. POST verifies the hardware functionality of various modules during boot-up. Based on the criticality of the test, the selftest module decides whether to go ahead with the boot-up sequence of a particular subsystem or interface during a POST failure.

POST runs memory built-in selftest (BISTs) and front-end port loopback tests. Memory BISTs verify the internal and external memory blocks present in the module. The memory tables are critical for proper functionality of the system so any failures in these tests results in the corresponding subsystem to be marked as "Failed" and thus that subsystem is not available for use.

Front-end port loopback tests verify the physical port front-end interface. These tests check if a particular interface can function properly. A test failure means that a particular interface has been marked as "Failed" and is now unavailable for use.

## Examples

Enabling fastboot:

```
switch# configure terminal
switch(config)# fastboot
switch(config)# end
switch# show running-config
Current configuration:
!
```

```
!Version ArubaOS-CX PL.10.06.0001
module 1/1 product-number j1677a
!
!
!
!
!
!
!
vlan 1
interface 1/1/1
 no shutdown
```

Disabling fastboot:

```
switch# configure terminal
switch(config)# no fastboot
switch(config)# end
switch(config)# write mem
Configuration changes will take time to process, please be patient.
switch# show running-config
Current configuration:
!
```

```
!Version ArubaOS-CX PL.10.06.0001
module 1/1 product-number j1677a
!
!
!
no fastboot
!
!
!
!
vlan 1
interface 1/1/1
 no shutdown
```

## show selftest

### Syntax

```
show selftest [brief]
show selftest line-module <SLOT-ID>
show selftest line-module <SLOT-ID> interface [brief]
show selftest interface [<PORT-NUM>]
```

### Description

Displays selftest results.

### Command context

Manager (#)

### Parameters

**[brief]**

Shows the selftest results as a brief description. Default.

## line-module

Shows the selftest results for a line module.

### <SLOT-ID>

Shows the selftest results for the slot ID of the line or fabric module.

### <PORT-NUM>

Shows the selftest results for the port number.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Displaying the output when fastboot is enabled on a 6100 switch:

```
switch# show selftest
Name Id Status ErrorCode LastRunTime

LineModule 1/1 passed 0x0
LineModule 1/2 passed 0x0

switch# show selftest line-module
Name Id Status ErrorCode LastRunTime

LineModule 1/1 passed 0x0
LineModule 1/2 passed 0x0

switch# show selftest line-module 1/1
Name Id Status ErrorCode LastRunTime

LineModule 1/1 passed 0x0
```

Displaying the output when fastboot is enabled:

```
switch# show selftest interface 1/1/2
Name Status ErrorCode LastRunTime

1/1/2 skipped 0x0

switch# show selftest line-module 1/1 interface
Name Status ErrorCode LastRunTime

1/1/1 skipped 0x0
1/1/2 skipped 0x0
1/1/3 skipped 0x0
1/1/31 skipped 0x0
```

Displaying the output when fastboot is disabled:

Testing to register read/write:



**NOTE:** This test is run irrespective of fastboot being enabled or disabled.

```
switch# show selftest
Name Id Status ErrorCode LastRunTime

LineModule 1/1 passed 0x0 2018-02-16 18:15:53
```

## sflow

### Syntax

```
sflow
```

```
no sflow
```

### Description

Enables the sFlow agent.

- In the `config` context, this command enables the sFlow agent globally on all interfaces.
- In an `config-if` context, this command enables the sFlow agent on a specific interface. sFlow cannot be enabled on a member of a LAG, only on the LAG.

The sFlow agent is disabled by default.

The `no` form of this command disables the sFlow agent and deletes all sFlow configuration settings, either globally, or for a specific interface.

### Command context

```
config
```

```
config-if
```

### Parameters

None.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling sFlow globally on all interfaces:

```
switch(config)# sflow
```

Disabling sFlow globally on all interfaces:

```
switch(config)# no sflow
```

Enabling sFlow on interface `1/1/1`:

```
switch(config)# interface 1/1/1
switch(config-if)# sflow
```

Disabling sFlow on interface `1/1/1`:

```
switch(config)# interface 1/1/1
switch(config-if)# no sflow
```

Enabling sFlow on interface **lag100**:

```
switch(config)# interface lag100
switch(config-if)# sflow
```

Disabling sFlow on interface **lag100**:

```
switch(config)# interface lag100
switch(config-if)# no sflow
```

## sflow agent-ip

### Syntax

```
sflow agent-ip <IP-ADDR>
```

```
no sflow agent-ip [<IP-ADDR>]
```

### Description

Defines the IP address of the sFlow agent to use in sFlow datagrams. This address must be defined for sFlow to function. HPE recommends that the address:

- can uniquely identify the switch
- is reachable by the sFlow collector
- does not change with time

The **no** form of this command deletes the IP address of the sFlow agent. This causes sFlow to stop working and no datagrams will be sent to the sFlow collector.

### Command context

config

### Parameters

**<IP-ADDR>**

Specifies an IP address in IPv4 format (**x.x.x.x**), where **x** is a decimal number from 0 to 255, or IPv6 format (**xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx**), where **x** is a hexadecimal number from 0 to F. The agent address is used to identify the switch in all sFlow datagrams sent to sFlow collectors. It is usually set to an IP address on the switch that is reachable from an sFlow collector.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the agent address to **10.10.10.100**:

```
switch(config)# sflow agent-ip 10.0.0.100
```

Setting the agent address to **2001:0db8:85a3:0000:0000:8a2e:0370:7334**:

```
switch(config)# sflow agent-ip 2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Removing the address configuration from the switch, which results in sFlow being disabled:

```
switch(config)# no sflow agent-ip
```

## sflow collector

### Syntax

```
sflow collector <IP-ADDR> [port <PORT>] [vrf <VRF>]
```

```
no sflow collector <IP-ADDR> [port <PORT>] [vrf <VRF>]
```

### Description

Defines a collector to which the sFlow agent sends data. Up to three collectors can be defined. At least one collector should be defined, and it must be reachable from the switch for sFlow to work.

### Command context

config

### Parameters

**collector <IP-ADDR>**

Specifies the IP address of a collector in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255, or IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

**port <PORT>**

Specifies the UDP port on which to send information to the sFlow collector. Range: 0 to 65536. Default: 6343.

**vrf <VRF>**

Specifies the VRF on which to send information to the sFlow collector. The VRF must be defined on the switch. If no VRF is specified, the default VRF (default) is used.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Defining a collector with IP address 10.10.10.100 on UDP port 6400:

```
switch(config)# sflow collector 10.0.0.1 port 6400
```

## sflow disable

### Syntax

```
sflow disable
```

### Description

Disables the sFlow agent, but retains any existing sFlow configuration settings. The settings become active if the sFlow agent is re-enabled.

## Command context

config

## Parameters

None.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Disabling sFlow support:

```
switch(config)# sflow disable
```

# sflow header-size

## Syntax

```
sflow header-size <SIZE>
```

```
no sflow header-size [<SIZE>]
```

## Description

Sets the sFlow header size in bytes.

The **no** form of this command sets the header size to the default value of 128.

## Command context

config

## Parameters

**header-size** <SIZE>

Specifies the sFlow header size in bytes. Range: 64 to 256. Default: 128.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the header size to 64 bytes:

```
switch(config)# sflow header-size 64
```

Setting the header size to the default value of 128 bytes:

```
switch(config)# no sflow header-size
```

## sflow max-datagram-size

### Syntax

```
sflow max-datagram-size <SIZE>
```

```
no sflow max-datagram-size [<SIZE>]
```

### Description

Sets the maximum number of bytes that are sent in one sFlow datagram.

The `no` form of this command sets maximum number of bytes to the default value of 1400.

### Command context

```
config
```

### Parameters

**max-datagram-size** <SIZE>

Specifies the maximum datagram size in bytes. Range: 1 to 9000. Default: 1400.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the datagram size to 1000 bytes:

```
switch(config)# sflow max-datagram-size 1000
```

Setting the header size to the default value of 1400 bytes:

```
switch(config)# no sflow max-datagram-size
```

## sflow polling

### Syntax

```
sflow polling <INTERVAL>
```

```
no sflow polling [<INTERVAL>]
```

### Description

Defines the global polling interval for sFlow in seconds.

The `no` form of this command sets the polling interval to the default value of 30 seconds.

### Command context

```
config
```

### Parameters

<INTERVAL>

Specifies the polling interval in seconds. Range: 10 to 3600. Default: 30.



## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the polling interval to 10:

```
switch(config)# sflow polling 10
```

Setting the polling interval to the default value.

```
switch(config)# no sflow polling
```

# sflow sampling

## Syntax

```
sflow sampling <RATE>
```

```
no sflow sampling [<RATE>]
```

## Description

Defines the global sampling rate for sFlow in number of packets. The default sampling rate is 4096, which means that one in every 4096 packets is sampled. A warning message is displayed when the sampling rate is set to less than 4096 and proceeds only after user confirmation.

The `no` form of this command sets the sampling rate to the default value of 4096.

## Command context

config

## Parameters

**sampling** <RATE>

Specifies the sampling rate. Range: 1 to 1000000000. Default: 4096.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the sampling rate to 5000:

```
switch(config)# sflow sampling 5000
```

Setting the sampling rate to the default:

```
switch(config)# no sflow sampling
```

Setting the sampling rate to 1000:

```
switch(config)# sflow sampling 1000
```

Setting the sFlow sampling rate lower than 4096 is not recommended and might affect system performance.

Do you want to continue [y/n]? **y**

```
switch(config)#
```

# show sflow

## Syntax

```
show sflow [interface <INTERFACE-NAME>] [vsx-peer]
```

## Description

Shows sFlow configuration settings and statistics for all interfaces, or for a specific interface

## Command context

Manager (#)

## Parameters

**interface** <INTERFACE-NAME>

Specifies the name of an interface on the switch.

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing sFlow information for all interfaces:

```
switch# show sflow
sFlow Global Configuration

sFlow enabled
Collector IP/Port/Vrf 10.0.0.2/6343/default
 10.0.0.3/6400/default
Agent Address 10.0.0.1
Sampling Rate 1024
Polling Interval 30
Header Size 128
Max Datagram Size 1400

sFlow Status

Running - Yes

sFlow Statistics

Number of Samples 200

- Agent address is not configured.
```

Showing sFlow information for interface 1/1/1:

```
switch# show sflow 1/1/1
sFlow configuration - Interface 1

sFlow enabled
```

|                   |      |
|-------------------|------|
| Sampling Rate     | 1024 |
| Number of Samples | 30   |

## rmon alarm

### Syntax

```
rmon alarm index <INDEX> snmp-oid <SNMP-OID> rising-threshold <RISING-THRESHOLD>
 falling-threshold <FALLING-THRESHOLD> [sample-interval <SAMPLE-INTERVAL>] [sample-type <ABSOLUTE|DELTA>]
no rmon alarm [index <INDEX>]
```

### Description

Stores configuration entries in an alarm table that defines the sample interval, sample-type, and threshold parameters for an SNMP MIB object. Only the SNMP MIB objects that resolve to an ASN.1 primitive type of INTEGER (INTEGER, Integer32, Counter32, Counter64, Gauge32, or TimeTicks) will be monitored.

The `no` form of this command removes all RMON alarms and allows you to specify an index to remove a particular RMON alarm.

### Command context

config

### Parameters

**index** <INDEX>

Specifies the RMON alarm index. Range: 1 to 20.

**snmp-oid** <SNMP-OID>

Specifies the SNMP MIB object to be monitored by RMON.

**rising-threshold** <RISING-THRESHOLD>

Specifies the upper threshold value for the RMON alarm.

**falling-threshold** <FALLING-THRESHOLD>

Specifies the falling threshold value for the RMON alarm. The falling threshold must be less than the rising threshold.

**sample-interval** <SAMPLE-INTERVAL>

Sample interval in seconds. Default: 30.

**sample-type** <ABSOLUTE|DELTA>

Specifies the method of sampling of the SNMP MIB object. Default: Absolute.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring RMON for the MIB object `ifOutErrors.15` with an index `1`, rising threshold of `2147483647` and falling threshold of `-2134` using `absolute` sampling for a sample interval of `100` seconds:

```
switch(config)# rmon alarm index 1 snmp-oid ifOutErrors.15 rising-threshold 2147483647
falling-threshold -2134 sample-type absolute sample-interval 100
```

Removing RMON alarm with the index 5:

```
switch(config)# no rmon alarm index 5
```

## show rmon alarm

### Syntax

```
show rmon alarm [index <INDEX>]
```

### Description

Displays the RMON alarm configurations.

### Command context

config

### Parameters

**index** <INDEX>

Specifies the RMON alarm index. Range: 1 to 20.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Showing all RMON alarm configurations:

```
switch# show rmon alarm
Index : 1
Enabled : true
Status : valid
MIB object : ifOutErrors.15
Sample type : delta
Sampling interval : 6535 seconds
Rising threshold : 100
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-09-21 05:58:11

Index : 3
Enabled : true
Status : invalid
MIB object : IF-MIB::ifDescr.19
Sample type : absolute
Sampling interval : 10000 seconds
Rising threshold : 4000
Falling threshold : 10
Last sampled value : 0
```

Showing RMON alarm with alarm index 1:

```
switch# show rmon alarm index 1
Index : 1
Enabled : true
Status : valid
```

```
MIB object : ifOutErrors.15
Sample type : delta
Sampling interval : 6535 seconds
Rising threshold : 100
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-06-21 05:58:11
```

Showing disabled RMON alarm information:

```
switch# show rmon alarm
Index : 1
Enabled : false
Status : valid
MIB object : ifOutErrors.15
Sample type : delta
Sampling interval : 6535 seconds
Rising threshold : 100
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-09-21 05:58:11

Index : 3
Enabled : false
Status : invalid
MIB object : IF-MIB::ifDescr.19
Sample type : absolute
Sampling interval : 10000 seconds
Rising threshold : 4000
Falling threshold : 10
Last sampled value : 0
```

## show snmp agent-port

### Syntax

```
show snmp agent-port
```

### Description

Displays SNMP agent UDP port number.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Displaying SNMP agent UDP port number:

```
switch# show snmp agent-port
SNMP agent port : 161
```

## show snmp community

### Syntax

```
show snmp community
```

### Description

Displays a list of all configured SNMPv1/v2c communities.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

When a user creates a custom community before enabling an SNMP agent, AOS-CX automatically removes the default `public` community from the system.

### Example

Displaying a list of all configured SNMPv1/v2c communities:

Before any community is created by user

```
switch# show snmp community

SNMP communities

public
```

After community is created by user

```
switch#show snmp community

SNMP communities

private
private2
```

## show snmp system

### Syntax

```
show snmp system
```

### Description

Displays SNMP description, location, and contact information.

### Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

Displaying SNMP description, location, and contact information:

```
switch# show snmp system
SNMP system information

System description : Aggregation router
System location : Main lab
System contact : John Smith, Lab Admin
```

## show snmp trap

### Syntax

```
show snmp trap
```

### Description

Displays all configured SNMP traps/informs receivers.

### Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

Displaying all configured SNMP trap and informs receivers:

```
switch# show snmp trap
```

| HOST        | PORT | TYPE   | VER | COMMUNITY/USER NAME | VRF     |
|-------------|------|--------|-----|---------------------|---------|
| 10.10.10.10 | 162  | trap   | v1  | public              | default |
| 10.10.10.10 | 162  | inform | v2c | public              | default |
| 10.10.10.10 | 162  | inform | v3  | name                | default |

## show snmp vrf

### Syntax

```
show snmp vrf
```

### Description

Displays the VRF on which the SNMP agent service is running.

### Command context

Operator (>) or Manager (#)



## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

Displaying SNMP services enabled on VRF:

```
switch#show snmp vrf
SNMP enabled VRF
```

```

default
```

## show snmpv3 context

### Syntax

```
show snmpv3 context
```

### Description

Displays all configured SNMP contexts.

### Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Displaying all configured SNMP contexts:

```
switch# show snmpv3 context
```

```

Name vrf Community ype[Instance_id]

A default public vrf
switch#
```

## show snmpv3 engine-id

### Syntax

```
show snmpv3 engine-id
```

### Description

Displays the configured SNMPv3 snmp engine-id.

If the SNMPv3 engine-id is not configured, by default a unique engine-id is created by the switch using a combination of the enterprise OID value and the switch's mac address.

## Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

Displaying the configured SNMPv3 engine-id:

```
switch# show snmpv3 engine-id
SNMP engine-id : 80:00:B8:5C:08:00:09:1d:de:a5
```

# show snmpv3 security-level

## Syntax

```
show snmpv3 security-level
```

## Description

Displays the configured SNMPv3 security level.

## Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Displaying the configured SNMPv3 security level:

```
switch# show snmpv3 security-level
SNMPv3 security-level : auth
```

# show snmpv3 users

## Syntax

```
show snmpv3 users
```

## Description

Displays all configured SNMPv3 users.

For more details on the user enabled status, see [snmpv3 security-level](#).

## Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

Displaying all configured SNMPv3 users:

```
switch# show snmpv3 users
```

| User  | AuthMode | PrivMode | Context | Enabled |
|-------|----------|----------|---------|---------|
| name  | md5      | none     | none    | False   |
| name2 | sha      | aes      | none    | True    |

# snmp-server agent-port

## Syntax

```
snmp-server agent-port <PORT>
```

```
no snmp-server agent-port [<PORT>]
```

## Description

Sets the UDP port number that the SNMP master agent uses to communicate. UDP port 161 is the default port.

The `no` form of this command sets the SNMP master agent port to the default value.

## Command context

```
config
```

## Parameters

**<PORT>**

Specifies the UDP port number that the SNMP master agent will use. Range: 1 to 65535. Default: 161.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the SNMP master agent port to 2000:

```
switch(config)# snmp-server agent-port 2000
```

Resetting the SNMP master agent port to the default value:

```
switch(config-schedule)# no snmp-server agent-port 2000
```

## snmp-server community

### Syntax

```
snmp-server community <STRING>
```

```
no snmp-server community <STRING>
```

### Description

Adds an SNMPv1/SNMPv2c community string. A community string is a password that controls read access to the SNMP agent. A network management program must supply this name when attempting to get SNMP information from the switch. A maximum of 10 community strings are supported. Once you create your own community string, the default community string (`public`) is deleted.

The `no` form of this command removes the specified SNMPv1/SNMPv2c community string. When no community string exists, a default community string with the value `public` is automatically defined.

### Command context

```
config
```

### Parameters

**<STRING>**

Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the SNMPv1/SNMPv2c community string to **private**:

```
switch(config)# snmp-server community private
```

Removing SNMPv1/SNMPv2c community string **private**:

```
switch(config)# no snmp-server community private
```

## snmp-server host

### Syntax

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>]
```

```
no snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>]
```

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>]
```

```
no snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>]
```

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [<VRF-NAME>]
```

```
no snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [vrf <VRF-NAME>]
```

## Description

Configures a trap/informs receiver to which the SNMP agent can send SNMP v1/v2c/v3 traps or v2c informs. A maximum of 30 SNMP traps/informs receivers can be configured.

The **no** form of this command removes the specified trap/inform receiver.

## Command context

config

## Parameters

### <IPv4-ADDR>

Specifies the IP address of a trap receiver in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. You can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100.

### <IPv6-ADDR>

Specifies the IP address of a trap receiver in IPv6 format (x:x::x:x).

### trap version <VERSION>

Specifies the trap notification type for SNMPv1, v2c or v3. Available options are: v1, v2c or v3.

### inform version v2c

Specifies the inform notification type for SNMPv2c.

### trap version v3

Specifies the trap notification type for SNMPv3.

### user <NAME>

Specifies the SNMPv3 user name to be used in the SNMP trap notifications.

### community <STRING>

Specifies the name of the community string to use when sending trap notifications. Range: 1 - 32 printable ASCII characters, excluding space and question mark. Default: public.

### <UDP-PORT>

Specifies the UDP port on which notifications are sent. Range: 1 - 65535. Default: 162.

### <VRF-NAME>

Specifies the VRF on which the SNMP agent listens for incoming requests.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config)# snmp-server host 10.10.10.10 trap version v1
switch(config)# no snmp-server host 10.10.10.10 trap version v1

switch(config)# snmp-server host a:b::c:d trap version v1
switch(config)# no snmp-server host a:b::c:d trap version v1

switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
```

```

switch(config)# snmp-server host a:b::c:d trap version v2c community public
switch(config)# no snmp-server host a:b::c:d trap version v2c community public

switch(config)# snmp-server host 10.10.10.10 trap version v2c community public port 5000
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public port 5000

switch(config)# snmp-server host 10.10.10.10 trap version v2c community public port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public port 5000 vrf default

switch(config)# snmp-server host a:b::c:d trap version v2c community public port 5000
switch(config)# no snmp-server host a:b::c:d trap version v2c community public port 5000

switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community public

switch(config)# snmp-server host a:b::c:d inform version v2c community public
switch(config)# no snmp-server host a:b::c:d inform version v2c community public

switch(config)# snmp-server host 10.10.10.10 inform version v2c community public port 5000
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community public port 5000

switch(config)# snmp-server host 10.10.10.10 inform version v2c community public port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community public port 5000 vrf default

switch(config)# snmp-server host a:b::c:d inform version v2c community public port 5000
switch(config)# no snmp-server host a:b::c:d inform version v2c community public port 5000

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin

switch(config)# snmp-server host a:b::c:d trap version v3 user Admin
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin port 2000
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin port 2000

switch(config)# snmp-server host a:b::c:d trap version v3 user Admin port 2000
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin port 2000

```

## snmp-server system-contact

### Syntax

```
snmp-server system-contact <INFO>
```

```
no snmp-server system-contact [<INFO>]
```

### Description

Sets SNMP contact information.

The `no` form of this command removes the SNMP contact information.

### Command context

```
config
```

### Parameters

<INFO>

Specifies SNMP contact information. Range: 1 to 128 printable ASCII characters, except for question mark (?).

### Authority

Administrators or local user group members with execution rights for this command.

## Examples

Defines SNMP contact information to be **John Smith, Lab Admin**:

```
switch(config)# snmp-server system-contact John Smith, Lab Admin
```

Removes SNMP contact information:

```
switch(config)# no snmp-server system-contact
```

## snmp-server system-description

### Syntax

```
snmp-server system-description <DESCRIPTION>
```

```
no snmp-server system-description
```

### Description

Sets the SNMP system description.

The **no** form of this command removes the SNMP system description.

### Command context

config

### Parameters

<DESCRIPTION>

Specifies the SNMP system description. Typical content to include would be the full name and version of the following:

- Hardware type of the system
- Software operating system
- Networking software

Range: 1 to 64 printable ASCII characters, except for the question mark (?).

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Defines the SNMP system description to be **mainSwitch**:

```
switch(config)# snmp-server system-description mainSwitch
```

Removes the SNMP system description:

```
switch(config)# no snmp-server system-description mainSwitch
```

## snmp-server system-location

### Syntax

```
snmp-server system-location <INFO>
```

```
no snmp-server system-location
```

### Description

Sets the SNMP location information.

The **no** form of this command removes the SNMP location information.

### Command context

```
config
```

### Parameters

**<INFO>**

Specifies the SNMP location information. Range: 1 to 128 printable ASCII characters, except for the question mark (?).

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Defines the SNMP location information to be **Main Lab**:

```
switch(config)# snmp-server system-location Main Lab
```

Removes the SNMP location information:

```
switch(config)# no snmp-server system-location
```

## snmp-server vrf

### Syntax

```
snmp-server vrf <VRF-NAME>
```

```
no snmp-server vrf <VRF-NAME>
```

### Description

Configures a VRF on which the SNMP agent listens for incoming requests. By default, the SNMP agent does not listen on any VRF. 6100 only supports default VRF.

The **no** form of this command stops the SNMP agent from listening for incoming requests on the specified VRF.

### Command context

```
config
```



## Parameters

**<VRF-NAME>**

Specifies the name of a VRF.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring the SNMP agent to listen on VRF default.

```
switch(config)# snmp-server vrf default
```

Stopping the SNMP agent from listening on VRF default.

```
switch(config)# no snmp-server vrf default
```

# snmpv3 context

## Syntax

```
snmpv3 context <NAME> vrf <VRF-NAME> [community <STRING>]
```

```
no snmpv3 context <NAME> [vrf <VRF-NAME>]
```

## Description

Creates an SNMPv3 context on the specified VRF.

The **no** form of this command removes the specified SNMP context.

## Command context

config

## Parameters

**<NAME>**

Specifies the name of the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark (?).

**vrf <VRF-NAME>**

Specifies the VRF associated with the context. Default: default.

**community <STRING>**

Specifies the SNMP community string associated with the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark. Default: public.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Creating an SNMPv3 context named **newContext**:

```
switch(config)# snmpv3 context newContext
```

## snmpv3 engine-id

### Syntax

```
snmpv3 engine-id <ENGINE-ID>
```

```
no snmpv3 engine-id <ENGINE-ID>
```

### Description

Configures the SNMPv3 SNMP engine-id allowing an administrator to configure a unique SNMP engine-id for the switch. This engine-id is used by the NMS management tool to identify and distinguish multiple switches on the same network.

The `no` form of this command restores the default engine-id, created by the switch using a combination of the enterprise OID value and the switch's mac address.

### Command context

```
config
```

### Parameters

**<ENGINE-ID>**

SNMPv3 SNMP engine-id in colon separated hexadecimal notation.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring the SNMPv3 engine-id:

```
switch(config)#
switch(config)# snmpv3 engine-id
WORD SNMPv3 snmp engine-id in colon seperated hexadecimal notation
switch(config)# snmpv3 engine-id 01:23:45:67:89:ab:cd:ef:01:23:45:67
```

Restoring the default SNMPv3 engine-id:

```
switch(config)# no snmpv3 engine-id
```

## snmpv3 security-level

### Syntax

```
snmpv3 security-level {auth | auth-privacy}
```

```
no snmpv3 security-level {auth | auth-privacy}
```

### Description

Configures the SNMPv3 security level. The security level determines which SMNPv3 users defined by the command `snmpv3 user` are able to connect.

The `no` form of this command changes the security level as follows:

- `no snmpv3 security-level auth`: Sets the security level to auth-privacy.
- `no snmpv3 security-level auth-privacy`: Sets the security level to no authentication or privacy, allowing any SNMP user to connect.

## Command context

config

## Parameters

### auth

SNMPv3 users that support authentication, or authentication and privacy are allowed.

### auth-privacy

Only SNMPv3 users with both authentication and privacy are allowed. This is the highest level of SNMPv3 security. Default.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the SNMPv3 security level to authentication and privacy:

```
switch(config)# snmpv3 security-level auth-privacy
```

Setting the SNMPv3 security level to authentication only:

```
switch(config)# snmpv3 security-level auth
```

Setting the SNMPv3 security level to no authentication and no privacy:

```
switch(config)# no snmpv3 security-level auth-privacy
```

Restoring the default SNMPv3 security level to authentication and privacy:

```
switch(config)# no snmpv3 security-level auth
```

# snmpv3 user

## Syntax

```
snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass {plaintext | ciphertext}
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass {plaintext | ciphertext} <PRIV-PWORD>]]
```

```
no snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass <PRIV-PWORD>]]
```

## Description

Creates an SNMPv3 user and adds it to an SNMPv3 context. The `no` form of this command removes the specified SNMPv3 user.

For more details on the user enabled status, see [snmpv3 security-level](#).

## Command context

config

## Parameters

**<NAME>**

Specifies the SNMPv3 username. Range 1 to 32 printable ASCII characters, excluding space and question mark (?).

**auth <AUTH-PROTOCOL>**

Specifies the authentication protocol used to validate user logins. Available options are: md5 or sha.

**auth-pass {plaintext | ciphertext} <AUTH-PWORD>**

Specifies the SNMPv3 user password. Range for `plaintext` is 8 to 32 printable ASCII characters, excluding space and question mark (?).

Range for `ciphertext` is 1 to 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.

**priv <PRIV-PROTOCOL>**

Specifies the SNMPv3 security protocol (encryption method). Available options are: aes or des.

**priv-pass {plaintext | ciphertext} <PRIV-PWORD>**

Specifies the SNMPv3 user privacy passphrase. Range for `plaintext` is 8 to 32 printable ASCII characters, excluding space and question mark (?).

Range for `ciphertext` is 1 to 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des priv-pass plaintext myprivpass
```

Removing an SNMPv3 user named **Admin**:

```
switch(config)# no snmpv3 user Admin
```

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des priv-pass plaintext myprivpass
```

Copying an SNMP user from switch 1 to switch 2.

On switch 1, configure a user called **Admin**, then issue the `show running-config` command to display switch configuration settings. The `snmpv3 user` command uses the `ciphertext` option to protect the users's passwords.

```
switch1(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword
priv des priv-pass plaintext myprivpass
switch1(config)# exit
switch1# show running-config
Current configuration:
```

```

!
!Version ArubaOS-CX XL.10.04.0001AD
!
!
!
snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
ssh server vrf default
!
!
!
!
interface vlan 1
 no shutdown
 ip dhcp
vlan 1

```

On switch 2, execute the `snmpv3 user` command that was displayed by `show running-config` on switch 1. This creates the user on switch 2 with the same configuration settings.

```

switch1(config)# snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=

```



**NOTE:** On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

## ssh (client login)

### Syntax

```
ssh [<USERNAME>@]{<IPV4> | <HOSTNAME>} [vrf <VRF-NAME>] [port <PORT-NUMBER>]
```

### Description

Establishes a client session with an SSH server which is typically another switch.

### Command context

Manager (#)

### Parameters

**<USERNAME>**

Specifies the username that the client uses to log in to an SSH server. When omitted, the username of the current session is used.

**{<IPV4> | <HOSTNAME>}**

Specifies the SSH server to which the SSH client will connect.

- **<IPV4>**: The IPv4 address.
- **<HOSTNAME>**: The host name.

**vrf <VRF-NAME>**

Specifies the VRF to be used for the SSH client session. When omitted, the default VRF named `default` is used.

**port <PORT-NUMBER>**

Specifies the SSH server TCP port number. When omitted, the default TCP port 22 is used.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Establishing an SSH client session (using the management VRF) with an SSH server:

```
switch# ssh admin@10.0.11.180 vrf mgmt
```

Establishing an SSH client session (using the default VRF and a specific port) with an SSH server:

```
switch# ssh admin@10.0.11.175 port 223
```

Configuring a test user on switch 1 and then connecting to switch 1 from switch 2 using the SSH client on the mgmt VRF:

```
** Configuring a test user on switch 1 **
switch(config)# user-group test
switch(config-usr-grp-test)# permit cli command ".*"
switch(config)# exit
switch(config)# user test-user group test password plaintext tst#9J%

** On switch 2, connecting to switch 1 using the SSH client **
switch# ssh test-user@10.0.11.177 vrf mgmt
```



**NOTE:** On the 6100 Switch Series, only the vrf named `default` is available. Replace any references to the `mgmt` or other VRFs with `default`.

## show ssh host-key

### Syntax

```
show ssh host-key [ecdsa | ed25519 | rsa]
```

### Description

Shows the public host keys for the SSH server. If the key type is not provided, all available host-keys are shown.

### Command context

Manager (#)

### Parameters

#### ecdsa

Selects the ECDSA host-key pair.

#### ed25519

Selects the ED25519 host-key pair.

#### rsa

Selects the RSA host-key pair.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Showing the ECDSA public host-key:

```
switch# show ssh host-key ecdsa
```

```
Key Type : ECDSA Curve : ecdsa-sha2-nistp256
```

```
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAhtuv5rABBBGs
...
04mjVFGMVKZ87RWkyrxeQa2fAGZZEp1902K33/k3q17fA4EivRzC75YvjDu8=
```

Showing all public host keys:

```
switch# show ssh host-key
```

```
Key Type : ECDSA Curve : ecdsa-sha2-nistp256
```

```
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAhtuv5rABBBGs
...
```



```
04mjVFGMVKZ87RWkyrxeQa2fAGZZEp1902K33/k3q17fA4EivRzC75YvjDu8=
```

```
Key Type : ED25519
```

```
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGb6910Jwoe8Hkl9K5YhqijrWI3yovNbiJVq6tw4WjJr4
```

```
Key Type : RSA Key Size : 2048
```

```
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDdVCXlw43h4n1bwg9jI6DSBMngymCdPD0JUG42Sn9IS
```

```
...
```

```
nGSXtrNy6OmlFDJTAY+zz5Kd8d21ZLuhf07IHNgF3pff65Xc8qNJBv
```

## show ssh server

### Syntax

```
show ssh server [vrf default]
```

### Description

Shows the SSH server configuration for the default VRF.

### Command context

Operator (>) or Manager (#)

### Parameters

**vrf default**

Specifies the default VRF.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing the SSH server configuration on the default VRF:

```
switch# show ssh server
```

```
SSH server configuration on VRF default :
```

|                   |                 |                     |       |
|-------------------|-----------------|---------------------|-------|
| IP Version        | : IPv4 and IPv6 | SSH Version         | : 2.0 |
| TCP Port          | : 22            | Grace Timeout (sec) | : 120 |
| Max Auth Attempts | : 6             |                     |       |

Ciphers:

chacha20-poly1305@openssh.com, aes128-ctr, aes192-cbc,  
aes128-cbc, aes192-ctr, aes256-gcm@openssh.com,  
aes128-gcm@openssh.com, aes256-ctr, aes256-cbc

Host Key Algorithms:

ecdsa-sha2-nistp256, ecdsa-sha2-nistp384, ecdsa-sha2-nistp521,  
ssh-ed25519, rsa-sha2-256, rsa-sha2-512, ssh-rsa

Key Exchange Algorithms:

curve25519-sha256, curve25519-sha256@libssh.org,  
ecdh-sha2-nistp256, ecdh-sha2-nistp384, ecdh-sha2-nistp521,  
diffie-hellman-group-exchange-sha256, diffie-hellman-group16-sha512,  
diffie-hellman-group18-sha512, diffie-hellman-group14-sha256,  
diffie-hellman-group14-sha1

MACs:  
hmac-sha1-etm@openssh.com, umac-64@openssh.com,  
umac-128@openssh.com, hmac-sha2-256, hmac-sha2-512, hmac-sha1

Public Key Algorithms:  
rsa-sha2-256, rsa-sha2-512ssh-rsa, ecdsa-sha2-nistp256,  
ecdsa-sha2-nistp384, ecdsa-sha2-nistp521, ssh-ed25519,  
x509v3-rsa2048-sha256, x509v3-ssh-rsa, x509v3-sign-rsa,  
x509v3-ecdsa-sha2-nistp256, x509v3-ecdsa-sha2-nistp384,  
x509v3-ecdsa-sha2-nistp521

## show ssh server sessions

### Syntax

```
show ssh server sessions [vrf default]
```

### Description

Shows the active SSH sessions on the default VRF.

### Command context

Operator (>) or Manager (#)

### Parameters

**vrf default**

Specifies the default VRF.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

If you provide the command with a VRF name, the command shows the active SSH session for the specified VRF. Any user can show sessions of all VRFs by using the `all-vrfs` parameter. The maximum number of sessions per VRF is five. The maximum SSH idle session timeout is 60 seconds.

### Examples

Showing the active SSH sessions on the default VRF:

```
switch# show ssh server sessions
```

```
SSH sessions on VRF default
```

```
IPv4 SSH Sessions
```

```
Server IP : 10.1.1.1
Client IP : 10.1.1.2
Client Port : 58835
```

```
IPv6 SSH Sessions
```

```
Server IP : FF01:0:0:0:0:0:0:FB
Client IP : FF01:0:0:0:0:0:0:FC
Client Port : 58836
```

# ssh ciphers

## Syntax

```
ssh ciphers <CIPHERS-LIST>
```

```
no ssh ciphers
```

## Description

Configures SSH to use a set of ciphers in the specified priority order. Ciphers in SSH are used for privacy of data being transported over the connection. The first cipher type entered in the CLI is considered a first priority. Each option is an algorithm that is used to encrypt the link and each name indicates the algorithm and cryptographic parameters that are used. Only ciphers that are entered by the user are configured.

The `no` form of this command removes the configuration of ciphers and reverts SSH to use the default set of ciphers.

## Command context

```
config
```

## Parameters

**<CIPHERS-LIST>**

Valid cipher types are:

- aes128-cbc
- aes192-cbc
- aes256-cbc
- aes128-ctr
- aes192-ctr
- aes256-ctr
- aes128-gcm@openssh.com
- aes256-gcm@openssh.com
- chacha20-poly1305@openssh.com

Default set of ciphers in priority order:

1. chacha20-1305@openssh.com
2. aes128-ctr
3. aes192-ctr
4. aes256-ctr
5. aes128-gcm@openssh.com
6. aes256-gcm@openssh.com

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring SSH to use only specified ciphers in the priority order:

```
switch(config)# ssh ciphers chacha20-poly1305@openssh.com aes256-ctr aes256-cbc
```

Reverting SSH to use the default set of ciphers:

```
switch(config)# no ssh ciphers
```

## ssh host-key

### Syntax

```
ssh host-key {ecdsa [ecdsa-sha2-nistp256 | ecdsa-sha2-nistp384 | ecdsa-sha2-nistp521] |
 ed25519 | rsa [bits {2048 | 4096}]] }
```

### Description

Generates an SSH host-key pair.

### Command context

config

### Parameters

#### ecdsa

Selects the ECDSA host-key pair type as `ecdsa-sha2-nistp256` (the default), `ecdsa-sha2-nistp384`, or `ecdsa-sha2-nistp521`.

#### ed25519

Selects the ED25519 host-key pair.

#### rsa

Selects the RSA host-key pair. Optionally, the key bit length is selected with either `bits 2048` (the default) or `bits 4096`.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

When an SSH server is enabled on a VRF for the first time, host-keys are generated.

If the host-key of the given type exists, a warning message is displayed with a request to overwrite the previous host-key with the new key.

## Examples

Overwriting an old ECDSA host-key with a new `ecdsa-sha2-nistp384` host-key:

```
switch(config)# ssh host-key ecdsa ecdsa-sha2-nistp384
ecdsa host-key will be overwritten.
Do you want to continue (y/n)?
```

Overwriting an old RSA host-key with a new RSA host-key with 2048 bits:

```
switch(config)# ssh host-key rsa bits 2048
rsa host-key will be overwritten.
Do you want to continue (y/n)?
```

Overwriting an ECDSA host-key with an ED25519 host-key pair:

```
switch(config)# ssh host-key ed25519
ed25519 host-key will be overwritten.
Do you want to continue (y/n)?
```

## ssh host-key-algorithms

### Syntax

```
ssh host-key-algorithms <HOST-KEY-ALGORITHMS-LIST>
```

```
no ssh host-key-algorithms
```

### Description

Configures SSH to use a set of host key algorithms in the specified priority order. Host key algorithms specify which host key types are allowed to be used for the SSH connection. The first host key entered in the CLI is considered a first priority. Each option represents a type of key that can be used. Host keys are used to verify the host that you are connecting to. This configuration allows you to control which host key types are presented to incoming clients, or which host key types to receive first from hosts. Only the host key algorithms that are specified by the user are configured.

The `no` form of this command removes the configuration of host key algorithms and reverts SSH to use the default set of algorithms.

### Command context

```
config
```

### Parameters

**<HOST-KEY-ALGORITHMS-LIST>**

Valid host key algorithms are:

- `ecdsa-sha2-nistp256`
- `ecdsa-sha2-nistp384`
- `ecdsa-sha2-nistp521`
- `rsa-sha2-256`
- `rsa-sha2-512`
- `ssh-ed25519`
- `ssh-rsa`

Default set of host key algorithms in priority order:

1. `ecdsa-sha2-nistp256`
2. `ecdsa-sha2-nistp384`
3. `ecdsa-sha2-nistp521`

4. ssh-ed25519
5. rsa-sha2-256
6. rsa-sha2-512
7. ssh-rsa

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring SSH to use only specified host key algorithms:

```
switch(config)# ssh host-key-algorithms ssh-rsa ssh-ed25519 ecdsa-sha2-nistp521
```

Reverting SSH to use the default set of host key algorithms:

```
switch(config)# no host-key-algorithms
```

## ssh key-exchange-algorithms

### Syntax

```
ssh key-exchange-algorithms <KEY-EXCHANGE-ALGORITHMS-LIST>
```

```
no ssh key-exchange-algorithms
```

### Description

Configures SSH to use a set of key exchange algorithm types in the specified priority order. The first key exchange type entered in the CLI is considered a first priority. Key exchange algorithms are used to exchange a shared session key with a peer securely. Each option represents an algorithm that is used to distribute a shared key in a way that prevents outside interference, manipulation, or recovery. Only the key exchange algorithms that are specified by the user are configured.

The `no` form of this command removes the configuration of key exchange algorithms and reverts SSH to use the default set of algorithms.

### Command context

config

### Parameters

<KEY-EXCHANGE-ALGORITHMS-LIST>

Valid key exchange algorithms are:

- curve25519-sha256
- curve25519-sha256@libssh.org
- diffie-hellman-group-exchange-sha1
- diffie-hellman-group-exchange-sha256
- diffie-hellman-group14-sha1

- diffie-hellman-group14-sha256
- diffie-hellman-group16-sha512
- diffie-hellman-group18-sha512
- ecdh-sha2-nistp256
- ecdh-sha2-nistp384
- ecdh-sha2-nistp521

Default set of key exchange algorithms in priority order:

1. curve25519-sha256
2. curve25519-sha256@libssh.org
3. ecdh-sha2-nistp256
4. ecdh-sha2-nistp384
5. ecdh-sha2-nistp521
6. diffie-hellman-group-exchange-sha256
7. diffie-hellman-group16-sha512
8. diffie-hellman-group18-sha512
9. diffie-hellman-group14-sha256
10. diffie-hellman-group-exchange-sha1

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring SSH to use a set of specified key exchange algorithms:

```
switch(config)# ssh key-exchange-algorithms ecdh-sha2-nistp256 curve25519-sha256
diffie-hellman-group-exchange-sha256
```

Reverting SSH to use the default set of key-exchange-algorithms:

```
switch(config)# no key-exchange-algorithms
```

## ssh known-host remove

### Syntax

```
ssh known-host remove {all | {<IPv4-ADDRESS> | <HOSTNAME> | <IPv6-ADDRESS>} }
```

### Description

Clears the list of trusted SSH servers for your user account. When you download or upload a file to or from a server using SFTP, you establish a trusted SSH relationship with that server. Each user account maintains its own set of SSH server host-keys for every server to which the user previously connected.

## Command context

config

## Parameters

**all**

Clears the trusted servers list.

**<IPv4-ADDRESS>**

Specifies the IPv4 address of the remote device.

**<HOSTNAME>**

Specifies the host name of the remote device. The length of the host name can be up to 255 characters.

**<IPv6-ADDRESS>**

Specifies the IPv6 address of the remote device.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Clearing the trusted server list:

```
switch(config)# ssh known-host remove all
```

Removing a specified server from the trusted server list:

```
switch(config)# ssh known-host remove 1.1.1.1
```

# ssh macs

## Syntax

ssh macs <MACS-LIST>

no ssh macs

## Description

Configures SSH to use a set of message authentication codes (MACs) in the specified priority order. The first MAC entered in the CLI is considered a first priority. MACs maintain the integrity of each message sent across an SSH connection. Each option represents an algorithm that can be used to provide integrity between peers. Only the MAC types that are specified by the user are configured.

The **no** form of this command removes the configuration of MACs and reverts SSH to use the default set of MACs.

## Command context

config

## Parameters

**<MACS-LIST>**

Valid MAC types are:



- `hmac-sha1`
- `hmac-sha1-96`
- `hmac-sha1-etm@openssh.com`
- `hmac-sha2-256`
- `hmac-sha2-512`
- `hmac-sha2-256-etm@openssh.com`
- `hmac-sha2-512-etm@openssh.com`

Default set of MACs in priority order:

1. `hmac-sha2-256-etm@openssh.com`
2. `hmac-sha2-512-etm@openssh.com`
3. `hmac-sha1-etm@openssh.com`
4. `hmac-sha2-256`
5. `hmac-sha2-512`
6. `hmac-sha1`

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring SSH to use a set of specified MACs:

```
switch(config)# ssh macs hmac-sha2-256 hmac-sha2-512
```

Reverting SSH to use the default set of MACs:

```
switch(config)# no ssh macs
```

## ssh maximum-auth-attempts

### Syntax

```
ssh maximum-auth-attempts <ATTEMPTS>
```

```
no maximum-auth-attempts
```

### Description

Sets the SSH maximum number of authentication attempts.

The `no` form of the command resets the maximum to its default of 6.

### Command context

```
config
```

## Parameters

### <ATTEMPTS>

Specifies the maximum number of SSH authentication attempts. Range: 1 to 10. Default: 6.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the maximum number of authentication attempts:

```
switch(config)# ssh maximum-auth-attempts 3
```

Resetting the maximum number of authentication attempts to its default of 6:

```
switch(config)# no maximum-auth-attempts
```

# ssh public-key-algorithms

## Syntax

```
ssh public-key-algorithms <PUBLIC-KEY-ALGORITHMS-LIST>
```

```
no ssh public-key-algorithms
```

## Description

Configures SSH to use a set of public key algorithms in the specified priority order. The first public key type entered in the CLI is considered a first priority. Public key algorithms specify which public key types can be used for public key authentication in SSH. Each option represents a public key type that the SSH server can accept or that the SSH client can present to a server. Only the public key algorithms that are chosen by the user are configured.

The `no` form of this command removes the configuration of public key algorithms and reverts SSH to use the default set.

## Command context

```
config
```

## Parameters

### <PUBLIC-KEY-ALGORITHMS-LIST>

Valid public key algorithm types are:

- ecdsa-sha2-nistp256
- ecdsa-sha2-nistp384
- ecdsa-sha2-nistp521
- ssh-ed25519
- ssh-rsa
- rsa-sha2-256
- rsa-sha2-512

- x509v3-ecdsa-sha2-nistp256
- x509v3-ecdsa-sha2-nistp384
- x509v3-ecdsa-sha2-nistp521
- x509v3-rsa2048-sha256
- x509v3-sign-rsa
- x509v3-ssh-rsa

Default set of public key algorithms in priority order:

1. rsa-sha2-256
2. rsa-sha2-512
3. ssh-rsa
4. ecdsa-sha2-nistp256
5. ecdsa-sha2-nistp384
6. ecdsa-sha2-nistp521
7. ssh-ed25519
8. x509v3-rsa2048-sha256
9. x509v3-ssh-rsa
10. x509v3-sign-rsa
11. x509v3-ecdsa-sha2-nistp256
12. x509v3-ecdsa-sha2-nistp384
13. x509v3-ecdsa-sha2-nistp521

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring SSH to use a set of specified public key algorithms:

```
switch(config)# ssh public-key-algorithms x509v3-ssh-rsa ssh-rsa rsa-sha2-256
```

Reverting SSH to use the default set of public key algorithms:

```
switch(config)# no ssh public-key-algorithms
```

## ssh server vrf

### Syntax

```
ssh server vrf <VRF-NAME>
```

```
no ssh server vrf <VRF-NAME>
```

## Description

Enables the SSH server on the specified VRF.

The `no` form of the command disables the SSH server on the specified VRF.

## Command context

`config`

## Parameters

**vrf** <VRF-NAME>

Specifies the VRF name.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling the SSH server on the management VRF:

```
switch(config)# ssh server vrf mgmt
```

Disabling the SSH server on the management VRF:

```
switch(config)# no ssh server vrf mgmt
```

## ip route

### Syntax

```
ip route <DEST-IPV4-ADDR>/<NETMASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>|reject|blackhole]
no ip route <DEST-IPV4-ADDR>/<NETMASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>|reject|blackhole]
```

### Description

Adds an IPv4 static route on the default VRF.

The `no` form of this command deletes a IPv4 static route.

### Command context

config

### Parameters

**<DEST-IPV4-ADDR>/<NETMASK>**

Specifies the IPv4 route destination.

**<NEXT-HOP-IP-ADDR>**

Specifies the next hop address for reaching the destination in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

**<INTERFACE>**

Specifies the next hop as an outgoing interface.

### blackhole

Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender.

### reject

Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch(config)# ip route 10.0.0.0/24 blackhole
switch(config)# ip route 10.0.1.0/24 reject
switch(config)# ip route 10.0.2.0/24 20.0.0.2
switch(config)# ip route 10.0.3.0/24 1/1/1
```

## ip route distance

### Syntax

```
ip route <DEST-IPV4-ADDR>/<NETMASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
no ip route <DEST-IPV4-ADDR>/<NETMASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
```

### Description

Configures the administrative distance for the IPv4 static route.

The **no** form of this command deletes the static route.

### Command context

config

### Parameters

**<DEST-IPV4-ADDR>**

Specifies an IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

**<MASK>**

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 32.

**<NEXT-HOP-IP-ADDR>**

Specifies the next hop IPv4 address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

**<INTERFACE>**

Specifies the next hop as an outgoing interface.

**distance <VALUE>**

Specifies the administrative distance to associate with this static route. Default: 1. Range: 1-255.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch(config)# ip route 10.0.2.0/24 20.0.0.2 distance 4
switch(config)# ip route 10.0.3.0/24 1/1/1 distance 6
```

```
switch(config)# no ip route 10.0.3.0/24 1/1/1 distance 6
```

## ip route tag

### Syntax

```
ip route <A.B.C.D/M> [A.B.C.D | INTERFACE] tag <1-4294967295>
no ip route <A.B.C.D/M> [A.B.C.D | INTERFACE] tag <1-4294967295>
```

## Description

Configures tag for IPv4 static route.

The `no` form of this command deletes tag for IPv4 static route.

## Command context

`config`

## Parameters

**tag**

Specifies and assigns tag for the route.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

```
switch(config)# ip route 10.1.1.1/32 20.1.1.2 tag 10
switch(config)# ip route 10.1.1.5/32 1/1/1 tag 20
```

```
switch(config)# no ip route 10.1.1.1/32 20.1.1.2 tag 10
switch(config)# no route 10.1.1.5/32 1/1/1 tag 20
```

# ipv6 route

## Syntax

```
ipv6 route <DEST-IPV6-ADDR>/<MASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>|reject|blackhole]
no ipv6 route <DEST-IPV6-ADDR>/<MASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>|reject|blackhole]
```

## Description

Adds an IPv6 static route.

The `no` form of this command deletes an IPv6 static route on the default VRF.

## Command context

`config`

## Parameters

**<DEST-IPV6-ADDR>**

Specifies the route destination in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

**<MASK>**

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.

**<NEXT-HOP-IP-ADDR>**

Specifies the next hop in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

### <INTERFACE>

Specifies the next hop as an outgoing interface.

### blackhole

Specifies that packets matching the destination route are silently discarded and no ICMP error notification is sent to the sender.

### reject

Specifies that packets matching the destination route are discarded and an ICMP error notification is sent to the sender.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

- Access platforms for IPv6 routes above /64 are not supported.
- IPv6 addresses with prefixes 65-127 will not be configured in the ASIC route table.

### Examples

```
switch(config)# ipv6 route 120::/124 blackhole
switch(config)# ipv6 route 121::/124 reject
switch(config)# ipv6 route 122::/124 1/1/1
switch(config)# ipv6 route 123::/124 120::1
```

```
switch(config)# no ipv6 route 122::/124 1/1/1
```

## ipv6 route distance

### Syntax

```
ipv6 route <DEST-IPV6-ADDR>/<MASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
```

```
no ipv6 route <DEST-IPV6-ADDR>/<MASK> [<NEXT-HOP-IP-ADDR>|<INTERFACE>] distance <VALUE>
```

### Description

Configures the administrative distance for the IPv6 static route

The **no** form of this command deletes the static route.

### Command context

config

### Parameters

#### <DEST-IPV6-ADDR>

Specifies the route destination address in IPv6 format  
(xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

#### <MASK>

Specifies the number of bits in the address mask in CIDR format (x), where x is a decimal number from 0 to 128.



#### <NEXT-HOP-IP-ADDR>

Specifies the next hop in IPv6 format (xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx), where x is a hexadecimal number from 0 to F.

#### <INTERFACE>

Specifies the next hop as an outgoing interface.

#### distance <VALUE>

Specifies the administrative distance to associate with this static route. Range: 1 to 255. Default: 1.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch(config)# ipv6 route 122::/124 1/1/1 distance 5
switch(config)# ipv6 route 123::/124 120::1 distance 6
```

```
switch(config)# no ipv6 route 123::/124 120::1 distance 6
```

## ipv6 route tag

### Syntax

```
ipv6 route <X:X::X:X/M> [X:X::X:X | INTERFACE] tag <1-4294967295>
```

```
no ipv6 route <X:X::X:X/M> [X:X::X:X | INTERFACE] tag <1-4294967295>
```

### Description

Configures tag for IPv6 static route.

### Command context

config

### Parameters

#### tag

Specifies and assigns tag for the route.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

```
switch(config)# ipv6 route 3001::1/128 1/1/1 tag 10
switch(config)# ipv6 route 3002::1/128 1000::2 tag 20
```

```
switch(config)# no ipv6 route 3001::1/128 1/1/1 tag 10
switch(config)# no ipv6 route 3002::1/128 1000::2 tag 20
```

# show ip route

## Syntax

```
show ip route [<A.B.C.D> | <A.B.C.D/M> | all-vrfs | bgp | connected | local | ospf | static | summary |
vrf<vrf-name>]
```

## Description

Displays IPv4 route tables.

## Command context

Operator (>) or Manager (#)

## Parameters

### <A.B.C.D>

Display longest prefix match.

### <A.B.C.D/M>

Display exact route match.

### all-vrfs

Display information for all VRFs.

### bgp

Display bgp routes only.

### connected

Display connected routes only.

### local

Display local routes only.

### ospf

Display ospf routes only.

### static

Display static routes only.

### summary

Display the aggregate count of routes per routing protocol.

### vrf <vrf-name>

Specify a VRF by VRF name (if no <VRF-NAME> is specified, the default VRF is implied).

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Showing IPv4 route tables:

```
switch# show ip route
```

```
Displaying ipv4 routes selected for forwarding
```

```
'[x/y]' denotes [distance/metric]

10.0.0.0/24, vrf default
 via vlan2, [0/0], connected
10.0.0.1/32, vrf default
 via vlan2, [0/0], local
10.100.11.0/24, vrf default
 via vlan1, [0/0], connected
10.100.11.82/32, vrf default
 via vlan1, [0/0], local
20.0.0.0/24, vrf default
 via 10.0.0.2, [1/0], static
20.0.1.0/24, vrf default
 via 10.0.0.2, [1/0], static
20.0.2.0/24, vrf default
 via vlan1, [1/0], static
20.0.4.0/24, vrf default
 blackhole route, [1/0], static
20.0.5.0/24, vrf default
 reject route, [1/0], static
```

Showing IPv4 route tables for the test VRF:

```
switch# show ip route vrf test

Displaying ipv4 routes selected for forwarding

'[x/y]' denotes [distance/metric]

30.0.0.0/24, 1 (null) next-hops
 via 30.0.0.2, [0/0], connected
90.0.0.0/24, 1 unicast next-hops
 via 30.0.0.1, [1/0], static
90.0.1.0/24, 1 unicast next-hops
 via 1/1/2, [1/0], static
90.0.3.0/24, blackhole, 1, [1/0], static
```

## show ipv6 route

### Syntax

```
show ipv6 route [<X.X.X.X> | <X.X.X.X/M> | all-vrfs | bgp | connected | local | ospf | static | summary |
vrf <vrf-name>]
```

### Description

Displays IPv6 route tables.

### Command context

Operator (>) or Manager (#)

### Parameters

**<X.X.X.X>**

Display exact route match.

**<X.X.X.X/M>**

Display exact route match.

**all-vrfs**

Display information for all VRFs.

**bgp**

Display bgp routes only.

**connected**

Display connected routes only.

**local**

Display local routes only.

**ospf**

Display ospf routes only.

**static**

Display static routes only.

**summary**

Display the aggregate count of routes per routing protocol.

**vrf <vrf-name>**

Specify a VRF by VRF name (if no *<VRF-NAME>* is specified, the default VRF is implied).

**Authority**

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

**Example**

Showing IPv6 route tables:

```
switch# show ipv6 route
```

Displaying ipv6 routes selected for forwarding

'[x/y]' denotes [distance/metric]

```
1000::/64, vrf default
 via vlan2, [0/0], connected
1000::1/128, vrf default
 via vlan2, [0/0], local
2000::/64, vrf default
 via vlan2, [1/0], static
2001::/64, vrf default
 via 1000::2, [1/0], static
3000:2301::/64, vrf default
 blackhole route, [1/0], static
4000:2301::/64, vrf default
 reject route, [1/0], static
```

## copy command-output

### Syntax

```
copy command-output "<COMMAND>" {<STORAGE-URL> | <REMOTE-URL> [vrf <VRF-NAME>]}
```

### Description

Copies the specified command output using TFTP, SFTP, or USB.

### Command context

Manager (#)

### Parameters

#### <COMMAND>

Specifies the command from which you want to obtain its output. Required.



**NOTE:** Users with auditor rights can specify these two commands only:

```
show accounting log
show events
```

{<STORAGE-URL> | <REMOTE-URL> [vrf <VRF-NAME>]}

Select either the storage URL or the remote URL for the destination of the copied command output. Required.

#### <STORAGE-URL>

Specifies the USB to copy command output.

Syntax: {usb} : /<FILE>

#### <REMOTE-URL>

Specifies the URL to copy the command output.

Syntax: {tftp:// | sftp://<USER>@}{<IP> | <HOST>}[:<PORT>][;blocksize=<VAL>]/<FILE>

#### vrf <VRF-NAME>

Specifies the VRF name. The default VRF name is default. Optional.

### Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

### Examples

Copying the output from the `show events` command to a remote URL:

```
switch# copy command-output "show events" tftp://10.100.0.12/file
```

Copying the output from the `show events` command to a file named `events` on a USB drive:

```
switch# copy command-output "show events" usb:/events
```

## copy diag-dump feature <FEATURE>

### Syntax

```
copy diag-dump feature <FEATURE> {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}
```

### Description

Copies the specified diagnostic information using TFTP, SFTP, or USB.

### Command context

Manager (#)

### Parameters

#### <FEATURE>

The name of a feature, for example `aaa`. Required.

{<REMOTE-URL> [vrf <VRF-NAME> | <STORAGE-URL>]}

Select either the remote URL or the storage URL for the destination of the copied command output. Required.

#### <REMOTE-URL>

Specifies the remote destination URL. Required. The syntax of the URL is the following:

Syntax: {tftp:// | sftp://<USER>@}{<IP> | <HOST>}[:<PORT>][;blocksize=<VAL>]/<FILE>

#### vrf <VRF-NAME>

Specifies the VRF name. If no VRF name is provided, the VRF named *default* is used. Optional.

#### <STORAGE-URL>

Specifies the USB to copy command output. Required.

Syntax: {usb}:/<FILE>

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Copying the output from the `aaa` feature to a remote URL with a specified VRF:

```
switch# copy diag-dump feature aaa tftp://10.100.0.12/diagdump.txt vrf mgmt
```

## copy diag-dump local-file

### Syntax

```
copy diag-dump local-file {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}
```

## Description

Copies the diagnostic information stored in a local file using TFTP, SFTP, or USB.

## Command context

Manager (#)

## Parameters

{<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}

Select either the storage URL or the remote URL for the destination of the copied command output. Required.

<REMOTE-URL>

Specifies the URL to copy the command output.

Syntax: {tftp:// | sftp://<USER>@}{<IP> | <HOST>}[:<PORT>][;blocksize=<VAL>]/<FILE>

vrf <VRF-NAME>

Specifies the VRF name. The default VRF name is default. Optional.

<STORAGE-URL>

Specifies the USB to copy command output.

Syntax: {usb}:/<FILE>

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

The copy diag-dump local-file command can be used only after the information is captured. Run the diag-dump <FEATURE-NAME> basic local-file command before you enter the copy diag-dump local-file command to capture the diagnostic information for the specified feature into the local file.

## Examples

Copying the output from the local file to a remote URL:

```
switch# diag-dump aaa basic local-file
switch# copy diag-dump local-file tftp://10.100.0.12/diagdump.txt
```

Copying the output from the local file to a USB drive:

```
switch# diag-dump aaa basic local-file
switch# copy diag-dump local-file usb:/diagdump.txt
```

## copy show-tech feature

### Syntax

copy show-tech feature <FEATURE> {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}

### Description

Copies show tech output using TFTP, SFTP, and USB.

## Command context

Manager (#)

## Parameters

{<REMOTE-URL> [vrf <VRF-NAME> | <STORAGE-URL>]}

Select either the remote URL or the storage URL for the destination of the copied command output. Required.

<REMOTE-URL>

Specifies the URL to copy the command output. Required.

Syntax: {tftp:// | sftp://<USER>@}{<IP> | <HOST>}[:<PORT>][;blocksize=<VAL>]/<FILE>

vrf <VRF-NAME>

Specifies the VRF name. The default VRF name is default. Optional.

<STORAGE-URL>

Specifies the USB to copy command output. Required.

Syntax: {usb}:/<FILE>

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Copying show tech output of the aaa feature using SFTP:

```
switch# copy show-tech feature aaa sftp://user@10.0.0.12/file.txt vrf mgmt
```

Copying show tech output of the config feature using SFTP on the mgmt VRF:

```
switch# copy show-tech feature config sftp://root@10.0.0.1/tech.txt vrf mgmt
```

## copy show-tech local-file

### Syntax

copy show-tech local-file {<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL>}

### Description

Copies show tech output stored in a local file.

## Command context

Manager (#)

## Parameters

{<REMOTE-URL> [vrf <VRF-NAME>] | <STORAGE-URL> ]}

Select either the remote URL or the storage URL for the destination of the copied command output. Required.



#### **<REMOTE-URL>**

Specifies the URL to copy the command output.

Syntax: {tftp:// | sftp://<USER>@}{<IP> | <HOST>}[:<PORT>][;blocksize=<VAL>]/<FILE>

#### **vrf <VRF-NAME>**

Specifies the VRF name. The default VRF name is default. Optional.

#### **<STORAGE-URL>**

Specifies the USB to copy command output.

Syntax: {usb}:/<FILE>

### **Authority**

Administrators or local user group members with execution rights for this command.

### **Usage**

Before entering the `copy show-tech local-file` command, run the `show tech` command with the `local-file` parameter for the specified feature.

### **Examples**

Copying the output to a remote URL:

```
switch# copy show-tech local-file tftp://10.100.0.12/file.txt
```

Copying the output to a remote URL with a VRF:

```
switch# copy show-tech local-file tftp://10.100.0.12/file.txt vrf mgmt
```

Copying the output to a USB:

```
switch# copy show-tech local-file usb:/file
```

## **copy support-files**

### **Syntax**

### **Description**

Copies a set of support files using TFTP, SFTP, or USB.

### **Command context**

Manager (#)

### **Parameters**

#### **<FEATURE-NAME>**

The feature name, for example, `aaa`.

{<REMOTE-URL> [vrf <VRF-NAME> | <STORAGE-URL>]}

Select either the remote URL or the storage URL for the destination of the copied command output. Required.

#### **<REMOTE-URL>**

Specifies the URL to copy the command output.

Syntax: {tftp:// | sftp://<USER>@}{<IP> | <HOST>}[:<PORT>][;blocksize=<VAL>]/<FILE>

#### **vrf <VRF-NAME>**

Specifies the VRF name. The default VRF name is default. Optional.

#### **<STORAGE-URL>**

Specifies the USB to copy command output.

Syntax: {usb}:/<FILE>

## **Usage**

If feature name is not provided, the command collects generic system-specific support information. If a feature name is provided, the command collects feature-specific support information.

## **Authority**

Administrators or local user group members with execution rights for this command.

## **Examples**

Copying the support files to a remote URL:

```
switch# copy support-files tftp://10.100.0.12/file.tar.gz
```

Copying the support files of the `lldp` feature to a remote URL with a specified VRF:

```
switch# copy support-files feature lldp tftp://10.100.0.12/file.tar.gz vrf mgmt
```

Copying the support files from the previous boot to a remote URL with a specified VRF:

```
switch# copy support-files previous-boot sftp://root@10.0.14.206/file.tar.gz vrf mgmt
```

Copying the support files to a USB:

```
switch# copy support-files usb:/file.tar.gz
```

Copying all the support files to a remote URL:

```
switch# copy support-files all sftp://root@10.0.14.216/file.tar.gz vrf mgmt
```

Copying the support files of the `config` feature to a USB:

```
switch# copy support-files feature config usb:/file.tar.gz
```

## **copy support-log**

### **Syntax**

### **Description**

Copies the specified support log for a daemon TFTP, SFTP, or USB.

### **Command context**

Manager (#)

## Parameters

**<DAEMON-NAME>**

Specifies the name of the daemon. Required.

**{<STORAGE-URL> | <REMOTE-URL> [vrf <VRF-NAME>]}**

Selects either the storage URL or the remote URL for the destination of the copied command output. Required.

**<STORAGE-URL>**

Specifies the USB to copy command output.

Syntax: {usb} : /<FILE>

**<REMOTE-URL>**

Specifies the URL to copy the command output.

Syntax: {tftp:// | sftp://<USER>@}{<IP> | <HOST>}[:<PORT>][;blocksize=<VAL>]/<FILE>

**vrf <VRF-NAME>**

Specifies the VRF name. If no VRF name is provided, the VRF named *default* is used. Optional.

## Usage

Fast log is a high performance, per-daemon binary logging infrastructure used to debug daemon level issues by precisely capturing the per daemon/module/functionalities debug traces in real time. Fast log, also referred to as support logs, helps users to understand the feature internals and its specific happenings. The fast logs from one daemon are not overwritten by other daemon logs because fast logs are captured as part of a daemon core dump. Fast logs are enabled by default.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Copying the support log from the daemon hpe-fand to a remote URL:

```
switch# copy support-log hpe-fand tftp://10.100.0.12/file
```

Copying the support log from the daemon hpe-fand to a remote URL with a VRF named mgmt:

```
switch# copy support-log hpe-fand tftp://10.100.0.12/file vrf mgmt
```

Copying the support log from the daemon hpe-fand to a USB:

```
switch# copy support-log hpe-fand usb:/support-log
```

## clear events

### Syntax

```
clear events
```

### Description

Clears up event logs. Using the `show events` command will only display the logs generated after the `clear events` command.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Clearing all generated event logs:

```
switch# show events

show event logs

2018-10-14:06:57:53.534384|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource utilization poll interval is changed to 27
2018-10-14:06:58:30.805504|lldpd|103|LOG_INFO|MSTR|1|Configured LLDP tx-timer to 36
2018-10-14:07:01:01.577564|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource utilization poll interval is changed to 49

switch# clear events

switch# show events

show event logs

2018-10-14:07:03:05.637544|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource utilization poll interval is changed to 34
```

## domain-name

### Syntax

```
domain-name <NAME>
```

```
no domain-name [<NAME>]
```

### Description

Specifies the domain name of the switch.

The `no` form of this command sets the domain name to the default, which is no domain name.

### Command context

```
config
```

## Parameters

**<NAME>**

Specifies the domain name to be assigned to the switch. The first character of the name must be a letter or a number. Length: 1 to 32 characters.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting and showing the domain name:

```
switch# show domain-name

switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
example.com
switch(config)#
```

Setting the domain name to the default value:

```
switch(config)# no domain-name
switch(config)# show domain-name

switch(config)#
```

# hostname

## Syntax

hostname <HOSTNAME>

no hostname [<HOSTNAME>]

## Description

Sets the host name of the switch.

The **no** form of this command sets the host name to the default value, which is **switch**.

## Command context

config

## Parameters

**<HOSTNAME>**

Specifies the host name. The first character of the host name must be a letter or a number. Length: 1 to 32 characters. Default: **switch**

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting and showing the host name:

```
switch# show hostname
switch
switch# config
switch(config)# hostname myswitch
myswitch(config)# show hostname
myswitch
```

Setting the host name to the default value:

```
myswitch(config)# no hostname
switch(config)#
```

## led locator

### Syntax

```
led locator {on | off | flashing}
```

### Description

Sets the state of the locator LED to `on`, `off` (default), or `flashing`.

### Command context

Manager (#)

### Parameters

#### **on**

Turns on the LED.

#### **off**

Turns off the LED, which is the default value.

#### **flashing**

Sets the LED to blink on and off repeatedly.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Setting the state of the locator LED:

```
switch# led locator flashing
```

## show capacities

### Syntax

```
show capacities <FEATURE>
```

### Description

Shows system capacities and their values for all features or a specific feature.

## Command context

Manager (#)

## Parameters

<FEATURE>

Specifies a feature. For example, aaa.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

Capacities are expressed in user-understandable terms. Thus they may not map to a specific hardware or software resource or component. They are not intended to define a feature exhaustively.

## Examples

Showing classifier-related capacities on the 6100:

```
switch# show capacities classifier
```

| System Capacities: Filter Classifier                              |       |
|-------------------------------------------------------------------|-------|
| Capacities Name                                                   | Value |
| Maximum number of Access Control Entries configurable in a system | 4096  |
| Maximum number of Access Control Lists configurable in a system   | 512   |
| Maximum number of class entries configurable in a system          | 4096  |
| Maximum number of classes configurable in a system                | 512   |
| Maximum number of entries in an Access Control List               | 1024  |
| Maximum number of entries in a class                              | 1024  |
| Maximum number of entries in a policy                             | 64    |
| Maximum number of classifier policies configurable in a system    | 512   |
| Maximum number of policy entries configurable in a system         | 4096  |

Showing all available capacities on the 6100:

```
switch# show capacities
```

| System Capacities:                                                                |       |
|-----------------------------------------------------------------------------------|-------|
| Capacities Name                                                                   | Value |
| Maximum number of Access Control Entries configurable in a system                 | 4096  |
| Maximum number of Access Control Lists configurable in a system                   | 512   |
| Maximum number of class entries configurable in a system                          | 4096  |
| Maximum number of classes configurable in a system                                | 512   |
| Maximum number of entries in an Access Control List                               | 1024  |
| Maximum number of entries in a class                                              | 1024  |
| Maximum number of entries in a policy                                             | 64    |
| Maximum number of classifier policies configurable in a system                    | 512   |
| Maximum number of policy entries configurable in a system                         | 4096  |
| Maximum number of clients supported for tracking the IP address in the system     | 128   |
| Maximum number of dynamic VLANs that can be allowed using MVRP                    | 256   |
| Maximum number of nexthops per IP ECMP group                                      | 1     |
| Maximum number of IP neighbors (IPv4+IPv6) supported in the system                | 1024  |
| Maximum number of IPv4 neighbors(# of ARP entries) supported in the system        | 1024  |
| Maximum number of IPv6 neighbors(# of ND entries) supported in the system         | 512   |
| Maximum number of L2 MAC addresses supported in the system                        | 8192  |
| Maximum number of L3 Groups for IP Tunnels and ECMP Groups                        | 1     |
| Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP groups | 1024  |
| Maximum number of configurable LAG ports                                          | 8     |
| Maximum number of members supported by a LAG port                                 | 8     |
| Maximum number of VLANs across ports allowed in loop-protect                      | 3328  |
| Maximum number of IGMP/MLD groups supported                                       | 512   |
| Maximum number of IGMP/MLD snooping groups supported                              | 512   |

|                                                                                   |      |
|-----------------------------------------------------------------------------------|------|
| Maximum number of Mirror Sessions configurable in a system                        | 4    |
| Maximum number of enabled Mirror Sessions in a system                             | 4    |
| Maximum number of mstp instances configurable in a system                         | 16   |
| Maximum number of Clients that can be authenticated on a port                     | 32   |
| Maximum number of Device Profiles allowed to be created on the system             | 8    |
| Maximum number of Port Access Roles allowed to be created on the system           | 32   |
| Maximum number of MAC Address that can be authorized on a port                    | 32   |
| Maximum number of Port Access Role VLAN IDs allowed to be created on the system   | 50   |
| Maximum number of Port Access Role VLAN names allowed to be created on the system | 50   |
| Maximum number of RBAC rules per user group                                       | 1024 |
| Maximum number of RPVST VLANs configurable on the system                          | 16   |
| Maximum number of RPVST VPORTs supported in a system                              | 512  |
| Maximum number of SVIs supported in the system                                    | 16   |
| Maximum number of routes (IPv4+IPv6) on the system                                | 512  |
| Maximum number of IPv4 routes on the system                                       | 512  |
| Maximum number of IPv6 routes on the system                                       | 512  |
| Maximum number of VLANs supported in the system                                   | 512  |

Showing all available capacities for mirroring:

```
switch# show capacities mirroring
```

| System Capacities: Filter Mirroring                        |       |
|------------------------------------------------------------|-------|
| Capacities Name                                            | Value |
| -----                                                      |       |
| Maximum number of Mirror Sessions configurable in a system | 4     |
| Maximum number of enabled Mirror Sessions in a system      | 4     |

Showing all available capacities for MSTP:

```
switch# show capacities mstp
```

| System Capacities: Filter MSTP                            |       |
|-----------------------------------------------------------|-------|
| Capacities Name                                           | Value |
| -----                                                     |       |
| Maximum number of mstp instances configurable in a system | 64    |

Showing all available capacities for VLAN count:

```
switch# show capacities vlan-count
```

| System Capacities: Filter VLAN Count            |       |
|-------------------------------------------------|-------|
| Capacities Name                                 | Value |
| -----                                           |       |
| Maximum number of VLANs supported in the system | 4094  |

## show capacities-status

### Syntax

```
show capacities-status <FEATURE>
```

### Description

Shows system capacities status and their values for all features or a specific feature.

### Command context

Manager (#)

### Parameters

<FEATURE>

Specifies the feature, for example `aaa` for which to display capacities, values, and status. Required.



## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing the system capacities status for all features:

```
switch# show capacities-status
```

| System Capacities Status                                                                       |  | Value | Maximum |
|------------------------------------------------------------------------------------------------|--|-------|---------|
| Capacities Status Name                                                                         |  |       |         |
| -----                                                                                          |  |       |         |
| Number of Access Control Entries currently configured                                          |  | 0     | 4096    |
| Number of Access Control Lists currently configured                                            |  | 0     | 512     |
| Number of class entries currently configured                                                   |  | 0     | 4096    |
| Number of classes currently configured                                                         |  | 0     | 512     |
| Number of policies currently configured                                                        |  | 0     | 512     |
| Number of policy entries currently configured                                                  |  | 0     | 4096    |
| Number of dynamic VLANs currently learnt using MVRP                                            |  | 0     | 256     |
| Number of IP neighbor (IPv4+IPv6) entries                                                      |  | 1     | 1024    |
| Number of IPv4 neighbor(ARP) entries                                                           |  | 1     | 1024    |
| Number of IPv6 neighbor(ND) entries                                                            |  | 0     | 512     |
| Number of L3 Groups for IP Tunnels and ECMP Groups currently configured                        |  | 0     | 1       |
| Number of L3 Destinations for Routes, Nexthops in ECMP groups and Tunnels currently configured |  | 0     | 1024    |
| Number of Mirror Sessions currently configured                                                 |  | 0     | 4       |
| Number of Mirror Sessions currently enabled                                                    |  | 0     | 4       |
| Number of mstp instances currently configured                                                  |  | 0     | 16      |
| Number of RPVST VLANs currently configured                                                     |  | 0     | 16      |
| Number of routes (IPv4+IPv6) currently configured                                              |  | 1     | 512     |
| Number of IPv4 routes currently configured                                                     |  | 1     | 512     |
| Number of IPv6 routes currently configured                                                     |  | 0     | 512     |
| Number of VLANs currently configured                                                           |  | 1     | 512     |

## show core-dump

### Syntax

```
show core-dump all
```

### Description

Shows core dump information about the specified module. When no parameters are specified, shows only the core dumps generated in the current boot of the management module. When the `all` parameter is specified, shows all available core dumps.

### Command context

Manager (#)

### Parameters

**all**

Shows all available core dumps.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

When no parameters are specified, the `show core-dump` command shows only the core dumps generated in the current boot of the management module. You can use this command to determine when any crashes are occurring in the current boot.

If no core dumps have occurred, the following message is displayed: `No core dumps are present`

To show core dump information for the standby management module, you must use the `standby` command to switch to the standby management module and then execute the `show core-dump` command.

In the output, the meaning of the information is the following:

**Daemon Name**

Identifies name of the daemon for which there is dump information.

**Instance ID**

Identifies the specific instance of the daemon shown in the `Daemon Name` column.

**Present**

Indicates the status of the core dump:

**Yes**

The core dump has completed and available for copying.

**In Progress**

Core dump generation is in progress. Do not attempt to copy this core dump.

**Timestamp**

Indicates the time the daemon crash occurred. The time is the local time using the time zone configured on the switch.

**Build ID**

Identifies additional information about the software image associated with the daemon.

**Examples**

Showing core dump information for the current boot of the active management module only:

```
switch# show core-dump
=====
Daemon Name | Instance ID | Present | Timestamp | Build ID
=====
hpe-fand | 1399 | Yes | 2017-08-04 19:05:34 | 1246d2a
hpe-sysmond | 957 | Yes | 2017-08-04 19:05:29 | 1246d2a
=====
Total number of core dumps : 2
=====
```

Showing all core dumps:

```
switch# show core-dump all
=====
Management Module core-dumps
=====
Daemon Name | Instance ID | Present | Timestamp | Build ID
=====
hpe-sysmond | 513 | Yes | 2017-07-31 13:58:05 | e70f101
hpe-tempd | 1048 | Yes | 2017-08-13 13:31:53 | e70f101
hpe-tempd | 1052 | Yes | 2017-08-13 13:41:44 | e70f101

Line Module core-dumps
=====
Line Module : 1/1
=====
dune_agent_0 | 18958 | Yes | 2017-08-12 11:50:17 | e70f101
dune_agent_0 | 18842 | Yes | 2017-08-12 11:50:09 | e70f101
=====
```

## show domain-name

### Syntax

show domain-name

### Description

Shows the current domain name.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

If there is no domain name configured, the CLI displays a blank line.

### Example

Setting and showing the domain name:

```
switch# show domain-name

switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
example.com
switch(config)#
```

## show environment fan



**NOTE:** show environment fan is not available for JL679A.

### Syntax

show environment fan

### Description

Shows the status information for all fans and fan trays (if present) in the system.

### Command context

Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Usage

For fan trays, *Status* is one of the following values:

### **ready**

The fan tray is operating normally.

### **fault**

The fan tray is in a fault event. The status of the fan tray does not indicate the status of fans.

### **empty**

The fan tray is not installed in the system.

For fans:

## Speed

Indicates the relative speed of the fan based on the nominal speed range of the fan.

### **N/A**

This value is not applicable to the 6100.

## Direction

The direction of airflow through the fan. Values are:

### **left-to-right**

Air flows from the left of the system to the right of the system.

### **N/A**

The fan is not installed.

## Status

Fan status. Values are:

### **uninitialized**

The fan has not completed initialization.

### **ok**

The fan is operating normally.

### **fault**

The fan is in a fault state.

### **empty**

The fan is not installed.

## Examples

Showing output for systems with fan trays for 6100 switch series:

```
switch# show environment fan
Fan information
```

| Mbr/Fan | Product Name | Serial Number | Speed | Direction     | Status | RPM |
|---------|--------------|---------------|-------|---------------|--------|-----|
| 1/1     | N/A          | N/A           | N/A   | left-to-right | ok     | N/A |

## show environment led

### Syntax

```
show environment led
```

### Description

Shows state and status information for all the configurable LEDs in the system.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Show state and status for the environment LED:

```
switch# show environment led
Mbr/Name State Status

1/locator off ok
```

## show environment power-supply

### Syntax

```
show environment power-supply
```

### Description

Shows status information about all power supplies in the switch.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Usage

The following information is provided for each power supply:

#### Mbr/PSU

Shows the member and slot number of the power supply.

#### Product Number

Shows the product number of the power supply.

#### Serial Number

Shows the serial number of the power supply, which uniquely identifies the power supply.

## PSU Status

The status of the power supply. Values are:

### OK

Power supply is operating normally.

### OK\*

Power supply is operating normally, but it is the only power supply in the chassis. One power supply is not sufficient to supply full power to the switch. When this value is shown, the output of the command also shows a message at the end of the displayed data.

### Absent

No power supply is installed in the specified slot.

### Input fault

The power supply has a fault condition on its input.

### Output fault

The power supply has a fault condition on its output.

### Warning

The power supply is not operating normally.

## Wattage Maximum

Shows the maximum amount of wattage that the power supply can provide.

## Example

Showing the output when only one power supply is installed in the switch:

| switch# show environment power-supply |                |               |            |                 |
|---------------------------------------|----------------|---------------|------------|-----------------|
| Mbr/PSU                               | Product Number | Serial Number | PSU Status | Wattage Maximum |
| 1/1                                   | N/A            | N/A           | OK         | 500             |

## show environment temperature

### Syntax

```
show environment temperature [detail]
```

### Description

Shows the temperature information from sensors in the switch that affect fan control.

### Command context

Operator (>) or Manager (#)

### Parameters

#### detail

Shows detailed information from each temperature sensor.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Usage

Temperatures are shown in Celsius.

Valid values for status are the following:

### **normal**

Sensor is within nominal temperature range.

### **min**

Lowest temperature from this sensor.

### **max**

Highest temperature from this sensor.

### **low\_critical**

Lowest threshold temperature for this sensor.

### **critical**

Highest threshold temperature for this sensor.

### **fault**

Fault event for this sensor.

### **emergency**

Over temperature event for this sensor.

## Examples

Showing current temperature information for a 6100 switch:

```
switch# show environment temperature
Temperature information
```

| Mbr/Slot-Sensor          | Module Type      | Current<br>temperature | Status    |
|--------------------------|------------------|------------------------|-----------|
| 1/1-COMe-Daughter-Boar   | line-card-module | 66.45 C                | normal    |
| 1/1-PCIE-Switch          | line-card-module | 95.82 C                | normal    |
| 1/1-Processor            | line-card-module | 00.00 C                | fault     |
| 1/1-Switch-ASIC          | line-card-module | 116.36 C               | emergency |
| 1/1-Switch-ASIC-Internal | line-card-module | 108.25 C               | critical  |
| 1/2-COMe-Daughter-Boar   | line-card-module | 67.29 C                | normal    |
| 1/2-PCIE-Switch          | line-card-module | 95.82 C                | normal    |
| 1/2-Processor-1          | line-card-module | 72.92 C                | normal    |
| 1/2-Processor-2          | line-card-module | 73.05 C                | normal    |
| 1/2-Switch-ASIC          | line-card-module | 97.41 C                | normal    |
| 1/2-Switch-ASIC-Internal | line-card-module | 97.62 C                | normal    |
| ...                      |                  |                        |           |

# show events

## Syntax

```
show events [-e <EVENT-ID> |
 -s {alert | crit | debug | emer | err | info | notice | warn} |
 -r | -a | -i <MEMBER-ID> | -n <count> |
 -c {lldp | ospf | ... | } |
 -d {lldpd | hpe-fand | ... |}]
```

## Description

Shows event logs generated by the switch modules since the last reboot.

## Command context

Manager (#)

## Parameters

**-e <EVENT-ID>**

Shows the event logs for the specified event ID. Event ID range: 101 through 99999.

**-s {alert | crit | debug | emer | err | info | notice | warn}**

Shows the event logs for the specified severity. Select the severity from the following list:

- **alert**: Displays event logs with severity alert and above.
- **crit**: Displays event logs with severity critical and above.
- **debug**: Displays event logs with all severities.
- **emer**: Displays event logs with severity emergency only.
- **err**: Displays event logs with severity error and above.
- **info**: Displays event logs with severity info and above.
- **notice**: Displays event logs with severity notice and above.
- **warn**: Displays event logs with severity warning and above.

**-r**

Shows the most recent event logs first.

**-a**

Shows all event logs, including those events from previous boots.

**-n <count>**

Displays the specified number of event logs.

**-c {lldp | ospf | ... | }**

Shows the event logs for the specified event category. Enter `show event -c` for a full listing of supported categories with descriptions.

**-d {lldpd | hpe-fand | ... | }**

Shows the event logs for the specified process. Enter `show event -d` for a full listing of supported daemons with descriptions.



## Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

## Examples

Showing event logs:

```
switch# show events

show event logs

2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to 70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to up for bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1 in Hardware
```

Showing the most recent event logs first:

```
switch# show events -r

show event logs

2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1 in Hardware
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to up for bridge_normal interface
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to 70:72:cf:51:50:7c
```

Showing all event logs:

```
switch# show events -a

show event logs

2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to 70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to up for bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1 in Hardware
```

Showing event logs related to IRDP:

```
switch# switch# show events -c irdp
2016-05-31:06:26:27.363923|hpe-rdiscd|111001|LOG_INFO|IRDP enabled on interface %s
2016-05-31:07:08:51.351755|hpe-rdiscd|111002|LOG_INFO|IRDP disabled on interface %s
```

Showing event logs related to LACP:

```
switch# show events -c lacp

show event logs

2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to 70:72:cf:51:50:7c
```

Showing event logs as per the specified slot ID:

```
switch# show events -i 1

Event logs from current boot

2020-04-22T05:36:14.430166+00:00 6300 vrfmgrd[3053]: Event[5401|LOG_INFO|MSTR|1|Created a vrf entity b1721d27-41c2-485d-9bae-2cfcbc9bd13d
2020-04-22T05:36:14.942597+00:00 6300 vrfmgrd[3053]: Event[5401|LOG_INFO|MSTR|1|Created a vrf entity 5eb532c9-5b4d-4d83-b34a-db24ae542d4e
2020-04-22T05:36:15.886252+00:00 6300 vsfd[710]: Event[9903|LOG_INFO|MSTR|1|Master 1 boot complete
```

Showing event logs as per the specified process:

```
switch# show events -d lacpd

show event logs

2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to 70:72:cf:51:50:7c
```

Displaying the specified number of event logs:

```
switch# show events -n 5

show event logs

```

```
2018-03-21:06:12:15.500603|arpmgrd|6101|LOG_INFO|AMM|-|ARPMGRD daemon has started
2018-03-21:06:12:17.734405|lldpd|109|LOG_INFO|AMM|-|Configured LLDP tx-delay to 2
2018-03-21:06:12:17.740517|lacpd|1307|LOG_INFO|AMM|-|LACP system ID set to 70:72:cf:d4:34:42
2018-03-21:06:12:17.743491|vrfgmr|5401|LOG_INFO|AMM|-|Created a vrf entity 42cc3df7-1113-412f-b5cb-e8227b8c22f2
2018-03-21:06:12:17.904008|vrfgmr|5401|LOG_INFO|AMM|-|Created a vrf entity 4409133e-2071-4ab8-adfe-f9662c06b889
```

## show hostname

### Syntax

```
show hostname
```

### Description

Shows the current host name.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Setting and showing the host name:

```
switch# show hostname
switch
switch# config
switch(config)# hostname myswitch
myswitch(config)# show hostname
myswitch
```

## show images

### Syntax

```
show images
```

### Description

Shows information about the software in the primary and secondary images.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Showing the primary and secondary images on a 6100 switch:

```
switch(config)# show images
```

```

ArubaOS-CX Primary Image

```

```
Version : PL.10.06.0002E
Size : 243 MB
Date : 2020-11-25 22:00:47 PST
SHA-256 : 61fe9233b2c842e8ac1731ad46949bd63e269c5c72d69290932ef19c1ebb0730
```

---

ArubaOS-CX Secondary Image

---

```
Version : PL.10.07.0000E-201-gba0c336
Size : 271 MB
Date : 2020-11-25 21:09:08 UTC
SHA-256 : 2fd6c646a8013efcca75729584bbb0fa54604086bad3f46e1c5d4e706b8b30ee
```

```
Default Image : primary
Boot Profile Timeout : 2 seconds
```

---

Management Module 1/1 (Active)

---

```
Active Image : primary
Service OS Version : PL.01.07.0003-internal
BIOS Version : PL.01.0001
```

## show module

### Syntax

```
show module
```

### Description

Shows information about installed line modules and management modules.



**NOTE:** Although this switch does not have removable modules, this command will still return information about the switch, referring to management modules and line modules.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Usage

Identifies and shows status information about the line modules and management modules that are installed in the switch.

To show the configuration information—if any—associated with that line module slot, use the `show running-configuration` command.

Status is one of the following values:

#### Active

This is the active management module.

#### Deinitializing

The switch is being deinitialized.

### Diagnostic

The switch is in a state used for troubleshooting.

### Down

The switch is physically present but is powered down.

### Failed

The moduleswitch has experienced an error and failed.

### Initializing

The switch is being initialized.

### Present

The switch hardware is installed in the chassis.

### Ready

The switch is available for use.

### Updating

A firmware update is being applied to the switch.

## Examples

Showing all installed modules:

```
switch# show module
```

Management Modules

=====

|      | Product |                      | Serial     |                |
|------|---------|----------------------|------------|----------------|
| Name | Number  | Description          | Number     | Status         |
| 1/1  | JL678A  | 6100F 24G 4SFP+ Swch | SG0ZKW600P | Active (local) |

Line Modules

=====

|      | Product |                      | Serial     |        |
|------|---------|----------------------|------------|--------|
| Name | Number  | Description          | Number     | Status |
| 1/1  | JL678A  | 6100F 24G 4SFP+ Swch | SG0ZKW600P | Ready  |

## show running-config

### Syntax

```
show running-config [<FEATURE>] [all]
```

### Description

Shows the current nondefault configuration running on the switch. No user information is displayed.

### Command context

Manager (#)

## Parameters

### <FEATURE>

Specifies the name of a feature. For a list of feature names, enter the `show running-config` command, followed by a space, followed by a question mark (?).

### **all**

Shows all default values for the current running configuration.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing the current running configuration:

```
switch> show running-config
Current configuration:
!
!Version ArubaOS-CX PL.10.06.0001-346-g56a12a8f4cf15
!export-password: default
ntp enable
!
!
!
!
ssh server vrf default
vlan 1
spanning-tree
spanning-tree instance 1 vlan 1,2,4-10
interface 1/1/1
 no shutdown
 vlan access 1
 portfilter 1/1/2-1/1/3

interface 1/1/2
 no shutdown
 vlan access 1
interface 1/1/3
 no shutdown
 vlan access 1
interface 1/1/4
 no shutdown
 vlan access 1
interface 1/1/5
 no shutdown
 vlan access 1
interface 1/1/6
 no shutdown
 vlan access 1
interface 1/1/7
 no shutdown
 vlan access 1
interface 1/1/8
 no shutdown
 vlan access 1
interface 1/1/9
 no shutdown
 vlan access 1
interface 1/1/10
 no shutdown
 vlan access 1
```

```

interface 1/1/11
 no shutdown
 vlan access 1
interface 1/1/12
 no shutdown
 vlan access 1
interface 1/1/13
 no shutdown
 vlan access 1
interface 1/1/14
 no shutdown
 vlan access 1
interface 1/1/15
 no shutdown
 vlan access 1
interface 1/1/16
 no shutdown
 vlan access 1
interface vlan 1
 ip dhcp
snmp-server vrf default
snmp-server community public
snmp-server host 1.1.1.1 inform version v2c
snmp-server host 1.1.1.1 trap version v2c
snmpv3 context A vrf default
!
!
!
!
!
https-server vrf default

```

Showing the current running configuration in json format:

```

switch> show running-config json
Running-configuration in JSON:
{
 "Monitoring_Policy_Script": {
 "system_resource_monitor_mm1.1.0": {
 "Monitoring_Policy_Instance": {
 "system_resource_monitor_mm1.1.0/system_resource_monitor_mm1.1.0.default": {
 "name": "system_resource_monitor_mm1.1.0.default",
 "origin": "system",
 "parameters_values": {
 "long_term_high_threshold": "70",
 "long_term_normal_threshold": "60",
 "long_term_time_period": "480",
 "medium_term_high_threshold": "80",
 "medium_term_normal_threshold": "60",
 "medium_term_time_period": "120",
 "short_term_high_threshold": "90",
 "short_term_normal_threshold": "80",
 "short_term_time_period": "5"
 }
 }
 }
 }
 },
 ...
 ...
 ...
 ...

```

Show the current running configuration without default values:

```

switch(config)# show running-config
Current configuration:
!

```

```

!Version ArubaOS-CX PL.10.06.0001-346-g56a12a8f4cf15
!export-password: default
!
!
!
ssh server vrf default
vlan 1
spanning-tree
interface 1/1/1
 no shutdown
 vlan access 1
interface 1/1/2
 no shutdown
 vlan access 1
interface 1/1/3
 no shutdown
 vlan access 1
interface 1/1/4
 no shutdown
 vlan access 1
interface 1/1/5
 no shutdown
 vlan access 1
interface 1/1/6
 no shutdown
 vlan access 1
interface 1/1/7
 no shutdown
 vlan access 1
interface 1/1/8
 no shutdown
 vlan access 1
interface 1/1/9
 no shutdown
 vlan access 1
interface 1/1/10
 no shutdown
 vlan access 1
interface 1/1/11
 no shutdown
 vlan access 1
interface 1/1/12
 no shutdown
 vlan access 1
interface 1/1/13
 no shutdown
 vlan access 1
interface 1/1/14
 no shutdown
 vlan access 1
interface 1/1/15
 no shutdown
 vlan access 1
interface 1/1/16
 no shutdown
 vlan access 1
interface vlan 1
 ip dhcp
!
!
!
!
!

```

```
https-server vrf default
switch#
switch#
```

## show running-config current-context

### Syntax

```
show running-config current-context
```

### Description

Shows the current non-default configuration running on the switch in the current command context.

### Command context

config or a child of config. See Usage.

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

You can enter this command from the following configuration contexts:

- Any child of the global configuration (config) context. If the child context has instances—such as interfaces—you can enter the command in the context of a specific instance.  
Support for this command is provided for one level below the config context. For example, entering this command for a child of a child of the config context not supported.  
If you enter the command on a child of the config context, the current configuration of that context and the children of that context are displayed.
- The global configuration (config) context.  
If you enter this command in the global configuration (config) context, it shows the running configuration of the entire switch. Use the show running-configuration command instead.

### Examples

Showing the running configuration for the current interface:

```
switch(config-if)# show running-config current-context
interface 1/1/1

 no shutdown
 description Example interface
vlan access 1
exit
```

Showing the current running configuration for the in-band management interface:

```
switch(config)# interface vlan 1
Switch(config-if-vlan)#description IN-BAND Management Interface
Switch(config-if-vlan)#ip dhcp
Switch(config-if-vlan)#no shutdown
Switch(config-if-vlan)#end
Switch#
```



Showing the current running configuration for the in-band management interface without DHCP:

```
switch(config)# interface vlan 1
Switch(config-if-vlan)#description IN-BAND Management Interface
Switch(config-if-vlan)#no ip dhcp
Switch(config-if-vlan)#ip address 192.168.10.1/24
Switch(config-if-vlan)#no shutdown
Switch(config-if-vlan)#end
Switch#
```

Showing the running configuration for the external storage share named `nasfiles`:

```
switch(config-external-storage-nasfiles)# show running-config current-context
external-storage nasfiles
 address 192.168.0.1
 vrf default
 username nasuser
 password ciphertext AQBapalKj+XMsZumHEwIc9OR6YcOw5Z6Bh9rV+9ZtKDKzvbaBAAAAB1CTrM=
 type scp
 directory /home/nas
 enable
switch(config-external-storage-nasfiles)#
```

## show startup-config

### Syntax

```
show startup-config [json]
```

### Description

Shows the contents of the startup configuration.



**NOTE:** Switches in the `factory-default` configuration do not have a startup configuration to display.

### Command context

Manager (#)

### Parameters

**json**

Display output in JSON format.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Showing the startup-configuration in JSON format:

```
switch# show startup-config json
Startup configuration:
{
 "AAA_Server_Group": {
 "local": {
 "group_name": "local"
 },

```

```
 "none": {
 "group_name": "none"
 }
},
...
```

## show system

### Syntax

```
show system
```

### Description

Shows general status information about the system.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Usage

CPU utilization represents the average utilization across all the CPU cores.

System Contact, System Location, and System Description can be set with the `snmp-server` command.

### Examples

Showing system information:

```
switch(config)# show system
Hostname : switch
System Description : PL.10.xx.xxxxx
System Contact :
System Location :

Vendor : Aruba
Product Name : JL678A 6100 24G 4SFP+ Swch
Chassis Serial Nbr : CN9ZKRD058
Base MAC Address : f860f0-c91160
ArubaOS-CX Version : PL.10.xx.xxxxx

Time Zone : UTC

Up Time : 9 hours, 10 minutes
CPU Util (%) : 3
Memory Usage (%) : 15
```

## show system resource-utilization

### Syntax

```
show system resource-utilization [daemon <DAEMON-NAME> | module <SLOT-ID>]
```

## Description

Shows information about the usage of system resources such as CPU, memory, and open file descriptors.

## Command context

Manager (#)

## Parameters

**daemon** <DAEMON-NAME>

Shows the filtered resource utilization data for the process specified by <DAEMON-NAME> only.



**NOTE:** For a list of daemons that log events, enter `show events -d ?` from a switch prompt in the manager (#) context.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing all system resource utilization data:

```
switch# show system resource-utilization
System Resources:
Processes: 147
CPU usage(%): 12
Memory usage(%): 13
Open FD's: 4128
mmc-type-a: Endurance utilization = 0-10%, Health = normal
mmc-type-b: Endurance utilization = 0-10%, Health = normal
switch#
```

Showing the resource utilization data for the pmd process:

```
switch# show system resource-utilization daemon pmd
Process CPU Usage Memory Usage Open FD's

pmd 2 1 14
```

Showing resource utilization data when system resource utilization polling is disabled:

```
switch# show system resource-utilization
System resource utilization data poll is currently disabled
```

Showing resource utilization data for a line module:

```
switch# show system resource-utilization module 1/1
System Resource utilization for line card module: 1/1
CPU usage(%): 0
Memory usage(%): 35
Open FD's: 512
```

## show tech

## Syntax

`show tech [basic | <FEATURE>] [local-file]`

## Description

Shows detailed information about switch features by automatically running the `show` commands associated with the feature. If no parameters are specified, the `show tech` command shows information about all switch features. Technical support personnel use the output from this command for troubleshooting.

## Command context

Manager (#)

## Parameters

### **basic**

Specifies showing a basic set of information.

### **<FEATURE>**

Specifies the name of a feature. For a list of feature names, enter the `show tech` command, followed by a space, followed by a question mark (?).

### **local-file**

Shows the output of the `show tech` command to a local text file.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

To terminate the output of the `show tech` command, enter **Ctrl+C**.

If the command was not terminated with **Ctrl+C**, at the end of the output, the `show tech` command shows the following:

- The time consumed to execute the command.
- The list of failed `show` commands, if any.

To get a copy of the local text file content created with the `show tech` command that is used with the `local-file` parameter, use the `copy show-tech local-file` command.

## Example

Showing the basic set of system information:

```
switch# show tech basic
=====
Show Tech executed on Wed Sep 6 16:50:37 2017
=====
[Begin] Feature basic
=====

Command : show core-dump all

no core dumps are present

...
=====
[End] Feature basic
=====
```

```
=====
1 show tech command failed
=====
```

Failed command:

```
1. show boot-history
```

```
=====
Show tech took 3.000000 seconds for execution
=====
```

Directing the output of the **show tech basic** command to the local text file:

```
switch# show tech basic local-file
```

```
Show Tech output stored in local-file. Please use 'copy show-tech local-file'
to copy-out this file.
```

## show usb

### Syntax

```
show usb
```

### Description

Shows the USB port configuration and mount settings.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

If USB has not been enabled:

```
switch> show usb
Enabled: No
Mounted: No
```

If USB has been enabled, but no device has been mounted:

```
switch> show usb
Enabled: Yes
Mounted: No
```

If USB has been enabled and a device mounted:

```
switch> show usb
Enabled: Yes
Mounted: Yes
```

## show usb file-system

### Syntax

```
show usb file-system [<PATH>]
```

## Description

Shows directory listings for a mounted USB device. When entered without the <PATH> parameter the top level directory tree is shown.

## Command context

Operator (>) or Manager (#)

## Parameters

### <PATH>

Specifies the file path to show. A leading "/" in the path is optional.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Usage

Adding a leading "/" as the first character of the <PATH> parameter is optional.

Attempting to enter '.' as any part of the <PATH> will generate an invalid path argument error. Only fully-qualified path names are supported.

## Examples

Showing the top level directory tree:

```
switch# show usb file-system
/mnt/usb:
'System Volume Information' dir1'

/mnt/usb/System Volume Information':
IndexerVolumeGuid WPSettings.dat

/mnt/usb/dir1:
dir2 test1

/mnt/usb/dir1/dir2:
test2
```

Showing available path options from the top level:

```
switch# show usb file-system /
total 64
drwxrwxrwx 2 32768 Jan 22 16:27 'System Volume Information'
drwxrwxrwx 3 32768 Mar 5 15:26 dir1
```

Showing the contents of a specific folder:

```
switch# show usb file-system /dir1
total 32
drwxrwxrwx 2 32768 Mar 5 15:26 dir2
-rwxrwxrwx 1 0 Feb 5 18:08 test1

switch# show usb file-system dir1/dir2
total 0
-rwxrwxrwx 1 0 Feb 6 05:35 test2
```

Attempting to enter an invalid character in the path:

```
switch# show usb file-system dir1/../../..
Invalid path argument
```

## show version

### Syntax

```
show version
```

### Description

Shows version information about the network operating system software, service operating system software, and BIOS.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

Showing version information:

```
switch(config)# show version

ArubaOS-CX
(c) Copyright 2017-2020 Hewlett Packard Enterprise Development LP

Version : PL.10.xx.xxxxx
Build Date : 2020-12-06 22:13:46 UTC
Build ID : ArubaOS-CX:PL.10.xx.xxxxx:9e8bf51170a6:202012062115
Build SHA : 9e8bf51170a602370f12e0bde814e8d8f49bf706
Active Image : secondary

Service OS Version : PL.10.xx.xxxxx-internal
BIOS Version : PL.01.0002
```

## system resource-utilization poll-interval

### Syntax

```
system resource-utilization poll-interval <SECONDS>
```

### Description

Configures the polling interval for system resource information collection and recording such as CPU and memory usage.

### Command context

```
config
```

## Parameters

<SECONDS>

Specifies the poll interval in seconds. Range: 10-3600. Default: 10.

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Configuring the system resource utilization poll interval:

```
switch(config)# system resource-utilization poll-interval 20
```

# top cpu

## Syntax

top cpu

## Description

Shows CPU utilization information.

## Command context

Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Showing top CPU information:

```
switch# top cpu
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
Tasks: 76 total, 2 running, 74 sleeping, 0 stopped, 0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total, 2487508 free, 897040 used, 661948 buff/cache
KiB Swap: 0 total, 0 free, 0 used, 2859196 avail Mem

 PID USER PRI NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
...
```

# top memory

## Syntax

top memory

## Description

Shows memory utilization information.



## Command context

Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Showing top memory:

```
switch> top memory
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
Tasks: 76 total, 2 running, 74 sleeping, 0 stopped, 0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total, 2487508 free, 897040 used, 661948 buff/cache
KiB Swap: 0 total, 0 free, 0 used, 2859196 avail Mem

 PID USER PRI NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
...
```

# usb

## Syntax

usb

no usb

## Description

Enables the USB ports on the switch. This setting is persistent across switch reboots and management module failovers. Both active and standby management modules are affected by this setting.

The `no` form of this command disables the USB ports.

## Command context

config

## Authority

Administrators or local user group members with execution rights for this command.

## Example

Enabling USB ports:

```
switch(config)# usb
```

Disabling USB ports when a USB drive is mounted:

```
switch(config)# no usb
```

# usb mount | unmount

## Syntax

usb {mount | unmount}

## Description

Enables or disables the inserted USB drive.

## Command context

Manager (#)

## Parameters

### **mount**

Enables the inserted USB drive.

### **unmount**

Disables the inserted USB drive in preparation for removal.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

A USB drive must be unmounted before removal.

The supported USB file systems are FAT16 and FAT32.

## Examples

Mounting a USB drive in the USB port:

```
switch# usb mount
```

Unmounting a USB drive:

```
switch# usb unmount
```

## terminal-monitor {notify | severity | filter}

### Syntax

```
terminal-monitor {notify <event|debug|all> | severity <level> | filter keyword}
```

```
no terminal-monitor
```

### Description

Enables the terminal monitor feature in the SSH session. It display all debug log or event log or both debug and event log messages. Terminal monitoring can be filtered with the severity options or with the help of keywords. Enabling terminal monitor without options displays both debug and event log with a severity error.

The `no` form of this command disables the terminal monitor configuration.

### Parameters

**notify <event|debug|all>**

Specifies the type of log notification.

- **Event:** Displays the event log messages.
- **Debug:** Displays the debug log messages.
- **All:** Displays both event and debug log messages.

**severity <level>**

Specifies the severity level for the logs. The different severity levels are emergency, critical, error (default), warning, notice, information, alert, and debug (shows all severities).

**filter <keyword>**

Specifies the filter by applying keyword for the logs.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring terminal monitor:

```
switch# terminal-monitor
Terminal-monitor is enabled successfully

switch# terminal-monitor notify all
Terminal-monitor is enabled successfully

switch# terminal-monitor notify event severity info
Terminal-monitor is enabled successfully
```

```
switch# terminal-monitor filter lldp
Terminal-monitor is enabled successfully
```

## show terminal-monitor

### Syntax

```
show terminal-monitor
```

### Description

Shows whether the terminal monitoring is enabled or disabled.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Displaying terminal monitor when enabled:

```
switch# show terminal-monitor
```

```
Terminal-monitor is enabled
```

```

Notify | Severity | Filter

event | debug | lldp

```

Displaying terminal monitor when disabled:

```
switch# show terminal-monitor
```

```
Terminal-monitor is disabled
```

## traceroute

### Syntax

```
traceroute {IPv4-address | hostname} [ip-option loosesourceroute <IPV4-ADDR>]
 [dstport <NUMBER> | maxttl <NUMBER> | minttl <NUMBER> |
 probes <NUMBER> | timeout <TIME>] [vrf {<VRF-NAME> | mgmt}]
```

### Description

Uses traceroute for the specified IPv4 address or hostname with or without optional parameters.

### Command context

Operator (>) or Manager (#)

### Parameters

#### IPv4-address

Specifies the IPv4 address of the device to use traceroute.

#### hostname

Specifies the hostname of the device to use traceroute.

#### ip-option

Specifies the IP option.

#### loosesourceroute <IPV4-ADDR>

Specifies the route for loose source record route. Enter one or more intermediate router IP addresses separated by ',' for loose source routing.

#### dstport <NUMBER>

Specifies the destination port, <1-34000>. Default: 33434

#### maxttl <NUMBER>

Specifies the maximum number of hops to reach the destination, <1-255>. Default: 30

#### minttl <NUMBER>

Specifies the Minimum number of hops to reach the destination, <1-255>. Default: 1

#### probes <NUMBER>

Specifies the number of probes, <1-5>. Default: 3

#### timeout <TIME>

Specifies the traceroute timeout in seconds, <1-60>. Default: 3 seconds

#### vrf <VRF-NAME>

Specifies the virtual routing and forwarding (VRF) to use, <VRF-NAME>.

## mgmt

Specifies use of the management interface.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Usage

Traceroute is a computer network diagnostic tool for displaying the route (path), and measuring transit delays of packets across an Internet Protocol (IP) network. It sends a sequence of User Datagram Protocol (UDP) packets addressed to a destination host. The time-to-live (TTL) value, also known as hop limit, is used in determining the intermediate routers being traversed towards the destination.

## Examples

```
switch# traceroute 10.0.10.1
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 3 sec. timeout, 3 probes
 1 10.0.40.2 0.002ms 0.002ms 0.001ms
 2 10.0.30.1 0.002ms 0.001ms 0.001ms
 3 10.0.10.1 0.001ms 0.002ms 0.002ms

switch# traceroute localhost
traceroute to localhost (127.0.0.1), 1 hops min, 30 hops max, 3 sec. timeout, 3 probes
 1 127.0.0.1 0.018ms 0.006ms 0.003ms

switch# traceroute 10.0.10.1 maxttl 20
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 20 hops max, 3 sec. timeout, 3 probes
 1 10.0.40.2 0.002ms 0.002ms 0.001ms
 2 10.0.30.1 0.002ms 0.001ms 0.001ms
 3 10.0.10.1 0.001ms 0.002ms 0.002ms

switch# traceroute 10.0.10.1 minttl 1
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 3 sec. timeout, 3 probes
 1 10.0.40.2 0.002ms 0.002ms 0.001ms
 2 10.0.30.1 0.002ms 0.001ms 0.001ms
 3 10.0.10.1 0.001ms 0.002ms 0.002ms

switch# traceroute 10.0.10.1 dstport 33434
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 3 sec. timeout, 3 probes
 1 10.0.40.2 0.002ms 0.002ms 0.001ms
 2 10.0.30.1 0.002ms 0.001ms 0.001ms
 3 10.0.10.1 0.001ms 0.002ms 0.002ms

switch# traceroute 10.0.10.1 probes 2
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 3 sec. timeout, 2 probes
 1 10.0.40.2 0.002ms 0.002ms
 2 10.0.30.1 0.002ms 0.001ms
 3 10.0.10.1 0.001ms 0.002ms

switch# traceroute 10.0.10.1 timeout 5
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 5 sec. timeout, 3 probes
 1 10.0.40.2 0.002ms 0.002ms 0.001ms
 2 10.0.30.1 0.002ms 0.001ms 0.001ms
 3 10.0.10.1 0.001ms 0.002ms 0.002ms

switch# traceroute localhost vrf red
traceroute to localhost (127.0.0.1), 1 hops min, 30 hops max, 3 sec. timeout, 3 probes
 1 127.0.0.1 0.003ms 0.002ms 0.001ms

switch# traceroute localhost mgmt
traceroute to localhost (127.0.0.1), 1 hops min, 30 hops max, 3 sec. timeout, 3 probes
 1 127.0.0.1 0.018ms 0.006ms 0.003ms

switch# traceroute 10.0.10.1 maxttl 20 timeout 5 minttl 1 probes 3 dstport 33434
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 20 hops max, 5 sec. timeout, 3 probes
 1 10.0.40.2 0.002ms 0.002ms 0.001ms
 2 10.0.30.1 0.002ms 0.001ms 0.001ms
 3 10.0.10.1 0.001ms 0.002ms 0.002ms

switch# traceroute 10.0.10.1 ip-option loosesourceroute 10.0.40.2
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 30 hops max, 3 sec. timeout, 3 probes
 1 10.0.40.2 0.002ms 0.002ms 0.001ms
 2 10.0.30.1 0.002ms 0.001ms 0.001ms
 3 10.0.10.1 0.001ms 0.002ms 0.002ms

switch# traceroute 10.0.10.1 ip-option loosesourceroute 10.0.40.2 maxttl 20 timeout 5 minttl 1 probes 3 dstport 33434
traceroute to 10.0.10.1 (10.0.10.1) , 1 hops min, 20 hops max, 5 sec. timeout, 3 probes
 1 10.0.40.2 0.002ms 0.002ms 0.001ms
```

|   |           |         |         |         |
|---|-----------|---------|---------|---------|
| 2 | 10.0.30.1 | 0.002ms | 0.001ms | 0.001ms |
| 3 | 10.0.10.1 | 0.001ms | 0.002ms | 0.002ms |

## traceroute6

### Syntax

```
traceroute6 {IPv6-address | hostname} [dstport <NUMBER> | maxttl <NUMBER> |
 probes <NUMBER> | timeout <TIME>] [vrf {<VRF-NAME> | mgmt}]
```

### Description

Uses traceroute for the specified IPv6 address or hostname with or without optional parameters.

### Command context

Operator (>) or Manager (#)

### Parameters

#### IPv6-address

Specifies the IPv6 address of the device to use traceroute.

#### hostname

Specifies the hostname of the device to use traceroute.

#### dstport <NUMBER>

Specifies the destination port, <1-34000>. Default: 33434

#### maxttl <NUMBER>

Specifies the maximum number of hops to reach the destination, <1-255>. Default: 30

#### probes <NUMBER>

Specifies the number of probes, <1-5>. Default: 3

#### timeout <TIME>

Specifies the traceroute timeout in seconds, <1-60>. Default: 3 seconds

#### vrf <VRF-NAME>

Specifies the virtual routing and forwarding (VRF) to use, <VRF-NAME>.

#### mgmt

Specifies use of the management interface.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Usage

Traceroute is a computer network diagnostic tool for displaying the route (path), and measuring transit delays of packets across an Internet Protocol (IP) network. It sends a sequence of User Datagram Protocol (UDP) packets addressed to a destination host. The time-to-live (TTL) value, also known as hop limit, is used in determining the intermediate routers being traversed towards the destination.

## Examples

```
switch# traceroute6 0:0::0:1
traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets
 1 localhost (::1) 0.117 ms 0.032 ms 0.021 ms

switch# traceroute6 localhost
traceroute to localhost (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets
 1 localhost (::1) 0.089 ms 0.03 ms 0.014 ms

switch# traceroute6 0:0::0:1 maxttl 30
traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets
 1 localhost (::1) 0.117 ms 0.032 ms 0.021 ms

switch# traceroute6 0:0::0:1 dsrport 33434
traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets
 1 localhost (::1) 0.117 ms 0.032 ms 0.021 ms

switch# traceroute6 0:0::0:1 probes 2
traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 2 probes, 24 byte packets
 1 localhost (::1) 0.117 ms 0.032 ms

switch# traceroute6 0:0::0:1 timeout 3
traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets
 1 localhost (::1) 0.117 ms 0.032 ms 0.021 ms

switch# traceroute6 localhost vrf red
traceroute to localhost (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets
 1 localhost (::1) 0.077 ms 0.051 ms 0.054 ms

switch# traceroute6 localhost mgmt
traceroute to localhost (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets
 1 localhost (::1) 0.089 ms 0.03 ms 0.014 ms

switch# traceroute6 0:0::0:1 maxttl 30 timeout 3 probes 3 dstport 33434
traceroute to 0:0::0:1 (::1) from ::1, 30 hops max, 3 sec. timeout, 3 probes, 24 byte packets
 1 localhost (::1) 0.117 ms 0.032 ms 0.021 ms
```



## clear udld statistics

### Syntax

```
clear udld statistics [interface <INTERFACE-NAME>]
```

### Description

Clears UDLD statistics for all interfaces or a specific interface.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Clearing all UDLD statistics on all interfaces:

```
switch# clear udld statistics
```

Clearing all UDLD statistics on interface 1/1/1:

```
switch# clear udld statistics interface 1/1/1
```

## show udld

### Syntax

```
show udld [interface <INTERFACE-NAME>]
```

### Description

Displays UDLD information for all interfaces or for a specific interface.

### Command context

Operator (>) or Manager (#)

### Parameters

**interface** <INTERFACE-NAME>

Specifies the name of a logical interface on the switch, which can be:

- An Ethernet interface associated with a physical port. Use the format `member/slot/port` (for example, `1/3/1`).
- UDLD runs only on physical interfaces. LAGs, tunnels, and the like are not supported. However, UDLD can be configured individually on each port of a LAG or trunk group. Configuring UDLD on a trunk group primary port enables UDLD on that port only.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Displaying all UDLD information:

```
switch# show udld
```

Abbreviations:

VTF - Verify-then-forward      FTV - Forward-then-verify  
NOR - RFC 5171 normal          AGG - RFC 5171 aggressive

| Interface | UDLD Config | UDLD State | UDLD Substate | UDLD Link | Mode | Interval |
|-----------|-------------|------------|---------------|-----------|------|----------|
| 1/1/1     | Disabled    | Inactive   | Undetermined  | Unblock   | FTV  | 8000     |
| 1/1/2     | Enabled     | Active     | Bidirectional | Unblock   | FTV  | 7000     |
| 1/1/3     | Enabled     | Active     | Blocked       | Block     | FTV  | 7000     |
| 1/1/4     | Enabled     | Inactive   | Uninitialized | Unblock   | NOR  | 7000     |
| 1/1/5     | Enabled     | Active     | ErrDisabled   | Block     | AGG  | 7000     |
| 1/1/6     | Disabled    | Active     | Detection     | Unblock   | NOR  | 7000     |

| Retries | Tx Pkts | Rx Pkts | Rx Pkts disc. | Rx Pkts drop. | Transitions |
|---------|---------|---------|---------------|---------------|-------------|
| 4       | 4       | 54      | 123           | 123           | 1           |
| 7       | 1234567 | 1548421 | 23214         | 1878981       | 3           |
| 4       | 3       | 77871   | 2157          | 81878         | 1           |
| 5       | 50      | 0       | 0             | 0             | 0           |
| 3       | 150     | 25      | 0             | 2             | 1           |
| 3       | 6       | 54      | 123           | 23            | 1           |

Displaying information for interface `1/1/1`:

```
switch# show udld interface 1/1/1
```

```
Interface 1/1/1
Config: Enabled
State: Active
Substate: Bidirectional
Link: Unblock
Version: Aruba OS
Mode: Forward then verify
Interval: 7000 milliseconds
Retries: 7
Tx: 1234567 packets
Rx: 1548421 packets, 23214 discarded packets, 1878981 dropped packets
Port transitions: 3
```

## Displaying the UDLD enable interfaces information:

```
switch# show udld enabled
```

### Abbreviations:

VTF - Verify-then-forward      FTV - Forward-then-verify  
NOR - RFC 5171 normal          AGG - RFC 5171 aggressive

| Interface | UDLD Config | UDLD State | UDLD Substate | UDLD Link | Mode | Interval | Retries | Tx Pkts | Rx Pkts | Rx Pkts disc. | Rx Pkts drop. | Transitions |
|-----------|-------------|------------|---------------|-----------|------|----------|---------|---------|---------|---------------|---------------|-------------|
| 2         | Enabled     | Active     | Bidirectional | Unblock   | FTV  | 7000     | 7       | 1234567 | 1548421 | 23214         | 1878981       | 3           |
| 3         | Enabled     | Active     | Blocked       | Block     | FTV  | 7000     | 4       | 3       | 77871   | 2157          | 81878         | 1           |
| 4         | Enabled     | Inactive   | Uninitialized | Unblock   | NOR  | 7000     | 5       | 50      | 0       | 0             | 0             | 0           |
| 5         | Enabled     | Active     | ErrDisabled   | Block     | AGG  | 7000     | 3       | 150     | 25      | 0             | 2             | 1           |

## udld

### Syntax

```
udld [disable]
```

```
no udld
```

### Description

Enables UDLD support on a physical interface. UDLD is disabled by default. UDLD is configured on a per-port basis and must be enabled at both ends of the link.

UDLD runs only on physical interfaces. LAGs, tunnels, and the like are not supported. However, UDLD can be configured individually on each port of a LAG or trunk group. Configuring UDLD on a trunk group's primary port enables UDLD on that port only.

The `no` form of this command disables UDLD support and resets all configuration values to their default settings.

### Command context

```
config-if
```

### Parameters

#### ***disable***

Disables UDLD on the interface but retains all UDLD configuration settings.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling UDLD on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# udld
```

Disabling UDLD on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no udld
```

# udld interval

## Syntax

```
udld interval <TIME>
```

```
no udld interval
```

## Description

Sets the packet transmission interval.

The `no` form of this command sets the packet transmission interval to the default value of 7000 ms.

The allowed values vary depending on the operation mode.

The default interval is 7000 ms (7 seconds) for both ArubaOS-Switch and RFC5171 operation modes.

Values must be specified as multiples of 10 ms (7000 ms is allowed but 7005 ms is not a valid setting).

Switch mode can be configured to support under 100ms (8325) unidirectional link detection times.



**NOTE:** Sessions under 100ms total detection time are susceptible to increasing processing load on the system. It is advisable to experiment with values that provide adequate detection times and system/protocol stability. Aruba recommends additional testing prior to configuring these sessions on a production environment.

However, these settings are recommended for specific deployments only, such as using UDLD for Ethernet Ring Protection Switching (ERPS) link-failure detection (ERPS is not supported on the 6100). The minimum detection time appropriate for your environment depends on the specific device family and configuration on which the protocol and system load is running. Aruba recommends additional testing for these configurations. During testing, monitor for unexpected false positive detections (i.e., UDLD records a failure when there was not any) on the interfaces running UDLD. Such false positive failures are an indication that the interval configuration requires tuning and that the system load might not allow such configuration.



**NOTE:** When configuring detection times under 100ms for LAG interfaces, consider adding the interface first to the LAG and then enabling UDLD in the interface, to avoid false positive link failure detections. Adding an interface to a LAG causes momentary control plane traffic interruption for up to 100ms, which UDLD detects as a link failure if the detection time is following the control traffic interruption interval.

## Command context

```
config-if
```

## Parameters

<TIME>

Specifies the packet transmission interval. Range: 200 ms to 90000 ms or 1000 ms to 90000 ms for the 6100 Switch Series (in increments of 10).

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting the packet transmission interval to 1000 ms on interface 1/1/1.

```
switch(config)# interface 1/1/1
switch(config-if)# udld interval 1000
```

Setting the packet transmission interval on interface 1/1/1 to the default value.

```
switch(config)# interface 1/1/1
switch(config-if)# no udld interval
```

Trying to set the packet interval to 1055 ms on interface 1 is rejected because the interval must be specified as a multiple of 10:

```
switch(config)# interface 1
switch(config-if)# udld interval 1055
Invalid interval. The interval value must be between 20ms and 90000ms and should be
specified as a multiple of 10, for example: 20, 100, 3000 or 90000.
```

Trying to set the packet interval to less than 7000 ms on interface 1 is rejected if using the RFC5171 mode.

```
switch(config)# interface 1
switch(config-if)# udld mode rfc5171 normal
switch(config-if)# udld interval 1000
Invalid interval. The interval must be equal or greater than 7000ms.
```

## udld mode

### Syntax

```
udld mode aruba-os {verify-then-forward | forward-then-verify}
```

```
udld mode rfc5171 <RFC5171-MODE>
```

```
no udld mode
```

### Description

Sets the operating mode.

The **no** form of this command sets the operating mode to the default value of **aruba-os** and **forward-then-verify**.

### Command context

```
config-if
```

### Parameters

**aruba-os {verify-then-forward | forward-then-verify}**

Selects the ArubaOS mode to use. Use this mode when interconnecting with HPE PVOS/Brocade/Foundry switches.

#### **verify-then-forward**

In this mode:

- Interfaces start as unblocked.
- Once an interface is determined to be bidirectional, it is blocked if the retry limit is reached without receiving any UDLD packets.

- Interfaces automatically unblock if a UDLD packet is received.
- On failover, the UDLD state does not change if the (interval \* retries) time is around 6 seconds.

### **forward-then-verify**

In this mode:

- Interfaces start as unblocked.
- Interfaces transition to the unblocked state when receiving UDLD packets.
- Once an interface is determined to be bidirectional, it is blocked if the retry limit is reached without receiving any UDLD packets.
- Interfaces automatically unblock if a UDLD packet is received.

### **rfc5171** <RFC5171-MODE>

Selects the RFC5171 mode to use. Use this mode when interconnecting with third-party switches.

#### **normal**

In this mode:

- Interfaces start as unblocked.
- Interfaces do not block when the retry limit is reached without receiving any UDLD packets (plus 8 extra packets sent to the peer). Instead, an event is generated.
- Interfaces automatically unblock if a UDLD packet is received.

#### **aggressive**

In this mode:

- Interfaces start as unblocked.
- Once an interface is determined to be bidirectional, an interface will block when the retry limit is reached without receiving any UDLD packets (plus 8 extra packets sent to the peer).
- Interfaces implement a limited/reduced errDisabled recovery mechanism. When the interface's state goes to errDisabled, a maximum of 3 attempts (5 minutes apart) are triggered to try and bring up the interface in case the remote endpoint is still sending UDLD packets. After these 3 retries, the interface will remain blocked even if UDLD packets are received. The only way to unblock the interface when this occurs is to disable (and optionally re-enable) UDLD on the interface. The retry limit is reset once the interface becomes unblocked.

### **Authority**

Administrators or local user group members with execution rights for this command.

### **Examples**

Setting the operating mode to **aruba-os** and **forward-then-verify** on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# udld mode aruba-os forward-then-verify
```

Setting the operating mode to **rfc5171** and **aggressive** on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# udld mode rfc5171 aggressive
```

Setting the operating mode on interface 1/1/1 to the default value:

```
switch(config)# interface 1/1/1
switch(config-if)# no udld mode
```

## udld retries

### Syntax

```
udld retries <COUNT>
```

```
no retries
```

### Description

Sets the UDLD retry count.

The `no` form of this command sets the retry count to the default value.

### Command context

```
config-if
```

### Parameters

<COUNT>

Specifies the UDLD retry count. Range: 3 to 10. Default: 4.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the UDLD retry count to 5 on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# udld retries 5
```

Setting the UDLD retry count on interface 1/1/1 to the default value:

```
switch(config)# interface 1/1/1
switch(config-if)# no udld retries
```

## ip udp-bcast-forward

### Syntax

```
ip udp-bcast-forward
```

```
no ip udp-bcast-forward
```

### Description

Enables UDP broadcast forwarding.

The `no` form of this command disables UDP broadcast forwarding.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling UDP broadcast forwarding:

```
switch(config)# ip udp-bcast-forward
```

Disabling UDP broadcast forwarding:

```
switch(config)# no ip udp-bcast-forward
```

## ip forward-protocol udp

### Syntax

```
ip forward-protocol udp <IPV4-ADDR> {<PORT-NUM> | <PROTOCOL>}
```

```
no ip forward-protocol udp
```

### Description

Defines the UDP server to which the interface forwards ingress UDP broadcast packets received on a specific UDP port. A maximum of 8 UDP broadcast servers can be configured per interface.

The `no` form of this command removes traffic forwarding for the specified server and port/protocol.

### Command context

```
config-if
```



## Parameters

### <IPV4-ADDR>

Specifies the UDP server IP address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255..

### <PORT-NUM>

Specifies the UDP port number for which traffic is forwarded.

### <PROTOCOL>

Specifies the protocol name for which traffic is forwarded. Supported protocols and their port numbers are:

- dns (53): Domain Name Service
- ntp (123): Network Time Protocol
- netbios-ns (137): NetBIOS Name Service
- netbios-dgm (138): NetBIOS Datagram Service
- radius (1812): Remote Authentication Dial-In User Service
- radius-old (1645): Remote Authentication Dial-In User Service
- rip (520): Routing Information Protocol
- snmp (161): Simple Network Management Protocol
- snmp-trap (162): Simple Network Management Protocol
- tftp (69): Trivial File Transfer Protocol
- timep (37): Time Protocol

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Forwarding DNS traffic to server 192.168.1.10 on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# ip udp-bcast-forward protocol udp 192.168.1.10 dns
```

Forwarding DNS traffic (port 53) to server 192.168.1.10 on interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# ip udp-bcast-forward protocol udp 192.168.1.10 53
```

## show ip forward-protocol udp

### Syntax

```
show ip forward-protocol udp [<INTERFACE-NAME>] [vsx-peer]
```

### Description

Shows the configured UDP forwarding settings for all interfaces or a specific interface.

## Command context

Manager (#)

## Parameters

**<INTERFACE-NAME>**

Specifies the name of an interface. Format `member/slot/port`.

**[vsx-peer]**

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing the configured UDP forwarding settings for all interfaces:

```
switch(config)# ip udp-bcast-forward
switch(config)# interface 1/1/1
switch(config-if)# ip forward-protocol udp 1.1.1.1 53
switch(config-if)# ip forward-protocol udp 8.1.1.1 ntp
switch(config-if)# ip forward-protocol udp 8.1.1.1 137
switch(config-if)# ip forward-protocol udp 4.4.4.4 138
switch(config-if)# ip forward-protocol udp 4.4.4.4 1812
switch(config-if)# ip forward-protocol udp 2.2.2.2 1645

switch(config)# interface 1/1/2
switch(config-if)# ip forward-protocol udp 2.2.2.2 rip
switch(config-if)# ip forward-protocol udp 2.2.2.2 161
switch(config-if)# ip forward-protocol udp 2.2.2.2 162
switch(config-if)# ip forward-protocol udp 2.2.2.2 69
switch(config-if)# ip forward-protocol udp 2.2.2.2 37
```

Showing the configured UDP forwarding settings for a specific interface:

```
switch# show ip forward-protocol udp
```

UDP Broadcast Forwarder : enabled

Interface: 1/1/1

| IP Forward Address | UDP Port |
|--------------------|----------|
| 2.2.2.2            | 1645     |
| 4.4.4.4            | 138      |
| 4.4.4.4            | 1812     |
| 1.1.1.1            | 53       |
| 8.1.1.1            | 123      |
| 8.1.1.1            | 137      |

Interface: 1/1/2

| IP Forward Address | UDP Port |
|--------------------|----------|
| 2.2.2.2            | 37       |
| 2.2.2.2            | 69       |
| 2.2.2.2            | 520      |
| 2.2.2.2            | 161      |
| 2.2.2.2            | 162      |

```
switch#show ip forward-protocol udp interface 1/1/1
```

UDP Broadcast Forwarder : enabled

Interface: 1/1/1

| IP Forward Address | UDP Port |
|--------------------|----------|
| 2.2.2.2            | 1645     |
| 4.4.4.4            | 138      |
| 4.4.4.4            | 1812     |
| 1.1.1.1            | 53       |
| 8.1.1.1            | 123      |
| 8.1.1.1            | 137      |

## user

### Syntax

```
user <USERNAME> group {administrators | operators | auditors | <USER-GROUP>}
 password [ciphertext <CIPHERTEXT-PASSWORD> | plaintext <PLAINTEXT-PASSWORD>]
```

```
no user <USERNAME>
```

### Description

Creates a user and adds the user to one of the user groups. Users are given the privileges of their group. For the three built-in user groups (`administrators`, `operators`, `auditors`), the privileges are fixed. For user-defined local user groups, the privileges are defined by the CLI command authorization rules of the group.

When entered without either optional `ciphertext` or `plaintext` parameters, the cleartext password is prompted for twice, with the characters entered masked with "\*" symbols.

The `no` form of this command removes a user account from the switch. The administrator cannot delete the user account from which they are logged in. The `admin` user cannot be deleted.

### Command context

`config`

### Parameters

**<USERNAME>**

Specifies the user name. Requirements:

- Must start with a lowercase letter.
- Can contain numbers and lowercase letters.
- Can include only these three special characters: hyphens ( - ), dots ( . ), and underscores ( \_ ).
- Can have a maximum of 32 characters.
- Cannot be empty.
- Cannot contain uppercase letters.
- Cannot be: `admin`, `root`, or `remote_user`.
- Cannot be Linux reserved names such as:

```
daemon, bin, sys, sync, proxy, www-data, backup, list, irc, gnats, nobody, systemd-bus-
proxy, sshd, messagebus, rpc, systemd-journal-gateway, systemd-journal-remote,
systemd-journal-upload, systemd-timesync, systemd-coredump, systemd-resolve,
rpcuser, vagrant, opsd, rdanet, _lldpd, rdaadmin, rdaweb, docker_container, tss.
```

## group

Selects the local user group to which the new user will be assigned.

**administrators | operators | auditors**

Selects one of three built-in local user groups.

**<USER-GROUP>**

Specifies an existing user-defined local user group.

## ciphertext <CIPHERTEXT-PASSWORD>

Specifies a ciphertext password. No password prompts are provided and the ciphertext password is validated before the configuration is applied for the user. The variable <CIPHERTEXT-PASSWORD> is Base64 and is typically copied from another switch using the `show running-config` command output and then pasted into this command.



**NOTE:** The administrator cannot construct ciphertext passwords themselves. The ciphertext is only created by an AOS-CX switch. The ciphertext is created by setting a password for a user with the `user` command. The ciphertext is available for copying from the `show running-config` output and pasting into the configuration on any other AOS-CX switch. The target switch must have the same export password (default or otherwise) as the source switch.

## plaintext <PLAINTEXT-PASSWORD>

Specifies the password without prompting. The password is visible as cleartext when entered but is encrypted thereafter. Command history does show the password as cleartext.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

- Up to 63 local users can be added, for a total of 64 users including the default user `admin`. A user can belong to only one group.
- The switch ships with the `admin` user account and three built-in local user groups: `administrators`, `operators`, and `auditors`. The `admin` account belongs to the `administrators` group. The Service OS also includes the administrator user `admin`. The two `admin` users are entirely distinct.
- When a local user account is removed, the user loses all active login/SSH sessions. Any calls on the existing REST session with that local user account fail with a permissions issue as soon as the user is deleted. Soon afterwards, the existing REST sessions with the deleted local user account become invalidated. If a user is viewing the GUI while their account is deleted, the user is redirected to the login page within 60 seconds. The home directory associated with the user is also removed from the switch.
- Cleartext passwords (whether entered with prompting or entered directly) must:
  - Contain only ASCII characters from hexadecimal 21 to hexadecimal 7E [`\x21-\x7E`] (decimal 33 to 126). Spaces are not allowed. When the password is entered directly without prompting, the "?" symbol (hexadecimal 3F [`\x3F`] (decimal 63)) is not permitted.
  - Contain at most 32 characters.
  - Contain at least the number of characters configured (optionally) for `minimum-password-length`.



**CAUTION:** Although empty passwords are supported, it is recommended that you use strong passwords for all production switches.



---

**NOTE:** Only an administrator can change the password of a user assigned to the operators role.

---

## Examples

Creating local user `jamie` in the `administrators` group with a prompted password:

```
switch(config)# user jamie group administrators password
Adding user jamie
Enter password:*****
Confirm password:*****
```

Creating user `chris` in the existing user-defined local user group `admuser2` with a cleartext password, using direct entry without prompting:

```
switch(config)# user chris group admuser2 password plaintext passWORDxJ|989
```

Creating user `alex` in the `operators` group with a ciphertext password (the ciphertext shown is a placeholder that must be replaced with actual ciphertext):

```
switch(config)# user alex group operators password ciphertext NDcDI2...8igJfA=
```

Removing user `jamie`:

```
switch(config)# no user jamie
User jamie's home directory and active sessions will be deleted.
Do you want to continue [y/n]?y
```

## user-group

### Syntax

```
user-group <GROUP-NAME>
```

```
no user-group <GROUP-NAME>
```

### Description

If `<GROUP-NAME>` does not exist, this command creates a local user group and then enters its context. If `<GROUP-NAME>` exists, this command enters the context for the specified `<GROUP-NAME>`. Within the `<GROUP-NAME>` context, several subcommands are available for working with rules that specify what CLI commands are permitted or denied for all members of the local group.

In addition to the three built-in user groups `administrators`, `operators`, and `auditors`, up to 29 user-defined local user groups can be defined. All users can be members of only one of the up to 32 groups.

The `no` form of this command deletes the specified user group. All members of the deleted group lose all command authorization privilege.



---

**CAUTION:** Do not causally delete user-defined local user groups without understanding the implications. Although user-defined local user groups can be deleted with the respective members losing all privileges, the three built-in groups `administrators`, `operators`, and `auditors` are always available and their privileges are unchangeable.

---

### Command context

```
config
```

## Authority

Administrators or local user group members with execution rights for this command.

## Subcommands

These subcommands are available within the user-defined local user group context (shown in the switch prompt as `config-usr-grp-<GROUP-NAME>`).

`[<SEQ-NUM>] {permit | deny} cli command "<REGEX>"`

`no <SEQ-NUM>`

Defines a CLI command privilege `permit` or `deny` rule. There is an implicit "`deny .*`" rule at the end of every user-defined group rule list. Members of a user-defined group without any `permit` rules have no CLI command privileges.

The `no` form of this subcommand deletes the specified (by sequence number) rule from the group.



**NOTE:** Rule evaluation proceeds from lowest to highest sequence number until the first successful match, resulting in either CLI command permission or denial. Rule evaluation ceases upon first match. Therefore, rules for related CLI commands must be defined in most restrictive to least restrictive order.

### <SEQ-NUM>

Specifies the CLI command rule sequence number. When omitted, a sequence number that is 10 greater than the highest existing sequence number is auto-assigned. When no rules exist, the first auto-assigned sequence number is 10.

### {permit | deny}

Sets the rule type as either `permit` or `deny`. Rule order is important. For example, these two related rules together authorize all `show` commands except for the `show aaa` commands.

```
switch(config-usr-grp-admuser2) #10 deny cli command "show aaa .*"
switch(config-usr-grp-admuser2) #20 permit cli command "show .*"

```

To achieve the wanted effect in this example, the `deny` rule must precede the `permit` rule. These two rules together achieve the following:

- All `show aaa` commands match on rule 10, triggering command denial, and the immediate cessation of further rule evaluation. Matching on rule 20 is never attempted.
- All other `show` commands (excluding `show aaa` commands) match on rule 20 and are therefore permitted.

### <REGEX>

Specifies the CLI command matching criteria of the rule. The criteria can be expressed as "`.*`" which matches all commands. Otherwise, the criteria is expressed as a POSIX-compliant regular expression (regex) string starting with an exact match command token (for example `show`) followed by a regex representing command arguments. The first word must be a string that contains only alphanumeric or hyphen characters.

For example, to allow all commands starting with the word `interface`, the regex must be "`interface .*`" or just "`interface`". Using "`interface.*`" (without the space) is not supported. For example, "`show .*`" matches every `show` command. Consult the Extended regular expression information available at: [https://pubs.opengroup.org/onlinepubs/9699919799/basedefs/V1\\_chap09.html#tag\\_09\\_04](https://pubs.opengroup.org/onlinepubs/9699919799/basedefs/V1_chap09.html#tag_09_04).

| Sample matching criteria | Sample matched CLI command or specifier | Matches                            |
|--------------------------|-----------------------------------------|------------------------------------|
| show .*                  | show accounting log                     | All show commands                  |
| bgp .*                   | bgp router-id 1.1.1.1                   | All bgp commands                   |
| interface .*             | interface 1/1/1                         | All interface specifiers           |
| vlan (3 4)               | vlan 3                                  | VLAN 3 or 4                        |
| vlan [1-9]               | vlan 5                                  | A single VLAN in the range 1 to 9  |
| vlan ([1-9] 1[0-9])      | vlan 19                                 | A single VLAN in the range 1 to 19 |

**[<SEQ-NUM>] comment <TEXT-STRING>**

**no <SEQ-NUM> comment**

Adds a comment to an existing rule. The no form of this subcommand removes an existing comment.

```
switch(config-usr-grp-admuser2)# 10 comment Deny all show aaa commands.
switch(config-usr-grp-admuser2)# 20 comment Permit all other show commands.
switch(config-usr-grp-admuser2)#
switch(config-usr-grp-admuser2)# show running-config current-context
user-group admuser2
 10 comment Deny all show aaa commands.
 10 deny cli command "show aaa .*"
 20 comment Permit all other show commands.
 20 permit cli command "show .*"

```

**include <GROUP-NAME>**

**[no] include <GROUP-NAME>**

Include all rules from the specified user-defined <GROUP-NAME>. Only one group can be included in the definition of another group. The content of the included group is effectively placed at the top of the rules list in the current group. If the specified <GROUP-NAME> does not exist, it is created.

The no form of this subcommand removes the specified included group from the current group. The specified included group must exist and must be included in the current group or else an error message is shown.

The name of the included group is shown at the top of the show user-group command for the group with the include.

In this example, group admuser1 is included in group admuser2. So the admuser1 rules are evaluated first and then the rules in the admuser2 group are only evaluated if no CLI command match occurs for the rules in group admuser1.

```
switch(config-usr-grp-admuser2)# include admuser1
switch(config-usr-grp-admuser2)# show user-group admuser2
User Group Summary
=====
Name : admuser2
Type : configuration
Included Group : admuser1
Number of Rules : 2
User Group Rules
=====
SEQUENCE NUM ACTION COMMAND COMMENT

10 deny show aaa .* Deny all show aaa commands.
20 permit show .* Permit all other show commands.

```



**resequence** [**<STARTING-SEQ-NUM>** **<INCREMENT>**]

Resequences the CLI command authorization rules. When entered without the optional parameters the rules are resequenced with a **<STARTING-SEQ-NUM>** of 10 and an **<INCREMENT>** of 10.

**<STARTING-SEQ-NUM>**

Specifies the starting sequence number.

**<INCREMENT>**

Specifies the sequence number increment.

Resequencing the rules to start at 100 with an increment of 20:

```
switch(config-usr-grp-admuser2)# resequence 100 20
switch(config-usr-grp-admuser2)# show running-config current-context
user-group admuser2
 100 comment Deny all show aaa commands.
 100 deny cli command "show aaa .*"
 120 comment Permit all other show commands.
 120 permit cli command "show .*"

```

Resequencing the rules to the default of starting at 10 with an increment of 10:

```
switch(config-usr-grp-admuser2)# resequence
switch(config-usr-grp-admuser2)# show running-config current-context
user-group admuser2
 10 comment Deny all show aaa commands.
 10 deny cli command "show aaa .*"
 20 comment Permit all other show commands.
 20 permit cli command "show .*"

```

**show running-config current-context**

Shows all the commands used to configure the rules in the current group context.

```
switch(config-usr-grp-admuser2)# show running-config current-context
user-group admuser2
 10 comment Deny all show aaa commands.
 10 deny cli command "show aaa .*"
 20 comment Permit all other show commands.
 20 permit cli command "show .*"

```

**list**

List the subcommands available within the user-defined group context.

**exit**

Exits the user-defined group context.

**end**

Exits the user-defined group context and then the config context.

## user password

### Syntax

**user** **<USERNAME>** **password** [**ciphertext** **<CIPHERTEXT-PASSWORD>** | **plaintext** **<PLAINTEXT-PASSWORD>**]

### Description

Changes a password for an account or enables the password for the admin account. When entered without either optional **ciphertext** or **plaintext** parameters, the cleartext password is prompted for twice, with the characters entered masked with "\*" symbols.

## Command context

config

## Parameters

**<USERNAME>**

Specifies the corresponding user name for the password you want to change.

**ciphertext <CIPHERTEXT-PASSWORD>**

Specifies a ciphertext password. No password prompts are provided and the ciphertext password is validated before the configuration is applied for the user. The variable **<CIPHERTEXT-PASSWORD>** is Base64 and is typically copied from another switch using the `show running-config` command output and then pasted into this command.



**NOTE:** The administrator cannot construct ciphertext passwords themselves. The ciphertext is only created by an AOS-CX switch. The ciphertext is created by setting a password for a user with the `user` command. The ciphertext is available for copying from the `show running-config` output and pasting into the configuration on any other AOS-CX switch. The target switch must have the same export password (default or otherwise) as the source switch.

**plaintext <PLAINTEXT-PASSWORD>**

Specifies the password without prompting. The password is visible as cleartext when entered but is encrypted thereafter. Command history does show the password as cleartext.

## Authority

Administrators or local user group members with execution rights for this command.

## Usage

The admin account is available on the switch without a password by default.

Cleartext passwords (whether entered with prompting or entered directly) must:

- Contain only ASCII characters from hexadecimal 21 to hexadecimal 7E [`\x21-\x7E`] (decimal 33 to 126). Spaces are not allowed. When the password is entered directly without prompting, the "?" symbol (hexadecimal 3F [`\x3F`] (decimal 63)) is not permitted.
- Contain at most 32 characters.
- Contain at least the number of characters configured (optionally) for `minimum-password-length`.



**CAUTION:** Although empty passwords are supported, it is recommended that you use strong passwords for all production switches.



**NOTE:** Only an administrator can change the password of a user assigned to the `operators` role.

## Examples

Enabling (or changing) a cleartext password for `admin`:

```
switch(config)# user admin password
Changing password for user admin
Enter password:*****
Confirm password:*****
```

Changing the cleartext password for user `chris`, using direct entry without prompting:

```
switch(config)# user chris password plaintext PASSwordZQ#@67
```

Changing the ciphertext password for user `alex` (the ciphertext shown is a placeholder that must be replaced with actual ciphertext):

```
switch(config)# user alex password ciphertext XqYJ36...W83D4Y=
```

## service export-password

### Syntax

```
service export-password
```

```
no service export-password
```

### Description

Configures a nondefault export password. The export password is used to transform critical security parameters (such as password hashes) into ciphertext suitable for exporting and showing by commands such as `show running-config`. This transformation enables safe switch configuration import and export.

The `no` form of this command reverts the export password to its factory default.



**NOTE:** All factory-default switches have identical default export passwords. For security, it is recommended that you set the same nondefault export password on every switch in a group that will exchange configuration information. Only switches with identical export passwords can exchange configuration information.

### Command context

```
config
```

### Authority

Administrators or local user group members with execution rights for this command.

### Usage

Prompts you twice for the new export password.

The export password must:

- Contain only ASCII characters from hexadecimal 21 to hexadecimal 7E [`\x21-\x7E`] (decimal 33 to 126). Spaces are not allowed.
- Contain at most 32 characters.
- Not be blank.

### Examples

Configuring a new export password:

```
switch(config)# service export-password
Enter password:*****
Confirm password:*****
```

Reverting the export password to its factory default:

```
switch(config)# no service export-password
```

## show user-group

### Syntax

```
show user-group [<GROUP-NAME>]
```

### Description

Shows user group information for the built-in groups plus any user-defined local user groups. When entered without *<GROUP-NAME>*, summary information is shown for all groups.

### Command context

Manager (#)

### Parameters

**<GROUP-NAME>**

Narrows the `show` command output to that of the specified group, and for local user groups, adds the User Group Rules list.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Show the list of all user groups, including built-in groups and local user groups.

```
switch# show user-group
```

| GROUP NAME     | GROUP TYPE    | INCLUDED GROUP | NUMBER OF RULES |
|----------------|---------------|----------------|-----------------|
| administrators | built-in      | n/a            | n/a             |
| admuser1       | configuration | --             | 5               |
| admuser2       | configuration | admuser1       | 2               |
| auditors       | built-in      | n/a            | n/a             |
| operators      | built-in      | n/a            | n/a             |

Show detailed information for local user group `admuser2`.

```
switch(config-usr-grp-admuser2)# show user-group admuser2
```

User Group Summary

```
=====
Name : admuser2
Type : configuration
Included Group : admuser1
Number of Rules : 2
User Group Rules
=====
```

| SEQUENCE NUM | ACTION | COMMAND     | COMMENT                         |
|--------------|--------|-------------|---------------------------------|
| 10           | deny   | show aaa .* | Deny all show aaa commands.     |
| 20           | permit | show .*     | Permit all other show commands. |

# show user information

## Syntax

```
show user information
```

## Description

Shows the following information for the logged-in user:

- User name.
- User authentication type: local, RADIUS, or TACACS+.
- User group: administrators, operators, or *<GROUP-NAME>*.
- User privilege level: For the built-in user groups and RADIUS or TACACS+, the role privilege level value is shown. For user-defined user groups, N/A is shown.

## Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing information for the admin user:

```
switch# show user information
Username : admin
Authentication type : Local
User group : administrators
User privilege level : 15
```

Showing information for a member of the user-defined local user group admuser2:

```
switch# show user information
Username : admin2-b
Authentication type : Local
User group : admuser2
User privilege level : N/A
```

Showing information for a member of operators:

```
switch# show user information
Username : operator
Authentication type : Local
User group : operators
User privilege level : 1
```

Showing information for remote RADIUS user rad\_user1 mapped to local user group administrators:

```
switch# show user information
Username : rad_user1
Authentication type : RADIUS
User group : administrators
User privilege level : 15
```

Showing information for remote RADIUS user `rad_user2` mapped to local user group `operators`:

```
switch# show user information
Username : rad_user2
Authentication type : RADIUS
User group : operators
User privilege level : 1
```

Showing information for remote TACACS+ `tac_user1` logged in with `priv-lvl 15` (mapped to user group `administrators`):

```
switch# show user information
Username : tac_user1
Authentication type : TACACS+
User group : administrators
User privilege level : 15
```

## show user-list

### Syntax

```
show user-list
```

### Description

Shows all configured users and their corresponding group names.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Show the user list from a switch with only the admin user defined.

```
switch# show user-list

USER GROUP

admin administrators
```

Show the user list after adding a user to the operators built-in group.

```
switch# show user-list

USER GROUP

admin administrators
oper1 operators
```

Show the user list after adding a user to the auditors built-in group.

```
switch# show user-list

USER GROUP

admin administrators
```

|        |           |
|--------|-----------|
| oper1  | operators |
| audit1 | auditors  |

Show the user list after adding a total of three users to two user-defined user groups.

```
switch# show user-list
```

| USER     | GROUP          |
|----------|----------------|
| admin    | administrators |
| oper1    | operators      |
| audit1   | auditors       |
| adm1a    | admuser1       |
| admin2-a | admuser2       |
| admin2-b | admuser2       |

## backup-controller ip

### Syntax

```
backup-controller ip <IP-ADDR>
```

```
no backup-controller ip <IP-ADDR>
```

### Description

Specifies the IP address of the backup controller for the zone.

The `no` form of this command deletes the IP address of the backup controller.

### Command context

```
config-ubt-<ZONE-NAME>
```

### Parameters

**<IP-ADDR>**

Specifies the IP address of the backup controller.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Specifying the backup controller ip address for `zone1`:

```
switch(config)# ubt zone zone1
switch(config-ubt-zone1)# backup-controller ip 10.116.51.11
```

Delete the configured backup controller IP address:

```
switch(config)# ubt zone zone1
switch(config-ubt-zone1)# no backup-controller ip 10.116.51.11
```

## enable

### Syntax

```
enable
```

```
no enable
```

### Description

Enables UBT.

The `no` form of this command disables UBT.



## Command context

config-ubt-<ZONE-NAME>

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling UBT for zone zone1:

```
switch(config)# ubt zone zone1
switch(config-ubt-zone1)# enable
```

Disabling UBT for zone1:

```
switch(config)# ubt zone zone1
switch(config-ubt-zone1)# no enable
```

# ip source-interface

## Syntax

```
ip source-interface {all | ubt} {interface <IFNAME> | <IPV4-ADDR>} [vrf <VRF-NAME>]
no ip source-interface {all | ubt} {interface <IFNAME> | <IPV4-ADDR>} [vrf <VRF-NAME>]
```

## Description

Sets a single source IP address for UBT. This ensures that all traffic sent by UBT has the same source IP address, regardless of how it egresses the switch.

This command provides two ways to set the source IP addresses: either by specifying a static IP address, or by using the address assigned to a switch interface. If you define both options, then the static IP address takes precedence.

The **no** form of this command deletes the single source IP address for UBT.

## Command contexts

config

## Parameters

**all**

When used no other parameters are required.

**interface <IFNAME>**

Specifies the name of the interface from which UBT obtains its source IP address. The interface must have a valid IP address assigned to it. If the interface has both a primary and secondary IP address, the primary IP address is used.

**<IPV4-ADDR>**

Specifies the source IP address to use for UBT. The IP address must be defined on the switch, and it must exist on the specified VRF, Default: default. Specify the address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

**vrf <VRF-NAME>**

Specifies the name of the VRF from which UBT sets its source IP address.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Setting interface 1/1/7 as the source address for UBT for VRF default:

```
switch(config)# ip source-interface ubt interface 1/1/7 vrf default
```

Deleting the configured source interface 1/1/7 as the source address for UBT for VRF default:

```
switch(config)# no ip source-interface ubt interface 1/1/7 vrf default
```

Specifying the static IP address 1.1.1.1 as the source address for UBT for VRF default:

```
switch(config)# ip source-interface ubt 1.1.1.1 vrf default
```

Deleting the configured ip address as the source address for UBT for VRF default:

```
switch(config)# no ip source-interface ubt 1.1.1.1 vrf default
```

# papi-security-key

## Syntax

```
papi-security-key {ciphertext <PASSKEY> | plaintext <PASSKEY>}
```

```
no papi-security-key
```

## Description

Specifies the shared key used to encrypt UBT PAPI messages exchanged between the switch and the controller cluster for the zone.

The `no` form of this command deletes the shared key.

## Command context

```
config-ubt-<ZONE-NAME>
```

## Parameters

**ciphertext** <PASSKEY>

Specifies an encrypted passkey. Length: 10 to 64 characters.

**plaintext** <PASSKEY>

Specifies a plaintext passkey. Length: 10 to 64 characters.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Specifying the PAPI security key `plaintext testingtesting` for UBT zone `zone1`:

```
switch(config)# ubt zone zone1
switch(config-ubt-zone1)# papi-security-key plaintext testingtesting
```

Removing the PAPI security key:

```
switch(config)# ubt zone zone1
switch(config-ubt-zone1)# no papi-security-key
```

## primary-controller ip

### Syntax

```
primary-controller ip <IP-ADDR>
```

```
no primary-controller ip <IP-ADDR>
```

### Description

Specifies the IP address of the primary controller IP address for the zone.

The `no` form of this command deletes the IP address of the primary controller.

### Command context

```
config-ubt-<ZONE-NAME>
```

### Parameters

**<IP-ADDR>**

Specifies the IP address of the primary controller.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Specify the primary controller IP address for `zone1`:

```
switch(config)# ubt zone zone1
switch(config-ubt-zone1)# primary-controller ip 10.116.51.10
```

Delete the configured primary controller IP address:

```
switch(config)# ubt zone zone1
switch(config-ubt-zone1)# no primary-controller ip 10.116.51.10
```

## sac-heartbeat-interval

### Syntax

```
sac-heartbeat-interval <TIME>
```

```
no sac-heartbeat-interval <TIME>
```

### Description

Specifies the SAC heartbeat refresh time interval in seconds.

The `no` form of this command sets the heartbeat interval to the default value.

### Command context

```
config-ubt-<ZONE-NAME>
```

## Parameters

<TIME>

Specifies the SAC heartbeat refresh time interval in seconds. Range: 1 to 8. Default: 1.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Specifying a heartbeat refresh interval of 1 for UBT zone1:

```
switch(config)# ubt zone zone1
switch(config-ubt-zone1)# sac-heartbeat-interval 1
```

Deleting the configured heartbeat refresh interval:

```
switch(config)# ubt zone zone1
switch(config-ubt-zone1)# no sac-heartbeat-interval
```

## show ip source-interface ubt

### Syntax

```
show ip source-interface ubt
```

### Description

Displays single source IP address configuration information for UBT.

### Command context

Operator (>) or Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Showing single source IP address configuration information:

```
switch# sh ip source-interface ubt
```

| Source-interface Configuration Information |               |               |         |
|--------------------------------------------|---------------|---------------|---------|
| Protocol                                   | Src-Interface | Src-IP        | VRF     |
| ubt                                        | vlan15        | 15.212.219.63 | default |

## show capacities ubt

### Syntax

```
show capacities ubt
```

### Description

Shows the maximum number of UBT clients and zones which can be configured in the system.

## Command context

Operator (>) or Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Showing maximum number of UBT clients and zones which can be configured:

```
switch# show capacities ubt

System Capacities: Filter UBT
Capacities
Name
 Value

Maximum number of UBT clients in a system 1017
Maximum number of UBT zones per VRF
1 8
```

## show ubt

### Syntax

```
show ubt [brief]
```

```
show ubt zone <ZONE-NAME> [brief]
```

### Description

Shows global configuration information for UBT in addition to detailed or brief information for a specific UBT zone.

### Command context

Operator (>) or Manager (#)

### Parameters

**zone <ZONE-NAME>**

Specifies the name of a zone. Length: 1 to 64 characters.

**brief**

Displays brief information.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Showing global UBT configuration information where local-VLAN mode has been configured:

```
switch# show ubt

Zone Name : zone1
UBT Mode : local-vlan
Primary Controller : 10.116.51.10
Backup Controller : 10.116.51.11
SAC HeartBeat Interval : 1
UAC KeepAlive Interval : 60
Reserved VLAN Identifier : 4094
VRF Name : default
Admin State : ENABLED
PAPI Security Key : AQBapdxySvGPvdTlkYn1/naKX4O3jKHrm28xLYfO6mLOK499BwAAAHdJp/bL4FE=
```

Showing global UBT configuration information where VLAN-extend mode has been configured:

```
switch# show ubt

Zone Name : zone1
UBT Mode : vlan-extend
Primary Controller : 10.116.51.10
Backup Controller : 10.116.51.11
SAC HeartBeat Interval : 1
UAC KeepAlive Interval : 60
Reserved VLAN Identifier : -NA-
VRF Name : default
Admin State : ENABLED
PAPI Security Key : AQBapdxySvGPvdTlkYn1/naKX4O3jKHrm28xLYfO6mLOK499BwAAAHdJp/bL4FE=
```

Showing brief global UBT configuration information where local-VLAN mode has been configured:

```
switch# show ubt brief
```

| Zone Name | UBT Mode   | Primary Controller Address | VRF Name | Status  |
|-----------|------------|----------------------------|----------|---------|
| zone1     | local-vlan | 10.116.51.10               | default  | Enabled |

Showing brief global UBT configuration information where VLAN-extend mode has been configured:

```
switch# show ubt brief
```

| Zone Name | UBT Mode    | Primary Controller Address | VRF Name | Status  |
|-----------|-------------|----------------------------|----------|---------|
| zone1     | vlan-extend | 10.116.51.10               | default  | Enabled |

Showing brief configuration for UBT zone1 where local-VLAN mode has been configured:

```
switch# show ubt zone zone1 brief
```

| Zone Name | UBT Mode   | Primary Controller Address | VRF Name | Status  |
|-----------|------------|----------------------------|----------|---------|
| zone3     | local-vlan | 30.116.51.10               | default  | Enabled |

Showing brief configuration for UBT zone1 where VLAN-extend mode has been configured:

```
switch# show ubt zone zone1 brief
```

| Zone Name | UBT Mode    | Primary Controller Address | VRF Name | Status  |
|-----------|-------------|----------------------------|----------|---------|
| zone1     | vlan-extend | 10.116.51.10               | default  | Enabled |

# show ubt information

## Syntax

```
show ubt information
```

```
show ubt information zone <ZONE-NAME>
```

## Description

Shows SAC and UAC information for UBT. Specifying a zone name displays UBT information for that zone.

## Command context

Operator (>) or Manager (#)

## Parameters

### **ZONE-NAME**

Specifies UBT zone name. Maximum characters: 64.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing SAC and UAC information for the tunneled node server:

```
switch# show ubt information
```

```
Zone zone1:
```

```
SAC Information
```

```
Active : 10.116.51.12
```

```
Standby : 10.116.51.13
```

```
Node List Information
```

```
Cluster Name : my-cluster
```

```
Node List
```

```
10.1.1.1
```

```
10.1.1.2
```

```
10.1.1.3
```

```
Bucket Map Information
```

```
Bucket Name : my-bucket
```

```
Bucket Map Active : [0...255]
```

| Bucket ID | A-UAC    | S-UAC    | Connectivity |
|-----------|----------|----------|--------------|
| 0         | 10.1.1.1 | 10.1.1.2 | L2           |
| 1         | 10.1.1.2 | 10.1.1.3 | L2           |
| 2         | 10.1.1.3 | 10.1.1.1 | L2           |

Showing SAC and UAC information for zone1:

```
switch# show ubt information zone zone1
```

```
Zone zone1:
SAC Information
```

```
Active : 10.116.51.12
Standby : 10.116.51.13
```

```
Node List Information
```

```
Cluster Name : my-cluster
```

```
Node List
```

```
10.1.1.1
10.1.1.2
10.1.1.3
```

```
Bucket Map Information
```

```
Bucket Name : my-bucket
Bucket Map Active : [0...255]
```

| Bucket ID | A-UAC    | S-UAC    | Connectivity |
|-----------|----------|----------|--------------|
| 0         | 10.1.1.1 | 10.1.1.2 | L2           |
| 1         | 10.1.1.2 | 10.1.1.3 | L2           |
| 2         | 10.1.1.3 | 10.1.1.1 | L2           |

## show ubt state

### Syntax

```
show ubt state
```

```
show ubt state zone <ZONE-NAME>
```

```
show ubt state zone <ZONE-NAME> uac-ip <UAC-ADDR>
```

### Description

Shows the global UBT state.

Specifying a zone shows the UBT state of that zone.

Specifying a UAC IP address shows the UBT state of that UAC.

### Command context

Operator (>) or Manager (#)

### Parameters

**zone** <ZONE-NAME>

Specifies UBT zone name. Maximum characters: 64.

**uac-ip** <UAC-ADDR>

Specifies the IP address of the user anchor controller for which to view user information. Specify the address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.



## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing the UBT state where local-VLAN mode has been configured:

```
switch# show ubt state
```

Zone zone1:

Local Master Server (LMS) State:

| LMS Type  | IP Address      | State               | Role                  |
|-----------|-----------------|---------------------|-----------------------|
| Primary   | : 15.212.219.55 | ready_for_bootstrap | operational_primary   |
| Secondary | : 15.212.219.57 | ready_for_bootstrap | operational_secondary |

Switch Anchor Controller (SAC) State:

|         | IP Address      | MAC Address       | State      |
|---------|-----------------|-------------------|------------|
| Active  | : 15.212.219.55 | 00:0b:86:b7:62:9f | Registered |
| Standby | : 15.212.219.57 | 00:0b:86:b7:64:0f | Registered |

User Anchor Controller (UAC): 15.212.219.55

| User              | Port   | State      | Bucket ID | Gre Key | VLAN |
|-------------------|--------|------------|-----------|---------|------|
| 00:00:00:00:00:05 | 1/1/19 | registered | 5         | 13      | 4000 |

User Anchor Controller (UAC): 15.212.219.57

| User              | Port   | State      | Bucket ID | Gre Key | VLAN |
|-------------------|--------|------------|-----------|---------|------|
| 00:00:00:00:00:04 | 1/1/20 | registered | 4         | 14      | 4000 |

Showing the UBT state where VLAN-extend mode has been configured:

```
switch# show ubt state
```

Zone zone1:

Local Master Server (LMS) State:

| LMS Type  | IP Address      | State               | Role                  |
|-----------|-----------------|---------------------|-----------------------|
| Primary   | : 15.212.219.55 | ready_for_bootstrap | operational_primary   |
| Secondary | : 15.212.219.57 | ready_for_bootstrap | operational_secondary |

Switch Anchor Controller (SAC) State:

|         | IP Address      | MAC Address       | State      |
|---------|-----------------|-------------------|------------|
| Active  | : 15.212.219.55 | 00:0b:86:b7:62:9f | Registered |
| Standby | : 15.212.219.57 | 00:0b:86:b7:64:0f | Registered |

User Anchor Controller (UAC): 15.212.219.55

| User              | Port   | State      | Bucket ID | Gre Key | VLAN |
|-------------------|--------|------------|-----------|---------|------|
| 00:00:00:00:00:05 | 1/1/19 | registered | 5         | 13      | 10   |

User Anchor Controller (UAC): 10.116.51.10

| User              | Port   | State      | Bucket ID | Gre Key | VLAN |
|-------------------|--------|------------|-----------|---------|------|
| 00:00:00:00:00:04 | 1/1/20 | registered | 4         | 14      | 20   |

Showing the UBT state of zone1:

```
switch# show ubt state zone zone1
```

Zone zone1:

Local Master Server (LMS) State:

| LMS Type  | IP Address      | State               | Role                  |
|-----------|-----------------|---------------------|-----------------------|
| Primary   | : 20.212.219.55 | ready_for_bootstrap | operational_primary   |
| Secondary | : 20.212.219.57 | ready_for_bootstrap | operational_secondary |

Switch Anchor Controller (SAC) State:

|         | IP Address      | MAC Address       | State      |
|---------|-----------------|-------------------|------------|
| Active  | : 20.212.219.55 | 00:ab:86:b7:62:9f | Registered |
| Standby | : 20.212.219.57 | 00:ab:86:b7:64:0f | Registered |

User Anchor Controller (UAC): 20.212.219.55

| User              | Port   | State      | Bucket ID | Gre Key | VLAN |
|-------------------|--------|------------|-----------|---------|------|
| 00:00:00:00:00:10 | 1/1/20 | registered | 5         | 16      | 30   |

User Anchor Controller (UAC): 20.212.219.57

| User              | Port   | State      | Bucket ID | Gre Key | VLAN |
|-------------------|--------|------------|-----------|---------|------|
| 00:00:00:00:00:20 | 1/1/30 | registered |           |         |      |

Showing the UBT state of a UAC with IP address 15.212.219.57 where local-VLAN mode has been configured:

```
switch# show ubt state zone zone1 uac-ip 15.212.219.57
```

User Anchor Controller (UAC): 15.212.219.57

| User              | Port   | State      | Bucket ID | Gre Key | VLAN |
|-------------------|--------|------------|-----------|---------|------|
| 00:00:00:00:00:04 | 1/1/20 | registered | 4         | 14      | 4000 |

Showing the UBT state of a UAC with IP address 15.212.219.55 where VLAN-extend mode has been configured:

```
switch# show ubt state zone zone1 uac-ip 15.212.219.55
```

User Anchor Controller (UAC): 15.212.219.55

| User              | Port   | State      | Bucket ID | Gre Key | VLAN |
|-------------------|--------|------------|-----------|---------|------|
| 00:00:00:00:00:07 | 1/1/10 | registered | 40        | 14      | 20   |
| 00:00:00:00:00:08 | 1/1/12 | registered | 28        | 14      | 30   |

# show ubt statistics

## Syntax

```
show ubt statistics
```

```
show ubt statistics zone <ZONE-NAME>
```

```
show ubt statistics zone <ZONE-NAME> uac-ip <UAC-ADDR>
```

## Description

Displays statistics for UBT.

Specifying a zone shows the UBT statistics for that zone.

Specifying a UAC IP address shows the UBT statistics for that UAC.

## Command context

Operator (>) or Manager (#)

## Parameters

**zone** <ZONE-NAME>

Specifies UBT zone name. Maximum characters: 64.

**uac-ip** <UAC-ADDR>

Specifies the IP address of the user anchor controller for which to view user information. Specify the address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

## Authority

Operators or Administrators or local user group members with execution rights for this command.

Operators can execute this command from the operator context (>) only.

## Examples

Showing UBT statistics where local-VLAN mode has been configured:

```
switch# show ubt statistics
```

```
UBT Statistics
```

```
Zone zone1:
```

```
Control Plane Statistics
```

```
Active : 10.1.1.1
```

```
Bootstrap Tx : 10
Nodelist Rx : 25
Bucketmap Rx : 21
Failover Tx : 4
Unbootstrap Tx : 7
Heartbeat Tx : 5
```

```
Bootstrap Rx : 10
Nodelist Ack Rx : 6
Bucketmap Ack Rx : 10
Failover Ack Rx : 3
Unbootstrap Ack Rx : 5
Heartbeat Rx : 3
```

```
Standby : 10.1.1.2
```

```
Bootstrap Tx : 10
Nodelist Rx : 25
Bucketmap Rx : 21
Failover Tx : 4
Unbootstrap Tx : 5
Heartbeat Tx : 7
```

```
Bootstrap Rx : 10
Nodelist Ack Rx : 6
Bucketmap Ack Rx : 12
Failover Ack Rx : 3
Unbootstrap Ack Rx : 3
Heartbeat Rx : 4
```

```

UAC : 10.1.1.3
 Bootstrap Tx : 10
 Unbootstrap Tx : 5
 Keepalive Tx : 2
 Bootstrap Ack Rx : 5
 Unbootstrap Ack Rx : 5
 Keepalive Ack Rx : 2

UAC : 10.1.1.4
 Bootstrap Tx : 5
 Unbootstrap Tx : 0
 Keepalive Tx : 0
 Bootstrap Ack Rx : 5
 Unbootstrap Ack Rx : 0
 Keepalive Ack Rx : 0

```

#### Data Plane Statistics

| UAC      | Packets Tx | Packets Rx |
|----------|------------|------------|
| 10.1.1.1 | 45678      | 23456      |
| 10.1.1.2 | 34567      | 23457      |

#### User Statistics

| UAC      | User Count |
|----------|------------|
| 10.1.1.1 | 1          |
| 10.1.1.2 | 2          |

Showing UBT statistics where VLAN-extend mode has been configured:

```
switch# show ubt statistics
```

#### UBT Statistics

Zone zone1:

#### Control Plane Statistics

```

Active : 10.1.1.1
 Bootstrap Tx : 10
 Nodelist Rx : 25
 Bucketmap Rx : 21
 Failover Tx : 4
 Unbootstrap Tx : 7
 Heartbeat Tx : 5
 Bootstrap Rx : 10
 Nodelist Ack Rx : 6
 Bucketmap Ack Rx : 10
 Failover Ack Rx : 3
 Unbootstrap Ack Rx : 5
 Heartbeat Rx : 3

Standby : 10.1.1.2
 Bootstrap Tx : 10
 Nodelist Rx : 25
 Bucketmap Rx : 21
 Failover Tx : 4
 Unbootstrap Tx : 5
 Heartbeat Tx : 7
 Bootstrap Rx : 10
 Nodelist Ack Rx : 6
 Bucketmap Ack Rx : 12
 Failover Ack Rx : 3
 Unbootstrap Ack Rx : 3
 Heartbeat Rx : 4

UAC : 10.1.1.3
 Bootstrap Tx : 10
 Unbootstrap Tx : 5
 Keepalive Tx : 2
 Bootstrap Ack Rx : 5
 Unbootstrap Ack Rx : 5
 Keepalive Ack Rx : 2

UAC : 10.1.1.4
 Bootstrap Tx : 5
 Unbootstrap Tx : 0
 Keepalive Tx : 0
 Bootstrap Ack Rx : 5
 Unbootstrap Ack Rx : 0
 Keepalive Ack Rx : 0

```

#### Data Plane Statistics

```

SAC tunnel Rx : 444
Standby-SAC tunnel Rx : 0

```

| UAC      | Packets Tx | Packets Rx |
|----------|------------|------------|
| 10.1.1.3 | 45678      | 23456      |
| 10.1.1.4 | 34567      | 23457      |

#### User Statistics

| UAC      | User Count |
|----------|------------|
| 10.1.1.3 | 1          |
| 10.1.1.4 | 2          |

Showing UBT statistics for zone1 where local-vlan mode has been configured:

```
switch# show ubt statistics zone zone1
```

#### UBT Statistics

Zone zone1:

#### Control Plane Statistics

```
Active : 10.1.1.1
 Bootstrap Tx : 10
 Nodelist Rx : 25
 Bucketmap Rx : 21
 Failover Tx : 4
 Unbootstrap Tx : 7
 Heartbeat Tx : 5
 Bootstrap Rx : 10
 Nodelist Ack Rx : 6
 Bucketmap Ack Rx : 10
 Failover Ack Rx : 3
 Unbootstrap Ack Rx : 5
 Heartbeat Rx : 3

Standby : 10.1.1.2
 Bootstrap Tx : 10
 Nodelist Rx : 25
 Bucketmap Rx : 21
 Failover Tx : 4
 Unbootstrap Tx : 5
 Heartbeat Tx : 7
 Bootstrap Rx : 10
 Nodelist Ack Rx : 6
 Bucketmap Ack Rx : 12
 Failover Ack Rx : 3
 Unbootstrap Ack Rx : 3
 Heartbeat Rx : 4

UAC : 10.1.1.3
 Bootstrap Tx : 10
 Unbootstrap Tx : 5
 Keepalive Tx : 2
 Bootstrap Ack Rx : 5
 Unbootstrap Ack Rx : 5
 Keepalive Ack Rx : 2

UAC : 10.1.1.4
 Bootstrap Tx : 5
 Unbootstrap Tx : 0
 Keepalive Tx : 0
 Bootstrap Ack Rx : 5
 Unbootstrap Ack Rx : 0
 Keepalive Ack Rx : 0
```

#### Data Plane Statistics

| UAC      | Packets Tx | Packets Rx |
|----------|------------|------------|
| 10.1.1.3 | 45678      | 23456      |
| 10.1.1.4 | 34567      | 23457      |

#### User Statistics

| UAC      | User Count |
|----------|------------|
| 10.1.1.3 | 1          |
| 10.1.1.4 | 2          |

Showing UBT statistics for zone1 where VLAN-extend mode has been configured:

```
switch# show ubt statistics zone zone1
```

#### UBT Statistics

Zone zone1:

#### Control Plane Statistics

```
Active : 10.1.1.1
 Bootstrap Tx : 10 Bootstrap Rx : 10
 Nodelist Rx : 25 Nodelist Ack Rx : 6
 Bucketmap Rx : 21 Bucketmap Ack Rx : 10
 Failover Tx : 4 Failover Ack Rx : 3
 Unbootstrap Tx : 7 Unbootstrap Ack Rx : 5
 Heartbeat Tx : 5 Heartbeat Rx : 3

Standby : 10.1.1.2
 Bootstrap Tx : 10 Bootstrap Rx : 10
 Nodelist Rx : 25 Nodelist Ack Rx : 6
 Bucketmap Rx : 21 Bucketmap Ack Rx : 12
 Failover Tx : 4 Failover Ack Rx : 3
 Unbootstrap Tx : 5 Unbootstrap Ack Rx : 3
 Heartbeat Tx : 7 Heartbeat Rx : 4

UAC : 10.1.1.3
 Bootstrap Tx : 10 Bootstrap Ack Rx : 5
 Unbootstrap Tx : 5 Unbootstrap Ack Rx : 5
 Keepalive Tx : 2 Keepalive Ack Rx : 2

UAC : 10.1.1.4
 Bootstrap Tx : 5 Bootstrap Ack Rx : 5
 Unbootstrap Tx : 0 Unbootstrap Ack Rx : 0
 Keepalive Tx : 0 Keepalive Ack Rx : 0
```

#### Data Plane Statistics

```
SAC tunnel Rx : 444
Standby-SAC tunnel Rx : 0
```

| UAC      | Packets Tx | Packets Rx |
|----------|------------|------------|
| 10.1.1.3 | 45678      | 23456      |
| 10.1.1.4 | 34567      | 23457      |

#### User Statistics

| UAC      | User Count |
|----------|------------|
| 10.1.1.3 | 1          |
| 10.1.1.4 | 2          |

Showing the UBT state of a UAC with IP address 123.222.212.90:

```
switch# show ubt statistics zone zone1 uac-ip 101.101.101.11
```

#### Data Plane Statistics

```
SAC tunnel Rx : 6457
Standby-SAC tunnel Rx : 0

UAC Packets Tx Packets Rx

101.101.101.11 : 145379605 145450113
```

# show ubt users

## Syntax

```
show ubt users [all | count | down | mac <MAC-ADDR> | {port <IF-NAME> | <IF-RANGE>} | up] zone <ZONE-NAME>
```

## Description

Displays user information for UBT.

## Command context

Operator (>) or Manager (#)

## Parameters

### all

Display information for all users.

### count

Display the total number of users configured to tunnel traffic.

### down

Display the users that are not able to tunnel traffic.

### mac <MAC-ADDR>

Display user information based on MAC address.

### port <IF-NAME> | <IF-RANGE>

Display user information for a specific interface or range of interfaces. For example, port 1/1/1 or port 1/1/1-1/1/10.

### up

Display user information that are active.

### zone <ZONE-NAME>

Specifies UBT zone name. Maximum characters: 64.

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Examples

Showing information for all users:

```
switch# show ubt users all
```

```
=====
Displaying All UBT Users for Zone: zone1
=====
Downloaded user roles are preceded by *
Port Mac Address Tunnel Status Secondary UserRole Failure Reason

1/25 00:00:00:11:12:03 activated authenticated ---/---
```

Showing information for users of zone1:

```
switch# show ubt users all zone zone1
```

```

=====
Displaying All UBT Users for Zone: zone1
=====
Downloaded user roles are preceded by *
Port Mac Address Tunnel Status Secondary UserRole Failure Reason

1/25 00:00:00:11:12:03 activated authenticated ---/---

```

Displaying the number of users that are tunneling traffic:

```

switch# show ubt users count

Total Number of Users using ubt Zone : zone1 is 2
=====
Total Number of Users in all the zones : 2
=====

```

Showing users that are down:

```

switch# show ubt users down

=====
Displaying UBT Users of Zone: zone1 having Tunnel Status DOWN
=====
Downloaded user roles are preceded by *
Port Mac Address Tunnel Status Secondary UserRole Failure Reason

1/25 00:00:00:11:12:03 activation_failed authenticated PBF Failure

```

Showing information for users of zone1 that are down:

```

switch# show ubt users down zone zone1

=====
Displaying UBT Users of Zone: zone1 having Tunnel Status DOWN
=====
Downloaded user roles are preceded by *
Port Mac Address Tunnel Status Secondary UserRole Failure Reason

1/25 00:00:00:11:12:03 activation_failed authenticated PBF Failure

```

Showing information for users on port 2/25:

```

switch# show ubt users port 2/25

=====
Displaying UBT Users of Zone: zone1
=====
Downloaded user roles are preceded by *
Port Mac Address Tunnel Status Secondary UserRole Failure Reason

2/25 00:00:00:11:12:03 activated authenticated ---/---

```

Showing information for users that are up:

```

switch# show ubt users up

=====
Displaying UBT Users of Zone: zone1 having Tunnel Status UP
=====
Downloaded user roles are preceded by *
Port Mac Address Tunnel Status Secondary UserRole Failure Reason

1/25 00:00:00:11:12:03 activated authenticated ---/---

```



Showing information for users of zone1 that are up:

```
switch# show ubt users up zone zone1

=====
Displaying UBT Users of Zone: zone1 having Tunnel Status UP
=====
Downloaded user roles are preceded by *
Port Mac Address Tunnel Status Secondary UserRole Failure Reason

1/25 00:00:00:11:12:03 activated authenticated ---/---
```

Showing information for the user with MAC address 00:00:00:11:12:03:

```
switch# show ubt users mac 00:00:00:11:12:03

Displaying UBT User of Zone: zone1 having MAC-Address: 00:00:00:11:12:03
Downloaded user roles are preceded by *
Port Mac Address Tunnel Status Secondary UserRole Failure Reason

1/25 00:00:00:11:12:03 activated authenticated ---/---
```

## uac-keepalive-interval

### Syntax

```
uac-keepalive-interval <TIME>
```

```
no uac-keepalive-interval <TIME>
```

### Description

Specifies the UAC keep alive refresh time interval in seconds.

The `no` form of this command sets the keep alive interval to the default value.

### Command context

```
config-ubt-<ZONE-NAME>
```

### Parameters

**<TIME>**

Specifies the UAC keep-alive refresh time interval in seconds. Range: 1 to 60. Default: 60.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Specifying a keepalive interval of 60 seconds for UBT zone1:

```
switch(config)# ubt zone zone1
switch(config-ubt-zone1)# uac-keepalive-interval 60
```

Deleting the configured UAC keepalive interval:

```
switch(config)# ubt zone zone1
switch(config-ubt-zone1)# no uac-keepalive-interval 60
```

## ubt

### Syntax

```
ubt zone <ZONE-NAME> vrf <VRF-NAME>
```

```
no ubt zone <ZONE-NAME> vrf <VRF-NAME>
```

### Description

Creates a user-based tunneling (UBT) zone with a specified zone name and VRF name. A UBT name is used to configure all tunneled node server properties advertised by the tunneled node feature.

The `no` form of this command removes the specified UBT zone.

### Command context

config

### Parameters

**<ZONE-NAME>**

Specifies a name for the UBT zone. Length: 1 to 64 characters.

**<VRF-NAME>**

Specifies the VRF on which to establish the UBT tunnel.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating UBT zone called `zone1` associated with a VRF called `default`:

```
switch(config)# ubt zone zone1 vrf default
```

Removing UBT zone `zone1` on VRF `default`:

```
switch(config)# no ubt zone zone1 vrf default
```

Deleting all UBT configurations:

```
switch(config)# no ubt
```

## ubt-client-vlan

### Syntax

```
ubt-client-vlan <VLAN-ID>
```

```
no ubt-client-vlan <VLAN-ID>
```

### Description

Specifies the UBT Client VLAN or local VLAN. This VLAN is used in local-VLAN mode only. If the UBT client VLAN is configured in VLAN-extend mode it is ignored. No other feature should be enabled on the UBT client VLAN.

The `no` form of this command removes the VLAN to use for tunneled clients.

### Command context

`config`

### Parameters

**<VLAN-ID>**

Specifies the VLAN ID to use for tunneled clients. Range: 1-4094.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating VLAN 4000:

```
switch(config)# vlan 4000
switch(config-vlan-4000)# no shutdown
```

Specifying UBT client VLAN 4000:

```
switch(config)# ubt-client-vlan 4000
```

Removing configured UBT client VLAN 4000:

```
switch(config)# no ubt-client-vlan 4000
```

## ubt mode vlan-extend

### Syntax

`ubt-mode vlan-extend`

`no ubt-mode`

### Description

Selects VLAN extended mode. When VLAN-extend mode is enabled clients are assigned to their UBT role-based VLAN in the hardware datapath.

The `no` form of the command selects local-VLAN mode. In local-VLAN mode clients are assigned to a local switch VLAN and associated with their UBT role-based VLAN when client traffic reaches the controller.

The default UBT mode is local-VLAN.

### Command context

`config`

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Setting the UBT mode to VLAN-extend:

```
switch(config)# ubt-mode vlan-extend
```

Setting the UBT mode back to the default of local-VLAN:

```
switch(config)# no ubt-mode
```

## description

### Syntax

```
description <DESCRIPTION>
```

### Description

Specifies a descriptive for a VLAN.

### Command context

```
config-vlan-<VLAN-ID>
```

### Parameters

**<DESCRIPTION>**

Specifies a description for the VLAN.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Assigning a description to VLAN 20:

```
switch(config)# vlan 20
switch(config-vlan-20)# description primary
```

## name

### Syntax

```
name <VLAN-NAME>
```

### Description

Associates a name with a VLAN.

### Command context

```
config-vlan-<VLAN-ID>
```

### Parameters

**<VLAN-NAME>**

Specifies a name for a VLAN. Length: 1 to 32 alphanumeric characters, including underscore (\_) and hyphen (-).

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Assigning the name **backup** to VLAN 20:

```
switch(config)# vlan 20
switch(config-vlan-20)# name backup
```

# show capacities svi-count

## Syntax

```
show capacities svi-count
```

## Description

Shows the maximum number of SVIs supported by the switch.

## Command context

Manager (#)

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Showing switch SVI capacity:

```
switch# show capacities svi-count
System Capacities: Filter SVI count
Capacities Name Value

Maximum number of SVIs supported in the system 16
```

# show vlan

## Syntax

```
show vlan [<VLAN-ID>]
```

## Description

Displays configuration information for all VLANs or a specific VLAN.

## Command context

Manager (#)

## Parameters

**<VLAN-ID>**

Specifies a VLAN ID.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Examples

Displaying configuration information for VLAN 2:

```
switch# show vlan 2
```

| VLAN | Name      | Status | Reason | Type   | Interfaces        |
|------|-----------|--------|--------|--------|-------------------|
| 2    | UserVLAN1 | up     | ok     | static | 1/1/1,1/1/3,1/1/5 |

Displaying configuration information for all defined VLANs:

```
switch# show vlan
```

| VLAN | Name           | Status | Reason     | Type   | Interfaces              |
|------|----------------|--------|------------|--------|-------------------------|
| 1    | DEFAULT_VLAN_1 | up     | ok         | static | 1/1/3-1/1/4             |
| 2    | UserVLAN1      | up     | ok         | static | 1/1/1,1/1/3,1/1/5       |
| 3    | UserVLAN2      | up     | ok         | static | 1/1/2-1/1/3,1/1/5-1/1/6 |
| 5    | UserVLAN3      | up     | ok         | static | 1/1/3                   |
| 10   | TestNetwork    | up     | ok         | static | 1/1/3,1/1/5             |
| 11   | VLAN11         | up     | ok         | static | 1/1/3                   |
| 12   | VLAN12         | up     | ok         | static | 1/1/3,1/1/6,lag1-lag2   |
| 13   | VLAN13         | up     | ok         | static | 1/1/3,1/1/6             |
| 14   | VLAN14         | up     | ok         | static | 1/1/3,1/1/6             |
| 20   | ManagementVLAN | down   | admin_down | static | 1/1/3,1/1/10            |

## show vlan port

### Syntax

```
show vlan port <INTERFACE-ID>
```

### Description

Displays the VLANs configured for a specific layer 2 interface.

### Command context

Manager (#)

### Parameters

**<INTERFACE-ID>**

Specifies an interface ID. Format: member/slot/port.

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

Displaying the VLANs configured on interface 1/1/3:

```
switch# show vlan port 1/1/3
```

| VLAN | Name           | Mode            |
|------|----------------|-----------------|
| 1    | DEFAULT_VLAN_1 | native-untagged |
| 2    | UserVLAN1      | trunk           |
| 3    | UserVLAN2      | trunk           |
| 5    | UserVLAN3      | trunk           |
| 10   | TestNetwork    | trunk           |
| 11   | VLAN11         | trunk           |
| 12   | VLAN12         | trunk           |
| 13   | VLAN13         | trunk           |
| 14   | VLAN14         | trunk           |
| 20   | ManagementVLAN | trunk           |

## show vlan summary

### Syntax

```
show vlan summary
```

### Description

Displays a summary of the VLAN configuration on the switch.

### Command context

Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Examples

Displaying a summary of the VLAN configuration on the switch:

```
switch# show vlan summary
Number of existing VLANs: 11
Number of static VLANs: 11
Number of dynamic VLANs: 0
```

## shutdown

### Syntax

```
shutdown
```

```
no shutdown
```

### Description

Disables a VLAN. (By default, a VLAN is automatically enabled when it is created with the `vlan` command.)

The `no` form of this command enables a VLAN.



## Command context

`config-vlan-<VLAN-ID>`

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling VLAN 20:

```
switch(config)# vlan 20
switch(config-vlan-20)# no shutdown
```

Disabling VLAN 20:

```
switch(config)# vlan 20
switch(config-vlan-20)# shutdown
```

# vlan

## Syntax

`vlan <VLAN-LIST>`

`no vlan <VLAN-LIST>`

## Description

Creates a VLAN and changes to the `config-vlan-id` context for the VLAN. By default, the VLAN is enabled. To disable a VLAN, use the `no shutdown` command.

If the specified VLAN exists, this command changes to the `config-vlan-id` context for the VLAN. If a range of VLANs is specified, the context does not change.

The `no` form of this command removes a VLAN. VLAN 1 is the default VLAN and cannot be deleted.

## Command context

`config`

## Parameters

**<VLAN-LIST>**

Specifies a single ID, or a series of IDs separated by commas (2, 3, 4), dashes (2-4), or both (2-4,6). Range: 1 to . A maximum of 512 (6100 Switch Series) VLANs are supported.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Creating VLAN 20:

```
switch(config)# vlan 20
switch(config-vlan-20)#
```

Removing VLAN 20:

```
switch(config)# no vlan 20
```

Creating VLANs 2 to 8 and 10:

```
switch(config)# vlan 2-8,10
```

Removing VLANs 2 to 8 and 10:

```
switch(config)# no vlan 2-8,10
```

## vlan access

### Syntax

```
vlan access <VLAN-ID>
```

```
no vlan access [<VLAN-ID>]
```

### Description

Creates an access interface and assigns an VLAN ID to it. Only one VLAN ID can be assigned to each access interface.

VLANs can only be assigned to a non-routed (layer 2) interface or LAG interface. By default, all interfaces are routed (layer 3) when created. Use the `no routing` command to disable routing on an interface and change the interface to a layer 2 interface.

The `no` form of this command removes an access VLAN from the interface in the current context and sets it to the default VLAN ID of 1.

### Command context

```
config-if
```

### Parameters

**<VLAN-ID>**

Specifies a single ID, or a series of IDs separated by commas (2, 3, 4), dashes (2-4), or both (2-4,6). Range: 1 to . A maximum of 512 (6100 Switch Series) VLANs are supported.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring interface 1/1/2 as an access interface with VLAN ID set to 20:

```
switch(config)# interface 1/1/2
```

```
switch(config-if)# vlan access 20
```

Removing VLAN ID 20 from interface 1/1/2:

```
switch(config)# interface 1/1/2
```

```
switch(config-if)# no vlan access 20
```

or:

```
switch(config)# interface 1/1/2
```

```
switch(config-if)# no vlan access
```

# vlan trunk allowed

## Syntax

```
vlan trunk allowed [<VLAN-LIST> | all]
```

```
no vlan trunk allowed [<VLAN-LIST>]
```

## Description

Assigns a VLAN ID to a trunk interface. Multiple VLAN IDs can be assigned to a trunk interface. These VLAN IDs define which VLAN traffic is allowed across the trunk interface.

VLANs can be assigned only to a non-routed (layer 2) interface or LAG interface. By default, all interfaces are routed (layer 3) when created. Use the `no routing` command to disable routing on an interface.

The `no` form of this command removes one or more VLAN IDs from a trunk interface. When the last VLAN is removed from a trunk interface, the interface continues to operate in trunk mode, and will trunk all the VLANs currently defined on the switch, and any new VLANs defined in the future. To disable the trunk interface, use the command **shutdown**.

## Command context

```
config-if
```

## Parameters

**<VLAN-LIST>**

Specifies a single ID, or a series of IDs separated by commas (2, 3, 4), dashes (2-4), or both (2-4,6). Range: 1 to .

**all**

Configures the trunk interface to allow all the VLANs currently configured on the switch and any new VLANs that are configured in the future.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Assigning VLANs 2, 3, and 4 to trunk interface 1/1/2:

```
switch(config)# interface 1/1/2
switch(config-if)# vlan trunk allowed 2,3,4
```

Assigning VLAN IDs 2 to 8 to trunk interface 1/1/2:

```
switch(config)# interface 1/1/2
switch(config-if)# vlan trunk allowed 2-8
```

Assigning VLAN IDs 2 to 8 and 10 to trunk interface 1/1/2:

```
switch(config)# interface 1/1/2
switch(config-if)# vlan trunk allowed 2-8,10
```

Removing VLAN IDs 2, 3, and 4 from trunk interface 1/1/2:

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan trunk allowed 2,3,4
```

Removing all VLANs assigned to trunk interface 1/1/2:

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan trunk allowed
```

## vlan trunk native

### Syntax

```
vlan trunk native <VLAN-ID>
```

```
no vlan trunk native [<VLAN-ID>]
```

### Description

Assigns a native VLAN ID to a trunk interface. By default, VLAN ID 1 is assigned as the native VLAN ID for all trunk interfaces. VLANs can only be assigned to a non-routed (layer 2) interface or LAG interface. Only one VLAN ID can be assigned as the native VLAN.



**NOTE:** When a native VLAN is defined, the switch automatically executes the `vlan trunk allowed all` command to ensure that the default VLAN is allowed on the trunk. To only allow specific VLANs on the trunk, issue the `vlan trunk allowed` command specifying only specific VLANs.

The `no` form of this command removes a native VLAN from a trunk interface and assigns VLAN ID 1 as its native VLAN.

### Command context

```
config-if
```

### Parameters

**<VLAN-ID>**

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Assigning native VLAN ID 20 to trunk interface 1/1/2:

```
switch(config)# interface 1/1/2
switch(config-if)# vlan trunk native 20
```

Removing native VLAN 20 from trunk interface 1/1/2 and returning to the default VLAN 1 as the native VLAN.

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan trunk native 20
```

or:

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan trunk native
```

Assigning native VLAN ID 20 to trunk interface 1/1/2 and then removing it from the list of allowed VLANs. (Only allow VLAN 10 on the trunk.)

```
switch(config)# interface 1/1/2
switch(config-if)# vlan trunk native 20
switch(config-if)# vlan trunk allowed 10
```

## vlan trunk native tag

### Syntax

```
vlan trunk native <VLAN-ID> tag
```

```
no vlan trunk native <VLAN-ID> tag
```

### Description

Enables tagging on a native VLAN. Only incoming packets that are tagged with the matching VLAN ID are accepted. Incoming packets that are untagged are dropped except for BPDUs. Egress packets are tagged.

The `no` form of this command removes tagging on a native VLAN.

### Command context

```
config-if
```

### Parameters

**<VLAN-ID>**

Specifies the number of a VLAN. Range: 1 to .

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling tagging on native VLAN 20 on trunk interface 1/1/2:

```
switch(config)# interface 1/1/2
switch(config-if)# vlan trunk native 20
switch(config-if)# vlan trunk native 20 tag
```

Removing tagging on native VLAN 20 assigned to trunk interface 1/1/2:

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan trunk native 20 tag
```

Enabling tagging on native VLAN 20 assigned to LAG trunk interface 2:

```
switch(config)# interface lag 2
switch(config-lag-if)# vlan trunk native 20
switch(config-lag-if)# vlan trunk native 20 tag
```

## voice

### Syntax

```
voice
```

```
no voice
```

## Description

Configures a VLAN as a voice VLAN.

The `no` form of this command removes voice configuration from a VLAN.

## Command context

```
config-vlan-<VLAN-ID>
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring VLAN 10 as a voice VLAN:

```
switch(config)# vlan 10
switch(config-vlan-10)# voice
```

Removing voice from VLAN 10:

```
switch(config-vlan-10)# no voice
```

VSF commands do not apply to the 6400 series switches.

## vsf member

### Syntax

```
vsf member <MEMBER-ID>
```

```
no vsf member <MEMBER-ID>
```

### Description

Creates VSF member context in the switch for the specified member.

The `no` form of this command removes the specified member from the stack. All configuration associated with the member, as well as the subsystems and interfaces of the member will also be removed.

If the member is physically present in the stack at the time it is removed, it will reboot with the default configuration and lose its identity as a member of the stack from which it was removed.



**NOTE:** When a physically present member is removed, it may cause the stack to split.

### Command context

```
config
```

### Parameters

**<MEMBER-ID>**

VSF member identifier. Required.

- Range for 6200F devices: 1-8.
- Range for 6300 devices: 1-10.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring a VSF member:

```
switch(config)# vsf member 2
switch(vsf-member-2)#
```

Removing a non-master member from the stack:

```
switch(config)# no vsf member 2
The specified switch will be unconfigured and rebooted
Do you want to continue (y/n)? y
```



---

**NOTE:** Removing the running master should be done with caution as it can make the stack unusable if there is no standby.

---

## member

### Syntax

member <MEMBER-ID>

### Description

Connects to the specified member in a VSF environment.

### Command context

Manager (#)

### Parameters

<MEMBER-ID>

VSF member ID. Required.

- Range for 6200F devices: 1-8.
- Range for 6300 devices: 1-10.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

VSF stack is formed with two members:

```
switch# member 2
admin@172.17.17.2's password:

Last login: 2019-09-30 11:42:17 from the console
User "admin" has logged in 1 time in the past 30 days
member#
```

Member to self:

```
switch# member 1
Already on member id 1
```

VSF stack is not formed and member not available:

```
switch# member 2
No stack role for member id 2
```

## type

### Syntax

type <TYPE>



## Description

Configures the part number of the VSF member being provisioned. After provisioning, the interfaces of the member are available for configuration.

When the member eventually joins the stack, it will boot up with the configuration made on the pre-provisioned interfaces.

To provision a member, the member number and the part number of the member must be specified.

## Command context

vsf-member-*<ID>*#

## Parameters

**<TYPE>**

The part number of the member being provisioned. Required.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring the part number of a VSF member:

```
switch(vsf-member-2)#
 type The part number of the member being provisioned

switch(vsf-member-2)# type ?
 jl658a 6300M 24SFP+ /4SFP56 Switch
 jl659a 6300M 48SR PoE CLS 6 /4SFP56 Switch
 jl660a 6300M 24SR PoE CLS 6 /4SFP56 Switch
 jl661a 6300M 48G PoE CLS 4 /4SFP56 Switch
 jl662a 6300M 24G PoE CLS 4 /4SFP56 Switch
 jl663a 6300M 48G /4SFP56 Switch
 jl664a 6300M 24G /4SFP56 Switch
 jl665a 6300F 48G PoE CLS 4 /4SFP56 Switch
 jl666a 6300F 24G PoE CLS 4 /4SFP56 Switch
 jl667a 6300F 48G /4SFP56 Switch
 jl668a 6300F 24G /4SFP56 Switch

switch(vsf-member-2)# type jl662a

switch(vsf-member-2)# show running-config
Current configuration:
!
!Version ArubaOS-CX
!
!
!
ssh maximum-auth-attempts 6
!
!
!
!
vlan 1
vsf member 1
 type jl661a
exit
vsf member 2
```

```
type j1662a
exit
```

## link

### Syntax

```
link <LINK-ID> [<IFRANGE>]
```

### Description

Creates or modifies a VSF link. The user can specify the physical interfaces that make up the VSF link.

Once an interface is part of a VSF link, all existing configuration on the interface is removed and the interface will operate as a VSF interface. At least one interface must be specified for the creation of a VSF link. VSF interfaces carry VSF traffic and can only be connected to other VSF interfaces.

The `no` form of the command can be used to remove interfaces from a link or remove configuration from the link completely.



**NOTE:** When configuration is removed from a link, it may cause the stack to split.

### Command context

```
vsf-member-<ID>#
```

### Parameters

**<LINK-ID>**

The VSF link number. Required. Range: 1-2.

**<IFRANGE>**

The interface identifier range. Required.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating and modifying VSF links:

```
switch(vsf-member-1)# link
<1-2> VSF Link number
switch(vsf-member-1)# link 1
IFRANGE Interface identifier range
<cr>
switch(vsf-member-1)# link 1 1/1/51
<cr>
switch(vsf-member-1)# link 1 1/1/49-1/1/50
<cr>
switch(vsf-member-1)# link 2 1/1/52
<cr>
switch(vsf-member-1)# link 1 1/1/49
<cr>

switch(vsf-member-5)# show running-config
Current configuration:
!
!Version ArubaOS-CX
!
!
```

```

!
!
ssh maximum-auth-attempts 6
!
!
!
!
vlan 1
interface 1/1/49
 no shutdown
interface 1/1/52
 no shutdown
vsf member 1
 type jl661a
 link 1 1/1/49
 link 2 1/1/52
exit

switch(vsf-member-1)# no link 1
<cr>
switch(vsf-member-1)# no link 1
This will cause the stack to split.
Do you want to continue (y/n)? y

switch(vsf-member-1)# no link 2
This will cause the stack to split and the residual stack fragment will become unusable.
Do you want to continue (y/n)? y

switch(vsf-member-1)# show running-config
Current configuration:
!
!Version ArubaOS-CX SL.10.02.0020-741-g11104d6~dirty
!
!
!
ssh maximum-auth-attempts 6
!
!
!
!
vlan 1
interface 1/1/52
 no shutdown
vsf member 1
 type jl661a
exit

```

Before removing an individual interface from the VSF link using the `no vsf link <x> <interface>` command, ensure that the interface is admin shutdown at both local and peer ends.

Example: Interface 1/1/25 on member 1 link 1 is connected to 2/1/25 on member 2 link 2. The user intends to remove 1/1/25 from link 1 of member 1. Interfaces 1/1/25 and 2/1/25 must be admin shutdown before removing them from the link configuration.

Before deleting the link using the `no link <x>` command, all individual interfaces in the VSF link must be admin shutdown both at local and peer ends.

## vsf split-detect

### Syntax

```
vsf split-detect <MGMT-INTERFACE>
```

## Description

Configures the VSF split detection method that specifies the mechanism used for stack fragment discovery when there is a stack split.

Once the stack fragments are discovered, the fragment having the primary member always wins. All non-VSF interfaces on the losing stack fragment will be brought down to minimize network disruption due to duplicate MAC/IP.

## Command context

config

## Parameters

**<MGMT-INTERFACE>**

Configures mgmt-interface as the split detection method. Connect the management interfaces of the primary and secondary members to the same L2 network. Optionally, the management interfaces of primary and secondary can be directly connected to each other.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring mgmt-interface as the split detection method:

```
switch(config)# vsf split-detect mgmt-interface
```

Removing split detection from the stack:

```
switch(config)# no vsf split-detect
```

# vsf secondary-member

## Syntax

vsf secondary-member <MEMBER-ID>

## Description

Configures a secondary member from the available members. The secondary member will normally operate as the Standby member of the stack.

The stack does not have a secondary member by default. If a secondary member is already configured, the user must un-configure the old secondary member and then configure new secondary member.



**NOTE:** Member 1 cannot be configured as the secondary member.

## Command context

config

## Parameters

**<MEMBER-ID>**

Secondary member number. Required.

- Range for 6200F devices: 2-8.
- Range for 6300 devices: 2-10.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Configuring and un-configuring a secondary member:

```
switch(config)# vsf secondary-member 3
This will save the configuration and reboot the specified switch.
Do you want to continue (y/n)? y

switch(config)# vsf secondary-member 4
The existing secondary member must be unconfigured first.

switch(config)# no vsf secondary-member
The secondary member will go for a reboot.
Do you want to continue (y/n)? y
```

## vsf renumber-to

### Syntax

```
vsf renumber-to <MEMBER-ID>
```

### Description

Renumbers VSF member 1 to a value from 2 through 10 (for 6300 devices) and 2 through 8 (for the 6200F device). Changing the member number causes the switch to reboot with the new member number. Only member 1 can be renumbered.



**NOTE:** VSF links must be configured before renumbering a switch. Renumbering will be disallowed if no links are configured or there are provisioned/physically present members.

### Command context

```
config
```

### Parameters

**<MEMBER-ID>**

Member number to which the member will be renumbered. Required.

- Range for 6200F devices: 2-8.
- Range for 6300 devices: 2-10.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Renumbering primary VSF member from 1 to 2:

```
switch(config)# vsf renumber-to 2
Member 1 cannot be renumbered until all other members are removed.

switch(config)# vsf renumber-to 2
Member 1 cannot be renumbered until a VSF link is configured.

switch(config)# vsf renumber-to 2
This will save the VSF configuration and reboot the switch.
Do you want to continue (y/n)? y
```

## vsf member reboot

### Syntax

```
vsf member <MEMBER-ID> reboot
```

### Description

Reboots the specified VSF member. Upon reboot, if the master is reachable, the member will rejoin the stack.

### Command context

Manager (#)

### Parameters

**<MEMBER-ID>**

Member number to be rebooted. Required.

- Range for 6200F devices: 1-8.
- Range for 6300 devices: 1-10.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Rebooting the primary switch of the stack:

```
switch# vsf member 1 reboot
Rebooting the master switch of the stack without a standby
will make the stack unusable.
Do you want to continue (y/n)? y

switch# vsf member 1 reboot
The master switch will reboot and the standby will become the master.
Do you want to continue (y/n)? y

switch# vsf member 2 reboot
This will reboot the specified switch.
Do you want to continue (y/n)? y
```

## interface

### Syntax

```
interface <IFRANGE>
```

## Description

Enters configuration context for one or more VSF link interfaces.

## Command context

config

## Parameters

<IFRANGE>

PORT identifier range. Required.



**NOTE:** VSF link interfaces cannot be included in a range with other interfaces.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Entering configuration context:

```
switch(config)# int 1/1/1
```

# shutdown

## Syntax

shutdown

## Description

Shuts down one or more VSF link interfaces.

## Command context

config-if-vsfc

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Shutting down a VSF link interface:

```
switch(config)# int 1/1/1-1/1/2
switch(config-if-vsfc-1/1/1-1/1/2)# shutdown
```



**NOTE:** Shutdown configuration for VSF interfaces is not persistent across reboots.

# show vsf

## Syntax

show vsf

## Description

Displays the list of provisioned VSF stack members.

## Command context

Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch# show vsf
MAC Address : 08:97:34:b0:0e:00
Secondary : 2
Topology : Ring
Status : No Split
Split Detection Method : None
```

| Mbr ID | MAC Address       | Type   | Status      |
|--------|-------------------|--------|-------------|
| 1      | 08:97:34:b0:0e:00 | JL666A | Master      |
| 2      | 08:97:34:b1:43:00 | JL665A | Standby     |
| 3      | 08:97:34:b7:cc:00 | JL663A | Member      |
| 4      |                   | JL662A | Not Present |

```
switch# show vsf
MAC Address : 08:97:34:b0:0e:00
Secondary : 2
Topology : Ring
Status : Active Fragment
Split Detection Method : mgmt-interface
```

| Mbr ID | MAC Address       | Type   | Status            |
|--------|-------------------|--------|-------------------|
| 1      | 08:97:34:b0:0e:00 | JL666A | Master            |
| 2      | 08:97:34:b1:43:00 | JL665A | In Other Fragment |
| 3      | 08:97:34:b7:cc:00 | JL663A | Member            |
| 4      |                   | JL662A | Not Present       |

## show vsf detail

## Syntax

```
show vsf detail
```

## Description

Displays detailed information related to the current state of the VSF stack and the stack members.

## Command context

Manager (#)



## Authority

Operators or Administrators or local user group members with execution rights for this command.  
Operators can execute this command from the operator context (>) only.

## Example

```
switch# show vsf detail
```

```
VSF Stack
 MAC Address : ec:eb:b8:d0:80:40
 Secondary : 2
 Topology : Ring
 Status : No Split
 Uptime : 0d 0h 23m
 Split Detection Method : None
 Software Version : SL.10.02.0000-7755

 Name : Aruba-VSF-6300F
 Contact :
 Location :

Member ID : 1
 MAC Address : ec:eb:b8:d0:80:40
 Type : JL666A
 Model : Aruba 6300F 24G PoE CLS 4 /4SFP56 Switch
 Status : Master
 ROM Version : SL.10.02.0000-7755
 Serial Number : CN7ZK90012
 Uptime : 0d 0h 23m
 CPU Utilization : 0%
 Memory Utilization : 20%
 VSF link 1 : Up, connected to peer member 2, link 1
 VSF link 2 : Down

Member ID : 2
 MAC Address : eb:ec:d8:e0:50:60
 Type : JL666A
 Model : Aruba 6300F 24G PoE CLS 4 /4SFP56 Switch
 Status : Standby
 ROM Version : SL.10.02.0000-7755
 Serial Number : CN7ZK90012
 Uptime : 0d 0h 23m
 CPU Utilization : 0%
 Memory Utilization : 15%
 VSF link 1 : Up, connected to peer member 1, link 1
 VSF link 2 : Down

Member ID : 3
 MAC Address :
 Type : JL666A
 Model : Aruba 6300F 24G PoE CLS 4 /4SFP56 Switch
 Status : Not Present
 ROM Version :
 Serial Number :
 Uptime :
 CPU Utilization :
 Memory Utilization :
 VSF link 1 :
 VSF link 2 :
```

## show vsf link

### Syntax

```
show vsf link
```

### Description

Displays the VSF link state for each member.

### Command context

Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Example

```
switch# show vsf link
```

```
VSF Member 1
```

| Link | Link State | Peer Member | Peer Link | Interfaces              |
|------|------------|-------------|-----------|-------------------------|
| 1    | Down       | 0           | 0         | 1/1/1,1/1/5,1/1/8,1/1/9 |
| 2    | Down       | 0           | 0         | 1/1/11-1/1/13,1/1/15    |

## show vsf member

### Syntax

```
show vsf member <MEMBER-ID>
```

### Description

Displays information about the specified VSF member.

### Command context

Manager (#)

### Parameters

<MEMBER-ID>

VSF member identifier. Required.

- Range for 6200F devices: 1-8.
- Range for 6300 devices: 1-10.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

```
switch# show vsf member 1
```

```
Member ID : 1
MAC Address : ec:eb:b8:d0:80:40
Type : JL557A
Model : Aruba JL557A 2930F-48G-740W-PoE+-4SFP Switch
Status : Master
ROM Version : SL.10.02.0000-7755
Serial Number : CN7ZK90012
Uptime : 0d 0h 18m
CPU Utilization : 0%
Memory Utilization : 15%
VSF link 1 : Down
VSF link 2 : Down
```

## show vsf topology

### Syntax

```
show vsf topology
```

### Description

Displays information about VSF stack member connections.

### Command context

Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

```
switch# show vsf topology
```

```
 Stby Master
+----+ +----+ +----+
| 3 |1==2| 2 |1==1| 1 |
+----+ +----+ +----+
```

## interface vxlan

### Syntax

```
interface vxlan 1
```

```
no interface vxlan 1
```

### Description

Creates VXLAN interface 1 and changes to the `config-vxlan-if` context. A maximum of one VXLAN interface is supported. By default, the VXLAN is disabled. To enable the VXLAN, use the command `no shutdown`.

The `no` form of this command removes VXLAN interface 1. This deletes the VXLAN tunnel, and all VNIs and VLAN-to-VNI mappings associated with it.

### Command context

```
config
```

### Parameters

1

Only one VXLAN interface is supported.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating VXLAN interface 1:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)#
```

Deleting VXLAN interface 1:

```
switch(config)# no interface vxlan 1
```

## routing

### Syntax

```
vni <VNI-NUMBER>
```

```
no vni <VNI-NUMBER>
```

## Description

Enables a layer 3 VNI for EVPN VXLAN distributed L3 gateways with symmetric IRB. The VNI is automatically assigned the default VRF. To assign another VRF, use the command `vrf`. Used with EVPN-based VXLANs.

The `no` form of this command disables symmetric routing on a VNI.

## Command context

```
config-vni-<VNI-NUMBER>
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling L3 VNI 1000 for EVPN VXLAN distributed L3 gateways with symmetric IRB using VRF `vrf-1`:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if) # vni 1000
switch(config-vni-1000) # routing
switch(config-vni-1000) # vrf vrf-1
```

Disabling L3 VNI 1000 for EVPN VXLAN distributed L3 gateways with symmetric IRB:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if) # vni 1000
switch(config-vni-1000) # no routing
```

# show interface vxlan1

## Syntax

```
show interface vxlan1
```

## Description

Shows detailed information about VXLAN interface 1.

## Command context

Manager (#)

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Showing settings for VXLAN interface 1.

```
switch# show interface vxlan1
Interface vxlan1 is up
Admin state is up
Description:VXLAN1
Underlay VRF: default
Destination UDP port: 4789
VTEP source IPv4 address: 1.1.1.1
```

| VNI | Routing | VLAN | VRF | VTEP Peers | Origin |
|-----|---------|------|-----|------------|--------|
|-----|---------|------|-----|------------|--------|

```

10 disabled 10 -- 2.2.2.2 static
10 disabled 10 -- 3.3.3.3 static
10 disabled 10 -- 4.4.4.4 static
20 disabled 20 -- 5.5.5.5 evpn
20 disabled 20 -- 6.6.6.6 evpn
30 disabled -- -- -- static
40 disabled 40 -- -- static
50 disabled -- -- 7.7.7.7 static
4000 enabled -- default 22.1.1.2 evpn
4001 enabled -- default 23.1.1.3 evpn

```

#### Aggregate Statistics

```

Decap:
 104222 input packets 15841744 bytes
 236 broadcast packets 26942 bytes
 0 drop packets
Encap:
 108527 output packets 11068728 bytes
 6 BUM packets 422 bytes
 0 drop packets

```

## show interface vxlan vni

### Syntax

```
show interface vxlan vni <VNI-NUMBER> vteps
```

### Description

Shows detailed VNI information for VXLAN interface 1.

### Command context

Manager (#)

### Parameters

**<VNI-NUMBER>**

Specifies the number for a VNI. Range: 1 to 16777214.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Usage

Status can be one of the following:

- operational: Virtual network ID is fully programmed on the switch hardware.
- configuration\_error: Virtual network ID programming in the switch hardware failed due to misconfiguration.
- no\_hw\_resources: Virtual network ID programming failed in the switch hardware due to insufficient resources.
- activating: Initial state of virtual network ID when it is configured.

When a tunnel endpoint is directly connected via nexthop, then nexthop reachability appears empty (--).

## Example

Showing VNI information:

```
switch# show interface vxlan vni
VNI : 1000 VLAN : 10
Routing : disabled VRF : --
VNI-Status : operational

VNI : 2000 VLAN : 20
Routing : enabled VRF : default
VNI-Status : activating
...
...

switch# show interface vxlan vni 1000
VNI : 1000 VLAN : 10
Routing : disabled VRF : --
VNI-Status : operational
```

```
switch# show interface vxlan vni vteps
VNI : 1000 VLAN : 10
Routing : disabled VRF : --
VNI-Status : operational
VTEPS
=====
 ORIGIN SOURCE DESTINATION VRF VTEP-STATUS
 -
 static 11.0.0.1 11.0.0.2 default operational

VNI : 2000 VLAN : 20
Routing : enabled VRF : default
VNI-Status : operational
VTEPS
=====
 ORIGIN SOURCE DESTINATION VRF VTEP-STATUS
 -
 evpn 11.0.0.1 14.0.0.2 default activating
```

## show interface vxlan vteps

### Syntax

```
show interface vxlan vteps [detail | <IPV4-ADDR>]
```

### Description

Shows information about the VTEPs on VXLAN interface 1.

### Command context

Manager (#)

### Parameters

#### detail

Show detailed information.

### <IPV4-ADDR>

Specifies the IP address of a VTEP peer in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Usage

Status can be one of the following:

- operational: Tunnel endpoint is fully programmed on the switch hardware.
- configuration\_error: Tunnel endpoint programming in the switch hardware failed due to misconfiguration.
- no\_hw\_resources: Tunnel endpoint programming failed in the switch hardware due to insufficient resources.
- activating: Initial state of tunnel endpoint when it is configured.

When a tunnel endpoint is directly connected via nexthop, then nexthop reachability appears empty (--).

### Example

Showing information for the VTEPs on VXLAN interface 1.

```
switch# show interface vxlan vteps
```

| Source   | Destination | Origin | Status      | VNI  | Routing  | VLAN | VRF  |
|----------|-------------|--------|-------------|------|----------|------|------|
| 11.0.0.1 | 11.0.0.2    | static | operational | 1000 | disabled | 10   | --   |
| 11.0.0.1 | 12.0.0.2    | static | activating  | 2000 | disabled | 20   | --   |
| 11.0.0.1 | 22.1.1.1    | evpn   | operational | 4000 | enabled  | --   | red  |
| 11.0.0.1 | 23.1.1.1    | evpn   | activating  | 4001 | enabled  | --   | blue |

Showing detailed information for the VTEPs on VXLAN interface 1.

```
switch# show interface vxlan vteps detail
```

Destination : 22.22.22.1  
Source : 21.21.21.1  
Origin : static  
VRF : default  
Status : operational  
Nexthops  
=====

| IP-ADDRESS | INTERFACE | NEXTHOP-MAC       |
|------------|-----------|-------------------|
| --         | 1/1/2     | 11:11:11:11:33:11 |

Destination : 33.33.33.1  
Source : 21.21.21.1  
Origin : evpn  
VRF : default  
Status : operational  
Nexthops  
=====

| IP-ADDRESS | INTERFACE | NEXTHOP-MAC       |
|------------|-----------|-------------------|
| 2.2.3.1    | 1/1/1     | 11:11:11:11:44:11 |



```

 2.2.2.1 lag1 11:11:11:11:22:11
 2.2.1.1 vlan21 11:11:11:11:11:11
 ...
 ...
switch# show interface vxlan vteps 33.33.33.1
Destination : 33.33.33.1
Source : 21.21.21.1
Origin : evpn
VRF : default
Status : operational
Nexthops
=====
 IP-ADDRESS INTERFACE NEXTHOP-MAC
 -
 2.2.3.1 1/1/1 11:11:11:11:44:11
 2.2.2.1 lag1 11:11:11:11:22:11
 2.2.1.1 vlan21 11:11:11:11:11:11

```

## shutdown

### Syntax

```
shutdown
```

```
no shutdown
```

### Description

Disables the VXLAN interface and deletes all VXLAN tunnels, segments, and members on the interface.

The `no` form of this command starts the VXLAN interface and creates all VXLAN tunnels and segments. If members are configured, they are added to the VXLAN segment.

### Command context

```
config-vxlan-if
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Disabling VXLAN interface 1:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# shutdown
```

Enabling VXLAN interface 1:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# no shutdown
```

## source ip

### Syntax

```
source ip <IPV4-ADDR>
```

## Description

Configures the source IP address for a VXLAN interface. All VXLAN encapsulated packets use this source IP address in the outer IP header.

If you change an existing source IP address, all tunnels with the old source IP address are deleted, and new tunnels are created with the new source IP address.

The `no` form of this command deletes the source IP address for the VXLAN interface and deletes all VXLAN tunnels using this source IP address.

## Command context

`config-vxlan-if`

## Parameters

**<IPV4-ADDR>**

Specifies the IP address to assign to the VXLAN interface in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. This must be an address assigned to an existing switch interface, either a loopback interface or a layer 3 interface.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Configuring the loopback IP address as the source IP address:

```
switch(config)# interface loopback 1
switch(config-loopback-if)# ip address 1.1.1.1/24
switch(config)# interface vxlan 1
switch(config-vxlan-if)# source ip 1.1.1.1
```

Configuring a layer 3 interface IP address as the source IP address:

```
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# routing (6300/6400 only)
switch(config-if)# ip address 11.10.10.1/24
switch(config)# interface vxlan 1
switch(config-vxlan-if)# source ip 10.10.10.1
```

Deleting the source IP address for VXLAN interface 1:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# no source ip 10.10.10.1
```

## system vlan-client-presence-detect

Not supported on the 6100 Switch Series.

## Syntax

`system vlan-client-presence-detect`

`no system vlan-client-presence-detect`

## Description

Enables VNI mapped VLANs when detecting the presence of a client. When enabled, VNI mapped VLANs are *up* only if there are authenticated clients on the VLAN, or if the VLAN has statically configured ports and those ports are *up*. When not enabled, VNI mapped VLANs are always *up*.

The `no` form of this command disables detection of clients on VNI mapped VLANs.

## Command context

```
config
```

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Enabling detection of clients:

```
switch(config)# system vlan-client-presence-detect
```

Disabling detection of clients:

```
switch(config)# no system vlan-client-presence-detect
```

# vlan

## Syntax

```
vlan <VLAN-ID>
```

```
no vlan <VLAN-ID>
```

## Description

Associates an existing VLAN with a VNI. Only one VLAN can be associated with a VNI and the VNI must have symmetric routing disabled. To change the VLAN associated with a VNI, execute the command `vlan` with a different VLAN ID.

The `no` form of this command removes the specified VLAN from a VNI. Traffic on the specified VLAN is no longer bridged on the VXLAN interface.

## Command context

```
config-vni-<VNI-NUMBER>
```

## Parameters

**<VLAN-ID>**

Specifies the number of an existing VLAN. Range: 2 to 4040.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Assigning VLAN 10 to VNI 1000:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)# vlan 10
```

Deleting VLAN 10 from VNI 1000:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)# no vlan 10
```

## vni

### Syntax

```
vni <VNI-NUMBER>
```

```
no vni <VNI-NUMBER>
```

### Description

Creates a VNI (Virtual Network Identifier) for the VXLAN interface and changes to the `config-vni-<VNI-NUMBER>` context. The VNI identifies a VXLAN segment, which acts as a logical network. The VNI can be associated with either a VLAN or VRF.

- When the VNI is associated with a VLAN, the VNI supports asymmetric routing.
- Enable support for asymmetric routing by executing the `routing` command. By default, the VNI is associated with the default VRF. To use another VRF, execute the `vrf` command.

The `no` form of this command deletes the specified VNI from the VXLAN interface. All VXLAN tunnels, VXLAN segments, and members associated with the VNI are deleted.

### Command context

```
config-vxlan-if
```

### Parameters

**<VNI-NUMBER>**

Specifies the number for a VNI. Range: 1 to 16777214.

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Creating VNI 1000:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)#
```

Deleting VNI 1000:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# no vni 1000
```

## vrf

### Syntax

```
vrf <VRF-NAME>
```

```
no vrf <VRF-NAME>
```

### Description

Changes the VRF associated with an L3 VNI after symmetric routing is activated using the `routing` command. The default VRF should not be configured as an EVPN-enabled VRF.

The `no` form of this command sets the VRF associated with an L3 VNI to the default VRF.

### Command context

```
config-vni-<VNI-NUMBER>
```

### Authority

Administrators or local user group members with execution rights for this command.

### Examples

Enabling L3 VNI 1000 using VRF `vrf-1`:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)# routing
switch(config-vni-1000)# vrf vrf-1
```

Setting the VRF on L3 VNI 1000 to the default VRF:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)# routing
switch(config-vni-1000)# no vrf vrf-1
```

## vtep-peer

### Syntax

```
vtep-peer <IPV4-ADDR>
```

```
no vtep-peer <IPV4-ADDR>
```

### Description

Adds a VTEP peer to a VNI. The VNI must not have routing enabled. The VTEP peer IP address must be reachable for a VXLAN tunnel to be established.

The `no` form of this command removes a VTEP peer from a VNI, which deletes the VXLAN tunnel to the peer.

### Command context

```
config-vni-<VNI-NUMBER>
```

## Parameters

**<IPV4-ADDR>**

Specifies the IP address of a VTEP peer in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

## Authority

Administrators or local user group members with execution rights for this command.

## Examples

Adding VTEP peer 10.10.10.1 to VNI 1000:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)# vlan 10
switch(config-vni-1000)# vtep-peer 10.10.10.1
```

Deleting VTEP peer 10.10.10.1 from VNI 1000:

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vni 1000
switch(config-vni-1000)# no vtep-peer 10.10.10.1
```

# vxlan-counters aggregate

*Supported on the 8325 and 8360 Switch Series only. Enabled by default in the 8400 Switch Series.*

## Syntax

vxlan-counters aggregate

no vxlan-counters aggregate

## Description

Attaches VXLAN counters to a VXLAN interface. The counters aggregate statistics for packets sent through the interface. Display statistics with the command `show interface vxlan`. Statistics are only displayed once a valid configuration is made on the interface.

## Command context

config-vxlan-if

## Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

## Example

Enabling counters for VXLAN interface 1.

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# vxlan-counters aggregate
```

Disabling counters for VXLAN interface 1.

```
switch(config)# interface vxlan 1
switch(config-vxlan-if)# no vxlan-counters aggregate
```

## erase all zeroize

### Syntax

```
erase all zeroize
```

### Description

Restores the switch to its factory default configuration. You will be prompted before the procedure starts. Once complete, the switch will restart from the primary image with factory default settings.



**NOTE:** Back up all data before running this command as all configuration settings will be lost.

### Command context

Manager (#)

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Restoring the switch to factory default configuration:

```

switch# erase all zeroize
This will securely erase all customer data and reset the switch
to factory defaults. This will initiate a reboot and render the
switch unavailable until the zeroization is complete.
This should take several minutes to one hour to complete.
Continue (y/n)? y
The system is going down for zeroization.

...

Preparing for zeroization

Storage zeroization
WARNING: DO NOT POWER OFF UNTIL
ZEROIZATION IS COMPLETE
This should take several minutes
to one hour to complete

Restoring files

...

We'd like to keep you up to date about:
* Software feature updates
* New product announcements

```

\* Special events  
Please register your products now at: <https://asp.arubanetworks.com>

switch login: admin  
Password:

Please configure the 'admin' user account password.  
Enter new password: \*\*\*\*\*  
Confirm new password: \*\*\*\*\*



## show ztp information

### Syntax

```
show ztp information
```

### Description

Shows information about Zero Touch Provisioning (ZTP) operations performed on the switch.

### Command context

Operator (>) or Manager (#)

### Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

### Usage

When a switch configured to use ZTP is booted from a factory default configuration, the switch contacts a DHCP server, which offers options for obtaining files used to provision the switch:

- The IP address of the TFTP server
- The name of the image file
- The name of the configuration file

The `show ztp information` command shows the options offered by the DHCP server and the status of the ZTP operation.

The status of the ZTP operation is one of the following:

### Success

The ZTP operation succeeded.

One of the following is true:

- Both the running configuration and the startup configuration were updated.
- The IP address of the TFTP server was received, but the offer did not include a configuration file or a firmware image file.

### Aborted - Custom startup configuration detected

The switch was booted from a configuration that is not the factory default configuration. For example, the administrator password has been set.

### **Aborted - Timed out while waiting to receive ZTP options**

The switch did not receive information from the DHCP server and it has been 10 minutes or longer since the switch was booted.

### **Aborted - Detected change in running configuration**

The running configuration was modified by a user while the ZTP operation was in progress.

### **Failed - TFTP server unreachable**

The TFTP server is not reachable at the specified IP address.

### **TFTP server information unavailable**

The switch did not receive the IP address of the TFTP server in the DHCP offer.

### **Failed - Invalid configuration file received**

Either the file transfer of the configuration file failed, or the configuration file is invalid (an error occurred while attempting to apply the configuration).

### **Failed - Invalid image file received**

Either the file transfer of the firmware image file failed, or the firmware image file is invalid (an error occurred while verifying the image).

## **Examples**

In the following example, the ZTP operation succeeded, and both an image file and a configuration file were provided.

```
switch# show ztp information
TFTP Server : 10.0.0.2
Image File : switchfile.swi
Configuration File : ztp.cfg
Status : Success
```

In the following example, the ZTP option succeeded. A configuration file was not provided, but an image file was provided.

```
switch# show ztp information
TFTP Server : 10.0.0.2
Image File : switchfile.swi
Configuration File : NA
ZTP Status : Success
```

In the following example, the ZTP operation failed because the TFTP server was unreachable.

```
switch# show ztp information
TFTP Server : 10.0.0.2
Image File : switchfile.swi
Configuration File : ztp.cfg
Status : Failed - TFTP server unreachable
```

In the following example, the ZTP operation was stopped because the DHCP server did not provide options within 10 minutes of initial switch boot.

```
switch# show ztp information
TFTP Server : NA
Image File : NA
Configuration File : NA
Status : Aborted - Timed out while waiting to receive ZTP options
```

In the following example, the ZTP operation was stopped because the switch was booted from a configuration that was not the factory default configuration.

```
switch# show ztp information
TFTP Server : 10.0.0.2
Image File : switchfile.swi
Configuration File : ztp.cfg
Status : Aborted - Custom startup configuration detected
```

## Accessing Aruba Support

|                         |                                                                                                                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aruba Support Services  | <a href="https://www.arubanetworks.com/support-services/">https://www.arubanetworks.com/support-services/</a>                                                                        |
| Aruba Support Portal    | <a href="https://asp.arubanetworks.com/">https://asp.arubanetworks.com/</a>                                                                                                          |
| North America telephone | 1-800-943-4526 (US & Canada Toll-Free Number)<br>+1-408-754-1200 (Primary - Toll Number)<br>+1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working) |
| International telephone | <a href="https://www.arubanetworks.com/support-services/contact-support/">https://www.arubanetworks.com/support-services/contact-support/</a>                                        |

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

### Other useful sites

Other websites that can be used to find information:

|                                           |                                                                                                                                       |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Airheads social forums and Knowledge Base | <a href="https://community.arubanetworks.com/">https://community.arubanetworks.com/</a>                                               |
| Software licensing                        | <a href="https://lms.arubanetworks.com/">https://lms.arubanetworks.com/</a>                                                           |
| End-of-Life information                   | <a href="https://www.arubanetworks.com/support-services/end-of-life/">https://www.arubanetworks.com/support-services/end-of-life/</a> |
| Aruba software and documentation          | <a href="https://asp.arubanetworks.com/downloads">https://asp.arubanetworks.com/downloads</a>                                         |

## Accessing updates

To download product updates:

## Aruba Support Portal

<https://asp.arubanetworks.com/downloads>

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

### My Networking

<https://www.hpe.com/networking/support>

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://support.hpe.com/portal/site/hpsc/aae/home/>



**IMPORTANT:** Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

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Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://asp.arubanetworks.com/notifications/subscriptions> (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

## Warranty information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

## Regulatory information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

### Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

## Documentation feedback

Aruba is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback ([docsfeedback-switching@hpe.com](mailto:docsfeedback-switching@hpe.com)). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.